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Mohieddine Jelali and Andreas Kroll

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# Hydraulic Servo-systems

**Modelling, Identification and Control**

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With 182 Figures



Springer

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*To Doris, Yasmin and Dunja*  
**M.J.**

*To Iris*  
**A.K.**

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## SERIES EDITORS' FOREWORD

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The series *Advances in Industrial Control* aims to report and encourage technology transfer in control engineering. The rapid development of control technology has an impact on all areas of the control discipline. New theory, new controllers, actuators, sensors, new industrial processes, computer methods, new applications, new philosophies..., new challenges. Much of this development work resides in industrial reports, feasibility study papers and the reports of advanced collaborative projects. The series offers an opportunity for researchers to present an extended exposition of such new work in all aspects of industrial control for wider and rapid dissemination.

Control engineering is a discipline of many different areas. Many standard control textbooks extract the core control engineering skills and sometimes leave behind real-world applications. In the *Advances in Industrial Control* monographs we often see control engineering from the engineer's perspective. This monograph by Mohieddine Jelali and Andreas Kroll is a perfect example of an industrial view of control engineering; this time for the field of hydraulic servo-systems.

Hydraulic servo-systems are essential actuators in many industrial sectors; the steel industry where hydraulic servo-systems actuate hot and cold rolling mills used to roll strip is a highly appropriate example. Jelali and Kroll manage in this monograph to convey the real intellectual and engineering challenge of designing controllers for such systems. The sheer art of whittling down extensive process information and data to provide a model which is just right for control studies is well described by the authors. Then there is the control engineering design problem: hydraulic servo-systems use compressible fluids leading to complex flow patterns and suffer from friction and stiction effects. Computer technology provides an opportunity to do better; to provide more advanced and better controller design. In this monograph, this is rounded out by the experience of the authors. What we use is what works well. We are pleased to welcome this authoritative contribution to the *Advances in Industrial Control* monograph series and we hope both industrial engineers and control engineering academics will find much of interest in the volume.

M.J. Grimble and M.A. Johnson  
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## PREFACE

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The main purpose of this book is to present the theoretical background, practical techniques and experimental results of the modelling, identification and control of hydraulic servo-systems. Hydraulic servo-systems are used in a wide variety of industrial fields. They provide many advantages over electrical motors, including high durability and the ability to produce large forces at high speeds. Unfortunately, the dynamic characteristics of these systems are highly non-linear and relatively difficult to control. The non-linearities arise from the compressibility of the hydraulic fluid, the complex flow properties of the servo-valve and the friction in the hydraulic cylinder. Many current industrial controllers achieve moderate bandwidth with fixed gain controllers by oversizing the cylinder diameter. This requires larger and more costly components and higher fluid flow rates in order to move a load at a given speed. The overall system is therefore much more expensive.

In writing this book an attempt is made to outline classical and advanced algorithms and the results of automatic control of hydraulic drives with the intention of providing the designer/engineer with all the approaches available. He/she will then be in the best possible position to develop a satisfactory design for the application at hand with minimal effort. It is also demonstrated that advanced strategies can provide significant improvements over conventional methods, of course at the expense of greater design and computational efforts. Nevertheless, the continual progress of computer technologies has greatly reduced the effort required to implement such advanced control strategies. The derivation and identification of suitable models is an important prerequisite for the design of advanced controllers and is thus treated in detail.

In order to make the book as self-contained as possible, yet readable for engineers, the necessary theoretical background of the methods presented is described – without rigorous mathematical theorems or proofs. Instead, the reader is referred to many well-written standard papers and books for proofs and detailed treatment of the theory. The emphasis is on procedures that in the experience of the authors proved to work well in practical implementations. Guidelines for the model derivation, controller commissioning and tuning are also provided.

The book is intended as a graduate text, as well as a reference and a practitioner's handbook to scientists and engineers involved in the analysis and design of hydraulic control systems. Inexhaustible sources of important foundations for hydraulic control systems are the excellent textbooks of Merritt (1967), Viersma (1980), Anderson (1988), and Walters (1991). However, all these books mainly cover the design of hydraulic systems in the sense of specifying the proper size of hydraulic elements, and are not directed toward the derivation, simulation and practical implementation of control laws (combined with the identification step) for

hydraulic servo-systems, not to mention the advanced control techniques developed in the last decade.

It is essential to have access to suitable software tools in order to simulate the models, to solve the problems related to the identification and control strategies presented in this book, and to gain experience of how the concepts actually work. This assumes that the reader has access to MATLAB/Simulink (or other similar packages), together with some optimisation, identification and control toolboxes. At the end of each chapter, implementation hints and links to available software packages (either commercially from TheMathworks or from the public domain) are given to allow the reader to simulate any of the concepts described in the book as fast as possible.

It is only fair to acknowledge the individual contributions of the authors. All fuzzy logic related sections of the book (Sections 5.4.2, 5.8, 6.7, 7.1.3 and 7.1.4) have been written by Andreas Kroll. He also contributed to the identification basics in Sections 5.1 and 5.2. The material for vibration damping control (Sections 6.9, 7.4 and 7.5) and for experimental results of modelling and control of differential cylinders (Sections 4.4.2, 7.2 and 7.3) has been provided by Werner Bernzen and Dirk Nissing, who were originally planned as co-authors of this book. This material has been edited and reworked by the principle author, who also has written the remaining sections of the book.

As “nobody is perfect”, errors may remain in the book. The authors would be grateful for reports of errors to be sent to one of the following email addresses: [mohieddine.jelali@web.de](mailto:mohieddine.jelali@web.de) or [andreas.a.kroll@de.abb.com](mailto:andreas.a.kroll@de.abb.com).

September 2002

*Mohieddine Jelali  
Andreas Kroll*

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As with all authors of technical works, we are indebted to the people who taught us the fundamentals of the subject, to the people who made the writing possible, and to the people who helped us to improve the quality of the textbook. We wish to acknowledge the vast array of contributors on whose work our presentation is based. The list of references gives some small measure of those to whom we are in debt. On a more personal level, we wish to express our appreciation to the people mentioned below.

First of all, we both wish to express our sincere gratitude to all the staff at Springer-Verlag London for their pleasant co-operation and for the quality production of the book. Special thanks go to Oliver Jackson for being very patient with us and for providing invaluable help and encouragement.

We are also very indebted to Samuel Ellis for his linguistic advice and for significantly improving the textual quality.

Kind permissions from many people and organisations (particularly VDI, IEEE and Bosch Rexroth), to republish material are appreciated. The original source and the copyright line appear prominently with each reprinted figure and table.

**M. J.  
A. K.**

My interest in modelling, identification and control of hydraulic servo-systems goes back to my years as a research assistant at the Department of Measurement and Control, Faculty of Mechanical Engineering of the University of Duisburg (Germany). Professor Helmut Schwarz introduced me to this subject, and his serious attitude to research has always been an example to me with which I judge my own efforts. Since then I have worked with many other people who augmented my knowledge of the subject. Part of the material included in the book is the result of research work funded by the German Research Foundation (DFG). I gratefully acknowledge this institution for its support.

Several colleagues and friends were kind enough to read and criticise parts of the manuscript. I am especially grateful to Werner Bernzen, Markus Bröcker, Dirk Nissing, Jan Polzer, and Andreas Wolff.

I am also deeply indebted to Professor Ralf Lemmen and Professor Heinrich Ratjen for reading parts of the manuscript and making corrections and suggestions to improve the book.

I also express my sincere gratitude to Werner Bernzen and Dirk Nissing for providing the material for the vibration damping control and for the modelling and control experimental results for differential cylinders.

I would like to thank Carston Otto for valuable hints concerning the neural-networks-based identification of hydraulic servo-systems.

Thanks also go to my co-author, Andreas, for his enthusiasm and encouragement during the writing of the book. Many of his corrections and suggestions helped to improve my own contribution.

Other parts of my presentation are based on material written by members of the Department of Measurement and Control, Faculty of Mechanical Engineering, University of Duisburg. May they forgive me for not always acknowledging it.

Writing a book takes a lot of time, and it was unavoidable that the writing (not to mention the thinking!) of the book had to be done on overtime at weekends and long nights. My heartfelt thanks go to my wife Doris and my daughters Yasmin and Dunja for letting me use their time. Without their never-ending support, patience and love I probably would never have been able to finish the book.

**M. J.**

Modelling and identification attracted my interest during my time as a postgraduate at the Department of Measurement and Control, Faculty of Mechanical Engineering of the University of Duisburg (Germany). The well-equipped department laboratory made fluid drives an interesting system for applying and validating new methods. Thanks are due to Professor Helmut Schwarz for providing the working environment and supervising this work. I am also grateful to Thyssen EnCoke (formerly Thyssen Still Otto Anlagentechnik GmbH) and the Federal Ministry of Education and Research (BMBF) as the major sponsors of this work.

The control methods presented in this book were developed after I left the University of Duisburg. A productive period with my former doctorate colleague Thomas Bernd helped to improve the fuzzy model identification methods further. Moreover, all the control concepts presented were jointly developed. Thomas also reviewed the fuzzy logic related chapters of this book. For all this, I am very grateful.

Many thanks are due to the principle author, Mohieddine, for driving this project and for many fruitful discussions.

Several students contributed with implementing, testing and validating methods in various case studies. I owe them many thanks.

Contributing to this book was a “private” project, carried out in long evenings, during weekends and holidays. Therefore, I am exceptionally grateful to my partner Iris for her patience and sacrificing our joint time in order to give me the freedom to contribute to this book.

**A. K.**



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# NOTATION

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## Arabic Notation and Symbols

|                       |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| $a$                   | acceleration                                                                                                  |
| $\text{ad}_f$         | Lie bracket                                                                                                   |
| $A, A_p$              | area, piston area                                                                                             |
| $\mathbf{A}$          | state-space system matrix                                                                                     |
| $\mathbf{b}$          | state-space input matrix                                                                                      |
| $c$                   | 1. sonic velocity in oil<br>2. spring rate/constant (stiffness)<br>3. number of fuzzy rules of a fuzzy system |
| $\mathbf{c}$          | state-space output matrix                                                                                     |
| $c_s$                 | parameter for static friction                                                                                 |
| $c_v$                 | valve flow coefficient                                                                                        |
| $C(s)$                | characteristic polynomial                                                                                     |
| $C_h$                 | hydraulic capacitance                                                                                         |
| $C_L, C_{Li}, C_{Le}$ | leakage flow coefficient (internal, external)                                                                 |
| $d$                   | 1. diameter<br>2. damper constant                                                                             |
| $d_i, d_o$            | inner diameter, outer diameter                                                                                |
| $d_h, d_v$            | hydraulic diameter, valve spool diameter                                                                      |
| $\mathbf{D}$          | form matrix of inner product norm                                                                             |
| $D_h$                 | hydraulic (dimensionless) damping ratio                                                                       |
| $e$                   | control error/deviation                                                                                       |
| $\mathbf{e}$          | unity vector                                                                                                  |
| $E(s)$                | Laplace-transformed control error/deviation                                                                   |
| $E, E'$               | bulk modulus, effective bulk modulus                                                                          |
| $f$                   | frequency [Hz]                                                                                                |
| $\mathbf{f}$          | search direction                                                                                              |
| $F, F_{ax}, F_{ext}$  | force/fluid force, axial flow force, external force                                                           |
| $F_{aux}, F_\Sigma$   | auxiliary forces                                                                                              |
| $F_t, F_{ref}$        | total force, reference force                                                                                  |
| $F_c, F_s$            | Coulomb friction, static friction                                                                             |
| $F_t, F_L$            | friction force, load force                                                                                    |
| $F_0$                 | spring pre-load reference force                                                                               |
| $G$                   | transfer function                                                                                             |
| $\mathbf{G}$          | gradient vector                                                                                               |
| $H$                   | Hessian matrix                                                                                                |

|                  |                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------|
| $I$              | current                                                                                                             |
| $\mathbf{I}$     | identity matrix                                                                                                     |
| $j$              | imaginary number; $j = \sqrt{-1}$                                                                                   |
| $J$              | moment of inertia                                                                                                   |
| $k$              | discrete time index                                                                                                 |
| $\mathbf{k}$     | state-space controller gain vector                                                                                  |
| $K$              | controller gain                                                                                                     |
| $K_p, K_v$       | position feedback gain, velocity feedback gain                                                                      |
| $K_{v,ff}$       | velocity feedforward gain                                                                                           |
| $K_a$            | acceleration feedback gain                                                                                          |
| $K_{msp}$        | main spool position sensor gain                                                                                     |
| $K_{pm}$         | spool position feedback gain                                                                                        |
| $K_{pL}$         | pressure-difference-force velocity feedback gain                                                                    |
| $K_{Qx}$         | valve flow gain                                                                                                     |
| $K_{Qp}$         | valve flow–pressure coefficient                                                                                     |
| $K_{px}$         | pressure sensitivity                                                                                                |
| $K_{xv}$         | valve gain                                                                                                          |
| $K_{xp}$         | velocity gain                                                                                                       |
| $K_{pL}$         | hydraulic cylinder gain                                                                                             |
| $K_{pL}$         | pressure-difference feedback gain                                                                                   |
| $K_{pL,ff}$      | load pressure feedforward gain                                                                                      |
| $K_{\dot{x},ff}$ | velocity feedforward gain                                                                                           |
| $l$              | 1. degree of (polynomial) non-linearity<br>2. length of single pipeline                                             |
| $L$              | 1. approximation of loss/cost function<br>2. inductance                                                             |
| $L_f$            | Lie derivative                                                                                                      |
| $M$              | number of model parameters (to be identified), $M = \dim(\boldsymbol{\theta})$                                      |
| $m$              | 1. mass<br>2. number of control inputs, $m = \dim(\mathbf{u})$                                                      |
| $m_p, m_t$       | piston mass, total mass                                                                                             |
| $n$              | 1. number of system states (system order), $n = \dim(\mathbf{x})$<br>2. number of inputs, which are delayed outputs |
| $n_d$            | dissipation number                                                                                                  |
| $n_{pu}$         | pump drive speed                                                                                                    |
| $N$              | number of data sets (measurements)                                                                                  |
| $p$              | 1. pressure<br>2. number of system outputs, $p = \dim(\mathbf{y})$<br>3. parameter of $L_p$ norm                    |
| $p_s, p_T, p_L$  | supply pressure, tank (return) pressure, load pressure                                                              |
| $\mathbf{p}$     | vector of parameter estimates of the orthogonalised model                                                           |
| $\mathbf{q}$     | vector field of zero dynamics                                                                                       |
| $Q, Q_L$         | (volume) flow, load flow                                                                                            |

|                                          |                                                                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $Q_{Li}, Q_{Le}$                         | internal leakage flow, external leakage flow                                                                                          |
| $r$                                      | 1. ratio<br>2. radius of a single pipeline<br>3. relative degree<br>4. reference trajectory                                           |
| $R$                                      | 1. outer radius,<br>2. correlation coefficient<br>3. resistance                                                                       |
| $Re, Re_{crit}$                          | Reynolds number, transition (or critical) Reynolds number                                                                             |
| $R_h$                                    | hydraulic resistance parameter                                                                                                        |
| $R$                                      | 1. covariance matrix<br>2. Gauss-Newton Hessian                                                                                       |
| $s$                                      | Laplace variable                                                                                                                      |
| $s_E$                                    | entropy                                                                                                                               |
| $\bar{s}$                                | transformed Laplace variable (see Equation 4.175)                                                                                     |
| $S$                                      | total stroke of actuator (cylinder)                                                                                                   |
| $t$                                      | (continuous) time                                                                                                                     |
| $T$                                      | 1. time constant<br>2. torque                                                                                                         |
| $T_0$                                    | sampling time/period                                                                                                                  |
| $u, \mathbf{u}$                          | input variable, valve input voltage, vector of input variables                                                                        |
| $\mathbf{u}_s$                           | vector of desired steady-state values of control signals                                                                              |
| $U(s)$                                   | Laplace-transformed input variable                                                                                                    |
| $V$                                      | 1. volume<br>2. loss/cost function                                                                                                    |
| $V_{pu}$                                 | pump geometrical volume (displacement)                                                                                                |
| $v$                                      | 1. velocity<br>2. external (new) input                                                                                                |
| $\mathbf{w}$                             | vector of input and output quantities (measures) in the orthogonalised model                                                          |
| $W$                                      | coefficient (weight) matrix                                                                                                           |
| $\mathbf{x}$                             | state vector                                                                                                                          |
| $x, \dot{x}, \ddot{x}$                   | position, velocity, acceleration                                                                                                      |
| $x_c, \dot{x}_c, \ddot{x}_c$             | cylinder position, velocity, acceleration                                                                                             |
| $x_p, \dot{x}_p, \ddot{x}_p$             | piston position, velocity, acceleration                                                                                               |
| $x_{ref}, \dot{x}_{ref}, \ddot{x}_{ref}$ | desired trajectory for actuator position, velocity, and acceleration                                                                  |
| $x_v, \dot{x}_v, \ddot{x}_v$             | spool valve position, velocity, acceleration                                                                                          |
| $x_{v,max}$                              | maximum spool displacement                                                                                                            |
| $y, \mathbf{y}$                          | output variable, vector of output variables                                                                                           |
| $y_{ref}$                                | reference signal (desired output)                                                                                                     |
| $Y(s)$                                   | Laplace-transformed output variable                                                                                                   |
| $\mathbf{Y}$                             | vector of recent output data (measurements)                                                                                           |
| $\mathbf{z}$                             | vector of output variables contaminated by noise $\mathbf{v}$<br>vector of transformed state variables for input–output linearisation |



|       |                          |
|-------|--------------------------|
| $Z_0$ | impedance constant       |
| $Z_c$ | characteristic impedance |

### Greek Notation and Symbols

|                                          |                                                                                                                   |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| $\alpha$                                 | 1. cubical expansion coefficient/viscosity–pressure coefficient<br>2. ratio of ring side area to piston side area |
| $\alpha_d$                               | discharge coefficient                                                                                             |
| $\alpha_i$                               | coefficients of characteristic polynomial                                                                         |
| $\beta$                                  | gain of torque motor                                                                                              |
| $\beta_i$                                | fuzzy basis function                                                                                              |
| $\beta_\theta$                           | heat expansion factor                                                                                             |
| $\delta$                                 | trust region radius                                                                                               |
| $\delta_{k,l}$                           | Kronecker delta (= 1 if $k = l$ , else = 0)                                                                       |
| $\delta(t)$                              | Dirac function                                                                                                    |
| $\epsilon$                               | residual/equation error/modelling error                                                                           |
| $\psi$                                   | basis function (of radial basis function network)                                                                 |
| $\Phi$                                   | matrix (stacked vectors) of input/output quantities (measures)                                                    |
| $\gamma$                                 | coefficients of state variable filter                                                                             |
| $\Gamma$                                 | propagation operator                                                                                              |
| $\eta$                                   | dynamic viscosity                                                                                                 |
| $\eta_{vol}$                             | (pump) volumetric efficiency                                                                                      |
| $\eta$                                   | state vector of zero dynamics                                                                                     |
| $\varphi, \dot{\varphi}, \ddot{\varphi}$ | rotation angle, (angular) speed, (angular) acceleration                                                           |
| $\varphi$                                | vector of input and output quantities (measures), regressors                                                      |
| $\kappa$                                 | isentropic exponent                                                                                               |
| $\kappa_p$                               | compressibility factor                                                                                            |
| $\lambda$                                | 1. eigenvalue<br>2. forgetting factor<br>3. parameter step size                                                   |
| $\nu$                                    | 1. kinematic viscosity/Poisson's number<br>2. fuzziness number                                                    |
| $\nu$                                    | reference point (vector)                                                                                          |
| $\nu(k), \nu(t_k)$                       | (white) noise process                                                                                             |
| $\mu$                                    | membership function                                                                                               |
| $\xi$                                    | constant term in conclusion of fuzzy model                                                                        |
| $\theta$                                 | temperature                                                                                                       |
| $\Theta$                                 | vector of model parameters (to be identified)                                                                     |
| $\rho$                                   | mass density                                                                                                      |
| $\sigma$                                 | 1. (co)variance<br>2. parameter for viscous friction                                                              |
| $\alpha(x)$                              | sigmoid function                                                                                                  |
| $\tau$                                   | 1. shear stress<br>2. discrete time delay (dead-time)                                                             |

|            |                                                               |
|------------|---------------------------------------------------------------|
| $\chi$     | feature vector                                                |
| $\omega$   | frequency ( $\omega = 2\pi f$ ) [ $s^{-1}$ or rad/s]          |
| $\omega_c$ | cut-off frequency of reference generator for actuator control |
| $\omega_h$ | hydraulic natural frequency                                   |
| $\zeta_i$  | parameters in linearised servo-system model                   |

### Abbreviations and Acronyms

|        |                                                    |
|--------|----------------------------------------------------|
| A/D    | analog to digital converter                        |
| AIC    | Akaike's information criterion                     |
| ANN    | artificial neural network                          |
| ARMAX  | autoregressive moving average with external input  |
| ARX    | autoregressive with external input                 |
| BIC    | Bayesian information criterion                     |
| BJ     | Box–Jenkins                                        |
| BP     | back-propagation                                   |
| BWF    | Butterworth filter                                 |
| CACSD  | computer-aided control system design               |
| D/A    | digital to analog converter                        |
| ERR    | error reduction rate                               |
| FCM    | fuzzy- <i>c</i> -means                             |
| FFC    | feedforward control                                |
| FIR    | finite impulse response                            |
| FNARX  | fuzzy network ARX                                  |
| FNOE   | fuzzy network OE                                   |
| FPE    | final prediction error                             |
| HM     | hydraulic motor                                    |
| HP     | hydraulic pump                                     |
| HSS    | hydraulic servo-system                             |
| LHP    | left half plane                                    |
| LILC   | Khinchin's law of the iterated logarithm criterion |
| LMI    | linear matrix inequality                           |
| MISO   | multiple-input single-output                       |
| MIMO   | multiple-input multiple-output                     |
| MLP    | multi-layer perceptron                             |
| MPC    | model predictive control(ler)                      |
| NARMAX | non-linear ARMAX                                   |
| NARX   | non-linear ARX                                     |
| NBJ    | non-linear BJ                                      |
| NFIR   | non-linear FIR                                     |
| NLS    | non-linear system                                  |
| NNARX  | neural network ARX                                 |
| NNOE   | neural network OE                                  |
| NOE    | non-linear OE                                      |
| NRSS   | normalised residual sum of squares                 |
| ODE    | ordinary differential equation                     |

|      |                                  |
|------|----------------------------------|
| OE   | output error                     |
| OLS  | orthogonal least squares         |
| PDE  | partial differential equation    |
| PE   | prediction error                 |
| PID  | proportional integral derivative |
| RBFN | radial basis function network    |
| RHP  | right half plane                 |
| RV   | relief valve                     |
| SISO | single-input single-output       |
| SIMO | single-input multiple-output     |
| SSIF | state-space innovations form     |

### Accents

|            |                                     |
|------------|-------------------------------------|
| $\dot{x}$  | time derivative of $x$ (velocity)   |
| $\ddot{x}$ | acceleration                        |
| $\hat{x}$  | estimated or simulated value of $x$ |
| $x^*$      | scaled (normalised) value of $x$    |
| $x_f$      | filtered value of $x$               |

### Indices

|      |                                                                               |
|------|-------------------------------------------------------------------------------|
| a    | armature                                                                      |
| ax   | axial                                                                         |
| A    | piston side of actuator                                                       |
| Al   | all-pass                                                                      |
| by   | bypass                                                                        |
| B    | ring side of actuator                                                         |
| c    | 1. cylinder<br>2. Coulomb<br>3. chamber<br>4. characteristic<br>5. controller |
| ca   | current amplifier                                                             |
| comp | compensated                                                                   |
| d    | discharge                                                                     |
| dyn  | dynamic                                                                       |
| el   | electrical                                                                    |
| en   | entry                                                                         |
| ex   | exit                                                                          |
| ext  | external                                                                      |
| f    | 1. friction<br>2. flapper                                                     |
| fb   | feedback                                                                      |
| ff   | feedforward                                                                   |
| fl   | flow                                                                          |

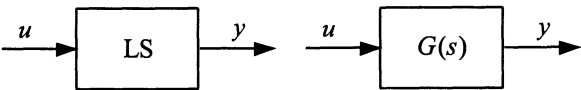
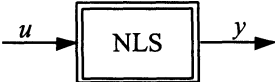
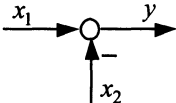
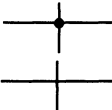
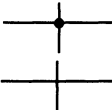
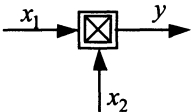
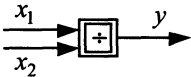
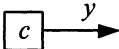
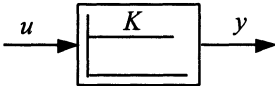
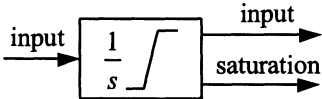
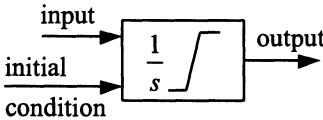
|       |                    |
|-------|--------------------|
| F     | force              |
| g     | (air-)gap          |
| h     | hydraulic          |
| ind   | induced            |
| L     | 1. load            |
|       | 2. leakage         |
|       | 3. laminar         |
| Li    | internal leakage   |
| Le    | (external) leakage |
| m     | 1. mechanical      |
|       | 2. main            |
| mo    | main orifice       |
| msp   | main spool         |
| mag   | magnetic           |
| max   | maximum            |
| min   | minimum            |
| n     | nozzle             |
| N     | nominal (rated)    |
| o     | orifice            |
| p     | piston             |
| paras | parasitic          |
| pi    | pilot              |
| pl    | pipeline           |
| pu    | pump               |
| re    | restrictor         |
| ref   | reference (target) |
| s     | 1. static/stiction |
|       | 2. spring          |
| sp    | spool              |
| ss    | spool side         |
| S     | system             |
| t     | 1. total           |
|       | 2. turbulent       |
| tm    | torque motor       |
| th    | theoretical        |
| T     | tank               |
| v     | 1. valve           |
|       | 2. viscous         |
|       | 3. velocity        |
| vol   | volumetric         |

## Operators and Standard Functions

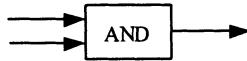
|                      |                            |
|----------------------|----------------------------|
| $\ \cdot\ $          | norm (of matrix or vector) |
| $\overline{(\cdot)}$ | mean value                 |
| $\forall$            | for all                    |

|                    |                                       |
|--------------------|---------------------------------------|
| $\in$              | element of                            |
| $\#$               | cardinal number                       |
| $\equiv$           | defined as                            |
| $:=$               | assignment operator                   |
| $\approx$          | roughly the same                      |
| $\ll$              | much lesser than                      |
| $\otimes$          | Kronecker product                     |
| $\text{ad}_f^i$    | Lie bracket                           |
| $\arg \min_x f(x)$ | minimising argument of $f(x)$         |
| $\dim(x)$          | dimension of vector $x$               |
| $\exp$             | exponential function                  |
| $E\{x\}$           | expected value of random variable $x$ |
| $\ln$              | natural logarithm                     |
| $\log$             | logarithm of basis 10                 |
| $L_f'$             | Lie derivative                        |

Elements of Block Diagrams

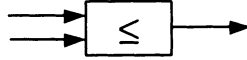
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|-------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------|
| Linear system (block)                                       |   |                       |
| Non-linear system (block)                                   |    |                       |
| Sum                                                         |    | $y = x_1 - x_2$       |
| Signal connection                                           |    |                       |
| Cross signals<br>(not connected)                            |    |                       |
| Product (multiplication)                                    |    | $y = x_1 x_2$         |
| Quotient (division)                                         |    | $y = \frac{x_1}{x_2}$ |
| Constant value                                              |   | $y = c$               |
| Proportional block (gain)                                   |  | $y = Ku$              |
| Integrator                                                  |  | $y = \int u(t) \, dt$ |
| Integrator with<br>saturation port                          |  |                       |
| Integrator with<br>saturation port<br>and initial condition |  |                       |

Logical operator

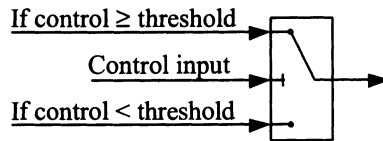


Set output to 1, if the  
operation is TRUE

Relational operator



Switch between  
two inputs



---

## CHAPTER 1

# INTRODUCTION

---

Although electrical drives have become increasingly popular for high-performance motion control, hydraulic servo-systems still find a wide variety of applications in present-day industrial motion systems, for instance in machining plants, robotics, motion simulators, fatigue testing systems, metal processing plants, mining, special purpose machines and so on.

Owing to increased computer power and ongoing developments in control theory, expectations regarding modelling of the non-linear dynamic behaviour of hydraulic servo-systems have increased. More detailed descriptions of dominant non-linear characteristics and relevant dynamics over wider frequency ranges have to be taken into account. The main non-linearities in hydraulic systems arise from the compressibility of the hydraulic fluid, the complex flow properties of the servo-valve and the friction in the hydraulic actuators. They depend on factors, which are difficult to measure or estimate online, such as oil bulk modulus, viscosity and temperature. Therefore, conventional feedback control, which is easily tuned by hand, is only sufficient for some cases. Advanced (model-based) control methods are necessary if high-performance motion control is required.

### 1.1 Historical View and Motivation for Hydraulic Systems

Brief but very interesting histories of hydraulic mechanisms have been compiled by Maskrey and Thayer (1978) and Backé and Murrenhoff (1994). In the following, we summarise the latter work and refer the reader to both references for more complete treatment.

Fluid power technology can be tracked back to the invention of the water clock by the Alexandrian inventor Ktesbios in about 250 B.C., but it was the work of Pascal, who introduced the principle of the hydraulic press in 1663, with which hydraulics gained importance. The first industrial application of the hydraulic press (using water as the working fluid) was performed later on by Josef Bramah (1749–1814), in 1795, in London.

Following the invention of the steam engine by James Watt (1736–1819), hydrostatic machines were widely used in England for energy exchange in the form of pressurised water networks. Pumps driven by steam engines were employed to generate pressurised water, which was in turn used to actuate pistons driving the processing machines (*e.g.*, mills).

In the second half of the 19th century, W.G. Armstrong (1810–1900) developed many hydrostatic machines and devices, primarily for use in shipbuilding (*e.g.*,



capstans, lifting jacks). Some of these control devices still resemble those used today.

Fluid power technology has then developed mainly from the beginning of the 20th century, where the first generation of hydraulic drives consisted of some flow control device driving the hydraulic actuator in an open loop manner. Common applications were therefore limited to hydraulic presses, jacks and winches.

The main advantage of fluid power, which led to its popularity, is the good ratio between force or torque delivered on the one hand and the actuator weight and size on the other hand. In many applications this allows the so-called direct drive construction, so that wear-sensitive gear-boxes can be avoided. Examples can be found in various types of hydraulic drive and transmission in mobile systems.

Whereas hydraulic drives were initially used for open loop actuation, the years following World War II saw an increase in the use of servo-control techniques, which allow accurate closed loop motion control. The development of hydraulic servo-techniques was pushed further by the contributions of Blackburn *et al.* (1960).

This opened a wide range of applications still found in industry today. Hydraulic servo-systems are used

- where relatively large forces or torques are required (industrial presses, mobile lifting, digging, material handling equipment, *etc.*),
- where fast, stiff response of resisting loads is needed (machine tool drives, flight simulators, rolling mills, *etc.*),
- where accurate control of response is necessary (control surfaces of aircraft, machine tool slides, industrial robots, *etc.*),
- where manual control of motion involving substantial forces is essential (heavy machinery, aircraft controls, automotive power steering, *etc.*), and
- as the final actuator subsystem in complex automatically controlled situations (electro-hydraulic flight simulators, industrial robots, fatigue and other programmable test rigs, theatre stage control, *etc.*).

Hydraulic control systems provide many advantages over other types of control (*e.g.*, electrical motors), some of which are (Merritt, 1967; Bauer, 1998; Will *et al.*, 1999):

- Hydraulic systems can produce larger forces/torques, and have higher load stiffness.
- The hydraulic fluid acts as a superb lubricant and avoids wear.
- Hydraulic actuators have higher speed of response with fast starts, stops, and speed reversals.
- Hydraulic systems may be operated under different conditions (continuous, intermittent, reversing, stalled).
- Both linear and rotary actuators are available and add to the great flexibility of hydraulic power elements.
- Overloading protection is easy.
- Smaller and lighter compact systems with long component life are available.

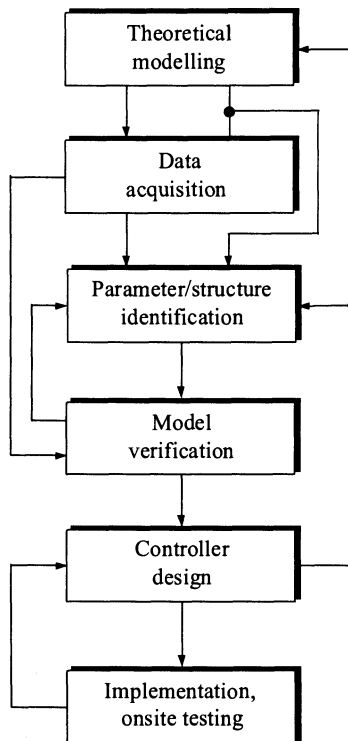
Besides the advantages of fluid power, there are obviously some disadvantages, some of which are (Merritt, 1967; Bauer, 1998; Will *et al.*, 1999):

- High costs of hydraulic components that result from small allowable tolerances.

- Fire and explosion hazards exist if a hydraulic system is used near a source of ignition.
- Dirty and contaminated fluids cannot be fully avoided. This may lead to clogging of valves and actuators, as well as permanent loss in performance and/or failure. Contaminated oil is actually the chief source of hydraulic control failures. Clean oil and reliability are synonymous terms in hydraulic control.
- The dynamic characteristics of hydraulic systems are highly non-linear and relatively difficult to control.
- Hydraulic sound or *noise*: the topic of noise is particularly critical today, as higher power density coupled with high operating pressures and lighter components result in increased noise level, but at the same time specifications and standards dictate the need for noise protection (to avoid health hazards) in many countries and applications.

## 1.2 Aims and Focus of the Book

This book focuses on the modelling, identification and control of hydraulic servo-systems. The modelling, control design, and implementation procedure is illustrated by Figure 1.1.



**Figure 1.1.** Flow diagram of the (grey-box) modelling and control design procedure

In order to obtain structural insight into the behaviour of hydraulic servo-systems, with respect to relevant dynamics as well as relevant non-linearities, the procedure usually starts with the theoretical, physically based modelling of the complete servo-system. This means that a mathematical model is constructed from basic physical laws (such as Newton's laws and equilibrium equations) using and combining available contributions on theoretical modelling of hydraulic systems found in relevant literature. The result is usually a non-linear dynamic (simulation) model of the hydraulic servo-system, including actuator, servo-valve, pipelines, and power supply. This model is used to perform various simulations, with realistic physical parameters for the given application, so that structural insight into the relevant dynamics and non-linearities of the system can be gained.

The insight obtained can be used to simplify and reduce the model where possible, and to neglect irrelevant non-linearities, as the model should not be too complex if it is to be practically useful. In this phase, linearisation of the model plays an important role, as it provides much insight in the dynamic characteristics of the system. After the model has been linearised and/or reduced, the model parameters can be identified from experimental data by applying suitable methods in an appropriate way. Experimental validation of the dynamic models identified is performed by comparing them with corresponding measurements.

So far, the grey-box modelling approach has been described, as it is the standard modelling route for hydraulic servo-systems. Alternatively, "black-box" models can be derived because of their convenience and wide usage for control design. Thus the emphasis of this book will be on the presentation of procedures to identify black-box models.

Using the results from the (grey-box or black-box) modelling step, the controller will be designed. Sometimes, the model is also changed in the controller design stage when the characteristics of the model are evaluated again. Once the controller design results are satisfactory, the controller can be implemented, tested and evaluated on the process. It may be necessary to repeat or modify the controller design step, making the whole modelling and controller design procedure iterative.

**What this book contains.** The following is a brief description of the main content of this book:

- The first main chapter briefly describes the structure and the main components of hydraulic servo-systems. Chapter 3 is basically a review of relevant topics in fluid mechanics so far as they are essential for the design of hydraulic control components and systems.
- The next two chapters of the book deal with the complete modelling approach for hydraulic servo-systems. Thereby, a subdivision is made with respect to the main phases in this modelling approach. Chapter 4 treats the detailed physically based modelling of the different subsystems, and derives simplified and reduced models for control purposes (first main topic of the book). The (experimental) identification including the validation of the models is discussed in Chapter 5 (second main topic).
- Chapter 6 treats the third main topic of this book, namely the control design for hydraulic servo-systems and the application of the models obtained to control design.

- Some of the modelling and control strategies described are demonstrated in different case studies in Chapter 7.

**What this book does not contain.** It is only fair to point out the areas of hydraulic control systems that are beyond the scope of this book:

- Constructive and operation (function) details of hydraulic servo-systems or their components are not within the scope of this book. The reader may refer to standard textbooks, such as Blackburn *et al.* (1960), Merritt (1967), Ivantysyn and Ivantysynova (1993), Backé and Murrenhoff (1994), Findeisen and Findeisen (1994), Rohner (1995), Merkle *et al.* (1997a, 1997b), Will *et al.* (1999).
- The aspect of system design of hydraulic servo-systems in the sense of specifying proper sizing of system components is not addressed here. This is well known and has been extensively discussed by Merritt (1967), Viersma (1980), Anderson (1988) and Walters (1991).

Note that from the viewpoint of servo-control, it may be emphasised, that good system design should be seen as a prerequisite to achieving good performance by means of closed loop control.

The relation between process design and control can be succinctly summarised by the following quotation from a paper by Ziegler and Nichols (1943): “In the application of automatic controllers, it is important to realize that controller and process form a unit; credit or discredit for results obtained are attributable to one as much as the other. A poor controller is often able to perform acceptably on a process, which is easily controlled. The finest controller made, when applied to a miserably designed process, may not deliver the desired performance. True, on badly designed processes, advanced controllers are able to eke out better results than older models, but on these processes there is a definite end-point which can be approached by the instrumentation and it falls short of perfection”.

## 1.3 Outline of the Chapters

The chapters of this book are organised in the following way:

**Chapter 2: General Description of Hydraulic Servo-systems.** Chapter 2 is devoted to the characterisation of hydraulic servo-systems and to the brief description of a number of subsystems, including the valves controlling the fluid flow to and from the actuator, the actuator itself, pumps and motors, the power supply, and the measurement and control principles and devices.

**Chapter 3: Physical Fundamentals of Hydraulics.** This chapter will discuss some general physical properties of the working fluid and cover equations of fluid motion, types of flow and flow through various passages (pipelines, orifices, and valves), which are needed in the subsequent sections.

**Chapter 4: Physically Based Modelling.** This chapter gives the theoretical model relations that constitute the non-linear dynamic model of the complete hydraulic servo-system using first principles. This involves

- *Comprehensive models* for valves, cylinders, pumps and motors, and pipelines.
- *Simplified and structured models*, as the models derived so far are too complex to be used for feedback control design. The simplified models may be achieved either by simplifications of some sub-models or by linearising the full model in a restricted area of the process variables.
- *Guidelines on how to measure or estimate the numerous parameters* and non-linearities involved in the physical models.

**Chapter 5: Experimental Modelling/Identification.** Accurate models or model parameters can be constructed from experimental input–output data. Different identification procedures are described and illustrated in this chapter that lead to:

- *Linear-in-parameters models* whose parameters can be estimated using classical least-squares analysis.
- *Conventional non-linear models* to be identified by the prediction error method and its subclasses.
- *Non-linear state-space models in canonical forms* using the orthogonal least-squares method based on delayed state variable filters.
- *Fuzzy models.* Algorithms for the representation of the plant by fuzzy models similar to Takagi–Sugeno-type models are described. A fuzzy state space representation is also introduced.
- *Neural-network-based models.* The most popular types of neural networks, multi-layer perceptrons and radial basis functions, are considered.

Whereas the first two approaches lead to “grey-box” models, the outcomes of the three remaining methods are “black-box” models. It should be noted here that the emphasis of this book concerning the identification is on “black-box” (and hybrid) modelling because of its convenience and wide usage for control design.

Furthermore, the experiment design and some practical aspects of the process identification will be discussed in this chapter.

**Chapter 6: Hydraulic Control Systems Design.** The purpose of Chapter 6 is to give a detailed survey of strategies in (model-based) control design for hydraulic servo-systems, including theoretical background. The algorithms covered encompass:

- *Conventional controllers.* Linear controllers (*e.g.*, PI, state space, *etc.*) that are widely used in hydraulic systems are reviewed.
- *Compensation of static non-linearities.* Conventional linear control structures are extended by compensating typical non-linearities.
- *Feedback linearising control.* The synthesis method based on an input–output (feedback) linearisation approach, which has become popular in recent years, is described and applied explicitly to models of hydraulic servo-systems derived in Chapter 3.
- *Fuzzy control.* Fuzzy control system design methods based on the fuzzy state space approach are explained. Model predictive controllers using fuzzy models are also presented.
- The approach of *active damping control* for flexible robots by means of suitable actuator (hydrostatic differential cylinder) control is introduced and discussed.

- The very important practical aspect of *state estimation*, especially *velocity*, *acceleration* and *load force* for hydraulic servo-systems is also addressed at the end of this chapter.

**Chapter 7: Case Studies and Experimental Results.** The experimental evaluation of different basic strategies in Chapter 7 completes the discussion on identification and model-based control design for hydraulic servo-systems. The four case studies described stem from different application fields of hydraulic servo-systems.

**Appendices:** Appendix A contains a collection of the industrial hydraulic symbols, which are used throughout the book. In Appendix B, the parameter sets for the experimental setups considered in Chapter 7, and some typical catalogue data sheets are given. Appendix C presents some basic concepts and definitions from matrix calculus and differential geometry, which are needed in some parts of the book.

## 1.4 Background of the Work and Bibliographical Notes

The main portion of the material used in this book comes from:

- the thesis “Nonlinear approximation and state estimation of time-continuous dynamic processes” (in German) written by M. Jelali (1997),
- the thesis “Fuzzy systems for modelling and control of complex technical systems” (in German) written by A. Kroll (1997),
- the thesis “Control of flexible robots with hydrostatic drives” (in German) written by W. Bernzen (1999a),
- the thesis “Identification, control and observer design of large manipulator systems” (in German) written by D. Nissing (2002),

as well as the numerous papers published by the authors in different journals and conference proceedings (see the reference section).

Most of these studies were carried out at the Department of Measurement and Control, Faculty of Mechanical Engineering, University of Duisburg, with Professor Dr. Helmut Schwarz as the main supervisor. Compared with these sources, the texts have been revised, some issues are no longer covered, different approaches have been presented in a joint framework, and recent ideas and results have been added.

Besides these studies, there are several books and other publications of general interest in the field of modelling and control of hydraulic servo-systems. An old, but still excellent textbook was written by Merritt (1967). More recently, Anderson (1988) has provided an in-depth look at electro-hydraulic components and systems with emphasis on (linear) control. For further reading, see the books by Viersma (1980), Walters (1991), Beater (1999) and Will *et al.* (1999). These books, however, are mainly concerned with the physical modelling and design of hydraulic servo-systems. There are no textbooks available that contain general approaches to the physical modelling and *experimental identification* of the non-linear dynamic behaviour of hydraulic servo-systems, *including the link to closed-loop control design and application*.

Interesting PhD theses that deal with some aspects of modelling and control of hydraulic systems include the following:

- Helduser (1977), Saffe (1986), Klein (1993) and Boes (1995) from the Institute of Hydraulic and Pneumatic Control, Technical University of Aachen, Germany.
- Köckemann (1989), Lausch (1990), Guo (1991), Lierschaft (1993), Ebner (1994), Pawlik (1994), Yin (1994), Lemmen (1995) and Vier (1999) from the Measurement and Control Department, Faculty of Mechanical Engineering, University of Duisburg, Germany.
- De Boer (1992), Lambrechts (1994), Heintze (1997) and van Schothorst (1997) from the Mechanical Engineering Systems and Control Group, Delft University of Technology, The Netherlands.

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## CHAPTER 2

# GENERAL DESCRIPTION OF HYDRAULIC SERVO-SYSTEMS

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This chapter contains a general characterisation of hydraulic servo-systems with some fundamental definitions and a brief description of a number of subsystems of hydraulic servo-systems. It is neither within the scope of this chapter nor our intention to go into the function/construction details of the different subsystems. For this, the reader should consult standard textbooks such as Blackburn *et al.* (1960), Backé (1992), Backé and Murrenhoff (1994), Findeisen and Findeisen (1994), Rohner (1995), Bauer (1998) and Will *et al.* (1999). See also the introductory monographs by Götz (1989, 1997), Kleman (1989), Merkle *et al.* (1997a,b), Scholz (1997) and Norvelle (2000).

## 2.1 Basic Structure of Hydraulic Servo-systems

A hydraulic servo-system is an arrangement of individual components, interconnected to provide a desired form of hydraulic transfer. The basic structure of hydraulic systems is shown in Figure 2.1 and consists of

- hydraulic power supply,
- control elements (valves, sensors, *etc.*),
- actuating elements (cylinder and/or motors), and
- other elements (pipelines, measuring devices, *etc.*).

The basic concept of a hydraulic system (a standard valve-controlled hydraulic system is used here as an example) is briefly described in the following:

- The pump converts the available (mechanical) power from the prime mover (electric or diesel motor) to hydraulic power at the actuator.
- Valves are used to control the direction of pump flow, the level of power produced, and the amount of fluid and pressure to the actuator. A linear actuator (cylinder) or a rotary actuator (motor) converts the hydraulic power to usable mechanical power output at the point required.
- The medium, which is a liquid, provides direct transmission and control, as well as lubrication of components, sealing in valves, and cooling of the system
- Connectors, which link the various system components, direct the power of the fluid under pressure, and fluid-flow return to tank (reservoir).
- Finally, fluid storage and conditioning equipment ensure sufficient quality and quantity, and cooling of the fluid.



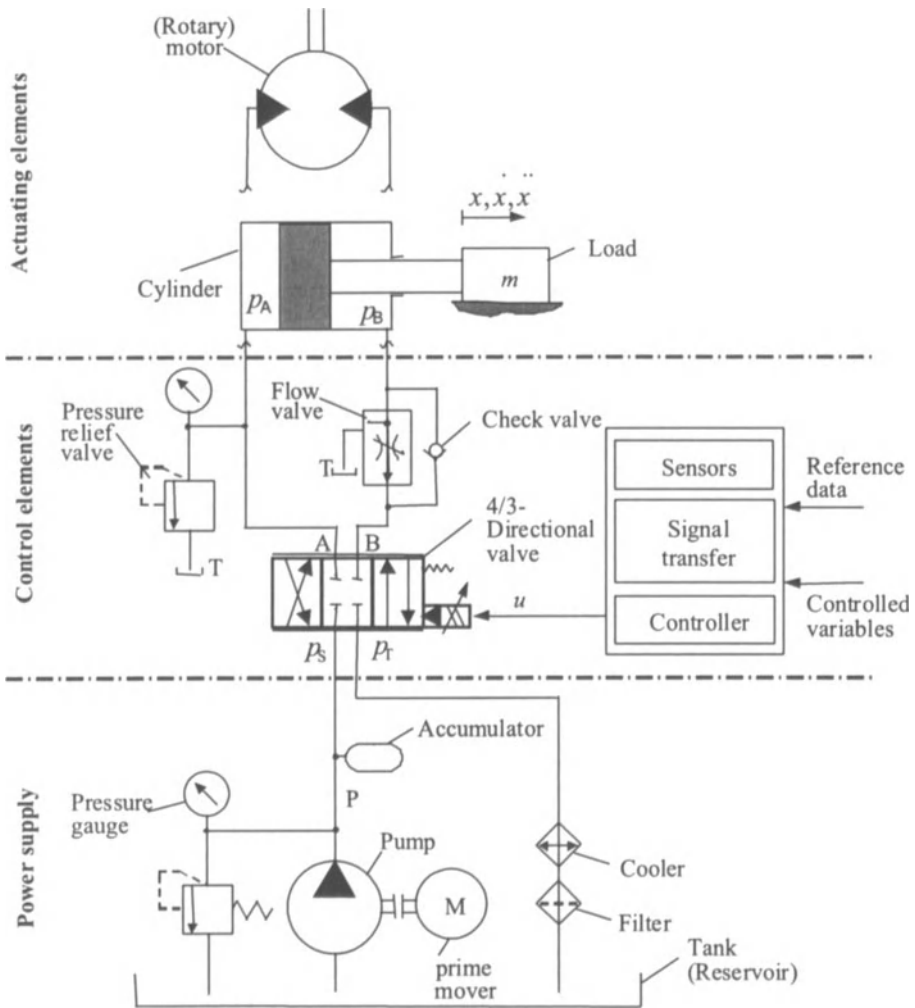


Figure 2.1. Basic structure of hydraulic systems (see Appendix A for hydraulic symbols)

## 2.2 Description of the Components

The descriptions of some hydraulic elements in this section are mainly based on the textbooks by Merritt (1967) and Rohner (1995). In particular, Rohner (1995) provides a detailed description of various hydraulic components (pumps, valves, actuators) and an explanation of their operation with an impartial evaluation divorced from the design and development philosophy of individual manufacturers.

## 2.2.1 Valves

Valves are the most important mechanical (or electrical) link to the fluid interface in hydraulic systems.

### 2.2.1.1 Valve Types

Basically there are four main categories of valves in hydraulics:

- (a) **Pressure valves** are used to control actuator force, and to determine and (pre)select pressure levels at which certain machine operations must occur:
  - *Pressure-relief valves* limit the maximum permissible system pressure, and divert some or all of the pump's flow to the tank when the pressure setting of the relief valve is reached. Pressure relief valves are "normally closed".
  - *Pressure-reducing valves* limit and maintain a constant downstream pressure (sub-circuit pressure) that is smaller than the system pressure regardless of pressure fluctuations in the main circuit upstream. Pressure reducing valves are "normally open".
- (b) **Check valves** are a very special type of directional control valve, as they only permit fluid flow in one direction while blocking flow in the reverse direction. They can be divided into unloaded or spring-loaded check valves, and check valves for logic operations (OR, AND).
- (c) **Flow valves** are used to control the rate of flow from one part of the hydraulic system to another, *i.e.*, they limit the maximum speed of cylinders and motors, limit the maximum power available to sub-circuits by controlling the flow of them, or proportionally divide or regulate the pump flow to various branches of the circuit.
- (d) **Directional valves** are used as multi-polar switches. Before the advent of servo and proportional valves they were used to control the direction of actuator motion, selected alternative control circuits, and performed logic control functions. Nowadays, however, proportionally variable controls allow infinitely adaptable and quickly variable setting of actuators with regard to force, speed and stroke position.

### 2.2.1.2 From Solenoid-valves to Servo-valves

The simplest variant of valves used for fluid power control is the solenoid-valve. It is only an on-off valve, *i.e.*, it is either actuated or un-actuated – fully open or fully closed, with no intermediate position. For almost a century, solenoid-valves have been used in fluid power applications with relatively few major changes to the original concept (Norvelle, 2000). Having recognised the great limit of the performance of hydraulic systems with solenoid-valves, control concepts integrating *servo-valves* have been developed (and thus *servo-hydraulic systems* or *servo-mechanisms*) since the 1950s. Servo-valves are faster responding directional, pressure and flow control valves, which are frequently used in closed-loop control arrangements to produce the highly sophisticated performance (in terms of high-frequency response) required by modern machines. Perhaps the first and most common applications of servo-valves are in aerospace vehicles, particularly in primary flight controls. Nowadays, servo-valves are widely used in many other industrial fields, such as mining, rolling, agriculture, transport and shipping.

### 2.2.1.3 From Servo-valves to “Refined” Proportional Valves

For long time there was just a huge gap between simple on-off valves and highly sophisticated servo-valves. That situation changed following the advent of the *proportional (control) valve* in the 1970s. Today, major hydraulic component manufacturers offer a host of proportional valves.

Proportional valves, as well as servo-valves have one major concept in common: usually the output variable, *i.e.*, spool valve position (directional valves), pressure (pressure valves) or flow (flow valves), is proportional to the input signal. Such an input signal may be manual, mechanical, pneumatic or electrical. Modern servo and proportional valve systems commonly use a low-power electrical analogue or digital signal.

As servo-valves are high-sophisticated control valves, frequently used in closed-loop control arrangements, they must have very accurate metering dimensions, and thus often involve high installation and purchasing costs. Even the slightest fluid contamination may cause the delicate mechanisms in the servo-valves to malfunction.

The ability of proportional valves to be used for directional, pressure and flow control, and to adjust them electronically to an infinite number of positions within their range, makes them extremely useful and popular for a wide range of hydraulic applications. The fundamental principle of operation of proportional valves is based on the use of a *proportional solenoid*, which moves the valve spool to the desired position. “Unrefined” proportional valves are usually less expensive with fewer performance demands, such as larger values of hysteresis and dead-bands. They are primarily used in open-loop applications. Summaries of the properties and the application fields of servo and proportional valves are given in Table 2.1 and Table 2.2.

Owing to further development of the proportional solenoids (inclusion of spool position feedback, improvement in the designs of the spools and electronics, better performance in accuracy, hysteresis, dead-band, etc.) in recent years, highly sophisticated proportional valves are also available today, which – despite high input power – show very good dynamic characteristics similar to those of servo-valves. These proportional valves are known as “*control valves*” or “*refined*” *proportional valves*, and can also be used for closed-loop controls. So, the definitions of “servo-valve” and “proportional valve” and the distinction between the two have become more and more vague and overlapping (since the 1980s). Even combinations of a two-stage servo-valve and a proportional valve (as main stage) are not unusual in many modern industrial applications (*e.g.*, mobile hydraulics), as such valves meet to high technical requirements and are simultaneously robust (Will *et al.*, 1999).

Servo-valves (and proportional valves) may be broadly classified as either single stage or multi-stage, *e.g.*, two- or three-stage. Single-stage servo-valves comprise a torque motor, which positions a spool valve (usually four-way) via a direct connection. Single-stage valves are simple and inexpensive, but have limited flow capacity (up to 20 dm<sup>3</sup>/min) due to the limited power capability of torque motors.

Multi-stage servo-valves have one or two hydraulic pre-amplifiers to multiply the force output of the torque motor substantially to a level sufficient to overcome considerable flow forces, friction forces, and forces resulting from acceleration or vibration. Flapper, jet pipe, and spool valves are used as a pilot (or master) stage valve, whereas the second (or slave) stage valve is almost universally spool type.

**Table 2.1.** Properties of servo and proportional valves

| Properties                                       | Servo-valve                                                    | Proportional valve                                                        |
|--------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| Electrical operator                              | Torque motor                                                   | Proportional solenoid                                                     |
| Input power                                      | ~ 0.02–1.0 W                                                   | 10–100 W                                                                  |
| Valve stroke                                     | Few cm                                                         | Around $\pm 1$ mm                                                         |
| Required filtration rating<br>contamination risk | Very high                                                      | Moderate                                                                  |
| Hysteresis                                       | ~ 0.1–0.5 %                                                    | ~ 0.3–7 %<br>~ 0.2–0.5 % (refined)                                        |
| Lap at middle position                           | Zero lap or little underlap                                    | Overlap (~ 5–20 %)<br>zero lap or little underlap<br>(refined)            |
| Flow-signal curve                                | Smooth due to zero lap or<br>underlap of all orifices          | Non-smooth due to positive<br>overlap of the main orifices<br>(dead band) |
| Flow loss at middle position                     | Relatively high due to lap zero<br>or underlap of all orifices | Less due to positive overlap<br>of the main orifices                      |
| Max. frequency                                   | ~ 100–200 Hz                                                   | ~ 10–70 Hz,<br>~ 50–150 Hz (refined)                                      |
| Allowable fabrication<br>tolerances              | Very small                                                     | Much higher                                                               |
| Production costs                                 | Very expensive                                                 | Moderately expensive                                                      |

**Table 2.2.** Application fields of servo and proportional valves

| Application fields                                                            | Servo-valve         | Proportional valve                                                                        |
|-------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------|
| Closed-loop/open-loop<br>control                                              | Closed-loop control | Open-loop control<br>Closed-loop control (refined)                                        |
| Position/angle and<br>force/torque control                                    | Well suited         | Not possible due to non-<br>smooth flow-signal curve<br>(dead band)<br>Possible (refined) |
| Velocity/speed control                                                        | Well suited         | Possible                                                                                  |
| Combination with pressure-<br>difference valves to act as<br>flow controllers | Not possible        | Well suited                                                                               |

#### 2.2.1.4 Classification of Spool Valves

Typical *spool valve* configurations are classified by

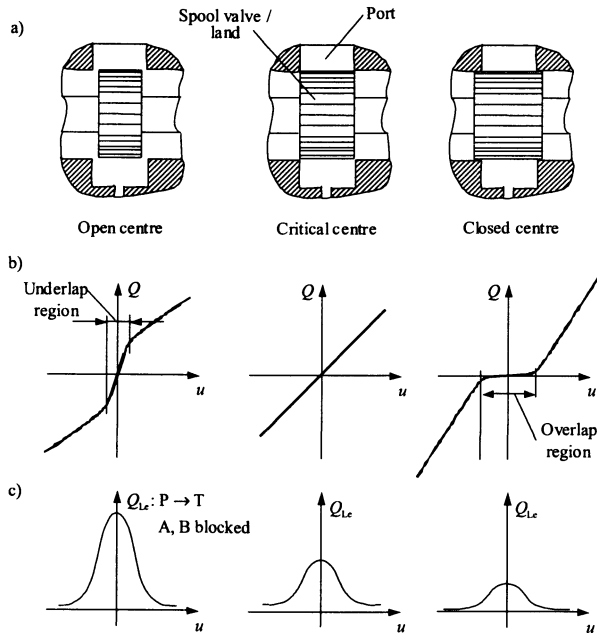
- The number of “ways” (or ports) flow can enter and leave the valve. Because all valves require a supply (P: pressure), a return (T: tank), and at least one line (A, B) to the load, common valves are either three-way or four-way. Of course, two-way valves also exist, and are used either to open or to close the flow path in a single line (on/off valves).
- The number of switching (or discrete) positions. The number of discrete positions on a spool varies from one in a primitive valve to the usual two or three. Special valves may have up to five positions or more. For instance, 5/3-valves are the most commonly used valves in mobile hydraulics.
- The method of valve actuation that causes the valve mechanism to move into an alternative position. External signal commands (electrical, manual, pilot

pressure), and internal signal commands (pilot pressure, spring force) may be applied to shift the valve mechanism.

- (d) The type of centre (open, closed, critical), when the valve spool is in neutral position.

### 2.2.1.5 Centre Types

Spool “lap” defines the width of the lands relative to the width of the ports in the valve bore. There are three possible lap configurations (see Figure 2.2a): If the width of the land is smaller than the port in the valve sleeve, the valve is said to have an *open centre* or *underlap*. For a short period all valve ports are connected to each other. This results in smooth, pressure-peak-free switching during the cross-over. However, undesirable actuator movements may occur with certain load conditions. Underlapped valves are normally preferred in (closed-loop) applications that require a continuous flow (see Figure 2.2b), such as in constant-flow systems or in order to maintain reasonable temperature. However, a disadvantage of such valves is the relatively high leakage flow at neutral position (so-called centre flow), see Figure 2.2c.



**Figure 2.2.** (a) Definition of centre types and their corresponding (b) flow-signal graphs and (c) leakage flow curves

*Closed centre* or *overlapped* valves have a land width greater than the port width when the spool is at neutral. For a short period all valve ports are sealed against each other. Thus, system pressure on the actuator is prevented from collapsing during the cross-over. The fully closed centre cross-over (dead band) may, however, cause backlash, and so undesirable pressure peaks occur in the system which vary with the amount of fluid flow and switching time.

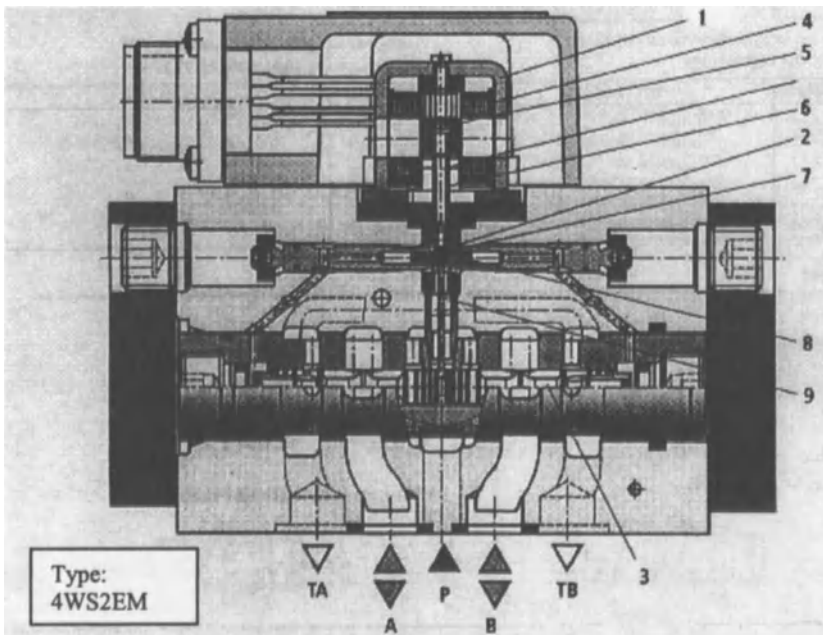
*Critical centre* or *zero lapped* valves have a land width identical to the port width and is a condition approached by practical machining. The vast majority of servo-valves commercially available are zero lapped, *i.e.*, critical centre valves, because of the emphasis on the linear flow-signal curve (absence of oil consumption in the neutral state).

### 2.2.1.6 Industrial Servo-valves

One of the most common servo-valve types used in closed-loop systems is the two-stage flapper-nozzle valve (see Figure 2.3). Such valves have two stages of power amplification:

- (a) The flapper-nozzle system converts the flapper motion, driven by a low-power electrical torque motor, into a hydraulically powered motion of the spool.
- (b) The small spool motions control relatively large oil flows through the spool ports, which is the second power amplification.

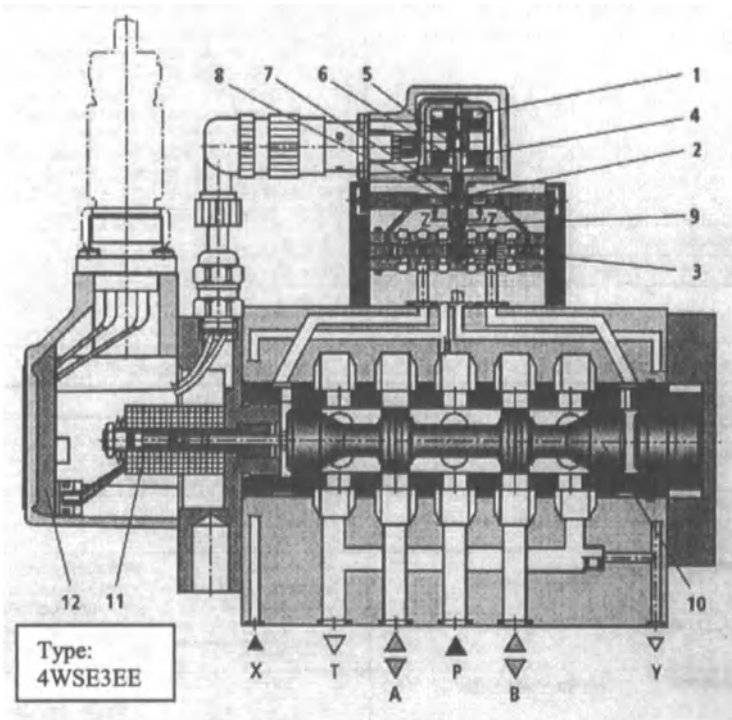
Flapper-nozzle servo-valves exist in several configurations, characterised by the electro-magnetic circuit, feedback mechanism, and function within the system. If high performance is required, an electrical feedback loop is provided, feeding the measured spool position back to the torque motor input.



**Figure 2.3.** Drawings of a two-stage flapper-nozzle servo-valve (Bosch Rexroth, 2000) [1: torque motor, 2: nozzle system, 3: pilot spool, 4: solenoid, 5: armature, 6: flexure tube, 7: flapper, 8: control nozzles]

When very large oil flows are required, say  $>100 \text{ dm}^3/\text{min}$ , it may even be necessary to have an additional stage in the servo-valve. In that case, a two-stage

flapper-nozzle servo-valve is typically used as a pilot-valve for the third stage, so that the configuration of Figure 2.4 is obtained.



**Figure 2.4.** Drawings of a three-stage flapper-nozzle servo-valve (Bosch Rexroth, 2000) [1: torque motor, 2: nozzle system, 3: pilot spool, 4: solenoid, 5: armature, 6: flexure tube, 7: flapper, 8: control nozzles, 9: spring (mechanical feedback), 10: main spool, 11: position sensor (LVDT-sensor), 12: electronic control unit]

## 2.2.2 Pumps and Actuators

Hydraulic pumps are used to convert mechanical energy into hydraulic energy, and actuators *vice versa*.

### 2.2.2.1 Pumps

Hydraulic *pumps* are devices that convert mechanical energy to hydraulic energy. Pumps are normally driven rotationally. Industrial hydraulic pumps are built in a variety of shapes, sizes, and pumping mechanisms, and are usually of the “positive displacement” type. Displacement (or flow output) is that volume of hydraulic fluid which gets transported through the pump by a single rotation of the pump drive shaft.

Positive (fixed or variable) displacement pumps are classified according to their pumping mechanism into gear pumps, vane pumps and piston pumps. Piston-type pumps are manufactured in both axial and radial designs. Generally, they are regarded as the true high-performance pumps, and thus are frequently used in

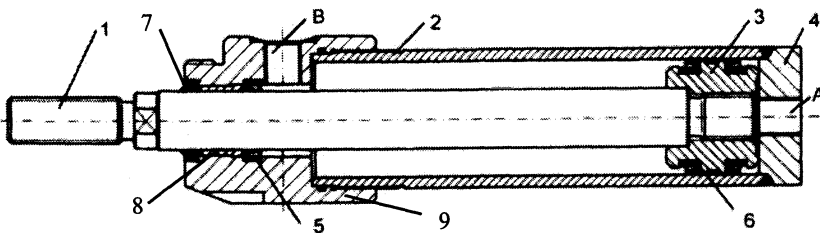
industrial and mobile hydraulics. Whereas in fixed displacement pumps (e.g., swash-plate fixed at a given angle) the amount of flow displaced by each pump shaft rotation cannot be changed, variable displacement pumps only deliver flow when and as required by the system.

### 2.2.2.2 Actuators (Cylinders, Motors)

Hydraulic *actuators* are devices to convert hydraulic energy provided by the pump and processed by the control elements (i.e., valves) into useful work (and consequently into power and mechanical energy respectively). Actuators have either a linear output (cylinders, rams or jacks) or rotary output (rotary actuators or motors). Motors very closely resemble hydraulic pumps in their construction. Indeed, many pumps can also be used as motors. Instead of pushing fluid into the system as pumps do, motors are pushed by the fluid and thus they develop torque and continuous rotary motion.

One of the classical components of hydraulic systems is the linear actuator, commonly known as the *hydraulic cylinder*.<sup>1</sup> Cylinders are used to convert hydraulic power into linear mechanical force or motion. Many types of linear actuator exist. Single-acting actuators permit the application of hydraulic force in one direction only.

Double-acting actuators allow hydraulic force in both directions to be applied. Double-ended actuators with rods on both actuator ends, also termed *symmetric* or *synchronising cylinders*, are used where the force developed must be equal in both directions (extension and retraction). Since the voids to be filled with hydraulic fluid are equal for extension and retraction, the resulting piston speeds are also equal for both strokes. Synchronising cylinders have greater overall length and are more expensive than double-acting cylinders. Thus, in practice, double-acting cylinders with one rod, also referred to as *asymmetric* or *differential cylinders*, are widely used (see Figure 2.5). They have the advantage that the space needed for the cylinder is considerably smaller, because there is only one piston rod.



**Figure 2.5.** Drawing of a hydraulic (differential) cylinder (Will *et al.*, 1999) [1: piston rod, 2: tube (barrel), 3: piston, 4: cylinder cap, 5: rod seal, 6: piston seal, 7: rod wiper, 8: guide bush, 9: cylinder head, A/B: oil ports]

As stated by Viersma (1980), an asymmetric four-way valve would then be desirable to obtain smooth system behaviour. However, Lemmen (1995) proved that the smoothness is *not always* achievable for all cases of application. On the other

<sup>1</sup>Hydraulic cylinders are sometimes called linear motors.



hand, asymmetric four-way valves are *not* commercially available, except on special request and for a special price. Thus, asymmetric cylinders are normally combined with symmetric valves. Consequently, this incompatibility leads to pressure jumps around null velocity, and smooth operation around this point is not possible (without velocity compensation; see Section 6.4.3). Although this fact makes the control of such valve–cylinder configurations more difficult, they are perfectly suitable for closed-loop systems, provided the effect of the asymmetry on performance and stability is taken into account when carrying out analysis and controller design and implementation for a hydraulic servo-system.

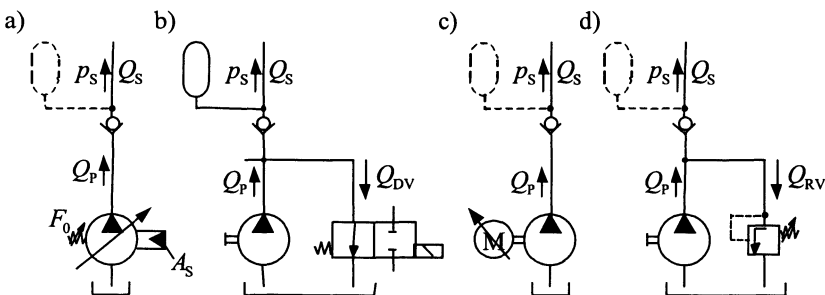
### 2.2.3 Power Supplies

In hydraulic servo-controls, the pump is part of an auxiliary unit that can be considered simply as a source of hydraulic power. Hydraulic servo-systems predominantly use pressure controllers in conjunction with variable-displacement pumps, but other controllers may be used to regulate flow levels, pressure and flow combined, or input power (as for mobile systems) (Rohner, 1985).

The basic configurations of hydraulic power supplies that provide constant supply pressure are (see Figure 2.6) (Will *et al.*, 1999):

- (a) variable-displacement pump with stroke regulator
- (b) switch-off pump
- (c) fixed-displacement pump with speed-controlled mover
- (d) fixed-displacement pump with pressure-relief valve.

The two classical pressure sources (a) and (d) are simple and safe (especially in conjunction with the use of an accumulator). Concept (c) is becoming more and more popular due to further development of electrical drive control. Switch-off pumps are not so common, since they can lead to persistent pressure fluctuations.



**Figure 2.6.** Basic configurations of pressure sources (Will *et al.*, 1999)

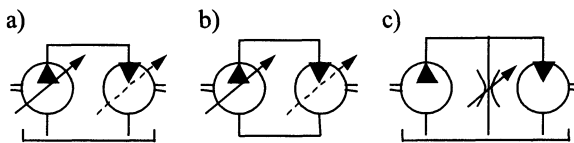
In the cases where the hydraulic system is subject to wide fluctuations in load (*i.e.*, pressure and/or flow), load-sensing systems (See Section 2.3) are predestined to save substantial energy. These systems have found wide application in mobile hydraulics and, of late, also in industrial hydraulics.

Other hydraulic control structures use power supplies in the form of flow sources, known as *hydrostatic transmissions* or *pump/motor circuits* (typically

applied in ship steering systems, antenna drives or vehicle drive systems), as shown in Figure 2.7.

The *open circuit* (Figure 2.7a) uses a pump to produce flow, which is utilised by a simple or complex system, with the return flow exhausting to the tank. The inlet section of the pump draws the fluid from the tank to complete the circuit.

The *closed circuit* (Figure 2.7b) is a continuous circuit between pump and motor, *i.e.*, the pump output flow is sent directly to the hydraulic motor and then returned in a continuous motion to the pump. The tank and heat exchanger take up leakage and auxiliary function flow. Closed circuits are typically used to maintain a given output speed proportional to a pump input command, and offer continuously variable output speed in all four quadrants.



**Figure 2.7.** Power supplies as flow sources (a) open circuit, (b) closed circuit, (c) principle of bypass control

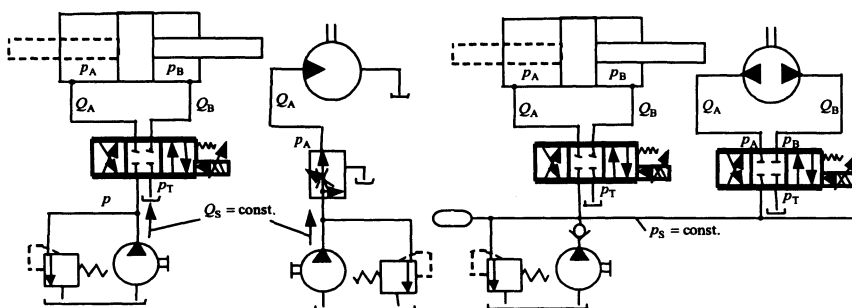
The principle of bypass control is to return the overproduced flow from the constant-delivery pump to the tank by means of an adjustable throttle valve.

A detailed description and properties of all power supply configurations mentioned in this section can be found in Will *et al.* (1999). Useful material in relation to hydrostatic transmissions and load-sensing systems is also provided by Rohner (1985).

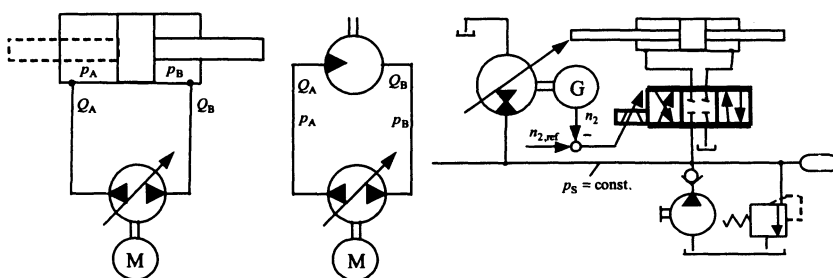
## 2.3 Classification of Hydraulic Servo-systems

Hydraulic servo-systems can principally be classified into three basic over-all systems:

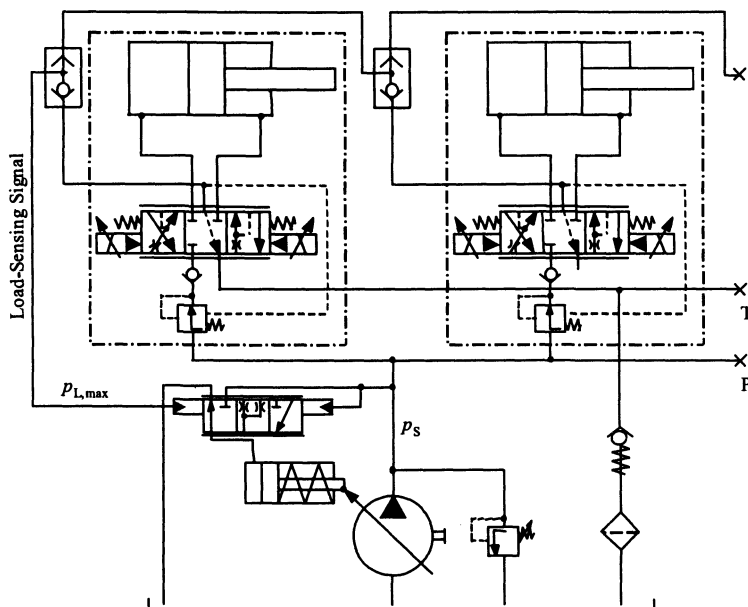
- (a) **Valve-controlled systems.** These consist of a valve controlling the flow from a hydraulic power supply to an actuation device (linear or rotary), regardless of the type of the valve (*i.e.* on/off or proportionally variable valves). The hydraulic power supply is either of a constant-pressure or a constant-flow type. See Figure 2.8.
- (b) **Pump/motor-controlled systems.** The hydraulic power control is performed by actuating a pump or a motor. Motor-controlled systems (or motor circuits) are also known as secondary controlled systems or hydrostatic transmission circuits. Again, the hydraulic power supply may be of a constant-pressure or a constant-flow type. See Figure 2.9.
- (c) **Load-sensing systems.** The concept of these systems ensures that the flow (power) is *automatically* adjusted to any load demand required by the actuators within the given pressure and flow boundaries. That means flow and pressure levels are simultaneously regulated. Many load-sensing systems are *combinations* of valve-controlled systems and pump/motor-controlled systems. See Figures 2.10 and 2.11.



**Figure 2.8.** Examples of valve-controlled servo-systems (Backé, 1992)

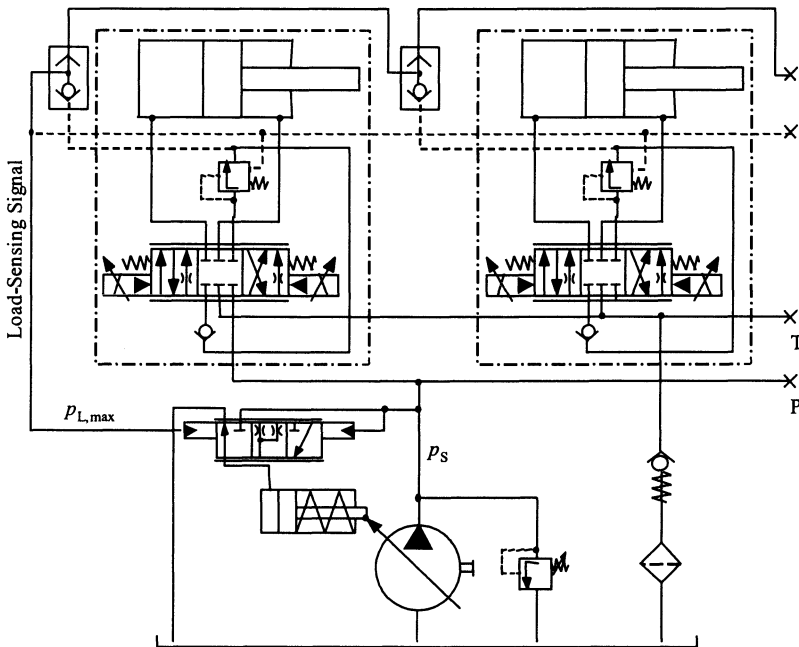


**Figure 2.9. Examples of pump/motor-controlled servo-systems (Backé, 1992)**



**Figure 2.10.** Circuit for load-sensing control of multiple actuators (Will *et al.*, 1999)

From the energy viewpoint, pump/motor-controlled systems are preferred owing to their higher efficiency. However, valve-controlled systems normally show better dynamic characteristics. Load-sensing systems are the best power-saving controls, and are very popular in mobile hydraulics. They are also becoming increasingly accepted in industrial hydraulics. However, load-sensing systems are usually complex and require high personnel costs (for maintenance and service). In general, it is the nature of the specific application that dictates the concept to be used.



**Figure 2.11.** Circuit for load-pressure-independent flow distribution (modified load sensing) of multiple actuators (Will *et al.*, 1999)

## 2.4 Measurement and Control Devices

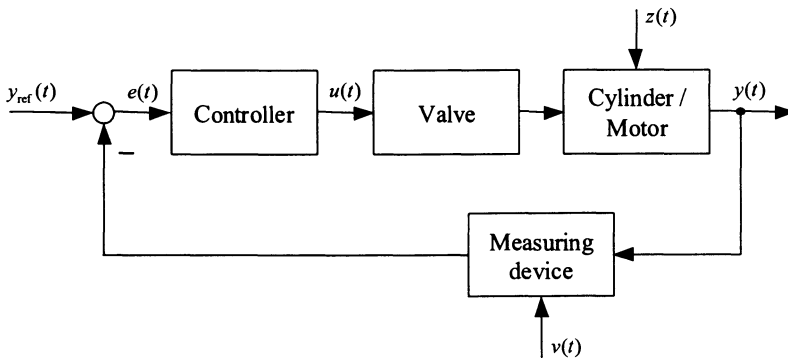
### 2.4.1 Control Loops

Hydraulic servo-systems, in general, are used to control one or more of the following actuator output variables: direction, velocity, acceleration, deceleration, position or force against a resisting load; see Table 2.3. They thus form a *closed loop* hydraulic control system, as shown in Figure 2.12 (Note that all blocks may, of course, be non-linear).

**Table 2.3.** Frequency of application of some controlled variables (Lemmen, 2002)

| Controlled variable(s)      | Application in mobile hydraulics | Application in industrial hydraulics |
|-----------------------------|----------------------------------|--------------------------------------|
| Position                    | Medium to seldom                 | Frequent                             |
| Pressure                    | Frequent                         | Medium                               |
| Pressure + Position + Power | Medium                           | Seldom                               |

The system output  $y(t)$ , usually sensed by electronic measuring devices, is to be forced to follow or track the reference signal  $y_{\text{ref}}(t)$ , despite the presence of disturbance inputs  $z(t)$  (e.g., load) to the system and despite errors in the sensor, represented by  $v(t)$  in Figure 2.12. Any deviation (error at the summing point) resulting from this comparison is fed back to the controller (whether analogue or digital, electrical or hydro-mechanical) to cause corrective action at the valve spool. Thus the control system is continually monitoring and correcting any minute deviation of load behaviour to maintain the desired reference signal. It is essential that the tracking is accurate, even if the dynamics of the plant should change somewhat during the operation.



**Figure 2.12.** Block diagram of a servo-hydraulic system in a closed control loop

The process of holding  $y(t)$  close to  $y_{\text{ref}}(t)$ , including the case where  $y_{\text{ref}} = 0$ , is referred to generally as the process of *regulation*. A system that has good regulation in the presence of disturbance signals is said to have good *disturbance rejection*. A system that has good regulation in the face of changes in the plant parameters is said to have low *sensitivity* to these parameters. A system that has both good disturbance rejection and low sensitivity is called *robust* (Franklin *et al.*, 1990).

In contrast to a closed loop control system, an *open-loop control* system has no feedback provision from the load behaviour and no corrections are made if the load behaviour falls short of the desired performance. Transient functions, such as accelerating or decelerating the load opposing the hydraulic actuator, are achieved and controlled by means of electronic ramp generators. Of course, there are numerous applications where an open-loop control system is sufficient and does not warrant the extra costs of a closed-loop system (Rohner, 1995).

### 2.4.2 Sensors/Transducers

Nowadays, hydraulic control systems usually require a high degree of accuracy<sup>2</sup> in system response and thus need precise devices to measure the controlled variables or other state variables like

- pressure,
- force,
- position (angle),
- velocity (speed),
- acceleration/deceleration, and
- flow.

Moreover, it is stressed that the accuracy of the parameters measured is of major importance, as the controlled parameters can never be more accurate than the measuring device's performance. This also concerns the frequency at which the measurements taken are transferred to the electronic controller, which in turn directly influences the system's response and dynamic behaviour. As sensors are generally expensive, it has to be stressed that the number of transducers must be minimised in practical solutions to keep the costs as low as possible.

In the following, the most used transducers<sup>3</sup> are briefly reviewed. For more explanations of the operation of the various sensor types, refer to Kleman (1989), and Norvelle (2000).

#### 2.4.2.1 Pressure Transducers

Devices that are used to measure *pressure* are (Rohner, 1995; Norvelle, 2000):

- Strain gauge principal transducers (unbonded, bonded, thin-film, diffused semiconductor), which are ideally suitable to measure static system pressure. Also, they are the only pressure transducers available that can stand high pressures.
- Piezoelectric pressure transducers. These produce an electrical signal through mechanical deformation of quartz crystals or special plastic foils. To make the piezoelectric effect useful to measure pressures, the charge obtained by the deformation must be amplified and transformed. Piezoelectric pressure sensors are mainly used to measure fast dynamic pressures, such as pressure spikes and pulsating pressures. They are not suitable for extended static pressure measurements.

To measure the pressures in the cylinder chambers, pressure sensors are usually placed in the pipelines that connect the valve with the cylinder. Care has to be taken if the pipelines are so long or soft (hose lines) that significant pipeline dynamics

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<sup>2</sup> *Accuracy* is the ability of an instrument to give results that are the true value of the measured parameter. *Precision* defines the instrument's ability to provide the same measurement of the same quantity under the same conditions repeatedly (Norvelle, 2000).

<sup>3</sup> Although the terms *sensor* and *transducer* are often used synonymously, they are completely different: A sensor is usually part of a transducer. Sensors are devices that acquire information (related to position, speed, pressure, etc.). Transducers are devices that convert a signal from one form to another (Norvelle, 2000).

occur. In this case, more than one pressure sensor should be installed in each pipeline.

Also, it is advantageous to measure the supply pressure, because fluctuations in the supply pressure are often observed in practical applications and these act as a disturbance on the servo-system. Tank pressure can often be neglected, because it is usually much smaller than the other pressures involved. Thus, measurement of tank pressure is not always necessary in practice.

#### 2.4.2.2 Force Transducers

The load force that acts as a disturbance on the servo-system should also be measured using force sensors (usually termed *load cells*) at the piston rod. Most commercially available load cells are designed to measure load acting in one direction only. However, to reduce costs, in most practical applications a force measurement is not provided. Instead, the load force is indirectly gathered from the pressure signals, multiplied by the corresponding area. See also Section 6.10.3 for more details about the estimation of load force. In principle, force sensors are identical to pressure sensors.

#### 2.4.2.3 Position Transducers

Beyond any doubt, the most important measurement signals are the *linear position* or *displacement* of pistons (or cylinders) and *rotary position (angle)* for motors and pumps. Devices used to measure position or angle are (Walters, 1991; Rohner, 1995):

- Potentiometers (linear, rotary),
- inductive voltage dividers,
- photoelectric digital sensors, and
- non-contacting transducers. These can produce either an analogue output signal (*i.e.*, providing an infinitely variable output related to an absolute datum) or a digital output signal (*i.e.*, providing a discrete incremental output without an absolute datum). Typical examples of non-contacting analogue transducers are linear variable displacement transducers (LVDTs), rotary variable displacement transducers (RVDTs), synchros and resolvers; examples of digital transducers are encoders and proximity sensors.

Of these, the LVDT (MTS Tempsonics) is probably the most widely used for measuring position. Note that the linearity of the LVDT makes it really only suitable for short stroke applications. The LVDT is thus also used to measure spool position within servo and proportional valves. For long stroke applications magnetostrictive devices (MTS Tempsonics) are well suited (basically for any conceivable length).

Other popularly used transducers are diffraction grating (Haidenhain), linear resolver (Farrand Inductosyn) and Sony Magnascale, which have their roots in the machine tool industry.

#### 2.4.2.4 Velocity (Speed) Transducers

Devices used to measure *velocity* or *angular speed* are (Kleman, 1989; Walters, 1991; Rohner, 1995):

- Rotational speed tachometers (usually referred as *tachos*) with permanently excited DC generators are used to measure rotational speed or revolutions. Tachos can have a wide operating range but have a lower threshold, which limits the minimum running speed. Rotational speed of hydraulic motors can also be measured by magnetic sensors (pickups), and optical encoders.
- Linear speed can be measured with a rod-shaped permanent magnet attached to the moving load. The permanent magnet slides within adjoining tube-like coils and generates velocity-dependent voltages in these coils.

Also, position signals from incremental encoder systems are usually used in practice to estimate velocity by differentiation in the computer software (see Section 6.10.1). This helps minimise sensor costs.

#### 2.4.2.5 Acceleration/Deceleration Transducers

Although acceleration sensors are on the market, they are not widely used in hydraulic control systems. As acceleration is the first time derivative of velocity and the second time derivative of displacement, both acceleration and deceleration are easily handled in motion control systems by controlling the rate of change of velocity (Walters, 1991). Acceleration is also often measured indirectly, in terms of force or pressure-drop measurement; see Section 6.10.2.

#### 2.4.2.6 Flow Transducers

Three types of flow transducer can be used (Walters, 1991):

- positive displacement units,
- piezo-electric venturi meters, and
- variable-resistance flow meters.

Of these, positive displacement units and variable-resistance flow meters are the most suitable flow transducers in hydraulic control applications. Note that the latter flow transducers are based on indirect measurement of the flow, in terms of the pressure drop across a resistance (*i.e.* an orifice restrictor), which is installed at the point where the flow is to be measured.

#### 2.4.2.7 Temperature Transducers

Temperature measurement is essential for accurate determination of viscosity. From a large number of measurement principles, the following are most used in hydraulic systems:

- thermocouples,
- thermistors, and
- metal-resistance detectors.

#### 2.4.2.8 Combined Transducers

In today's practice, it is desirable to have access to several measurement variables using a single measurement system. For instance, combined pressure and temperature sensors are available.



## 2.5 Application Examples

Thus far, we have introduced basic structures of single hydraulic systems. Usually, however, hydraulic drive systems consist of several hydraulic lines with mutual interactions between them. A *hydraulic line* is defined as an arrangement of individual components, interconnected to provide a desired form of hydraulic power transfer, and to fulfil a specified task. Such a system is referred to as a *multi-line* hydraulic system. The interactions or couplings arise from the existence of common components and/or of power or signal connections between the different lines; see Vier (1999) for more details.

Typical examples for the application of hydraulic servo-systems are hydraulically actuated manipulators and rolling mills.

### 2.5.1 Hydraulically Actuated Manipulators

Figure 2.13 shows a concrete pump manipulator for distributing concrete in the construction industry; this is an example for a very wide operating range (*i.e.*, reach). Such applications demand actuators with a significant power density able to produce high forces with a fast response, which can be achieved profitably by the use of *hydraulic actuators*.



**Figure 2.13.** A hydraulically operated concrete pumping manipulator (Schwing GmbH)

### 2.5.2 Hydraulic Automatic Gauge Control for Rolling Mills

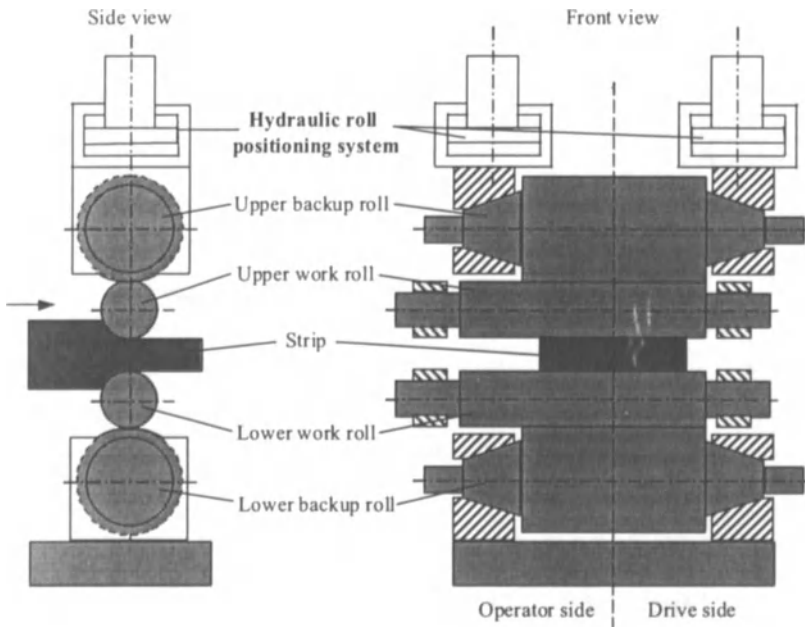
In the rolling of metals, especially the final cold rolling of sheet materials, the variation of gauge (centreline thickness) along the length of the strip must be

controlled within very tight tolerances (typically 0.5-2.0%). In order to achieve these tolerances, fast-acting hydraulic servo-systems, with bandwidths of up to few hundreds of rad/s, have become the standard in modern rolling mills.

In a typical single-stand reversing strip mill, the strip is paid off a coiler at one side of the stand, reduced in thickness as it passes between the work rolls, and recoiled by a coiler at the other side of the stand. In the next pass, the roll gap is reduced, and the process is repeated in the reverse direction. This sequence continues until the strip is of the desired final thickness. The rolls are typically supported by neck bearings, running in chocks that can move vertically in the mill housing for the purpose of altering the roll gap.

The thickness of the rolled strip is predominantly determined by the gap that is set between the work rolls (although the tension in the strip, incoming material hardness variations, hardening of the material during rolling, mill housing stiffness and other parameters also contribute in a complicated manner). Typically, the roll gap is initially set by electric screw-down drives. However, once the strip is threaded, any changes necessary to control the roll gap during rolling are made by extending or contracting the two hydraulic systems (see Figure 2.14).

In principle, these are simply large diameter, short stroke cylinders, operated at high pressures, and therefore able to move very rapidly. There are two identical hydraulic systems, one for the operator side and one for the drive side.



**Figure 2.14.** Schematic arrangement of a four-high mill stand seen from side and front. The main issue to note is the functionality of the hydraulic systems controlling the upper rolls, which adjust the roll gap.

*Hydraulic automatic gauge control* (HAGC, or simply AGC) schemes control the hydraulic cylinders in one of the two basic modes, pressure (load) control or

position control. In practice, there may be more than one servo-valve, and more than one transducer, to improve accuracy; see Figure 2.15. For a pressure-control scheme, the position transducers are replaced by pressure transducers. Usually, both are fitted, and the type of control is therefore selectable. For the purpose of this chapter, the position control loop is considered. Detailed modelling and evaluation of dynamic characteristics of HAGC systems is provided by Guo (1991) and Totz and Jelali (1999).

The cylinder position reference signal will have contributions from several other (higher situated) control loops (*e.g.*, thickness feedback and thickness feedforward controls). The exit strip thickness is usually measured and any deviation from the required value suggests a trim to the capsule position reference signal (so-called monitor control loop).

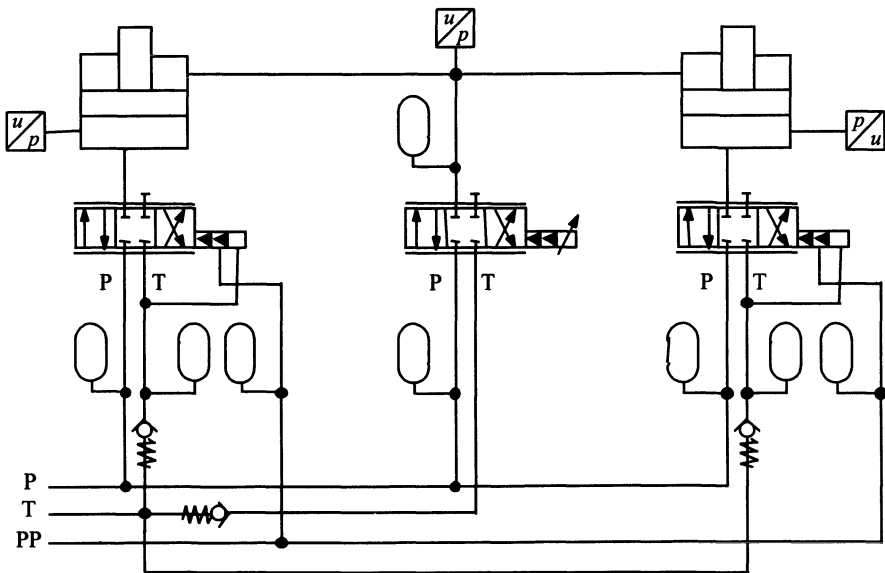


Figure 2.15. Typical (simplified) hydraulic layout of hydraulic roll gap adjusting systems

Until recently, because of the fast dynamics of hydraulic AGC loops, control was implemented using standard P and PI designs, sometimes with a static servo-valve load compensation, realised by analogue electronics. Now, suppliers can provide AGC loops in direct-digital-control form. The presence of digital computers in these control loops provides the opportunity to investigate the use of more sophisticated control methods.

Dutton and Groves (1996), for instance, proposed an adaptive controller with a simple pole-placement algorithm. Kugi *et al.* (1999) derived and successfully applied a non-linear controller based on input-output linearisation, combined with an active eccentricity compensation technique. Also, a position controller based on feedback linearisation has been designed by Pedersen (1999) for special hydraulic systems (combination of servo-valve with an oil cylinder and a grease cylinder) still used for controlling the roll gap of some heavy plate mills.

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## CHAPTER 3

# PHYSICAL FUNDAMENTALS OF HYDRAULICS

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The purpose of this chapter is to define some physical properties of hydraulic fluids, and to discuss the fundamental laws and equations of fluid motion, types of flow, and the flow through orifices and valves. It should be mentioned that the intention of this chapter is not to present the complete theoretical basics of fluid motion in hydraulics (or fluid mechanics); it rather summarises those equations and concepts which will be required in the subsequent chapters. For more details and information the reader is always referred to the corresponding literature given throughout the whole chapter.

The writing of this chapter was largely influenced by the extensive discussions on flow through orifices and valves in the standard textbooks by Merritt (1967), and more recently the fine work by Beater (1999). Analyses and discussions on the material of this chapter can also be found, *e.g.*, in Findeisen and Findeisen (1994) and in the recent work by Will *et al.* (1999).

### 3.1 Physical Properties of Fluids

*Fluids* (liquids and gases) are bodies without their own shape; they can flow, *i.e.*, they can undergo great variations of shape under the action of forces; the weaker the force, the slower the variation (Lencastre, 1987).

The normal tension on the surface element of a fluid is called *pressure*. It is, at a given point, identical in all directions. Pressure can be calculated as

$$p = \frac{\text{Force}}{\text{Area}} = \frac{F}{A} \quad (3.1)$$

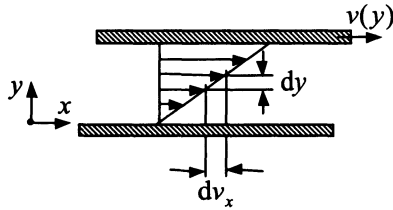
and thus has the dimensions of force per unit area ( $\text{N/m}^2$ ).

#### 3.1.1 Viscosity and Related Quantities

The coefficient of dynamic viscosity,  $\eta$ , is the parameter that represents the existence of tangential forces in liquids in movement. Suppose two plates (or fluid layers) are moving at a distance apart of  $dy$ , and at a relative velocity  $dv_x$  (see Figure 3.1.), then the shear stress

$$\tau = \frac{\text{Shear Force}}{\text{Area}} = \eta \frac{dv_x}{dy} \quad (\text{Newton}) \quad (3.2)$$

arises. Thereby,  $\eta$  is a proportionality factor and is called *dynamic viscosity*.



**Figure 3.1.** Couette flow; definition of shear stress

The coefficient of *kinematic viscosity*,  $\mu$ , is the ratio of the coefficient of dynamic viscosity to the fluid density, i.e.,

$$\mu = \frac{\text{Dynamic Viscosity}}{\text{Density}} = \frac{\eta}{\rho} \quad (3.3)$$

The dynamic viscosity of liquids varies considerably with the temperature:

$$\eta_\theta = \eta_0 e^{-\lambda_1(\theta - \theta_0)} \quad (3.4)$$

where  $\eta_0$  is the dynamic viscosity at reference temperature  $\theta_0$ . The viscosity–temperature coefficient  $\lambda_1$  should be determined by experiments for the fluid considered. For mineral oils, it lies between  $0.036$  and  $0.057 \text{ K}^{-1}$  (Ivantysyn and Ivantysynova, 1993).

The influence of pressure is given by

$$\eta = \eta_0 e^{\alpha p} \quad (3.5)$$

where  $\alpha$  is the viscosity–pressure coefficient that depends on the temperature; see Table 3.1 for the mineral oil HLP 32. For HFC fluids and HFD fluids the values of  $\alpha = 0.35 \text{ Pa}^{-1}$  and  $\alpha = 2.2 \text{ Pa}^{-1}$  can be used respectively (Ivantysyn and Ivantysynova, 1993). The effect of pressure on viscosity is not so important in practice.

**Table 3.1.** Viscosity–pressure coefficient for the mineral oil HLP 32 (Ivantysyn and Ivantysynova, 1993)

| $\theta [^\circ\text{C}]$ | $\alpha [10^{-2} \text{ Pa}^{-1}]$ |
|---------------------------|------------------------------------|
| 0                         | 3.268                              |
| 10                        | 2.900                              |
| 20                        | 2.595                              |
| 30                        | 2.339                              |
| 40                        | 2.121                              |
| 50                        | 1.933                              |
| 60                        | 1.770                              |
| 70                        | 1.626                              |
| 80                        | 1.499                              |
| 90                        | 1.385                              |
| 100                       | 1.283                              |

### 3.1.2 Mass Density, Bulk Modulus and Related Quantities

Mass density,  $\rho$ , or simply *density* is the mass contained in a unit volume:

$$\rho = \frac{\text{Mass}}{\text{Volume}} = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV} \quad (3.6)$$

The density of hydraulic fluids normally lies between 0.85 and 0.91 kg/dm<sup>3</sup>. Indeed, the density of hydraulic fluids is a function of both pressure and temperature, *i.e.*,  $\rho = \rho(p, \theta)$ . The first three terms of a Taylor's series for two variables may be used as an approximation (Merritt, 1967):

$$\begin{aligned} \rho &\approx \rho_0 + \left( \frac{\partial \rho}{\partial p} \right)_\theta (p - p_0) + \left( \frac{\partial \rho}{\partial \theta} \right)_p (\theta - \theta_0) \\ &= \rho_0 \left[ 1 + \frac{1}{E} (p - p_0) - \alpha (\theta - \theta_0) \right] \end{aligned} \quad (3.7)$$

where  $\rho$ ,  $p$  and  $\theta$  are respectively the mass density, pressure, and temperature of the fluid about the initial values  $\rho_0$ ,  $p_0$  and  $\theta_0$ . Equation 3.7 is the linearised equation of state for fluids. In hydraulic phenomena, the usual assumption of constant temperatures reduces the linearised state Equation 3.7 of fluids to the simple form (Merritt, 1967)

$$\rho = \rho_i + \frac{\rho_i}{E} p \quad (3.8)$$

where  $\rho_i$  is the mass density at zero pressure.

The quantity

$$E \equiv \rho_0 \left( \frac{\partial p}{\partial \rho} \right)_\theta = -V_0 \left( \frac{\partial p}{\partial V} \right)_\theta \quad (3.9)$$

is the change in pressure divided by the fractional change in volume at a constant temperature. It is called the modulus of elasticity, also termed the *isothermal bulk modulus* or simply *bulk modulus* of the liquid. It significantly influences the dynamics of hydraulic systems. For mineral oils, and for common pressures and temperatures ( $\theta \in [-40, 120]^\circ\text{C}$ ,  $p \leq 450$  bar), one may assume a mean value for the bulk modulus, typically

$$E_{\text{mineral oil}} = (14-16) \times 10^3 \text{ bar} = (1.4-1.6) \times 10^9 \text{ N/m}^2 = (1400-1600) \text{ MPa} \quad (3.10)$$

However, from a practical viewpoint, this is a very rough approximation, as the bulk modulus varies considerably with pressure, for instance, according to

$$E_{\text{isen}} = E + K_p p \quad (3.11)$$

Typical values are  $E = 16500$  bar and  $K_p = 9.558$ . The influence of temperature is negligible.

The quantity

$$\alpha \equiv -\frac{1}{\rho_0} \left( \frac{\partial \rho}{\partial \theta} \right)_p = \frac{1}{V_0} \left( \frac{\partial V}{\partial \theta} \right)_p \quad (3.12)$$

is the fractional change in volume due to a change in temperature at constant pressure. It is called the *cubical expansion coefficient* of the liquid.

In the literature, there are many formulae for the calculation of the density of hydraulic fluids as a function of temperature and pressure. For example, the density at atmospheric pressure ( $1 \text{ bar} = 10^5 \text{ N/m}^2$ ) and variable temperature  $\theta$  is given by

$$\rho_\theta = \frac{\rho_0}{1 + \beta_\theta(\theta - \theta_0)} \quad (3.13)$$

where  $\rho_0$  is the density at reference temperature  $\theta_0$  (say  $15^\circ\text{C}$ ), and  $\beta_\theta$  denotes the heat expansion factor, e.g.,  $0.65 \times 10^{-3} \text{ K}^{-1}$  for mineral oils,  $0.7 \times 10^{-3} \text{ K}^{-1}$  for HFC-fluids, and  $0.75 \times 10^{-3} \text{ K}^{-1}$  for HFD-fluids (Matthies, 1995). The density of hydraulic fluids following a change in pressure can be expressed as

$$\rho_p = \frac{\rho_0}{1 - \kappa_p(p - p_0)} \quad (3.14)$$

where  $\kappa_p$  is the compressibility factor

$$\kappa_p = \frac{1}{V_0} \left( \frac{\partial V}{\partial p} \right)_\theta \quad (3.15)$$

For variable temperature *and* variable pressure the density can be calculated by

$$\rho = \rho_\theta (1 + \kappa_p \Delta p) \quad (3.16)$$

Typical values for  $\kappa_p$  are:  $0.7 \times 10^{-4} \text{ bar}^{-1}$  for mineral oils,  $0.3 \times 10^{-4} \text{ bar}^{-1}$  for HFC-fluids, and  $0.35 \times 10^{-4} \text{ bar}^{-1}$  for HFD-fluids (Matthies, 1995). From these values, it can be concluded that the effect of pressure on the fluid density is minor, and thus negligible in practice.

### 3.1.3 Effective Bulk Modulus

The bulk modulus of a liquid is substantially lowered by entrained gas and mechanical compliance. According to Merritt (1967), estimates of entrapped air in hydraulic systems run as high as 20% when the fluid is at atmospheric pressure. As the pressure is increased, much of this air dissolves into the liquid and does not affect the bulk modulus. The major source of mechanical compliance may be the hydraulic lines connecting valves and pumps to actuators.

#### 3.1.3.1 Influence of Entrained Air

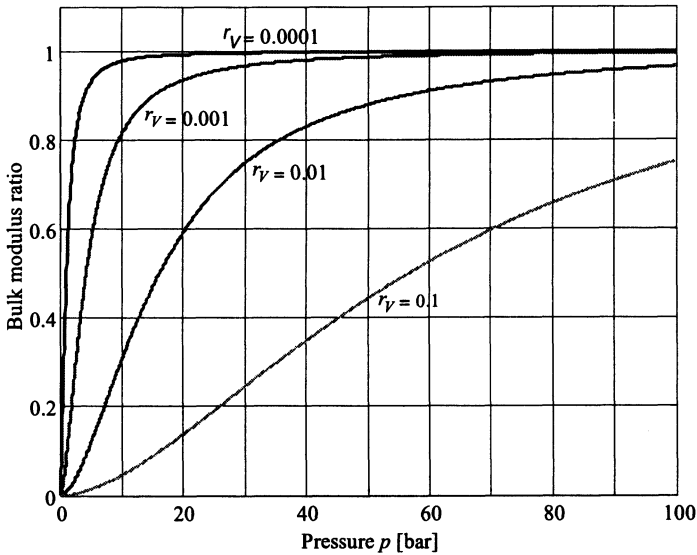
Some work has been done on determining the bulk modulus of liquid–air mixtures and that of containers due to mechanical compliance. Backé und Murrenhoff (1994:103) proposed the following formulae for the isentropic bulk modulus of liquid–air mixtures (see also Beater, 1999:26):

$$E'_{\text{isen}}(p) = E_{\text{isen}} \frac{1 + r_v}{1 + \left(\frac{p_0}{p}\right)^{1/\kappa} r_v \frac{E}{\kappa p}} \quad r_v \equiv \frac{V_{G0}}{V_{L0}} \quad (3.17)$$

with

- $E_{\text{isen}}$  isentropic bulk modulus of the liquid (without entrained air),
- $V_{G0}$  volume of gas entrained in the liquid at atmospheric pressure,
- $V_{L0}$  volume of the liquid at atmospheric pressure,
- $p_0$  atmospheric pressure ( $p_0 = 1$  bar),
- $p$  liquid pressure, and
- $\kappa$  isentropic exponent ( $\kappa = 1.4$ ).

Figure 3.2 shows traces of the ratio  $E'_{\text{isen}}(p)/E_{\text{isen}}$  for some values of the volume ration  $r_v$ . In Figure 3.3, specific pressure-dependence of bulk modulus is plotted.



**Figure 3.2.** Influence of entrained air volume on the isentropic bulk modulus

Especially in low-pressure regions (say,  $p \leq 100$  bar), the influence of entrained gas on the bulk modulus is substantial. At a pressure of about 0.6 bar, entrained air can explode (so-called Diesel effect). This effect cause highly undesired erosion defects, power losses, pressure peaks, and noise. This phenomenon is better known as cavitation (sudden implosion of gas bubbles, when fluid pressure decreases under vapour pressure) (Lemmen, 2002).

Note that the expressions given above require the accurate determination of many quantities (for example, of the volume of gas entrained in the liquid), and thus may be difficult to use in practice.



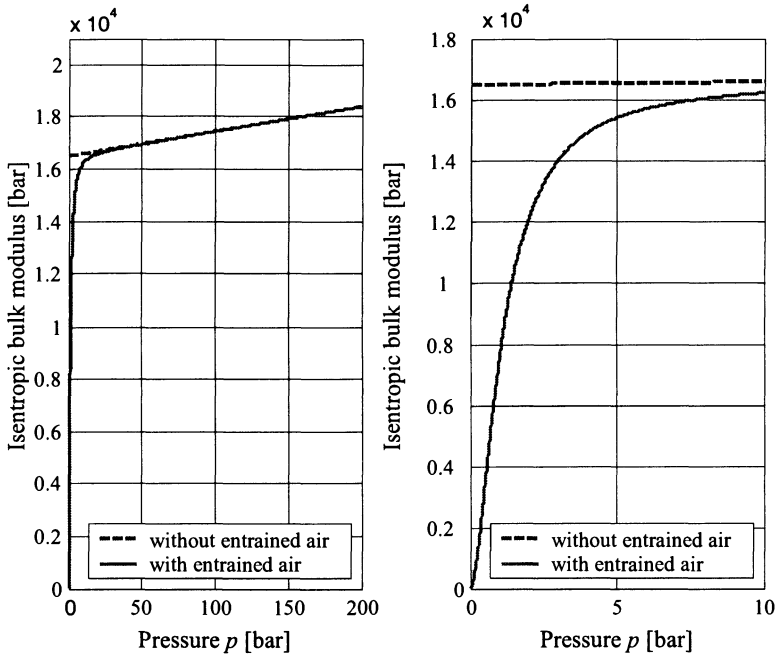


Figure 3.3. Typical trace of isentropic bulk modulus ( $r_v = 0.0001$ )

### 3.1.3.2 Influence of Mechanical Compliance

The bulk modulus of cylindrical pipelines can be calculated as (Theißen, 1983)

$$E' = E \frac{1}{1 + \frac{E}{E_p} w} \quad (3.18)$$

where  $E_p$  is the bulk modulus of the (steel) pipeline. For thick-walled pipelines the coefficient  $w$  is given by

$$w = \frac{2 \left( \frac{d_o}{d_i} \right)^2 (1 + \nu) + 3(1 - 2\nu)}{\left( \frac{d_o}{d_i} \right)^2 - 1} \quad (3.19)$$

with

- $d_o$  outer diameter,
- $d_i$  inner diameter, and
- $\nu$  Poisson's number,  $\nu = 0.3$  for steel.

For thin-walled pipelines with the wall thickness  $s$ , ( $s/d_o < 0.1$ ), Equation 3.19 approximates to

$$w = \frac{d_i}{s} \quad (3.20)$$

Table 3.2 shows some results found by experiment. It can be seen that especially in the case of high-pressure rubber-based hoses, reinforced with interwoven metal threads, the influence of elasticity is considerable.

**Table 3.2.** Values for bulk modulus  $E'$  (Viersma, 1980)

| Nominal pressure [Mpa] | $E'$ [MPa] for steel pipeline<br>$R_i = 6.25$ mm; $R_o = 8$ mm | $E'$ [MPa] for high-pressure<br>hose $R_i = 6.25$ mm |
|------------------------|----------------------------------------------------------------|------------------------------------------------------|
| 5                      | 1460                                                           | 500                                                  |
| 9                      | 1510                                                           | 537                                                  |
| 13                     | 1570                                                           | 568                                                  |
| 22.5                   | 1890                                                           |                                                      |

### 3.1.3.3 Empirical Effective Bulk Modulus

Other researchers have derived empirical formulae for the calculation of the *effective bulk modulus*  $E'$ , including the effects of entrained air and mechanical compliance, based on direct measurements. The commonly used equation for calculation of the bulk modulus  $E'$  for hydraulic cylinders in German literature is that of Lee (1977):

$$E'(p) = a_1 E_{\max} \log \left( a_2 \frac{p}{p_{\max}} + a_3 \right) \quad (3.21)$$

with the parameters  $a_1 = 0.5$ ,  $a_2 = 90$ ,  $a_3 = 3$ ,  $E_{\max} = 18000$  bar, and  $p_{\max} = 280$  bar.

Hoffmann (1981) proposed the formula

$$E'(p) = E_{\max} \left[ 1 - \exp(-0.4 - 2 \times 10^{-7} p) \right] \quad (3.22)$$

with the pressure  $p$  in pascals.

According to Eggerth (1980), the effective bulk modulus can be expressed as

$$E'(p) = \frac{1}{k_1 + k_2 (p/p_0)^{-\lambda}} \quad (3.23)$$

with the parameters  $k_1$  and  $k_2$  in Table 3.3;  $p_0$  is assumed to be 10 bar.

**Table 3.3.** Parameters of Eggerth's formula (Beater, 1999)

| Temperature [°C] | $k_1 [10^{-10} \text{ m}^2/\text{N}]$ | $k_2 [10^{-10} \text{ m}^2/\text{N}]$ | $\lambda$ |
|------------------|---------------------------------------|---------------------------------------|-----------|
| 20               | 4.943                                 | 1.9540                                | 1.480     |
| 50               | 5.469                                 | 3.2785                                | 1.258     |
| 90               | 5.762                                 | 4.7750                                | 1.100     |

The relations for effective bulk modulus  $E'$  are plotted in Figure 3.4. Although these formulae are approximate, they are sufficient for design purposes. However, experimental data are always preferable.

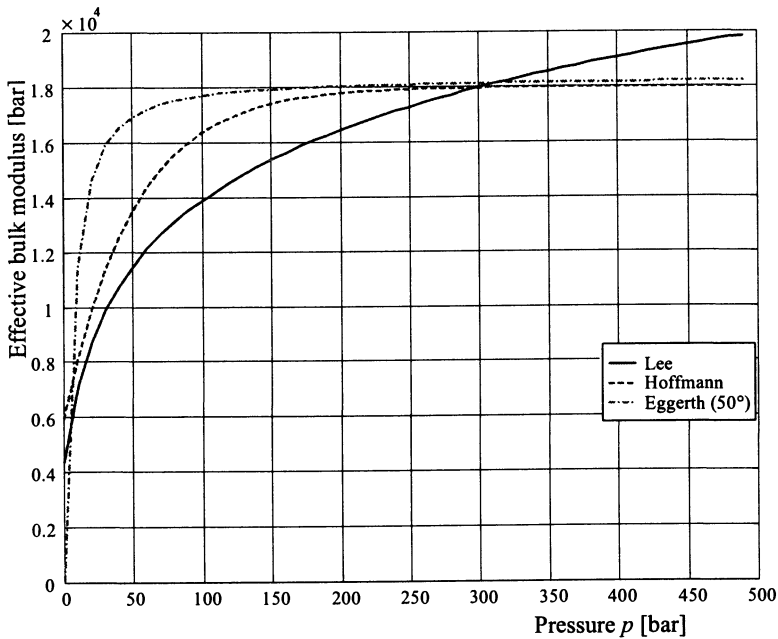


Figure 3.4. Comparison of different formulae for the calculation of  $E'$

### 3.1.4 Section Summary

The most important three physical properties (viscosity, density and bulk modulus) have been introduced and discussed. The following concluding statements are important from a practical viewpoint:

- Density can be considered constant.
- Viscosity of fluids varies markedly with temperature (Equation 3.4), and to a much lesser degree with pressure.
- Bulk modulus essentially depends on pressure, entrained air and mechanical compliance. Empirical formulae, such as Equations 3.21, 3.22, and 3.23, are recommended for the calculation of the effective bulk modulus. However, some parameter adjustments may be necessary in practice.

## 3.2 General Equations of Fluid Motion

In this section, the basic principles of conservation and laws governing fluid flow and associated phenomena will be briefly summarised. More detailed derivations can be found in a number of standard textbooks on fluid mechanics (e.g., Slattery, 1972; White, 1986; Lencastre, 1987; Spurk, 1996; Oertel, 1999). Detailed theoretical development and discussions about the conservation laws in fluid mechanics with a special “nice” section (4.3) on pipeline hydraulics can be found in the excellent book by Truckenbrodt (1996).

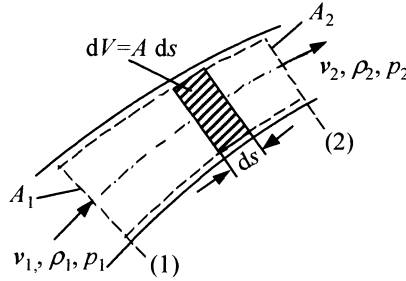
Conservation laws can be derived by considering a given quantity of matter, control mass or control volume, and its extensive properties, such as mass, momentum and energy. In fluid mechanics, there are several ways to present the conservation equations, such as the control mass approach, the control volume approach and the control tube approach.

### 3.2.1 Continuity Equation and Pressure Transients

Consider a control tube as depicted in Figure 3.5. The integral form of the mass conservation (continuity) equation can be formulated as (Truckenbrodt, 1996)

$$\int_{(1)}^{(2)} \frac{\partial(\rho A)}{\partial t} ds + \rho_2 v_2 A_2 - \rho_1 v_1 A_1 = 0 \quad (\text{control tube}) \quad (3.24)$$

where the density  $\rho = \rho(t, s)$  is, in general, not constant.



**Figure 3.5.** Definition of control tube

For incompressible fluids, *i.e.*,  $\rho = \text{const.}$  (which is a standard assumption in hydraulics), Equation 3.24 can be reduced to

$$v_1(t) A_1 = v_2(t) A_2 \quad (\rho = \text{const.}) \quad (3.25)$$

or more generally

$$Q = v(t) A = \text{const.} \quad (\text{volume flow}) \quad (3.26)$$

For steady flow, the continuity equation can be expressed as

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2 \quad (\rho \neq \text{const.}) \quad (3.27)$$

or in the general form

$$\dot{m} = \rho v A = \text{const.} \quad (\text{mass flow rate}) \quad (3.28)$$

Next, the mass conservation equation is written in the differential coordinate-free form (for a control tube element of length  $ds$ ; see Figure 3.5)

$$\frac{\partial \rho}{\partial t} + \text{div}(\rho v) = 0 \quad (3.29)$$

Again, two special cases can be given:

$$\operatorname{div} \mathbf{v} = 0 \quad (\rho = \text{const.}) \quad (3.30)$$

$$\operatorname{div}(\rho \mathbf{v}) = 0 \quad (\text{steady flow}) \quad (3.31)$$

Given a coordinate system (Cartesian, cylindrical or spherical), Equation 3.29 can take a specific form by providing the expression for the divergence ( $\operatorname{div}$ ) operator in that coordinate system. The expression of the continuity equation in a cylindrical coordinate system  $(r, \varphi, x)$  is given by (Truckenbrodt, 1996)

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \left( \frac{\partial(\rho r v_r)}{\partial r} + \frac{\partial(\rho v_\varphi)}{\partial \varphi} \right) + \frac{\partial(\rho v_x)}{\partial x} = 0 \quad (3.32)$$

Consider again the mass conservation equation for a control volume  $V$  and let the accumulated or stored mass of fluid inside be  $m$  with a mass density of  $\rho$ . Since all fluid must be accounted for, as the medium is assumed continuous, the rate at which mass is stored must equal the incoming mass flow rate minus the outgoing mass flow rate. Therefore, we can write

$$\sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} = \frac{d(\rho V)}{dt} = \rho \dot{V} + V \dot{\rho} \quad (3.33)$$

Taking into account Equation 3.8 and dividing Equation 3.33 by  $\rho$  leads to

$$\sum Q_{\text{in}} - \sum Q_{\text{out}} = \dot{V} + \frac{V}{E} \dot{p} \quad (3.34)$$

If the volume is fixed ( $V = V_0$ ), Equation 3.33 becomes

$$\dot{p} = \frac{E}{V_0} (\sum Q_{\text{in}} - \sum Q_{\text{out}}). \quad (3.35)$$

This equation is fundamental for the description of the pressure dynamics in hydraulic compartments.

### 3.2.2 Navier–Stokes Equation

The momentum conservation equation is known as the *Navier–Stokes equation* (in differential form) (Lencastre, 1987)

$$\rho \frac{d\mathbf{v}}{dt} = \rho \mathbf{g} - \operatorname{grad} p + \eta \left[ \operatorname{div}(\operatorname{grad} \mathbf{v}) + \frac{1}{3} \operatorname{grad}(\operatorname{div} \mathbf{v}) \right] \quad (3.36)$$

where

- $\rho \mathbf{g}$  represents the body forces.
- $\rho d\mathbf{v}/dt$  the inertial forces.
- $\operatorname{grad} p$  is the vector of components  $\partial p / \partial x_i$ . It corresponds to the derivative or inclination of the pressure in the direction of the flow.
- The term  $\eta \operatorname{div}(\operatorname{grad} \mathbf{v})$  represents diffusion of the vector  $\mathbf{v}$  within the flow. It represents the action of one particle on the others owing to the effect of viscosity.
- The term  $1/3 \eta \operatorname{grad}(\operatorname{div} \mathbf{v})$  represents the influence of compressibility and vanishes in the case of incompressible liquids.

Letting the fluid be incompressible and dividing Equation 3.36 throughout by  $\rho$  leads to

$$\frac{d\mathbf{v}}{dt} = \mathbf{g} - \frac{1}{\rho} \text{grad } p + \mu \text{div}(\text{grad } \mathbf{v}) \quad (3.37)$$

Given the hypothesis of the external forces being derived from a potential  $\xi$ , then  $\mathbf{g} = \text{grad } \xi$ . Thus, in the case of incompressible liquids, Equation 3.37 becomes

$$-\text{grad } \xi + \frac{1}{\rho} \text{grad } p = -\frac{d\mathbf{v}}{dt} + \mu \text{div}(\text{grad } \mathbf{v}) \quad (3.38)$$

If the potential is that of gravity, *i.e.*,  $\xi = -gz$ , then dividing throughout by  $g$  gives:

$$\text{grad} \left( z + \frac{p}{\gamma} \right) = -\frac{1}{g} \frac{d\mathbf{v}}{dt} + \frac{\eta}{\gamma} \text{div}(\text{grad } \mathbf{v}) \quad (3.39)$$

with  $\gamma = \rho g$ . In the case of a perfect or ideal liquid, *i.e.*,  $\eta = 0$ , which does not exist in reality, Equation 3.39 becomes

$$\text{grad} \left( z + \frac{p}{\gamma} \right) = -\frac{1}{g} \frac{d\mathbf{v}}{dt} \quad (3.40)$$

For a fluid element along the path line, Equation 3.39 can be rearranged, taking into account  $\eta = \rho\mu$  and  $\gamma = \rho g$ , to give *Euler's equation*

$$\frac{d}{ds} \left( z + \frac{p}{\gamma} \right) = -\frac{1}{g} \frac{dv}{dt} + \frac{\mu}{g} \text{div}(\text{grad } \mathbf{v}) \quad v = v_s \quad (3.41)$$

Let us now return to the Navier–Stokes Equation 3.37 and write the system in cylindrical coordinates (Truckenbrodt, 1996):

$$\frac{dv_r}{dt} = g_r - \frac{1}{\rho} \frac{\partial p}{\partial r} + \mu \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial v_r}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \varphi^2} + \frac{\partial^2 v_r}{\partial x^2} - \frac{2}{r^2} \frac{\partial v_\varphi}{\partial \varphi} - \frac{v_r}{r^2} \right] \quad (3.42)$$

$$\frac{dv_\varphi}{dt} = \frac{1}{r} \left( g_\varphi - \frac{1}{\rho} \frac{\partial p}{\partial \varphi} \right) + \mu \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial v_\varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_\varphi}{\partial \varphi^2} + \frac{\partial^2 v_\varphi}{\partial x^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \varphi} - \frac{v_\varphi}{r^2} \right] \quad (3.43)$$

$$\frac{dv_x}{dt} = g_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \mu \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial v_x}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_x}{\partial \varphi^2} + \frac{\partial^2 v_x}{\partial x^2} \right] \quad (3.44)$$

### 3.2.3 Bernoulli's Theorem

Considering Equation 3.41, and bearing in mind that

$$\frac{dv}{dt} = \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s} \quad (\text{substantial acceleration})$$

$$v \frac{\partial v}{\partial s} = \frac{\partial (v^2 / 2)}{\partial s} \quad (3.45)$$

it follows that

$$\frac{\partial}{\partial s} \left( z + \frac{p}{\gamma} + \frac{v^2}{2g} \right) = -\frac{1}{g} \frac{dv}{dt} + \frac{\mu}{g} \operatorname{div}(\operatorname{grad} v) \quad (3.46)$$

The first element of the equation has an essentially global energy significance. It represents the variation in the total energy discharged per unit weight of a particle along its trajectory.

If the viscosity terms are removed from the equation, *i.e.*, the flow may resemble a perfect fluid, we have

$$\frac{\partial}{\partial s} (gz + gp / \gamma + v^2 / 2) = -\frac{1}{g} \frac{\partial v}{\partial t} \quad (3.47)$$

In the case of steady flow,  $\partial v / \partial t = 0$ , energy conservation holds:

$$E = z + \frac{p}{\gamma} + \frac{v^2}{2g} = \text{const.} \quad (3.48)$$

this being the expression that represents *Bernoulli's theorem* for one-dimensional steady flows. In the case of an incompressible liquid in steady flow, in which the friction forces and, consequently, energy losses can be disregarded, the total energy of a particle is maintained along its trajectory.

### 3.2.4 Section Summary

From the variety of formulae presented in this section, those most important and those most often needed in practice are the continuity equations for incompressible fluids (Equations 3.25–3.28) and Bernoulli's Equation 3.48 (or other equivalent variants of it). Another often-used equation is the fundamental Equation 3.35 for the description of the pressure dynamics in hydraulic compartments.

The general continuity equation and Navier–Stokes equations are only interesting for the analysis of pipeline dynamics, see Section 4.2.5.

## 3.3 Flow Through Passages

Two distinct types of fluid flow through passages can occur:

- *Laminar* or *viscous* flow, in which each fluid particle describes a well-defined trajectory, with a velocity only in the direction of the flow.
- *Turbulent* or *hydraulic* flow (this being the most usual in hydraulic phenomena), in which each particle, apart from the velocity in the direction of the flow, is animated by fluctuating cross-current velocities.

The Reynolds number, *Re*, defined by

$$\text{Re} = \frac{\rho v d_h}{\eta} = \frac{v d_h}{\mu} \quad (3.49)$$

is the characteristic parameter: for lower values of  $\text{Re}$ , the flow is laminar; for higher values the flow is turbulent. Thereby,  $v$  is the average velocity of flow.  $d_h$  represents the hydraulic diameter, which is defined by

$$d_h = \frac{4A}{S} \quad (3.50)$$

where  $A$  is the flow section area and  $S$  is the flow section perimeter. For each flow case, the characteristic length is agreed upon and empirical values are obtained for the Reynolds number which describes transition from that viscosity- to inertia-dominated flows.

### 3.3.1 Flow Establishment in Pipelines

One basic element of hydraulic systems is cylindrical pipelines, in which flow may be *laminar* or *turbulent*. The characteristic length to be used for the Reynolds number is inside pipeline diameter  $d$ , *i.e.*,

$$\text{Re} = \frac{v d}{\mu} \quad (3.51)$$

The transition from laminar to turbulent flow has been observed experimentally to occur in the range  $2000 < \text{Re}_{\text{crit}} < 4000$ , typically  $\text{Re}_{\text{crit}} = 2300$ . Below  $\text{Re} = 2300$  the flow is always laminar; above  $\text{Re} = 4000$  the flow is usually, but not always, turbulent. It is possible to have laminar flow at Reynolds number considerably above 4000 if extreme care is taken to avoid disturbances which would lead to turbulence. However, these instances are exceptional, and the upper limit of 4000 is a good rule (Viersma, 1980).

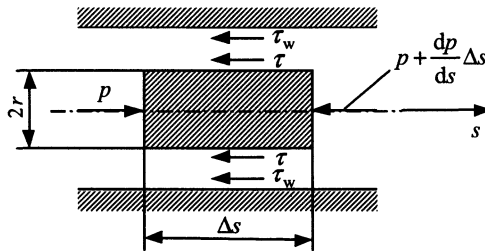


Figure 3.6. Force equilibrium of fluid elements in cylindrical pipelines

#### 3.3.1.1 Hagen–Poiseuille Law

Consider a cylindrical pipeline of radius  $r \leq R$  and let the flow be steady and laminar. The starting point is the force equilibrium in the axial direction (Figure 3.6); that means



$$\tau = -\frac{r}{2} \frac{dp}{dx} \quad \tau_w = -\frac{R}{2} \frac{dp}{dx} \quad \frac{\tau}{\tau_w} = \frac{r}{R} \quad (3.52)$$

where  $\tau_w$  is the shear stress at the pipeline wall (*i.e.*, at  $r = R$ ). On the other hand, the shear stress Equation 3.2 can be written as ( $v \equiv v_x$ ,  $dy \equiv -dr$ )

$$\tau = -\eta \frac{dv}{dr} = -\frac{r}{2} \frac{dp}{dx} \quad (3.53)$$

Equations 3.52 and 3.53 can then be combined to obtain

$$\frac{dv}{dr} = \frac{r}{2\eta} \frac{dp}{dx} \quad (3.54)$$

the relationship for calculating the velocity profile for laminar flows in cylindrical pipelines. In fact, the integration of Equation 3.54 (with  $dp/dx = \text{const.}$ , and  $v(R) = 0$ ) yields the velocity profile (Figure 3.7)

$$v(r) = -\frac{1}{4\eta} (R^2 - r^2) \frac{dp}{dx} \quad (3.55)$$

and leads to the maximum velocity and the mean velocity

$$v_{\max} = v(0) = -\frac{R^2}{4\eta} \frac{dp}{dx} \quad \bar{v} = -\frac{R^2}{8\eta} \frac{dp}{dx} \quad (3.56)$$

respectively.

Finally, the continuity Equation 3.26 and Equation 3.56 are combined to give the so-called *Hagen–Poiseuille equation*

$$Q = A\bar{v} = -\frac{\pi R^4}{8\eta} \frac{dp}{dx} \quad (3.57)$$

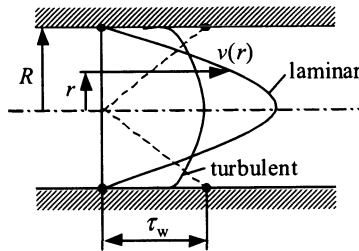


Figure 3.7. Velocity profiles for laminar and turbulent flows in a cylindrical pipeline

### 3.3.2 Flow Through Orifices

Orifices are sudden restrictions of short length (ideally zero length for a sharp-edged orifice) in the flow passage and may have a fixed or variable area (see Figure 3.8). Orifices are generally used to control flow, or to create a pressure differential

(valves). Two types of flow regime exist, depending on whether inertia or viscous forces dominate. The flow velocity through an orifice must increase above that in the upstream region to satisfy the law of continuity. At high Reynolds numbers, the pressure drop across the orifice is caused by the acceleration of the fluid particles from the upstream velocity to the higher jet velocity. At low Reynolds numbers, the pressure drop is caused by the internal shear forces resulting from fluid viscosity.

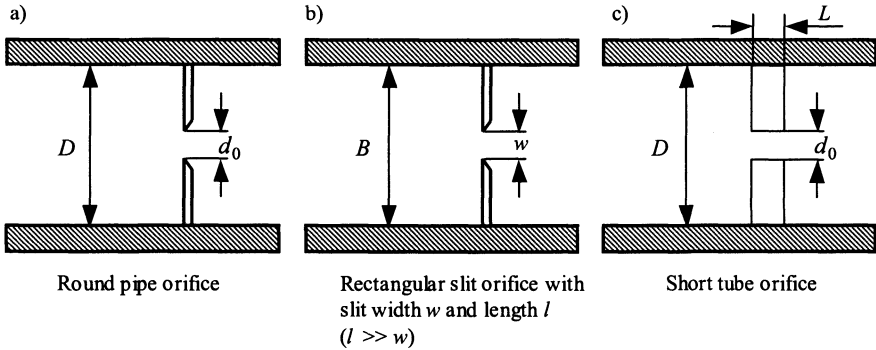


Figure 3.8. Round, slit-type and short tube orifices

### 3.3.2.1 Orifice Equations for Turbulent Flow

Since most orifice flows occur at high Reynolds numbers, this region is of major importance. Such flows are often referred to as “turbulent” (Figure 3.9b), but the term does not have quite the same meaning as in pipeline flow (Merritt, 1967). Referring to Figure 3.9a, the fluid particles are accelerated up to the jet velocity between sections 1 and 2. The flow between these sections is streamline or potential flow, and experience justifies the use of Bernoulli’s theorem in this region.

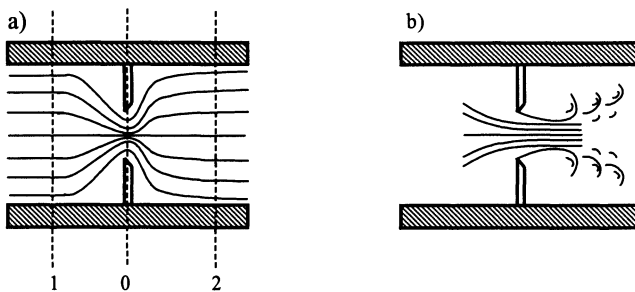


Figure 3.9. Flow through an orifice: (a) laminar flow; (b) turbulent flow

According to Bernoulli’s theorem (Equation 3.48), the total energy losses of the hydraulic flow are derived from the energy degraded into heat by friction of the particles against one another and by friction of the particles against the walls of the conduit. The energy dissipated due to friction between sections 1 and 2 will be equal to

$$\Delta p_{ls} = (p_1 + \rho v_1^2 / 2 + \rho g z_1) - (p_2 + \rho v_2^2 / 2 + \rho g z_2) \quad (3.58)$$

It is common to use the dimensionless pressure loss factor  $\zeta$ , which is defined as

$$\zeta = \frac{\Delta p_{ls}}{\rho v_1^2 / 2} \quad \text{or} \quad \Delta p_{ls} = \zeta \rho v_1^2 / 2 \quad (3.59)$$

The factor  $\zeta$  depends on the geometry of the conduit and on the Reynolds number which can be approximated by

$$\zeta(\text{Re}) = \frac{k_1}{\text{Re}} + k_2 \quad (3.60)$$

Taking into account that at a point far from the orifice

$$v_1 = v_2 = v \quad \text{and} \quad A_1 = A_2 = A = \frac{\pi}{4} d^2 = \text{const.} \quad (3.61)$$

we get the flow as the product of conduit area and the speed, *i.e.*,

$$Q = Av = A \sqrt{\frac{2}{\rho \zeta} (p_1 - p_2)} \quad (3.62)$$

Instead of Equation 3.62, it is common in the field of hydraulics to use the modified orifice equation

$$Q = \alpha_d A \sqrt{\frac{2}{\rho} \Delta p} \quad (3.63)$$

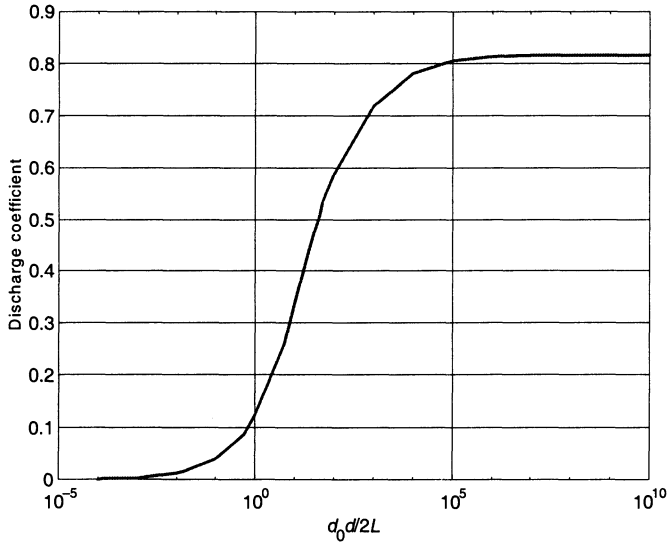
where  $\alpha_d$  is the discharge coefficient. Theoretically,  $\alpha_d = \pi/(\pi+2) = 0.611$  (von Mises, 1917). This can be used for all sharp-edged orifices regardless of the particular geometry, if the flow is turbulent and  $A_0 \ll A$ .

### 3.3.2.2 Discharge Coefficient for Turbulent Flow

Sharp-edged orifices (Figure 3.8) are desirable for their predictable characteristics and insensitivity to temperature changes. However, cost frequently prohibits their use, especially as fixed restrictors, and orifices with length (Figure 3.8c) are often employed instead. An average discharge coefficient for such short tube orifices can be expressed as (Merritt, 1967); see Figure 3.10:

$$\alpha_d = \left( 2.28 + 64 \frac{2L}{d_0 d} \right)^{-1/2} \quad \text{for} \quad \frac{d_0 d}{2L} \leq 50$$

$$\alpha_d = \left[ 1.5 + 13.74 \left( \frac{2L}{d_0 d} \right)^{1/2} \right]^{-1/2} \quad \text{for} \quad \frac{d_0 d}{2L} > 50 \quad (3.64)$$



**Figure 3.10.** Discharge coefficient  $\alpha_d$  for short tube orifices according to Equation 3.64

According to Lichtarowicz *et al.* (1965), the average discharge coefficient can be estimated using (see Figure 3.11)

$$\frac{1}{\alpha_d} = \sqrt{1 - \beta^4} \left[ \frac{1}{\alpha_{d,\max}} + \frac{20}{\text{Re}_h} \left( 1 + 2.25 \frac{L}{d_h} \right) - \frac{0.005 \frac{L}{d_h}}{1 + 7.5 (\log \text{Re}_h - 3.824)^2} \right] \quad (3.65)$$

where

$$\text{Re}_h = \sqrt{\frac{2\Delta p}{\rho}} \frac{d_h}{\mu} = \text{Re} \frac{\sqrt{1 - \beta^4}}{\alpha_d} \quad \beta = \frac{d_0}{d} \quad (3.66)$$

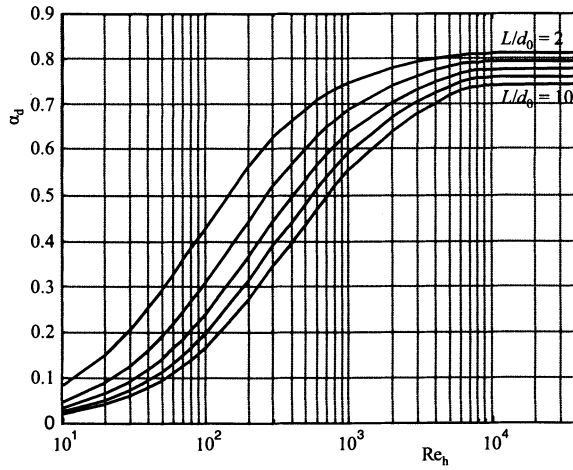
$$\alpha_{d,\max} = 0.827 - 0.0085 \frac{L}{d_h} \quad (3.67)$$

### 3.3.2.3 Discharge Coefficient for Turbulent–Laminar Flow

The formulae proposed above are only valid if turbulent flow occurs. Turbulence is ensured only at “large enough” Reynolds numbers:

$$\text{Re} = \frac{2vh}{\mu} \quad (3.68)$$

where  $h$  is the smallest dimension of the (rectangular) orifice. At low temperatures, low orifice pressure drops, and/or small orifice openings, the Reynolds number may become sufficiently low to permit laminar flow.



**Figure 3.11.** Discharge coefficient  $\alpha_d$  for short tube orifices according to Equation 3.65

Experiments carried out by Viersma (1980) proved that at very sharp edges in narrow orifices the critical value  $Re_{crit}$  is as low as 20, whereas slightly rounded off edges increased  $Re_{crit}$  to 80 or higher. Thus, at very sharp edges,  $\alpha_d$  may be assumed to be constant at  $Re > Re_{crit} \approx 20$ .

Although the analysis leading to Equation 3.63 is not valid at low Reynolds numbers, many attempts have been made to extend this equation to the laminar region by plotting the discharge coefficient as a function of Reynolds number, *i.e.*,

$$\alpha_d = \delta \sqrt{Re} \quad (3.69)$$

as pointed out by many investigators (Wuest, 1954; Viersma, 1980). The quantity  $\delta$  depends on geometry and is called the laminar flow coefficient. Viersma (1980) found that

$$\delta = \frac{\alpha_{d,turb}}{\sqrt{Re_{crit}}} = \frac{0.611}{\sqrt{10}} \approx 0.1932 \quad \text{hypothetical} \quad (3.70)$$

$$\delta = \frac{\alpha_{d,turb}}{\sqrt{Re_{crit}}} = \frac{0.611}{\sqrt{20}} \approx 0.1366 \quad \text{for sharp edges} \quad (3.71)$$

$$\delta = \frac{\alpha_{d,turb}}{\sqrt{Re_{crit}}} = \frac{0.611}{\sqrt{80}} \approx 0.0683 \quad \text{for slightly rounded-off edges} \quad (3.72)$$

The discharge coefficient can then be represented by the asymptotes shown in Figure 3.12 in the laminar region and  $\alpha_d = 0.611$  in the turbulent region.

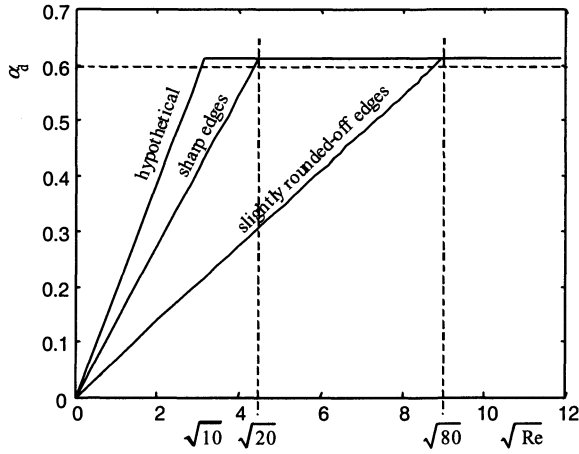


Figure 3.12.  $\alpha_d = f(\sqrt{Re})$  according to Viersma (1980)

### 3.3.2.4 Orifice Equations for Laminar Flow

Expressing the Reynolds number as

$$Re = \frac{(Q/A_0)d_h}{\mu} \quad (3.73)$$

and substituting Equations 3.73 and 3.69 into Equation 3.63 yields

$$Q = \frac{2\delta^2 d_h}{\mu\rho} A_0 \Delta p \quad (3.74)$$

for low Reynolds numbers.

Wuest (1954) has theoretically determined expressions for laminar flow through sharp-edged circular orifices (in an infinite plane, *i.e.*,  $d_0 \ll d$  in Figure 3.8a) as

$$Q = \frac{\pi d_0^3}{50.4\mu\rho} \Delta p \quad (3.75)$$

and through sharp-edged rectangular slits (of height  $b_0$  and width  $w$  in an infinite plane, *i.e.*,  $b_0 \ll B$  in Figure 3.8b, with  $w \gg b_0$ )

$$Q = \frac{\pi b_0^2 w}{32\mu\rho} \Delta p \quad (3.76)$$

Equating Equation 3.74 to Equations 3.75 and 3.76 gives  $\delta = 0.2$  for a sharp-edged round orifice and  $\delta = 0.157$  for a sharp-edged slit orifice.

### 3.3.3 Flow Through Valves

Flows through valving orifices (Figure 3.13) are usually described by the orifice Equation 3.63 with a linear relationship between the valve spool position  $x_v$  and the flow area (critical centre), *i.e.*,

$$Q = Q(x_v, \Delta p) = c_v x_v \sqrt{p_1 - p_2} = c_v x_v \sqrt{\Delta p} \quad (3.77)$$

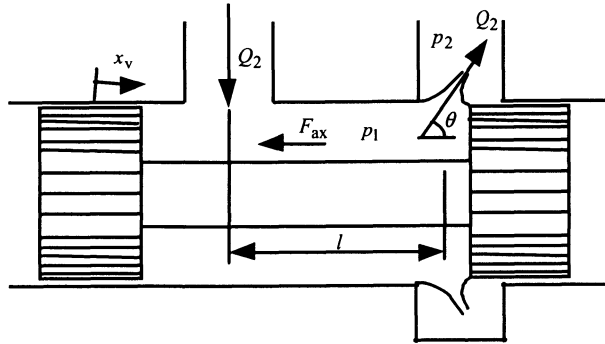
with the flow coefficient

$$c_v = \pi d_v \alpha_d \sqrt{\frac{2}{\rho}} \quad (3.78)$$

for servo-valves ( $d_v$ : diameter of the valve spool), and

$$c_v = 4|x_v| \tan(\alpha/2) \alpha_d \sqrt{\frac{2}{\rho}} \quad (3.79)$$

for special proportional valves with triangular valve seats having groove angle  $\alpha$  (see Köckemann *et al.*, 1991).



**Figure 3.13.** Axial flow force on spool due to unequal jet angles

Note that  $c_v$  is usually given in

$$\frac{\text{dm}^3}{\text{min} \sqrt{\text{bar mm}}} \quad \text{or equivalently} \quad \frac{\text{m}^3}{\text{s} \sqrt{\text{N}}} \quad (3.80)$$

Note also that Equation 3.77 can be written using the valve voltage  $u_v$  as

$$Q = Q(u, \Delta p) = c_v \frac{x_{v,\max}}{u_{\max}} u \sqrt{p_1 - p_2} \equiv c_{vu} u \sqrt{\Delta p} \quad (3.81)$$

This means that the value and the dimension of  $c_v$  have to be adapted to the signal used (which can be the valve stroke  $x_v$ , the valve voltage  $u_v$ , or the valve current  $I_v$ ). In the rest of this book, only the symbol  $c_v$  will be used without regard to the nature of the valve signal (normalised or not).

In practice, the flow coefficient may best be determined experimentally, or it may be calculated using the catalogue data ( $Q_N$ ,  $\Delta p_N$ , and  $x_{v,\max}$ ) of the valve manufacturer

$$c_v = \frac{Q_N}{\sqrt{\Delta p_N / 2}} \frac{1}{x_{v,\max}} \quad (3.82)$$

where  $Q_N$  is the nominal flow,  $\Delta p_N$  the nominal pressure drop, and  $x_{v,\max}$  the maximum stroke of the valve. The corresponding discharge coefficient is

$$\alpha_d = \frac{Q_N}{A(x_{v,\max}) \sqrt{\Delta p_N / \rho}} \quad (3.83)$$

Since Equation 3.77 is not valid for low Reynolds numbers, Feigel (1987a) derived the following flow equation to be used for laminar–turbulent valve flow cases

$$Q = c_v x_v \left[ \sqrt{\left( \frac{c_{lt}}{x_v} \right)^2 + \Delta p} - \frac{c_{lt}}{x_v} \right] \quad (3.84)$$

It is assumed that  $A = \pi d x_v$ , and the introduced laminar–turbulent flow coefficient is calculated by

$$c_{lt} = \frac{\delta \mu \sqrt{\rho \text{Re}_{\text{crit}}}}{4\sqrt{2}\delta^2} \quad (3.85)$$

where  $\delta$  is the slope of the curve  $\alpha_d = f(\sqrt{\text{Re}})$  according to Equation 3.69. As  $x_v \rightarrow \infty$  Equation 3.84 becomes Equation 3.77, and thus may also be used for turbulent valve flow. A typical value of  $c_{lt}$  is 0.6 (Saffe, 1986).

Yet another approximation formula for the discharge coefficient that has been used by many researchers (e.g., Klein, 1993), and which considers the dependence on the valve spool position  $x_v$ , is given by

$$\alpha_d = \alpha_d(x_v) = \alpha_{d0} \left( 1 - K_{d,\text{corr}} \frac{|x_v|}{x_{v,\max}} \right) \quad (3.86)$$

where  $\alpha_{d0}$  is the basic discharge coefficient,  $K_{d,\text{corr}}$  a correction factor, and  $x_{v,\max}$  the maximum spool displacement. Typical values are  $\alpha_{d0} = 0.65$ , and  $K_{d,\text{corr}} = 0.32$ .

Finally, a generalised expression for the flow through valve orifices reads:

$$Q = Q(x_v, \Delta p) = \alpha_d A(x_v) \sqrt{2/\rho} \sqrt{\Delta p} \quad (3.87)$$

where  $A(x_v)$  is the area of the valve orifice.  $A(x_v)$  depends on the orifice geometry (i.e., geometrical form of the orifice and centre type), which varies from one manufacturer to another, especially for proportional valves.



### 3.3.4 Section Summary

Orifice flow is laminar for  $Re < Re_{crit}$  with flow rates directly related to pressure drop as given by Equation 3.74. In the vicinity of  $Re_{crit}$ , both inertia and viscosity are important. For  $Re > Re_{crit}$ , the flow can be treated as turbulent and is described by the orifice Equation 3.63. Commonly, the orifice Equation 3.63 is most used for all situations with a total disregard for the types of flow that can be encountered. This is justified in the majority of cases, but it can lead to gross errors in certain instances. Typical and realistic values of  $\alpha_d$  lie between 0.65 and 0.75.

The more practical way is to apply the orifice Equation 3.77 with the calculation of the flow coefficient according to Equation 3.82 (or equivalently Equation 3.83).

## 3.4 Spool Port Forces

Closely related to the flows through the spool ports is the axial force on the spool. This flow force is caused by the change of momentum of the flow, due to a difference in jet angles for the inlet flows and outlet flows, as depicted in Figure 3.13.

The steady-state axial flow force on the spool can be calculated by (Merriitt, 1967; Lausch, 1990)

$$F_{ax,steady} = 2\alpha_d^2 A(x_v) \cos\theta \Delta p \quad (3.88)$$

The jet angle  $\theta$  can be assumed constant, namely  $\theta \approx 69^\circ$ , leading to  $\cos\theta = 0.358$ , which corresponds to the theoretical value (if there is no radial clearance between the valve spool and sleeve) derived by von Mises (1917).

Feigel (1992) proposed the following formulae for the calculation of steady-state flow forces on uncompensated spool valves:

$$F_{ax,steady} = K_f Q \sqrt{\Delta p} \quad (3.89)$$

with  $K_f = 0.077$  [N min/(dm<sup>3</sup> bar)] for one-edge valves,  $K_f = 0.054$  for two-edge valves, and  $K_f = 0.109$  for tetragonal valves. Equation 3.89 is combined with Equation 3.77 to give

$$F_{ax,steady} = K_f c_v \Delta p x_v \quad (3.90)$$

Thus far the discussion has considered only the steady-state flow force. If the slug of fluid in the valve chamber is accelerated, then a force is produced which reacts on the face of the spool valve lands. The magnitude of dynamic flow force is given by Newton's second law as

$$F_{ax,dyn} = ma = \rho l \dot{Q} \quad (3.91)$$

With Equation 3.77, the dynamic flow force becomes

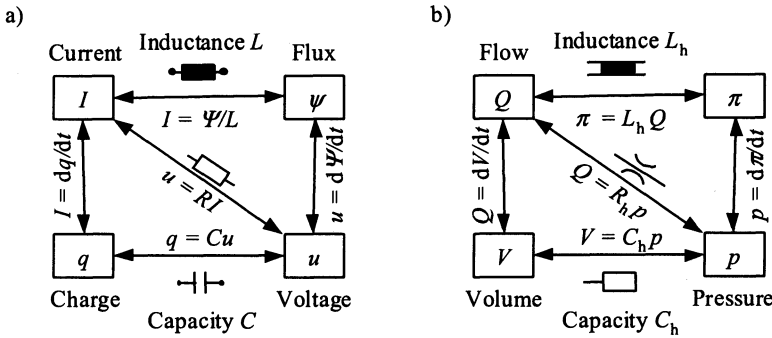
$$F_{ax,dyn} = \rho l c_v \Delta p \dot{x}_v + \rho l c_v x_v \frac{\Delta \dot{p}}{2\sqrt{\Delta p}} \quad (3.92)$$

Therefore, the dynamic flow force is proportional to spool velocity and pressure changes. The velocity term is the more significant because it represents a damping force; the pressure rate term is usually neglected.

In practice, the axial spool forces do not seem to play any significant role for the valve manufacturers. Although several compensation techniques to reduce or eliminate these forces have been investigated (see Merritt, 1967, and Feigel, 1992), none has found wide acceptance by practitioners. The practical solution to this problem is to use a two-stage servo-valve, in which the pilot stage, usually a flapper-nozzle valve, provides an appropriate force to stroke the main-stage spool valve.

### 3.5 Electro-hydraulic Analogy

The principles of electro-hydraulic analogy are summarised in Figure 3.14.



**Figure 3.14.** Relationships between variables in (a) electrical and (b) hydraulic systems (Beater, 1999)

#### 3.5.1 Hydraulic Capacitance

Equation 3.35 can be written as

$$Q = \frac{V_0}{E} \dot{p} = C_h \dot{p} \quad (3.93)$$

The proportionality factor  $C_h$  is referred to as the *hydraulic capacitance*

$$C_h \equiv - \left( \frac{\partial V}{\partial p} \right)_\vartheta = \frac{V_0}{E} \quad (3.94)$$

in analogy to the capacitance of a capacitor in electrical circuits.

#### 3.5.2 Hydraulic Resistance

The *hydraulic resistance*  $R_{h,L}$  for laminar flow can, for instance, be determined from the Hagen–Poiseuille Equation (3.57)

$$Q = \frac{\pi R^4}{8\eta l} \Delta p \quad (3.95)$$

for cylindrical pipelines of radius  $R$  and length  $l$ . Equation 3.95 gives

$$R_{h,l} = \frac{\Delta p}{Q} = \frac{8\eta l}{\pi R^4} \quad (3.96)$$

Since in this case the pressure drop is directly proportional to the flow, a resistance given by this equation is called a linear resistance to motion.

In general, however, the hydraulic resistance is non-linear to motion, *e.g.*, due to the square-root function in Equation 3.63, which can be rearranged to yield:

$$R_{h,l} = \frac{\Delta p}{Q^2} = \frac{\rho}{2\alpha_d^2 A^2} \quad (3.97)$$

or from Equation 3.77:

$$R_{h,l} = \frac{\Delta p}{Q^2} = \frac{1}{c_v^2 x_v^2} \quad (3.98)$$

In practice, it is preferable to work with the  $\Delta p$  vs.  $Q$  characteristic (or flow–pressure function) established by measurements on the actual valve in question. The manufacturers of standardised valves usually present this characteristic in their catalogues.

### 3.5.3 Hydraulic Inductance

The combination of Newton's law

$$F_a = ma = A l \rho a \quad (3.99)$$

the continuity equation

$$Q = Av \Rightarrow a = \dot{v} = \frac{\dot{Q}}{A} \quad (3.100)$$

and

$$F_a = A \Delta p \quad (3.101)$$

leads to the hydraulic inductance

$$L_h = \frac{\Delta p}{\dot{Q}} = \frac{l\rho}{A} \quad (3.102)$$

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## CHAPTER 4

# PHYSICALLY BASED MODELLING

---

This chapter is devoted to the derivation of physical models for hydraulic servo-systems and includes the most relevant dynamic and non-linear effects that are involved in hydraulic servo-systems. The model derivation is based on the physical fundamentals described in Chapter 3 and on first principles. Many assumptions and simplifications must be made to keep the model complexity as low as possible.

First, Section 4.2 introduces the elementary non-linear models of hydraulic servo-systems. A typical overall model is summarised in Section 4.3, and it is formulated as a state-space model. Model simplifications are proposed and linearised models are derived in Section 4.4. Finally, Section 4.5 reviews some principles for determining the main physical model parameters.

The development of this chapter is partly based on the work by Merritt (1967), Saffe (1986), Lierschaft (1993), Pawlik (1994), Ebner (1995), Lemmen (1995), Heintze (1997), van Schothorst (1997), Bernzen (1999a,b), as well as a large number of papers cited throughout the text.

## 4.1 Introduction

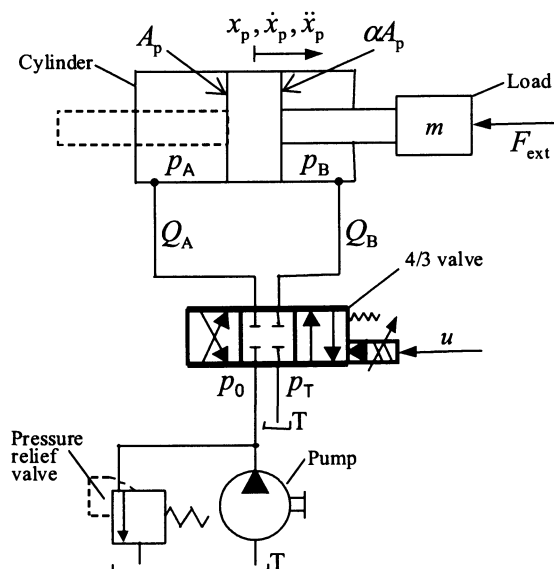
Among the wide variety of hydraulic actuator and valve combinations (see Chapter 2), the valve–cylinder configuration shown in Figure 4.1 is considered. This mainly consists of a proportional valve or servo-valve and a differential cylinder (also known as a single rod; solid lines only in Figure 4.1), referred to as an  $A+\alpha A$  configuration (Backé, 1992), and is probably the most widely used valve–actuator combination due to its compact size.

As a special case for  $\alpha = A_B/A_A = 1$  the synchronising cylinder (mostly known as a double-rod or dual-rod cylinder) is represented by dashed and solid lines in Figure 4.1. From a modelling point of view, almost any flow-controlled hydraulic servo-system can be reduced to this basic valve–cylinder configuration.

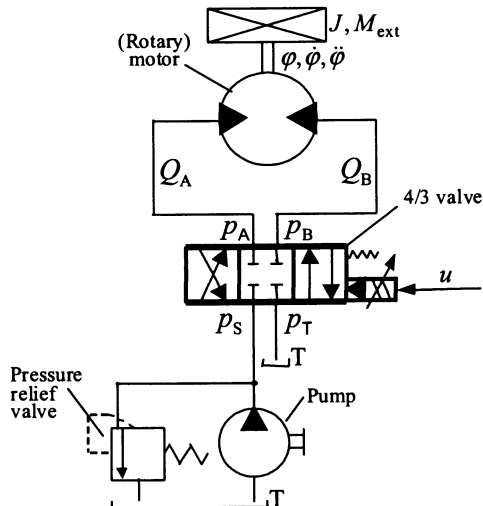
Note that the model derivation for valve–(rotary) motor configurations closely parallels that given for valve–cylinder combinations; see Table 4.1 and Figure 4.2.

**Table 4.1.** Analogy between linear and rotary devices

| Linear actuator (cylinder) | Rotary device (motor, pump)                 |
|----------------------------|---------------------------------------------|
| Piston position $x_p$      | Angular position $\varphi$                  |
| Piston area $A$            | Volumetric displacement of motor $V/(2\pi)$ |
| Mass $m$                   | Inertia $J$                                 |
| Chamber volumes $V_{A,B}$  | Compressed volume $V_0$                     |
| Force $F$                  | Torque $T$                                  |



**Figure 4.1.** Valve–cylinder combination with power supply



**Figure 4.2.** Valve–motor combination with power supply

### 4.1.1 Characterisation of Subsystems

Although a general characterisation of hydraulic servo-systems has already been given in Chapter 2, a more precise description, in view of the mathematical modelling of this system, is to be given, including the system boundary.

The control input  $u$  provided by the valve (actuating element) is generally either a voltage or current, and can be considered as an ideal input signal, *i.e.*, the input impedance for the electrical signal is infinitely large. The control input  $u$  of the (servo-)valve is used to control the oil flow through the valve orifices. Oil is supplied by a power supply unit under presumably constant supply pressure  $p_s$ , while the return flow is fed to a tank under the (low) return pressure  $p_T$ .

The actuator (cylinder) consists of two oil chambers, separated by the piston. The resulting oil flows  $Q_A$  and  $Q_B$  moving into and out of the chambers drive the piston, thereby generating the required pressures  $p_A$  and  $p_B$ , respectively, to move the load of the actuator. In this way, the piston motion (expressed in terms of piston velocity) depends on the load of the actuator. Actually, for motion systems with free moving bodies, this load can be seen as an inertia  $m_p$  plus some external force  $F_{ext}$ , which might include gravity forces for example.

A constant valve opening and a constant pressure drop over the valve result in a constant oil flow which will generate a linear movement (translation) of constant speed. This explains the basic integrating behaviour of a hydraulic actuator between the input  $u$  and the output  $y$  (piston position). Because the oil in the two chambers is compressible, the two oil columns act as two springs. The load is clamped between these “springs” via the piston. This causes the second-order behaviour, which is always found in series with the integrating character of a hydraulic actuator.

In order to reduce the complexity of the modelling of the system inside the defined system boundary, it is useful to distinguish a number of subsystems (Figure 4.3):

1. There is the (servo-)valve (Section 4.2.1), which transfers the control input into the oil flows, depending on the actuator pressures. Although this device is designed to be fast and to show linear input–output behaviour, its actual behaviour is generally not ideal. Because the valve flow drives the actuator, any non-ideal behaviour of the valve propagates through the complete servo-system, for which reason the valve is explicitly considered in the modelling as a separate subsystem.
2. There is the hydraulic actuator including load mass (Section 4.2.2), with the driving oil flows  $Q_A$  and  $Q_B$  and the external force  $F_{ext}$  as inputs, and correspondingly the actuator pressures  $p_A$  and  $p_B$ , and position  $x_p$  or velocity  $\dot{x}_p$  as outputs. The hydraulic actuator as subsystem forms the kernel of the complete hydraulic servo-system.

The actuator base should always be designed to be as stiff as possible, with an impedance (*e.g.*, inertia) that is large enough to avoid parasitic energy exchange. In other words, no parasitic motions of the base should occur. Whenever parasitic motions occur in an application, one should be aware of the fact that the system dynamics are affected by the impedance of the base. In that case, the actuator cannot be assumed to be rigidly connected to a base with infinite impedance; see Section 4.2.3.5.

3. The set of pipelines between the valve and the actuator is to be considered as a third subsystem (Section 4.2.5), which is especially important, for instance, when the actuator has a long stroke. Because of the compressibility and the inertia of the oil, the relatively long pipelines cannot be seen as static devices; pressure waves travel with a finite velocity through the line and are almost ideally reflected at the end of the line. Owing to the pipeline dynamics, pressures and

flows at the cylinder side have to be distinguished from those at the valve side. This is indicated in Figure 4.3, where the pressures and the flows have been given indices, depending on whether the valve-side variables or the cylinder-side variables are concerned; the index Av denotes the valve-side of pipeline A, and the index Ac denotes the cylinder-side; the indices Bv and Bc denote the valve- and cylinder-side of pipeline B respectively.

4. Next, the set of pipelines between the power supply system and the valve, as well as between the tank and the valve, is to be considered as a fourth subsystem. This is especially important when the distance between main line and accumulator/valve is large (Section 4.2.5). In many hydraulic servo-control applications, where the valve is placed very close to the cylinder, the dynamics of the return line (or case drain line for motors) may be essential for the function and especially for the durability of the hydraulic system.
5. In most hydraulic servo-applications, the power supply unit is designed in such a way that the system maintains a constant supply pressure for a certain range of operation, *i.e.*, oil flow demand. An effective way to do this, is the application of hydraulic accumulators in combination with a pressure-controlled flow pump (Viersma, 1980). Especially for high-performance demands on the hydraulic servo-system with respect to piston velocity, one should be aware that the limits of the range of operation may be reached, resulting in pressure drops. Another example is the use of load-sensing systems, where flow and pressure levels are simultaneously regulated. The resulting fluctuations may be taken into account by modelling the supply pressure  $p_s$ , as well as the return pressure  $p_T$ , to be dependent on the flow delivered (Section 4.2.4). If an ideal oil supply is assumed (which is not realistic in practice), the pressures  $p_s$  and  $p_T$  are constant.

Thus, taking pipeline dynamics into account, the complete hydraulic servo-system can be represented by its five subsystems, with interconnections as shown in Figure 4.3. However, for low-frequency behaviour, where pipeline effects do not play a role in the input-output behaviour, and for an ideal oil supply, the model of a hydraulic servo-system reduces to one with only two subsystems (valve + cylinder); see Figure 4.4.

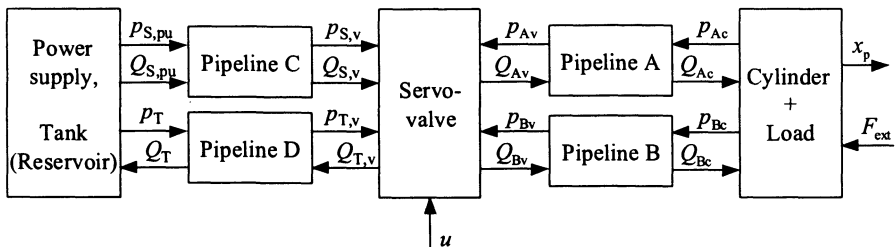
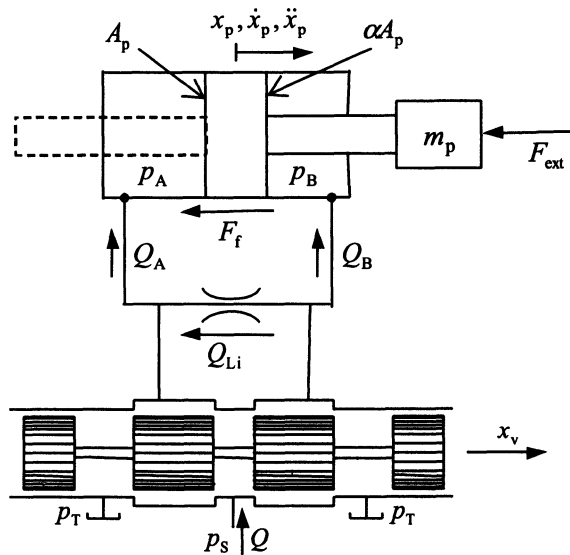


Figure 4.3. Subsystems of hydraulic servo-systems with interconnections

### 4.1.2 Model Complexity and Applications

In order to obtain insight into the various physical phenomena that play a role in the behaviour of a hydraulic servo-system, the modelling approach starts with extensive theoretical modelling of the complete system. Thereby, the models are based on

basic physical laws such as mass balances for oil volumes, equations of motion for moving parts, equations for turbulent flow through small restrictions, and so on, as described in Chapter 3. In the theoretical modelling, all effects are included that are expected to influence the (dynamic) behaviour of the system. The contents of this section are based on earlier work in the area of hydraulic servo-techniques (Merritt, 1967; Köckemann, 1989; Lausch, 1990; Lierschaft, 1993; Pawlik, 1994; Heintze, 1997; van Schothorst, 1997).



**Figure 4.4.** Schematic representation of valve-cylinder combination with variable definitions

A problem with such complex non-linear models is that it is difficult to choose the large number of physical parameters involved in the model such that quantitatively valid simulation results are obtained. Although many of parameter values are known *a priori* with reasonable accuracy, a large number of parameters are only known within a certain range, and some are even completely unknown. This may be due to manufacturing tolerances, or due to the fact that manufacturers do not provide parameter values, because they consider them as proprietary information.

The consequence of this problem is that the theoretical model is often not useful for quantitative analysis of the hydraulic servo-system behaviour. Nevertheless, a lot of qualitative insight can be obtained from simulations, which can be utilised to reduce the hydraulic servo-system model such that only relevant dynamics are taken into account. Furthermore, the simulation results provide the necessary insight to decide which non-linearities of the hydraulic servo-system are relevant. Thus, a relatively simple model for the hydraulic servo-system can be derived, which includes relevant non-linearities and dynamics and which forms a good basis for experimental parameter estimation of the system properties.

Whereas the model of the hydraulic servo-system is also to be used for control design, the model should not only be qualitatively correct, but should also have predictive value in a quantitative sense. This means that the model should accurately



represent the dynamic and non-linear behaviour of a real system. This is possible if there is a clear relation between the input–output behaviour of the model and the parameters of the model, so that the parameters can be chosen or adjusted such that the model fits the behaviour of some real servo-system. In other words, it should be possible to identify the model parameters from experimental input–output data.

For this purpose, it is at least required that the model is of the same order as the relevant dynamics of the real system, and that only the dominant non-linearities of the real system are included in the model. In other words, the model should not include irrelevant dynamics and/or non-linearities. Actually, the theoretical model, including “all” physical phenomena that possibly play a role in the system behaviour, does not fulfil these requirements on the model; it is too complex to be used directly for identification and control design purposes. It should therefore be simplified, as discussed in Section 4.4.

However, in order not to disregard the advantages of the theoretical model, the simplification should be such, that:

- the (dominant) dynamic behaviour described by the theoretical model is preserved,
- and the physical structure of the model is not changed, so that (dominant) non-linear effects related to certain physical phenomena can be taken into account.

## 4.2 Elementary Models

Several physically based models have been derived and illustrated for hydraulic servo-systems or subsystems. However, the models are dispersed in different publications. The aim of this section is to collect some of the most fundamental models of hydraulic servo-systems, including valves, cylinders, pipelines and power supplies, by unifying, improving and completing previous studies in this field.

### 4.2.1 Valves

Because a valve is a complicated device, the theoretical modelling is likewise rather complicated. In line with other research on this topic, reported in the literature (Merritt, 1967; Lin and Akers, 1989, 1991; Lausch, 1990; van Schothorst 1997), many dynamic and non-linear effects are included, resulting in a complex non-linear model.

In fact, most of the non-linearities present in valves are:

- *Dead band* (for valves with overlap).
- *Square-root function* of the flow equations.
- *Saturations* (of stroke, velocity, flow).
- *Valve hysteresis*, which results from frictional forces between spool and valve body not being evenly distributed over the entire spool stroke.
- *Response sensitivity*, i.e., the amount of electrical input signal change needed to obtain a measurable valve flow change in the same direction from which stopping occurred.

- *Reversal error*, i.e., the amount of electrical input signal change needed to obtain a measurable valve flow change in the opposite direction from which stopping occurred.
- *Repeatability*, i.e., the valve's ability to produce an identical flow rate when the same input signal is repeated.
- Complicated flow-induced forces and *friction forces*.
- Different non-linearities in the torque motor model.

#### 4.2.1.1 Pressure–Flow Equations for Spool Valves

Flows through valve orifices are described by the orifice Equation 3.77, which takes the direction of the pressure drop (flow direction) into account, i.e.,

$$Q = Q(x_v, \Delta p) = c_v x_v \sqrt{|\Delta p|} \operatorname{sign}(\Delta p) \quad (4.1)$$

Consider the four-way spool valve with an ideal critical centre as shown in Figure 4.5. From this figure, the flow equations can then be written as

$$Q_A = Q_1 - Q_2 = c_{v1} \operatorname{sg}(x_v) \operatorname{sign}(p_s - p_A) \sqrt{|p_s - p_A|} - c_{v2} \operatorname{sg}(-x_v) \operatorname{sign}(p_A - p_T) \sqrt{|p_A - p_T|} \quad (4.2)$$

$$Q_B = Q_3 - Q_4 = c_{v3} \operatorname{sg}(-x_v) \operatorname{sign}(p_s - p_B) \sqrt{|p_s - p_B|} - c_{v4} \operatorname{sg}(x_v) \operatorname{sign}(p_B - p_T) \sqrt{|p_B - p_T|} \quad (4.3)$$

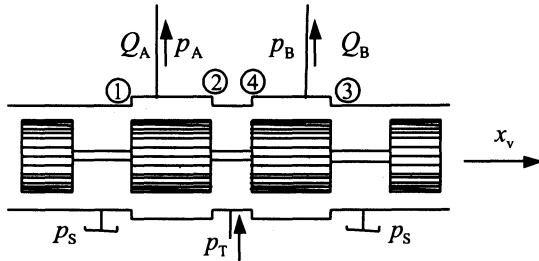


Figure 4.5. Zero lapped 4/3 (four ports/three switching positions) spool valve

The function  $\operatorname{sg}(x)$  is defined by

$$\operatorname{sg}(x) = \begin{cases} x & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases} \quad (4.4)$$

The flow direction is determined by the sign of  $x_v$ .  $c_{vi}$ ,  $i = 1, \dots, 4$  are discharge coefficients<sup>1</sup> of the valve orifices (also termed as valve constants), which will be equal if all orifices are identical.

The corresponding flow equations for underlapped valves can be written as

<sup>1</sup> Note again that the value and the dimension of  $c_v$  have to be adapted to the valve signal used in the flow equation (see Section 3.3.3).

$$Q_A = c_{v1} \operatorname{sg}(x_{u1} + x_v) \operatorname{sign}(p_s - p_A) \sqrt{|p_s - p_A|} \\ - c_{v2} \operatorname{sg}(x_{u2} - x_v) \operatorname{sign}(p_A - p_T) \sqrt{|p_A - p_T|} \quad (4.5)$$

$$Q_B = c_{v3} \operatorname{sg}(x_{u3} - x_v) \operatorname{sign}(p_s - p_B) \sqrt{|p_s - p_B|} \\ - c_{v4} \operatorname{sg}(x_{u4} + x_v) \operatorname{sign}(p_B - p_T) \sqrt{|p_B - p_T|} \quad (4.6)$$

where  $x_{ui} \geq 0$ ,  $i = 1, \dots, 4$  are the underlaps of the valve orifices, which will be equal if all orifices are geometrically identical. For overlapped valves, the flow relations can be expressed as

$$Q_A = c_{v1} \operatorname{sg}(-x_{o1} + x_v) \operatorname{sign}(p_s - p_A) \sqrt{|p_s - p_A|} \\ - c_{v2} \operatorname{sg}(-x_{o2} - x_v) \operatorname{sign}(p_A - p_T) \sqrt{|p_A - p_T|} \quad (4.7)$$

$$Q_B = c_{v3} \operatorname{sg}(-x_{o3} - x_v) \operatorname{sign}(p_s - p_B) \sqrt{|p_s - p_B|} \\ - c_{v4} \operatorname{sg}(-x_{o4} + x_v) \operatorname{sign}(p_B - p_T) \sqrt{|p_B - p_T|} \quad (4.8)$$

with the overlaps  $x_{oi} \geq 0$ ,  $i = 1, \dots, 4$ .

All flow relations given above also include the special case when the pressure on any chamber volume exceeds the system pressure, *e.g.*, when sudden and rapid reversing of an actuator with a large load occurs. In this situation the actuator will act as a pump.

**Radial Clearance and Leakage Flows.** So far we have been concerned with the pressure–flow behaviour of critical centre valves. However, because the production of an exact critical centre valve is hardly possible, the “practical critical centre valve” has radial clearance and just a little underlap. Consequently, there always exist some extra leakage flows in the valve–actuator system which influence the performance and behaviour of such valves and the associated pressure–flow curves for small openings. Thus, accurate models take into account the leakage flows in the case of underlap or overlap, and of the fact that rounded edges and radial clearance cause laminar flow for small port openings (Merritt, 1967; McCloy and Martin, 1980; van Schothorst, 1997).

As outside this small region the theoretical Equations 4.2 and 4.3 fit very well, these effects are not considered here. Yet, it is better from a practical viewpoint to identify and compensate for these effects using measured data (see also Sections 4.5.5 and 6.4.3.1), rather than trying to model these phenomena theoretically and thus introduce more uncertain parameters into the model.

#### 4.2.1.2 Dynamic Characteristics of Servo-valves

The dynamic behaviour of valves involves a large number of parameters. Thereby, many parameters may only be known within some (wide) range, or even completely unknown. Parameter sets obtained from different information sources (manufacturers’ catalogues and drawings, literature, heuristic/manual optimisation) do not reflect very realistic (dynamic) behaviour (van Schothorst, 1997). An accurate analytical description would be time consuming and extremely difficult to

validate, *e.g.*, see Merritt (1967) and Dahm (1978) and more recently van Schothorst (1997) for details. This could only be useful for design engineers if they are interested in the influence of all elements of the valve on the transient behaviour.

Nowadays, manufacturers' catalogue information usually provides the well-known step responses and/or frequency responses for various sizes and types of valve. Therefore, it is useful to use this information to develop simple model approximations of valves.

The inspection of step responses and frequency diagrams; see Appendix B.2, suggests an approximation of the servo-valve by a second-order model of the form (Helduser, 1977; Lee, 1977; Köckemann, 1989; Lausch, 1990; Backé, 1992;):

$$\frac{1}{\omega_v^2} \ddot{x}_v^* + \frac{2D_v}{\omega_v} \dot{x}_v^* + x_v^* + f_{hs} \text{sign}(\dot{x}_v^*) = K_v u^* \quad (4.9)$$

with the normalised valve position, velocity, acceleration, and the valve input signal with respect to the maximum stroke and maximum valve voltage:

$$x_v^* = \frac{x_v}{x_{v,\max}} \quad -1 \leq x_v^* \leq 1 \quad (4.10)$$

$$\dot{x}_v^* = \frac{\dot{x}_v}{x_{v,\max}} \quad -\dot{x}_{v,\lim} \leq \dot{x}_v^* \leq \dot{x}_{v,\lim} \quad (4.11)$$

$$\ddot{x}_v^* = \frac{\ddot{x}_v}{x_{v,\max}} \quad (4.12)$$

The parameters valve gain  $K_v$ , natural frequency  $\omega_v$  and damping coefficient  $D_v$  in Equation 4.9 can be gained from identification or normally extracted from the manufacturer's catalogue information; see Section 4.5.2. The parameter  $f_{hs}$  takes into account the valve hysteresis and response sensitivity.

Equations 4.2, 4.3 and 4.9 can be combined to form the block diagram of the servo-valve in Figure 4.6.

#### 4.2.1.3 Model of Multi-stage Servo-valves

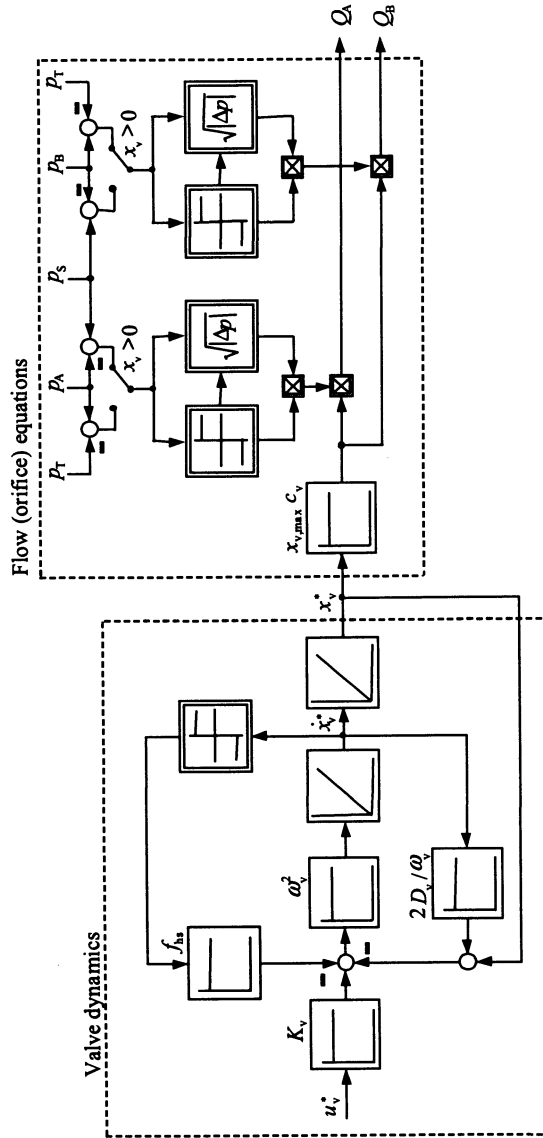
Referring to Figure 2.4 and to the schematic representation in Figure 4.7, the structure of the model of a three-stage servo-valve, including feedback of the main-spool position, is shown in Figure 4.8. Both valve spools are assumed to have a critical centre. The model presented in the following is widely adapted from Hayase *et al.* (1993, 2000) and van Schothorst (1997).

**Model of the Pilot-valve.** The pilot-valve consists of a flapper-nozzle system driven by a torque motor and a spool valve.

**Torque Motor Dynamics.** The electro-magnetic torque motor, which drives the flapper, is controlled by an electrical current  $I$ . This current is generated by a current amplifier, which converts the valve control input  $u$  (voltage) into a current, with

$$\frac{L_{tm}}{R_{tm}} \dot{I} + I = K_{ca} u \quad (4.13)$$

where  $L_{tm}$  is the inductance of the torque motor,  $R_{tm}$  the resistance of the torque motor, and  $K_{ca}$  the current amplifier gain.



**Figure 4.6.** A simplified block diagram of servo-valves (critical centre, neglected flow forces)

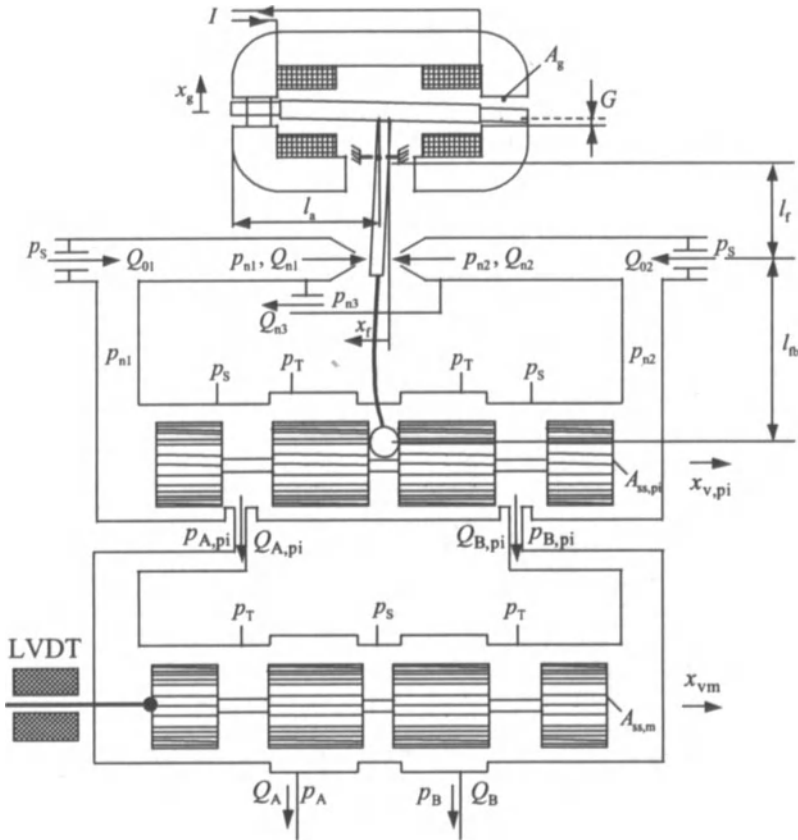


Figure 4.7. Schematic representation of the three-stage servo-valve

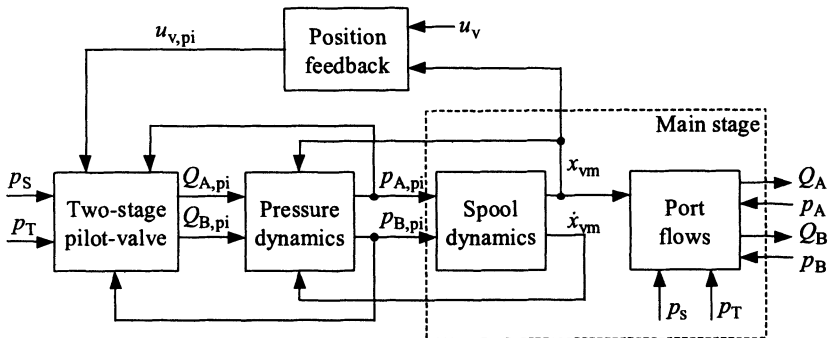


Figure 4.8. Model structure of three-stage servo-valves

The total net torque developed on the armature is theoretically described by

$$T_s = \frac{\mu_0 A_g l_g}{4} \left[ \left( \frac{M_0 + IN}{G - x_g} \right)^2 - \left( \frac{M_0 - IN}{G + x_g} \right)^2 \right] \quad (4.14)$$

where  $\mu_0$  stands for the permeability of a particular portion of the magnetic circuit,  $A_g$  is the cross-sectional area of the air gap,  $l_a$  is the length of armature,  $M_0$  is the magnetomotive force of the permanent magnets,  $N$  is the number of coil windings and  $G$  is the air-gap length at the neutral position of the armature.

However, it is easier and often assumed that the armature torque is linear in the input current for small rotations of the armature, *i.e.*,

$$T_a = \beta I \quad (4.15)$$

where  $\beta$  is known as the gain of the torque motor.

*Flapper–nozzle Dynamics.* The variation of the gap distance due to armature rotations is expressed in the displacement of the armature tip  $x_g$ , which is related to the deflection,  $x_f$ , of the flapper between the nozzles by the armature rotation and the flapper length  $l_f$  as follows:

$$x_g = \frac{l_a}{l_f} x_f \quad (4.16)$$

As the flapper rotates only over small angles ( $\approx 0.01$  rad), the equation of motion can be expressed in terms of the flapper deflection:

$$\frac{J_a}{l_f} \ddot{x}_f + \sigma_f \dot{x}_f + K_a x_f = T_a + T_n - T_b \quad (4.17)$$

where  $J_a$  is the inertia of the flapper armature,  $\sigma_f$  is the viscous friction coefficient of the flapper,  $K_a$  is the stiffness of the flexure tube that connects the flapper to the housing, and  $T_n$  is the torque due to the flow forces.  $T_b$  represents the feedback spring torque, which only applies in the case where mechanical spool position feedback is used.

The resulting torque due to flow forces on the flapper can be calculated by the theoretical expression

$$T_n = \frac{\pi}{4} d_n^2 l_f \left[ 1 + \frac{16\alpha_{dn}^2}{d_n^2} (x_{n0}^2 + x_f^2) \right] (p_{n1} - p_{n2}) + 8\pi l_f \alpha_{dn}^2 x_{n0} x_f (p_{n1} - p_{n2} - 2p_{n3}) \quad (4.18)$$

*Continuity in the Nozzles.* Applying the continuity Equation 3.35 to the valve chambers and to the volume between the nozzles and the outlet orifice yields

$$\dot{p}_{n1} = \frac{E'}{V_{n1}} (Q_{01} - Q_{n1} + A_{ss,pi} \dot{x}_{v,pi}) \quad (4.19)$$

$$\dot{p}_{n2} = \frac{E'}{V_{n2}} (Q_{02} - Q_{n2} - A_{ss,pi} \dot{x}_{v,pi}) \quad (4.20)$$

$$\dot{p}_{n3} = \frac{E}{V_{n3}} (Q_{n1} + Q_{n2} - Q_{n3}) \quad (4.21)$$

Thereby,  $V_{ni}$ ,  $i = 1, 2, 3$  are the valve chamber volumes, and  $A_{ss,pi}$  and  $\dot{x}_{v,pi}$  the spool side area and the spool velocity respectively.

The flows through the inlet restrictions are expressed as

$$Q_{01} = A_0 \alpha_{dn} \sqrt{\frac{2}{\rho} (p_s - p_{n1})} \quad (4.22)$$

$$Q_{02} = A_0 \alpha_{dn} \sqrt{\frac{2}{\rho} (p_s - p_{n2})} \quad (4.23)$$

where  $A_0$  is the orifice area of the inlet restrictions. The nozzle flows  $Q_{n1}$  and  $Q_{n2}$  can be determined by applying the orifice Equation 3.77 (for turbulent flows) to give

$$Q_{n1} = \alpha_{dn} \pi d_n (x_{f0} + x_f) \sqrt{p_{n1} - p_{n3}} \quad (4.24)$$

$$Q_{n2} = \alpha_{dn} \pi d_n (x_{f0} - x_f) \sqrt{p_{n2} - p_{n3}} \quad (4.25)$$

where  $p_{ni}$ ,  $i = 1, 2, 3$  are the nozzle pressures,  $x_f$  the flapper displacement,  $x_{f0}$  the flapper-nozzle distance in neutral position,  $d_n$  the nozzle diameter and  $\alpha_{dn}$  the discharge coefficient of the nozzle for turbulent flows. The nozzle flow  $Q_{n3}$  through the outlet orifice (leakage flow) is calculated by

$$Q_{n3} = A_{n3} \alpha_{dn} \sqrt{\frac{2}{\rho} (p_{n3} - p_T)} \quad (4.26)$$

where  $A_{n3}$  is the outlet orifice area.

*Pilot-spool Dynamics.* Applying Newton's second law to the forces on the spool yields

$$m_{s,pi} \ddot{x}_{v,pi} + F_f(\dot{x}_{v,pi}) = A_{ss,pi} (p_{n1} - p_{n2}) - \frac{T_{fb}}{l_f + l_{fb}} - F_{ax} \quad (4.27)$$

where  $m_{s,pi}$  is the mass of the pilot spool,  $F_f(\dot{x}_{v,pi})$  is the velocity dependent friction force (e.g.,  $\sigma_s \dot{x}_{v,pi}$  for simplicity),  $l_{fb}$  is the length of the feedback spring (if present), and  $F_{ax}$  the axial flow force on the spool. The latter force can be calculated by

$$F_{ax} = 2\alpha_{dn} \cos \theta \left[ A_{s1} (p_s - p_{A,pi}) - A_{s2} (p_{A,pi} - p_T) + A_{s3} (p_{B,pi} - p_T) - A_{s4} (p_s - p_{B,pi}) \right] \quad (4.28)$$

where  $A_{si}$ ,  $i = 1, 2, 3, 4$  are the spool port opening areas. The jet angle  $\theta$  can be assumed constant, namely  $\theta \approx 69^\circ$ , leading to  $\cos \theta = 0.358$ .

*Position Feedback.* If a mechanical feedback of the spool to the flapper position is present, the corresponding feedback spring torque acting on the flapper can be related to virtual deformations at the end of the spring, using the feedback spring constant  $K_{fb}$  as follows (van Schothorst, 1997):



$$T_{fb} = K_{fb} \left( \frac{l_f + l_{fb}}{l_f} x_f + x_{v,pi} \right) \quad (4.29)$$

Obviously, in the case where the spool position feedback is not mechanical, but electrical, the feedback spring torque has to be set to zero. At the same time, Equation 4.13 for the torque motor input has to be modified to

$$\frac{L_{tm}}{R_{tm}} \dot{I} + I = K_{ca} (u - K_{sp} x_{v,pi}) \quad (4.30)$$

where  $K_{sp}$  is the feedback gain, including the spool position transducer gain.

The servo-valve flows are determined by the following equations (assuming zero lapping, and neglecting leakage flows):

$$Q_{A,pi} = c_{v1} \operatorname{sg}(x_{v,pi}) \operatorname{sign}(p_s - p_{A,pi}) \sqrt{|p_s - p_{A,pi}|} \\ - c_{v2} \operatorname{sg}(-x_{v,pi}) \operatorname{sign}(p_{A,pi} - p_T) \sqrt{|p_{A,pi} - p_T|} \quad (4.31)$$

$$Q_{B,pi} = c_{v3} \operatorname{sg}(-x_{v,pi}) \operatorname{sign}(p_s - p_{B,pi}) \sqrt{|p_s - p_{B,pi}|} \\ - c_{v4} \operatorname{sg}(x_{v,pi}) \operatorname{sign}(p_{B,pi} - p_T) \sqrt{|p_{B,pi} - p_T|} \quad (4.32)$$

**Model of the Main Stage.** Similar to the pilot spool, the dynamics of the main-spool are

$$\dot{p}_{A,pi} = \frac{E'}{V_{A,pi}} (Q_{A,pi} + A_{ss,m} \dot{x}_{vm}) \quad (4.33)$$

$$\dot{p}_{B,pi} = \frac{E'}{V_{B,pi}} (Q_{B,pi} - A_{ss,m} \dot{x}_{vm}) \quad (4.34)$$

$$m_{sm} \ddot{x}_{vm} + F_f(\dot{x}_{vm}) = A_{ss,m} (p_{A,pi} - p_{B,pi}) - F_{sx,m} \quad (4.35)$$

Owing to the facts that the side areas of the main-spool are relatively large with respect to the chamber volumes at both sides of the spool and that the acceleration and friction forces are generally very small relative to the maximum driving force on the spool, namely  $p_s A_{ss,pi}$ , the pressure dynamics of the main stage can be neglected in the frequency range of interest. Then, Equations 4.33–4.35 can be simplified into the two static relations

$$Q_{B,pi} = A_{ss,m} \dot{x}_{vm} = -Q_{A,pi} \quad (4.36)$$

$$A_{ss,m} (p_{A,pi} - p_{B,pi}) = 0 \quad (4.37)$$

Finally, the static relations describing the actuator flows are given by

$$Q_A = c_{v1} \operatorname{sg}(x_{vm}) \operatorname{sign}(p_s - p_A) \sqrt{|p_s - p_A|} - c_{v2} \operatorname{sg}(-x_{vm}) \operatorname{sign}(p_A - p_T) \sqrt{|p_A - p_T|} \quad (4.38)$$

$$Q_B = c_{v3} \operatorname{sg}(-x_{vm}) \operatorname{sign}(p_s - p_B) \sqrt{|p_s - p_B|} - c_{v4} \operatorname{sg}(x_{vm}) \operatorname{sign}(p_B - p_T) \sqrt{|p_B - p_T|} \quad (4.39)$$

To stabilise the integrator, the position of the main spool  $x_{vm}$  is fed back to the pilot stage by an electrical feedback loop of the form

$$u_{v,pi}^* = K_{pm} [u_v^* - (1 + K_{dm}s)x_{vm}^*] \quad (4.40)$$

where  $K_{pm}$  is the proportional gain,  $K_{dm}$  the differential gain and  $u_v$  the reference signal. The analogue main-spool position feedback loop is realised in the electronics by the manufacturer, who sets the values for  $K_{pm}$  and  $K_{dm}$ . In fact, Equations 4.36 and 4.37, together with the flow Equations 4.31 and 4.32, give the main spool the character of a non-linear integrator. Depending on the port geometry of the pilot spool, the main-spool velocity  $\dot{x}_{vm}$  is related non-linearly to the pilot-spool position  $x_{v,pi}$ .

Note that the modelling of the main spool by Equations 4.36 and 4.37 introduces algebraic loops in the non-linear simulation model, which may cause problems during simulation. The reader should consult van Schothorst (1997) for more details.

**Simplified Empirical Model.** It is usually possible to use a simple dynamic model according to Equation 4.9 with its five parameters (in conjunction with the flow Equations 4.38 and 4.39) being determined from the manufacturer's catalogue and/or by means of identification (using the manufacturer's parameters as initial guesses), regardless of the complexity of the servo-valve. However, the parameters of the simplified model no longer have a direct physical interpretation, *i.e.*, they cannot be used to reconstruct any physical parameters of the servo-valve. It might be noted here that, even for the substantially more complicated servo-valve model described above, the *a priori* knowledge of physical parameters is insufficient to determine the physical parameters in the theoretical model uniquely from identified parameters, as stated by van Schothorst (1997). This means that a kind of gap between the servo-valve model and the identified model may remain.

However, this fact is not of serious importance, as the two models are used separately for the reasons given. For control design purposes, an accurate (and often real-time capable) characterisation of the dynamics is required, which makes the simplified model, once validated, useful. On the other hand, the complicated model provides insight that can be used for system design, *i.e.*, in the component design stage, where exact knowledge of the model parameters is not necessary.

#### 4.2.1.4 Modelling of Pressure Valves

So far, servo-valves have been discussed as the principle component in controlling flow and/or pressure in hydraulic servo-systems. (The analysis holds widely for proportional valves.) However, other valves are also required for auxiliary functions in the generation and utilisation of hydraulic power. For instance, pressure valves are very often used in hydraulic systems and have multiple functions (see Chapter 2). Common to almost all hydraulic systems is the *pressure-relief* valve. A schematic representation of such a valve is shown in Figure 4.9.

The equation describing the spool motion is

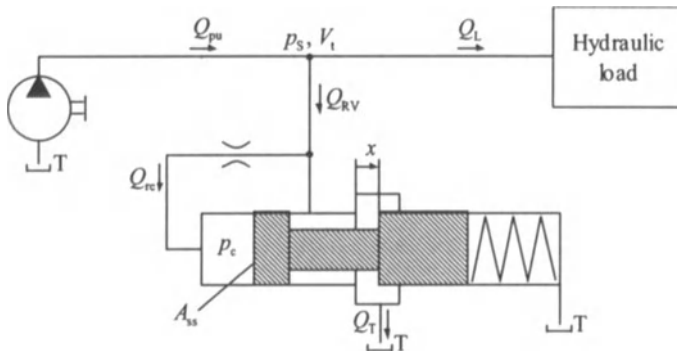
$$m_e \ddot{x} + F_f(\dot{x}) + K_s x + F_{ax}(x, p_s) = A_{ss} p_c - F_0 \quad (4.41)$$

where  $m_e$  is the spool mass plus one-third of the mass of the spring,  $F_f$  is the friction force,  $x$  is the spool displacement,  $K_s$  is the spring stiffness,  $F_{ax}$  is the axial flow force,  $A_{ss}$  is the spool side area,  $p_c$  is the sensed pressure (in the left side-control-chamber), and  $F_0$  is the spring pre-load force.

The flow into the sensed pressure chamber links the supply pressure  $p_s$  to the control chamber pressure  $p_c$  and spool motion in the following manner:

$$\dot{p}_c = \frac{V_c}{E'} (Q_{re} - A_{ss} \dot{x}) = \frac{V_c}{E'} \left[ \alpha_d A_{re} \text{sign}(p_c - p_s) \sqrt{\frac{2}{\rho} |p_c - p_s|} - A_{ss} \dot{x} \right] \quad (4.42)$$

where  $V_c$  is the volume of the left side-control-chamber,  $\alpha_d$  is the discharge coefficient, and  $A_{re}$  is the area of the (usually fixed) restrictor.



**Figure 4.9.** Schematic representation of a pressure-relief valve

Applying the continuity equation to the chamber of the supply pressure (being controlled) leads to

$$\begin{aligned} \dot{p}_s &= \frac{V_t}{E'} (Q_{pu} - Q_{re} - Q_L - Q_{Le} + Q_{mo}) \\ &= \frac{V_t}{E'} \left[ Q_{pu} - \alpha_d A_{re} \text{sign}(p_c - p_s) \sqrt{\frac{2}{\rho} |p_c - p_s|} \right. \\ &\quad \left. - C_L p_s + \alpha_d A_{mo}(x) \text{sign}(p_s - p_T) \sqrt{\frac{2}{\rho} |p_s - p_T|} \right] \end{aligned} \quad (4.43)$$

where  $V_t$  is the total volume of the chamber where pressure is controlled,  $Q_{pu}$  the ideal pump flow,  $Q_L$  an arbitrary load flow,  $Q_{Le}$  the leakage flow (including load leakage and pump leakage),  $Q_{mo}$  the flow through the main orifice,  $C_L$  the leakage coefficient, and  $A_{mo}$  the area of the main orifice.

Equations 4.41, 4.42, and 4.43 define the complete relief-valve system and can simultaneously be solved to analyse its dynamics. As two-stage pressure-relief valves are often used in hydraulic systems, their model can be derived in a similar

way. See also the recent paper by Zung and Perng (2002), where a handy non-linear dynamic model is described.

### 4.2.2 Hydraulic Cylinders

The theoretical modelling of the hydraulic actuator is less involved than that of the servo-valve. Basically, the principal model relations have been given earlier by, among others, Merritt (1967) and Viersma (1980). Other aspects of the basic actuator model have also been addressed in the literature, *e.g.*, Coulomb friction (Southward *et al.*, 1989; Heintze *et al.*, 1993; Armstrong-Helouvry *et al.*, 1994), and hydrostatic bearings (Blok, 1976).

Although most parameters of the theoretical actuator model are rather accurately known *a priori*, the quantitative validity of the model can generally be improved by experimental estimation of the parameters. For this purpose, the theoretical model is slightly simplified, neglecting irrelevant dynamics and non-linearities, resulting in a compact model for the actuator, which is easily identified from experiments.

The most important (non-linear) effects, which contribute to the cylinder model, are listed below:

- *Geometrical asymmetry* due to the difference in piston and ring side area.
- Pressure dependent *effective bulk modulus* in conjunction with the fluid elasticity as well as the elasticity of mechanical compliance.
- *Position dependent “actuator stiffness”*, *i.e.*, the natural frequency, and therefore the damping ratio of the transient dynamics, varies with cylinder position.
- *Friction forces* opposing the piston velocity that encompasses (linear) viscous friction and highly non-linear Coulomb friction or dry friction.

In particular, the oil compressibility influences the dynamics of hydraulic servo-systems significantly. The hydraulic fluid acts like a spring and, therefore, introduces a second-order mass–spring system whose natural frequency limits the bandwidth of any hydraulic servo-system abruptly. The damping of the system due to leakage flow and viscous friction is relatively small.

#### 4.2.2.1 Pressure Dynamics in Cylinder Chambers

Applying the continuity Equation 3.35 to each of the cylinder chambers (see Figure 4.4) yields

$$Q_A - Q_{Li} = \dot{V}_A + \frac{V_A}{E' (p_A)} \dot{p}_A \quad (4.44)$$

$$Q_B + Q_{Li} - Q_{Le} = \dot{V}_B + \frac{V_B}{E' (p_B)} \dot{p}_B \quad (4.45)$$

where  $V_A$  is the volume of piston chamber and  $V_B$  is the volume of ring chamber, both including the valve connecting line and chamber volume.  $Q_{Li}$  and  $Q_{Le}$  denote the internal leakage flow and the external leakage flow respectively. The volumes of the chambers may be written<sup>2</sup> as

<sup>2</sup> The piston is assumed to be initially centred.

$$V_A = V_{pl,A} + \left( \frac{S}{2} + x_p \right) A_p = V_{A0} + x_p A_p \quad (4.46)$$

$$V_B = V_{pl,B} + \left( \frac{S}{2} - x_p \right) \alpha A_p = V_{B0} - x_p \alpha A_p \quad (4.47)$$

where  $V_{pl,A}$  and  $V_{pl,B}$  are the pipeline volumes at the A-side and B-side respectively. The initial chamber volumes  $V_{A0}$  and  $V_{B0}$  consist of an “efficient” part (*i.e.*, the volume required to fill only the chambers) and an “inefficient” part (*i.e.*, mainly the volume of pipelines between the valve and the actuator). The initial chamber volumes are not necessarily equal, but it is usually assumed that the piston is centred such that these are equal. That is:

$$V_{A0} = V_{B0} = V_0 \quad (4.48)$$

The time derivatives of Equations 4.46 and 4.47 are given by

$$\dot{V}_A = A_p \dot{x}_p \quad (4.49)$$

$$\dot{V}_B = -\alpha A_p \dot{x}_p \quad (4.50)$$

Equations 4.44 and 4.45 can be rearranged to yield the pressure dynamics equations

$$\dot{p}_A = \frac{1}{C_{hA}} (Q_A - A_p \dot{x}_p + Q_{Li} - Q_{LeA}) \quad (4.51)$$

$$\dot{p}_B = \frac{1}{C_{hB}} (Q_B + \alpha A_p \dot{x}_p - Q_{Li} - Q_{LeB}) \quad (4.52)$$

The hydraulic capacitance of each chamber is given by

$$C_{hA} = C_h(p_A, x_p) = \frac{V_A(x_p)}{E'_A(p_A)} = \frac{V_{pl,A}(x_{p0} + x_p)A_p}{E'_A(p_A)} \quad (4.53)$$

$$C_{hB} = C_h(p_B, x_p) = \frac{V_B(x_p)}{E'_B(p_B)} = \frac{V_{pl,B}(x_{p0} - x_p)\alpha A_p}{E'_B(p_B)} \quad (4.54)$$

The internal or cross-chamber leakage flow (*i.e.*, leakage from one chamber to the other) can be calculated by (provided the flow is laminar)

$$Q_{Li} = C_{Li}(p_B - p_A) \quad (4.55)$$

where  $C_{Li}$  is the internal leakage flow coefficient. External leakage (*i.e.*, leakage from each cylinder chamber to case drain or to tank) is neglected, *i.e.*,  $Q_{LeA} = Q_{LeB} = 0$ .

#### 4.2.2.2 Equations of Piston Motion

The equation of piston motion governing the load motion arises by applying Newton's second law to the forces on the piston. The resulting force equation is

$$m_t \ddot{x}_p + F_f(\dot{x}_p) = (p_A - \alpha p_B) A_p - F_{\text{ext}} \quad (4.56)$$

The total mass  $m_t$  consists of the piston mass  $m_p$  and the mass of hydraulic fluid in the cylinder chambers and in the pipelines,  $m_{A,\text{fl}}$  and  $m_{B,\text{fl}}$  respectively:

$$m_t = m_p + m_{A,\text{fl}} + m_{B,\text{fl}} \quad (4.57)$$

The mass of fluid can be calculated by

$$m_{A,\text{fl}} = \rho [V_{p1,A} + (x_{p0} + x_p) A_p] \quad (4.58)$$

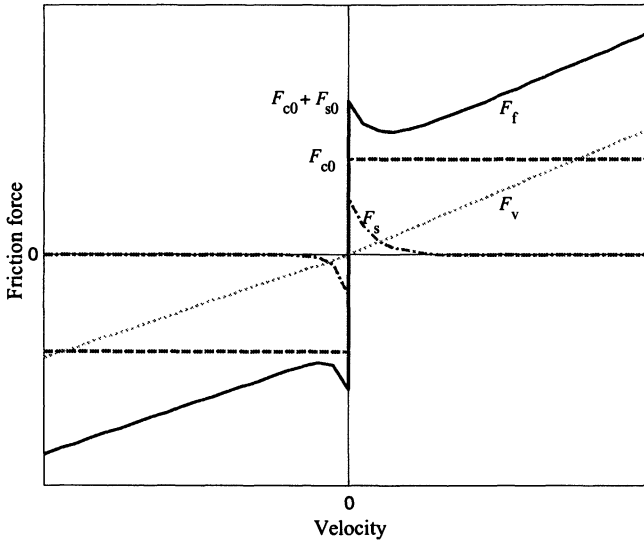
$$m_{B,\text{fl}} = \rho [V_{p1,B} + (x_{p0} - x_p) \alpha A_p] \quad (4.59)$$

However, the mass of fluid can usually be neglected compared with the piston mass.

One standard method is to model friction as a function of velocity, *e.g.*, Equation 4.60, which is referred to as the Stribeck friction curve (after Stribeck, 1902); see Figure 4.10:

$$\begin{aligned} F_f(\dot{x}_p) &= F_v(\dot{x}_p) + F_c(\dot{x}_p) + F_s(\dot{x}_p) \\ &= \sigma \dot{x} + \text{sign}(\dot{x}_p) \left[ F_{c0} + F_{s0} \exp\left(-\frac{|\dot{x}_p|}{c_s}\right) \right] \end{aligned} \quad (4.60)$$

The three characteristic parts of this curve are: viscous friction  $F_v$ , static friction  $F_s$  and Coulomb friction  $F_c$ .  $\sigma$  is the parameter for viscous friction,  $F_{c0}$  is the parameter for Coulomb friction,  $F_{s0}$  and  $c_s$  (known as Stribeck velocity) are the parameters for static friction. A typical Stribeck friction curve is shown in Figure 4.10. See Section 4.5.4 for more details.



**Figure 4.10.** Velocity-dependent friction force (Stribeck curve)

Equations 4.51, 4.52 and 4.56 can be combined to form the block diagram of the hydraulic cylinder in Figure 4.11.

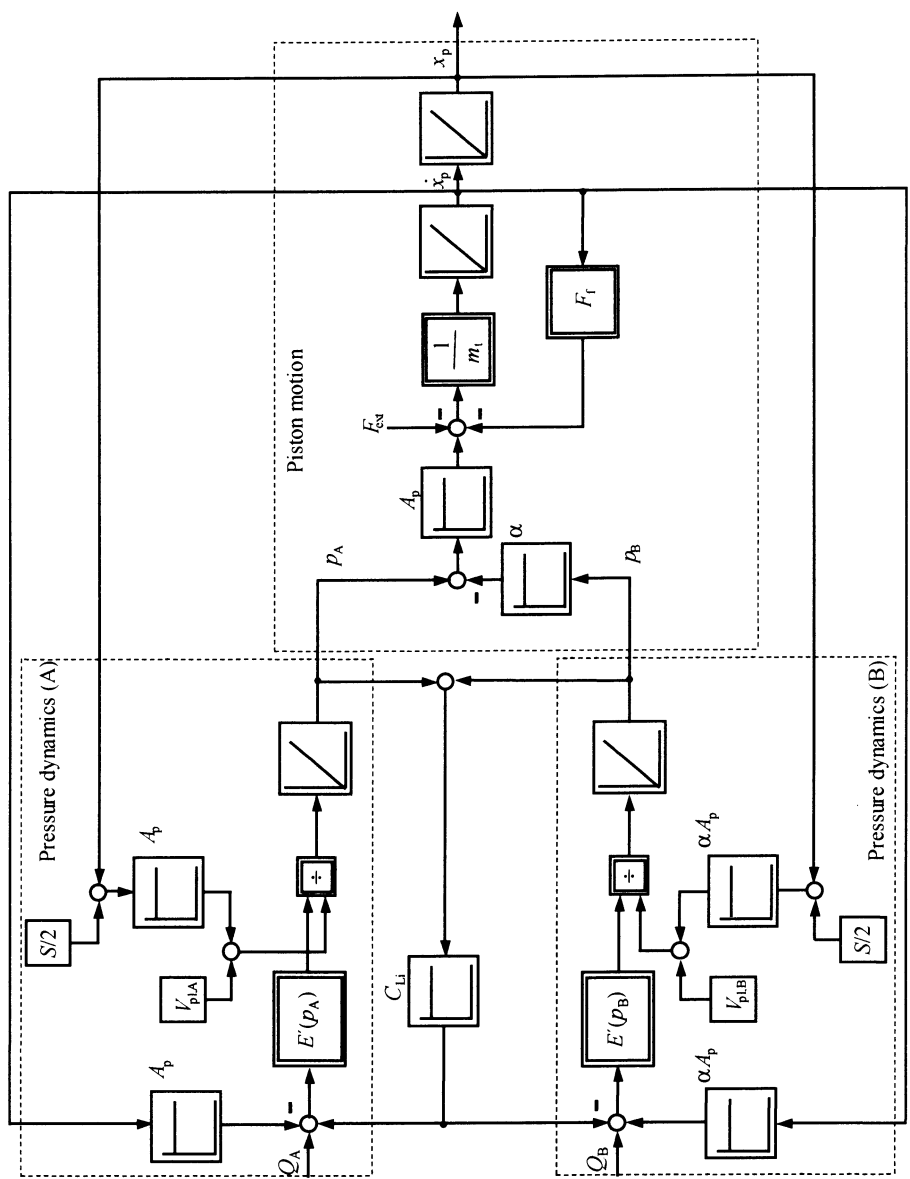


Figure 4.11. A block diagram of differential cylinders

### 4.2.3 Hydraulic Pumps and Motors

As already mentioned at the beginning of this chapter, the model derivation for rotary devices (pump/motor) closely parallels that given for cylinders; see Table 4.1.

This analogy will be exploited merely to outline the main model relations for pumps and motors.

#### 4.2.3.1 Torque Balance for Hydraulic Pumps

Newton's second law is used to get the torque balance equation for the pump

$$J_{\text{HP}} \ddot{\phi}_{\text{HP}} + T_{\text{f,HP}}(\dot{\phi}_{\text{HP}}) = T_{\text{mover}} - \eta_{\text{HP,vol}} T_{\text{HP,th}} - T_{\text{aux}} \quad (4.61)$$

where  $J_{\text{HP}}$  is the moment of inertia of the pump,  $T_{\text{mover}}$  the drive torque of the pump, and  $T_{\text{aux}}$  the torque required for the auxiliary sets. The frictional torque  $T_{\text{f}}$  can be modelled as

$$T_{\text{f}}(\dot{\phi}_{\text{HP}}) = T_{\text{v}} \dot{\phi}_{\text{HP}} + \text{sign}(\dot{\phi}_{\text{HP}}) \left[ T_{\text{c0}} + T_{\text{s0}} \exp\left(-\frac{|\dot{\phi}_{\text{HP}}|}{c_{\text{s}}}\right) \right] \quad (4.62)$$

The ideal pump flow  $Q_{\text{HP,th}}$  is determined by

$$Q_{\text{HP,th}} = \frac{V_{\text{HP,th}}}{2\pi} \dot{\phi}_{\text{HP}} \quad (4.63)$$

where  $V_{\text{HP,th}}$  is the theoretical displacement of the pump. The calculation of the pump displacement depends on the pumping mechanism. For the widely used axial piston pumps, the following relationship applies:<sup>3</sup>

$$V_{\text{HP,th}} = z d_{\text{c}} A_{\text{p}} \tan \alpha_{\text{HP}} \quad (4.64)$$

where  $\alpha_{\text{HP}}$  is the tilt angle,  $z$  the number of axial pistons,  $A_{\text{p}}$  the effective piston area, and  $d_{\text{c}}$  the pitch circle diameter of the cylinder block. However, the real output of the pump is less than the theoretical displacement due to internal leakage. Hence, the effective displacement can be calculated using the estimated volumetric efficiency<sup>4</sup>  $\eta_{\text{HP,vol}}$ :

$$V_{\text{HP,eff}} = \eta_{\text{HP,vol}} V_{\text{HP,th}} \quad (4.65)$$

As the pump is integrated in a hydraulic circuit, a pressure load acts on the axial pistons and is then transformed into a force. The resulting theoretical torque is approximated<sup>5</sup> by

$$T_{\text{HP,th}} = \frac{V_{\text{HP,th}}}{2\pi} \Delta p_{\text{HL}} \quad (4.66)$$

where  $\Delta p_{\text{HL}}$  is the pressure difference between the high-pressure line and low-pressure line.

<sup>3</sup> For other pumping mechanisms see Will *et al.* (1999).

<sup>4</sup> Gear pumps have a volumetric efficiency of approximately 85–96%, vane pumps 85–93%, and piston pumps 95–98% (Rohner, 1985).

<sup>5</sup> The piston inertia term  $J_{\text{p}} \ddot{\phi}_{\text{HP}}$  is neglected; see Pawlik (1994).



### 4.2.3.2 Torque Balance for Hydraulic Motors

Newton's second law is applied to obtain the torque balance equation for the motor

$$J_{\text{HM}} \ddot{\phi}_{\text{HM}} + T_{\text{f, HM}} (\dot{\phi}_{\text{HM}}) = \frac{1}{\eta_{\text{HM}}} T_{\text{HM, th}} + T_{\text{HM, L}} \quad (4.67)$$

where  $J_{\text{HM}}$  is the moment of inertia of the motor and  $T_{\text{HM, L}}$  is the load torque of the motor. The frictional torque can be modelled as

$$T_{\text{f}} (\dot{\phi}_{\text{HM}}) = T_{\text{v}} \dot{\phi}_{\text{HM}} + \text{sign}(\dot{\phi}_{\text{HM}}) \left[ T_{\text{c0}} + T_{\text{s0}} \exp \left( - \frac{|\dot{\phi}_{\text{HM}}|}{c_{\text{s}}} \right) \right] \quad (4.68)$$

The ideal motor flow is determined by

$$Q_{\text{HM, th}} = \frac{V_{\text{HM, th}}}{2\pi} \dot{\phi}_{\text{HM}} \quad (4.69)$$

where  $V_{\text{HM}}$  is the displacement of the motor. The (axial piston) motor displacement is given by

$$V_{\text{HM, th}} = z d_{\text{c}} A_{\text{p}} \tan \alpha_{\text{HM}} \quad (4.70)$$

The effective displacement can be calculated using the estimated overall (volumetric + mechanical) efficiency  $\eta_{\text{HM}}$ :

$$V_{\text{HM, eff}} = \frac{1}{\eta_{\text{HM}}} V_{\text{HM, th}} \quad (4.71)$$

The theoretical torque is approximated by

$$T_{\text{HM, th}} = \frac{V_{\text{HM, th}}}{2\pi} \Delta p_{\text{HL}} \quad (4.72)$$

### 4.2.3.3 Pressure Dynamics in Motor Chambers

The continuity Equation 3.35 for the motor chambers can be written as (see also Section 4.2.2.1)

$$\dot{p}_{\text{A}} = \frac{E'_{\text{A}}}{V_0} \left[ Q_{\text{A}} - \frac{V_{\text{HM}}}{2\pi} \dot{\phi} + C_{\text{Li}} (p_{\text{B}} - p_{\text{A}}) \right] \quad (4.73)$$

$$\dot{p}_{\text{B}} = \frac{E'_{\text{B}}}{V_0} \left[ Q_{\text{B}} + \frac{V_{\text{HM}}}{2\pi} \dot{\phi} - C_{\text{Li}} (p_{\text{B}} - p_{\text{A}}) \right] \quad (4.74)$$

where  $Q_{\text{A}}$  and  $Q_{\text{B}}$  are the flows provided by the hydraulic control element (*i.e.*, valve or pump). Thereby, both chamber volumes have been assumed to be equal  $V_0$  (*i.e.*,  $V_{\text{A}} = V_{\text{B}} = V_0$ ), and the external leakage (*i.e.* leakage to case drain line) has been neglected. Introducing the load pressure

$$p_L = p_A - p_B \quad (4.75)$$

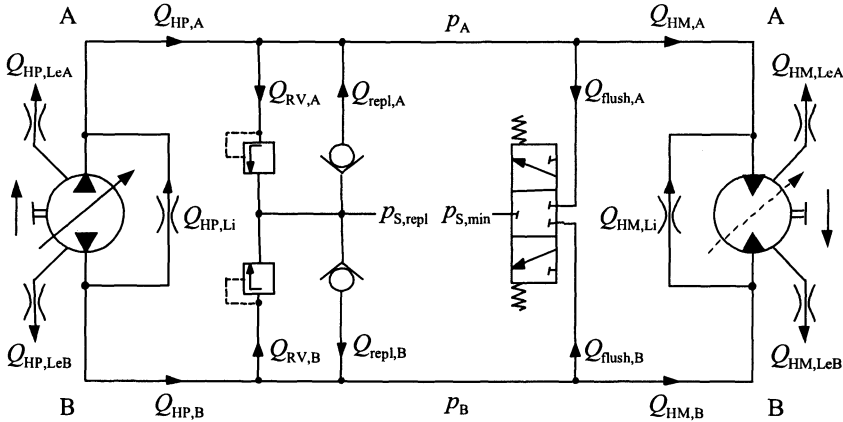
leads to the simplified pressure dynamic equation

$$\dot{p}_L = \frac{E'}{V_0} \left( Q_L - \frac{V_{HM}}{2\pi} \dot{\phi} - C_{L_i} p_L \right) \quad (4.76)$$

where  $Q_L$  is the load flow.

#### 4.2.3.4 Pressure Dynamics in Closed-circuit Hydrostatic Transmissions

In a hydrostatic transmission designed as a closed-circuit drive, the hydraulic fluid from the hydraulic motor returns directly to the pump inlet, not to a reservoir. Usually, the pump has variable displacement and is designed for both flow directions. Additional fluid is pumped into the low-pressure side of the circuit by means of a replenishing pump to replace leakage losses and establish a minimum pressure in each line. Safety relief valves provide protection for the system from damage due to pressure peaks, they absorb shocks during motor reversal, and they limit maximum system pressure. Moreover, a flushing valve is normally used to select the pressure switch and the system relief valve into the high-pressure circuit side, and to divert excessive hot and contaminated fluid from the motor discharge through a filter and cooler back to the tank. A schematic representation of a closed hydrostatic drive is depicted in Figure 4.12.



**Figure 4.12.** A schematic representation of a closed hydrostatic transmission

The compressed flow in each chamber (A/B) is determined by

$$Q_{c,A/B} = Q_{HP,A/B} - Q_{HM,A/B} - Q_{RV,A/B} + Q_{repl,A/B} - Q_{flush,A/B} \quad (4.77)$$

where  $Q_{RV}$  is the flow into the relief valve,  $Q_{repl}$  the flow from the replenishing pump, and  $Q_{flush}$  the flow into the flushing valve. The flow balance equations for the hydraulic pump and the hydraulic motor read as

$$\mathcal{Q}_{HP,A} = \mathcal{Q}_{HP,th} - \mathcal{Q}_{HP,LcA} + \mathcal{Q}_{HP,Li} \quad (4.78)$$

$$\mathcal{Q}_{HM,A} = \mathcal{Q}_{HM,th} + \mathcal{Q}_{HM,LcA} - \mathcal{Q}_{HM,Li} \quad (4.79)$$

respectively. The flows for the B-side are straightforward:

$$\mathcal{Q}_{HP,B} = -\mathcal{Q}_{HP,th} - \mathcal{Q}_{HP,LcB} - \mathcal{Q}_{HP,Li} \quad (4.80)$$

$$\mathcal{Q}_{HM,B} = -\mathcal{Q}_{HM,th} + \mathcal{Q}_{HM,LcB} + \mathcal{Q}_{HM,Li} \quad (4.81)$$

The (laminar) leakage flows can be approximated by

$$\mathcal{Q}_{LcA} = C_{Lc} p_A \quad (4.82)$$

$$\mathcal{Q}_{LcB} = C_{Lc} p_B \quad (4.83)$$

$$\mathcal{Q}_{Li} = K_{Li} (p_B - p_A) \quad (4.84)$$

Finally, the pressure dynamics in each chamber can be calculated by

$$\dot{p}_{A/B} = \frac{E'_{A/B}}{V_{A/B}} \mathcal{Q}_{c,A/B} \quad (4.85)$$

#### 4.2.3.5 Compliant Structures

Thus far we have derived models for hydraulic servo-systems under the assumption that the actuator and the floor are rigid. However, the dynamic properties of hydraulic servo-systems can be changed dramatically if part of the actuated mass is connected more or less flexibly to the piston, and/or if the actuator stands on a floor which has its own dynamics. Such a situation frequently arises in hydraulically actuated machine tools, *e.g.*, in rolling mills where the hydraulic actuator is completely integrated in the mill stand, *i.e.*, between the backup roll chocks and the mill housing; see Section 2.5.2 for more details.

Another example is flexible robots for heavy loads and/or wide operating ranges. The elastic deformations of such robots, which are normally driven by hydrostatic differential cylinders, have to be taken into account; see Sections 2.5.1 and 6.9.

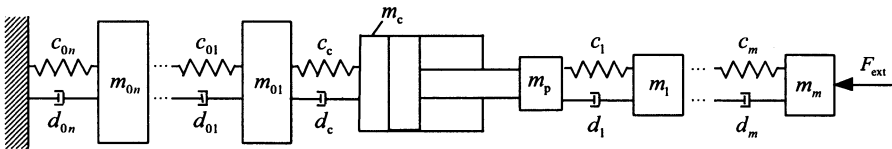


Figure 4.13. Hydraulic servo-systems with “parasitic” mass-spring systems

Because compliant structures could deteriorate the dynamic properties of servo-systems completely, they are sometimes indicated as “parasitic” mass-spring systems; see Figure 4.13. To limit its influence, the natural frequency of a parasitic system itself should be designed as high as possible, preferably ten times higher (or even more) than the designed bandwidth of the complete system (Viersma, 1980).

Applying Newton’s second law to the forces on the piston and the cylinder leads to

$$m_t \ddot{x}_p + F_f(\dot{x}_p - \dot{x}_c) = (p_A - \alpha p_B) A_p - F_{\text{paras},p} \quad (4.86)$$

$$m_c \ddot{x}_c - F_f(\dot{x}_p - \dot{x}_c) = -(p_A - \alpha p_B) A_p + F_{\text{paras},c} \quad (4.87)$$

with the “parasitic” forces

$$F_{\text{paras},p} = F_{\text{ext}} + \sum_{i=1}^m m_i \ddot{x}_i \quad (4.88)$$

$$F_{\text{paras},c} = -(c_{0n} x_{0n} + d_{0n} \dot{x}_{0n}) + \sum_{j=1}^n m_{0j} \ddot{x}_{0j} \quad (4.89)$$

The pressure dynamics can be expressed as

$$\dot{p}_A = \frac{E'_A(p_A)}{V_A(x_p - x_c)} [Q_A - A_p(\dot{x}_p - \dot{x}_c) + Q_{Li} - Q_{LeA}] \quad (4.90)$$

$$\dot{p}_B = \frac{E'_B(p_B)}{V_B(x_p - x_c)} [Q_B + \alpha A_p(\dot{x}_p - \dot{x}_c) - Q_{Li} - Q_{LeB}] \quad (4.91)$$

#### 4.2.4 Power Supplies

As mentioned in Chapter 2, there are many configurations of hydraulic power supply. In practice, the assumption of a constant supply pressure is usually justifiable, provided some strict conditions are met (Viersma, 1980), *i.e.*,

- The accumulator is placed very close (within 0.3 m) to the pump.
- The connection line between main line and accumulator is as short as possible (preferably less than 0.05 m).
- A sufficiently large gas-accumulator (1 dm<sup>3</sup>) is placed very close to the servo-valve, provided the steady-state losses between the pump and the servo-valve are not too low. Such an accumulator acts like an effective wide band filter.

Moreover, the effect of the pressure control valve can be neglected compared to that of the accumulator. Thus, Viersma (1980) proposed a very simple model for controlled pumps:

$$G_{pu}(s) = \frac{P_s(s)}{P_{s,\text{ref}}(s)} = \frac{K_{pu}}{1 + T_{pu}s} \quad (4.92)$$

However, the variation of supply pressure  $p_s$  depending on the flow delivered should be taken into account in the simulation model. This is especially important for load-sensing systems or for high-performance demands on the hydraulic servo-system with respect to piston velocity, where the limits of the range of operation may be reached, resulting in pressure drops.

The pressure control mechanisms vary greatly from one manufacturer to another. The methods used are: hand wheels, manual servo-, remote hydraulic controls, remote electrical controls, and direct internal hydraulic controls; see Rohner (1985). It is beyond the scope of this section to present models for all types of power supply.

Only a simplified model of the classical combination of a variable-displacement pump and stroke regulator is described.

Typically, variable delivery pumps use a built-in hydraulically operated mechanism, essentially a three-way valve-controlled piston with spring return, which varies the tilt angle of the swash plate (or yoke), and thus the pump displacement, to maintain constant supply pressure; see Figure 4.14.

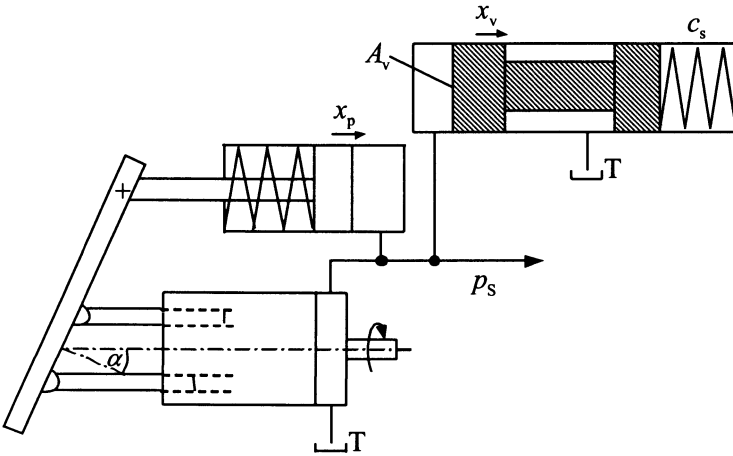
As described in Section 4.2.3.1, the ideal flow of, say, axial piston pumps depends on their (angular) speed  $\dot{\phi}_{HP}$  and tilt angle  $\alpha_{HP}$ , *i.e.*,

$$Q_{HP,th} = \frac{z d_c A_p}{2\pi} \dot{\phi}_{HP} \tan \alpha_{HP} \quad (4.93)$$

The relation between the stroke control position  $x_p$  and the tilt angle  $\alpha_{HP}$  is given by

$$x_p = l_A \sin \alpha_{HP} \quad (4.94)$$

where  $l_A$  is the length of the stroke arm. As the swash plate angle increases, so does the cylinder stroke, resulting in increased pump displacement. When the swash plate is vertical to the drive shaft (*i.e.*, zero swash plate angle), the piston stroke also becomes zero, and pump displacement is theoretically zero. In practice, however, axial piston pumps always have some internal leakage, which must be drained to the tank through the case drain port.



**Figure 4.14.** Stroke-regulated constant-pressure hydraulic power supply

The force balance equation of the pilot spool is

$$p_s A_v - F_0 = m \ddot{x}_v + F_f(\dot{x}_v) + c_s x_v \quad (4.95)$$

where  $F_0$  is the spring pre-load reference force (adjusted with the adjustment screw),  $A_v$  is the pilot-spool side area,  $x_v$  is the pilot-spool position, and  $F_f$  is the friction force.

The pilot-spool position and stroke-control piston position can be (for simplicity)<sup>6</sup> approximately related by (Merritt, 1967:340)

$$\frac{x_p}{x_v} = \frac{\frac{K_{Qx,L}}{A_p}}{s \left( \frac{s^2}{\omega_h^2} + \frac{2D_h}{\omega_h} s + 1 \right)} \quad (4.96)$$

where  $K_{Qx,L}$  is the flow gain of the pilot-spool valve (Section 4.4.4.1),  $\omega_h$  is the hydraulic natural frequency due to the mass of trapped oil, spring and swash plate, and  $D_h$  is the hydraulic damping ratio (Section 4.4.4.3). Neglecting the flow to the pilot spool, the continuity equation for flows in the supply pressure chamber can be written

$$\frac{V_t}{E} \dot{p}_s + C_L p_s = \frac{V_{HP,th}}{2\pi} \dot{\phi}_{HP} - Q_L \quad (4.97)$$

with the load flow  $Q_L$  and the coefficient  $C_L$  of internal pump leakage.

Equations 4.93–4.97 can be combined to build the overall model of a stroke-regulated hydraulic power supply.

## 4.2.5 Pipelines

The hydraulic system's components are usually connected by pipelines (or hose lines). As long as the length of pipeline does not exceed a certain limit (Beater, 1999:34), *i.e.*,

$$l < \frac{c}{10f_{\max}} \quad (4.98)$$

the whole pipeline volume can be added (as an inefficient volume) to the corresponding chamber volume of the cylinder model, and the pipeline dynamics can be neglected. Thereby,  $c$  is the sonic velocity (or wave speed) in oil.  $f_{\max}$  denotes the maximum value of the interesting frequency. Otherwise, the dynamic behaviour of hydraulic pipelines, having distributed parameters, must be considered, and pipeline modelling has to be performed.

The wave speeds in rigid lines ( $c_s$ ) and in hoses ( $c_w$ ) are calculated from (Goodson and Leonard, 1972)

$$c_s = \sqrt{\frac{E}{\rho}}, \quad c_w = c_s \left( 1 + \frac{E}{E_h} \frac{2r_h}{s_h} \right)^{-1/2} \quad (4.99)$$

where  $r_h$  is the hose inner radius and  $s_h$  the hose wall thickness. Generally, the wave speed is determined using

<sup>6</sup> Of course, a complicated model for valve–piston combination can be inserted instead of Equation 4.96.

$$c = \sqrt{\frac{E'}{\rho}} \quad (4.100)$$

with the effective bulk modulus, which takes the effects of entrained air and mechanical compliance into account; see Section 3.1.3.

In many industrial applications, the inefficient volumes appear to be significantly large. This is due to installation of several hydraulic safety features between the valve and the actuator. These “safety manifolds” make it constructionally difficult to place the valve directly upon the actuator and, therefore, introduce a significant amount of inefficient volume (Heintze, 1997). Another aspect is the visco-chaotic behaviour of hose lines, which are used in many applications.

More specifically, there are many situations in which pipeline effects may play an important role in the hydraulic control systems behaviour:

1. There are many applications, in which the oil supply unit is separated from the hydraulic servo-system, often requiring long supply lines. The dynamics due to these long lines may, in practice, lead to undesired oscillations, possibly excited by the pump-flow-pulsations. Research performed by Viersma (1980) and Ham (1982) has provided a lot of insight into these dynamics, and has also led to practical means to suppress ripples in the supply pressure due to these dynamics (van Schothorst, 1997).
2. In some other applications, pipeline dynamics have to be considered because the servo-valve cannot directly be placed on the actuator chambers, so that relatively long pipelines between the valve and the actuator chambers are required. This typically applies for long-stroke cylinders and some (heavy duty) rotary drive applications, where it is practically impossible to place the servo-valve directly on the actuator (van Schothorst, 1997).
3. Last but not least, there are some applications, such as hydraulic power assisted car steering and aircraft brake hydraulic systems, where the modelling of supply and return lines is one of the key problems in the modelling of the whole system. On the one hand, these hydraulic lines consist of quite different parts, such as pipes and rubber hoses. On the other hand, they may have great effect on the generation of considerable pressure peaking and oscillations, and thus hydraulic noise. It is pointed out that the return line pressure in hydraulic power assisted car steering goes down to about  $-1$  bar. As the pressure is taken relative to atmosphere,  $-1$  bar means that the vapour pressure of the fluid must be reached and therefore cavitation likely occurs (Ratjen, 2001b).

When considering the effect of pipelines between valve and actuator, the system configuration can be schematically depicted as in Figure 4.15, where a clear distinction is made between the states at both sides A and B, as indicated earlier in Figure 4.3.

To describe the physical effects of the presence of pipelines properly in a hydraulic servo-system, a modular set-up of the model, in which the pipeline models are easily included or omitted, is highly desirable. For this reason, the basis for pipeline modelling lies in the theoretical, physical modelling of a single pipeline, extensively reported in the literature (d'Souza and Oldenburger, 1964; Goodson and Leonard, 1972).

It might be noted here that the servo-valve dynamics are actually not involved in this coupling, because the spool position of the servo-valve is basically not affected by the pressures  $p_{Av}$  and  $p_{Bv}$  in Figure 4.15. So, except for a non-linearity in the expression for the servo-valve flow, which is to be taken into account explicitly, the servo-valve can be considered as an ideal flow source. This means, from a modelling viewpoint, that the investigation of the effect of pipeline dynamics actually involves the inclusion of pipeline dynamics in the basic model of the hydraulic actuator.

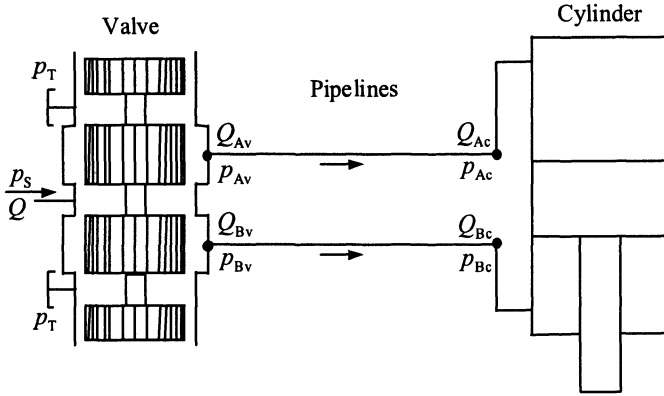


Figure 4.15. System configuration considering pipeline dynamic effects

#### 4.2.5.1 Basic Model Equations

Simplified (yet sufficient) models for pipeline dynamics have been derived by many researchers under the following assumptions:

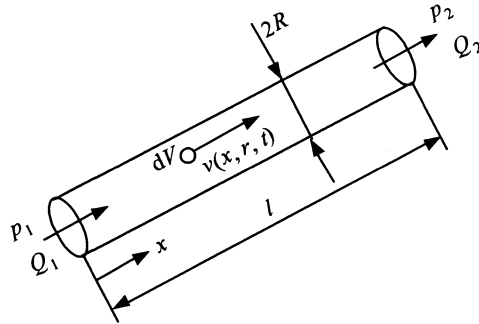
- rigid pipe (finite wall stiffness can be considered by corrected oil bulk modulus  $E$ ); see Figure 4.16;
- laminar flow;
- constant temperature and negligible heat transfer effects;
- negligible tangential flow;
- negligible differences of  $p$  and  $\rho$  in the radial direction and;
- negligible radial speed (Hagen–Poiseuille flow);

The presentation given below largely follows Pawlik (1994).

For each pipeline volume element  $dV$ , consider the Navier–Stokes Equations 3.42–3.44. Now, a major simplification can be made on the basis that the fluid mainly flows into axial ( $x$ ) direction, and the fluid motion in the other directions is negligible. Moreover, differences of  $p$  in the radial direction and gravitation terms are also neglected. Thus we can write the Navier–Stokes Equation in the simplified form

$$\rho \frac{dv_x}{dt} = -\frac{dp}{dx} + \eta \left( \frac{1}{r} \frac{\partial v_x}{\partial r} + \frac{\partial^2 v_x}{\partial r^2} \right) \quad (4.101)$$





**Figure 4.16.** Flow through a single (cylindrical) pipeline

From the relationship

$$\frac{dv_x}{dx} = \frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} \quad (4.102)$$

Equation 4.101 becomes

$$\rho \left( \frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} \right) = -\frac{dp}{dx} + \eta \left( \frac{1}{r} \frac{\partial v_x}{\partial r} + \frac{\partial^2 v_x}{\partial r^2} \right) \quad (4.103)$$

Since  $v_x \ll c \Rightarrow v_x \partial v_x / \partial x \ll \partial v_x / \partial t$ , and  $\partial v_x / \partial x \ll \partial v_x / \partial r$ , the Navier–Stokes Equation 4.103 can be further simplified as

$$\rho \frac{\partial v_x}{\partial t} = -\frac{dp}{dx} + \eta \left( \frac{1}{r} \frac{\partial v_x}{\partial r} + \frac{\partial^2 v_x}{\partial r^2} \right) \quad (4.104)$$

In this equation, the velocity  $v_x$  and the pressure  $p$  have to be determined. For this, the continuity Equation 3.29 must be considered. With the simplifications mentioned above, the continuity equation takes the form

$$\frac{1}{\rho} \frac{d\rho}{dt} = -\frac{\partial v_x}{\partial x} \quad (4.105)$$

The introduction of sonic velocity (Goodson and Leonard, 1972; Truckenbrodt, 1996)

$$c^2 = \frac{dp}{d\rho} = \left( \frac{\partial p}{\partial \rho} \right)_{s_E = \text{const.}} = \frac{E'}{\rho} \quad (4.106)$$

and the pressure differential leads to

$$\frac{\partial p}{\partial t} + v_x \frac{\partial p}{\partial x} = -E' \frac{\partial v_x}{\partial x} \quad (4.107)$$

Since  $v_x \partial p / \partial x \ll \partial p / \partial t$ , Equation 4.107 can be simplified as

$$\frac{\partial p}{\partial t} = -E' \frac{\partial v_x}{\partial x} \quad (4.108)$$

#### 4.2.5.2 Transformation in the Frequency Domain

The introduction of the Laplace transformation and rearrangements for Equation 4.104 yield

$$r^2 \frac{\partial^2 v_x(x, r, s)}{\partial r^2} + r \frac{\partial v_x(x, r, s)}{\partial r} - \frac{r^2}{\mu} s \left( v_x(x, r, s) + \frac{1}{s\rho} \frac{\partial p(x, s)}{\partial x} \right) = 0 \quad (4.109)$$

The variable substitution

$$u(x, r, s) \equiv v_x(x, r, s) + \frac{1}{s\rho} \frac{\partial p(x, s)}{\partial x} \quad (4.110)$$

and the normalisation

$$\tilde{r}^2 = -\frac{s}{\mu} r^2 \quad (4.111)$$

bring Equation 4.109 into the following form:

$$\tilde{r}^2 \frac{\partial^2 u(x, r, s)}{\partial \tilde{r}^2} + \tilde{r} \frac{\partial u(x, r, s)}{\partial \tilde{r}} + \tilde{r}^2 u(x, r, s) = 0 \quad (4.112)$$

Thus, Bessel's differential equation of the form

$$x^2 y'' + xy' + (x^2 - n^2)y = 0 \quad (4.113)$$

is obtained.

Using, for instance, the handbook of mathematical formulas and tables by Bronstein and Semendjajew (1991), the general solution of Equation 4.112 reads

$$u(x, r, s) = C_1 J_0(\tilde{r}) + C_2 Y_0(\tilde{r}) \quad (4.114)$$

In this,  $J_0$  is a Bessel function of the first kind and  $Y_0$  a Bessel function of the second kind (or Weber function).  $C_1$  and  $C_2$  are constants to be found by boundary conditions. At  $r = 0$ , we have

$$u \neq \infty \quad J_0(0) = 1 \quad Y_0(0) = -\infty \quad (4.115)$$

and so

$$C_2 = 0 \quad (4.116)$$

At  $r = R$ , it follows from Equations 4.110 and 4.114

$$C_1 = \frac{1}{s\rho} \frac{\partial p(x, s)}{\partial x} \frac{1}{J_0(\tilde{R})} \quad (4.117)$$

So, the complete solution becomes

$$v_x(x, \tilde{r}, s) = \frac{1}{s\rho} \frac{\partial p(x, s)}{\partial x} \left( \frac{J_0(\tilde{r})}{J_0(\tilde{R})} - 1 \right) \quad (4.118)$$

Now the flow can be determined:

$$\begin{aligned} Q(x, s) &= \int_0^R 2\pi r v_x(x, r, s) dr = \int_0^{\tilde{R}} 2\pi \tilde{r} v_x(x, \tilde{r}, s) \left( -\frac{v}{s} \right) d\tilde{r} \\ &= -\frac{2\pi\mu}{s^2\rho} \frac{\partial p(x, s)}{\partial x} \frac{1}{J_0(\tilde{R})} \int_0^{\tilde{R}} J_0(\tilde{r}) \tilde{r} d\tilde{r} - \frac{\pi R^2}{s\rho} \frac{\partial p(x, s)}{\partial x} \end{aligned} \quad (4.119)$$

With the relationships (Bronstein and Semendjajew, 1991)

$$\begin{aligned} \frac{2n}{x} J_n(x) &= J_{n-1}(x) + J_{n+1}(x) \\ \frac{d}{dx} (x^n J_n(x)) &= x^n J_{n-1}(x) \end{aligned} \quad (4.120)$$

the flow can finally be expressed as

$$Q(x, s) = \frac{\pi R^2}{s\rho} \frac{\partial p(x, s)}{\partial x} \left( \frac{2}{\tilde{R}} \frac{J_1(\tilde{R})}{J_0(\tilde{R})} - 1 \right) = \frac{\pi R^2}{s\rho} \frac{\partial p(x, s)}{\partial x} \frac{J_2(\tilde{R})}{J_0(\tilde{R})} \quad (4.121)$$

From this equation, the mean velocity can be calculated:

$$\bar{v}_x(x, s) = \frac{1}{A_{pl}} Q(x, s) = \frac{1}{s\rho} \frac{\partial p(x, s)}{\partial x} \frac{J_2(\tilde{R})}{J_0(\tilde{R})} \quad (4.122)$$

Now the Laplace transformation can also be applied to the continuity Equation 4.108 to give

$$p(x, s) = -E' \frac{\partial \bar{v}_x(x, s)}{\partial x} \quad (4.123)$$

Substituting Equation 4.122 into Equation 4.123 brings the description

$$p(x, s)s = -\frac{E'}{A_{pl}} \frac{\partial Q(x, s)}{\partial x} \quad (4.124)$$

and with Equation 4.121

$$\frac{s^2\rho}{E'} Q(x, s) + \frac{\partial^2 Q(x, s)}{\partial x^2} \frac{J_2(\tilde{R})}{J_0(\tilde{R})} = 0 \quad (4.125)$$

Taking into account Equation 4.106 leads to

$$\frac{\partial^2 Q(x, s)}{\partial x^2} - \frac{s^2}{c^2} N(s) Q(x, s) = 0 \quad (4.126)$$

with the dynamic viscosity function

$$N(s) = -\frac{J_0(\tilde{R})}{J_2(\tilde{R})} \quad (4.127)$$

Combining Equations 4.124 and 4.126, we have the Navier–Stokes equation in the following form:

$$\frac{A_{pt}}{E'} \frac{\partial p(x, s)}{\partial x} + \frac{s}{c^2} N(s) Q(x, s) = 0 \quad (4.128)$$

In similar way, combining Equations 4.106, 4.121 and 4.123 yields

$$\frac{\partial^2 p(x, s)}{\partial x^2} - \frac{s^2}{c^2} N(s) p(x, s) = 0 \quad (4.129)$$

#### 4.2.5.3 Approximation of the Dynamic Viscosity Function

The dynamic viscosity function in Equation 4.127 (bearing in mind Equation 4.111) can be expressed by the Bessel functions  $J_n$  (of the first kind)

$$N(s) = -\frac{J_0(\tilde{R})}{J_2(\tilde{R})} = \frac{\mu'}{s} \frac{\sum_{k=0}^{\infty} \frac{2^k}{(k!)^2} \left(\frac{s}{\mu'}\right)^k}{\sum_{k=0}^{\infty} \frac{2^{k+1}}{k!(k+2)!} \left(\frac{s}{\mu'}\right)^k} \quad (4.130)$$

with the so-called “viscosity factor”

$$\mu' \equiv \frac{8\mu}{R^2} \quad (4.131)$$

The function  $N(s)$  plays an important role in (one-dimensional) pipeline flows as it includes the friction and inertia effects, using the coefficient of viscosity  $\mu$  of the fluid. A major problem is the simulation of the Bessel functions with *infinite* terms. For practical simulations, the approximation of  $N(s)$  has to be considered.

A well-known good approximation (via transformation and optimisation in the time domain followed by re-transformation into the frequency domain) is (Trikha, 1975; Ham, 1982)<sup>7</sup>

$$N(s) = 1 + \frac{\mu'}{s} + \frac{K_1}{1 + \frac{\beta_1}{\mu'} s} + \frac{K_2}{1 + \frac{\beta_2}{\mu'} s} + \frac{K_3}{1 + \frac{\beta_3}{\mu'} s} \quad (4.132)$$

with the parameters listed in Table 4.2.

<sup>7</sup> Reliable up to  $\omega \approx 1000\mu'$  (Trikha, 1975) and  $\omega \approx 300\mu'$  (Ham, 1982).

**Table 4.2.** Parameters of approximations of  $N(s)$

|        | $K_1$  | $K_2$  | $K_3$  | $\beta_1$ | $\beta_1$ | $B_1$  |
|--------|--------|--------|--------|-----------|-----------|--------|
| Trikha | 0.1918 | 0.0948 | 0.0407 | 0.2496    | 0.0352    | 0.0024 |
| Ham    | 0.1515 | 0.1620 | 0.020  | 0.3030    | 0.04      | 0.001  |

Two well-known special cases are:

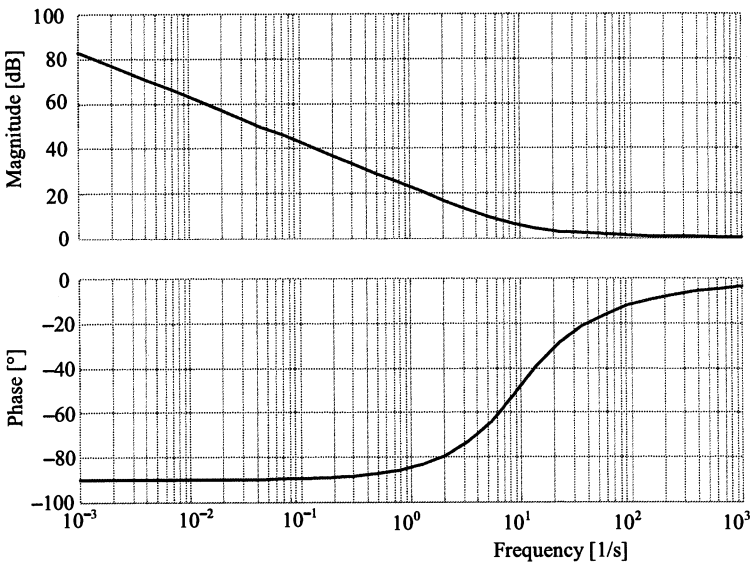
1. The linear friction model, in which viscous friction is included by adding a linear friction term to the momentum balance, *i.e.*, the dynamic viscosity function takes the form

$$N(s) = 1 + \frac{\mu'}{s} \quad (4.133)$$

Equation 4.133 represents a rough approximation over the whole frequency range, and includes the asymptotes for low and high frequencies.

2. By setting the coefficient of viscosity  $\mu$  to zero, the so-called frictionless fluid line model is obtained, *i.e.*,  $N(s) = 1$ .

As can be seen from the Bode diagram of the dynamic viscosity function  $N(s)$  in Figure 4.17, high-frequency disturbances are more damped than low-frequency ones.



**Figure 4.17.** Bode diagram of the viscosity function  $N(s)$

This fact can also be explained by considering the velocity profile in the pipeline, which is determined from Equations 4.118 and 4.122 as

$$v^*(x, r, s) = \frac{v(x, r, s)}{\bar{v}(x, s)} = \frac{J_0(\tilde{r}) - J_0(\tilde{R})}{J_2(\tilde{R})} \quad (4.134)$$

This relationship is shown in Figure 4.18. At low frequencies, a laminar flow profile occurs, but at high frequencies a turbulent flow arises. The latter is characterised through high velocity gradients near the pipe wall and smaller velocity gradients at the middle of the pipe. For laminar flows, velocity gradients are smaller, and hence lower damping for low frequencies results.

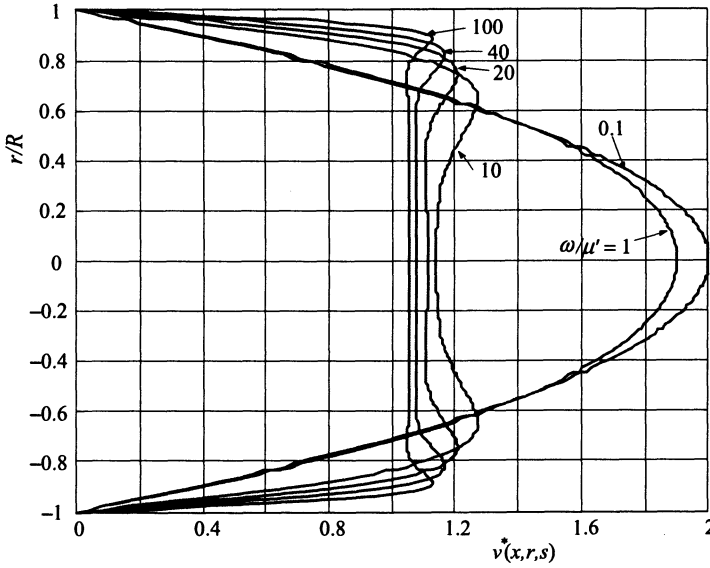


Figure 4.18. (Scaled) velocity profile in the pipeline for different friction situations

#### 4.2.5.4 Frequency Domain Solutions

The solution of the second-order differential Equation 4.126 is

$$Q(x, s) = C_1 \exp\left(\frac{x}{c} s \sqrt{N(s)}\right) + C_2 \exp\left(-\frac{x}{c} s \sqrt{N(s)}\right) \quad (4.135)$$

Thus (with Equation 4.124)

$$p(x, s) = -Z_c \sqrt{N(s)} \left[ C_1 \exp\left(\frac{x}{c} s \sqrt{N(s)}\right) - C_2 \exp\left(-\frac{x}{c} s \sqrt{N(s)}\right) \right] \quad (4.136)$$

with the line characteristic impedance

$$Z_c = \frac{\rho c}{A_{pl}} = \frac{\sqrt{\rho E'}}{A_{pl}} \quad (4.137)$$

Inserting the boundary conditions leads to the solution of the constants  $C_1$  and  $C_2$ . Letting  $p = p_1$  and  $Q = Q_1$  at the entry of the pipeline (i.e.,  $x = 0$ ), we find

$$C_1 = \frac{1}{2} \left( Q_1(s) - \frac{p_1(s)}{Z_c \sqrt{N(s)}} \right) \quad C_2 = \frac{1}{2} \left( Q_1(s) + \frac{p_1(s)}{Z_c \sqrt{N(s)}} \right)$$

Letting  $p = p_2$  and  $Q = Q_2$  at the end of the pipeline (i.e.,  $x = l$ ), the solutions take the form

$$Q_2(s) = -\frac{1}{Z_c \sqrt{N(s)}} \left[ \frac{\exp(\Gamma(s)) - \exp(-\Gamma(s))}{2} \right] p_1(s) + \left[ \frac{\exp(\Gamma(s)) + \exp(-\Gamma(s))}{2} \right] Q_1(s) \quad (4.138)$$

$$p_2(s) = -Z_c \sqrt{N(s)} \left[ \frac{\exp(\Gamma(s)) - \exp(-\Gamma(s))}{2} \right] Q_1(s) + \left[ \frac{\exp(\Gamma(s)) + \exp(-\Gamma(s))}{2} \right] p_1(s) \quad (4.139)$$

$\Gamma(s)$  is the propagation operator

$$\Gamma(s) = \frac{ls}{c} \sqrt{N(s)} \quad (4.140)$$

which models the time  $T = l/c$  that the pressure wave takes to travel along the line.

The introduction of hyperbolic functions

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{1 + e^{-2x}}{2e^{-x}} \quad \sinh x = \frac{e^x - e^{-x}}{2} = \frac{1 - e^{-2x}}{2e^{-x}}$$

leads to the so-called *four-pole equation*

$$\begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} \cosh \Gamma(s) & -\frac{1}{Z_c \sqrt{N(s)}} \sinh \Gamma(s) \\ -Z_c \sqrt{N(s)} \sinh \Gamma(s) & \cosh \Gamma(s) \end{bmatrix} \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} \quad (4.141)$$

The four-pole equation, relating the signals at the ends of pipelines, is proved to be an extremely powerful tool for the analysis, synthesis and design of hydraulic pipelines. With the notation

$$\begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} A(s) & B(s) \\ C(s) & D(s) \end{bmatrix} \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} \quad (4.142)$$

the four-pole equation can be easily rearranged for the other flow direction (from the end to the entry of the pipeline):

$$\begin{aligned}
 \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} &= \frac{1}{A(s)D(s) - B(s)C(s)} \begin{bmatrix} D(s) & -B(s) \\ -C(s) & A(s) \end{bmatrix} \begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix} \\
 &= \begin{bmatrix} D(s) & -B(s) \\ -C(s) & A(s) \end{bmatrix} \begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix}
 \end{aligned} \tag{4.143}$$

since the determinant of the system transfer function equals unity.

**Summary.** The equivalent relationships between  $Q_1$ ,  $p_1$ ,  $Q_2$  and  $p_2$  can be presented quite differently. The four *causal* representations (see also Goodson and Leonard, 1972) of a single pipeline using the four-pole equations are given as follows:

$$\text{Config. 1: } \begin{bmatrix} p_1(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} \frac{Z_c \sqrt{N(s)} \cosh \Gamma(s)}{\sinh \Gamma(s)} & -\frac{Z_c \sqrt{N(s)}}{\sinh \Gamma(s)} \\ \frac{Z_c \sqrt{N(s)}}{\sinh \Gamma(s)} & -\frac{Z_c \sqrt{N(s)} \cosh \Gamma(s)}{\sinh \Gamma(s)} \end{bmatrix} \begin{bmatrix} Q_1(s) \\ Q_2(s) \end{bmatrix} \tag{4.144}$$

$$\text{Config. 2: } \begin{bmatrix} Q_1(s) \\ Q_2(s) \end{bmatrix} = \begin{bmatrix} \frac{\cosh \Gamma(s)}{Z_c \sqrt{N(s)} \sinh \Gamma(s)} & -\frac{1}{Z_c \sqrt{N(s)} \sinh \Gamma(s)} \\ \frac{1}{Z_c \sqrt{N(s)} \sinh \Gamma(s)} & -\frac{\cosh \Gamma(s)}{Z_c \sqrt{N(s)} \sinh \Gamma(s)} \end{bmatrix} \begin{bmatrix} p_1(s) \\ p_2(s) \end{bmatrix} \tag{4.145}$$

$$\text{Config. 3: } \begin{bmatrix} Q_2(s) \\ p_1(s) \end{bmatrix} = \begin{bmatrix} \frac{1}{\cosh \Gamma(s)} & -\frac{\sinh \Gamma(s)}{Z_c \sqrt{N(s)} \cosh \Gamma(s)} \\ \frac{Z_c \sqrt{N(s)} \sinh \Gamma(s)}{\cosh \Gamma(s)} & \frac{1}{\cosh \Gamma(s)} \end{bmatrix} \begin{bmatrix} Q_1(s) \\ p_2(s) \end{bmatrix} \tag{4.146}$$

$$\text{Config. 4: } \begin{bmatrix} Q_1(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} \frac{1}{\cosh \Gamma(s)} & \frac{\sinh \Gamma(s)}{Z_c \sqrt{N(s)} \cosh \Gamma(s)} \\ -\frac{Z_c \sqrt{N(s)} \sinh \Gamma(s)}{\cosh \Gamma(s)} & \frac{1}{\cosh \Gamma(s)} \end{bmatrix} \begin{bmatrix} Q_2(s) \\ p_1(s) \end{bmatrix} \tag{4.147}$$

For the modelling of pipelines between valve and actuator, Configuration 3 (*i.e.*, Equation 4.146) will be applied in this section. For the modelling of the distributed pressure dynamics in the actuator chambers, Configuration 1 (*i.e.*, Equation 4.144) can be used. However, the latter is not relevant in practice, as pointed out by van Schothorst (1997), and thus will not be taken into account here.

#### 4.2.5.5 Finite Dimensional (Simulation) Models

The problem with the description of the pipeline dynamics in the frequency domain using a four-pole equation is that it represents an infinite-order model, corresponding to the distributed parameter character of the system. This makes the theoretical frequency domain model less suitable for system analysis, especially in the time domain. In order to overcome this problem, approximation of the



theoretical model, Equation 4.141, is required. Many approximation techniques are reported in the literature (Feldmann, 1971; Sidell and Wormley, 1977; Viersma, 1980; Margolis and Yang, 1985; Pawlik, 1994). Three of these approximation approaches are summarised here. For more details, the reader should consult the original work.

**Viersma's Discrete Model.** In this approach, the pipeline is cut into  $n_s$  segments of length  $\Delta x$ , and the four-pole equation is applied to each of the segments (see Figure 4.19). The continuity Equation 4.124 and the Navier–Stokes Equation 4.128 are written in the form

$$\frac{s}{c} p(s) + Z_c \frac{dQ(s)}{dx} = 0 \quad (4.148)$$

$$\frac{dp(s)}{dx} + \frac{s}{c} Z_c Q(s) N(s) = 0 \quad (4.149)$$

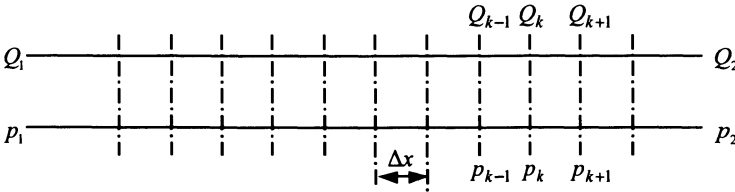
with  $N(s)$  as in Equation 4.132. Using finite difference derivatives

$$\frac{dQ}{dx} = \frac{Q_k - Q_{k-1}}{\Delta x} \quad \frac{dp}{dx} = \frac{p_{k+1} - p_k}{\Delta x} \quad (4.150)$$

for the elements  $k-1$ ,  $k$ , and  $k+1$  leads directly to

$$p_k = \frac{c}{\Delta x} \frac{Z_c (Q_{k-1} - Q_k)}{s} \quad k = 1, 2, \dots, n_s \quad (4.151)$$

$$Z_c Q_k = \frac{c}{N(s) \Delta x} \frac{p_k - p_{k+1}}{s} \quad (4.152)$$



**Figure 4.19.** Segmentation of pipelines

Equation 4.152 (taking into account Equation 4.132) can also be rewritten as

$$Z_c Q_k = \frac{c}{s \Delta x} \left[ (p_k - p_{k+1}) - Z_c Q_k \left( \mu' + \frac{K_1 s}{1 + \tau_1 s} + \frac{K_2 s}{1 + \tau_2 s} + \frac{K_3 s}{1 + \tau_3 s} \right) \right] \quad (4.153)$$

Figure 4.20 shows the block diagram to simulate one pipeline segment.

For the implementation of this block diagram in simulation environments, some remarks have to be made (Viersma, 1980):

- The length  $\Delta x$  of each segment should be sufficiently small to guarantee accurate simulations at higher frequencies. A rule of thumb is  $\Delta x = \lambda/10$ , where  $\lambda = c/f_{\max}$  is the wavelength of the highest frequency  $f_{\max}$  to be relevant.

- The time scale could be adjusted by changing the sonic velocity  $c$  and the viscosity  $\mu$  proportional to the time scale factor.
- An implication of the simulation is that, at any instance at any location, pressures and flows can be determined.

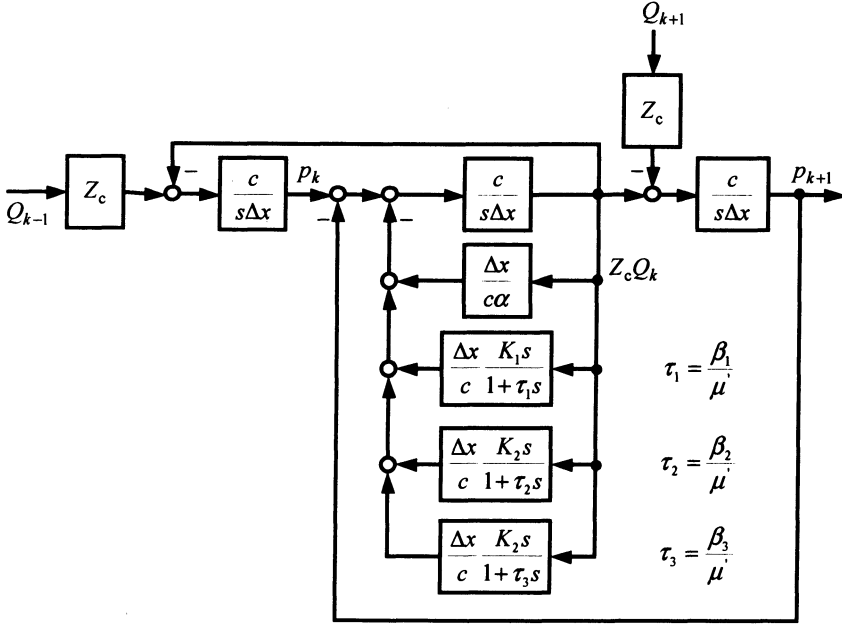


Figure 4.20. Viersma's simulation model for one pipeline segment

**Pawlik's Discrete Model.** In this approach, the pipeline is also cut into  $n_s$  segments, and the electro-hydraulic analogy principles in Section 3.5 are exploited to derive the discrete model for the pipeline; see Figure 4.21. The four-pole equations can then be written for the whole pipeline system

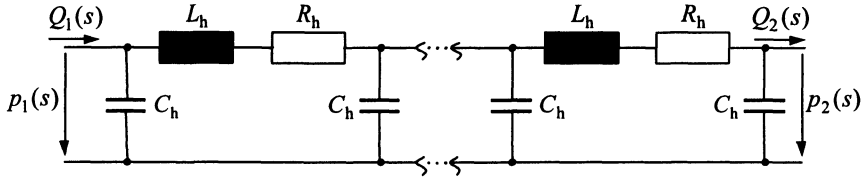
$$\begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} A(s) & B(s) \\ C(s) & A(s) \end{bmatrix}^{n_s} \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} = V^{n_s}(s) \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} \quad (4.154)$$

with

$$A(s) = 1 + R_h C_h s + L_h C_h s^2 \quad (4.155)$$

$$B(s) = -C_h s (2 + R_h C_h s + L_h C_h s^2) \quad (4.156)$$

$$C(s) = -(L_h s + R_h) \quad (4.157)$$



**Figure 4.21.** Pawlik's discrete pipeline model

The parameters hydraulic capacitance  $C_h$ , hydraulic resistance  $R_h$  and hydraulic inductance  $L_h$  can be calculated from those of the total pipeline system<sup>8</sup>

$$C_h = \frac{C_{h,tot}}{2n_s} \quad R_h = \frac{R_{h,tot}}{n_s} \quad L_h = \frac{L_{h,tot}}{n_s} \quad (4.158)$$

$C_h$  and  $L_h$  are given by (Equations 3.94 and 3.102 respectively)

$$C_{h,tot} = \frac{A_{pl} l}{E'} \quad (4.159)$$

$$L_{h,tot} = \frac{\rho l}{A_{pl}} \quad (4.160)$$

For the determination of  $R_h$ , Equation 3.54 is written as

$$\left. \frac{\partial p_{fric}(x, s)}{\partial x} = -\frac{2\eta}{R} \frac{\partial v_x(x, \tilde{r}, s)}{\partial r} \right|_{r=R} \quad (4.161)$$

Second, the velocity gradient in this equation can be derived from Equation 4.118 as

$$\left. \frac{\partial v_x(x, \tilde{r}, s)}{\partial r} \right|_{r=R} = j \frac{1}{\pi R^2} \sqrt{\frac{s}{\mu}} \frac{\tilde{R} J_1(\tilde{R})}{2 J_1(\tilde{R}) - \tilde{R} J_0(\tilde{R})} Q(x, s) \quad (4.162)$$

The integration (over the pipeline length  $l$ , bearing in mind Equation 4.161) leads to the pressure drop due to friction

$$\Delta p_{fric}(s) = -\frac{\rho L s}{\pi R^2} \left[ \frac{J_0(\tilde{R})}{J_2(\tilde{R})} + 1 \right] Q(x, s) \quad (4.163)$$

Finally, the total hydraulic resistance can be determined from

$$R_{h,tot} = \frac{\Delta p_{fric}(s)}{Q(x, s)} = \frac{\rho l s}{A_{pl}} [N(s) - 1] \quad (4.164)$$

It can easily be proven that, for the linear friction model (Equation 4.133), Equation 4.164 exactly reduces to the hydraulic resistance derived from the Hagen–Poiseuille equation (*i.e.* Equation 3.96).

Using the transformation matrix

<sup>8</sup> A very simple pipeline model consists of one resistance, one inductance and one capacitance for the whole pipeline. Such a model approximates the dynamic behaviour of pipelines very well in the low-frequency region as reported by Ratjen (2001b).

$$T_v(s) = -\frac{B(s)}{2\sqrt{A^2(s)-1}} \begin{bmatrix} 1 & 1 \\ \frac{\sqrt{A^2(s)-1}}{B(s)} & -\frac{\sqrt{A^2(s)-1}}{B(s)} \end{bmatrix} \quad (4.165)$$

and

$$T_v^{-1}(s) = \begin{bmatrix} -\frac{\sqrt{A^2(s)-1}}{B(s)} & -1 \\ -\frac{\sqrt{A^2(s)-1}}{B(s)} & 1 \end{bmatrix} \quad (4.166)$$

the transfer matrix  $V(s)$  is brought into the form

$$V(s) = T_v(s)A(s)T_v^{-1}(s) \quad (4.167)$$

with

$$A(s) = \begin{bmatrix} \lambda_1(s) & 0 \\ 0 & \lambda_2(s) \end{bmatrix} \quad (4.168)$$

and

$$\lambda_{1/2}(s) = A(s) \pm \sqrt{A^2(s)-1} \quad (4.169)$$

Thus, the transfer function of the whole pipeline system can finally be calculated by

$$V_{\text{tot}}(s) = V^{n_s}(s) = [T_v(s)A(s)T_v^{-1}(s)]^{n_s} = T_v(s)A^{n_s}(s)T_v^{-1}(s) \quad (4.170)$$

This leads to the four-pole equations

$$\begin{bmatrix} Q_2(s) \\ p_2(s) \end{bmatrix} = \begin{bmatrix} \frac{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)}{2} & \sqrt{\frac{B(s)}{C(s)}} \frac{\lambda_1^{n_s}(s) - \lambda_2^{n_s}(s)}{2} \\ \sqrt{\frac{C(s)}{B(s)}} \frac{\lambda_1^{n_s}(s) - \lambda_2^{n_s}(s)}{2} & \frac{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)}{2} \end{bmatrix} \begin{bmatrix} Q_1(s) \\ p_1(s) \end{bmatrix} \quad (4.171)$$

or equivalently

$$\begin{bmatrix} Q_2(s) \\ p_1(s) \end{bmatrix} = \begin{bmatrix} 1 - \frac{[\lambda_1^{n_s}(s) - \lambda_2^{n_s}(s)]^2}{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)} & \sqrt{\frac{B(s)}{C(s)}} \frac{\lambda_1^{n_s}(s) - \lambda_2^{n_s}(s)}{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)} \\ -\sqrt{\frac{C(s)}{B(s)}} \frac{\lambda_1^{n_s}(s) - \lambda_2^{n_s}(s)}{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)} & \frac{2}{\lambda_1^{n_s}(s) + \lambda_2^{n_s}(s)} \end{bmatrix} \begin{bmatrix} Q_1(s) \\ p_2(s) \end{bmatrix} \quad (4.172)$$

**Modal Representation of Yang and Tobler.** The modal approximation according to Yang and Tobler (1991) is also very appealing for simulation and control testing purposes, because it preserves the zeros of Equation 4.141, reproducing transmission delays faithfully. The input–output behaviour of the theoretical model, *i.e.*, the three distinct causal transfer functions, is represented by infinite product series models of the form

$$\frac{1}{\cosh \Gamma} \approx \sum_{i=1}^{\infty} \frac{(-1)^{i+1} (2/n_d) \lambda_{ci}}{\bar{s}^2 + 8\bar{s} + \lambda_{ci}^2} \quad (4.173)$$

$$\frac{Z_c \sqrt{N} \sinh \Gamma}{\cosh \Gamma} \approx \sum_{i=1}^{\infty} \frac{(2Z_c/n_d)(\bar{s} + 8)}{\bar{s}^2 + 8\bar{s} + \lambda_{ci}^2} \quad (4.174)$$

$$\frac{\sinh \Gamma}{Z_c \sqrt{N} \cosh \Gamma} \approx \sum_{i=1}^{\infty} \frac{(2Z_c/n_d)\bar{s}}{\bar{s}^2 + 8\bar{s} + \lambda_{ci}^2} \quad (4.175)$$

The normalised Laplace operator  $\bar{s}$ , the dissipation number  $n_d$ , and the root indexes  $\lambda_{ci}$  are calculated as follows:

$$\bar{s} = \frac{8s}{\mu}, \quad n_d = \frac{L\mu'}{8c}, \quad \lambda_{ci} = \frac{\pi(2i-1)}{2n_d} \quad (4.176)$$

Each of these models then represents a single resonance mode of pipeline dynamics. The infinite product expansions are then truncated (*i.e.*, taking  $n$  modes only) to give a finite dimensional linear system. The detailed procedures for obtaining the coefficients of these models are presented in Yang and Tobler (1991) and the references included therein. As far as the pipelines between the valve and the actuator are concerned, the modal approximation of the theoretical model Equation 4.141 becomes the model (of order  $2n$ ):

$$\begin{aligned} \dot{\mathbf{x}}_{pl} &= \mathbf{A}_{pl} \mathbf{x}_{pl} + \mathbf{B}_{pl} \mathbf{u}_{pl} \\ \mathbf{y}_{pl} &= \mathbf{C}_{pl} \mathbf{x}_{pl} \end{aligned} \quad (4.177)$$

with  $(i = 1, 2, \dots, n)$

$$\begin{aligned} \mathbf{x}_{pl} &= [p_{1,1} \quad Q_{2,1} \quad p_{1,2} \quad Q_{2,2} \quad \dots \quad p_{1,n} \quad Q_{2,n}]^T \\ \mathbf{u}_{pl} &= [p_2 \quad Q_1]^T, \quad \mathbf{y}_{pl} = [p_1 \quad Q_2]^T \\ \mathbf{A}_{pl} &= \text{diag}[\mathbf{A}_{pl,1} \quad \mathbf{A}_{pl,2} \quad \dots \quad \mathbf{A}_{pl,n}] \\ \mathbf{B}_{pl} &= [\mathbf{B}_{pl,1} \quad \mathbf{B}_{pl,1} \quad \dots \quad \mathbf{B}_{pl,n}] \\ \mathbf{C}_{pl} &= [\mathbf{C}_{pl,1} \quad \mathbf{C}_{pl,1} \quad \dots \quad \mathbf{C}_{pl,n}] \end{aligned} \quad (4.178)$$

and

$$\begin{aligned}
 A_{pli} &= \begin{bmatrix} 0 & \frac{(-1)^i}{8} \mu' Z_c \lambda_{ci} \\ \frac{(-1)^{i+1} \mu' \lambda_{ci}}{8 Z_c} & 0 \end{bmatrix} \\
 B_{pli} &= \begin{bmatrix} 0 & -\frac{\mu'}{4 Z_c n_d} \\ \frac{Z_c \mu'}{4 n_d} & 0 \end{bmatrix} \\
 C_{pl,i} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
 \end{aligned}$$

The parameters of this model actually refer to the linear friction model. For the more general model (also called the dissipative model), some parameters will be frequency dependent (due to the dynamic viscosity function  $N(s)$ ), and can be determined as indicated by Yang and Tobler (1991). The model would then be unnecessarily complicated.

As already mentioned in Section 4.2.5.4, the subsystem model for the pipelines between valve and actuator should have the input–output variant denoted by Configuration 3 (*i.e.*, Equation 4.146). The reason can be explained as follows: From the valve flow Equations 4.38 and 4.39, it is obvious that the flows  $Q_{Av}$  and  $Q_{Bv}$  at the valve side have to be defined as inputs of the pipeline models, and correspondingly the pressures  $p_{Av}$  and  $p_{Bv}$  as outputs. On the other hand, the pressure dynamics Equations 4.51 and 4.52 need the pressures  $p_{Ac}$  and  $p_{Bc}$  at the cylinder side to be defined as inputs of the pipeline models, and the cylinder flows  $Q_{Ac}$  and  $Q_{Bc}$  as outputs.

With the model in Equations 4.176–4.178, a useful low-order approximate model for the pipeline effects in hydraulic servo-systems is then available, in which the physical structure and the physical parameters are preserved. For simulation purposes, the first two to three modes of each pipeline may generally be expected to be relevant (*i.e.*,  $n = 2$ , or  $n = 3$ ). For control design purposes, the fundamental mode of the approximation (*i.e.*,  $n = 1$ ), which captures the essential low-frequency behaviour of the pipelines, is usually sufficient. In the case of the pipelines between the servo-valve and the actuator, this means:

$$\begin{bmatrix} \dot{p}_{A/B,v} \\ \dot{Q}_{A/B,c} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{\mu' Z_c \lambda_{ci}}{8} \\ \frac{\mu' \lambda_{ci}}{8 Z_c} & 0 \end{bmatrix} \begin{bmatrix} p_{A/B,v} \\ Q_{A/B,c} \end{bmatrix} + \begin{bmatrix} 0 & -\frac{\mu'}{4 Z_c n_d} \\ \frac{Z_c \mu'}{4 n_d} & 0 \end{bmatrix} \begin{bmatrix} p_{A/B,c} \\ Q_{A/B,v} \end{bmatrix} \quad (4.179)$$

### 4.3 Typical Non-linear State-space Models

If the state variables and the input variables are defined as

$$\begin{aligned} x_1 &\equiv x_p & x_2 &\equiv \dot{x}_p & x_3 &\equiv p_A & x_4 &\equiv p_B & x_5 &\equiv x_v^* & x_6 &\equiv \dot{x}_v^* \\ u_1 &\equiv u_v^* \text{ (valve input)} & u_2 &\equiv F_{\text{ext}} \text{ (external disturbance)} \end{aligned} \quad (4.180)$$

then the completely non-linear model of hydraulic servo-systems, which consists of a (two-stage) servo-valve and a (differential) cylinder can be written as

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{m_t(x_1)} \left[ (x_3 - \alpha x_4) A_p - F_f(x_2) - u_2 \right] \\ \dot{x}_3 &= \frac{E'_A(x_3)}{V'_A(x_1)} \left[ Q_A(x_3, x_5) - A_p x_2 + Q_{Li}(x_3, x_4) \right] \\ \dot{x}_4 &= \frac{E'_B(x_4)}{V'_B(x_1)} \left[ Q_B(x_4, x_5) + \alpha A_p x_2 - Q_{Li}(x_3, x_4) \right] \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= \omega_v^2 \left[ u_1 - \frac{2D_v}{\omega_v} x_6 - x_5 - f_{hs} \text{sign}(x_5) \right] \end{aligned} \quad (4.181)$$

The total mass is given by

$$m_t(x_1) = m_p + \rho [V_{LA} + (x_{10} + x_1) A_A] + \rho [V_{LB} + (x_{10} - x_1) \alpha A_p] \quad (4.182)$$

The friction forces are given by the equation

$$F_f(x_2) = \sigma x_2 + \text{sign}(x_2) \left[ F_{c0} + F_{s0} \exp\left(-\frac{|x_2|}{c_s}\right) \right] \quad (4.183)$$

The effective bulk modulus is expressed as

$$E'_A(x_3) = a_1 E_{\max} \log \left[ 100 \left( a_2 \frac{x_3}{p_{\max}} + a_3 \right) \right] \quad (4.184)$$

$$E'_B(x_4) = a_1 E_{\max} \log \left[ 100 \left( a_2 \frac{x_3}{p_{\max}} + a_3 \right) \right] \quad (4.185)$$

The cylinder chamber volumes are given by

$$V_A(x_1) = V_{pl,A} + (x_{10} + x_1) A_p \quad (4.186)$$

$$V_B(x_1) = V_{pl,B} + (x_{10} - x_1) \alpha A_p \quad (4.187)$$

The flow equations are characterised by

$$\begin{aligned} Q_A = & c_{v1} \operatorname{sg}(u_1) \operatorname{sign}(p_s - x_3) \sqrt{|p_s - x_3|} \\ & - c_{v2} \operatorname{sg}(-u_1) \operatorname{sign}(x_3 - p_T) \sqrt{|x_3 - p_T|} \end{aligned} \quad (4.188)$$

$$\begin{aligned} Q_B = & c_{v3} \operatorname{sg}(-u_1) \operatorname{sign}(p_s - x_4) \sqrt{|p_s - x_4|} \\ & - c_{v4} \operatorname{sg}(u_1) \operatorname{sign}(x_4 - p_T) \sqrt{|x_4 - p_T|} \end{aligned} \quad (4.189)$$

The (internal) leakage flow can be calculated by

$$Q_{Li}(x_3, x_4) = K_{Li}(x_4 - x_3) \quad (4.190)$$

Note that a number of standard assumptions that are typically satisfied in practice have been made for the derivation of this non-linear model:

- The hydraulic pump delivers a constant supply pressure, irrespective of the oil flow.
- The pressure in the tank is constant.
- A symmetrical critical-centre 4-way valve is modelled.
- The flow through the valve is turbulent.
- The leakage flow is laminar.
- The inefficient volumes of *e.g.* the pipelines between valve and actuator can be modelled as additive inefficient strokes.
- Coulomb, static and viscous friction are acting upon the actuator.
- The surroundings of the actuator and the load are rigid.
- Possible dynamic behaviour of the pressure in the pipelines between valve and actuator is assumed to be negligible (*i.e.*, the valve is placed directly upon the actuator). These dynamics can, however, simply be added if they appear to be essential, as explained below.

The pipeline dynamics between valve and actuator chambers are now taken into account in form of the Yang–Tobler modal representation, and integrated with the valve flow equations and the actuator model.

The equation of motion, the pressure dynamics in the cylinder chambers and the valve dynamics are still described by Equation 4.181. The valve flow is taken from the Equations 4.38 and 4.39 to be

$$\begin{aligned} Q_{Av} = & c_{v1} \operatorname{sg}(u_1) \operatorname{sign}(p_s - y_{pIA,1}) \sqrt{|p_s - y_{pIA,1}|} \\ & - c_{v2} \operatorname{sg}(-u_1) \operatorname{sign}(y_{pIA,1} - p_T) \sqrt{|y_{pIA,1} - p_T|} \end{aligned} \quad (4.191)$$

$$\begin{aligned} Q_{Bv} = & c_{v3} \operatorname{sg}(-u_1) \operatorname{sign}(p_s - y_{pIB,1}) \sqrt{|p_s - y_{pIB,1}|} \\ & - c_{v4} \operatorname{sg}(u_1) \operatorname{sign}(y_{pIB,1} - p_T) \sqrt{|y_{pIB,1} - p_T|} \end{aligned} \quad (4.192)$$

The pipeline models are given in the form of Equation 4.177

$$\begin{aligned} \dot{\mathbf{x}}_{pIA} &= \mathbf{A}_{pIA} \mathbf{x}_{pIA} + \mathbf{B}_{pIA} \mathbf{u}_{pIA} & \dot{\mathbf{x}}_{pIB} &= \mathbf{A}_{pIB} \mathbf{x}_{pIB} + \mathbf{B}_{pIB} \mathbf{u}_{pIB} \\ \mathbf{y}_{pIA} &= \mathbf{C}_{pIA} \mathbf{x}_{pIA} & \mathbf{y}_{pIB} &= \mathbf{C}_{pIB} \mathbf{x}_{pIB} \end{aligned} \quad (4.193)$$



with

$$\begin{aligned} \mathbf{u}_{\text{plA}} &= [p_{\text{Ac}} \quad Q_{\text{Av}}]^T & \mathbf{u}_{\text{plB}} &= [p_{\text{Bc}} \quad Q_{\text{Bv}}]^T \\ \mathbf{y}_{\text{plA}} &= [p_{\text{Av}} \quad Q_{\text{Ac}}]^T & \mathbf{y}_{\text{plB}} &= [p_{\text{Bv}} \quad Q_{\text{Bc}}] \end{aligned} \quad (4.194)$$

and the corresponding matrices  $\mathbf{A}_{\text{plA}}$ ,  $\mathbf{A}_{\text{plB}}$ ,  $\mathbf{B}_{\text{plA}}$ ,  $\mathbf{B}_{\text{plB}}$ ,  $\mathbf{C}_{\text{plA}}$ ,  $\mathbf{C}_{\text{plB}}$ , and state vectors  $\mathbf{x}_{\text{plA}}$ ,  $\mathbf{x}_{\text{plB}}$  as in Equations 4.178. The extended state vector is defined as

$$\begin{aligned} x_1 &\equiv x_p & x_2 &\equiv \dot{x}_p & x_3 &\equiv p_{\text{Ac}} & x_4 &\equiv p_{\text{Bc}} & x_5 &\equiv x_v^* & x_6 &\equiv \dot{x}_v^* \\ x_7 &\equiv p_{\text{Av},1} & x_8 &\equiv Q_{\text{Ac},1} & \dots & x_{7+2n-1} &\equiv p_{\text{Av},n} & x_{7+2n} &\equiv Q_{\text{Ac},n} \\ x_{7+2n+1} &\equiv p_{\text{Bv},1} & x_{7+2n+2} &\equiv Q_{\text{Bc},1} & \dots & x_{7+4n-1} &\equiv p_{\text{Bv},n} & x_{7+4n} &\equiv Q_{\text{Bc},n} \end{aligned} \quad (4.195)$$

## 4.4 Structured and Simplified Models of Valve-controlled Systems

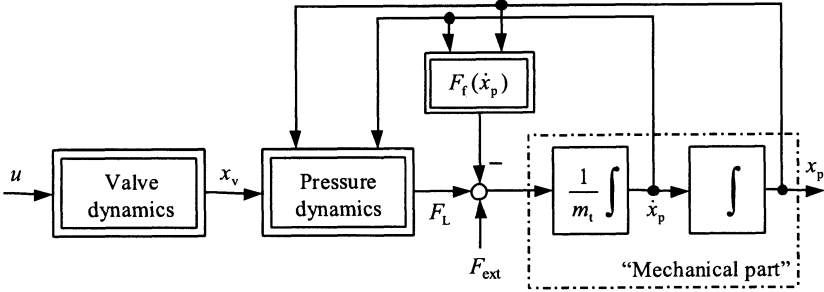
In the theoretical model presented until now, most dynamic and non-linear effects that may be relevant in the hydraulic servo-system are included. By means of simulations with this model, the admissibility to neglect or to simplify some effects can be investigated. Furthermore, under some assumptions that are satisfied in practice, and for some points of operation, simplified and linearised models of the hydraulic servo-system can be derived. As a result, a reasonably simple model will be obtained. It describes the relevant dynamic behaviour of hydraulic servo-systems, while the physical structure of the model is widely preserved.

### 4.4.1 Relevance of Valve and Pipeline Dynamics

For low-frequency behaviour, which is the frequency range of interest for most applications, the pipeline effects do not play a role in the input–output behaviour. However, in the high-frequency behaviour of hydraulic servo-systems, the pipelines between valve and actuator chambers introduce badly damped resonance effects (*i.e.* pressure peaking and oscillations). Essentially, pipeline dynamics contain RHP zeros in the complex plane, making the system non-minimum phase. This may not only cause poor response, but may lead to instability if a high-gain controller is used.

Appendix B.2 shows examples of step responses and frequency responses of servo-valves provided by the manufacturer (Bosch Rexroth, 2000). In most cases, where the closed-loop system bandwidth is not too high, the control input applied to the spool valve is directly proportional to the spool position, *i.e.*, the dynamics of the valve is fast enough to be neglected – at least in the controller design. However, the servo-valve dynamics has a marked filtering effect on the transient pressures. The resulting pressure ripples have been measured by Watton (1987) and Sohl and Bobrow (1999) and showed a higher degree of significance at lower temperature due to the introduction of significant line resistance. Thus the valve dynamics should always be included in the simulation model for testing the controller.

If the servo-valve is now modelled as a proportional system with a gain  $K_v$ , the system order is reduced to  $n = 4$ . The model can then be split up into a non-linear “hydraulic part” model of the order  $n = 2$  with the states  $p_A = x_3$  and  $p_B = x_4$ , and a linear “mechanical part” model of the same order, provided the friction forces are neglected (or only viscous friction is taken into account); see Figure 4.22. Otherwise, the non-linear friction force term can be considered as a disturbance term with the velocity  $x_2$  as the disturbance variable.



**Figure 4.22.** Simplified and structured model of hydraulic servo-systems ( $F_L$  is the load force; see Sections 4.4.3 and 4.4.4) (Taken from Bernzen (1999), p 22 (original publication). Reprinted with kind permission of VDI Verlag.)

For positive control input signal  $u \geq 0$  the “hydraulic part” model takes the form

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'_A(x_3)}{V_A(x_1)} [-A_p x_2 + K_{Li}(x_3 - x_4)] \\ \frac{E'_B(x_4)}{V_B(x_1)} [\alpha A_p x_2 - K_{Li}(x_3 - x_4)] \end{bmatrix} + \begin{bmatrix} \frac{E'_A(x_3)}{V_A(x_1)} c_{v1} \sqrt{p_s - x_3} \\ -\frac{E'_B(x_4)}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \end{bmatrix} \quad (4.196)$$

and for negative control input signal  $u < 0$

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'_A(x_3)}{V_A(x_1)} [-A_p x_2 + K_{Li}(x_3 - x_4)] \\ \frac{E'_B(x_4)}{V_B(x_1)} [\alpha A_p x_2 - K_{Li}(x_3 - x_4)] \end{bmatrix} + \begin{bmatrix} \frac{E'_A(x_3)}{V_A(x_1)} c_{v2} \sqrt{x_3 - p_T} \\ -\frac{E'_B(x_4)}{V_B(x_1)} c_{v3} \sqrt{p_s - x_4} \end{bmatrix} \quad (4.197)$$

Assuming a constant bulk modulus for both cylinder chambers and neglecting leakage flows results in the simple model for positive input  $u > 0$

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'}{V_A} c_v \sqrt{p_s - x_3} \\ -\frac{E'}{V_B} c_v \sqrt{x_4 - p_T} \end{bmatrix} u + \begin{bmatrix} -\frac{E'}{V_A} A_p \\ \frac{E'}{V_B} \alpha A_p \end{bmatrix} x_2 \quad (4.198)$$

and for negative input  $u < 0$

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'}{V_A} c_v \sqrt{x_3 - p_T} \\ -\frac{E'}{V_B} c_v \sqrt{p_s - x_4} \end{bmatrix} u + \begin{bmatrix} -\frac{E'}{V_A} A_p \\ \frac{E'}{V_B} \alpha A_p \end{bmatrix} x_2 \quad (4.199)$$

#### 4.4.2 Approximation of Pressure Dynamics

Relationships will now be derived to estimate the time constant of pressure dynamics in hydraulic volumes. This will help to decide whether pipeline and valve dynamics may be neglected.

##### 4.4.2.1 Experimental Step Responses

Figures 4.23–4.25 show the valve position step response, and the corresponding pressures in the cylinder chambers A and B respectively. These experimental results have been gathered from experiments with the blocked (*i.e.*,  $\dot{x}_p = 0$ ) small differential cylinder of Section 7.2.1.

From these figures, it can clearly be seen that the rise times of the pressure are small. A rough estimation formula for the time constant of the pressure dynamics in a simple volume  $V$  (*i.e.*, cylinder chamber) will be derived in the next section.

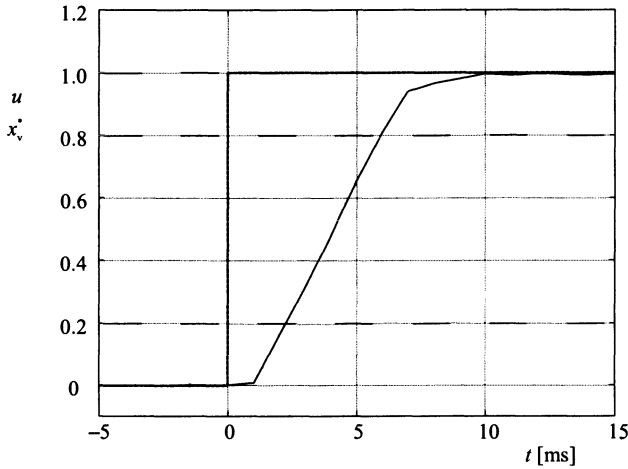


Figure 4.23. Valve spool position step response (dotted line:  $u$ ; solid line:  $x_v^*$ )

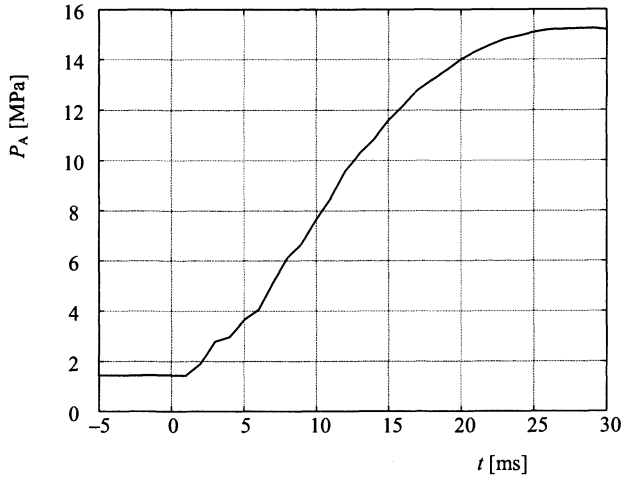
##### 4.4.2.2 Estimation of Time Constant

The differential equation for the pressure dynamics in a cylinder chamber of volume  $V$  is given by

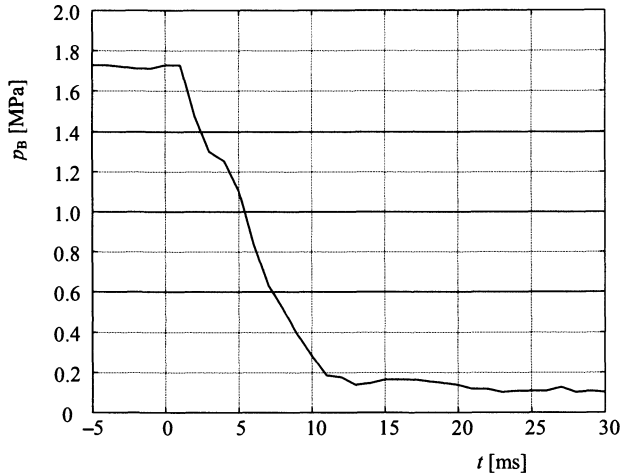
$$\dot{p}(t) = \frac{E'}{V} c_v \sqrt{p_s - p(t)} \dot{x}_v^*(t) \quad (4.200)$$

with the oil bulk modulus  $E'$ , the valve coefficient  $c_v^*$  and the normalised valve position  $-1 < x_v^* < 1$ . With the normalised pressure

$$p^*(t) \equiv \frac{p(t)}{p_s} \quad (4.201)$$



**Figure 4.24.** Pressure in chamber B (blocked cylinder)



**Figure 4.25.** Pressure in chamber A (blocked cylinder)

Equation 4.200 can be expressed as

$$\dot{p}^*(t) = \frac{E' c_v^*}{V \sqrt{p_s}} \sqrt{1 - p^*(t)} x_v^*(t) \quad (4.202)$$

or in linearised form

$$\Delta \dot{p}^*(t) = \frac{E' c_v^*}{V \sqrt{p_s}} \left( \frac{-x_{v0}^*}{2\sqrt{1-p_0^*}} \Delta p^*(t) + \sqrt{1-p_0^*} \Delta x_v^*(t) \right) \quad (4.203)$$

Let  $x_v^* = 0$  and write the last equation as

$$T_p \Delta \dot{p}^*(t) = \Delta x_v^*(t) \quad (4.204)$$

where the time constant is determined by

$$T_p = \frac{V \sqrt{p_s}}{E' c_v^*} \frac{1}{\sqrt{1-p_0^*}} \quad (4.205)$$

Hence  $T_p$  is directly proportional to the volume  $V$ , and thus to the cylinder dimension. This means that the time constant will be low for small cylinders and high for big cylinders. The value of  $T_p$  can serve as a criterion as to whether pipeline or pressure dynamics have to be taken into account or not.

#### 4.4.3 Introduction of Load Pressure

In this section we have to distinguish between hydraulic servo-systems with synchronising cylinders and those with differential cylinders (see Figure 4.1).

##### 4.4.3.1 Symmetric Actuator

For a symmetric valve combined with a symmetric cylinder (where the pressure areas in each chamber are equal), it is common practice to introduce the pressure drop across the load, or simply the load pressure

$$p_L = p_A - p_B \quad (4.206)$$

which corresponds to the flow through the load, or simply the load flow

$$Q_L = Q_A - Q_B \quad (4.207)$$

Therefore, it is possible to express load flow as a function of valve position and load pressure, *i.e.*,

$$Q_L = Q_L(x_v, p_L) \quad (4.208)$$

Using the continuity equation

$$Q_A = -Q_B \quad (4.209)$$

and the relation (provided leakage flows are negligible)

$$p_s + p_T = p_A + p_B \quad (4.210)$$

yields

$$\begin{aligned}
p_A &= \frac{p_S + p_T + p_L}{2} \\
p_B &= \frac{p_S + p_T - p_L}{2}
\end{aligned} \tag{4.211}$$

and

$$Q_L = c_v x_v \sqrt{2} \sqrt{p_S - p_T - \text{sign}(x_v) p_L} \tag{4.212}$$

Then the following equation for the pressure dynamics results:

$$\begin{aligned}
\dot{p}_L &= \frac{E'}{V} \left[ (Q_A - Q_B) - 2A_p \dot{x}_p \right] \\
&= \frac{E'}{V} \left( c_v x_v \sqrt{2} \sqrt{p_S - p_T - \text{sign}(x_v) p_L} - 2A_p \dot{x}_p \right)
\end{aligned} \tag{4.213}$$

Thereby it has been assumed that

$$\begin{aligned}
E'_A &= E'_B = E' \\
V_A &= V_B = V
\end{aligned} \tag{4.214}$$

#### 4.4.3.2 Asymmetric Actuator

For a symmetric valve combined with an asymmetric cylinder, however, such load flow cannot be introduced because the inlet flow and the outlet flow are always different due to the continuity equation, *i.e.*,

$$\begin{aligned}
Q_B &= \alpha A_p \dot{x}_p \\
Q_A &= A_p \dot{x}_p
\end{aligned} \tag{4.215}$$

(even if the internal leakage flow is negligible).

Instead of the load pressure in Equation 4.206, it is now more convenient to define the load pressure

$$p_L \equiv p_A - \alpha p_B \tag{4.216}$$

which can be seen as the “virtual” pressure (Feigel, 1987a) required to counterbalance the friction and load forces

$$p_L = \frac{F_f + F_L}{A_p} = \frac{F}{A_p} \tag{4.217}$$

Equations 4.2, 4.3, 4.215 and 4.216 can be combined to give the following relations:

$$\begin{aligned}
p_B &= p_T + \alpha^2(p_S - p_A) \\
p_A &= \frac{1}{1+\alpha^3}(\alpha^3 p_S + \alpha p_T + p_L) \\
p_B &= \frac{1}{1+\alpha^3}(\alpha^2 p_S + p_T - \alpha^2 p_L) \\
\dot{x}_p &= \frac{1}{A_p \sqrt{(1+\alpha^3) R_h}} \sqrt{p_S - \alpha p_T - p_L}
\end{aligned} \tag{4.218}$$

for  $\dot{x}_p > 0$ , and

$$\begin{aligned}
p_B &= p_S + \alpha^2(p_T - p_A) \\
p_A &= \frac{1}{1+\alpha^3}(\alpha p_S + \alpha^3 p_T + p_L) \\
p_B &= \frac{1}{1+\alpha^3}(p_S + \alpha^2 p_T - \alpha^2 p_L) \\
\dot{x}_p &= \frac{1}{A_p \sqrt{(1+\alpha^3) R_h}} \sqrt{\alpha p_S - p_T + p_L}
\end{aligned} \tag{4.219}$$

for  $\dot{x}_p < 0$ , where  $R_h$  is the so-called hydraulic (turbulent) resistance

$$R_h = \frac{1}{(c_v x_v)^2} \tag{4.220}$$

It is clear from these equations that with  $p_S > p_T$ ,  $\alpha < 1$ ,  $p_L > 0$  smaller pressures in both chamber volumes always occur when the piston is running out (*i.e.*,  $\dot{x}_p > 0$ ) than those occurring when the piston is running in (*i.e.*,  $\dot{x}_p < 0$ ), *i.e.*,  $p_{A,out} < p_{A,in}$  and  $p_{B,out} < p_{B,in}$ .

Moreover, for equal spool valve positions, *i.e.*,  $R_{h,out} = R_{h,in}$  (symmetric valve), we have the velocity ratio

$$r_v(p_S, p_T, p_L) = \frac{\dot{x}_{p,in}}{\dot{x}_{p,out}} = \sqrt{\frac{\alpha p_S - p_T + p_L}{p_S - \alpha p_T - p_L}} \tag{4.221}$$

This function is continuously ascending for all  $\alpha p_S - p_T + p_L \geq 0$  and  $p_S - \alpha p_T - p_L \geq 0$ , and it thus has no extrema. In the case of no load ( $p_L = 0$ ), the piston running-out velocity is usually higher than the running-in velocity.

#### 4.4.4 Linearised Models

A linearisation of the algebraic non-linear equations which describe the pressure-flow characteristics of valves is often necessary before (linear) identification or control design techniques can be applied.

#### 4.4.4.1 Valve Sensitivity Coefficients

The expression of Equations 4.2 and 4.3 as a Taylor series about a particular operating point  $P_0 = (x_{v0}, p_{A0}, p_{B0})$ <sup>9</sup>

$$\Delta Q = \left. \frac{\partial Q}{\partial x_v} \right|_{P_0} \Delta x_v + \left. \frac{\partial Q}{\partial p} \right|_{P_0} \Delta p + \dots \quad (4.222)$$

leads to

$$\Delta Q_A = K_{Qx,A} \Delta x_v + K_{Qp,A} \Delta p_A \quad (4.223)$$

$$\Delta Q_B = K_{Qx,B} \Delta x_v + K_{Qp,B} \Delta p_B \quad (4.224)$$

Thereby, the *flow gains* are defined by

$$K_{Qx,A} = \left. \frac{\partial Q_A}{\partial x_v} \right|_{P_0} = \begin{cases} c_v \sqrt{(p_S - p_{A0})} & \text{for } x_v > 0 \\ c_v \sqrt{(p_{A0} - p_T)} & \text{for } x_v < 0 \end{cases} \quad (4.225)$$

$$K_{Qx,B} = \left. \frac{\partial Q_B}{\partial x_v} \right|_{P_0} = \begin{cases} -c_v \sqrt{(p_{B0} - p_T)} & \text{for } x_v > 0 \\ -c_v \sqrt{(p_S - p_{B0})} & \text{for } x_v < 0 \end{cases} \quad (4.226)$$

The flow–pressure coefficients are expressed as

$$K_{Qp,A} = \left. \frac{\partial Q_A}{\partial p_A} \right|_{P_0} = \begin{cases} \frac{-c_v x_{v0}}{2\sqrt{(p_S - p_{A0})}} & \text{for } x_v < 0 \\ \frac{c_v x_{v0}}{2\sqrt{(p_{A0} - p_T)}} & \text{for } x_v > 0 \end{cases} \quad (4.227)$$

$$K_{Qp,B} = \left. \frac{\partial Q_B}{\partial p_B} \right|_{P_0} = \begin{cases} \frac{-c_v x_{v0}}{2\sqrt{(p_{B0} - p_T)}} & \text{for } x_v > 0 \\ \frac{c_v x_{v0}}{2\sqrt{(p_S - p_{B0})}} & \text{for } x_v < 0 \end{cases} \quad (4.228)$$

Other useful quantities are the *pressure sensitivities*, defined by

$$K_{px,A} = \left. \frac{\partial p_A}{\partial x_v} \right|_{P_0} \quad (4.229)$$

$$K_{px,B} = \left. \frac{\partial p_B}{\partial x_v} \right|_{P_0} \quad (4.230)$$

which are related to the other quantities by the well-known relation from calculus

<sup>9</sup> The  $\Delta$  operator stands for the deviation from the operating point about which the model is linearised. This operator is often omitted for simplicity, especially there, where it is clear that a linear representation is meant.



$$\frac{\partial p}{\partial x_v} = \frac{\partial Q / \partial x_v}{\partial Q / \partial p} \quad \text{or} \quad K_{px} = \frac{K_{Qx}}{K_{Qp}} \quad (4.231)$$

The coefficients  $K_{Qx}$ ,  $K_{Qp}$  and  $K_{px}$  are referred to as *valve sensitivity coefficients* and are extremely important in determining stability, frequency response, and other dynamic characteristics (Merritt, 1967): the flow gain directly affects the open loop gain constant in a system and, therefore, has a direct influence on system stability. The flow–pressure coefficient directly affects the damping ratio of valve–cylinder combinations. The pressure sensitivity of valves is quite large, which accounts for the ability of valve–cylinder combinations to breakaway large friction loads with little error.

**Symmetric Actuator.** The same procedure with Equation 4.212 leads to the flow gain

$$K_{Qx,L} = \left. \frac{\partial Q_L}{\partial x_v} \right|_{p_0} = \begin{cases} c_v \sqrt{2} \sqrt{p_s - p_T + p_L} & \text{for } x_v < 0 \\ c_v \sqrt{2} \sqrt{p_s - p_T - p_L} & \text{for } x_v > 0 \end{cases} \quad (4.232)$$

and the flow–pressure coefficient

$$K_{Qp_L} = \left. \frac{\partial Q_L}{\partial p_L} \right|_{p_0} = \begin{cases} \frac{c_v x_{v0}}{\sqrt{2} \sqrt{p_s - p_T + p_{L0}}} & \text{for } x_v < 0 \\ \frac{-c_v x_{v0}}{\sqrt{2} \sqrt{p_s - p_T - p_{L0}}} & \text{for } x_v > 0 \end{cases} \quad (4.233)$$

#### 4.4.4.2 Linear State-space Model

The linearisation of the non-linear Equations 4.181–4.190 yields the state-space description (see also Appendix C.4)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\zeta_1 & \zeta_2 & -\alpha\zeta_2 & 0 & 0 \\ 0 & -\zeta_3\zeta_5 & \zeta_5(\zeta_7 - \zeta_4) & \zeta_4\zeta_5 & \zeta_8\zeta_5 & 0 \\ 0 & \alpha\zeta_3\zeta_6 & \zeta_4\zeta_6 & -\zeta_6(\zeta_9 + \zeta_4) & \zeta_{10}\zeta_6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\zeta_{11} & -\zeta_{12} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \zeta_{14} \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ -\zeta_{13} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_2 \quad (4.234)$$

with the parameters

$$\begin{aligned}
\varsigma_1 &= \frac{\sigma}{m_p}, \quad \varsigma_2 = \frac{A_p}{m_p}, \quad \varsigma_3 = A_p, \quad \varsigma_4 = K_{Li}, \quad \varsigma_5 = \frac{E'(x_{30})}{V_A(x_{10})} \\
\varsigma_6 &= \frac{E'(x_{40})}{V_B(x_{10})}, \quad \varsigma_7 = K_{Qp,A}, \quad \varsigma_8 = K_{Qx,A}, \quad \varsigma_9 = K_{Qp,B}, \quad \varsigma_{10} = K_{Qx,B} \\
\varsigma_{11} &= \omega_v^2, \quad \varsigma_{12} = 2D_v\omega_v, \quad \varsigma_{13} = \frac{1}{m_p}, \quad \varsigma_{14} = \frac{\omega_v^2}{u_{\max}}
\end{aligned} \tag{4.235}$$

Thereby, the influence of the piston position change  $\Delta x_p$  in  $V_A(x_p)$  and  $V_B(x_p)$ , as well as pressure changes  $\Delta p_A$  and  $\Delta p_B$  in  $E'(p_A)$  and  $E'(p_B)$ , has been neglected.  $\Delta Q_A$  and  $\Delta Q_B$  have been replaced by Equations 4.223 and 4.224.

#### 4.4.4.3 Reduced State-space Model and Frequency Response Characteristics

In this section, first let the valve dynamics be negligible. Then, a linear model of reduced order will be derived to analyse the frequency response characteristics of hydraulic systems; see also Feigel (1987b). To eliminate the pressures  $p_A$  and  $p_B$ , the static relationships, Equations 4.218 or 4.219, are linearised to give

$$\Delta p_A = \frac{1}{1+\alpha^3} \Delta p_L \tag{4.236}$$

$$\Delta p_B = -\frac{\alpha^2}{1+\alpha^3} \Delta p_L \tag{4.237}$$

Next, the differentiation of Equation 4.216 delivers

$$\dot{p}_A - \alpha \dot{p}_B = \dot{p}_L \tag{4.238}$$

Now, linearising, combining Equations 4.51 and 4.52 and substituting them into Equation 4.238, and taking into account Equations 4.223, 4.224, 4.236 and 4.237, yields the load pressure dynamics equation

$$\Delta \dot{p}_L = K_Q \Delta x_v - \frac{1}{T_h} \Delta p_L - K_d \Delta \dot{x}_p \tag{4.239}$$

Thereby, the parameters  $K_Q$ ,  $K_d$  and  $T_h$  may be expressed as

$$K_Q = \frac{E'_A}{V_A} K_{Qx,A} - \alpha \frac{E'_B}{V_B} K_{Qx,B} \tag{4.240}$$

$$K_d = \frac{A_p}{C_h}, \quad C_h = \left( \frac{E'_A}{V_A} + \alpha^2 \frac{E'_B}{V_B} \right) \tag{4.241}$$

$$T_h = \frac{1}{\alpha \frac{E'_B}{V_B} \left[ \frac{K_{Qp,B} \alpha^2 + C_{Li} (1 + \alpha^2)}{1 + \alpha^3} \right] - \frac{E'_A}{V_A} \left[ \frac{K_{Qp,A} - C_{Li} (1 + \alpha^2)}{1 + \alpha^3} \right]} \tag{4.242}$$

Finally, combining Equation 4.239 with the linearised Equation 4.56

$$\begin{aligned}
\Delta \ddot{x}_p &= -\frac{\sigma}{m_p} \Delta \dot{x}_p + \frac{A_p}{m_p} \Delta p_L - \frac{1}{m_p} \Delta F_{\text{ext}} \\
&\equiv -\frac{1}{T_m} \Delta \dot{x}_p + \frac{K_m}{T_m} \Delta p_L - \frac{1}{m_p} \Delta F_{\text{ext}}, \quad T_m = \frac{m_p}{\sigma}, \quad K_m = \frac{A_p}{\sigma}
\end{aligned} \tag{4.243}$$

Laplace transformed, leads (after some algebraic manipulations) to

$$\begin{aligned}
\Delta x_p(s) &= \frac{\frac{A_p}{m_p} K_Q \Delta x_v(s) - \frac{1+T_h s}{m_p T_h} \Delta F_{\text{ext}}(s)}{s \left[ s^2 + \left( \frac{1}{T_m} + \frac{1}{T_h} \right) s + \frac{1}{T_m} \left( \frac{1}{T_h} + K_m K_d \right) \right]} \\
&\equiv \frac{\frac{A_p}{m_p} K_Q \Delta x_v(s) - \frac{1+T_h s}{m_p T_h} \Delta F_{\text{ext}}(s)}{s (s^2 + 2D_h \omega_h s + \omega_h^2)} \\
&= \frac{K_m T_h K_Q \Delta x_v(s) - \frac{1+T_h s}{m_p} T_m \Delta F_{\text{ext}}(s)}{T_m T_h s^3 + (T_m + T_h) s^2 + (1 + K_m K_d T_h) s}
\end{aligned} \tag{4.244}$$

Thus, the hydraulic natural frequency is given by

$$\omega_h = \sqrt{\frac{\sigma}{m_p T_h} + \frac{A_p}{m_p} K_d} \tag{4.245}$$

and the (dimensionless) damping ratio by

$$D_h = \frac{\frac{1}{T_m} + \frac{1}{T_h}}{2\omega_h} \tag{4.246}$$

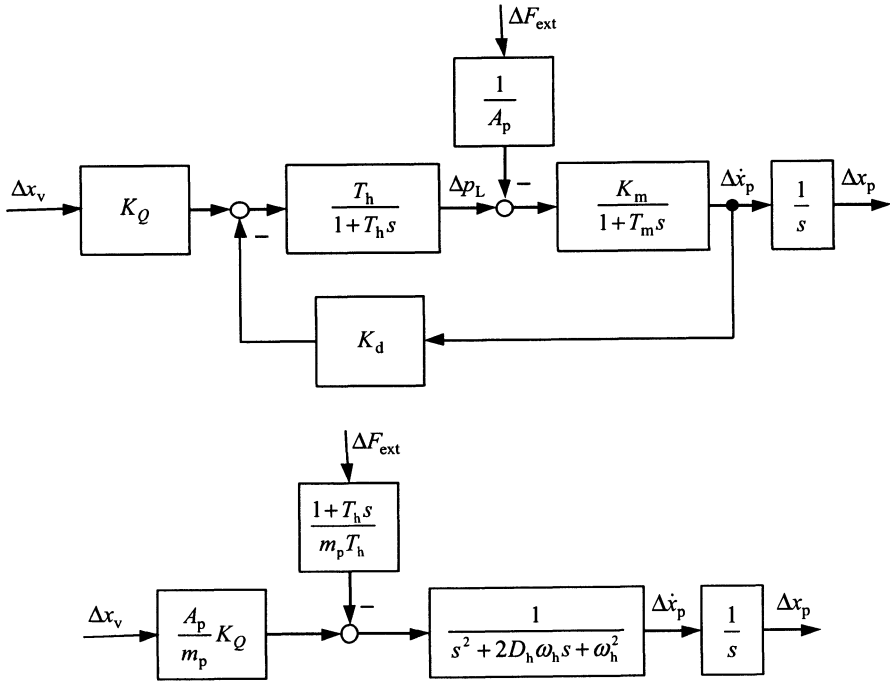
If friction effects and leakage flows are negligible,  $\omega_h$  and  $D_h$  become<sup>10</sup>

$$\omega_h = \sqrt{\frac{A_p^2}{m_p} \left( \frac{E'_A}{V_A} + \alpha^2 \frac{E'_B}{V_B} \right)} \tag{4.247}$$

$$D_h = \frac{1}{\frac{2\alpha}{1+\alpha^3} \left( -\alpha^2 \frac{E'_B}{V_B} K_{Qp,B} - \frac{E'_A}{V_A} K_{Qp,A} \right) \sqrt{\frac{A_p^2}{m_p} \left( \frac{E'_A}{V_A} + \alpha^2 \frac{E'_B}{V_B} \right)}} \tag{4.248}$$

Equation 4.244 gives the dynamic response of the cylinder to both valve position and load force inputs; see also Figure 4.26.

<sup>10</sup> Equation 4.247 (with  $E'_A = E'_B = E$ ) is widely used for calculating the natural frequency of hydraulic cylinders; see Götz (1989:103), and Norvelle (2000:165), for example.



**Figure 4.26.** Two equivalent block diagrams of the simplified and linearised model

If now the state variables and the input variables are defined as

$$\begin{aligned} x_1 &\equiv \Delta x_p & x_2 &\equiv \Delta \dot{x}_p & x_3 &\equiv \Delta p_L \\ u_1 &\equiv \Delta u_v^* & u_2 &\equiv \Delta F_{\text{ext}} \end{aligned} \quad (4.249)$$

then the reduced linear model of hydraulic cylinders can be written as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{1}{T_m} & \frac{A_p}{m_p} \\ 0 & -K_d & -\frac{1}{T_h} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ K_Q^* \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ -\frac{1}{m_p} \\ 0 \end{bmatrix} u_2$$

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{b}_1 u_1 + \mathbf{b}_2 u_2 \quad (4.250)$$

The model structure of Equation 4.250 is a compact state-space description of the dominant servo-system dynamics, with physical quantities constituting the model states. This model has three poles, one of which lies at the origin, representing the integrating behaviour of the system. The two other poles form a conjugate complex pole pair.

The valve input non-linearity (which is a function of the pressure and valve input) has two effects: an extra term comparable to leakage ( $-1/T_h$ ) is introduced and the input gain of the system varies with the pressures (*i.e.*, pressure difference) in the actuator and the valve control signal term ( $K_Q$ ). The natural frequency of the system is determined by the load mass ( $A_p/m_p$ ) and the stiffness of the oil chambers, which is position dependent ( $C_h$ ). The damping of the resonance frequency is determined by the viscous friction ( $-1/T_m = -\sigma/m_p$ ) and the leakage ( $-1/T_h$ ).

In most cases, the model in Equation 4.250 will be sufficient to describe the dynamics of a hydraulic actuator driving a stiff inertial load. If the load is not stiff, but constituted by some flexible mechanical system, the model will have to be extended with some states, describing the dynamics of the load, as already treated in Section 4.2.3.5.

**Remarks.** Some important remarks concerning this linearised model are given:

- The leakage to the tank is assumed to be negligible.
- Only the parameter  $\sigma$  is considered for the parameterisation of the friction effects, because the other (Coulomb) friction parameters have been excluded from the model as a result of the linearisation process (only smooth non-linearities are relevant).
- Leakage and (viscous) friction introduce damping in the system. Suppose  $\sigma = 0$  and  $-1/T_h = 0$ , the second-order subsystem of Equation 4.244 would have poles on the imaginary axis, *i.e.*,  $D_h = 0$ , and

$$\lambda_{1,2} = \pm j \sqrt{\left| \frac{A_p}{m_p} \right| - \frac{A_p}{C_h}} = \pm j \sqrt{\frac{A_p^2}{m_p} \left( \frac{E'_A}{V_A} + \alpha^2 \frac{E'_B}{V_B} \right)} \quad (4.251)$$

**Symmetric Actuator.** For this case, the state-space model can more easily be derived from Equations 4.213 and 4.232, and thus takes the simple form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\sigma}{m_p} & \frac{A_p}{m_p} \\ 0 & -2A_p \frac{E'}{V} & K_{QpL} \frac{E'}{V} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ K_{Qx,L} \frac{E'}{V} \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ -\frac{1}{m_p} \\ 0 \end{bmatrix} u_2$$

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{b}_1 u_1 + \mathbf{b}_2 u_2 \quad (4.252)$$

**Extension with Respect to Valve and Pipeline Dynamics.** According to Heintze (1997), the behaviour of the pipeline dynamics can also be observed in the transfer function<sup>11</sup>  $P_L^*(s)/X_V^*(s)$ ; see Figure 4.27. On measuring the pressure difference near the valve, the measured pipeline dynamics will show lightly damped zeros in

<sup>11</sup> Capital letters (*e.g.*,  $X$ ,  $Y$ ) represent for Laplace-transformed variables (*e.g.*,  $x$ ,  $y$ ).

addition to lightly damped poles. A second-order system for the description of the pipeline dynamics can then be formulated as

$$\frac{X_{va}^*(s)}{X_{vv}^*(s)} = \frac{s^2 + 2D_z\omega_z s + \omega_z^2}{s^2 + 2D_p\omega_p s + \omega_p^2} \quad (4.253)$$

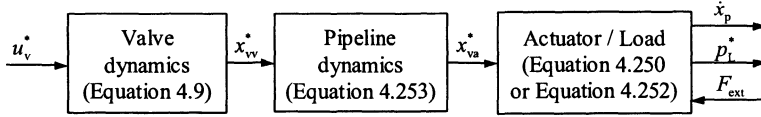


Figure 4.27. Simplified hydraulic servo-system model

#### 4.4.4.4 Classification of Linear Models

As described in Section 2.4, feedback measuring devices can be obtained to sense almost any physical quantity, such as position, velocity, acceleration, force, pressure, *etc.*, and hydraulic servo-systems may be designed to control any of these quantities. Position is probably most often controlled. For each controlled variable, a simplified (linear) transfer function of the hydraulic system (valve + actuator) can be given, and used for analysis and controller design. The control input is the normalised valve voltage  $u = u_v^*$ .

**Position Control.** Combining the servo-valve dynamics model (Equation 4.9) with the reduced linear actuator model (Equation 4.250, Figure 4.26), the transfer function for position control design is obtained (IT<sub>4</sub>-system):

$$\begin{aligned} G_x(s) &= \frac{X_p(s)}{U(s)} = \frac{K_v K_o^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{\frac{A_p}{m_p}}{s^2 + 2D_h\omega_h s + \omega_h^2} \frac{1}{s} \\ &= \frac{K_v K_o^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{K_m T_h}{T_m T_h s^2 + (T_m + T_h)s + (1 + K_m K_d T_h)} \frac{1}{s} \end{aligned} \quad (4.254)$$

**Velocity Control.** Since velocity is integrated to get position, the transfer function for velocity control design can be easily deduced from Equation 4.254 as (PT<sub>4</sub>-system)

$$G_v(s) = \frac{\dot{X}_p(s)}{U(s)} = \frac{K_v K_o^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{\frac{A_p}{m_p}}{s^2 + 2D_h\omega_h s + \omega_h^2}$$

$$= \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{K_m T_h}{T_m T_h s^2 + (T_m + T_h)s + (1 + K_m K_d T_h)} \quad (4.255)$$

**Acceleration Control.** As velocity is differentiated to yield acceleration, the transfer function for acceleration control is written as (DT<sub>4</sub>-system)

$$\begin{aligned} G_a(s) &= \frac{\ddot{X}_p(s)}{U(s)} = \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{\frac{A_p}{m_p} s}{s^2 + 2D_h \omega_h s + \omega_h^2} \\ &= \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{K_m T_h s}{T_m T_h s^2 + (T_m + T_h)s + (1 + K_m K_d T_h)} \end{aligned} \quad (4.256)$$

**Load Pressure (difference-pressure) Control.** From Equation 4.243 it follows that

$$P_L(s) = \left( \frac{m_p}{A_p} + \frac{\sigma}{A_p} \frac{1}{s} \right) \ddot{X}(s) \quad (4.257)$$

Thus, the transfer function for load pressure control can be expressed as (PDT<sub>4</sub>-system)

$$\begin{aligned} G_{pL}(s) &= \frac{P_L(s)}{U(s)} = \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{s + \frac{\sigma}{m_p}}{s^2 + 2D_h \omega_h s + \omega_h^2} \\ &= \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{K_m T_h (1 + T_m s)}{T_m T_h s^2 + (T_m + T_h)s + (1 + K_m K_d T_h)} \end{aligned} \quad (4.258)$$

**Load Force (difference-pressure Force) Control.** From the relation  $F_L = A_p p_L$ , we obtain a similar transfer function of the PDT<sub>4</sub>-type

$$\begin{aligned} G_F(s) &= \frac{F_L(s)}{U(s)} = \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{A_p \left( s + \frac{\sigma}{m_p} \right)}{s^2 + 2D_h \omega_h s + \omega_h^2} \\ &= \frac{K_v K_\rho^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \frac{K_m T_h (1 + T_m s)}{T_m T_h s^2 + (T_m + T_h)s + (1 + K_m K_d T_h)} \end{aligned} \quad (4.259)$$

**Flow Control.** For flow control the transfer function is simplified to (PT<sub>2</sub>-system)

$$G_Q(s) = \frac{Q(s)}{U(s)} = \frac{K_v K_Q^*}{\frac{1}{\omega_v^2} s^2 + \frac{2D_v}{\omega_v} s + 1} \quad (4.260)$$

## 4.5 Determination of Specific Model Parameters

The parameters required for the analytical description and simulation of the system can be partly gathered from the manufacturer's catalogue information and drawings. Other data, such as discharge coefficients, friction and effective bulk modulus, could be determined either by experiments or by expressions given in the literature.

The physical parameters as used in Equations 4.181–4.190 can be determined with several specific measurements. Whereas it is impossible to perform actuator measurements without a valve, it is possible to obtain measurements that purely reflect valve properties. Mounting the valve on a specific manifold, for instance as illustrated in Figure 4.28, some characteristics can be measured. Nowadays, a manufacturer's catalogue information usually provides some characteristic graphs for the valves.

### 4.5.1 Static Valve Characteristics

In the following, those valve steady-state characteristics that are best known and understood are presented.

#### 4.5.1.1 Flow–Signal Function

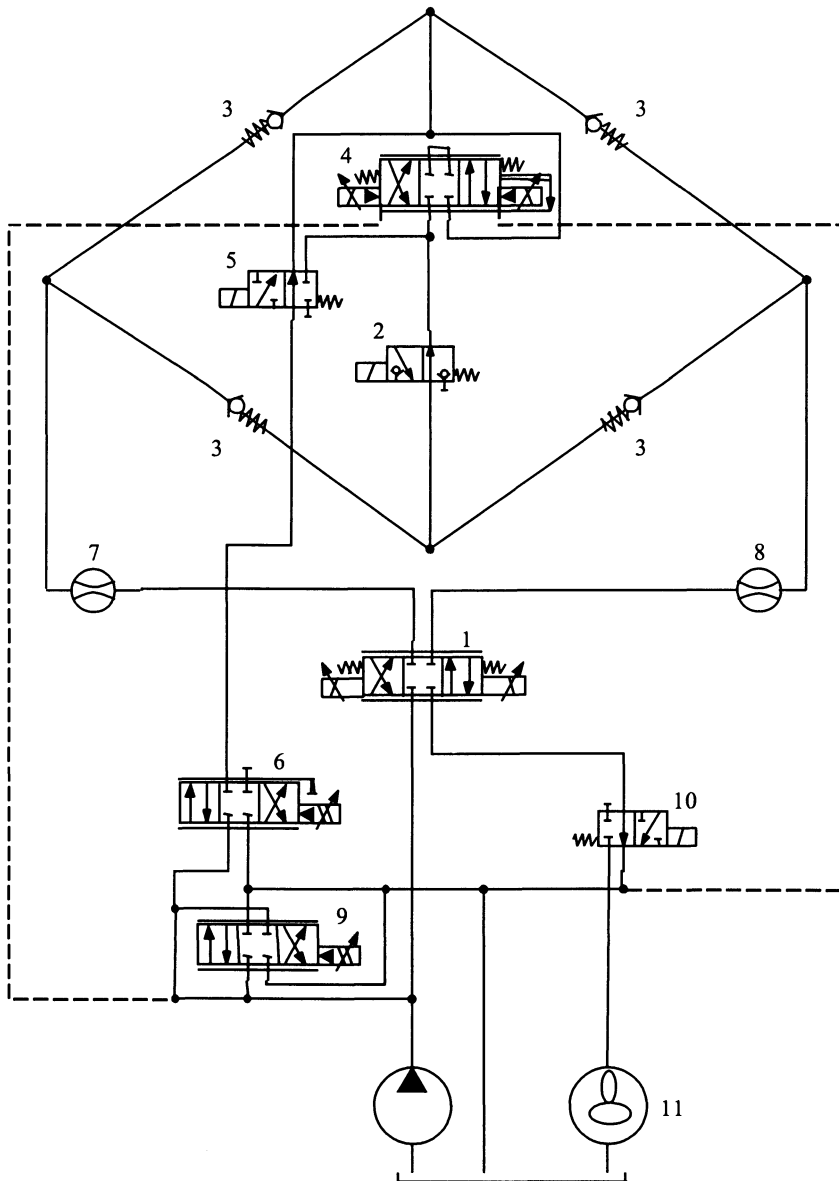
A valve is partly characterised by the relationship between valve input  $u^*$  and valve flow  $Q$  in a steady-state situation and at a certain supply pressure  $p_s$ . This relationship can be measured and is called a flow–signal function  $Q(u^*, p_s = \text{const.}, p_L = 0) \equiv Q(u^*)$ . Ideally, the flow–signal graph is a straight line. However, effects like saturation for larger valve inputs, valve offsets due to underlap or overlap, hysteresis due to internal friction effects, and some non-smooth behaviour around zero input are often seen in flow–signal curves measured in practice. Qualitative examples are shown in Figure 4.29.

The flow–signal curve can be measured at the manifold shown in Figure 4.28. The required pressure control is ensured using a servo-valve (9), which keeps the system pressure constant. This valve is connected to the supply and the tank simultaneously, and thus is able to respond very quickly to system state changes.

#### 4.5.1.2 Flow–Pressure Function

If the valve input is held at a maximum (*i.e.*,  $u_v = u_{v,\max}$ ), and the valve flow  $Q$  in a steady-state situation is measured at a certain supply pressure  $p_s$  under different load pressures, the graph  $Q(u_v = u_{v,\max}, p_s = \text{const.}, p_L) \equiv Q(p_L)$  is obtained and referred to as a flow–pressure function. Typical flow–pressure curves are depicted in Figure 4.30.





**Figure 4.28.** A manifold to perform measurements of valve characteristics (Taken from Lierschaft (1993), p 84 (original publication). Reprinted with kind permission of VDI Verlag.)

#### 4.5.1.3 Pressure–Signal Function

Another well-known static characteristic of a valve is the *pressure–signal function* (or *stiffness graph*), which consists of the measured relation between the valve input and the pressures or the pressure drop over the piston. See also Section 4.5.5.

In order to measure the pressure–signal function at the manifold of Figure 4.28, the ports A and B of the test valve (1) have to be closed. This can be done by means of an air-tight seat-valve (2), which is embedded in a GRAETZ-circuit (3). A load valve (4) is placed in series with the valve (2) to simulate different actuators and loads, *i.e.*, to vary the load pressure  $p_L$ . Another valve (6) is placed in the supply and return lines of the load valve (4) via a switch-valve (5). With the valve (6), non-symmetric actuators can be simulated. The oil flows  $Q_A$  and  $Q_B$  are measured using the flow meters (7) and (8).

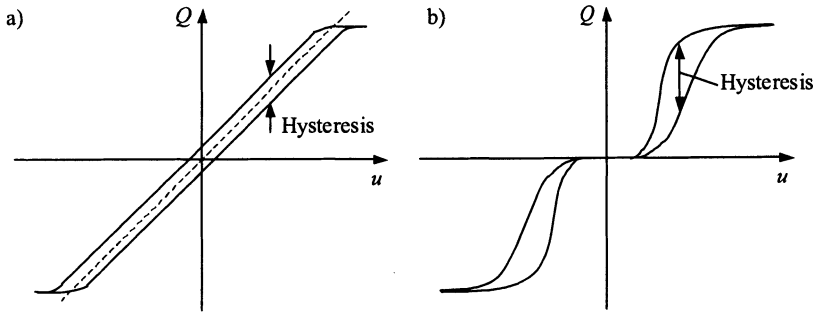


Figure 4.29. Typical flow–signal curves for (a) servo-valve and (b) proportional valve

## 4.5.2 Dynamic Valve Characteristics

It is, of course, possible to determine the dynamic characteristics of a valve by special tests on the manifold in Figure 4.28, *i.e.*, excitation by defined input signals and subsequent parameter identification. An alternative easy way is to trust the manufacturer's catalogue information; see Appendix B.2. Then the parameters valve gain  $K_v$  and damping coefficient  $D_v$  (Equation 4.9) can normally be assumed to be very close to unity.

In contrast, the valve frequency  $\omega_v$  shows considerable input amplitude and pressure dependence, *i.e.*,  $\omega_v = \omega_v(u_v, p)$ ; see the example in Figure 4.31. This diagram can be integrated in the model in the form of look-up tables, whose values are extracted from the manufacturer's catalogue information; see Appendix B.2. Note that there is a dependence on the valve nominal flow.

The maximum (allowable) speed (maximum utilisation) of the valve (Equation 4.9) (Schulte, 1994)

$$\dot{x}_{v,lim} = 4 \frac{s_{lim}}{s_N} f_{lim} \quad (4.261)$$

can be estimated using the frequency response provided by the valve manufacturer.  $f_{lim}$  denotes the frequency at  $\varphi = -45^\circ$ ,  $s_{lim}$  is the corresponding magnitude and  $s_N$  represents the magnitude at  $\varphi = -45^\circ$  and  $u = 100\%$ .

Regarding the main static non-linearities, it is usually sufficient to consider the valve hysteresis  $f_{hys}$  and the response sensitivity  $f_{sens}$ , which can be combined to give the constant friction (Schulte, 1994)

$$f_{hs} = 0.5f_{hys} + f_{sens} \quad (4.262)$$

acting in the opposite direction to the velocity. Both  $f_{hys}$  and  $f_{sens}$  are normally given by the valve manufacturer; otherwise, they may be taken from the literature. For more accurate modelling of friction effects; see Section 4.5.4.

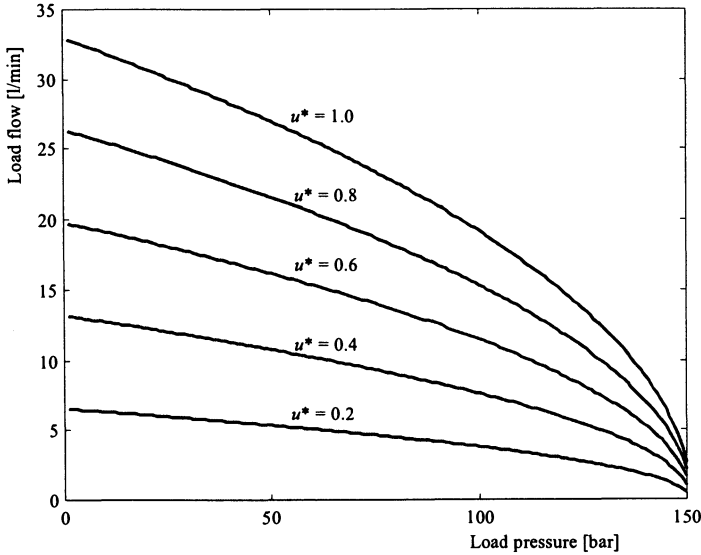


Figure 4.30. Typical flow–pressure graphs

### 4.5.3 Actuator Dimensions and Mass

The piston areas and the stroke are often accurately specified by the manufacturer, or can easily be calculated from the construction drawings of the actuator.

A difficult point is the determination of the inefficient volumes. Often, rather imprecise estimations can be obtained from construction drawings of the actuator and the hydraulic layout (pipelines between valve and actuator, use of safety manifolds).

An estimate of the mass distribution of the load attached to the actuator can be calculated from the construction drawings or by weighing the object.

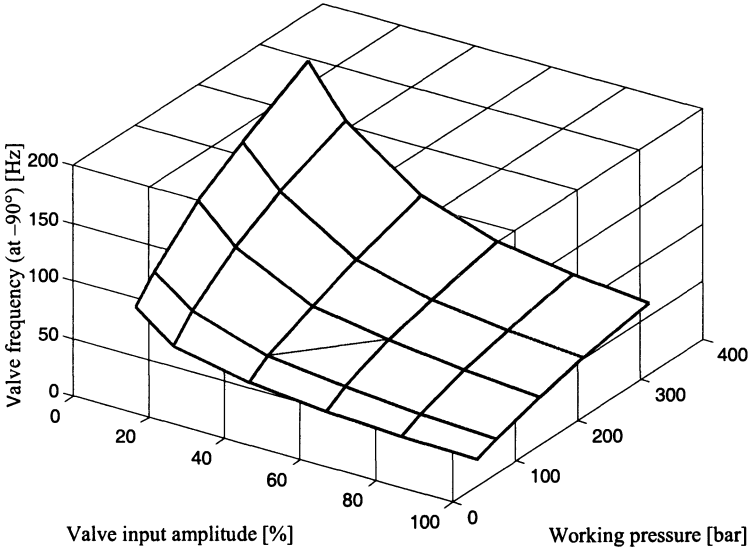
### 4.5.4 Friction Forces

One of the main non-linearities of the cylinder model is non-linear friction. Depending on the application, Coulomb friction and stiction phenomena may have a particularly significant effect on the controller's performance, as predicted by experiments and simulations.

There are many different ways of modelling Coulomb friction and related effects (Armstrong-Helouvry *et al.*, 1994). In the most general case, the non-linear friction force on the piston depends not only on the actuator velocity  $\dot{x}_p$ , but also on the

pressure difference  $\Delta p$  across the piston (when the piston is sticking), and possibly on the piston position  $x_p$ , and the temperature  $\theta$ . Thus, the general non-linear friction model reads:<sup>12</sup>

$$F_f = F_f(x_p, \dot{x}_p, \Delta p, \theta, t). \quad (4.263)$$



**Figure 4.31.** Typical valve frequency look-up field for a servo-valve (type: Bosch Rexroth 4WSE3EE16)

A more practical (Stribeck) model, however, divides friction forces into static friction, Coulomb friction and viscous friction. Viscous friction is proportional to the velocity and contributes to stability and damping in hydraulic systems. Although viscous friction cannot be predicted sufficiently, it should never be neglected in dynamic analysis. “Coulomb friction is the most magic non-linearising factor in hydraulic servo-systems and is always suspect when something irregular is happening” (Viersma, 1980). This friction type may lead to instability problems in practice, *i.e.*, stick-slip phenomena at low speeds.

One empirical friction function, already introduced in Section 4.2.2.2 (Equation 4.60), is

$$F_f(\dot{x}_p) = \sigma \dot{x} + \text{sign}(\dot{x}_p) \left[ F_{c0} + F_{s0} \exp \left( -\frac{|\dot{x}_p|}{c_s} \right) \right] \quad (4.264)$$

<sup>12</sup> It is assumed that the cylinder stands on a rigid floor and does not move. Otherwise, the friction force will be a function of the relative velocity between the piston and the cylinder, *i.e.*,  $F_f = F_f(x_p - x_c, \dot{x}_p - \dot{x}_c, \Delta p, \theta, t)$ ; see Section 4.2.3.4.

Another (slightly different) function adopted from Armstrong-Helouvry *et al.* (1994) is given by

$$F_f(\dot{x}_p) = \sigma \dot{x} + \text{sign}(\dot{x}_p) \left\{ F_{c0} + (F_s - F_{c0}) \exp \left[ - \left( \frac{\dot{x}_p}{v_s} \right)^2 \right] \right\} \quad (4.265)$$

The parameters  $\sigma$ ,  $F_{c0}$ ,  $F_{s0}$ , and  $c_s$  must be identified from experimental data. For this, the total actuator force is expressed as

$$F_t = F_L(p_A, p_B) = m\ddot{x} + F_f \quad (4.266)$$

By measuring the friction force

$$F_f = F_L(p_A, p_B) - m\ddot{x} \quad (4.267)$$

required to move the piston (without load, *i.e.*,  $F_{ext} = 0$ ), we can develop such a model  $F_f(\dot{x})$ . At a constant velocity, Equation 4.267 becomes

$$F_f = F_L(p_A, p_B) \quad (4.268)$$

The acceleration  $\ddot{x}_p$ , if needed, is calculated by the twofold difference quotient of the position sensor. The pressure difference force is determined by

$$F_L(p_A, p_B) = (p_A - \alpha p_B) A_p \quad (4.269)$$

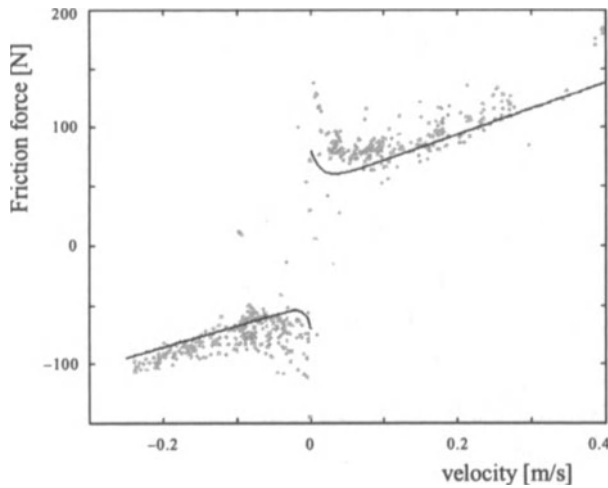
with the pressures  $p_A$  and  $p_B$  in chambers A and B. By measuring the friction force at regular intervals as the cylinder is extended and retracted, it is then possible using the force equilibrium Equation 4.267 (or better 4.268) to obtain  $\hat{F}_f(\dot{x}_p)$ .

However the friction models proposed above cannot capture the asymmetry of the friction forces as it usually occurs in differential cylinders. To deal properly with this, Equation 4.60 should be extended to (Nissing, 2002)

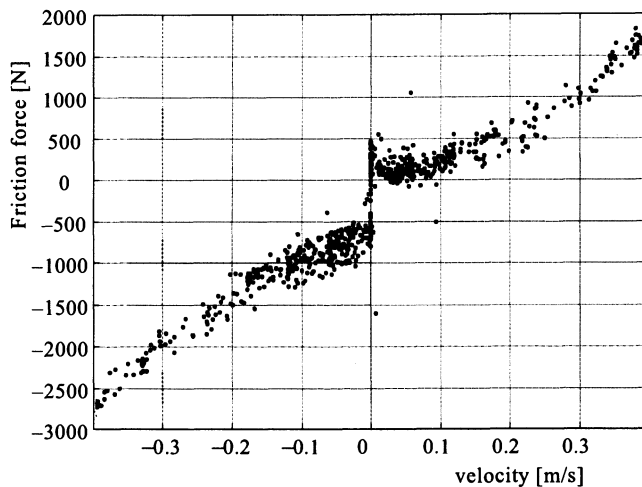
$$F_f(\dot{x}_p) = \begin{cases} \sigma^+ \dot{x}_p + \text{sign}(\dot{x}) \left[ F_{c0}^+ + F_{s0}^+ \exp \left( - \frac{|\dot{x}_p|}{c_s^+} \right) \right] & \forall \dot{x}_p \geq 0 \\ \sigma^- \dot{x}_p + \text{sign}(\dot{x}) \left[ F_{c0}^- + F_{s0}^- \exp \left( - \frac{|\dot{x}_p|}{c_s^-} \right) \right] & \forall \dot{x}_p < 0 \end{cases} \quad (4.270)$$

A typical measured result for the curve is depicted in Figure 4.32 (Polzer and Nissing, 2000a). From this plot, it is possible to obtain the parameters of Equation 4.270 (using an identification algorithm; see Chapter 5), and the approximation is also shown in the same figure with the parameters  $\sigma = 220$  N s/m,  $F_{c0} = 50$  N,  $F_{s0} = 30$  N and  $c_s = 0.015$  m/s for  $\dot{x}_p > 0$  and  $\sigma = 180$  N s/m,  $F_{c0} = 50$  N,  $F_{s0} = 20$  N and  $c_s = 0.007$  m/s for  $\dot{x}_p < 0$ . As one can clearly see, the friction force graph is

non-symmetric. The curve would be slightly different for different cycles, which is due to the random characteristics of the friction.



**Figure 4.32.** Measured and approximated friction force for cool oil (small cylinder with  $d = 25$  mm,  $S = 200$  mm) (Taken from Nissing (2002), p 17 (original publication). Reprinted with kind permission of VDI Verlag.)



**Figure 4.33.** Measured friction forces (big cylinder with  $d = 63$  mm,  $S = 500$  mm) (Taken from Bernzen (1999), p 19 (original publication). Reprinted with kind permission of VDI Verlag.)

It should be noted that there is a significant difference in the parameters of  $F_f(\dot{x}_p)$  depending on the oil/cylinder temperature. Here, the friction force is plotted

for cool oil ( $\sim 30^\circ\text{C}$ ). Furthermore, the data in each plot are highly dispersed, because of measurement errors and noise. The friction also depends strongly on the position of the piston rod, which is not considered in this investigation. However, for practical implementation these simplifications are valid. Finally, the Stribeck curve will obviously vary for different cylinder dimensions; see Figure 4.33 (Bernzen, 1999a).

Further investigations on the estimation of friction forces for friction compensation purposes can be found in Section 6.10.2. Also, recent analysis of this subject is provided by Ratjen (2001a) and Klotzbach and Henrichfreise (2002).

#### 4.5.5 Leakage Coefficients and Valve Underlap

Although leakage has only been modelled in the actuator, the valve also shows some leakage effects (due to radial clearance and underlap). Moreover, as the external leakage has been neglected, only parameter  $C_{Li}$  remains to describe these effects. With this parameter, it is expected to find a reasonable measure for the leakage effects within the complete actuator and valve system.

When the actuator has been *fixed rigidly* (Figure 4.34, left) in one position (for instance, the piston is positioned at one side of the stroke), the smallest valve opening would result in supply pressure in one chamber of the actuator, for the case where no leakage occurs. However, leakage introduces an oil flow (over the piston), which causes a pressure drop.

Consequently, the pressure in the specific chamber will be less than  $p_s$ . Note that the pressure in the other chamber will be equivalent to the pressure in the tank, as the valve is opened and permits a free flow of small amounts of oil. Typical pressure–signal curves for different values of leakage coefficient are shown in Figure 4.34 (right).

To estimate  $C_{Li}$  from a stiffness graph, the relations (taking into account Equation 4.2 for  $x_v < 0$ , and Equations 4.215 and 4.55)

$$Q_B = \alpha Q_A = -\alpha c_v x_v \sqrt{p_A - p_T} \quad (4.271)$$

$$Q_B = Q_{Li} = C_{Li} (p_B - p_A) \quad (4.272)$$

are used to give

$$\alpha c_v x_v \sqrt{p_A - p_T} = C_{Li} (p_A - p_B) \quad (4.273)$$

Differentiation of both sides of this equation with respect to  $x_v$  and rearrangement leads to

$$C_{Li} = \frac{\alpha c_v}{K_{pA,0} (1 + \alpha^2)} \sqrt{p_{A|x_v=0} - p_T} \quad (4.274)$$

$K_{pA,0}$  denotes the slope of the stiffness graph at neutral to be determined experimentally.  $p_{A|x_v=0}$  is the pressure in chamber A at  $x_v = 0$ , which can be extracted from the stiffness graph, or approximated by

$$p_{A|x_v=0} = \frac{\alpha p_s + \alpha^3 p_T}{1 + \alpha^3} \quad (4.275)$$

The stiffness graph can also best be used to identify the underlap of the valve, and thus improve the model quality, as demonstrated by Saffe (1986) and Feigel (1987a). This takes into account the fact that exact critical centre valves cannot be produced, and the majority of valves met in practice have radial clearance and little underlap. Consequently, accurate models should take these effects into account.

To estimate the average underlap, the valve flow Equation 4.5 is written as

$$Q_1 = c_v [x_v + x_{u1/2} (1 + \delta_{1/2})] \sqrt{p_s - p_A} \quad (4.276)$$

$$Q_2 = c_v [-x_v + x_{u1/2} (1 - \delta_{1/2})] \sqrt{p_A - p_T} \quad (4.277)$$

Setting  $Q_1 = Q_2$ , and rearranging with respect to  $p_A$  leads to

$$p_A = \frac{p_T + K^2 p_s}{1 + K^2} \quad K = \frac{x_v + x_{u1/2} (1 + \delta_{1/2})}{-x_v + x_{u1/2} (1 - \delta_{1/2})} \quad (4.278)$$

and with  $x_v = 0$  to

$$\delta_{1/2} = \frac{-1 + \sqrt{\frac{p_{A|x_v=0} - p_T}{p_s - p_{A|x_v=0}}}}{1 + \sqrt{\frac{p_{A|x_v=0} - p_T}{p_s - p_{A|x_v=0}}}} \quad (4.279)$$

Differentiation of Equation 4.278 with respect to  $x_v$ , and rearranging yields

$$x_{u1/2} = \frac{4(1 + \delta_{1/2})(p_s - p_T)}{K_{pA,0} \left[ 1 + \left( \frac{1 + \delta_{1/2}}{1 - \delta_{1/2}} \right)^2 \right]^2 (1 - \delta_{1/2})^3} \quad (4.280)$$

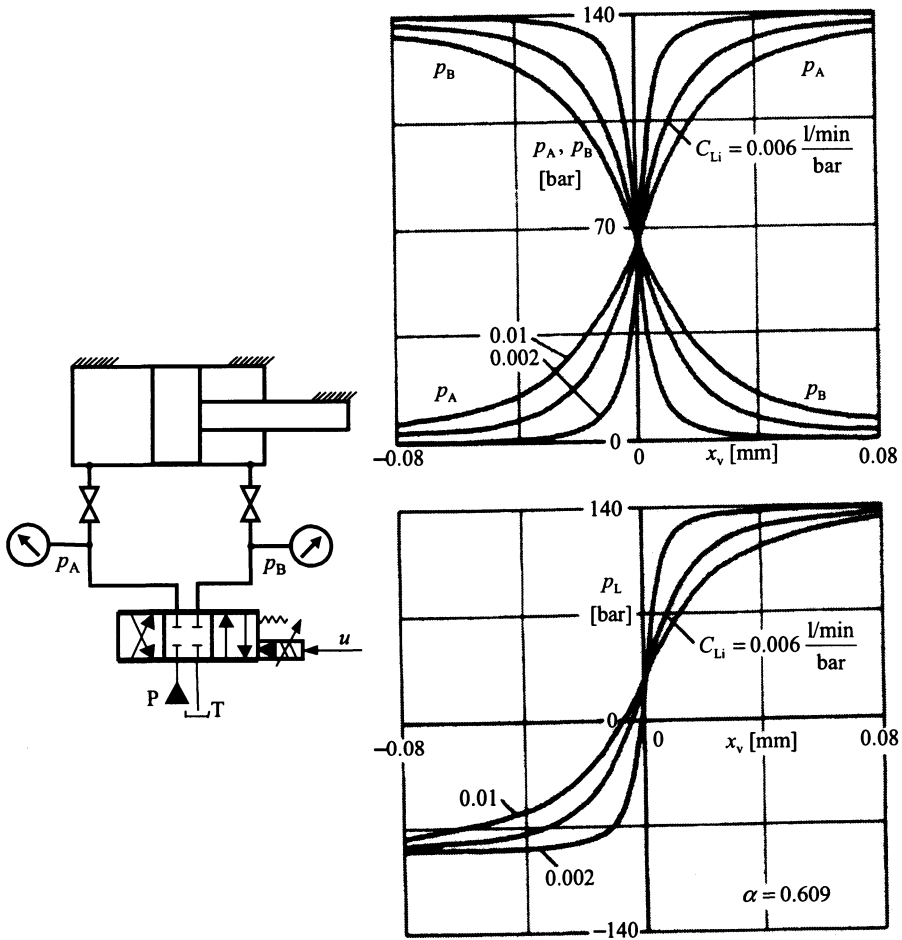
with  $K_{pA/B,0}$  extracted from the measured pressure–signal graph.

The same procedure for the B-side (Equation 4.6) gives the estimates

$$\delta_{3/4} = \frac{-1 + \sqrt{\frac{p_{B|x_v=0} - p_T}{p_s - p_{B|x_v=0}}}}{1 + \sqrt{\frac{p_{B|x_v=0} - p_T}{p_s - p_{B|x_v=0}}}} \quad (4.281)$$

$$x_{u3/4} = -\frac{4(1 + \delta_{3/4})(p_s - p_T)}{K_{pB,0} \left[ 1 + \left( \frac{1 + \delta_{3/4}}{1 - \delta_{3/4}} \right)^2 \right]^2 (1 - \delta_{3/4})^3} \quad (4.282)$$





**Figure 4.34.** Fixed actuator (left) and example (right) of measured pressure-signal graph (Feigel, 1987a)

## 4.6 Implementation and Software Tools

### 4.6.1 Simulation of Friction Forces

The discontinuity of the friction force function  $F_f(\dot{x})$  for  $\dot{x} = 0$  and of  $F_c$  for  $x = 0$  yields non-unique relations between  $\dot{x}$  and  $F_f$  and between  $x$  and  $F_c$ , i.e.,  $F_f$  and  $F_c$  cannot be calculated for these particular system conditions. However, by introducing an auxiliary force  $F_{aux}$  such that Equation 4.60 has the form (Lausch, 1988, 1990)

$$F_r(\dot{x}) = \sigma \dot{x} + \text{sign}(\dot{x}) \left[ F_{c0} + |F_s^*(\dot{x})| \right] + F_{aux}(F_\Sigma, F_{r0}, \dot{x}) \quad (4.283)$$

with

$$|F_s^*(\dot{x})| = F_{s0} \exp\left(-\frac{|\dot{x}|}{c_s}\right) \quad \text{for } \dot{x} \neq 0 \quad (4.284)$$

it is possible to satisfy the condition for the balance of forces for the states  $\dot{x} = 0$  and  $x = 0$ . The auxiliary force is expressed as

$$F_{aux}(F_\Sigma, F_{r0}, \dot{x}) = \begin{cases} F_{r0} \text{sg}(\dot{x}) \text{sign}(F_\Sigma) & \text{for } |F_\Sigma| \geq F_{r0} \text{sg}(\dot{x}) \\ F_\Sigma & \text{for } |F_\Sigma| < F_{r0} \text{sg}(\dot{x}) \end{cases} \quad (4.285)$$

with

$$F_\Sigma = (p_A - \alpha p_B) A_p - F_{ext} - \sigma \dot{x} - \text{sign}(\dot{x}) (F_{c0} + |F_s^*(\dot{x})|) \quad (4.286)$$

$F_{r0}$  is the maximum frictional force at standstill:

$$F_{r0} = F_{c0} + F_{s0} \quad (4.287)$$

With the auxiliary force it is now possible that the acceleration of the mass is zero if the external forces are not greater than the sum of the forces of static and Coulomb friction (and of the spring pre-load, if any).

Then, Equation 4.56 can be rewritten as

$$m_i \ddot{x} + \sigma \dot{x} = (p_A - \alpha p_B) A_p - F_{ext} - \text{sign}(\dot{x}) (F_{c0} + |F_s^*(\dot{x})|) - F_{aux}(F_\Sigma, F_{r0}, \dot{x}), \quad (4.288)$$

and the block diagram in Figure 4.11 has to be modified accordingly.

## 4.6.2 Simulation of Mechanical Saturations

Valve spool position and velocity, as well as cylinder position and velocity, are usually physically limited. This kind of saturation cannot be taken into account by including a double integrator with saturation blocks (or a double integrator with output saturation limits) into the model, as it does not realistically describe the real physical behaviour of the system. Figure 4.36 shows this incorrectly simulated behaviour. Although the position has reached its upper saturation limit ( $x = 1$  at  $t \approx 0.155$  s), the velocity is still internally integrated.

Instead, non-linear elements (logical operations) have to be added to ensure that velocity is immediately set to zero as soon as the valve (or cylinder) reaches its upper or lower saturation limit (see Figures 4.35 and 4.37). This means that the saturation is considered as a “hard” non-elastic endpoint, and therefore the spool (or piston) will abruptly come to a standstill.

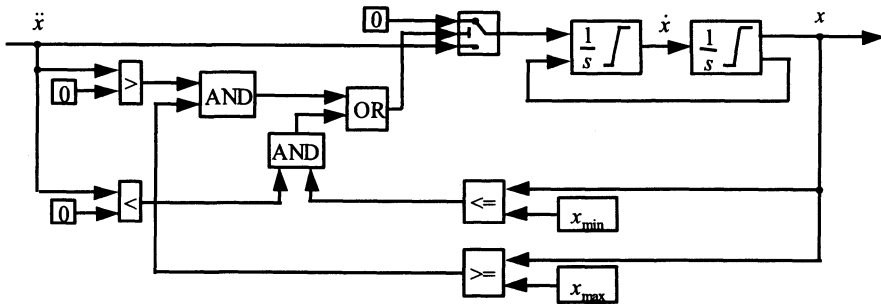


Figure 4.35. Model of mechanical saturations (double reset-integrator)

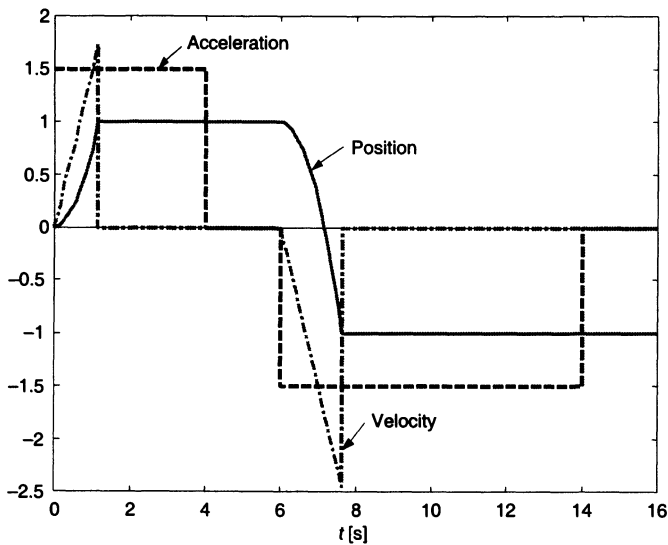
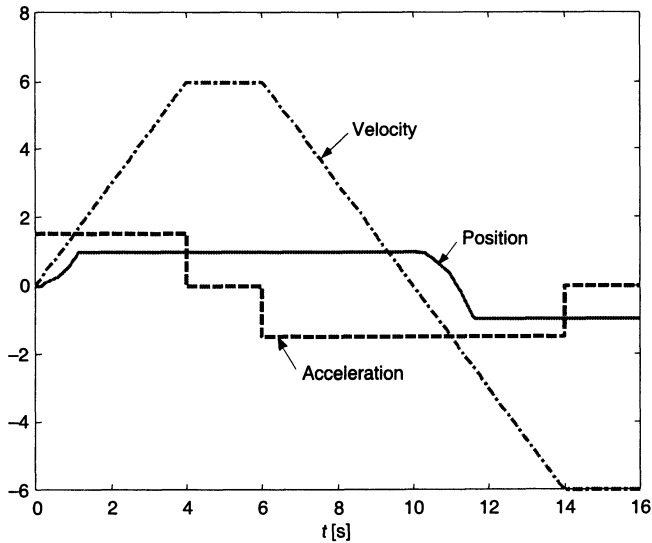


Figure 4.36. Simulation result using a double integrator with output saturation limits

### 4.6.3 Simulation Packages

There are quite a number of simulation packages for hydraulic components and systems, which can be used for design and analysis of the dynamic characteristics of hydraulic circuits. Some of the hydraulic simulation packages are briefly described based on prospectus from their suppliers.

- *DSH* or *DSHplus* (Fluidon): user-friendly program for the simulation of complex fluid–technical systems. It provides the user with modules that support him from the first conception to the actual start-up of the system. DSHplus is perhaps the oldest and most known hydraulic simulation package available. <http://www.fluidon.com>.



**Figure 4.37.** Simulation result using the double reset-integrator

- *Hydraulic Blockset* (The MathWorks): extension of The MathWorks' Simulink environment for design and simulation of hydraulic systems. This blockset provides a collection of more than 50 Simulink blocks, including hydraulic cylinders, pumps, and valves. <http://www.mathworks.com>.
- *Hylib* (Dynasim): the hydraulics library of Dymola by Peter Beater provides basic model classes for the modelling of pumps, motors and cylinders, restrictions and valves, hydraulic lines, lumped volumes and sensors. Hylib contains more than 90 models. <http://www.dynasim.se> or e-mail [info@beater.de](mailto:info@beater.de).
- *AMESim* (Imagine): offers a complete and unique simulation environment, providing the most adaptive solutions of fluid power (gas and liquid), mechanical, thermo-fluid and control systems. The hydraulic library contains a set of classical hydraulic components (such as accumulators, relief valves, check valves, orifices, pumps, motors, jacks, *etc.*), as well as different sources and transducer models. <http://www.amesim.com>.
- *HOSPAN* (Linköping University): package suited to simulate fluid power systems with automated parameter variation and frequency analysis. <http://hydra.ikp.liu.se>.
- *Hydro Analyst* (Flotron): program for the analysis of open and closed loop electro-hydraulic systems operating in both the frequency and time domains. Separate programs are provided for flow and pressure control. A *supplementary* program for power efficiency and dissipation is included. <http://www.flotron.co.uk>.

## 4.7 Section Summary

As the approach of physical modelling is the most suitable method to gain insight into the relevant dynamics and non-linear effects of technical systems, comprehensive models of different elements (valves, actuators, pumps) of hydraulic servo-systems have been derived from first principles (Section 4.2).

For the example of valve-controlled hydraulic cylinders, Section 4.4 has demonstrated in which cases and under which realistic assumptions the detailed model equations can be simplified. The aim was to present reduced and physically structured models that are well suited for control design purposes. It must be stressed that the judgement of whether model simplification is useful depends substantially on the application at hand and can be best taken based on simulations of the non-linear models. Special attention has to be paid to the decision of whether the valve dynamics and/or pipeline dynamics are relevant. If the dynamics are sufficiently fast, they may be ignored in the controller design. However, these effects should be included in the simulation model for testing the controllers. The description and parameterisation of friction and bulk modulus are very important, although difficult, issues (Sections 4.5.4 and 3.1).

In the case where a linear description is sufficient, it is more efficient to derive it from a reduced/simplified non-linear model. Such linear models are advantageous, as they are relatively simple, *i.e.*, include only a few parameters, and preserve the physical structure of the model. Moreover, the application of linear control design methods is straightforward. Some linear models have been derived in Section 4.4.4 for valve-controlled systems. The reader should also consult Anderson (1988) and Will *et al.* (1999), who provide linear models of a large number of hydraulic components and systems.

The results of Section 4.4 can formally be extended to valve-controlled motors by consideration of Table 4.1. Models of more complicated hydraulic circuits can be derived based on the combination of the sub-models presented in this chapter. The reader should also consult Anderson (1988), Guo (1991), Ebner (1995), and Will *et al.* (1999). Highly recommended for system designers is the use of one of the simulation packages mentioned in Section 4.6.3.

The measurements described in Section 4.5 can provide in a short time a reasonable insight into specific properties of the system, or even an estimation of all parameters required for a model of the system (see also Section 3.1). However, when all the parameters are required, the execution of the above experiments is a time-consuming activity. In addition, the estimates of, for example, the inefficient volumes or the bulk modulus are inaccurate. Application of identification methods, as described in Chapter 5, is an option for finding almost all the required parameters from one experiment.

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## CHAPTER 5

# EXPERIMENTAL MODELLING (IDENTIFICATION)

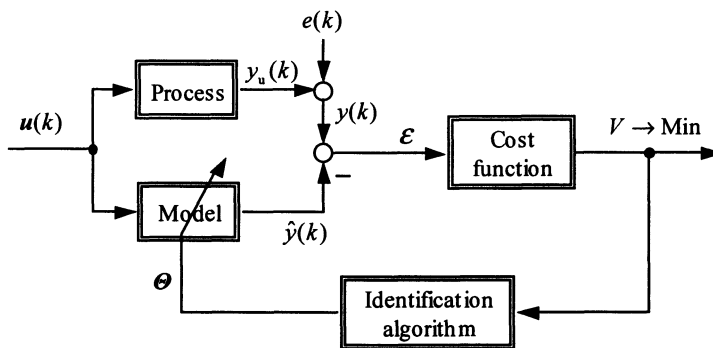
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In Chapter 4, theoretical models of hydraulic servo-systems have been presented and analysed. The subsystem models of hydraulic servo-systems have been brought into representations, which are suitable for controller design. This finally led to simple models, describing the relevant dynamics and non-linearities of the systems, which form the basis for grey-box modelling. This additionally requires experimental modelling, *i.e.*, derivation of models from input–output data. The modelling approach is also referred to as statistical or empirical modelling, the derivation process as identification.

The model identification and validation for hydraulic servo-systems is the topic of this chapter. Besides classical identification approaches, selected methods for statistical (fuzzy and neural-network-based) modelling will also be treated.

### 5.1 Introduction

The task of identification is to determine the model parameters  $\Theta$ , such that the model predictions differ as little as possible from measured values according to a specified criterion; see Figure 5.1.



**Figure 5.1.** Identification task: determine model such that residual  $\varepsilon$  is minimised

Typically, the mean-squared error

$$V(\Theta) = \frac{1}{2N} \sum_{k=1}^N (y(k) - \hat{y}(k))^2 \quad (5.1)$$

is used as a criterion (for MISO systems).  $V(\Theta)$  is also referred to as the “objective”, “loss function” or “performance criterion”. In Equation 5.1,  $k$  denotes discrete time

samples ( $k = t/T_0$ ,  $T_0$ : sampling period),  $\hat{y}$  is the estimated output, and  $N$  is the number of data points. The true system output  $y_u$  is usually disturbed by the noise  $e$ , resulting in the measurable output  $y$ .

An identification algorithm reduces the cost  $V$  regarding the identification data. In parallel, or during model validation, the cost  $V$  regarding independent data is determined to test generalisation properties and avoid parameter over-fitting. The model evaluation concept (serial-parallel or parallel) has to be specified for a well-defined task (see Section 5.8.5 for a thorough discussion).

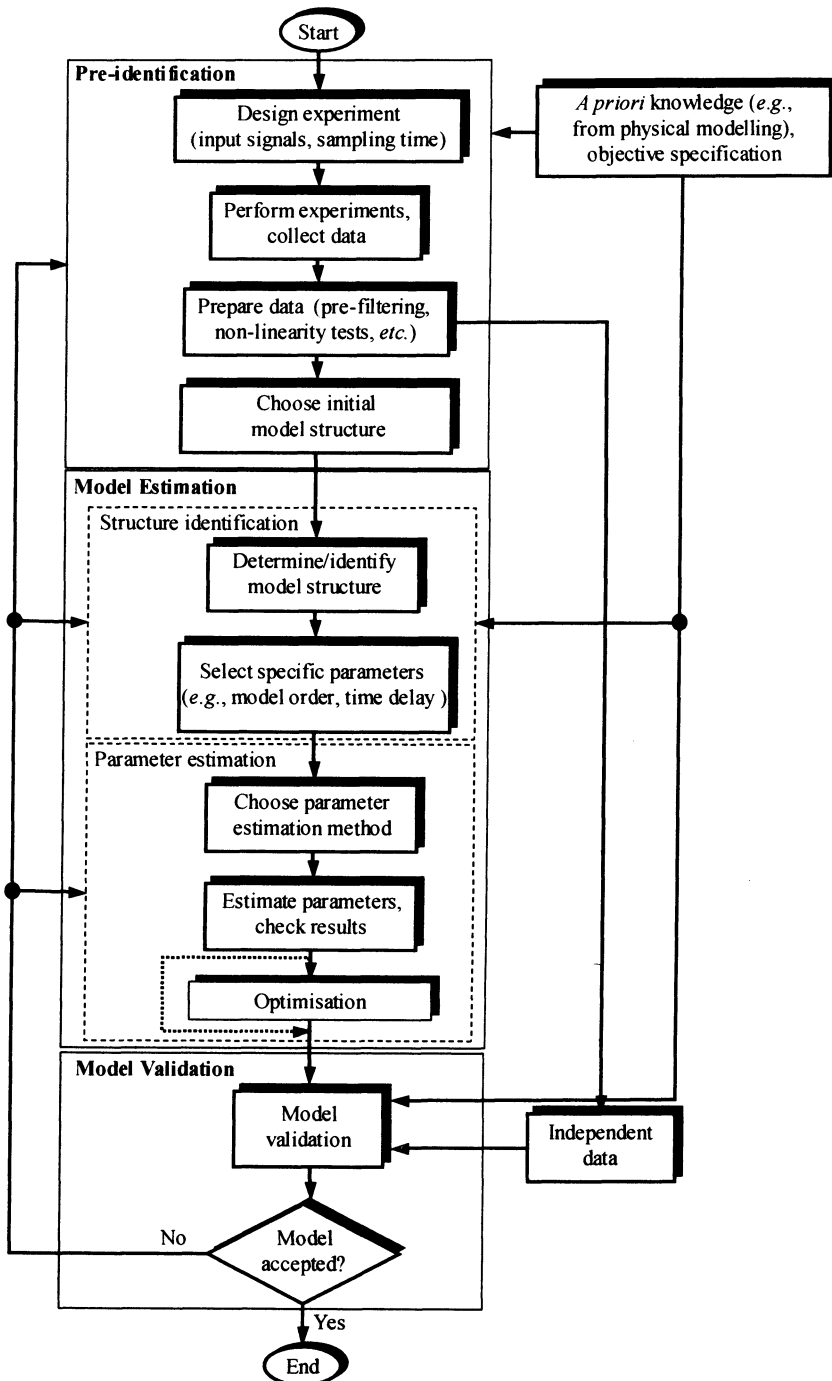
### 5.1.1 Generic Identification Procedure

When attempting to identify a model of a dynamic system, here of a hydraulic servo-system, it is common practice to follow the procedure illustrated in Figure 5.2, which consists of the following main steps:

1. Design and perform the experiments, which involves choice of excitation signal, selection of sampling period, measurement and recording of the necessary data sets for identification, and pre-treatment of data. See Section 5.2.
2. Selection of an appropriate model structure (or model structure set), typically a differential or difference equation of a certain order. In this stage, a reasonable choice should be made by the user, based on prior knowledge about the system. Much engineering ingenuity is often necessary to make this choice. See Sections 5.3 and 5.4.
3. Usage of some mathematical methods (minimising the error between the output of the model and the measurements of the process) to identify the model structure and to estimate the unknown parameters of the model. See Sections 5.5, 5.6, 5.8, and 5.9. Sometimes, the model quality can be enhanced by performing an additional parameter optimisation. This is, however, not a default identification step.
4. Validate, *i.e.*, test, the model obtained to see whether it is an appropriate representation of the system with respect to the requirements; see Section 5.10. If this is not the case, some of the steps 1–3 have to be repeated.

The paths going from the validation block and back to the previous stages indicate that the identification process is executed in an iterative manner:

- *Path leading back to parameter estimation.* In general, it is difficult to guarantee that the estimation algorithm will converge to a global optimum for a non-linear estimation problem. This is particularly valid for the numerically very efficient derivative-based methods, which are often used. This may be caused, for example, by local minima. The parameter identification step should be repeated with different (random) choices of initial values for the parameters to reduce the risk of being caught in a (bad) local minimum of the loss function. Other than that, global search strategies, such as simulated annealing, evolutionary strategies, and genetic algorithms, may be applied. Although such approaches promise to find the global optimum, they require huge computational burden.



**Figure 5.2.** Schematic flowchart of system identification process



- *Path leading back to model structure selection.* Since a model structure is defined as a set of candidate models (or model family), and one is interested in a model with a minimum number of model parameters, model selection tools can be used. The optimum model structure can be extracted in an automated way by comparing different candidate structures in terms of some performance measures; see Sections 5.5.5 and 5.9.1.
- *Path leading back to experiment.* If, after many trials, it seems to be impossible to get an acceptable model regardless of how the model structure is chosen, this may imply that data material is insufficient. This may be caused either by insufficient data processing or by lack of information in the data set. The former reason is very common and typically indicates that the input signal does not sufficiently excite the system so that the operating range is covered well enough. Therefore, new data have to be acquired.

As pointed out by Söderström and Stoica (1989), there are no foolproof identification methods that always and directly lead to a correct result. Instead, there are a number of theoretical results, which are useful from a practical point of view. Even so, the user must combine the application of such theory with *common sense and intuition* to get the most appropriate result. Some practical issues are discussed in the following that could help when dealing with *system identification in practice*, especially in hydraulics.

### 5.1.2 Linear vs. Non-linear Identification

The approaches taken in the rest of this chapter are very much along the lines of the authoritative treatments of linear system identification found in the textbooks by Goodwin and Payne (1977), Ljung and Söderström (1987), Söderström and Stoica (1989), Isermann (1992) and Ljung (1999). Identification of *non-linear models* is, however, only marginally treated in these books. Only recently, Haber and Keviczky (1999), and Nelles (2001) published monographs on non-linear system identification.

In this work, an attempt has been made to describe (linear and non-linear) methods that have actually been successfully applied on hydraulic systems without any mathematical proofs of the statements.

Besides classical identification techniques, fuzzy models and artificial neural networks can be used to describe non-linear dynamic systems of hydraulic servo-systems and are therefore addressed in this section. The sections on fuzzy systems and neural networks in this book are not intended to be exhaustive, but focus on a selected few concepts out of the many proposed. Aspects of practical application are addressed in addition to methods.

The reader is referred to some textbooks for a wider or deeper study:

- Babuška (1998), Kroll (1997) for fuzzy identification and modelling (*recommended references for this book*) and Driankov *et al.* (1993) for an introduction into fuzzy logic and control.
- Hirota (1993) and Zimmermann and von Altrock (1994) for collections of successful applications of fuzzy logic, modelling and control.

- Nørgaard *et al.* (2000) for neural-networks-based modelling and control (*recommended reference for this book*), and Hertz *et al.* (1991) and Haykin (1999) for an introduction to neural computation.
- Hunt *et al.* (1995) for a collection of useful contributions about neural networks methods and applications.

### 5.1.3 Online vs. Offline Identification

Since most parameters in hydraulic servo-systems do not change during operation, it is self-suggesting to perform the identification *offline* and to adjust the servo-control before the operation accordingly. This approach is frequently taken and has the merit that one can avoid many of the problems of identifiability of the parameters: a series of experiments can be carefully designed, such that single properties are clearly observable in each experiment, until all interesting parameters can be accurately inferred from accumulated information.

Although it is recommended here to avoid *online* identification algorithms when other solutions are available and sufficient, there are some reasons that may motivate one to tackle the problems that are implied with online identification; for instance:

- Some dynamic parameters in hydraulic servo-systems are in fact not constant, e.g., temperature-dependent bulk modulus.
- The problems of online identification are mainly solvable.
- An online identification scheme finally presents a more general approach which can also be specialised to the task of offline identification.

Adaptive control is an example of such an application where it is useful to identify the model online, simultaneously with the acquisition of measurements. This subject is treated extensively by Isermann *et al.* (1992), Åström and Wittenmark (1995) and Landau *et al.* (1998).

Moreover, there are many textbooks dealing with the subject of online (recursive) identification, e.g., Goodwin and Payne (1977), Söderström and Stoica (1989), Johansson (1993) and Ljung (1999). A special textbook for recursive identification has been written by Ljung and Söderström (1987).

## 5.2 Pre-identification Process

For the identification of (non-linear) systems, it is important to define the type of system to be identified properly. Besides some pre-computations and the adopted system structure, the characterisation of the input-output behaviour is very important. Thereby, the types of signal to be used during experiments play an important role. Therefore, the excitation of non-linear system dynamics is also discussed. It is obvious in the pre-identification stage that not much can be done without a reasonable and good data record.

### 5.2.1 Design of Input Signals

In the field of linear system identification, a *pseudo-random binary signal* (PRBS) that only contains two amplitude levels is widely used; see Figure 5.3. However,

Leontaritis and Billings (1987a) show, that for non-linear systems identifiability can be lost with this type of excitation signal.

For non-linear systems, it is important that all (interesting) amplitudes and frequencies and all their combinations are represented in the input signal (this is termed the *curse of dimensionality*). Therefore, other types of random excitation signal are recommended, such as

- Independent sequences with a Gaussian or uniform distribution.
- Pseudo-random multilevel signals (PRMS), where the level is changed at each  $N_s$ -th sampling instant or at random instance.
- Chirp (or multi-sinusoidal) signals.

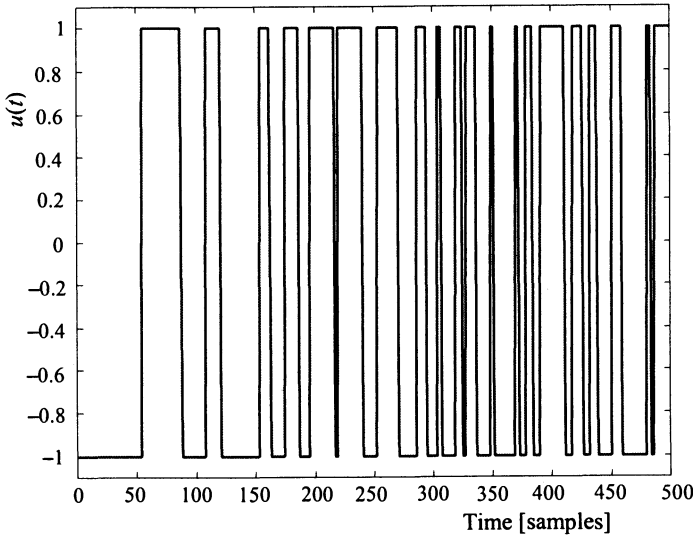


Figure 5.3. An example of a PRBS

### 5.2.1.1 Pseudo-random Multi-level Signals

Let  $v(k)$  be a white noise process with variance  $\sigma_v^2$ . A PRMS (also known as amplitude-modulated PRBS) is obtained by keeping the same (amplitude) value for  $N_s$  steps (increasing the clock period), *i.e.*,

$$u(k) = v \left[ \text{int} \left( \frac{k-1}{N_s} \right) + 1 \right] \quad k = 1, 2, \dots \quad (5.2)$$

where  $\text{int}(x)$  is the integer part of  $x$ .

A generalisation of this signal is obtained by introducing an additional (random) variable for deciding when to change the level (Nørgaard *et al.*, 2000):

$$u(k) = \begin{cases} u(k-1) & \text{with probability } \alpha \\ u(k) & \text{with probability } 1-\alpha \end{cases} \quad (5.3)$$

It is clear that if  $\alpha$  is chosen close to unity, the input will remain constant over long intervals and hence be of low-frequency character. An example of a PRMS is depicted in Figure 5.4. For detailed information about the generation and characteristics of PRMSs, refer to Haber and Keviczky (1999:639–710).

Practical experiences have shown that the best model performance will be achieved if the duration of a change (or hold time) in amplitude is selected as a trade-off between two facts. On the one hand, the duration has to be long enough to cause an effect on the output. On the other hand, too long a time will not give any more dynamic information about the system. Thus, a rule of thumb is: a suitable hold time has to lie close to the settling time of the step response of the system.

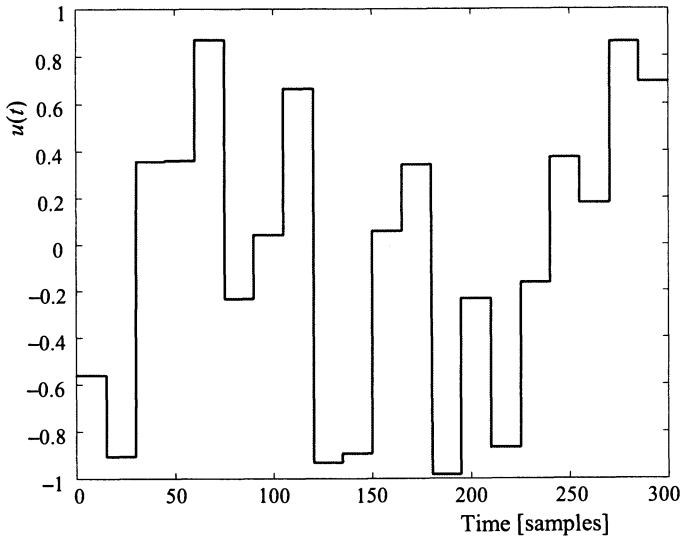


Figure 5.4. An example of a PRMS

### 5.2.1.2 Chirp Signals

A chirp signal is a sinusoid with a gradually increasing frequency (starting from  $\omega_{\text{start}}$  and ending by  $\omega_{\text{end}}$ ) (Franklin *et al.*, 1990):

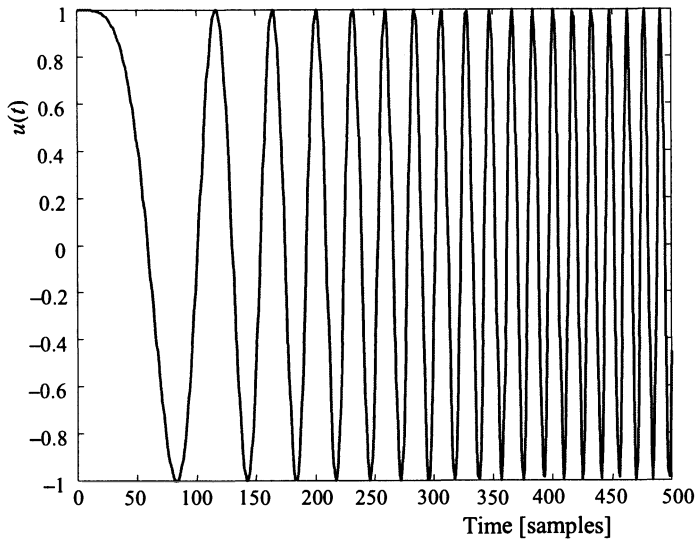
$$u(k) = u_0 + a \cdot \sin(\omega_k k T_0) \quad 1 \leq k \leq N \quad (5.4)$$

where

$$\omega_k = \omega_{\text{start}} + \frac{k}{N}(\omega_{\text{end}} - \omega_{\text{start}}) \quad (5.5)$$

Figure 5.5 illustrates such a chirp signal. When applying this signal to a non-linear system, it should be repeated for a number of combinations of different offset values ( $u_0$ ) and amplitudes ( $a$ ).

Experience in the identification for hydraulic systems shows that a PRMS is the most suitable choice of input signal. They simply lead to better models than for example using chirp signals; see Senger and Jelali (1996) for more details.



**Figure 5.5.** A chirp signal

## 5.2.2 Pre-computations

### 5.2.2.1 Choice of Sampling Interval

The selection of the best sampling interval for system identification is the result of *compromising* many factors:

- If the total time interval for an identification experiment is fixed, it is useful to choose a short sampling interval, since then more working points (*i.e.* amplitude–frequency combinations) of the system can be covered. The sampling interval should not be too short compared with the dynamics of the system under consideration in order to avoid serious problems with numerical ill-conditioning of the parameter estimation due to only minor changes between succeeding measurements.
- If the total number of data points is fixed, then the sampling interval must be selected by a trade-off: if it is very large, then the data will contain very little information about the high-frequency dynamics; if it is very small, then the disturbances may have a relatively large influence, and the data will contain little information about the low-frequency dynamics.
- If the identification is performed with control system design as the ulterior motive, certain other aspects will enter. For instance, a low sampling interval allows a rapid tracking and a smoother control input. However, a fast sampled model will often be non-minimum phase (Åström and Wittenmark, 1997).
- A very short sampling interval may cause systems with a pole excess of two or more to become non-minimum phase. This effect leads to special problems when designing regulators.

As a rough rule of thumb one can choose the sampling interval to be 10–20% of the settling time of the step response of the system (or of the time constant of interest). And it may often be much worse to choose the sampling interval too large than too small (Söderström and Stoica, 1989).

Some rules of thumb are available for linear systems; by utilising characteristic values of a step response, it is proposed to select the sampling time out of the following intervals:

$$\frac{T_0}{T_{63}} = \frac{1}{10} \dots \frac{1}{6} \quad (\text{Unbehauen, 1995; parametric models}) \quad (5.6)$$

$$\frac{T_0}{T_{95}} = \frac{1}{15} \dots \frac{1}{5} \quad (\text{Isermann, 1992}) \quad (5.7)$$

$$\frac{T_0}{T_{95}} = \frac{1}{15} \dots \frac{1}{4} \quad (\text{Isermann, 1987; digital control}) \quad (5.8)$$

$$\frac{T_0}{T_{63}} = 0.18 \dots 0.36 \quad (\text{Wernstedt, 1989; discrete models}) \quad (5.9)$$

where  $T_{63}$  and  $T_{95}$  are the time periods after which the step responses achieve 63% and 95% of the steady-state values respectively.

Reuter (1995) proposed an optimal sampling interval for a hydraulic servo-system (considered in Section 7.1) between 0.86 and 1.3 ms as a result of an entropy analysis. Therefore, for hydraulic servo-systems, the sampling interval should be selected in the range of 1–2 ms, which is common practice today. However, the right value of the sampling time depends on the size and dynamics of the hydraulic system at hand.

### 5.2.2.2 Preparation of Data

When the data have been collected from the identification experiment using the selected input sequence, they are not likely to be in shape for immediate use in identification algorithms. Indeed, the raw data will often contain measurement errors and other information, which should not be incorporated into the model.

- *Prefiltering* is often necessary to avoid aliasing, and remove noise, periodic disturbances, offsets and drifts from the measured signals. Analog anti-aliasing (low-pass) filters should be used prior to the sampling. The bandwidth of filters should be smaller than the sampling frequency.
- *Removing of outliers* (or bad data) from the data set (or alternatively insertion of interpolated values of the output signal) may be necessary. If omitted, the outliers will influence the estimated model considerably. They tend to appear as spikes in the sequence of prediction error, and will hence give large contributions to the loss function. However, removing certain points or time intervals from the data set is possible only for static data, otherwise it will give rise to transients. Therefore, it may be preferable for dynamic systems to carry out the experiment again in order to get a better data set.
- *Removing the mean and scaling* of all signals to the same variance is highly recommended. Indeed, scaling makes the estimation algorithm numerically

robust, leads to faster convergence, and simply tends to give better model quality. Typically, the measured input  $u(k)$  and output  $y(k)$  sequences are scaled to zero mean and unity variance by (Haber and Keviczky, 1999:399; Nørgaard, 2000a)

$$u^*(k) = \frac{u(k) - E\{u(k)\}}{\sigma_u} = \frac{u(k) - \bar{u}}{\sigma_u} \quad (5.10)$$

$$y^*(k) = \frac{y(k) - E\{y(k)\}}{\sigma_y} = \frac{y(k) - \bar{y}}{\sigma_y} \quad (5.11)$$

prior to identification.  $(\cdot)^*$ ,  $E\{\cdot\}$ ,  $\bar{(\cdot)}$  stand for the normalised, the expected, and the mean value.  $\sigma$  is the standard deviation of the corresponding signal.

### 5.3 Overview of Model Structures

The key step in the identification process is to decide upon a model structure, within which a good model is to be found. If a model structure is given, the system identification process reduces to a parameter estimation problem that is a minor problem in most cases. In practice, typically several structures are tested and the process of identification really becomes the process of evaluating and choosing between the resulting models in these different structures (Ljung, 1993). On the other hand, it is trivial that a bad model structure cannot yield a good model, regardless of the amount and quality of the available data and the chosen parameter estimation method.

The first problem is thus to decide whether the system admits a standard *linear* model description (e.g., ARX model; see Section 5.3.2), or whether a tailor-made (*non-linear*) model set must be constructed. The decision can be taken whether based on prior knowledge about the system or based on the application of *non-linearity test methods*; see Billings and Fakhouri (1978), Billings and Voon (1983), Haber (1995), and Haber and Keviczky (1999:399–483) for surveys of this subject.

In the linear case, the chances of fast success are good due to the simplicity of the problem and the availability of a number of standard algorithms and software packages. In the non-linear case, some physical insight has to be gained before a model can be estimated, or hoping that the non-linear dynamics can be picked up by a non-linear black-box structure. This problem is clearly application dependent and, therefore, not often discussed in the literature. Thus, this book mainly focuses on this topic and provides a contribution to solving this difficult problem.

It cannot, however, be sufficiently stressed that the key to success in system identification (in particular of non-linear models) lies in thinking, intuition, and insight, which can never be made obsolete by automated model construction.

For the sake of simplicity, the most following equations are formulated for SISO systems. However, the extension to MIMO systems is straightforward.

#### 5.3.1 Introductory Remarks and Definitions

In the field of identification of non-linear systems, two principal tendencies can be recognised (Stoica and Söderström, 1982). The first is to use a *very general*

description to represent the non-linear system, and the second shows the use of some *particular* characterisations for certain classes of non-linear systems.

Besides the distinction between general and particular descriptions of non-linear systems, there are three different ways of tackling the problem of constructing a mathematical model of a given process (Sohlberg, 1998). Commonly, a system model is based on either complete knowledge of the process, termed a “*white-box*” model, or based principally on experimental data, *i.e.*, a “*black-box*” model. White-box models are typically used as a simulation model, and are generally continuous and non-linear in nature. Between black-box modelling and white-box modelling there is grey zone, in which some physical insight is available, but several parameters remain to be determined from observed data. In this case “*grey-box*” model approaches are used.

The difference between these modelling principles and “grey-box” modelling lies in the systematic use of partial *a priori* knowledge of the process and data from an experiment. That means that black-box modelling needs only identification and white-box modelling requires only physical modelling, but grey-box modelling needs both physical modelling and parameter estimation. In practice, it is the actual purpose and context of the modelling that decides what kind of model should be chosen.

### 5.3.1.1 White-box Structures

A natural way to construct a mathematical model for a given process is white-box modelling. It is based on the pre-assumption that the process can be described entirely from prior knowledge and physical insight, *i.e.*, by mathematical equations, such as differential equations, algebraic equations, logical relationships and similar types of equation. White-box models are widespread in the field of hydraulic servo-systems, as described in Chapter 4. One is often led to the state-space form (set of first order ODEs)

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\Theta}) & \mathbf{x}(t_0) &= \mathbf{x}_0 \\ \mathbf{y}(t) &= \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\Theta})\end{aligned}\tag{5.12}$$

in which  $\mathbf{x}$  denotes the state vector,  $\mathbf{u}$  the input vector,  $\boldsymbol{\Theta}$  the (unknown) parameter vector, and  $\mathbf{y}$  the output vector. Generally, the parameter vector consists of the unknown model parameter vector  $\boldsymbol{\beta}$  and the unknown initial state vector  $\mathbf{x}_0$ , *i.e.*,

$$\boldsymbol{\Theta} \equiv \begin{bmatrix} \boldsymbol{\beta}^T & \mathbf{x}_0^T \end{bmatrix}^T\tag{5.13}$$

The advantages of model-building from chemical and physical principles are that it gives insight into the system behaviour, that the different parameters and variables have physical interpretations, that system variations can comparatively easily be tested, and that it can be carried out even in the design phase as it does not require operational data. The drawback is that it is expensive (difficult and time-consuming) to construct such models. As a trade-off, mathematical model-building is often combined with experiments to increase model accuracy and gain confidence in the models that will be used later for control design. This leads directly to grey-box modelling.



### 5.3.1.2 Black-box Structures

Another way to construct mathematical models for a given process is based on the idea that the process is regarded as completely unknown and that it is not necessary to use any model structure which reflects the physical structure of the process. Instead, a model from a family of standard models is used to describe the process. Such model families are known to have good flexibility and have been successfully used in the past. The (unknown) model parameters are then estimated using data from an experiment with the process, and the model provides only for an input–output relation of the process. Black-box modelling is normally used for processes where the underlying chemo-physical description is complicated and/or uncertain, such as servo-valves in hydraulic servo-systems. Moreover, black-box models are commonly used to model industrial processes in conjunction with adaptive control and model predictive control.

Some surveys of black-box modelling of non-linear systems have been given by Haber and Keviczky (1976), Mehra (1979), Billings (1980) and Sjöberg *et al.* (1995).

In line with Sjöberg *et al.* (1995), Ljung (1999) and many other researchers, the output of any (SISO) dynamic system at discrete time instances can mathematically be described as a function of past inputs and outputs, *i.e.*,

$$y(k) = \Gamma(\phi, \Theta) + e(k) \quad (5.14)$$

where  $y(k)$  is the observed output,  $\phi$  a vector of past quantities (usually referred to as a regression vector), and  $\Theta$  a vector of unknown parameters (usually termed as a parameter vector).  $e(k)$  is a noise term, which accounts for the fact that the next output  $y(t)$  will not be an exact function of past data. The goal of identification is to approximate the true input–output relationship by a predictor (model)

$$\hat{y}(k+j|k) = g(\phi, \Theta) = g(y(k), y(k-1), \dots, u(k), u(k-1), \dots, \Theta) \quad (5.15)$$

which gives rise to a minimum error between the observed outputs and the predicted ones. Using Equation 5.15, future values of the output beyond  $(k+1)$  are predicted based on the information available up to the instant  $k$ , assuming in the deterministic case that disturbances will not affect the system in the future. To simplify the notation,  $\hat{y}(k+j|k)$  will be replaced by  $\hat{y}(k+j)$ .

The model selection problem can then, in general, be regarded as composed of two design tasks:

1. selection of the regression vector  $\phi$  from past information,
2. selection of the non-linear function  $g$ .

The possibilities for regressor choice are discussed later in Sections 5.3.2 and 5.3.3. Concerning the nature of the non-linear function  $g$ , several popular choices are possible (*e.g.*, classical polynomials, Volterra expansions,<sup>1</sup> kernel estimators, wavelet networks, radial basis networks,  $B$ -splines, multi-layer networks, fuzzy models). All these belong to the parameterised function family  $g(\phi, \Theta)$  that can be written as function expansions

<sup>1</sup> Volterra-series models are only suitable for small-sized problems, as the number of parameters/regressors is usually exorbitant.

$$g(\varphi, \Theta) = \sum_i \alpha_i f_i(\varphi, \beta, \gamma) \quad (5.16)$$

where the basis functions  $f_i(\varphi, \beta, \gamma)$  can be constructed in different ways. Indeed, the expansion in Equation 5.16 plays the role of a *unified framework* for investigating most non-linear black-box model structures (Ljung, 1999).

### 5.3.1.3 Grey-box Structures

For many industrial processes, *e.g.*, hydraulic servo-systems as mentioned in Chapter 4, only incomplete knowledge about the structure and/or the parameters of the process is available. Thus, grey-box models are suitable for this type of process.

Since grey-box model design requires both physical modelling and parameter estimation, it may be more expensive to develop than a black-box model, *i.e.*, it may take longer to construct a grey-box than a black-box model; see Sohlberg (1998) for more details about the grey-box construction procedure. On the other hand, grey-box modelling leads to better models of the process, which are more transparent/physically interpretable and have better generalisation properties than black-box models. The process can then be controlled more efficiently, and the quality of the product produced can be improved. Further discussion of this topic can be found in Kroll (2000).

Actually, the non-linear identification problem posed in hydraulic servo-systems is a typical example of grey-box modelling, since physical insight is available, but several parameters remain to be determined from observed data. Here, two sub-cases should be distinguished:

1. **Complementing physical modelling.** A first interesting issue of the grey-box modelling approach is the reconstruction of underlying physical quantities from the parameters identified, since the parameters identified have a physical interpretation.
2. **Semi-physical modelling.** Physical insight is used to suggest certain (non-linear) combinations and/or transformations of measured data signals (*i.e.*, regressors). These new signals are then subjected to model structures of black-box character.

### 5.3.1.4 Hybrid Models

One of the most popular approaches to the representation of non-linear systems is that of *hybrid models* (Ljung, 1993), *i.e.*, models that combine linear dynamic blocks and one or more static non-linear block(s), such as the Hammerstein model (a static non-linear element followed by a linear system) and the Wiener model (a linear system followed by a static non-linear element). Note that the term “hybrid model” is not well defined and used for significantly varying concepts in the literature.

The Hammerstein model is the most attractive, from a system identification point of view. In contrast, the identification of Wiener models is more difficult, because the input of the non-linearity is not measurable and is therefore unknown, and the linear dynamics are still to be identified.

Restricting attention for a while to the dominant static non-linearities, it can easily be seen that the servo-valve model (see Figure 4.6) is of the Wiener type, the actuator model (see Figure 4.11) of the Hammerstein type, and the three-stage servo-valve model (see Figure 4.8) is a typical example of a more general model structure

that can be characterised by the Describing Function theory (Atherton, 1975; Föllinger, 1993). An in-depth treatment of the Describing Function method and its application to non-linear hydraulic system identification in the frequency domain can be found in van Schothorst (1997).

Although this method has been successfully applied to the identification of hydraulic servo-system components (servo-valve, actuator, pipelines), we do not believe that it will be of use in practical implementations. It is a very special and relatively complicated identification routine, and thus will not be considered further.

Of particular recent interest in system identification is to conceive hybrid model structures that are capable of dealing with dynamic effects, described by differential/difference equations and with logical constraints, “the ifs and the buts” of the system, *i.e.*, *fuzzy systems*, or by *artificial neural networks*. In the last years, quite intensive work has been done in this area and many concrete results are now available for these kinds of hybrid model structure. Some of these results will be presented in Sections 5.4.2 and 5.4.3.

### 5.3.1.5 Linear Parameterisability

Whatever the model structure should be, it is very advantageous if it is *linear in the parameters*. The linearity of the model in its parameters has the consequence that the parameters can be found in one step by solving a system of linear equations rather than using non-linear iterative optimisation algorithms. Such problems are very attractive, in that the estimation problem is much simplified, the solution is unique (*i.e.*, there is no problem with local minima), and can be found using the classical least-squares solver (see Section 5.5.2). Such linear-in-the-parameters model structures can be written as

$$y(k) = \sum_{i=1}^M \varphi_i(k) \theta_i = \boldsymbol{\theta}^T \boldsymbol{\varphi}(k) \quad (5.17)$$

Usually, physically based models of hydraulic servo-systems are actually non-linear in the parameters, but some simplifications can lead to linear-in-the-parameters sub-models, as explained in Section 5.7.

## 5.3.2 Review of Linear Model Structures

To get some guidance about the choice of model structures (especially of regressors), the linear case is reviewed, following Sjöberg *et al.* (1995) and Ljung (1999).

### 5.3.2.1 Input–Output Models

The linear black-box structures frequently used in practice are all variants of the general family

$$A(q)y(k) = q^{-\tau} \frac{B(q)}{F(q)} u(k) + \frac{C(q)}{D(q)} e(k) \quad (5.18)$$

using different ways of picking up “poles” of the system and different ways of describing the noise characteristics. In Equation 5.18,  $q$  denotes the backward shift (or delay) operator that works on a signal  $x(k)$  in the following way:

$$q^{-\tau} x(k) = x(k - \tau) \quad (5.19)$$

where  $\tau$  is an integer number of sampling periods<sup>2</sup> (i.e., the dynamics contain a delay of  $\tau$  samples).  $A(q)$ ,  $B(q)$ ,  $C(q)$ ,  $D(q)$ , and  $F(q)$  are polynomials<sup>3</sup> in  $q^{-1}$  of order  $n$ ,  $m$ ,  $p$ ,  $l$ , and  $r$  respectively:

$$\begin{aligned} A(q) &= 1 + a_1 q^{-1} + a_2 q^{-2} + \dots + a_n q^{-n} \\ B(q) &= b_1 q^{-1} + b_2 q^{-2} + \dots + b_m q^{-m} \\ C(q) &= 1 + c_1 q^{-1} + c_2 q^{-2} + \dots + c_p q^{-p} \\ D(q) &= 1 + d_1 q^{-1} + d_2 q^{-2} + \dots + d_l q^{-l} \\ F(q) &= 1 + f_1 q^{-1} + f_2 q^{-2} + \dots + f_r q^{-r} \end{aligned} \quad (5.20)$$

Equation 5.18 can be rewritten as

$$y(k) = q^{-\tau} \underbrace{\frac{B(q)}{A(q)F(q)}}_{=:G(q)} u(k) + \underbrace{\frac{C(q)}{A(q)D(q)}}_{=:H(q)} e(k) \quad (5.21)$$

The optimal predictor<sup>4</sup> associated with Equation 5.21 is given by

$$\begin{aligned} \hat{y}(k) &= H^{-1}(q)G(q)u(k) + [1 - H^{-1}(q)]y(k) \\ &= q^{-\tau} \frac{D(q)B(q)}{C(q)F(q)} u(k) + \left[ 1 - \frac{A(q)D(q)}{C(q)} \right] y(k) \end{aligned} \quad (5.22)$$

which can be represented in a “pseudo-linear” regression form (see Ljung (1999:88–90) for derivation)<sup>5</sup>

$$\hat{y}(k) = \hat{y}(k | k-1) = \Theta^T \varphi(k-1) \quad (5.23)$$

with the parameter vector

$$\Theta^T = [-a_1, \dots, -a_n, b_1, \dots, b_m, -f_1, \dots, -f_r, c_1, \dots, c_p, -d_1, \dots, -d_l] \quad (5.24)$$

<sup>2</sup> In this presentation, a delay-free system is defined to have  $\tau = 0$ . Many other authors use  $\tau = 1$  for such systems, with a corresponding modification of the polynomial  $B(q)$ .

<sup>3</sup> Following Ljung (1999),  $q$  is chosen as an argument of the polynomials rather than  $q^{-1}$  (which perhaps would be more natural in view of the right side) in order to be in formal agreement with  $z$ -transform and Fourier-transform expressions.

<sup>4</sup> The predictor is said to be optimal, as it extracts all the information contained in the signals up to the unpredictable part, the noise. Thus,  $e(k) = y(k) - \hat{y}(k | \Theta)$  is used to eliminate  $e(k)$  from Equation 5.21 which leads to Equation 5.22.

<sup>5</sup> A large number of authors use  $\hat{y}(k | \Theta) = \Theta^T \varphi(k)$  for Equation 5.23. Since the regression vector only contains past information (up to  $k-1$ ), the description in Equation 5.23 is preferred here. However, both representations are formally equivalent.

and the regression vector

$$\begin{aligned} \boldsymbol{\varphi}(k-1) = & [y(k-1), \dots, y(k-n), u(k-\tau-1), \dots, u(k-\tau-m), w(k-1), \dots, w(k-r), \\ & \varepsilon(k-1), \dots, \varepsilon(k-p), v(k-1), \dots, v(k-l)]^T \end{aligned} \quad (5.25)$$

$w(k)$  and  $v(k)$  are defined as

$$\begin{aligned} w(k) & \equiv q^{-\tau} \frac{B(q)}{F(q)} u(k) \\ v(k) & \equiv A(q)y(k) - w(k) \end{aligned} \quad (5.26)$$

Note that Equation 5.23 is mathematically equivalent to

$$\hat{y}(k+1) = \hat{y}(k+1 | k) = \boldsymbol{\Theta}^T \boldsymbol{\varphi}(k) \quad (5.27)$$

which will often be used below in this chapter.

The special cases of the description in Equation 5.22 are known as

- The *autoregressive with external input* (ARX) model ( $C = D = F = 1$ ). This is a “complete” linear model from the black-box perspective, *i.e.*, it is capable of describing all linear systems, including their noise characteristics, as polynomial orders tend to infinity. The only drawback is that the polynomial orders may have to be chosen larger than the dynamics require, in order to accommodate the noise description.
- The *autoregressive moving average with external input* (ARMAX) model ( $D = F = 1$ ). This is the best known modification of the ARX models, where the noise model is given “parameters on its own”.
- The *finite impulse response* (FIR) model ( $A = C = D = F = 1$ ). This is the simplest dynamic model, which is a linear regression on delayed input signals. Note, however, that  $m$  may be quite large and is not directly related to the system’s order.
- The *output-error* (OE) model ( $A = C = D = 1$ ). The OE model is capable of describing all linear systems (as polynomial orders tend to infinity), but not the character of additive noise. The main advantage is that fewer regressors are normally required to get a good approximation. The minimisation over  $\boldsymbol{\Theta}$ , however, becomes more complicated, and must be solved iteratively.
- The *Box–Jenkins* (BJ) model ( $A = 1$ ) has been proposed and treated by Box and Jenkins (1970). It is an OE model with additional degrees of freedom for the noise model, *i.e.*, the additive disturbance can be any coloured disturbance (and not only a white noise). BJ models are very flexible, but require the estimation of a large number of parameters and, therefore, are seldom used in practice.
- When  $A(q)$  contains the factor  $1 - q^{-1}$ , the model is called *autoregressive integrated moving average (with external input)* (ARIMA(X)). Such models are useful for describing drifting and non-stationary disturbances (or processes which even have an integrator in the dynamics) (Söderström and Stoica, 1989)

The regressors, *i.e.*, the elements of the regression vector  $\boldsymbol{\varphi}$ , are, in this general case, given by

1.  $u(k-l)$ : past inputs (associated with the  $B$ -polynomials).
2.  $y(k-l)$ : past measured outputs (associated with the  $A$ -polynomials).

3.  $\hat{y}_u(k-l|\Theta)$ : simulated outputs from past  $u$  only (associated with the  $F$ -polynomials).
4.  $\varepsilon(k-l|\Theta) = y(k-l) - \hat{y}(k-l|\Theta)$ : past prediction errors (associated with the  $C$ -polynomials).
5.  $\varepsilon_u(k-l|\Theta) = y(k-l) - \hat{y}_u(k-l|\Theta)$ : past simulation errors (associated with the  $D$ -polynomials).  $\hat{y}_u(k)$  is the model output when none of the observed outputs is involved in the regression vector in any form.

Note that in case  $A \neq 1$ , “simulated output” refers to the quantity  $A(q)y(k)$ .

**Example 5.1.** The predictor of the popular ARX<sup>6</sup> model takes the form

$$\hat{y}(k) = q^{-\tau} B(q)u(k) + [1 - A(q)]y(k) \quad (5.28)$$

Bearing in mind the polynomials in Equation 5.20, the ARX model can equivalently be written in the form of a one-step-ahead predictor

$$\hat{y}(k+1) = \Theta^T \phi(k) \quad (5.29)$$

with

$$\begin{aligned} \Theta^T &= [-a_1, \dots, -a_n, b_1, \dots, b_m] \\ \phi(k) &= [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T \end{aligned} \quad (5.30)$$

□

### 5.3.2.2 State-space Models

In a state-space form, the relationship between the input, noise, and output is written as a system of first-order differential or difference equations using an auxiliary state vector  $x$ . The description of linear systems in state-space form became an increasingly dominating approach after the pioneering work of Kalman (1960) on prediction and linear quadratic control.

The (general) linear state-space description of the form (stochastic model)

$$\begin{aligned} x(k+1) &= Ax(k) + bu(k) + v(k) \\ y(k) &= c^T x(k) + w(k) \end{aligned} \quad (5.31)$$

is a widely used alternative to the input–output model structures discussed up to now.  $v(k)$  and  $w(k)$  are assumed to be sequences of independent random variables (usually known as “process noise” and “measurement noise”, respectively) with zero mean values and the covariances

$$\begin{aligned} E\{v(k)v^T(l)\} &= R_v \delta_{k,l} \\ E\{w(k)v^T(l)\} &= E\{v(k)w^T(l)\} = R_{vw} \delta_{k,l} \end{aligned}$$

<sup>6</sup> The notation ARX( $n, m, d$ ) is sometimes used to specify the polynomial degrees (of  $A(q)$  and  $B(q)$ ) and time delay assumed in the model structure.

$$E\{w(k)w^T(l)\} = R_w \delta_{k,l} \quad (5.32)$$

The optimal one-step-ahead predictor for such systems is the Kalman filter

$$\begin{aligned} \hat{x}(k+1) &= A\hat{x}(k) + bu(k) + L(k)[y(k) - c^T \hat{x}(k)] \\ \hat{y}(k) &= c^T \hat{x}(k) \end{aligned} \quad (5.33)$$

with the filter gain matrix

$$L(k) = (AS(k)c + R_{vw})(c^T S(k)c + R_w)^{-1} \quad (5.34)$$

where  $S$  represents the positive semi-definite solution of the Ricatti equation

$$\begin{aligned} S(k+1) &= AS(k)A^T + R_v \\ &\quad - (AS(k)c + R_{vw})(c^T S(k)c + R_w)^{-1}(AS(k)c + R_{vw})^T \end{aligned} \quad (5.35)$$

The predictor filter of Equation 5.33 can be rewritten as in the so-called *state-space innovations form* (SSIF):

$$\begin{aligned} \hat{x}(k+1) &= A\hat{x}(k) + bu(k) + L(k)\varepsilon(k) \\ y(k) &= c^T \hat{x}(k) + \varepsilon(k) \end{aligned} \quad (5.36)$$

As the predictor filter can be written as (Söderström and Stoica, 1989; Ljung, 1999)

$$\hat{y}(k | \Theta) = c^T (qI - A + Lc^T)^{-1} bu(k) + c^T (qI - A + Lc^T)^{-1} Ly(k) \quad (5.37)$$

the relationship between the SSIF and the general input–output form (ARMAX, Equation 5.22) holds<sup>7</sup>

$$\begin{aligned} G(q) &= c^T (qI - A)^{-1} b \\ H(q) &= c^T (qI - A)^{-1} L + 1 \quad H^{-1}(q) = 1 - c^T (qI - A + Lc^T)^{-1} L \end{aligned} \quad (5.38)$$

Insights into physical mechanisms of the system can more easily be incorporated into state-space models than into input–output models introduced in Section 5.3.2.1. Moreover, state-space regressors are less restricted in their internal structure. This implies again that it might be possible to obtain a more efficient model with a smaller number of regressors by using a state-space model (Sjöberg *et al.*, 1995). In other words, the state-space model can represent a wider range of dynamic systems than the input–output model.

Also, state-space models have the following nice advantages (Chou and Maciejowski, 1997):

- (i) Causality is built into the model structure.
- (ii) In general, the best and most reliable numerical algorithms for system transformation, simplification, and analysis, as well as the exploitation of models

<sup>7</sup> Note the matrix inversion lemma  $(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$  (with general matrices  $A, B, C, D$  of corresponding dimensions).

in the synthesis of observers/filters and controllers, provided that the system model is state-space form.

### 5.3.3 Non-linear Input–output Models

Following the nomenclature in (non-linear) system identification literature (Chen *et al.*, 1990a,b; Chen and Billings, 1992; Sjöberg *et al.*, 1995), the most widely known (input–output) structures are presented in the following.

**NARMAX models** represent non-linear versions of the ARMAX model and are general forms of the input–output models (Chen and Billings, 1992). NARMAX models can be mathematically described as

$$\begin{aligned}\hat{y}(k+1) &= g(\boldsymbol{\varphi}, \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1), \\ &\quad \varepsilon(k), \dots, \varepsilon(k-r+1)]^T\end{aligned}\quad (5.39)$$

**NARX models** involve no stochastic input, unlike the NARMAX models. NARX models can be mathematically expressed as

$$\begin{aligned}\hat{y}(k+1) &= g(\boldsymbol{\varphi}, \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T\end{aligned}\quad (5.40)$$

**NFIR models** are simplified versions of NARMAX models. They contain only the observed inputs as shown below:

$$\begin{aligned}\hat{y}(k+1) &= g(\boldsymbol{\varphi}, \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= [u(k-\tau), \dots, u(k-\tau-m+1)]^T\end{aligned}\quad (5.41)$$

**NOE models** use the output of the model as a regressor instead of the observed output. NOE models are governed by

$$\begin{aligned}\hat{y}(k+1) &= g(\boldsymbol{\varphi}, \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= [\hat{y}_u(k), \dots, \hat{y}_u(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T\end{aligned}\quad (5.42)$$

**NBJ models** can be mathematically described as

$$\begin{aligned}\hat{y}(k+1) &= g(\boldsymbol{\varphi}, \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= [\hat{y}(k), \dots, \hat{y}(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1), \\ &\quad \varepsilon(k), \dots, \varepsilon(k-r+1), \varepsilon_u(k), \dots, \varepsilon_u(k-r+1)]^T\end{aligned}\quad (5.43)$$

Since the regressors of NOE, NBJ and NARMAX models contain past information from the model itself, they are also classified as (externally) *recurrent networks*.

**Example 5.2.** The widely used *polynomial* NARX model (also known as the Kolmogorov–Gabor polynomial model) can be expressed as



$$\begin{aligned}
\hat{y}(k) = & \sum_{i=1}^n a_i y(k-i) + \sum_{i=1}^n \sum_{j=i}^n a_{i,j} y(k-i) y(k-j) + \dots \\
& + \sum_{i=1}^n \dots \sum_{v=p}^n \sum_{w=v}^n a_{i,\dots,v,w} y(k-i) \dots y(k-v) y(k-w) \\
& + \sum_{i=0}^{n-1} b_i u(k-i) + \sum_{i=0}^{n-1} \sum_{j=i}^{n-1} b_{i,j} u(k-i) u(k-j) + \dots \\
& + \sum_{i=0}^{n-1} \dots \sum_{v=p}^{n-1} \sum_{w=v}^{n-1} b_{i,\dots,v,w} u(k-i) \dots u(k-v) u(k-w) \\
& + \sum_{i=0}^{n-1} \sum_{j=1}^n c_{i,j} u(k-i) y(k-j) + \dots \\
& + \sum_{i=0}^{n-1} \dots \sum_{v=p}^{n-1} \sum_{w=1}^n c_{i,\dots,v,w} u(k-i) \dots u(k-v) y(k-w) + \dots \\
& + \sum_{i=0}^{n-1} \sum_{j=1}^n \dots \sum_{v=p}^n \sum_{w=v}^n c_{i,j,\dots,v,w} u(k-i) y(k-j) \dots y(k-v) y(k-w)
\end{aligned} \tag{5.44}$$

with the polynomial coefficients as the unknown (to be estimated) parameters. Owing to the high complexity of such models, they should only be used in combination with structure selection algorithms, such as those proposed by Kortmann (1989).  $\square$

### 5.3.3.1 Serial-parallel vs. Parallel Model

The dynamic behaviour of model is essentially influenced by the way the output is fed back into the network. Two approaches, often discussed in the literature, are the serial-parallel model (of NARX type) and the parallel model (of NOE type); see Figure 5.6. In the parallel model, there is complete parallelism (for both input and output) between model and system, whereas parallelism for the serial-parallel model is only given for the input.

If one is specifically interested in a simulation model (or a long-horizon prediction model), it would, in principle, be better to consider NOE models, as these are identified as providing an optimal simulator. However, the identification of such models is more difficult and time-consuming than identifying an NARX model. (In some references the model is identified as a serial-parallel model, though simulated as a parallel model; see also the discussion in Sections 5.4.2.3 and 5.8.5.1.)

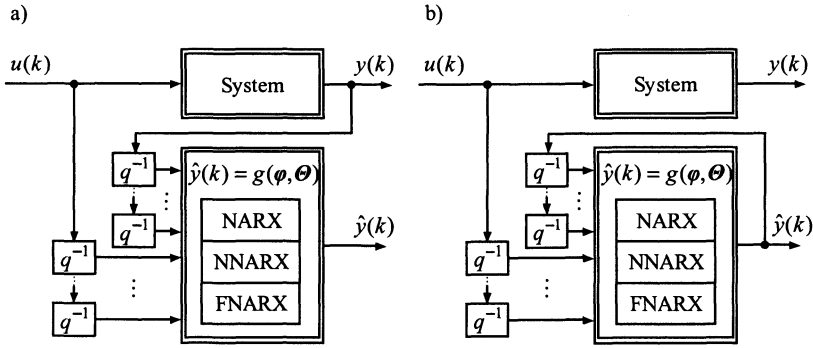


Figure 5.6. (a) Serial-parallel model structures; (b) parallel model structures

### 5.3.3.2 Determination of Lag Space

The selection of the right lag space, *i.e.*, the dynamic orders ( $n$  and  $m$ ) of delayed signals used as regressors, is decisive for the model quality and a difficult problem as well. Besides trial-and-error and prior knowledge about the system, there are very few strategies for reliable automatic determination of the lag space. The most prominent method is to use the so-called Lipschitz quotients

$$l_{ij}^{(n)} = \frac{|y(i) - y(j)|}{\sqrt{[\varphi_1(i) - \varphi_1(j)]^2 + \dots + [\varphi_n(i) - \varphi_n(j)]^2}} \quad (5.45)$$

for  $i = 1, \dots, N$ ,  $j = 1, \dots, N$  and  $i \neq j$ , as proposed by He and Asada (1993). The subscript “(n)” stands here for the number of regressors. The Lipschitz order index is defined as the geometric average of the  $c$  (with  $c \in [0.01N, 0.02N]$ ) largest Lipschitz quotients

$$\bar{l}^{(n)} = \left[ \prod_{k=1}^c \sqrt{n} l^{(n)}(k) \right]^{1/c} \quad l^{(n)}(k) \equiv \{c \text{ largest quotients } l_{ij}^{(n)}\} \quad (5.46)$$

Evaluating the order index for different lag choices ( $n$  or  $m$ ) and plotting it as a function of each parameter enables one to select the optimal number of regressors as the “knee-point” of the curve.

### 5.3.4 Non-linear State-space Models

In analogy to the continuous-time description in Equation 5.12, the state-space model of MIMO non-linear dynamic systems can be expressed as

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{f}(\mathbf{x}(k), \mathbf{u}(k), \boldsymbol{\Theta}) + \mathbf{v}(k) & \mathbf{x}(0) &= \mathbf{x}_0 \\ \mathbf{y}(k) &= \mathbf{h}(\mathbf{x}(k), \mathbf{u}(k), \boldsymbol{\Theta}) + \mathbf{w}(k) \end{aligned} \quad (5.47)$$

where  $\mathbf{v}(k)$  and  $\mathbf{w}(k)$  are zero-mean disturbance and noise vectors of appropriate dimensions. In this case the model is defined by the functions  $\mathbf{f}$  and  $\mathbf{h}$ .

In a similar way to the input–output model, the system described by the state-space model of this equation can be identified after having selected a model structure. A well-known representation is the SSIF model, which involves innovations  $\varepsilon(k)$  in their regression vector and can be described as (Habtom, 1999)

$$\begin{aligned}\hat{\mathbf{x}}(k+1) &= \mathbf{f}(\boldsymbol{\varphi}(k), \boldsymbol{\Theta}) \\ \boldsymbol{\varphi}(k) &= \begin{bmatrix} \hat{\mathbf{x}}^T(k) & \mathbf{u}^T(k) & \boldsymbol{\varepsilon}^T(k) \end{bmatrix}^T \\ \hat{\mathbf{y}}(k) &= \mathbf{C}\hat{\mathbf{x}}(k)\end{aligned}\tag{5.48}$$

The optimal one-step-ahead prediction can be constructed using approximations  $\hat{\mathbf{f}}$  and  $\hat{\mathbf{h}}$ , and the extended Kalman filter approach (Ljung, 1999), *i.e.*,

$$\begin{aligned}\hat{\mathbf{x}}(k+1) &= \hat{\mathbf{f}}(\hat{\mathbf{x}}(k), \mathbf{u}(k)) + \mathbf{L}(k)\boldsymbol{\varepsilon}(k) \\ \hat{\mathbf{y}}(k | \boldsymbol{\Theta}) &= \hat{\mathbf{h}}(\hat{\mathbf{x}}(k))\end{aligned}\tag{5.49}$$

where  $\mathbf{L}(k)$  is the Kalman filter gain matrix. This will depend explicitly on the time, the functions  $\hat{\mathbf{f}}$  and  $\hat{\mathbf{h}}$ , and the covariance matrices of the disturbance and noise sequences.

## 5.4 Description of Selected Non-linear Model Structures

In this section, selected non-linear model structures, which have been developed and/or successfully applied by the authors of this book to hydraulic servo-systems, will be treated in more detail.

### 5.4.1 Continuous-time Special (Canonical) Models

Although continuous-time dynamic systems can be sufficiently approximated by non-linear discrete-time models and numerous (discrete-time) model structures have been proposed, it is generally not straightforward to relate the estimated parameters to the physical behaviour or to a continuous-time description of the system. In continuous-time systems, many characteristics can be directly related to some of the system parameters. In discrete-time forms, such relationships are often less obvious. Hence, identification of continuous-time models from sampled data records is sometimes desirable to enhance the interpretation of the system behaviour and to aid controller design.

#### 5.4.1.1 Linear-in-the-parameters Polynomial Models

In the following approach, the non-linear models are assumed to be differential equations linear in the parameters in order to apply the well-known least-squares estimation methods. The possible model components (*i.e.*, regressors) can be linear and non-linear – usually polynomial – functions of the input and output signals, as well as the associated higher order derivatives. The task of the structure identification is then to select the significant components among all possible ones.

In analogy to the NARX model for discrete-time identification (Equation 5.44), the continuous-time non-linear polynomial input–output model of the form

$$\begin{aligned}
 \hat{y}(t) = & \sum_{i=0}^n a_i y^{(i)}(t) + \sum_{i=0}^n \sum_{j=i}^n a_{i,j} y^{(i)}(t) y^{(j)}(t) + \dots \\
 & + \sum_{i=0}^n \dots \sum_{v=p}^n \sum_{w=v}^n a_{i,\dots,v,w} y^{(i)}(t) \dots y^{(v)}(t) y^{(w)}(t) + \sum_{i=0}^{n-1} b_i u^{(i)}(t) \\
 & + \sum_{i=0}^{n-1} \sum_{j=i}^{n-1} b_{i,j} u^{(i)}(t) u^{(j)}(t) + \dots + \sum_{i=0}^{n-1} \dots \sum_{v=p}^{n-1} \sum_{w=v}^{n-1} b_{i,\dots,v,w} u^{(i)}(t) \dots u^{(v)}(t) u^{(w)}(t) \\
 & + \sum_{i=0}^{n-1} \sum_{j=0}^n c_{i,j} u^{(i)}(t) y^{(j)}(t) + \dots + \sum_{i=0}^{n-1} \dots \sum_{v=p}^{n-1} \sum_{w=1}^n c_{i,\dots,v,w} u^{(i)}(t) \dots u^{(v)}(t) y^{(w)}(t) \\
 & + \dots + \sum_{i=0}^{n-1} \sum_{j=1}^n \dots \sum_{v=p}^n \sum_{w=v}^n c_{i,j,\dots,v,w} u^{(i)}(t) y^{(j)}(t) \dots y^{(v)}(t) y^{(w)}(t) \quad (5.50)
 \end{aligned}$$

or in compact representation

$$\hat{y}(t) = - \sum_{i=2}^{n+1} \alpha_i y^{(i-1)}(t) - {}^l P \left[ y, \dot{y}, \dots, y^{(n)}, u, \dot{u}, \dots, u^{(n-1)} \right] + \sum_{j=1}^n \beta_j u^{(j-1)}(t) \quad (5.51)$$

can be used as a basis for continuous-time system identification.  $n$  is the order of dynamics of the system,  $u^{(i)}(t)$  and  $y^{(i)}(t)$  are the higher order derivatives of the input and output signals respectively.  ${}^l P[\cdot]$ , ( $l \geq 2$ ), is a polynomial non-linear function of degree  $l$  (referred to as degree of non-linearity), and can be defined by means of the reduced Kronecker product notation (see Appendix C.1) as follows (Jelali, 1997):

$${}^l P[\zeta] = \sum_{i=2}^{n+1} \rho_k^T \zeta^{[k]} \quad (5.52)$$

with the vectors

$$\rho_k^T = [\rho_{k,1} \quad \rho_{k,2} \quad \dots \quad \rho_{k,s_k}] \quad , \quad s_k = \binom{2n+k}{k} = \frac{(2n+k)!k!}{(2n)!} \quad (5.53)$$

and

$$\zeta = \begin{bmatrix} y & \dot{y} & \dots & y^{(n)} & u & \dot{u} & \dots & u^{(n-1)} \end{bmatrix}^T \in \mathbb{R}^{2n+1} \quad (5.54)$$

In other words,  ${}^l P[\cdot]$  encompasses the inputs, the outputs, the associated higher order derivatives and higher powers, and cross products between all these signals, *i.e.* the regressors.

It is then easy to show that Equation 5.51 can be rewritten in the (linear) regression form

$$\hat{y}(t) = \sum_{i=1}^M \varphi_i(t) \Theta_i = \boldsymbol{\varphi}^T(t) \boldsymbol{\Theta} \quad (5.55)$$

where

$$\boldsymbol{\varphi}^T = - \begin{bmatrix} \dot{y} & -\ddot{y} & \dots & -y^{(n)} & (\zeta^{[2]})^T & \dots & (\zeta^{[l]})^T & u & \dot{u} & \dots & u^{(n-1)} \end{bmatrix} \quad (5.56)$$

is the regression vector, and

$$\boldsymbol{\Theta} = [\boldsymbol{\alpha}^T \quad \boldsymbol{\rho}_2^T \quad \dots \quad \boldsymbol{\rho}_l^T \quad \boldsymbol{\beta}^T]^T \in \mathbb{R}^M, \quad M = 2n + \sum_{k=1}^l s_k \quad (5.57)$$

is the parameter vector. The parameter vector  $\boldsymbol{\Theta}$ , therefore, has to be identified from input–output measurements of the plant. This could be done by least-squares-based estimation methods (see Section 5.5.2). However, by reformulating the problem such that each parameter  $\Theta_i$  can be estimated independently using an orthogonal algorithm, considerable advantages can be achieved (see Section 5.5.3).

#### 5.4.1.2 Approximate Polynomial Continuous-time Canonical Forms

The model structures presented below (introduced by Jelali and Schwarz (1995a–c) and Jelali (1997)) permit to avoid the complex physical model structure with a lot of unknown parameters. It has been proven that they improve linear approximations. Furthermore, the identification of the model directly in these canonical structures has the advantage that state non-linear observers/filters and controllers can be designed more easily, as already mentioned in Section 5.3.2. Difficult calculations, which mainly concern the integration of linear or non-linear partial differential equations (Zeitz, 1990; Schwarz, 1991), can then be avoided. Yet, such canonical model structures are well-suited for identification as they have minimal number of parameters – a highly desirable property. The consequence of this property is that the parameter estimation problem is well-conditioned, and thus the accuracy of parameter estimation (measured by the variance of the parameter estimates) will be high.

**Observer Canonical Form.** For the non-linear observer canonical form (NOCF) (Zeitz, 1990; Schwarz, 1991)

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} -a_1(x_n) \\ x_1 - a_2(x_n) \\ \vdots \\ x_{n-1} - a_n(x_n) \end{bmatrix} + \begin{bmatrix} b_1(x_n) \\ b_2(x_n) \\ \vdots \\ b_n(x_n) \end{bmatrix} u(t) \\ y(t) &= x_n(t) \end{aligned} \quad (5.58)$$

the approximations

$$a_i(x_n) = \sum_{j=1}^l \alpha_{j,i} x_n^j \quad i = 1, 2, \dots, n$$

$$b_i(x_n) = \sum_{j=0}^l \alpha_{j,i} x_n^j \quad i = 1, 2, \dots, n \quad (5.59)$$

are used to obtain (Jelali, 1997)

$$\begin{aligned} \hat{y}(t) &= \frac{1}{\alpha_{1,1}} \left[ -y^{(n)} - \sum_{j=2}^l \alpha_{j,1} y^j - \sum_{i=2}^n \sum_{j=1}^l \alpha_{j,i} (y^j)^{(i-1)} + \sum_{j=1}^n \sum_{k=0}^l \beta_{k,j} (y^k u)^{(j-1)} \right] \\ &\equiv \sum_{i=1}^M \varphi_i(t) \theta_i = \boldsymbol{\varphi}^T(t) \boldsymbol{\theta} \end{aligned} \quad (5.60)$$

where

$$M = n(1 + 2l) \quad (5.61)$$

$$\boldsymbol{\varphi}^T = \begin{bmatrix} -y^{(n)} & -y^2 & \dots & -y^l & -\dot{y} & \dots & -(y^l)^{(1)} & \dots & -y^{(n-1)} & \dots & -(y^l)^{(n-1)} \\ u & \dots & y^l u & \dots & (y^l u)^{(n-1)} \end{bmatrix} \quad (5.62)$$

$$\boldsymbol{\theta} = \begin{bmatrix} 1 & \alpha_{2,1} & \dots & \alpha_{l,1} & \alpha_{1,2} & \dots & \alpha_{l,2} & \dots & \alpha_{1,n} & \dots & \alpha_{l,n} \\ \beta_{0,1} & \dots & \beta_{l,1} & \dots & \beta_{l,n} \end{bmatrix} \quad (5.63)$$

The NOCF, Equation 5.58, is characterised by the property that a canonical form observer can be designed by pole placement like in the linear case.

**Observability Canonical Form.** Consider the non-linear observability canonical form (NOBCF) described by the state-space model (Zeitz, 1990; Schwarz, 1991)

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} x_2 \\ \vdots \\ x_n \\ -f_n(\mathbf{x}) \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ g_n(\mathbf{x}) \end{bmatrix} u(t) \\ y(t) &= x_1(t) \end{aligned} \quad (5.64)$$

Then, the non-linear functions  $f_n(\mathbf{x})$  and  $g_n(\mathbf{x})$  are approximated by the polynomial functions

$$\begin{aligned} f_n(\mathbf{x}) &= \sum_{i=1}^l \boldsymbol{\alpha}_i^T \mathbf{x}^{[i]} \\ g_n(\mathbf{x}) &= \beta_{0,1} + \sum_{j=1}^l \boldsymbol{\beta}_j^T \mathbf{x}^{[j]} \end{aligned} \quad (5.65)$$

To reduce the number of model parameters, one could use also the approximations

$$\begin{aligned}
f_n(x) &= \sum_{i=1}^n \sum_{j=1}^l \alpha_{j,i}^* \left(x_1^j\right)^{(i-1)} \\
g_n(x) &= \beta_{0,1}^* + \sum_{j=1}^n \sum_{k=1}^l \beta_{k,j}^* \left(x_1^k\right)^{(j-1)}
\end{aligned} \tag{5.66}$$

instead of Equation 5.65.

Some calculations lead to the input–output relationship (Jelali, 1997)

$$\begin{aligned}
\hat{y}(t) &= \frac{1}{\alpha_{1,1}} \left[ -\sum_{j=2}^n \alpha_{1,j} y^{(j-1)} - y^{(n)} - \sum_{i=2}^l \alpha_i^T \phi^{[i]} + \sum_{i=1}^l \beta_i^T \phi^{[i]} u + \beta_{0,1} u \right] \\
&\equiv \sum_{i=1}^M \phi_i(t) \Theta_i = \phi^T(t) \Theta
\end{aligned} \tag{5.67}$$

where

$$\zeta = \begin{bmatrix} y & \dot{y} & \dots & y^{(n-1)} \end{bmatrix}^T \tag{5.68}$$

$$M = n+1 + \sum_{j=2}^l \binom{n+j-1}{j} + \sum_{j=1}^l \binom{n+j-1}{j} \tag{5.69}$$

$$\begin{aligned}
\phi^T &= \begin{bmatrix} -\dot{y} & \dots & -y^{(n-1)} & -y^{(n)} & -(\zeta^{[2]})^T & \dots \\ & & -(\zeta^{[l]})^T & \zeta^T u & \dots & (\zeta^{[l]})^T u & u \end{bmatrix}
\end{aligned} \tag{5.70}$$

$$\Theta = \frac{1}{\alpha_{1,1}} \begin{bmatrix} \alpha_{1,2} & \dots & \alpha_{1,n-1} & 1 & -\alpha_2^T & \dots & -\alpha_l^T & \beta_1^T & \dots & \beta_l^T & \beta_{0,1} \end{bmatrix}^T \tag{5.71}$$

The resulting NOCF and NOBCF will be called approximate polynomial NOCF and NOBCF, respectively. One can prove that Equations 5.60 and 5.67 are special forms of Equation 5.51.

## 5.4.2 Fuzzy Models

### 5.4.2.1 Introduction to Fuzzy Modelling

Zadeh (1965) proposed the concept of “fuzzy logic” to offer a mathematically exact theory to model uncertain information. The resulting basic concept can be interpreted, for example, as an extension of binary logic, which led to the introduction of “*fuzzy sets*”: whereas the membership of an element of the universe of discourse to a classical set is restricted to a binary “yes/no” or “1/0” decision, it is a continuous number  $\mu \in [0; 1]$  in the case of fuzzy sets. The “*membership function*”  $\mu_A(x): \mathbb{R} \rightarrow [0; 1]$  assigns the membership  $\mu$  to the fuzzy set  $A$  of its argument  $x$ . Human logic can be interpreted to work in the form of non-crisp causal relationships. In order to model these, Zadeh introduced “*fuzzy rules*”: their premise

(IF section) classifies the validity or applicability of a rule, yielding again a (non-crisp) continuous number  $\mu \in [0; 1]$ . The consequence (THEN section) can take different forms as discussed in the following. Both, continuous- and discrete-time fuzzy models have been proposed in the literature. Only the latter will be discussed in this book.

Two different fuzzy system types are typically distinguished. Relational (or “Mamdani”) type fuzzy systems (Tong 1978) feature linguistic premises and consequences:

$$\text{IF } (Z(k) \text{ IS } Z_p) \text{ AND } (W(k) \text{ IS } W_h) \text{ THEN } Y(k) \text{ IS } Y_l \quad (5.72)$$

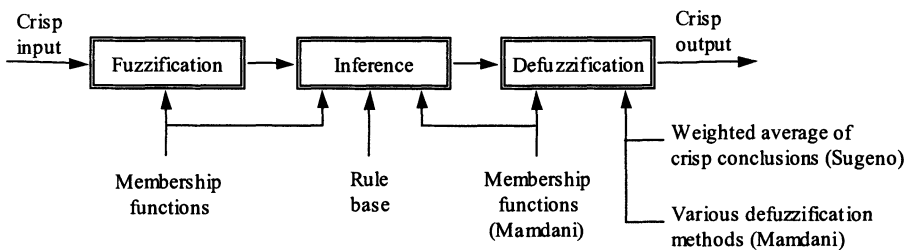
This rule can be interpreted as: dependent on how close the (fuzzy) input  $Z(k)$  is to its reference  $Z_p$  and how close the (fuzzy) input  $W(k)$  is to its reference  $W_h$ , the reference output  $Y_l$  is weighted in order to determine the contribution of this rule in a set of rules, which is evaluated. The result of the evaluation is a fuzzy set.

The other type, functional (or “Takagi–Sugeno”) type fuzzy systems, uses a crisp function in the consequences, in contrast to relational systems (Takagi and Sugeno, 1983):

$$\text{IF } (Z(k) \text{ IS } Z_p) \text{ AND } (W(k) \text{ IS } W_h) \text{ THEN } \eta_i(k+1) = f_i(z(k), w(k)) \quad (5.73)$$

This rule can be interpreted as the relational rule, except that the result of the evaluation is a crisp number (from the conclusion) plus the degree of applicability (from the premise).

A fuzzy system (model or controller) consists of a fuzzification, an inference and a defuzzification module (see Figure 5.7). The fuzzification module converts the crisp inputs into fuzzy quantities by using the defined reference membership functions for the premise (inputs). The inference engine evaluates all fuzzy rules for the given fuzzy input. The defuzzification module converts the fuzzy conclusion back into a crisp number. Mamdani-type systems deliver each a fuzzy conclusion. In contrast, the conclusions of Sugeno systems each deliver a crisp conclusion. Hence, Mamdani-type systems also have membership functions for the conclusions, though Sugeno-type systems have not. In fact, while Sugeno-type systems derive the global conclusion as a simple weighted average, there are many methods for Mamdani-type systems.



**Figure 5.7.** Processing steps of a fuzzy system: fuzzification, inference and defuzzification

Generally, fuzzy models can be gained through (Harris and Moore, 1989):

- Verbalisation. The model synthesis is carried out by knowledge acquisition from human experts (only for Mamdani-type systems).



- Fuzzification. A crisp mathematical model is transformed into a fuzzy model.
- Identification. A (parametrical) fuzzy model is derived from input–output data of a system.

The reasons for the attractiveness of fuzzy models include their good approximation capability and their flexibility of application. In fact, some classes have been proven to be universal approximators (*e.g.*, Kosko, 1992; Buckley, 1993; Wang, 1994; Rovatti, 1998; Ying, 1998).

The modelling of human operator heuristics as a (Mamdani-type) fuzzy controller permitted automatic control of processes resisting conventional automatic control, while not requiring process modelling. This drove a boom in fuzzy control in the late 1970s with focus on process industry applications, where modelling is often difficult and costly.

In the following, the path of process modelling followed by model-based controller design is considered. The Sugeno-type modelling approach is utilised in this book. Sugeno-type systems received significant attention in the area of non-linear system analysis, identification, modelling and control. This is believed to be due to some new key results in transferring *formal* methods to these systems providing the user with a toolbox with matured theory behind it. Examples include system stability, controller design with guaranteed stability using LMI methods, controller design as a parallel compensator with pole placement and similar techniques. Some recent monographs dedicate a significant share to Sugeno-type systems (Babuška, 1998; Leenaerts and Bokhoven, 1998; Murray-Smith and Johanson, 1998; Nelles, 2001).

The discrete-time models will be gained by identification. Sugeno-type models typically yield higher numerical prediction quality than Mamdani-type models. This is advantageous, *e.g.*, in case of model utilisation within a model predictive controller (Kroll and Bernd, 2000). Moreover, Sugeno-type models permit one to transfer methods from linear control system engineering for fuzzy-model-based control, *e.g.*, in the form of a “parallel distributed compensator” (Wang *et al.*, 1995).

The extended Sugeno-type model introduced further on can be understood as a generalization of radial (and hyper) basis function networks, which are described in Section 5.4.3.2; see Kroll (1996).

The reader is referred to Kroll (1997) and Babuška (1998) for a deeper study regarding fuzzy identification and modelling. Driankov *et al.* (1993) is one of many textbooks that provide an introduction to fuzzy logic and control. Readers interested in an overview of successful industrial applications of fuzzy logic, modelling and control, besides the servo-hydraulic systems considered in this book, are referred to Hirota (1993) and Zimmermann and von Altroch (1994).

#### 5.4.2.2 Representation of Dynamics in Fuzzy Models

Each fuzzy rule represents a *static* mapping of inputs to outputs, as a rule has no memory. Dynamic behaviour is achieved by augmentation with an external memory: the latter stores the most recent values of inputs and outputs back to a specified finite horizon and supplies them as inputs to the rule. This is referred to as an “external recurrent model”. Figure 5.8 sketches such an approach for a single-input single-output model. It also indicates handling of a dead time  $\tau$ , and scaling factors  $S_u$  and  $S_y$  for model inputs and outputs respectively. A normalised quantity is indicated by

“\*”. For readability, we will not differentiate between scaled and non-scaled quantities in the following.

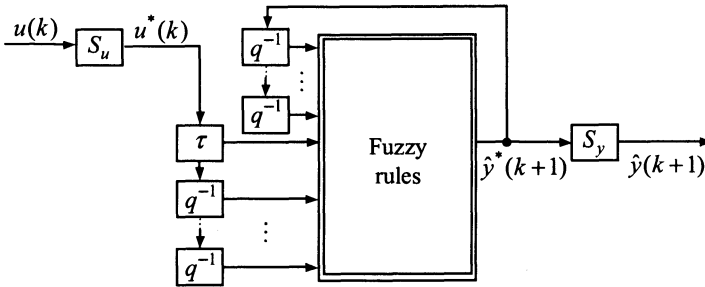
### 5.4.2.3 Mathematical Description of Sugeno-type Fuzzy Models

**Premise (Antecedent): Membership Functions, Distance Norm and Fuzziness.** Sugeno-type fuzzy models consist of  $c$  rules, each with a linguistic premise and a crisp conclusion:

$$\text{IF } (Z(k) \text{ IS } Z_p) \text{ AND } (W(k) \text{ IS } W_h) \text{ THEN } \eta_i(k+1) = f_i(z(k), w(k)) \quad (5.74)$$

Instead of multiple scalar expressions in a premise, a single vector type expression is used:

$$\text{IF } (\chi(k) \text{ IS } \chi_i) \text{ THEN } \eta_i(k+1) = f_i(z(k), w(k)) \quad (5.75)$$



**Figure 5.8.** Dynamical fuzzy models are obtained by externally memorising  $n$  historical output values and  $m$  historical input values and using them as inputs of the fuzzy rules (applicable for both Mamdani- and Sugeno-type fuzzy models)

The degree of fulfilment of the rule (a crisp number  $\mu \in [0; 1]$ ), given the current feature vector  $\chi(k)$ , is derived by comparing it with the rule's reference  $\chi_i$ . In fact, multivariate (multidimensional) fuzzy sets are used instead of univariate (scalar) ones to permit one to overcome the otherwise enforced grid-type partition. Practically, the degree of fulfilment is computed by evaluating the according membership function  $\mu_i(k)$  associated with the rule  $i$ . (In case a premise consists of several scalar comparisons, each comparison (...IS...) has its associated membership function and is evaluated separately. Then, the results are combined using a T-norm to provide for the total degree of fulfilment of a rule.) Figure 5.9 illustrates the difference between grid-type and multivariate membership functions and the resulting partitioning.

One interesting, multivariate membership function originates from (probabilistic) fuzzy classification (Bezdek, 1981): each membership function  $\mu_i(\chi)$  compares the distance of its argument  $\chi$  from its assigned reference point (vector)  $v_i$  with the distances to all  $c-1$  other reference points in order to compute the membership value. A distance of zero (“singularity”) is handled separately. Mathematically the  $\mu_i(\chi)$  can be written as:

$$\mu_i(\chi) = \begin{cases} \frac{1}{\sum_{j=1}^c \left[ \frac{\|\chi - v_i\|}{\|\chi - v_j\|} \right]^{\frac{2}{\nu-1}}} & \text{for } \chi \notin \{v_1, \dots, v_c\} \quad (\text{no singularities}) \\ 1 & \text{for } \chi = v_i \\ 0 & \text{for } \chi \in \{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_c\} \end{cases} \quad (\text{for singularities}) \quad (5.76)$$

This membership function is also referred to as “probabilistic”, as the summation of all memberships to the  $c$  partitions yields unity for any argument:  $\sum_{i=1}^c \mu_i(\chi) \equiv 1 \forall \chi$ . The parameter  $\nu$  adjusts the fuzziness of  $\mu_i(\chi)$ : the fuzziness increases with  $\nu$ . This membership function is steady, non-convex and steadily differentiable for  $1 < \nu < 3$  (Kroll, 1997).

Possibilistic membership functions represent alternative multivariate membership functions. They originate from possibilistic fuzzy clustering (Krishnapuram and Keller, 1993), but also multivariate Gaussian membership functions belong to this type. Possibilistic membership functions are steady, convex and typically steadily differentiable. In contrast to probabilistic membership functions, the summation of all memberships to the  $c$  partitions does not yield unity for any argument.

Further on, the probabilistic membership function type will be used, owing to its advantages (steady differentiability, transferability of methods from pattern recognition to identification) for identification and modelling. Note that univariate possibilistic membership functions can be derived from the multivariate probabilistic membership functions, *e.g.*, by projection (Babuška, 1998).

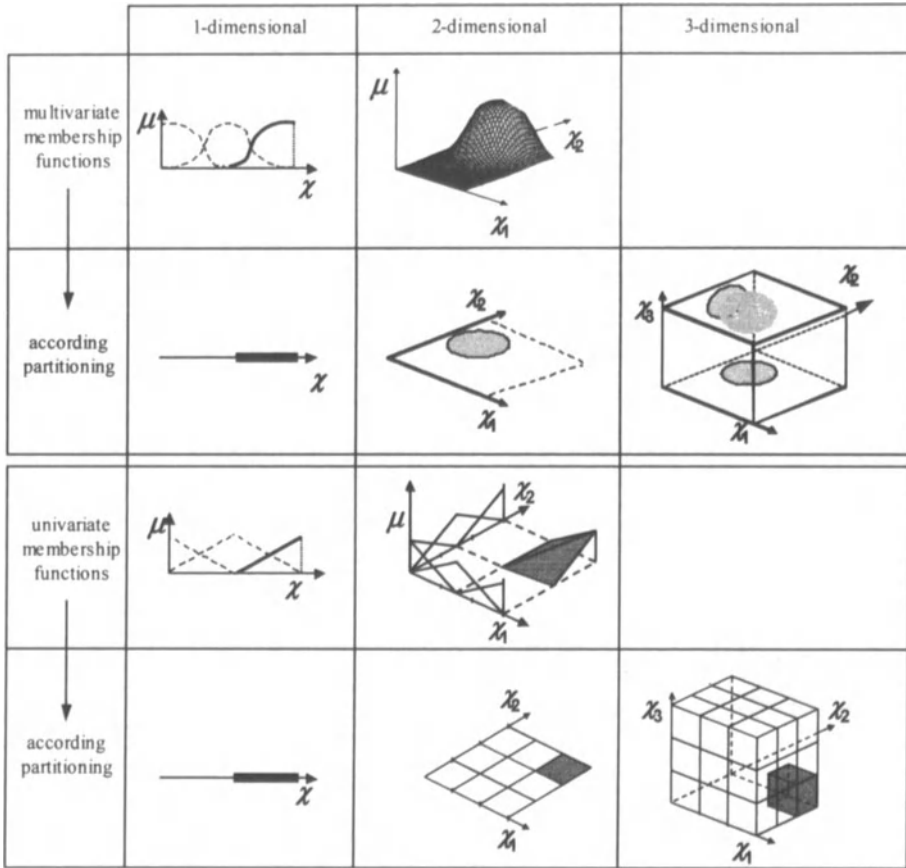
The membership function uses a vector norm  $\|\cdot\|_\nu$  in order to quantify the distances occurring. The distance norm should be chosen according to the structures that occur in the data.

Prominent norms are the  $L_p$  norm (Hartung and Elpelt, 1992):

$$\|\chi\|_p = \sqrt[p]{\sum_{j=1}^M |\chi_j|^p} \quad p \geq 1 \quad (5.77)$$

which results in structures of rhomb to sphere and then to rectangle. It includes the 1-norm for  $p = 1$ , the Euclidian norm for  $p = 2$ , and the maximum norm for  $p \rightarrow \infty$ ; see Figure 5.10.

Another norm frequently used is the inner product norm  $\|\chi\|_D^2 = \chi^T D \chi$  with the form matrix  $D$  (Bronstein and Semendjajew, 1991). It generates ellipsoidal iso-normal lines; see Figure 5.11.



**Figure 5.9.** Grid- and multivariate-type membership functions and partitioning (sketched crisp for simplicity in second and fourth rows) for an 1-, 2- or 3-dimensional features space

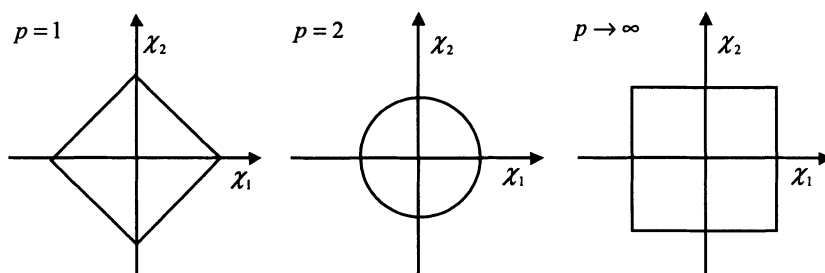
It includes, for example, the Euclidian norm for  $D = I$  and the Mahalonobis norm as special cases. The Mahalonobis norm is defined by its symmetrical form matrix

$$D = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,M} \\ \sigma_{2,1} & \sigma_2^2 & & \vdots \\ \vdots & & \ddots & \vdots \\ \sigma_{M,1} & \cdots & \cdots & \sigma_M^2 \end{bmatrix}^{-1} \quad (5.78)$$

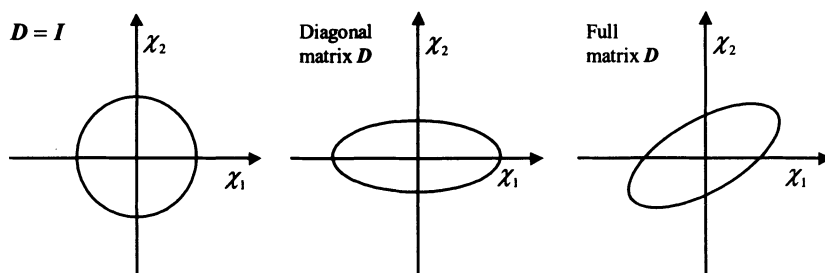
with the (co)variances

$$\sigma_{j,l} = \frac{1}{N} \sum_{k=1}^N (\chi_j(k) - \bar{\chi}_j)(\chi_l(k) - \bar{\chi}_l); \quad \sigma_j = \sigma_{j,j} \quad \forall j, l \in \{1; M\} \quad (5.79)$$

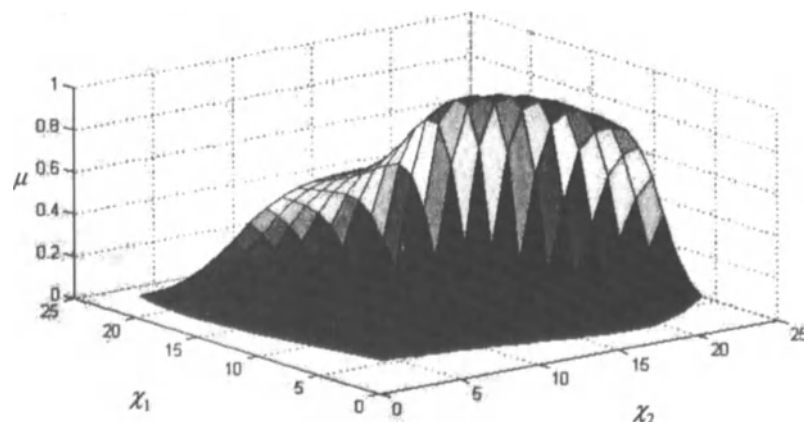
Figure 5.12 shows an example of a membership function, which employs the Mahalonobis norm to assess distances.



**Figure 5.10.** Iso-normal lines around the origin for a  $L_p$  norm and different realisations of  $p$



**Figure 5.11.** Iso-normal lines around the origin for an inner product norm and different realisations of the form matrix  $D$



**Figure 5.12.** Example of a membership function according to Equation 5.76 with a two-dimensional argument (four prototypes, Mahalonobis norm,  $\nu = 1.5$ )

**Conclusion (Consequent).** The conclusions contain dynamic local models  $f_i$ . They can be of any type, although in the following the local models are restricted to linear models with an additional offset term. The offset term typically permits a significant

improvement of the prediction quality (Kroll, 1997:75; Babuška, 1998:29). The local models are either of state-space or input–output type. An example for the latter is the following fuzzy rule:

$$\text{IF } (x(k) \text{ IS } \mathcal{X}) \text{ THEN} \\ \hat{y}_i(k+1) = [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)] \Theta_i + \xi_i \quad (5.80)$$

$y$  is the system output,  $u$  the system input,  $\xi$  a constant,  $\Theta$  the parameter vector, and  $i$  the number of the rule.

Another common restriction, which is also assumed here, is that all conclusions of a model have the same structure.

A model can be evaluated in two ways (see also Section 5.3.3 with Figure 5.6):

- serial–parallel, and
- parallel.

In the case of a serial–parallel evaluation, a prediction is made from measured input and measured historical output values. On the contrary, stored predicted values are used as model inputs for the historical output values in the case of a parallel evaluation. The latter also means that an incorrect prediction error is possibly amplified through the model. Parallel evaluation is important for long-range prediction purposes (as possibly in MPC) and within simulation applications; see also Henson and Seborg (1996). Figure 5.13 sketches the differences between these two evaluation modes (see also Figure 5.6).

Mathematically, for an SISO model the conclusion of model  $i$  can be written as:

$$\hat{y}_i(k+1) = \Theta_i^T \varphi(k) + \xi_i \quad (5.81)$$

with the regression vector

$$\varphi(k) = \varphi_{\text{SP,SISO}}(k) = [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T \quad (5.82)$$

for serial–parallel evaluation. For parallel evaluation the regression vector is given as

$$\varphi(k) = \varphi_{\text{P,SISO}}(k) = [\hat{y}(k), \dots, \hat{y}(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T \quad (5.83)$$

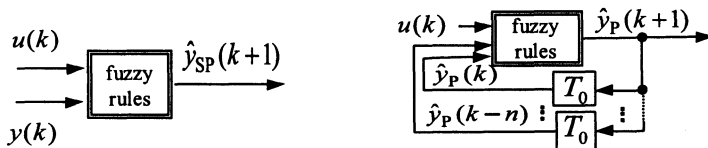


Figure 5.13. Serial–parallel (left) and parallel model evaluation (right)

The parameter vector is in both cases given as (the offset parameter  $\xi_i$  is kept separately):

$$\boldsymbol{\theta}_i^T = [-a_{1,i}, \dots, -a_{n,i}, b_{1,i}, \dots, b_{m,i}] \quad (5.84)$$

The extension to an MISO model with  $g$  inputs  $u_1, \dots, u_g$  for a serial-parallel evaluation is straightforward:

$$\begin{aligned} \boldsymbol{\varphi}(k) &= \boldsymbol{\varphi}_{\text{SP,MISO}}(k) \\ &= [y(k), \dots, y(k-n+1), u_1(k-\tau_1), \dots, u_1(k-\tau_1-m_1+1), \dots, \\ &\quad u_g(k-\tau_g), \dots, u_g(k-\tau_g-m_g+1)]^T \\ \boldsymbol{\theta}_i^T &= [-a_{1,i}, \dots, -a_{n,i}, b_{1,i,1}, \dots, b_{m,i,1}, \dots, b_{1,i,g}, \dots, b_{m,i,g}] \end{aligned} \quad (5.85)$$

The extension for parallel evaluation follows accordingly.

**Composing the Overall Model from the Set of Local Models.** The  $c$  local predictions, *i.e.*, the outputs of the  $c$  models/rules, are superposed fuzzily: each local prediction is weighted with its degree of fulfilment  $\mu_i$  and then summed up to form the global prediction. In the case of input-output-type models:

$$\hat{y}(k+1) = \frac{\sum_{i=1}^c \mu_i(k) \hat{y}_i(k+1)}{\sum_{j=1}^c \mu_j(k)} \quad (5.86)$$

is obtained. The introduction of fuzzy basis functions (Wang, 1994)

$$\beta_i(k) := \frac{\mu_i(k)}{\sum_{j=1}^c \mu_j(k)} \quad (5.87)$$

simplifies the writing to

$$\hat{y}(k+1) = \sum_{i=1}^c \beta_i(k) \hat{y}_i(k+1) \quad (5.88)$$

Note that  $\beta_i(k) \equiv \mu_i(k)$  is valid for probabilistic membership functions.

The mathematical description of the fuzzy model is summarised in Figure 5.14.

#### 5.4.2.4 Fuzzy Input-output vs. State-space Models

Up to now, fuzzy models with conclusions of input-output type have been considered. This form typically follows from identification. Typical utilisations include prediction applications such as simulation, forecasting, or as a prediction model within an input-output-type model predictive controller (MPC)<sup>\*</sup>. In contrast, fuzzy models with conclusions of the state-space type are advantageous if the model structure is exploited, as in system analysis and controller design.

Conclusions of the input-output type can be transformed into the state-space type. As in linear theory, many equivalent state-space models can be generated. In

<sup>\*</sup> State-space model based MPC recently get significant attention in academia and industry.

the following, a non-minimal realisation is introduced, as its handling is more transparent. The difficulty in the transformation lies in the handling of the constant  $\xi_i$  of each rule. As mentioned before, the offset term is considered, as it typically permits a significant improvement of the prediction quality. It can either be handled explicitly (parametrical approach) or as an additional input quantity (variable approach).

$$\begin{aligned}
 & \text{IF } (X(k) \text{ IS } \chi_i) \text{ THEN } \eta_i(k+1) = f_i(z(k), w(k)) \\
 & \mu_i(X) = \begin{cases} \frac{1}{\sum_{j=1}^c \left[ \frac{\|X - v_i\|}{\|X - v_j\|} \right]^{\frac{2}{\nu-1}}} & \text{for } X \in \{v_1, \dots, v_c\} \quad (\text{no singularities}) \\ 1 & \text{for } X = v_i \\ 0 & \text{for } X \in \{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_c\} \end{cases} \quad (\text{for singularities}) \\
 & \hat{y}_i(k+1) = \Theta_i^T \varphi(k) + \xi_i \\
 & \varphi_{\text{SP,SISO}}(k) = [y(k), \dots, y(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T \quad (\text{serial-parallel case}) \\
 & \varphi_{\text{P,SISO}}(k) = [\hat{y}(k), \dots, \hat{y}(k-n+1), u(k-\tau), \dots, u(k-\tau-m+1)]^T \quad (\text{parallel case}) \\
 & \Theta_i^T = [-a_{1,i}, \dots, -a_{n,i}, b_{1,i}, \dots, b_{m,i}] \\
 & \hat{y}(k+1) = \frac{\sum_{i=1}^c \mu_i(k) \hat{y}_i(k+1)}{\sum_{j=1}^c \mu_j(k)} = \sum_{i=1}^c \beta_i(k) \hat{y}_i(k+1)
 \end{aligned}$$

**Figure 5.14.** Summarised mathematical SISO fuzzy model description (input–output form)

An equivalent state-space model can be gained as follows. The  $c$  local models are given as:

$$\begin{aligned}
 \hat{x}_i(k+1) &= A_i x(k) + b_i u(k) + e_{n+m+\tau-1,1} \xi_i \\
 \hat{y}_i(k+1) &= c^T \hat{x}_i(k+1)
 \end{aligned} \tag{5.89}$$

in matrix–vector notation or as:



$$\begin{bmatrix} \hat{y}_i(k+1) \\ y(k) \\ \vdots \\ y(k-n+2) \\ \hline u(k) \\ \vdots \\ u(k-\tau-m+2) \end{bmatrix} = \begin{bmatrix} -a_{1,i} & -a_{2,i} & \cdots & -a_{n,i} & (0) & b_{1,i} & \cdots & b_{m,i} \\ 1 & 0 & \cdots & 0 & & & & \\ & \ddots & \ddots & 0 & & & & \\ & & & 1 & & & & \\ (0) & & & & & & & \\ \hline & & & & 0 & \cdots & \cdots & 0 \\ & & & & 1 & \ddots & & \vdots \\ & & & & & \ddots & \ddots & \vdots \\ (0) & & & & & & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} y(k) \\ y(k-1) \\ \vdots \\ \frac{y(k-n+1)}{u(k-1)} \\ \vdots \\ \vdots \\ u(k-\tau-m+1) \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \\ \vdots \\ \vdots \\ 0 \end{bmatrix} u(k) + \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix} \xi_i \quad (5.90)$$

$$\hat{y}_i(k+1) = [1 \ 0 \ \cdots \ 0] \hat{\mathbf{x}}_i(k+1) \quad (5.91)$$

The  $c$  local states can be composed to a global state, which is then used to predict the output signal:

$$\begin{aligned} \hat{\mathbf{x}}(k+1) &= \sum_{i=1}^c \beta_i(k) \hat{\mathbf{x}}_i(k+1) \\ \hat{y}(k+1) &= \mathbf{c}^T \hat{\mathbf{x}}(k+1) \end{aligned} \quad (5.92)$$

Alternatively, the  $c$  local output signals can be composed to form the global output prediction:

$$\hat{y}(k+1) = \sum_{i=1}^c \beta_i(k) \hat{y}_i(k+1) \quad (5.93)$$

The state-space representation will be used in Section 6.7.1 for developing a fuzzy state control algorithm.

### 5.4.3 Artificial Neural Networks

Artificial neural networks (or simply neural networks) have been widely used in many areas of signal processing. A particular network structure, namely the feedforward neural network, has been employed in non-linear systems identification (Chen *et al.*, 1990a,b, 1992; Nørgaard *et al.*, 2000), owing to their excellent approximation properties (Cybenko, 1989; Funahashi, 1989). In a feedforward network, the neurons are generally grouped into layers. The signal flow is from the input layer through hidden layers to the output layer via unidirectional connections.

The neurons are connected from one layer to the next, but not within the same layer. Prominent examples of feedforward networks include the multi-layer perceptron (MLP) network (Rumelhart *et al.*, 1986) and the radial basis function (RBF) network (Poggio and Girosi, 1989; Chen *et al.*, 1990a,b).

Extensive overviews of neural networks, ranging from the biological background to the most ANN architectures and learning algorithms to hardware implementations, can be found by Zell (1994) and Haykin (1999). The use of ANN, especially MLP and RBF networks, for the analysis of pattern recognition and classification problems is described by Bishop (1997), and Ripley (1997). A special book about neuro-fuzzy modelling and control has been written by Brown and Harris (1994).

Generally speaking, RBF networks are best suited for low- and medium-dimensional problems. Owing to the need for complex non-linear optimisation algorithms (which are complex and tedious) to estimate the parameters of MLP networks, these are generally more recommended for high-dimensional problems.

#### 5.4.3.1 Multi-layer Perceptron Networks

A neural network generally consists of a set of processing elements, the neurons, that are connected by *weights* (or *synapses*) into a special architecture. The basic model of a *neuron* (or *node* or *unit*) is depicted in Figure 5.15a. All output signals  $y_{l-1,i}$  of the preceding layer are weighted, summed up, and the result is processed by the activation function  $f(\cdot)$  to give the output  $y_{l,i}$  of the layer:

$$\hat{y}(k+1) = w_{0i}y_{l-1,0} + \sum_{j=1}^p y_{l-1,j}w_{ji} \quad (5.94)$$

with the weights  $w_{ji}$ . An offset is applied to the unit, represented by a constant signal  $y_{l-1,1} = 1$  and its weight  $w_{0i}$ .

The MLP is the most common type of neural network architecture, despite its problems due to slow weight convergence through non-linear learning (estimation). Figure 5.15b shows an MLP network with one hidden layer and an output layer. The input (layer) acts only as a buffer for distributing the input signals to the first hidden layer. A popular version of the activation function is the sigmoid function

$$\sigma(x) = \frac{1}{1 + \exp(-x)} \quad (5.95)$$

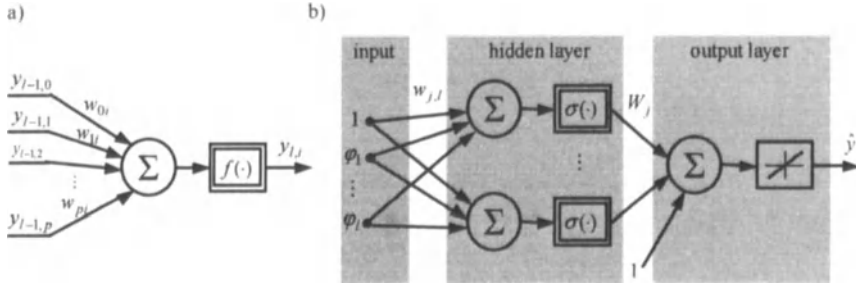
which gives a smooth model with the advantage that gradient-based parameter estimation methods can be used.

Thus, the mathematical formula expressing what is going on in the two-layer MLP network with sigmoid hidden units and linear output units reads as

$$\hat{y}(k+1) = g_{\text{MLP}}(\boldsymbol{\varphi}, \boldsymbol{\Theta}) = \sum_{j=1}^h W_j \sigma \left[ \sum_{l=1}^r w_{j,l} \varphi_l(k) + w_{j,0} \right] + W_0 \quad (5.96)$$

where the parameter vector  $\boldsymbol{\Theta}$  contains all the adjustable parameters of the network, *i.e.*, the weights and offsets  $\{w_{j,l}, W_j\}$ ,  $r$  is the number of input units, and  $h$  is the number of neurons in the hidden layer.

From Equation 5.96 (compared with Equation 5.16), it is clear that an MLP network is just a particular instance of the general model structure in Equation 5.16 with the sigmoid choice as the mother basis function (Ljung, 1999). Of course, any other activation function, such as the tangent hyperbolic function<sup>9</sup>  $\sigma(x) = \tanh(x)$ , can be used for MLP networks.



**Figure 5.15.** (a) Details of a neuron; (b) multi-layer perceptron with single hidden layer

#### 5.4.3.2 Radial Basis Function Networks

RBF networks are special feedforward networks, which are non-linear functions that are built up into one function that can partition the pattern space successfully. Figure 5.16 sketches the architecture of an RBF network.

Whereas the MLP network builds its classification from hyperplanes defined by the weighted sums as arguments to non-linear functions, an RBF network uses hyperellipsoids to partition the pattern space. These are defined by functions of the form  $\Psi(\|\phi - \mathbf{v}\|)$ , where  $\|\cdot\|$  denotes some distance metric (or norm). The output of the network is found from

$$\hat{y}(k+1) = g_{\text{RBF}}(\phi, \Theta) = \sum_{j=1}^h \lambda_j \Psi_j(\|\phi - \mathbf{v}_j\|) \quad (5.97)$$

where  $\mathbf{v}_j$  is the centre vector, which represents the position of the  $j$ th basis function. The hidden layer is fully connected to the output layer by weights  $\lambda_j$ .

It is standard, but not necessary, for the function  $\Psi$  to be a common function type, typically (Warwick *et al.*, 1995b)

$$(i) \quad \Psi(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (\text{Gaussian}) \quad (5.98)$$

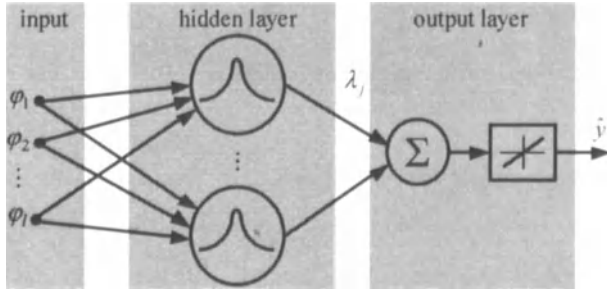
$$(ii) \quad \Psi(x) = x^2 \log(x) \quad (\text{thin plate spline}) \quad (5.99)$$

$$(iii) \quad \Psi(x) = (x^2 + \alpha^2)^{1/2} \quad (\text{multiquadratic}) \quad (5.100)$$

$$(iv) \quad \Psi(x) = (x^2 + \alpha^2)^{-1/2} \quad (\text{inverse multiquadratic}) \quad (5.101)$$

<sup>9</sup> It makes no difference which of the two functions is used since they are related to each other by a simple transformation.

where  $\sigma$  is a given standard deviation (width of the Gaussian function), and  $\alpha$  a scaling or “width” parameter.



**Figure 5.16.** RBF network

Of the possible selections, the Gaussian function is the most intuitive and thus widely used. The norm usually employed is the Euclidean distance governed by

$$\|\boldsymbol{\varphi} - \mathbf{v}_j\| = \sqrt{\sum_{l=1}^{n+m} (\varphi_l - v_{j,l})^2} \quad (5.102)$$

If, however, it is preferred to use different width for each input, then the output of the network can be described by

$$\hat{y}(k+1) = g_{\text{RBF}}(\boldsymbol{\varphi}, \boldsymbol{\Theta}) = \sum_{j=1}^h \lambda_j \exp \left[ -\frac{1}{2} \sum_{l=1}^r \frac{(\varphi_l - v_{j,l})^2}{\sigma_{j,l}} \right] \quad (5.103)$$

where  $\sigma_{j,l}$  is the standard deviation corresponding to the  $j$ th hidden unit and  $l$ th input.

Comparison of Equations 5.97 and 5.103 with Equation 5.16 shows that RBF networks are also just a particular instance of the general model structure with the Gaussian function (or any other functional form) as the mother basis function.

The RBF networks can be understood as a special case of the fuzzy models described in Section 5.4.2: the weight of each basis function can be understood as a constant local model. This represents a special case of the dynamic local models of Sugeno-type fuzzy models. The fuzzy model could use an RBF as membership function, which is sometimes the case, though other functions are preferred. The similarity is obvious from comparing Equations 5.93 and 5.97. The RBF  $\Psi_j$  corresponds to the fuzzy basis function  $\beta_j$ , and the weight  $\lambda_j$  to the prediction  $\hat{y}_j$  of the dynamic local model.

### 5.4.3.3 Representation of Dynamics in Artificial Neural Networks

If, now, the  $\varphi_r$  are chosen to be the  $n$  past outputs and the  $m$  past inputs (*i.e.*,  $r = n + m$ ), then a dynamic MLP or RBF network – more precisely an NNARX – is obtained, which is the acronym for neural network ARX. Similarly, all the regression vector types introduced in Section 5.3 can be used to construct NNFIR, NNARMAX, NNOE, and NNSSIF structures.

Now, a special, but very interesting model structure consisting of two neural networks (NN1:  $\hat{f}[\cdot]$  and NN2:  $\hat{g}[\cdot]$ ) is introduced (Nørgaard *et al.*, 2000):

$$\begin{aligned}\hat{y}(k) = & \hat{f}(y(k-1), \dots, y(k-n), u(k-2), \dots, u(k-m), \Theta_f) \\ & + \hat{g}(y(k-1), \dots, y(k-n), u(k-2), \dots, u(k-m), \Theta_g)u(k-1)\end{aligned}\quad (5.104)$$

or equivalently

$$\begin{aligned}\mathbf{x}(k+1) = & \begin{bmatrix} x_2(k) \\ x_3(k) \\ \vdots \\ \hat{f}(\mathbf{x}(k), u(k-1), \dots, u(k-m+1), \Theta_f) \end{bmatrix} \\ & + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \hat{g}(\mathbf{x}(k), u(k-1), \dots, u(k-m+1), \Theta_g) \end{bmatrix} u(k) \\ y(k) = & x_n(k)\end{aligned}\quad (5.105)$$

with the state vector being defined by

$$\mathbf{x}(k) = [y(k-n+1), \dots, y(k-1), y(k)]^T \quad (5.106)$$

The main motivation for this model structure is that it becomes easier to develop non-linear controllers in the framework of feedback linearisation using neural networks (see Sections 6.5.1 and 6.8.1).

Besides the activation or basis functions, the model structure of a neural network consists of the number of hidden layers and the number of units to be included in each layer; see Sections 5.9.1 and 5.9.3 for further details.

## 5.5 Parameter Estimation Methods

Assume now that suitable data sets have been acquired and that some model structures have been selected. According to the procedure in Figure 5.2, the next stage is then to apply the (identification) data set to pick the “best” model among the candidates contained in the model structure. This stage is referred to as model estimation, and it includes the structure identification and *model parameter estimation*.

Again, only selected methods are described based on the research work and experience of the authors of this book. For the sake of simplicity, the methods (except the maximum likelihood method) will be presented for SISO systems, but these can be extended without great modifications to MIMO systems.

### 5.5.1 Prediction Error Methods

In classical (*i.e.*, “Ljungian”) system identification theory, the general term *prediction-error identification methods* (PEM) is used for the family of approaches that correspond to the minimisation problem (Ljung, 1999)

$$\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\mathbf{Z}^N) = \arg \min_{\boldsymbol{\theta}} V(\boldsymbol{\theta}, \mathbf{Z}^N), \quad V(\boldsymbol{\theta}, \mathbf{Z}^N) \equiv \frac{1}{N} \sum_{k=1}^N \ell(\varepsilon(k, \boldsymbol{\theta})) \quad (5.107)$$

where  $\ell(\cdot)$  is a scalar-valued (typically positive) function of the prediction error

$$\varepsilon(k, \boldsymbol{\theta}) = y(k) - \hat{y}(k) \quad (5.108)$$

$\mathbf{Z}^N$  is a set of recorded inputs and outputs over the time interval  $1 \leq k \leq N$

$$\mathbf{Z}^N = \{u(1), y(1), \dots, u(N), y(N)\} \quad (5.109)$$

Depending on the choice of  $\ell(\cdot)$ , specific well-known parameter estimation methods are obtained as the special cases of Equation 5.107, such as *least-squares* (LS) method (as well as its extensions) and *maximum likelihood* (ML) method. Both methods are formulated in Sections 5.5.2 and 5.5.4, respectively.

### 5.5.2 Classical Least-squares Analysis

The linear (in the parameters) LS (also referred to as least-mean-squares) approach is the simplest type of parametric model estimator. This technique was originally developed independently by Gauss and Legendre in the early 19th century (Goodwin and Payne, 1977).

The identification problem is to find an estimate  $\hat{\boldsymbol{\theta}}$  of the parameter vector  $\boldsymbol{\theta}$  from (sampled) output data  $y(1), \dots, y(N)$  and appropriately selected regressors  $\varphi(1), \dots, \varphi(N)$ . Given these data, a system of linear equations (linear regression) is assumed, namely

$$\hat{y}(k) = \boldsymbol{\theta}^T \boldsymbol{\varphi}(k-1) \quad (5.110)$$

With this equation the prediction error becomes

$$\varepsilon(k) = y(k) - \boldsymbol{\theta}^T \boldsymbol{\varphi}(k-1) \quad (5.111)$$

which is, of course, linear in the parameter vector  $\boldsymbol{\theta}$ .

#### 5.5.2.1 Least-squares Estimate

The LS estimate of  $\boldsymbol{\theta}$  is defined as the vector  $\hat{\boldsymbol{\theta}}$  that minimises the loss function

$$V(\boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \varepsilon^2(k) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} [y(k) - \boldsymbol{\theta}^T \boldsymbol{\varphi}(k-1)]^2 \quad (5.112)$$

which corresponds to the PE minimisation problem in Equation 5.107 with the choice

$$\ell(\varepsilon) = \frac{1}{2} \varepsilon^2 \quad (5.113)$$

The (unique) solution of the optimisation problem stated above is given by

$$\hat{\boldsymbol{\theta}} = \left[ \frac{1}{N} \sum_{k=1}^N \boldsymbol{\varphi}(k-1) \boldsymbol{\varphi}^T(k-1) \right]^{-1} \left[ \frac{1}{N} \sum_{k=1}^N \boldsymbol{\varphi}(k-1) y(k) \right] \quad (5.114)$$

The estimated parameters  $\boldsymbol{\theta}$  are optimal in the LS sense, though often have a bias, depending on the disturbance properties (Isermann, 1992; Reuter, 1995; Ljung, 1999). More specifically, the LS estimates are unbiased if the noise is white. In the case that the process does not meet this (unrealistic) assumption, the parameters will be *biased* and *non-consistent*. Note that a bias means that the parameters systematically deviate from their optimal values, and non-consistency implies that this bias does not even approach zero as the number of data points  $N$  goes to infinity.

Note also that some strategies, such as the instrumental variables (IV) method, are available that avoid the non-consistency of the LS method. For details about the IV method, the optimal choice of the instrumental variables, and its relationship to the PE method, the reader is referred to Söderström and Stoica (1989) and Ljung (1999).

### 5.5.2.2 Matrix Formulation

Equations 5.110–5.114 can be written more conveniently in matrix form. Define the  $N \times 1$  column vector  $Y$  and  $N \times M$  regression matrix

$$Y = \begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}, \quad \Phi = \begin{bmatrix} \boldsymbol{\varphi}^T(1) \\ \vdots \\ \boldsymbol{\varphi}^T(N) \end{bmatrix} \quad (5.115)$$

Then the criterion in Equation 5.112 can be expressed as

$$V(\boldsymbol{\theta}) = \frac{1}{2N} (Y - \Phi \boldsymbol{\theta})^T (Y - \Phi \boldsymbol{\theta}) \quad (5.116)$$

and the estimate becomes

$$\hat{\boldsymbol{\theta}} = (\Phi^T \Phi)^{-1} \Phi^T Y \quad (5.117)$$

Note that Equation 5.117 gives the pseudo-inverse solution to the over-determined ( $N > M$ ) system of linear equations

$$Y = \Phi \boldsymbol{\theta} \quad (5.118)$$

### 5.5.2.3 Weighted Least-Squares Estimate

The performance criterion of Equation 5.112 is essentially based on the view that all the errors are given equal importance. This is not always the case, and a very simple addition can take known variations of the significance of the PE in each sample into account. The weighting can be used to cut out gross errors. It is also sometimes used for the identification of local models in case of fuzzy identification: a data set has generally a different membership to each local model, which can be reflected in the according weight during identification. Such a scheme is referred to as *weighted LS* and is based on the performance measure

$$V(\boldsymbol{\theta}) = \sum_{k=1}^N w(k) \varepsilon^2(k) = (\mathbf{Y} - \boldsymbol{\Phi}\boldsymbol{\theta})^T \mathbf{W} (\mathbf{Y} - \boldsymbol{\Phi}\boldsymbol{\theta}) \quad (5.119)$$

In this equation, the weighting function  $w(k)$  has to be positive and, presumably, to be small where the errors are expected to be large, and *vice versa*. Subject to the coefficient matrix being diagonal and non-singular, the estimate becomes

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\boldsymbol{\Phi}^T \mathbf{W} \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{W} \mathbf{Y} \\ &= \left[ \sum_{k=1}^N w(k) \boldsymbol{\varphi}(k-1) \boldsymbol{\varphi}^T(k-1) \right]^{-1} \left[ \sum_{k=1}^N w(k) \boldsymbol{\varphi}(k-1) y(k) \right] \end{aligned} \quad (5.120)$$

Equation 5.120 reduces to ordinary LS for  $\mathbf{W} = \mathbf{I}$ , the identity matrix.

A common choice for  $w(k)$  is  $w(k) = (1 - \gamma) \gamma^{N-k}$  for  $\gamma < 1$ . This choice weights more recent ( $k$  near  $N$ ) observations more than past ( $k$  near unity) observations and corresponds to a first-order filter operating on the squared error. The factor  $1 - \gamma$  causes the gain of the equivalent filter to be unity for constant errors. For a  $\gamma$  near to unity, the filter memory becomes long, and noise effects are reduced. For smaller  $\gamma$ , the memory is short, and the estimate can track changes that can occur in  $\boldsymbol{\theta}$  if the computation is done over and over again, as  $N$  increases – a process called *recursive LS*. The past is weighted geometrically with weighting  $\gamma$ , but a rough estimate of the memory length is given by  $1/(1 - \gamma)$ ; so, for example, a  $\gamma = 0.99$  corresponds to a memory of about 100 samples (Franklin *et al.*, 1990).

### 5.5.2.4 Recursive Least-squares Estimate

Alternatively, the recursive variant of the algorithm can be used, which is useful for online applications (as in adaptive control):

$$\hat{\boldsymbol{\theta}}(k) = \hat{\boldsymbol{\theta}}(k-1) + \mathbf{L}(k) \left[ y(k) - \hat{\boldsymbol{\theta}}^T(k-1) \boldsymbol{\varphi}(k-1) \right] \quad (5.121)$$

with

$$\mathbf{L}(k) = \frac{\mathbf{P}(k-1) \boldsymbol{\varphi}(k-1)}{\lambda(k) + \boldsymbol{\varphi}^T(k-1) \mathbf{P}(k-1) \boldsymbol{\varphi}(k-1)} \quad (5.122)$$

and



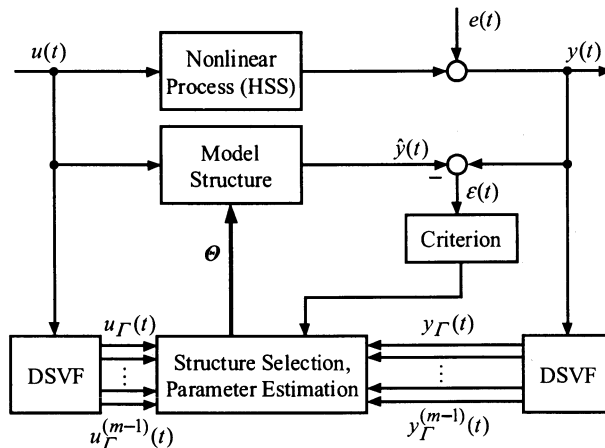
$$\begin{aligned}
 \mathbf{P}(k) &= \frac{1}{\lambda(k)} \left[ \mathbf{P}(k-1) - \frac{\mathbf{P}(k-1)\boldsymbol{\varphi}(k-1)\boldsymbol{\varphi}^T(k-1)\mathbf{P}(k-1)}{\lambda(k) + \boldsymbol{\varphi}^T(k-1)\mathbf{P}(k-1)\boldsymbol{\varphi}(k-1)} \right] \\
 &= \frac{1}{\lambda(k)} \left[ \mathbf{I} - \mathbf{L}(k)\boldsymbol{\varphi}^T(k-1) \right] \mathbf{P}(k-1)
 \end{aligned} \tag{5.123}$$

Lacking further knowledge, the covariance matrix  $\mathbf{P}(0)$  is typically chosen as  $\mathbf{P}(0) = k_{LS}\mathbf{I}$  with  $k_{LS} = 100, \dots, 10000$  and  $\boldsymbol{\Theta} = \mathbf{0}$  (Wernstedt, 1989; Isermann, 1992; Reuter, 1995). The most common choice of forgetting factor  $\lambda(k)$  is to take a constant factor  $\lambda$ , that is slightly less than unity, typically in the range between 0.98 and 0.995 (Ljung, 1999).

### 5.5.3 Orthogonal Least-squares Estimator

The continuous-time identification scheme used here to estimate the parameters of model structures introduced in Section 5.4.1 is shown in Figure 5.17. For handling time derivatives of measurements, delayed state variable filters (DSVF) as a two-stage second-order all-pass filter followed by a Butterworth filter are included. The delayed filtered signals, *i.e.*, inputs and outputs, and associated higher order derivatives collected from the DSVFs are used for the identification of the unknown model parameters applying an orthogonal LS estimator.

The whole procedure has been proposed by Tsang and Billings (1994), where a detailed description of the algorithm and its properties is given. In the following sections, only the main points and results are briefly quoted.



**Figure 5.17.** Scheme of the identification of SISO non-linear models using an orthogonal LS method combined with DSVFs (Taken from Jelali (1997), p 40 (original publication). Reprinted with kind permission of VDI Verlag)

#### 5.5.3.1 Delayed State Variable Filters

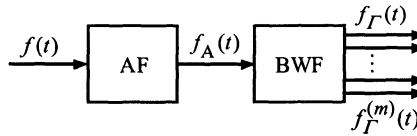
The DSVF consists of a two-stage second-order all-pass filter

$$G_{Al}(s) = \prod_{i=1}^{\nu_1} \left( \frac{s - a_i}{s + a_i} \right) \prod_{k=1}^{\nu_2} \frac{(s - b_k)^2 + c_k^2}{(s + b_k)^2 - c_k^2} \quad (5.124)$$

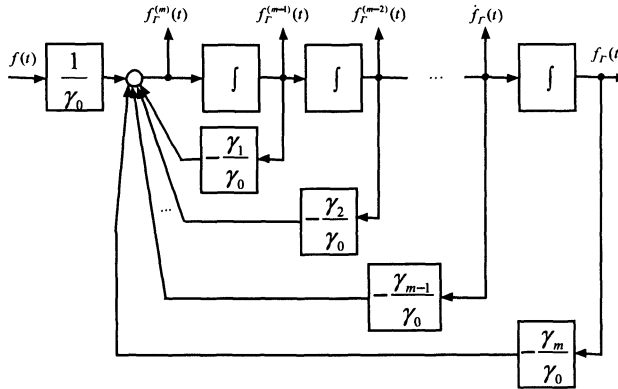
followed by a low-pass Butterworth filter (see Figure 5.18). The block diagram of a Butterworth filter as an SVF is shown in Figure 5.19. Estimates of the all-pass filter parameters  $a_1$ ,  $b_1$ ,  $a_2$ , and  $b_2$  are listed in Table 5.1 for  $m = 2-8$  (order of the Butterworth filter).

The DSVF can be implemented digitally. For this, the transfer functions of both components (all-pass filter and Butterworth filter) are transformed in the state-space form. Firstly,  $G_A(s)$  in Equation 5.124 is rewritten as

$$G_{Al}(s) = \frac{\sum_{i=1}^{\nu} (-1)^i \alpha_i (Ts)^i}{\sum_{i=1}^{\nu} \alpha_i (Ts)^i} = \frac{\sum_{i=0}^{\nu} \beta_i s^i}{\sum_{i=0}^{\nu} \rho_i s^i + s^{\nu}} \quad (5.125)$$



**Figure 5.18.** Cascaded all-pass filter (AF) and Butterworth filter (BWF)



**Figure 5.19.** Block diagram of a state variable filter (Taken from Jelali (1997), p 40 (original publication). Reprinted with kind permission of VDI Verlag)

The corresponding controller canonical form is given by

$$\begin{aligned} \dot{\mathbf{x}}_A(t) &= \mathbf{A}_A \mathbf{x}_A(t) + \mathbf{b}_A f(t) \\ f_A(t) &= \mathbf{c}_A^T \mathbf{x}_A(t) + \beta_v f(t) \end{aligned} \quad (5.126)$$

where

$$A_A = \begin{bmatrix} 0 & & & \\ \vdots & & I_{v-1} & \\ 0 & & & \\ -\rho_0 & -\rho_1 & \cdots & -\rho_{v-1} \end{bmatrix} \quad b_A = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad c_A^T = \begin{bmatrix} \beta_0 - \beta_v \rho_0 \\ \beta_1 - \beta_v \rho_1 \\ \vdots \\ \beta_{v-1} - \beta_v \rho_{v-1} \end{bmatrix} \quad (5.127)$$

**Table 5.1.** Parameters of the two-stage second-order all-pass filter (Tsang and Billings, 1994)

| $m$ | $a_1$  | $b_1$  | $a_2$  | $b_2$  |
|-----|--------|--------|--------|--------|
| 2   | 0.9211 | 0.2725 | 0.5863 | 1.2337 |
| 3   | 1.1995 | 0.2970 | 2.4141 | 5.1284 |
| 4   | 0.7302 | 0.3971 | 0.4705 | 1.9801 |
| 5   | 0.7353 | 0.5472 | 1.1936 | 0      |
| 6   | 0.5809 | 0.6250 | 0.8370 | 0      |
| 7   | 0.4807 | 0.6692 | 0.5710 | 0.2170 |
| 8   | 0.4104 | 0.6741 | 0.4646 | 0.2240 |

The state-space model of the Butterworth filter is given as (see Figure 5.19)

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{b}f_A(t)$$

$$f_r(t) = \begin{bmatrix} f_r(t) \\ \vdots \\ f_r^{(m-1)}(t) \\ f_r^{(m)}(t) \end{bmatrix} = \mathbf{C} \begin{bmatrix} \mathbf{x}(t) \\ f_A(t) \end{bmatrix} \quad (5.128)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & & & \\ \vdots & & I_{m-1} & \\ 0 & & & \\ -\frac{\gamma_m}{\gamma_0} & -\frac{\gamma_{m-1}}{\gamma_0} & \cdots & -\frac{\gamma_1}{\gamma_0} \end{bmatrix}$$

$$\mathbf{b}_A = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \frac{1}{\gamma_0} \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} & & & 0 \\ & & I_{m-1} & \vdots \\ & & & 0 \\ -\frac{\gamma_m}{\gamma_0} & \cdots & -\frac{\gamma_1}{\gamma_0} & 1 \end{bmatrix} \quad (5.129)$$

The coefficients  $\gamma_i$ ,  $i = 1, 2, \dots, m$ , of the Butterworth filter have to be chosen such that the corresponding poles are given by

$$P_i = -\sin\left[(2i-1)\frac{\pi}{2m}\right] + j\cos\left[(2i-1)\frac{\pi}{2m}\right] \quad (5.130)$$

The SVF equations can be implemented using, *e.g.*, the Runge–Kutta method.

The application of the DSVF to Equation 5.51 yields

$$\begin{aligned}\hat{y}_r(t) &= -\sum_{i=2}^{n+1} \alpha_i y_r^{(i-1)}(t) - {}^t P_r [\cdot] + \sum_{j=1}^n \beta_j u_r^{(j-1)}(t) + \varepsilon(t) \\ &= \sum_{i=1}^M \Theta_i \varphi_{i,r}(t) + \varepsilon(t) \equiv \Theta^T \boldsymbol{\varphi}_r(t) + \varepsilon(t)\end{aligned}\quad (5.131)$$

where  $\varepsilon(t)$  is the modelling error,  $\Theta_i$  represent the unknown parameters, and  $\varphi_{i,r}$  are the delayed filtered model terms.

### 5.5.3.2 Basic Algorithm

The orthogonal LS estimator transforms Equation 5.131 into an equivalent orthogonal equation of the form

$$\hat{y}_r(t) = \sum_{i=1}^M p_i w_i(t) + e(t) \quad (5.132)$$

The orthogonalisation can be performed using the Gram–Schmidt method, which yields (Billings *et al.*, 1988; Korenberg *et al.*, 1988; Tsang and Billings, 1994)

$$w_1(t) = \varphi_{1,r}(t) \quad (5.133)$$

$$w_i(t) = \varphi_{i,r}(t) - \sum_{r=1}^{i-1} \eta_{ri} w_r(t), \quad i = 2, 3, \dots, M \quad (5.134)$$

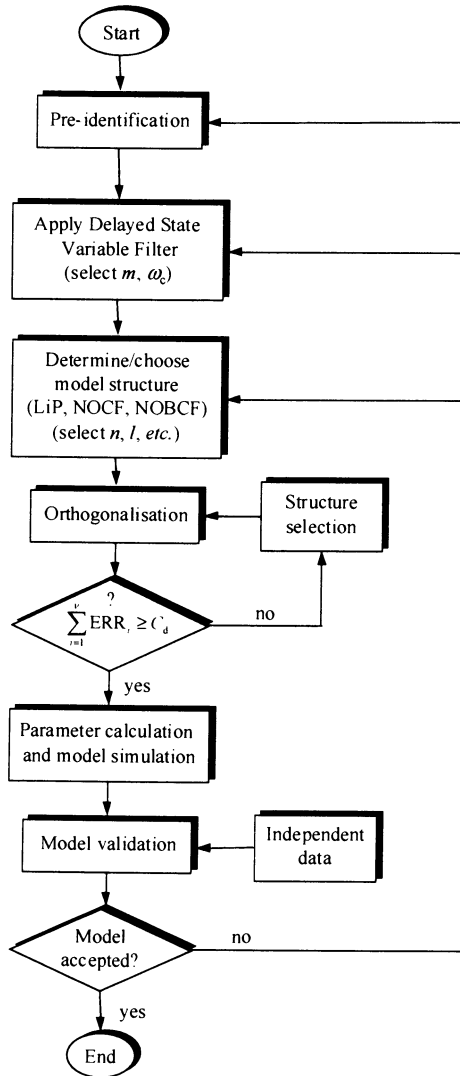
$$\eta_{ri} = \frac{\frac{1}{N} \sum_{j=1}^N w_r(t_j) \varphi_{i,r}(t_j)}{\frac{1}{N} \sum_{j=1}^N w_r^2(t_j)} \quad \begin{array}{l} r = 1, 2, \dots, i-1 \\ i = 2, 3, \dots, M \end{array} \quad (5.135)$$

The estimates of the parameters in Equation 5.132 are then given by

$$\hat{p}_i = \frac{\frac{1}{N} \sum_{j=1}^N y_r(t_j) w_i(t_j)}{\frac{1}{N} \sum_{j=1}^N w_i^2(t_j)} \quad i = 1, 2, \dots, M \quad (5.136)$$

The estimation of the parameters of the orthogonalised model is easy and independent from each other. Thus, entering a new model component does not alter the values of the previously estimated parameters. The estimates of the original model parameters  $\Theta_i$  can be computed as

$$\begin{aligned}\hat{\Theta}_M &= \hat{p}_M, \\ \hat{\Theta}_i &= \hat{p}_i - \sum_{j=i+1}^M \eta_{ij} \hat{\Theta}_j, \quad i = M-1, M-2, \dots, 1\end{aligned}\quad (5.137)$$



**Figure 5.20.** Flow chart of the identification procedure with automatic structure selection (Taken from Jelali (1997), p 58 (original publication). Reprinted with kind permission of VDI Verlag)

### 5.5.3.3 Automatic Structure Selection

Because the error reduction ratio (ERR)

$$ERR_i \equiv \frac{\frac{1}{N} \sum_{j=1}^N \hat{p}_i^2 w_i^2(t_j)}{\frac{1}{N} \sum_{j=1}^N y_i^2(t_j)} \times 100, \quad i = 1, \dots, M \quad (5.138)$$

determines the relative contribution of each individual orthogonal data set  $w_i$ , the sum of the ERRs can be used as a structure selection tool (Tsang and Billings, 1994).

If the sum of the ERRs is close to 100, the estimated model should be an adequate representation of the system under consideration. The orthogonalisation has to be terminated if the sum of the ERRs of the selected terms reaches the threshold  $C_d$  or the ERRs of the remaining candidate terms are less than  $C_e$ , *i.e.*, the remaining candidate terms do not show any improved performance.

In practice,  $C_d$  and  $C_e$  are chosen in the ranges  $90\% \leq C_d \leq 99.99\%$  and  $0.001\% \leq C_e \leq 0.1\%$ . The flow chart of the identification procedure is given in Figure 5.20. This procedure has been used as basis for the CONTIDNS Toolkit (Jelali and Spielmann, 1996; Jelali, 1997); see also Section 5.11.

### 5.5.4 Maximum Likelihood Method

The maximum likelihood estimator (Åström and Eykhoff, 1971; Goodwin and Payne, 1977) represents one prominent method to identify *general non-linear* (physically based) models (even in state-space form), and is thus described in the following. The derivation of the method is adapted from Goodwin and Payne (1977). Refer also to Söderström and Stoica (1989) and Ljung (1999).

The task of the maximum likelihood identification is to find an estimate  $\hat{\boldsymbol{\theta}}$  of the parameter vector  $\boldsymbol{\theta}$  such that

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} p_{\text{LF}}(\mathbf{Z}^N, \boldsymbol{\theta}) \quad (5.139)$$

where  $p_{\text{LF}}(\cdot)$  denotes the probability density of the prediction errors, and is known as the likelihood function on the parameters:

$$p_{\text{LF}}(\mathbf{Z}^N, \boldsymbol{\theta}) \equiv \left[ (2\pi)^p \det \mathbf{R} \right]^{-N/2} \exp \left( -\frac{1}{2} \sum_{k=1}^N \boldsymbol{\varepsilon}^T(k, \boldsymbol{\theta}) \mathbf{R}^{-1} \boldsymbol{\varepsilon}(k, \boldsymbol{\theta}) \right) \quad (5.140)$$

Since the minimum, not the maximum, of the loss function is sought, the negative logarithm of the likelihood function (referred to as the maximum likelihood loss function) is utilised:

$$V(\boldsymbol{\theta}, \mathbf{R}) = \frac{N}{2} \sum_{k=1}^N \boldsymbol{\varepsilon}^T(k, \boldsymbol{\theta}) \mathbf{R}^{-1} \boldsymbol{\varepsilon}(k, \boldsymbol{\theta}) + \frac{N}{2} \ln \det \mathbf{R} + \frac{N}{2} p \ln(2\pi) \quad (5.141)$$

The prediction errors are assumed to be Gaussian with zero mean values and covariance matrix  $\mathbf{R}$ . In the case where  $\mathbf{R}$  is given, the last two terms in Equation 5.141 are constant (which do not affect the minimisation), and the optimisation problem is equivalent to the minimisation of the loss function

$$V(\boldsymbol{\theta}, \mathbf{R}) = \frac{1}{2N} \sum_{k=1}^N \boldsymbol{\varepsilon}^T(k, \boldsymbol{\theta}) \mathbf{R}^{-1} \boldsymbol{\varepsilon}(k, \boldsymbol{\theta}) \quad (5.142)$$

With this, we have actually been led to the LS criterion (a special case of PE criterion) for the multivariable case (see Ljung, 1999:206).

For the case where  $\mathbf{R}$  has to be estimated, straightforward differentiation of Equation 5.141 gives

$$\frac{\partial V(\boldsymbol{\Theta}, \mathbf{R})}{\partial \mathbf{R}} = \frac{N}{2} \mathbf{R}^{-1} - \frac{1}{2} \mathbf{R}^{-1} \left( \sum_{k=1}^N \boldsymbol{\varepsilon}(k, \boldsymbol{\Theta}) \boldsymbol{\varepsilon}^T(k, \boldsymbol{\Theta}) \right) \mathbf{R}^{-1} \quad (5.143)$$

Setting  $\partial V / \partial \mathbf{R}$  equal to zero leads to the estimate  $\hat{\mathbf{R}}$  of the covariance matrix  $\mathbf{R}$

$$\hat{\mathbf{R}}(\boldsymbol{\Theta}) = \frac{1}{N} \sum_{k=1}^N \boldsymbol{\varepsilon}(k, \boldsymbol{\Theta}) \boldsymbol{\varepsilon}^T(k, \boldsymbol{\Theta}) \quad (5.144)$$

Substituting Equation 5.144 into 5.141 yields

$$V(\boldsymbol{\Theta}) = \frac{N}{2} \ln \det \hat{\mathbf{R}}(\boldsymbol{\Theta}) + \frac{N}{2} p \ln(2\pi + 1) \quad (5.145)$$

The minimisation of the loss function, Equation 5.145, which is also equivalent to the minimisation of

$$V(\boldsymbol{\Theta}) = \frac{N}{2} \ln \det \hat{\mathbf{R}}(\boldsymbol{\Theta}) \quad (5.146)$$

leads to the maximum likelihood estimation  $\hat{\boldsymbol{\Theta}}$  of the parameter vector. In this particular case, we may simply use

$$V(\boldsymbol{\Theta}) = \det \left[ \frac{1}{N} \sum_{k=1}^N \boldsymbol{\varepsilon}(k, \boldsymbol{\Theta}) \boldsymbol{\varepsilon}^T(k, \boldsymbol{\Theta}) \right] \quad (5.147)$$

which again corresponds to the PE criterion for the multivariable case.

From the above analysis, it can be concluded that the ML method becomes a special case of the prediction error method corresponding to the loss function of Equation 5.147 (under the assumption that the prediction errors are Gaussian with zero mean values).

### 5.5.5 Bias/Variance Dilemma and Regularisation Concepts

It is well-known that the expected value of the prediction error can be split into a bias term and a variance term, *i.e.*, (Nelles, 2001)

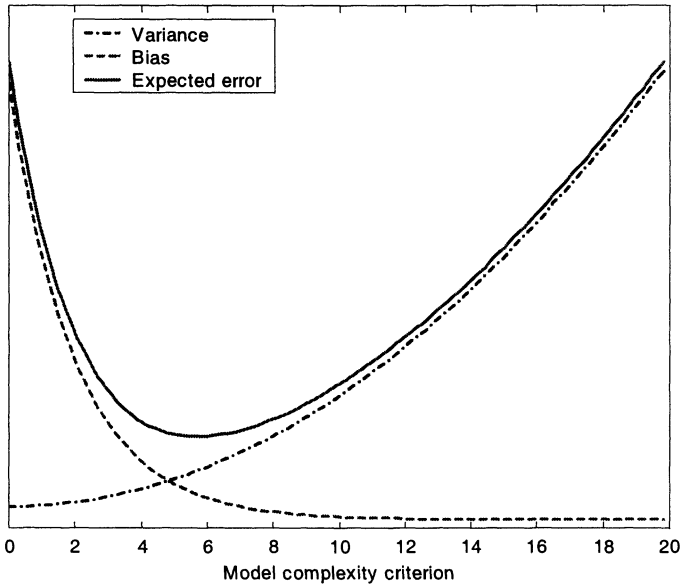
$$\begin{aligned} E\{(y_u - \hat{y})^2\} &= [y_u - E\{\hat{y}\}]^2 + E\{[\hat{y} - E\{\hat{y}\}]^2\} \\ (\text{model error})^2 &= (\text{bias error})^2 + \text{variance error} \end{aligned} \quad (5.148)$$

$y_u$  is the true system output, which is usually disturbed by the noise  $e$ , resulting in the measurable output  $y$ . See Sjöberg (1995), Ljung (1999), Nørgaard *et al.* (2000), Nelles (2001) for deeper discussions.

The bias term is due to the model structure not being sufficiently general to represent the underlying system, and the variance term is due to the variability in the estimated parameters, *i.e.*, the optimal parameters for the structure cannot be identified on the basis of a finite data set. Thus, with a more general model structure

the bias term will be reduced, but only at the expense of an increased variance term (as a more complicated model structure involves more parameters and hence the variability of the parameter estimates will increase); see Figure 5.21. The challenge, therefore, is to balance the terms against each other, and this is known as the *bias/variance dilemma*.

Related to bias/variance are the expressions over- and under-fitting. *Over-fitting* means that the model not only represents the features of the system, but to an undesired extent also adapts to the noise in the identification data set. Likewise, the term *under-fitting* is used when the bias error is dominating. It must be stressed that over-fitting can only be observed in test data, but not in identification data. This makes over-fitting very critical, since the user has to look at the identification error and the test error simultaneously.



**Figure 5.21.** Typical relationship between bias, variance and model structure complexity

There are a number of concepts to deal with the bias/variance dilemma:

1. **Explicit Regularisation.** Adding a *regularisation* term to the cost function  $V$  (Equation 5.107 or 5.112)

$$V_{\text{MSE,reg}}(\boldsymbol{\theta}, \mathbf{Z}^N) = \frac{1}{2N} \sum_{k=1}^N [y(k) - \hat{y}(k)]^2 + \frac{\delta}{2N} \|\boldsymbol{\theta} - \boldsymbol{\theta}^\# \|^2 \quad (5.149)$$

where  $\delta$  is referred to as the regularisation parameter (known as weight decay in the ANN context).  $\boldsymbol{\theta}^\#$  denotes some nominal fixed point in the parameter space, and is often taken as the origin, *i.e.*,  $\boldsymbol{\theta}^\# = \mathbf{0}$  (Ljung, 1999). The regularisation parameter  $\delta$  serves as a tuning knob, essentially to determine which model parameters are effective. In other words, the benefit of regularisation is that



*superfluous* parameters no longer have a bad influence on the model performance (Sjöberg and Ljung, 1995). For this, the fact that each eigenvalue in the Hessian corresponds to one parameter is exploited, and each parameter with an eigenvalue larger than  $\delta$  is considered to be superfluous. Moreover, regularisation has a smoothing effect on the criterion, and it significantly reduces the number of local minima. The challenge is then to select  $\delta$  such that the average test error is minimised. The most reliable method for choosing the optimal regularisation parameter is typically through a trial-and-error experiment with a representative validation data set.

2. **Implicit Regularisation.** Terminating the parameter estimation algorithm before the true minimum of the criterion has been found. It is shown by Sjöberg and Ljung (1995) that an “unfinished” search for the minimisation of a criterion function has the same effects as (explicit) regularisation towards the nominal value  $\Theta^\#$ , and thus is termed “implicit regularisation”. However, it is usually preferred to apply the explicit regularisation technique over early stopping, as the derivation of many important quantities related to optimal model structure selection and model validation (in Sections 5.9 and 5.10) assumes that the minimum of the cost function is achieved (Nørgaard *et al.*, 2000). Another point is that early stopping should only be used if the model structure can be assumed to be flexible enough (*i.e.* over-parameterised). This is the reason why poorly converging optimisation, such as steepest descents, can still work reasonably well in connection with large neural networks and early termination (Nelles, 2001). Otherwise, *i.e.*, if the model complexity is close to the optimal one, no over-fitting will occur, and the convergence of all parameters is strongly recommended.
3. **Search of Optimal Model Structure.** The aim is to find a model structure  $M$  (within a model structure set) that allows the network to best minimise a given cost function  $J$ , *i.e.*,

$$\hat{M} = \hat{M}(Z^N) = \arg \min_M J(M, Z^N) \quad (5.150)$$

Such procedures for optimisation of the network structure are important, but they result in difficult non-convex optimisation problems. Several criteria, such as the statistical criteria FPE and AIC, have been proposed in system identification literature to serve as a cost function  $J$  in Equation 5.150. These criteria usually contain two terms: one involving the prediction errors computed on the identification data set (bias) and a term involving the number of parameters in the model (variance). The overall goal is to determine a model with *best bias/variance trade-off*. See Section 5.9.1 and 5.9.3 for further details.

Common strategies for model complexity optimisation can be split up into two main categories: *forward selection* (or growing in the ANN context) algorithms and *backward selection* (or pruning in the ANN context) algorithms; see also Section 5.9.3. Forward selection starts with a very simple model, and increases its complexity in each iteration until no performance improvement can be achieved. The prominent approach for forward selection is the orthogonal least-squares (OLS) algorithm, as explained in Section 5.5.3. Backward selection works in the opposite direction to forward selection.

## 5.6 Optimisation Algorithms

In most cases the minimum of  $V(\boldsymbol{\theta})$  cannot be found analytically. For such cases the minimisation must be performed using numerical (iterative) search routines. This is called non-linear optimisation, and classical treatment of the problem is to minimise the sum of squares (*non-linear LS problem*)

$$\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\mathbf{Z}^N) = \arg \min_{\boldsymbol{\theta}} V(\boldsymbol{\theta}, \mathbf{Z}^N), \quad V(\boldsymbol{\theta}, \mathbf{Z}^N) \equiv \frac{1}{2N} \sum_{k=1}^N \varepsilon^2(k, \boldsymbol{\theta}) \quad (5.151)$$

given in many standard textbooks (Hoffmann and Hofmann, 1971; Fletcher, 1987; Dennis and Schnabel, 1996). See also Walter and Pronzato (1997). The compact representation and notation given in this section generally follows Dennis and Schnabel (1996).

### 5.6.1 Newton's Method

Newton's method applied to Equation 5.151 means<sup>10</sup>

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \mathbf{H}^{-1}(\boldsymbol{\theta}_k) \mathbf{G}(\boldsymbol{\theta}_k) \quad (5.152)$$

where  $\boldsymbol{\theta}_k$  denotes the (current)  $k$ th iteration point in the search. Rather than using this formula, which requires inverting a matrix, it is more economical to solve the equivalent set of linear equations

$$\mathbf{H}(\boldsymbol{\theta}_k) \boldsymbol{\rho}_k = -\mathbf{G}(\boldsymbol{\theta}_k) \quad (5.153)$$

for the search direction  $\boldsymbol{\rho}_k$ . The gradient  $\mathbf{G}$  and the Hessian (second-order derivative matrix)  $\mathbf{H}$  are defined by

$$\mathbf{G}(\boldsymbol{\theta}) = \mathbf{V}'(\boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^N \psi(k, \boldsymbol{\theta}) \boldsymbol{\varepsilon}(k, \boldsymbol{\theta}) \quad (5.154)$$

$$\mathbf{H}(\boldsymbol{\theta}) = \mathbf{V}''(\boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^N \psi(k, \boldsymbol{\theta}) \boldsymbol{\psi}^T(k, \boldsymbol{\theta}) - \frac{1}{N} \sum_{k=1}^N \psi'(k, \boldsymbol{\theta}) \boldsymbol{\varepsilon}(k, \boldsymbol{\theta}) \quad (5.155)$$

respectively, with the notation of gradient

$$\psi(k, \boldsymbol{\theta}) \equiv \frac{d\hat{y}(k | \boldsymbol{\theta})}{d\boldsymbol{\theta}} \quad (5.156)$$

and  $\psi'$  the  $(M \times M)$  Hessian of  $\varepsilon$ .

The full Newton approach (Equations 5.152, 5.154 and 5.155) is a fast local (quadratically convergent) method for the non-linear LS problem under standard assumptions. A problem with it, however, is that the second term of the Hessian is usually either unavailable or inconvenient to obtain, and it is too expensive to approximate by finite differences (Dennis and Schnabel, 1996).

<sup>10</sup> The simplest choice of  $\mathbf{H} = \mathbf{I}$  leads to the *gradient* or *steepest-descent* method, which is fairly inefficient close to the minimum.

### 5.6.2 Damped Gauss–Newton Method

The Gauss–Newton algorithm differs from Newton's method only in the Hessian, from which the second term is omitted (compare with Equation 5.155), *i.e.*,

$$R(\boldsymbol{\theta}) = V^*(\boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^N \psi(k, \boldsymbol{\theta}) \psi^T(k, \boldsymbol{\theta}) \quad (5.157)$$

the so-called Gauss–Newton Hessian. This reduces the computational effort and improves the convergence property of the algorithm. The Gauss–Newton update is

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - R^{-1}(\boldsymbol{\theta}_k) G(\boldsymbol{\theta}_k) \quad (5.158)$$

or equivalently

$$R(\boldsymbol{\theta}_k) \boldsymbol{\rho}_k = -G(\boldsymbol{\theta}_k) \quad (5.159)$$

Two major drawbacks of the method are the slow convergence on problems that are sufficiently non-linear or have reasonably large residuals, and the not necessarily global convergence (Dennis and Schnabel, 1996).

There are two ways to improve the Gauss–Newton method: using it with a line search or with a trust region strategy. These two approaches lead to two algorithms (referred to as the *damped Gauss–Newton* method and the *Levenberg–Marquardt* method, respectively) that are frequently used in practice. The former algorithm is described in the following, and the latter in the next section.

The damped Gauss–Newton method (*i.e.*, Gauss–Newton method with a line search) is simply

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \lambda_k R^{-1}(\boldsymbol{\theta}_k) G(\boldsymbol{\theta}_k) \quad (5.160)$$

The sequence of scalars  $\lambda_k$  is used as a means to control the step length (and thus improve the convergence of Equation 5.160). One may choose  $\lambda_k$  so that

$$\lambda_k = \arg \min_{\lambda} V(\boldsymbol{\theta}_k - \lambda R^{-1}(\boldsymbol{\theta}_k) G(\boldsymbol{\theta}_k)) \quad (5.161)$$

which is a one-dimensional optimisation problem, or by other methods described by Dennis and Schnabel (1996:Chapter 6.3).

Since the damped Gauss–Newton method always takes descent steps that satisfy the line-search criteria, it is locally convergent on almost all non-linear LS problems, including large-residual or very non-linear problems. In fact, it is usually globally convergent, but may still be very slowly convergent on some problems (Dennis and Schnabel, 1996).

### 5.6.3 Levenberg–Marquardt Algorithm

A better alternative to the (damped) Gauss–Newton algorithm, especially for ill-conditioned problems, is to apply the *Levenberg–Marquardt algorithm* (or simply the *Marquardt algorithm*),

$$[R(\boldsymbol{\theta}_k) + \mu_k I] \boldsymbol{\rho}_k = -G(\boldsymbol{\theta}_k) \quad (5.162)$$

The idea of adding a diagonal matrix to the Gauss–Newton Hessian (called *regularisation*) can be tracked back to Levenberg (1944), who originally suggested taking

$$\mu_k = \frac{2\mathbf{G}^T(\boldsymbol{\theta}_k)\mathbf{G}(\boldsymbol{\theta}_k)}{V(\boldsymbol{\theta}_k)} \quad (5.163)$$

This approach (called the Levenberg algorithm) was then extended by Marquardt (1963), and hence the name Levenberg–Marquardt algorithm is widely used.

This algorithm works very well in practice and has become the standard of *non-linear LS* routines (Press *et al.*, 1992). It searches for the minimum of the loss function only inside some neighbourhood around the current iterate, *i.e.*, within a ball with radius  $\delta$

$$\boldsymbol{\theta}_{k+1} = \arg \min_{\boldsymbol{\theta}} V_k(\boldsymbol{\theta}) \quad \text{subject to} \quad |\boldsymbol{\theta} - \boldsymbol{\theta}_k| \leq \delta_k \quad (5.164)$$

The ball mentioned is to be interpreted as the region around  $\boldsymbol{\theta}_k$  inside which the approximation can be trusted to be a valid approximation of the true criterion. The method with reference to this interpretation of the principle is known as (model-) the trust region method (Dennis and Schnabel, 1996).

The parameters  $\mu$  and  $\delta$  must be calculated in each iteration. Many versions of the Levenberg–Marquardt algorithm have been coded using various strategies to choose these parameters. The recommended Marquardt recipe is shown in Algorithm 5.1.

**Algorithm 5.1.** Levenberg–Marquardt algorithm

- Step 1. Select an initial guess  $\boldsymbol{\theta}_0$  for the set of fitted parameters, and pick a modest value for  $\mu$ , say  $\mu = 0.001$ .
- Step 2. Solve Equation 5.162 for  $\boldsymbol{\rho}_k$  and evaluate  $V(\boldsymbol{\theta}_{k+1})$ .
- Step 3. If  $V(\boldsymbol{\theta}_{k+1}) \geq V(\boldsymbol{\theta}_k)$ , increase  $\mu$  by a factor of 10 (or any other substantial factor), and go back to Step 2.
- Step 4. If  $V(\boldsymbol{\theta}_{k+1}) < V(\boldsymbol{\theta}_k)$ , decrease  $\mu$  by a factor of 10, update the trial solution  $\boldsymbol{\theta}_{k+1} \leftarrow \boldsymbol{\theta}_k + \boldsymbol{\rho}_k$ , and go back to Step 2.
- Step 5. Terminate the algorithm if the stopping criterion is fulfilled.

Another Levenberg–Marquardt algorithm proposed by Fletcher (1987) makes use of the ratio between the actual decrease and the predicted decrease

$$r_k = \frac{V(\boldsymbol{\theta}_k) - V(\boldsymbol{\theta}_k + \boldsymbol{\rho}_k)}{V(\boldsymbol{\theta}_k) - L(\boldsymbol{\theta}_k + \boldsymbol{\rho}_k)} \quad (5.165)$$

where

$$L(\boldsymbol{\theta}_k + \boldsymbol{\rho}_k) = V(\boldsymbol{\theta}_k) + \boldsymbol{\rho}_k^T \mathbf{G}(\boldsymbol{\theta}_k) + \frac{1}{2} \boldsymbol{\rho}_k^T \mathbf{R}(\boldsymbol{\theta}_k) \boldsymbol{\rho}_k \quad (5.166)$$

to adjust the size of the elements of the diagonal matrix added to the Gauss–Newton Hessian. Equation 5.166 measures the accuracy to which the predicted reduction approximates the actual decrease, in the sense that the closer  $r_k$  is to unity, the better is the agreement, since it is in fact this criterion used for deriving the Gauss–Newton algorithm by approximating the Hessian. Hence, it is logical to let the search

direction approach the Gauss–Newton search direction (by decreasing the value of  $\mu$ ) when the predicted decrease is close to the actual one. On the other hand, the search direction is allowed to approach the gradient direction if the predicted decrease is far from the actual one. The procedure for  $k$ th iteration takes the form of Algorithm 5.2.

**Algorithm 5.2.** Levenberg–Marquardt algorithm (Fletcher’s recipe)

- Step 1. Select an initial parameter guess  $\boldsymbol{\theta}_0$  and an initial value  $\mu_0$ .  
 Step 2. Determine the search direction by solving Equation 5.162.  
 Step 3. If the predicted decrease is close to the actual decrease let the search direction approach the Gauss–Newton search direction while increasing the step size, *i.e.*,  
 $r_k > 0.75 \Rightarrow \mu_k = \mu_k / 2$ .  
 Step 4. If the predicted decrease is far from the actual decrease let the search direction approach the gradient direction while decreasing the step size, *i.e.*,  
 $r_k < 0.25 \Rightarrow \mu_k = 2\mu_k$ .  
 Step 5. If  $V(\boldsymbol{\theta}_k + \boldsymbol{\rho}_k) < V(\boldsymbol{\theta}_k)$  then accept  $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \boldsymbol{\rho}_k$  as a new iterate and let  $\mu_{k+1} = \mu_k$  and  $k = k + 1$ .  
 Step 6. If the stopping criterion is not satisfied got to Step 2.

The values 0.001 and 10 in the first version of the Levenberg–Marquardt algorithm, as well as 0.25, 0.75 and 2 in the second version, are arbitrary and can be substituted for other values without loss of the convergence properties.

## 5.6.4 Computational Aspects

### 5.6.4.1 Computing the Search Direction

A main subtask of the optimisation algorithms presented above is the determination of the search direction (or Newton direction)  $\boldsymbol{\rho}_k$  through solving Equation 5.162. As the Gauss–Newton Hessian (Equation 5.157) is always positive definite (and symmetric), Equation 5.162 can be efficiently solved with the help of different methods, such as Householder transformation and Cholesky factorisation. Detailed Levenberg–Marquardt algorithms permitting practical implementation can be found in Hoffmann and Hofmann (1971), Press *et al.* (1992), Bernd (1996) and Nørgaard *et al.* (2000). See also The MathWorks’ Optimisation Toolbox (For use with MATLAB).

### 5.6.4.2 Stopping Criterion

A stopping criterion is necessary to answer the questions “Have we solved the problem?”, “Have we reason to a stop?”, or “Have we run out of money, time, or patience?” (Dennis and Schnabel, 1996). The stopping criterion can be chosen from different criteria, such as maximum number of iterations, upper bound for the gradient, upper bound for parameter change; see Dennis and Schnabel (1996) and Nørgaard *et al.* (2000) for more details.

### 5.6.4.3 Initial Parameter Guess

As mentioned earlier, an initial estimate  $\Theta_0$  is required to start the damped Gauss–Newton algorithm (Equation 5.160) or the Levenberg–Marquardt algorithm (Equation 5.162). The choice of initial value will often significantly influence the number of iterations needed to find the minimum point, and is a highly problem-dependent question. In some cases there might be *a priori* information available, which then of course should be taken into account. For some model structures there are special methods that can be applied for finding an appropriate initial estimate, as described by Söderström and Stoica (1989).

### 5.6.4.4 Local Minima

Another fundamental problem linked to the identification based on numerical search algorithms is the potential risk that the algorithm gets stuck in a local minimum before the global minimum point of the loss function is determined. The theoretical analysis of the possible existence of local minima is hard, and there are only a few general results available. For aspects related to this problem, the reader should consult, for example, Söderström and Stoica (1989).

Spending some time coming up with a good initial value  $\Theta_0$  for where to start the iterations is usually time well spent. This is, however, not a realistic option in most non-linear black-box problems, where little *a priori* knowledge is available. The best approach in such cases is to identify the model parameters a couple of times, each time starting with a different initialisation of the parameter vector (known as a multi-start approach). Other than that, only various global search strategies are left, such as simulated annealing, genetic algorithms or evolutionary strategies; these, however, are very time-consuming, as they need a long time to converge to the global minimum, if this is ever possible. Yet, a combination of local and global optimisation techniques may be useful. In such combination, a global method is first applied to find a region where the global optimum could lie, and a local method is then used to find a good local minimum (which may be the global optimum itself) in that region.

### 5.6.4.5 Evaluation of Gradients

As already mentioned in Sections 5.6.1–5.6.3, the expressions for gradient and Hessian matrix are usually complex, and an extensive computational burden is required. Thus, they are normally approximated by numerical rather than by analytical expressions. However, practical experience in system identification has proven that it is far better to use analytical expressions at least for the calculation of gradients. The computational burden depends greatly on the model structure; and sometimes numerical differentiation has to be applied.

From Equation 5.108 (in MIMO form) it can be seen that

$$\frac{d\mathbf{e}^T(t)}{d\Theta} = -\frac{d\hat{\mathbf{y}}^T(t)}{d\Theta} \quad (5.167)$$

Hence, the calculation of the output sensitivities  $\partial\hat{\mathbf{y}}^T/\partial\Theta$  is required at each time instant to evaluate the Equations 5.154 and 5.155.

Provided  $f(x, u, \Theta)$  and  $g(x, u, \Theta)$  in Equation 5.12 are smooth functions, the output sensitivities  $\partial \hat{y}^T / \partial \Theta$  can be computed from

$$\frac{d\hat{y}(t)}{d\Theta^T} = \frac{\partial g(x, u, \Theta)}{\partial x^T} \frac{dx(t)}{d\Theta^T} + \frac{\partial g(x, u, \Theta)}{\partial \Theta^T} \quad (5.168)$$

The state sensitivities  $dx(t)/d\Theta^T$  can be calculated by solving the equation

$$\frac{d\dot{x}(t)}{d\Theta^T} = \frac{\partial f(x, u, \Theta)}{\partial x^T} \frac{dx(t)}{d\Theta^T} + \frac{\partial f(x, u, \Theta)}{\partial \Theta^T} \quad (5.169)$$

numerically, in general. However, for some subclasses of non-linear systems, explicit formulae for the computation can be obtained; for instance, for bilinear systems details can be found by Yin (1992, 1994) and Yin and Schwarz (1992).

Generally, the approximation of the output sensitivities is performed by means of finite difference quotients:

$$\frac{\partial \hat{y}(k)}{\partial \Theta_i} \approx \frac{\hat{y}(k, \Theta + \Delta \Theta_i \cdot e_i) - \hat{y}(k, \Theta)}{\Delta \Theta_i} \quad \begin{array}{l} i = 1, 2, \dots, M \\ k = 1, 2, \dots, N \end{array} \quad (5.170)$$

$e_i$  stands for the  $i$ th unit vector. The step size  $\Delta \Theta_i$  should be small enough to ensure a small approximation error. However, too small a step size would lead to large quantisation errors resulting from the subtraction of two nearly equal values on a digital computer.

A more accurate, but more complicated approximation of the output sensitivities is achieved by

$$\frac{\partial \hat{y}(k)}{\partial \Theta_i} \approx \frac{\hat{y}(k, \Theta + \Delta \Theta_i \cdot e_i) - \hat{y}(k, \Theta - \Delta \Theta_i \cdot e_i)}{2\Delta \Theta_i} \quad (5.171)$$

## 5.7 Grey-box Identification of Non-linear Hydraulic Servo-system Models

As mentioned at the beginning of Chapter 4, experience and simulations have identified many physical model parameters as critical (*e.g.* flow coefficients  $c_{vi}$ , fluid bulk modulus  $E$ , friction coefficients, total mass  $m_i$ ), such that they should at least be checked for accuracy using experimental parameter identification. This leads directly to the grey-box identification. In this section, some sub-models will be rewritten in a form appropriate for system identification. This implies, for instance, a transformation to a suitable discrete-time form, if necessary.

### 5.7.1 Identification of Pressure Dynamics Model

Consider the sub-model of the pressure dynamics from Section 4.2.2.1, and assume that test data provide us with sampled values for  $x_p$  and  $p_A, p_B$ . Finite difference derivatives can then be used to obtain values for  $\dot{x}_p$  and  $\dot{p}_A, \dot{p}_B$  (see Sections 5.6.4 and 6.10.1.1).

If we first rewrite Equations 4.44 and 4.45 – neglecting the leakage flows and the pressure dependence of the bulk modulus – as

$$-V_A(x_p)\dot{p}_A \frac{1}{E'_A} = A_p \dot{x}_p - Q_A \quad (5.172)$$

$$-V_B(x_p)\dot{p}_B \frac{1}{E'_B} = Q_B - \alpha A_p \dot{x}_p \quad (5.173)$$

and then cast the data

$$\begin{bmatrix} -V_A(1)\dot{p}_A(1) & w_1(1) & w_2(1) \\ -V_A(2)\dot{p}_A(2) & w_1(2) & w_2(2) \\ \vdots & \vdots & \vdots \\ -V_A(N)\dot{p}_A(N) & w_1(N) & w_2(N) \end{bmatrix} \begin{bmatrix} 1/E'_A \\ c_{v1} \\ c_{v2} \end{bmatrix} = \begin{bmatrix} A_p \dot{x}_p(1) \\ A_p \dot{x}_p(2) \\ \vdots \\ A_p \dot{x}_p(N) \end{bmatrix} \quad (5.174)$$

i.e., in the form of Equation 5.118, where (for  $k = 1, 2, \dots, N$ )

$$\begin{aligned} w_1(k) &= u(k)\sqrt{p_S - p_A(k)} \quad \text{for } u(k) \geq 0 \\ w_2(k) &= 0 \end{aligned} \quad (5.175)$$

and

$$\begin{aligned} w_1(k) &= 0 \\ w_2(k) &= u(k)\sqrt{p_A(k) - p_T} \quad \text{for } u(k) < 0 \end{aligned} \quad (5.176)$$

LS analysis can be used to identify the parameters  $1/E'_A$ ,  $c_{v1}$  and  $c_{v2}$  directly. A similar equation is derived for the parameters  $1/E'_B$ ,  $c_{v3}$  and  $c_{v4}$ :

$$\begin{bmatrix} V_B(1)\dot{p}_B(1) & w_3(1) & w_4(1) \\ V_B(2)\dot{p}_B(2) & w_3(2) & w_4(2) \\ \vdots & \vdots & \vdots \\ V_B(N)\dot{p}_B(N) & w_3(N) & w_4(N) \end{bmatrix} \begin{bmatrix} 1/E'_B \\ c_{v3} \\ c_{v4} \end{bmatrix} = \begin{bmatrix} \alpha A_p \dot{x}_p(1) \\ \alpha A_p \dot{x}_p(2) \\ \vdots \\ \alpha A_p \dot{x}_p(N) \end{bmatrix} \quad (5.177)$$

where (for  $k = 1, 2, \dots, N$ )

$$\begin{aligned} w_3(k) &= u(k)\sqrt{p_B(k) - p_T} \quad \text{for } u(k) \geq 0 \\ w_4(k) &= 0 \end{aligned} \quad (5.178)$$

and

$$\begin{aligned} w_3(k) &= 0 \\ w_4(k) &= u(k)\sqrt{p_S - p_B(k)} \quad \text{for } u(k) < 0 \end{aligned} \quad (5.179)$$

## 5.7.2 Identification of Load Dynamics Model

The same procedure can be applied to Equation 4.56 – considering only viscous friction –



$$m_i \ddot{x}_p + \sigma \dot{x}_p = (p_A - \alpha p_B) A_p - F_{\text{ext}} \quad (5.180)$$

for the estimation of the total mass  $m_i$ , the viscous friction coefficient  $\sigma$  and the external force  $F_{\text{ext}}$ . This leads to

$$\begin{bmatrix} \ddot{x}_p(1) & \dot{x}_p(1) & 1 \\ \ddot{x}_p(2) & \dot{x}_p(2) & 1 \\ \vdots & \vdots & \vdots \\ \ddot{x}_p(N) & \dot{x}_p(N) & 1 \end{bmatrix} \begin{bmatrix} m_i \\ \sigma \\ F_{\text{ext}} \end{bmatrix} = \begin{bmatrix} A_p(p_A(1) - \alpha p_B(1)) \\ A_p(p_A(2) - \alpha p_B(2)) \\ \vdots \\ A_p(p_A(N) - \alpha p_B(N)) \end{bmatrix} \quad (5.181)$$

### 5.7.3 Online Identification for Adaptive Control

As mentioned earlier, in Section 5.1.3, some dynamic parameters in HSSs, such as the bulk modulus, actually vary during operation. This motivates the application of online identification. Moreover, the operating condition and load disturbance of the hydraulic system vary in an unknown fashion, such that any approach (conventional or modern) with constant gains cannot guarantee the desired performance. In particular, the parameter estimates gained online can be used in an adaptive control scheme. The recursive version of the LS algorithm, described in Section 5.5.2, is suitable to track the parameter changes in the model Equations 5.172, 5.173 and 5.180; see Figure 5.22.

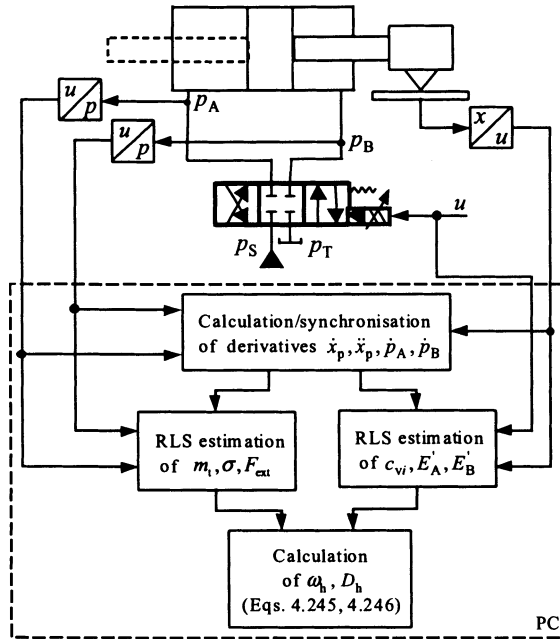


Figure 5.22. Online identification of grey-box models of HSSs

Once the parameters  $E'_A$ ,  $E'_B$ ,  $c_{vi}$ ,  $m_i$  and  $\sigma$  have been identified, the natural frequency and the damping ratio of the hydraulic servo-system can now be determined more accurately, exploiting again the relationships introduced in Section 4.4.4.3.

However, care has to be taken of the fact that the finite differentiation of the position and pressure signals will cause a (phase) delay in the resulting signals that has to be corrected for, *e.g.*, by introducing hold or time delay elements.

### 5.7.4 Identification of General Models

Although the HSS behaviour is highly non-linear, there is much less attention to the identification of non-linear HSS models in the literature; see Yin (1994), who proposed to use the maximum likelihood method.

## 5.8 Fuzzy Identification

The identification of fuzzy models consists of several phases. The basic identification procedure is similar to classical identification, though some augmentations are required due to the particular model structure. Figure 5.23 shows a lumped version of Figure 5.2, which will guide the identification workflow in this section.

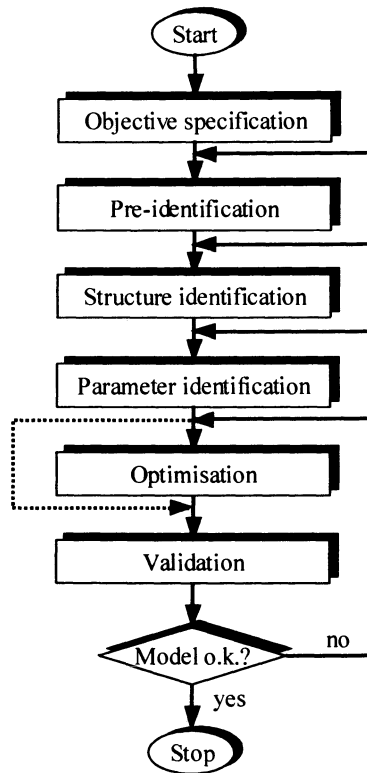
In the following, only the fuzzy-specific identification tasks are discussed. The following first two steps of the identification procedure are assumed to have already been accomplished:

- Modelling objective specification (model use, scope of modelling: system boundaries, operational scenarios, key characteristics).
- Pre-identification (choice of model type, identification of possible grey-box elements, determination of model inputs and outputs, selection of sample time, design of experiments to gain data with required information content, experiment execution and data logging, data conditioning: handling of outliers, scaling).

The parameter identification step is executed several times with different (random) choices of initial values for the prototypes: the algorithm, which is used for their placement, does not provide for globally optimal results. A classical LS algorithm is used to estimate the remaining linear parameters (estimation bases on serial-parallel model evaluation). Non-linear parameters (*e.g.*, related to the membership functions) are provided by the user and kept fixed.

A sufficiently parameterised model is typically already obtained after the parameter identification step. The resulting model is used to initialise the optimisation step. The application of the non-linear optimisation step permits one to optimise non-linear parameters. Even more important, the model can be optimised for parallel model evaluation. The latter typically provides for a more robust parameterisation and improved multi-step-ahead predictions.

The validation of the fuzzy model is carried out with the same methods as in other identification approaches; see Section 5.10.



**Figure 5.23.** Fuzzy identification procedure

The reader is referred to the following references for studying basic aspects of identification (references to special issues are embedded in the appropriate text section):

- Kroll (1997) and Babuška (1998) for fuzzy identification and modelling (recommended references for this book)
- Ljung (1999) for theoretical basics of identification.

The following subsections discuss the identification steps according to Figure 5.23 in detail.

### 5.8.1 Introduction and Model Parameter Overview

The identification approach follows the special structure of the fuzzy rules: their composition of premise and their conclusion parts require one to perform dedicated tasks.

The premises define the fuzzy partition of the model space and therefore the (fuzzy) validity range and the superposition of the local models. Regarding identification, firstly some design decisions have to be made and the structural parameters have to be identified: the feature vector has to be defined, the degree of fuzziness of the partitioning has to be chosen and the distance assessment has to be selected. Either grid-type or multidimensional partitioning has to be selected. (It is

assumed that the type of membership function has already been selected to be of the probabilistic type.) The partitioning itself is computed in the second parameter estimation step. The number of partitions is typically determined by comparing the model quality for different choices, if no expert knowledge is available. These tasks are not typical for conventional identification. Fuzzy clustering algorithms and statistical methods will be employed to accomplish them.

The conclusions contain the dynamic local models, typically of ARX type, and they define the major dynamic behaviour. Regarding identification, first of all their structure has to be determined, *i.e.*, the composition of the regression vector. Second, the parameters have to be estimated.

Summarising, the following parameters refer to the model structure (see also the overview in Figure 5.24):

1. Parameters defining the structure of the fuzzy partition:
  - choice of axis-parallel (“grid-type”) or free-form (“multidimensional”) partitioning
  - number of rules  $c$ , *i.e.*, number of partitions/local models
  - composition of the feature vector  $\chi$
  - fuzziness  $\nu$  of partition (shape of membership function)
  - distance norm type  $\|\cdot\|_\nu$  (shape of membership function).
2. Parameters defining the structure of the local dynamical models:
  - dead time  $\tau$  of each input quantity
  - number  $m$  of considered historical inputs (for each input quantity)
  - number  $n$  of considered historical outputs.

| Parameter location | Parameter type                                        |                                                      |
|--------------------|-------------------------------------------------------|------------------------------------------------------|
|                    | Structure                                             | Detail                                               |
|                    | Premise                                               | Conclusion                                           |
|                    | $c, \ \cdot\ _\nu, \nu$                               | $\tau, m, n$                                         |
|                    | $\nu_1, \dots, \nu_c$<br>Norm parameter<br>( $D, p$ ) | $\theta_1, \dots, \theta_c$<br>$\xi_1, \dots, \xi_c$ |

Figure 5.24. Overview of fuzzy model parameters

## 5.8.2 Structure Identification

### 5.8.2.1 Selection of the Number of Partitions and Features

One central issue is the choice of the number  $c$  of partitions, which is equivalent to the number of membership functions and fuzzy rules for the chosen fuzzy model type. Within pattern classification, several methods have been developed for partitioning static data with the correct number of partitions, *e.g.*,

- (i) select  $c$  to minimise the objective function used within fuzzy clustering (Bezdek, 1981) or a modified version (Sugeno and Yasukawa, 1993),

- (ii) select  $c$  to minimise the partition entropy (Bezdek, 1981),
- (iii) use of separation indices (Dunn, 1974), and
- (iv) merging of similar clusters according to eigenvectors, eigenvalues and covariance (Krishnapuram and Freg, 1992).

However, the data from dynamic systems commonly do not feature the well-separated structures required for a successful application of these methods. In practice, the determination of the number of partitions is part of a combined structure search, which includes partitioning, feature and regression vector selection. It is not possible to solve these tasks independently of each other and obtain optimal results.

Figure 5.25 shows a basic algorithm. The underlying idea is to start with a rough partition with the related model and increase the granularity until the predictions deteriorate. This idea is common for both, grid-type and multidimensional partitioning (*e.g.*, Sugeno and Kang, 1988; Sugeno and Tanaka, 1991; Kroll and Agte, 1997; Nelles *et al.*, 1998). The major difference is that grid-type partitioning demands a more complex combinatorial search in each iteration step, whereas multidimensional partitioning can be carried out by easier continuous learning/optimisation. On the other hand, the grid-type partitioning is less abstract and, therefore, easier to interpret.

To simplify the problem, all local models have the same structure and, therefore, all use the same regression vector  $\phi(k)$  (conclusion). The task of the structure identification regarding the conclusion is to define the number  $n$  and  $m$  of considered historical outputs and inputs respectively, as well as a possible dead time  $\tau$ . An advanced version even determines the single significant terms. In the case of MISO models with  $g$  inputs,  $m$  and  $\tau$  have to be determined for each input.

The determination of the most significant conclusion terms of SISO models can be carried out using the OLS-based algorithm of Kortmann (1989). Though Kortmann applies the algorithm to identify a globally valid model, the algorithm is only used to identify the model structure in the fuzzy model context (Kroll and Agte, 1997). The actual parameter values are not used.

Alternatively, basic physical considerations and correlation methods can be used to determine dead times. Simple heuristics regarding the dynamical process behaviour often permit a good first estimation of the model structure: aperiodic non-linear behaviour is dealt with by  $m = 1$ ,  $n = 1$ , and oscillatory behaviour with  $m = 2$ ,  $n = 2$ . The horizon regarding  $y$  is experienced to be more significant than regarding  $u$ : mostly  $m = 1$  turns out to be sufficient; sometimes  $m = 2$  increases the model quality, but seldom does  $m$  take a higher value. Having such heuristics to restrict the search space, commonly the according candidate models are calculated and assessed. When an automatic (OLS-based) regression vector candidate selection is not possible or wanted, the structure selection can be implemented by a major loop running through all regression vector candidates as a sub-loop of the number of partitions. The basic algorithm can be advanced further *e.g.* by carrying out a regression vector structure search for each partition (Nelles *et al.*, 1998).

The feature vector (premise)  $\chi(k)$  is typically selected by expert choice or the full regression vector is used. In the case of expert choice, the guiding idea is that one (or more) key auxiliary variable is known, which indicates different behavioural regimes, *e.g.*, as the process non-linearity appears clearly along its universe of

discourse. Then it remains to determine the partitions either by expert choice or automatically, *e.g.*, by divide-and-assess or by clustering methods.

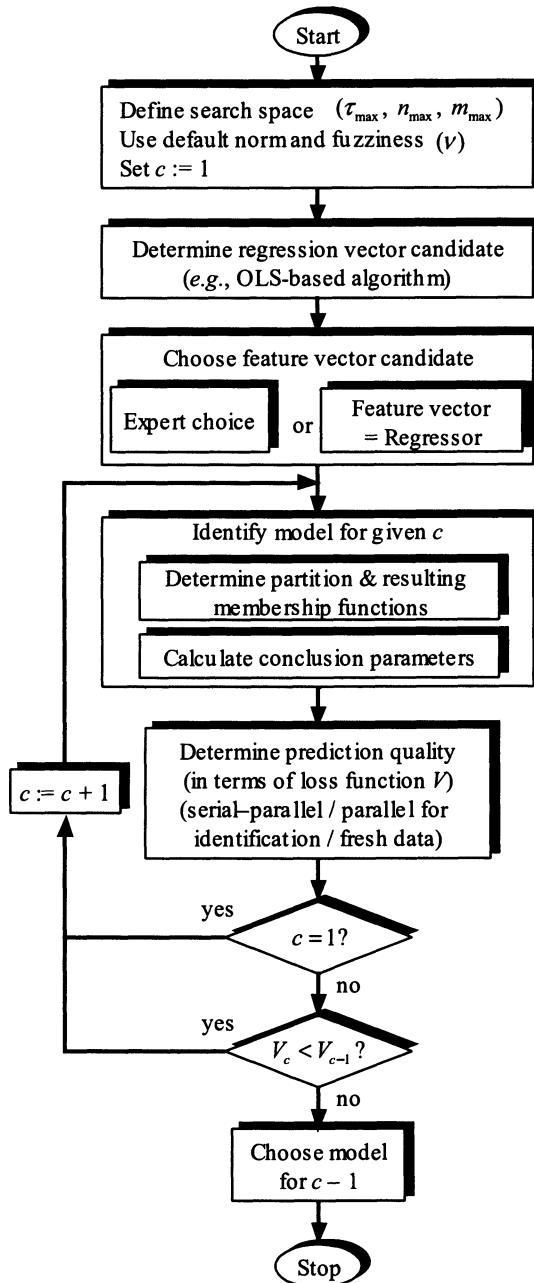


Figure 5.25. Flow chart of structure search

Use of the full regression vector as candidate feature vector is driven by the typical lack of knowledge by which to select the features in context of black-box modelling. The full regression vector is used as a feature candidate pool in the case of structure search algorithms for grid-type partitions. The structure search algorithm follows a divide-and-assess strategy and eliminates non-significant features (e.g., Sugeno and Kang, 1988; Sugeno and Tanaka, 1991; Nelles *et al.*, 1998). The *a priori* defined regression vector is used for the conclusion or a more advanced regression structure search is also carried out for each new partition candidate.

In the case of the multidimensional partitioning, the feature vector is typically kept to be the regression vector: the partitioning algorithm distributes the partitions not just along the features (in the sense of co-ordinate axis), but along the patterns prevalent in the data (which are somehow spread in the multidimensional space) themselves.

As already mentioned, data from dynamic systems commonly do not feature well-separated patterns. Hence, increasing the granularity of the partitioning typically improves the numerical fit of the predictions to the training data, but will finally result in over-fitting. In order to check for over-fitting, the prediction quality for parallel model evaluation regarding the training data, and in particular regarding fresh data, is calculated. Both are used in the termination criterion for finer partitioning.

### 5.8.2.2 Distance Norms and Fuzziness Choice for Prototype Oriented Metrical Cluster Algorithms

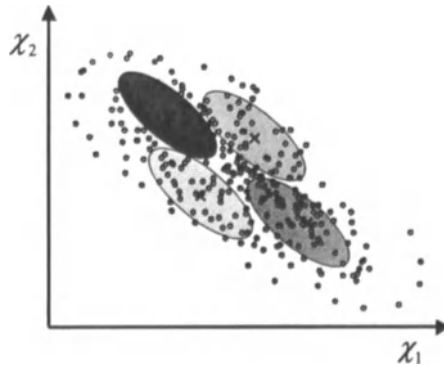
Cluster algorithms are used to identify the distance-based partitioning. A particular class of algorithms, the prototype oriented metric fuzzy cluster algorithms, is selected. Most metrical fuzzy cluster algorithms require the user to choose a distance norm, which is typically valid uniformly for all clusters. In contrast, e.g., the Gustafson and Kessel (or fuzzy covariance) algorithm (Gustafson and Kessel, 1979) calculates a form matrix  $D$  for each cluster.

These cluster algorithms minimise an objective function. In the case of badly scaled data, the features with the largest magnitude may dominate the clustering; ideally, though, the features that are most significant for the non-linear behaviour should steer the clustering. If known, they can be stressed by introducing weights in the objective function. Typically this insight is not available, and normalised features are recommended. Some distance norms have a self-scaling effect (e.g., Mahanolobis norm); others require the user to take care of the scaling (e.g., Euclidian norm).

The norm should be chosen according to the patterns prevalent in the data. This knowledge is seldom available in practice, and an according analysis may be too costly. Good experiences were gained using the covariance-matrix-based Mahanolobis norm: as the equi-normal lines follow the data distribution, it has the capability to yield ellipsoidal rotated clusters, as shown in the example in Figure 5.26. The Euclidian norm is recommended as a simpler and less powerful alternative. (The feature space can be transformed (rotation) and scaled to yield the same result, though the Mahanolobis norm is felt to be easier to use.) The Mahanolobis norm's form matrix is calculated *a priori* from the identification data, once the composition of the feature vector has been decided upon (see Equations 5.78 and 5.79).

There is no generally optimal choice of  $\nu$ , as that depends on the structures and noise contained in the data. The effect of  $\nu$  becomes more obvious from studying some particular choices (refer to Kroll (1997) for a detailed discussion):

- For  $\nu \rightarrow 1$  the fuzzy partitioning becomes crisp.
- The membership functions are steadily differentiable for  $\nu \in (1; 3)$ . This is important for the application of gradient-based optimisation methods. Moreover, this selection supports the cluster forming during fuzzy clustering.
- Choosing  $\nu$  close to one (say  $\nu = 1.1$ ) may yield models better fitting the identification data. Often this is due to over-fitting, which becomes visible during cross-validation. Larger values of  $\nu$  (say  $\nu > 2$ ) yield a significant overlapping of the rules, which typically reduces the achievable prediction quality.
- Choosing  $\nu \rightarrow \infty$  yields unwanted “spot-type” clusters. This means membership functions that provide a value of unity for the prototype location and approximately the asymptotic value  $1/c$  even for short distances. Moreover, values close to unity, *e.g.*,  $\nu = 1.05$ , may cause numerical problems in the context of fuzzy identification.



**Figure 5.26.** Example for data and partitioning, which is not aligned with the coordinate axis.

Good practical experiences were gained with  $\nu \in [1.2; 1.6]$ ; a good default value is 1.5. Note that the experiences with  $\nu$  mainly refer to FCM-type partitions.

### 5.8.3 Parameter Identification (Premise)

A variety of prototype oriented metric fuzzy cluster algorithms has been developed that minimises an objective function by unsupervised learning. Some of the key issues to be considered when selecting an algorithm in the context of fuzzy identification are:

- Does it use locally varying or globally uniform distance norms?
- Will it provide for probabilistic or possibilistic membership functions?
- How easy is its application (complexity, probability to be caught in an unfavourable local minimum, robustness of operation, required difficult manual choices of design parameters)?



- What number of model parameters will result from its use (due to impact on the bias/variance-dilemma and the probability to be caught in an unfavourable local minimum)?

Some algorithm examples are: fuzzy-*c*-means (FCM), fuzzy-*c*-varieties (FCV), fuzzy-*c*-elliptotypes (FCE), fuzzy-covariance matrix (FKM), also referred to as Gustafson and Kessel algorithm, Gath and Geva, possibilistic FCM (PCM) and fuzzy-*c*-shells (FCS). Other types of cluster algorithm include the earlier crisp (non-fuzzy) ones or the not prototypical oriented ones. Krishnapuram (1993) and Mourot *et al.* (1993) provide good qualitative overviews on the manifold fuzzy cluster algorithms. Bezdek (1981) records details on most. Babuška (1998) discusses different algorithms in the context of identification.

The Gustafson and Kessel (1979) algorithm calculates a form matrix  $D$  for each cluster in its probabilistic and possibilistic variant. In practice, this means a significant increase of the number of model parameters, which is commonly not justified by an according increase of prediction quality. The simple FCM algorithm depicts a better compromise. It uses the same distance norm for all clusters. The best experiences were obtained with the probabilistic original version, the FCM, also referred to as fuzzy ISODATA (Bezdek, 1981). These experiences are based on a detailed comparison of algorithms within an application study (Stappen, 1996).

In the following, the FCM algorithm will be used to determine the prototypes of the membership functions. The original algorithm (Bezdek, 1981) was developed for classification purposes; its objective was to determine  $c$  clusters such that the membership values for the given finite set of  $N$  data points to  $c$  clusters minimises the “within-group sum-of-squared-error (WGSS) criterion”:

$$V_{\text{FCM}}(\Theta_{\text{premise}}) = \sum_{k=1}^N \sum_{i=1}^c \mu(\chi(k))^i \|v_i - \chi(k)\|^2 \quad (5.182)$$

The FCM calculates the “fuzzy partition matrix”  $U$ :

$$U = [\mu_{i,k}] = \left[ \begin{array}{ccc} \mu_{1,1} & \cdots & \mu_{1,N} \\ \vdots & \ddots & \vdots \\ \mu_{c,1} & \vdots & \mu_{c,N} \end{array} \right] \left. \vphantom{\begin{array}{ccc} \mu_{1,1} & \cdots & \mu_{1,N} \\ \vdots & \ddots & \vdots \\ \mu_{c,1} & \vdots & \mu_{c,N} \end{array}} \right\} \begin{array}{l} c \text{ clusters} \\ N \text{ data sets} \end{array} \quad (5.183)$$

Obviously, the objective of the fuzzy clustering is a definition of point-wise memberships. In contrast, the objective of the fuzzy identification is to yield a parametric membership function and therefore different. In order to accomplish this different objective, a modified version of the algorithm (Algorithm 5.3) is used (Kroll, 1997): this focuses on the cluster prototype and uses the fuzzy partition matrix as auxiliary variable, whereas the original algorithm does this the other way round. Hence, it yields the prototypes and the according membership functions of all clusters, which can be evaluated for any argument.

**Algorithm 5.3. Modified FCM algorithm****Step 1. Choosing design parameters**

Specify the number of clusters  $c$  ( $2 \leq c \leq N$ ), fuzziness parameter  $v \in (1; \infty)$ , distance norm  $\|\cdot\|_V$  (for feature assessment), matrix norm  $\|\cdot\|_M$  and termination threshold  $\varepsilon_T$  (for algorithm termination condition).

**Step 2. Initialisation**

Specify initial values for the  $c$  prototypes  $v_1^{(0)}, \dots, v_c^{(0)}$ : determine the minimal and maximal values occurring in the identification data for each feature. Randomly place the  $c$  prototypes within these intervals. Set the iteration step to  $l = 0$ .

**Step 3. Update of discrete fuzzy partition**

- a. Determine all singularities, *i.e.* all data points with distance 0 to one of the prototypes ( $d_{i,k} := \|\mathbf{x}_k - v_i^{(l)}\|_V = 0$ ;  $i = 1, \dots, c$ ,  $k = 1, \dots, N$ ).

Let  $I_k^{(l)} := \{i \in \{1; c\} \mid d_{i,k}^{(l)} := \|\mathbf{x}_k - v_i^{(l)}\|_V = 0\}$  be the set of all clusters, where  $\mathbf{x}_k$  has zero distance to the according prototype (*i.e.* is a singularity). Let  $\tilde{I}_k^{(l)} := I_c - I_k^{(l)}$  be the set of clusters, where  $\mathbf{x}_k$  depicts no singularity.

- b. Calculate the memberships  $\mu_{i,k}^{(l+1)}$  of all data points  $k = 1, \dots, N$  for all clusters ( $i = 1, \dots, c$ ):

$$\mu_{i,k}^{(l+1)} = \begin{cases} \left[ \sum_{j=1}^c \left( \frac{d_{i,k}^{(l)}}{d_{j,k}^{(l)}} \right)^{\frac{2}{v-1}} \right]^{-1} & \text{if no singularities occur } (I_k^{(l)} = \emptyset) \\ \begin{cases} 0 \forall i \in \tilde{I}_k^{(l)} \\ a_{i,k}^{(l)} \forall i \in I_k^{(l)} \end{cases} & \text{if singularities occur } (I_k^{(l)} \neq \emptyset) \end{cases}$$

with  $a_{i,k}^{(l)}$  subject to:  $\sum_{i \in I_k^{(l)}} a_{i,k}^{(l)} = 1$ . Choose, *e.g.*,  $a_{i,k}^{(l)} = 1/\#(I_k^{(l)})$ .

**Step 4. Calculate the new prototypes**

$$v_i^{(l+1)} = \frac{\sum_{k=1}^N (\mu_{i,k}^{(l+1)})^v \mathbf{x}_k}{\sum_{k=1}^N (\mu_{i,k}^{(l+1)})^v}; \quad i = 1, \dots, c$$

**Step 5. Check the termination condition**

IF  $\|v_1^{(l+1)} - v_1^{(l)}, \dots, v_c^{(l+1)} - v_c^{(l)}\|_M \leq \varepsilon_T$

THEN Stop

ELSE  $l = l + 1$ , back to Step 3.

The algorithm is quite simple. What may cause a smell of complexity is the handling of seldom occurring singularities. Regarding terminology, let  $\mu_{i,k}$  be the

membership of data point  $\mathbf{x}(k)$  to cluster number  $i$ . Let  $(\cdot)^{(l)}$  denote the iteration step of the algorithm.  $I_c := \{1; c\}$  depicts the set of all cluster numbers.

Extending the Hölder norm (Lancaster and Tismenetsky, 1985) provides for a matrix norm, which can be used in the termination criterion:

$$\left\| \mathbf{v}_1^{(l+1)} - \mathbf{v}_1^{(l)}, \dots, \mathbf{v}_c^{(l+1)} - \mathbf{v}_c^{(l)} \right\|_M = \left( \sum_{i=1}^c \sum_{j=1}^{\#(\mathbf{v}_i)} \left| \mathbf{v}_{i,j}^{(l+1)} - \mathbf{v}_{i,j}^{(l)} \right|^p \right)^{1/p} \quad (5.184)$$

where the index  $j$  runs over all elements of the sets  $(\mathbf{v}_i)$ , which contain all indices of those data samples (out of all  $N$ ) that belong to the cluster  $i$ , i.e., which are nearest to the cluster centre  $\mathbf{v}_i$ . Typical design parameter choices are  $p = 2$  and  $\varepsilon_T = 1$ .

The result of the clustering depends on the initialisation. The initialisation can be carried out in other ways than proposed, but superior performance was gained this way (Kroll, 1995). Local convergence of the FCM was proven in Hathaway and Bezdek (1988). Global convergence characteristics are discussed in Kim *et al.* (1988).

Having executed the FCM algorithm, the fuzzy partitioning is obtained. All prototypes, and therewith all membership functions, have been fully determined.

### 5.8.4 Parameter Identification (Conclusion)

The task remaining is to identify the parameters of the local models. Either the local models are identified independently from each other using the weighted LS algorithms, or the global model is identified as a whole. The latter is followed here, as it typically provides for better prediction quality. The disadvantage of this approach, though, is that this improved global performance may sometimes go with reduced local model behaviour due to the “blind” numerical optimisation of the global error.

The global prediction of the fuzzy model is given as:

$$\begin{aligned} \hat{y}(k+1) &= \sum_{i=1}^c \mu_i(k) \left[ \xi_i + \boldsymbol{\theta}_i^T \boldsymbol{\varphi}(k) \right] \\ &= \left[ \xi_1, \dots, \xi_c \mid \boldsymbol{\theta}_1^T, \dots, \boldsymbol{\theta}_c^T \right] \begin{bmatrix} \mu_1(k) \\ \vdots \\ \mu_c(k) \\ \frac{\mu_1(k)\boldsymbol{\varphi}(k)}{\mu_1(k)\boldsymbol{\varphi}(k)} \\ \vdots \\ \mu_c(k)\boldsymbol{\varphi}(k) \end{bmatrix} \\ &= \boldsymbol{\theta}^{*T} \boldsymbol{\varphi}^*(k) \end{aligned} \quad (5.185)$$

As the identification problem is *linear in the parameters*, an estimate of the parameter vector  $\boldsymbol{\theta}^*$  can be computed by using the (non-recursive) ordinary LS method (see Section 5.5.2)

$$\hat{\Theta}^* = (\Phi^T \Phi)^{-1} \Phi^T Y \quad (5.186)$$

with

$$Y = \begin{bmatrix} y(2) \\ \vdots \\ y(N+1) \end{bmatrix} \quad \Phi = \begin{bmatrix} \phi_{SP}^{*T}(1) \\ \vdots \\ \phi_{SP}^{*T}(N) \end{bmatrix} \quad (5.187)$$

The membership values can also be represented as weighting factors and kept separate from the regression vector. Then the parameter estimation is carried out using the weighted LS method.

Alternatively, the recursive variant of the algorithm discussed in Section 5.5.2 can be used, which is useful for online applications. The parameterisation of the algorithm is discussed in Section 5.5.2. Note that this estimation method calculates the parameters for serial-parallel, but not for parallel model evaluations. The later can only be implemented by non-linear optimisation methods, as discussed in the next section.

### 5.8.5 Optimisation

Up to now the fuzzy model has been fully parameterised. The underlying objectives were: (i) minimising a particular objective function from fuzzy clustering for the prototype locations and (ii) minimising least-mean-square prediction errors of the serial-parallel model evaluation regarding the separately carried out estimation of the conclusion function parameters. While the single steps produce optimal results on their own, the resulting non-linear fuzzy model parameterisation is not optimal. This section discusses what has been achieved with the simple methods up to now and what could be gained by applying non-linear optimisation methods.

On the one hand, the optimisation methods are by a magnitude more complex to implement and can take magnitudes more computational time. Moreover, the model quality already obtained is typically remarkable. On the other hand, significant improvements of model quality can be achieved using non-linear optimisation methods. Therefore, an additional parameter optimisation is not a default identification step.

#### 5.8.5.1 Perspectives and Pitfalls of Optimisation

The model parameterisation determined up to now is not optimal in the LS sense: the partitioning is optimal regarding the objective function of the clustering algorithm

$$V_{FCM}(\Theta_{\text{premise}}) = \sum_{k=1}^N \sum_{i=1}^c \mu(\chi(k))^v \|\mathbf{v}_i - \chi(k)\|^2 \quad (5.188)$$

The conclusion function parameters have been calculated using the “least-mean-squares”. This means, firstly, that they are generally not bias free and, secondly, they have been determined for serial-parallel model evaluation:

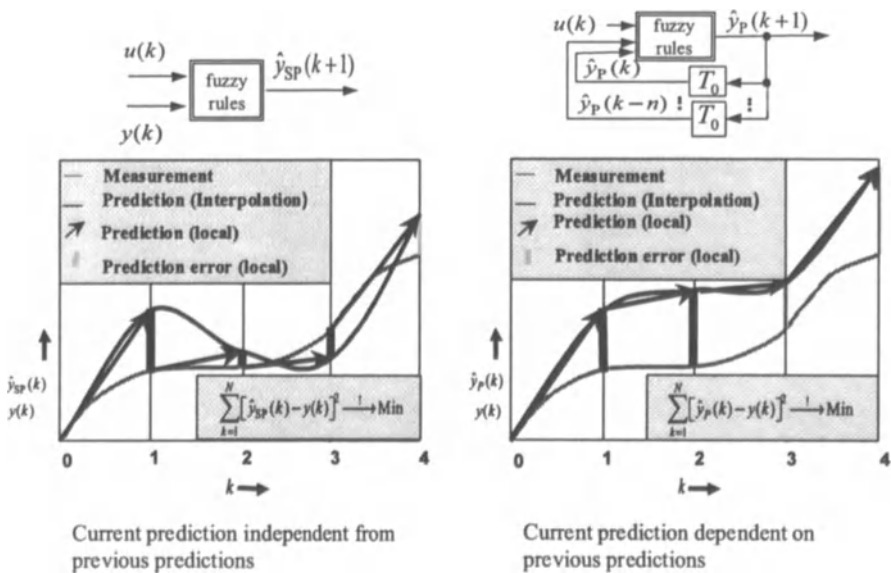
$$V(\theta_{\text{conclusion}}) = \frac{1}{N} \sum_{k=1}^N (y(k) - \hat{y}_{\text{serial-parallel}}(k))^2 \quad (5.189)$$

Nevertheless, the model obtained by now is typically quite good and may already be sufficient in many cases. However, models that have been identified by minimising costs of serial-parallel predictions often feature limited prediction quality for parallel evaluation; see also Henson and Seborg (1996). In the case of parallel model evaluation, the predicted outputs are fed back as model inputs. In serial-parallel evaluation a single prediction step is made from inputs and measured historical outputs. This rationale is illustrated in Figure 5.27.

Good parallel evaluation properties are important for using the model for simulation or for long-range predictions, *e.g.*, in the context of model-based predictive control. The reason why models are not trained for parallel evaluation by default is the resulting complex non-linear recursive optimisation problem.

Non-linear optimisation can be used to achieve an optimal parameterisation for parallel model evaluation:

$$V(\theta_{\text{total}}) = \frac{1}{N} \sum_{k=1}^N (y(k) - \hat{y}_{\text{parallel}}(k))^2 \quad (5.190)$$



**Figure 5.27.** Concept and interpretation of serial-parallel and parallel model evaluation

All continuous parameters of the model, including distance norm and prototypes for instance, can be adjusted. Cost reduction of a magnitude was experienced, while good generalisation properties were kept (Bernd *et al.*, 1997). The experience shows that it is typically advantageous to restrict this pure numerical optimisation to the prototype locations and the conclusion function parameters.

The same methods can be used to optimise all model parameters regarding serial–parallel model evaluation. The experience shows that this often causes over-fitting with, as a consequence, worse generalisation properties and worse parallel evaluation quality even regarding the identification data. This is valid even if methods like regularisation are utilised to reduce over-fitting. Therefore, if the focus is on serial–parallel model evaluation, it is not recommended to implement an optimisation step.

### 5.8.5.2 Implementation

From the global solution and complexity point of view, derivative-free optimisation approaches, such as genetic algorithms or evolutionary strategies, are attractive. From a computational perspective, though (including what can be computed in a tolerable amount of time), gradient-based methods have been selected, even if they only result in local optima.

Methods using second-order derivatives cause too high an analytical effort. Therefore, quasi-second-order methods, which approximate the Hessian, are of particular interest. Of these, the Levenberg and the Levenberg–Marquardt algorithm are especially interesting (see Section 5.6.3). The Levenberg–Marquardt method was found to be the most efficient (Nelles, 1995; Bernd, 1996; Bernd *et al.*, 1997). Sometimes numerical problems can occur due to large parameters adjustment per optimisation cycle. The next best experiences were found with Levenberg’s method. (There is an even better-performing method for serial–parallel evaluation, which is not considered here; see Kroll (1996).)

The cost function is steadily differentiable in the case of  $v \in (1; 3)$  and analytical derivatives are recorded in Kroll (1997). Still, the implementation is challenging, as parallel evaluation means that the sensitivities depend not only on the last data set, but on the entire evolution since initialisation.

The result of the optimisation depends on its application. Bernd (1996) showed that the identification steps proposed before will not always provide for initial values in the attractive region of the global cost optimum. Therefore, it is proposed to carry out an alternative optimisation with a different initialisation and to select the best resulting model. In the alternative case, the initial prototypes are computed by clustering exactly as before. The initial conclusion parameters are all the same for all fuzzy rules and are taken from a linear-affine model (*i.e.*, a fuzzy model with  $c = 1$ ), which was optimised.

## 5.9 Identification with Artificial Neural Networks

The basic procedure for the identification of neural network models (see Figure 5.28) is similar to classical and fuzzy identification (Figures 5.2 and 5.23). In the following, only some neural-network-specific identification tasks are briefly discussed.

The reader is referred to Sjöberg (1995) and Nørgaard *et al.* (2000) for a detailed study. Also, a good collection of papers about this topic is provided by Hunt *et al.* (1995).

Note that, in the neural network literature, the *estimation* is called *training* or *learning*, and the *validation* is called *test* or *generalisation*.

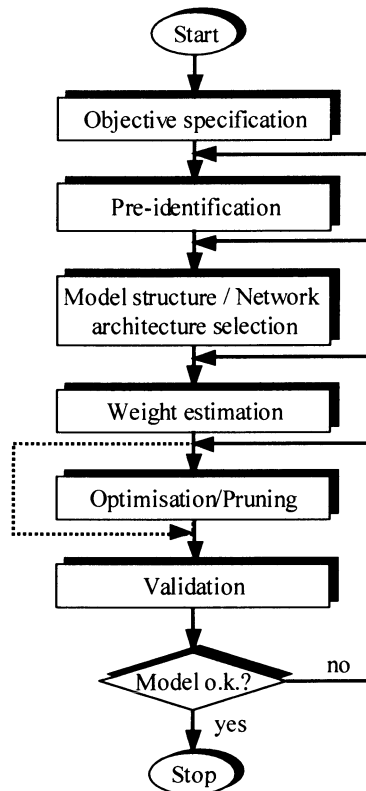
### 5.9.1 Selection of Artificial Neural Network Architectures

In the following, it is assumed that the network type has already been chosen. Before the training of a neural network can be performed, the model structure (network) selection stage has to make appropriate choices of:

1. the components of the regression vector,
2. the number of hidden layers the ANN should have, and the number of units to be included in each layer, and
3. the activation function (or radial basis function for RBF networks).

The possibilities for the choice of activation functions/basis functions and for the selection of the regression vector have already been given in Section 5.4.3.

The question of how many layers to use is not easy. In principle, with many basis functions, one hidden layer is sufficient for modelling most practically reasonable systems (Cybenko, 1989; Barron, 1993). Sontag (1993) provides many useful and interesting insights into the importance of second hidden layers in non-linear structures. The main drawback of using ANNs with more than one hidden layer is that implementation, training, and statistical analysis become much more complicated.



**Figure 5.28.** Artificial neural network identification procedure

Restricting the attention to single hidden layer, fully connected networks, the common (trial-and-error) working procedure to determine an adequate number of hidden units is to increase their number gradually while evaluating the test error. When a number of hidden units has been reached above which the gain in generalisation is insignificant, the ANN is accepted.

Sometimes (e.g., if the training is limited), however, the ANN should be selected to contain only the most essential weights. In this case, the ANN architecture selection has to be automated.

Such (automatic) network structure selection approaches typically result in optimisation problems of the type in Equation 5.150. As already mentioned, it is important that a model structure with an appropriate number of degrees of freedom is found, in the sense that it balances bias and variance. Model structure identification algorithms are therefore usually based on statistical criteria that have this property. The common criteria used in structure selection are the Akaike's final prediction error (FPE) criterion and the Akaike's information criterion (AIC); see Section 5.10.3.

## 5.9.2 Estimation of Weights

The parameter (*i.e.*, weight) estimation problem for neural networks, described in Section 5.4.3, can be posed and solved by (exactly) the same methods as for classical identification presented in Section 5.5, especially in relation with the optimisation algorithms in Section 5.6.

### 5.9.2.1 Parameter Estimation for Multi-layer Perceptron Networks

For MLP networks, which represent non-linear functions of the parameters to be identified, the prediction error method based on optimisation algorithms (see Section 5.6) is the suggested *training* or *learning* (*i.e.*, parameter estimation) approach. The *Levenberg–Marquardt algorithm* is, owing to its suitability for ill-conditioned problems, generally recommended and thus widely used. See also Nørgaard *et al.* (2000).

For two-layer MPL networks, *i.e.*, networks with a single hidden layer, (see Section 5.4.3, Figure 5.15, Equation 5.96 )

$$\begin{aligned}\hat{y}(k) &= W_0 + \sum_{j=1}^h W_j \frac{1}{1 + \exp\left(-\sum_{l=1}^{n+m} w_{j,l} \varphi_l(k) - w_{j,0}\right)} \\ &\equiv W_0 + \sum_{j=1}^h W_j \frac{1}{1 + \sigma_j(k)}\end{aligned}\quad (5.191)$$

the calculation of output sensitivities, and therewith of the gradient, is relatively easy to perform:

$$\frac{\partial \hat{y}(k)}{\partial W_j} = \begin{cases} 1 & \text{for } j = 0 \\ \frac{1}{1 + \sigma_j(k)} & \text{for } j > 0 \end{cases} \quad (5.192)$$



$$\frac{\partial \hat{y}(k)}{\partial w_{j,l}} = \begin{cases} \frac{\sigma_j(k)}{(1 + \sigma_j(k))^2} & \text{for } l = 0 \\ \frac{\varphi_i(k) \sigma_j(k)}{(1 + \sigma_j(k))^2} & \text{for } l > 0 \end{cases} \quad (5.193)$$

Using these analytical expressions, the gradient and Hessian matrix can be built in an appropriate way.

### 5.9.2.2 Back-propagation Algorithm

This algorithm has gained significant attention and widespread use in connection with neural networks, *e.g.*, see Rumelhardt *et al.* (1986) and Hertz *et al.* (1991). It is nothing else than a convenient algorithm for computing the gradients of the predictor (or output sensitivities),<sup>11</sup> *i.e.*, a chain rule for differentiation of Equation 5.96, for general multi-layer feedforward networks with arbitrary activation functions. The BP algorithm is usually combined with gradient descent algorithms, and is also referred to as the generalised delta rule. However, with MLP networks the parameters (weights) appear in a non-linear manner, and hence the BP algorithm in combination with gradient descent algorithms is very slow in determining the optimum weights. So, (pseudo-) second-order training algorithms such as the Broyden–Fletcher–Goldfarb–Shanno algorithm and the Levenberg–Marquardt algorithm have to be employed. See McLoone and Irwin (1995) for a comparative study of the BP algorithm with some second-order estimation algorithms.

### 5.9.2.3 Parameter Estimation for Radial Basis Function Networks

The use of the RBF approach has the advantage that, once the RBF has been chosen (*i.e.*, the basis function  $\Psi$  together with its centres and widths), then all that is left is the determination of the coefficients  $\lambda_{j,l}$  using the LS algorithm. More recently, RBF networks have therefore been gaining popularity as an alternative to MLP networks. By choosing centres appropriately, it is possible that simple RBF network models can offer similar approximation capabilities to MLP networks.

The most difficult task when identifying an RBF network is the selection of basis function centres (Chen *et al.*, 1990a,b, 1992). These should suitably sample the network input domain and reflect the data distribution (Chen *et al.*, 1992). The parameters, centres and standard deviations of the Gaussian functions, can be chosen randomly or in a reasonable way, for example, with the help of standard clustering or lattice methods. Typically, the nearest neighbour method is employed. More suitable is the use of a clustering algorithm, such as the mean-tracking algorithm (Warwick *et al.*, 1995a,b), which gives a statistically best selection of centre quantity and position.

Another suitable approach, based on the *c*-means clustering algorithm to adjust the centres and combined with an LS algorithm to estimate the connection weights

<sup>11</sup> In the NN community, the term back-propagation is often used for the entire search algorithm. It is, however, more consistent to keep this notation just for the algorithm used to calculate the gradient (Ljung, 1999).

(called a hybrid recursive algorithm), has been developed by Chen *et al.* (1992) to fit RBF models. See also Dubois *et al.* (1995) for a simplified version of this algorithm. It has been shown that this hybrid approach offers a powerful online identification algorithm for RBF networks, and that the reduction in the mean-squares error was much faster compared with that of MLP networks. The reader is encouraged to consult Chen *et al.* (1992) for details.

Note here that the use of fuzzy clustering algorithms in an RBF structure leads to very similar models to those described in Section 5.4.2, *i.e.*, Takagi–Sugeno-type fuzzy models. Consequently, the same parameter estimation techniques as in Section 5.8 can be applied.

### 5.9.3 Optimisation of Network Architecture (Growing/Pruning)

Up to now, the overall network architecture (with a particular structure of the regression vector, and fixed numbers of layers/units) is assumed to be decided *a priori*, and the training algorithm is used just to estimate the weights. However, owing to the curse of dimensionality problem, this is not always feasible, especially when the input space is high-dimensional. The identification usually ends up with a model with too many parameters, and thus results in a high variance of the estimated weights. This again directly leads to the bias/variance dilemma already discussed in Section 5.5.5. Besides the regularisation techniques, the approach of searching for optimal model structure has been mentioned as another way to approach the bias/variance problem.

In the ANN literature, a number of iterative model refinement procedures have been proposed; see Murray-Smith (1994) and Johansen and Foss (1995) for reviews of the subject. One common approach is to grow a relatively simple initial model to include more modelling capability. Another one is to train and then prune a relatively complex model in order to produce a more parsimonious model (*i.e.*, to remove superfluous model flexibility). Principally, model *growing* and *pruning* can be done concurrently at each refinement step by making competing candidates, or the model can first be refined and then pruned in a repeated fashion. In other words, pruning starts with a fully connected network architecture, which is large enough to describe the system, and we then remove connections (*i.e.*, to eliminate the according weights) one at a time until the optimal architecture has been obtained. Growing works in the opposite direction, *i.e.*, it starts with a small network architecture, and then gradually increases it.

Before applying a pruning algorithm, it is highly recommended first to perform a rough explicit regularisation with a regularisation parameter, which is somewhat smaller than the optimal one (about a factor 2–100) (Nørgaard *et al.*, 2000). This generally results in a smoothness of the criterion, which is an important requirement for pruning algorithms to be successful. Two prominent pruning algorithms, the *Optimal Brain Surgeon* (Hassibi and Stork, 1993; Hansen and Pedersen, 1994) and the *Optimal Brain Damage* (Le Cun *et al.*, 1990), have been extensively described (with some modifications) by Nørgaard *et al.* (2000).

Note finally that neither the parameter estimation nor the structure optimisation algorithms guarantee to find the global optimum. Besides the global minimum, many close-to-optimal (good) local minima may exist (Johansen and Foss, 1995).

For a review of structure identification algorithms refer to Murray-Smith (1994) and Johansen and Foss (1995).

## 5.10 Model Validation and Comparison of Model Structures

“It is not enough to come up with a model of the process under consideration – we must also have a measure of its reliability. *Model validation* is the process of examining the model, assessing its quality and possibly rejecting its use for the purpose in question. In a sense, this could be viewed as the *essential process of identification* – the estimation phase is really just a means to provide candidate models that might pass the needle's eye of validation” (Ljung, 1993). Indeed, it is seldom a problem to fit some identification data set accurately. Trying to validate the model with a test data set, though, typically provides for a less flattering result.

Model validation is related to questions such as (Söderström and Stoica, 1989):

- Is a given model flexible enough?
- Is a given model too complex?
- Which model structure of two or more candidates should be chosen?
- Should the model be accepted or rejected taking the specified objective into account?

In the field of system identification, a host of different techniques of (non-linear) model validation is offered (Leontaritis and Billings, 1987a, 1987b; Söderström and Stoica, 1989; Haber and Unbehauen, 1990; Wigren, 1990; Ljung, 1999), such as statistical tests on residuals (correlation tests), evaluation of the loss function as a function of the model order, and simulation of the model identified. Some of these tests and criteria are described in the following. It might be stressed here that for the determination of an appropriate model structure it is generally recommended to use a combination of statistical tests and visual inspection of plots of relevant signals/predictions.

### 5.10.1 Prediction, Simulation and Cross-validation

A natural start is to test how well the model is able to reproduce “fresh” data, *i.e.*, data that were not used to estimate the model, in simulation. This is indeed common practice and is perhaps the most useful, pragmatic way to gain confidence in (or reject) a model (Ljung, 1993). Simple plots in the same diagram that compare predictions with actual measurements in identification and validation (or test) data sets can provide a prime but good understanding of the deviations.

It is important not to rely too heavily on a visual inspection of one-step-ahead predictions (*i.e.*, serial-parallel model evaluation as shown in Figures 5.6 and 5.13),  $\hat{y}(k | \hat{\theta})$ . These “naïve” predictions will generally be close to the measured outputs, especially when the sampling frequency is high relative to the dynamics of the systems. In other words, a small prediction error clearly does not imply that the model is good. Instead, one should inspect the  $l$ -step-ahead predictions ( $l > 1$ ) or make a pure simulation of the model (*i.e.*, parallel model evaluation as shown in Figures 5.6 and 5.13), corresponding to an infinite prediction horizon. Refer also to the discussion in Section 5.8.5.1.

This type of validation will be of practical interest in relation to model predictive control strategies, as will be discussed in Section 6.7.2.

### 5.10.2 Residual Tests

The model validation using residual tests is based on the assumption that all information has likely been extracted from the identification data set if there is no correlation between the residuals and all (linear and non-linear) combinations of past data. The residuals are the prediction errors ( $\varepsilon$  in Equation 5.108), and reflect unmodelled parts and disturbances. Because it is practically impossible to check all possible correlations, one commonly chooses only a few auto- and cross-correlation functions.

It is shown by Billings and co-workers that the following conditions for the sampled correlation functions should ideally hold (Billings and Voon, 1986; Billings *et al.*, 1992; Billings and Zhu, 1994):

*Linear correlation function tests:*

$$\hat{R}_{\varepsilon\varepsilon}(l) := \frac{\sum_{k=1}^{N-l} (\varepsilon(k) - \bar{\varepsilon})(\varepsilon(k-l) - \bar{\varepsilon})}{\sum_{k=1}^N (\varepsilon(k) - \bar{\varepsilon})^2} = \begin{cases} 1, & l = 0 \\ 0, & l \neq 0 \end{cases} \quad (5.194)$$

$$\hat{R}_{u\varepsilon}(l) := \frac{\sum_{k=1}^{N-l} (u(k) - \bar{u})(\varepsilon(k-l) - \bar{\varepsilon})}{\sqrt{\sum_{k=1}^N (u(k) - \bar{u})^2 \sum_{k=1}^N (\varepsilon(k) - \bar{\varepsilon})^2}} = 0 \quad \forall l \quad (5.195)$$

*Non-linear correlation function tests:*

$$\hat{R}_{u^2\varepsilon^2}(l) := \frac{\sum_{k=1}^{N-l} (u^2(k) - \bar{u^2})(\varepsilon^2(k-l) - \bar{\varepsilon^2})}{\sqrt{\sum_{k=1}^N (u^2(k) - \bar{u^2})^2 \sum_{k=1}^N (\varepsilon^2(k) - \bar{\varepsilon^2})^2}} = 0 \quad \forall l \quad (5.196)$$

$$\hat{R}_{u^2\varepsilon}(l) := \frac{\sum_{k=1}^{N-l} (u^2(k) - \bar{u^2})(\varepsilon(k-l) - \bar{\varepsilon})}{\sqrt{\sum_{k=1}^N (u^2(k) - \bar{u^2})^2 \sum_{k=1}^N (\varepsilon(k) - \bar{\varepsilon})^2}} = 0 \quad \forall l \quad (5.197)$$

$$\hat{R}_{\varepsilon\beta}(l) := \frac{\sum_{k=1}^{N-l} (\varepsilon(k) - \bar{\varepsilon})(\beta(k-l-1) - \bar{\beta})}{\sqrt{\sum_{k=1}^N (\varepsilon(k) - \bar{\varepsilon})^2 \sum_{k=1}^N (\beta(k) - \bar{\beta})^2}} = 0 \quad l \geq 0 \quad (5.198)$$

$$\hat{R}_{\alpha\epsilon^2}(l) := \frac{\sum_{k=1}^{N-l} (\alpha(k-l) - \bar{\alpha}) (\epsilon^2(k-l) - \bar{\epsilon^2})}{\sqrt{\sum_{k=1}^N (\alpha(k) - \bar{\alpha})^2 \sum_{k=1}^N (\epsilon^2(k) - \bar{\epsilon^2})^2}} = \begin{cases} \gamma, & l=0 \\ 0, & l \neq 0 \end{cases} \quad (5.199)$$

$$\hat{R}_{\alpha u^2}(l) := \frac{\sum_{k=1}^{N-l} (\alpha(k-l) - \bar{\alpha}) (u^2(k-l) - \bar{u^2})}{\sqrt{\sum_{k=1}^N (\alpha(k) - \bar{\alpha})^2 \sum_{k=1}^N (u^2(k) - \bar{u^2})^2}} = 0 \quad \forall l \quad (5.200)$$

where

$$\alpha(k) = y(k)\epsilon(k) \quad (5.201)$$

$$\beta(k) = u(k)\epsilon(k) \quad (5.202)$$

and

$$\gamma = \frac{\sqrt{\sum_{k=1}^N (\epsilon^2(k) - \bar{\epsilon^2})}}{\sqrt{\sum_{k=1}^N (\alpha(k) - \bar{\alpha})^2}} \quad (5.203)$$

In practice, however, the correlation coefficients will never be exactly zero for all lags. Thus, a common way to analyse the correlation functions is to compute and plot them versus the time lag (typically) in the interval  $l \in [-25, 25]$  in a diagram (see Figure 5.29) to indicate if the estimated correlations are significant or not. The diagram is then complemented with a confidence interval (e.g., 95% or 99%) for the asymptotic distribution to check if the functions are within this interval, i.e., if  $-1.96/\sqrt{N} < \hat{R} < 1.96/\sqrt{N}$  or  $-2.58/\sqrt{N} < \hat{R} < 2.58/\sqrt{N}$  respectively.

### 5.10.3 Model Structure Test Criteria

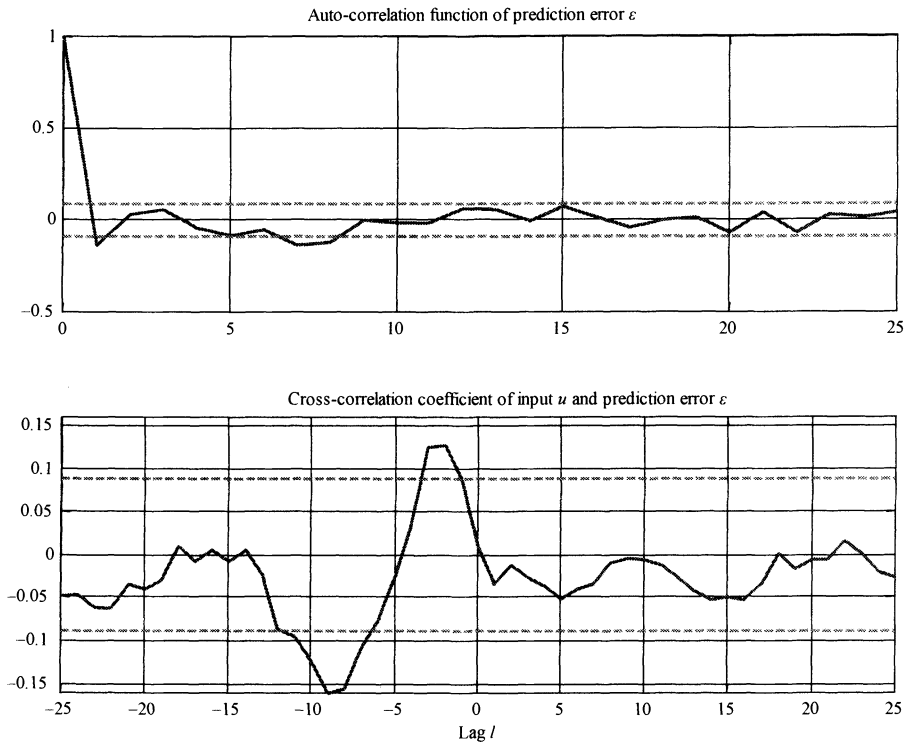
A paramount idea during modelling and identification is to keep the model as simple as possible. Thus, the quality of models should be evaluated or compared using a performance index (in terms of a scalar quantity), e.g., the *normalised residual sum of squares* (NRSS)

$$\text{NRSS} \equiv \frac{1}{N} \sum_{k=1}^N [\hat{y}(k) - y(k)]^2 \quad (5.204)$$

or *Sachs multiple correlation coefficient* ( $0 < R_s < 1$ ) (Sachs, 1978)

$$R_s \equiv \frac{\sum_{k=1}^N [\hat{y}(k) - \bar{y}(k)]^2}{\sum_{k=1}^N [y(k) - \bar{y}(k)]^2} \quad (5.205)$$

The model giving the smallest value of NRSS or the greatest value of  $R_S$  is best, and thus selected. The problem of using these and other criteria is that they point out the best model with too many – often with all possible – model components. In other words, minimising such criteria (for structure selection) will lead to a parsimonious model structures.



**Figure 5.29.** Examples of auto-correlation ( $\hat{R}_{\varepsilon\varepsilon}(l)$ ) test and cross-correlation ( $\hat{R}_{ue}(l)$ ) test ( $N = 500$ , 95% confidence band:  $\pm 0.0877$ )

To avoid this problem, performance indices (or structure selection criteria) which take the complexity of the model, *i.e.*, the number of parameters, into account have been introduced, *e.g.*, *Theil's adjusted multiple correlation coefficient* ( $0 < R_T^2 < 1$ ) (Theil, 1971)

$$R_T^2 = 1 - (1 - R_S^2) \frac{N-1}{N-M} \quad (5.206)$$

and *Amemiya's correlation coefficient* ( $0 < R_A^2 < 1$ ) (Amemiya, 1980)

$$R_A^2 = 1 - (1 - R_S^2) \frac{N+M}{N-M} \quad (5.207)$$

A model with a higher  $R_t^2$  or  $R_A^2$  is better than one with a smaller value. By means of these performance indices, a model that has small residuals and few parameters simultaneously is optimal. This is quite natural for engineers.

Other prominent model performance indices are the so-called information criteria, such as

- Final prediction error criterion (FPE; Akaike, 1970)

$$\text{FPE} = N \ln(\text{NRSS}) + N \ln \frac{N+M}{N-M} \quad (5.208)$$

- Akaike's information criterion (AIC; Akaike, 1972; Sakamoto *et al.*, 1986)

$$\text{AIC}(\phi_1) = N \ln(\text{NRSS}) + \phi_1 M \quad \text{with } \phi_1 > 0, \text{ e.g., } \phi_1 = 2 \quad (5.209)$$

- Khinchin's law of the iterated logarithm criterion (LILC; Hannan and Quinn, 1979; Hannan and Rissanen, 1982)

$$\text{LILC}(\phi_2) = N \ln(\text{NRSS}) + 2M\phi_2 \ln(\ln(N)) \quad \text{with } \phi_2 \geq 1 \quad (5.210)$$

- Bayesian information criterion (BIC; Kashyap, 1977)

$$\text{BIC} = N \ln(\text{NRSS}) + M \ln N \quad (5.211)$$

In all these criteria, the more parameters that are included in the model structure, the smaller the first term will be. However, each parameter carries a variance penalty that will contribute (with a corresponding factor) to the expected mean-square-error fit.

## 5.11 Implementation and Software Tools

Needless to say, the identification of HSS models poses the possibilities of a number of model structures and estimation algorithms. Therefore, it is necessary to execute many experiments in order to gather good knowledge about the various existing possibilities and eventually to make improvements on them. This requires tedious work and is hardly possible without tools which comprise the various possibilities and which obey modifications. For this purpose, many toolboxes for identification of dynamic systems have already been developed, and are running in the MATLAB/Simulink environment.

Commercially available MATLAB toolboxes (<http://www.mathworks.com/products>) are:

- *Curve Fitting Toolbox* for performing model fitting and analysis.
- *Optimisation Toolbox* for solving standard and large-scale optimisation problems.
- *Signal Processing Toolbox* for performing signal processing, analysis, and algorithm development.
- *System Identification Toolbox* for the identification of general (black-box) linear models (implemented algorithms: OE, ROE, PEM).

- *Frequency Domain Identification Toolbox* for identifying linear SISO systems from time responses or measurements of the system's frequency response.
- *IDCON Classic Toolbox* and *IDCON Nonlinear Toolbox* for identification of continuous-time linear and (state-space) non-linear models respectively.
- *Fuzzy Logic Toolbox* for design and simulation of fuzzy logic systems.
- *Neural Network Toolbox* for design and simulation of neural networks.
- *MMLE3 Identification Toolbox* for identification of the parameters of continuous-time state-space models, using either the maximum likelihood method or the output error method.
- *Tai-Ji ID: Automatic Closed-loop Identification Toolbox* for automatic closed-loop identification from data pre-treatment to model validation, especially developed for model-based process control, such as MPC and linear robust control.

Free of charge MATLAB toolboxes are:

- *CONTIDNS Toolkit* for continuous-time identification of non-linear systems in special canonical forms (Jelali and Spielmann, 1996; Jelali, 1997); contact the first author of this book.
- *FREQID Toolbox* for frequency domain identification of (multivariable) linear models, <http://www.ocp.tudelft.nl/sr/downloads/software.htm>.
- *NNSYSID: Neural Network Based System Identification Toolbox* (Nørgaard, 2000a), <http://www.iau.dtu.dk/research/control/nnsysid.html> for the identification of (MISO) neural network models (MLP) using different optimisation algorithms.
- *FMID: Fuzzy Modelling and Identification Toolbox* (Babuška, 1998), Collection of MATLAB functions for the construction of Takagi–Sugeno fuzzy models from data, <http://dutera.et.tudelft.nl/~crweb/software> or <http://lcewww.et.tudelft.nl/~babuska>.
- *State Space Identification (SMI) Toolbox* for the identification of multivariable linear time-invariant (LTI) and multivariable Wiener systems from sampled input–output sequences, <http://dutera.et.tudelft.nl/~crweb/software>.
- *FIMO: Fuzzy Identification and Modelling Toolkit* for identification of an enhanced version of discrete-time Sugeno-type fuzzy models (Bernd and Kroll, 1998); contact the second author of this book.

Another MATLAB tool reported in the literature (Nelles and Isermann, 1996; Nelles, 1997; Nelles *et al.*, 1997; Nelles *et al.*, 2000) is

- *LOLIMOT: Local Linear Model Tree* for the identification of Sugeno-type fuzzy models; e-mail: [mvogt@iat.tu-darmstadt.de](mailto:mvogt@iat.tu-darmstadt.de).

## 5.12 Section Summary

In this chapter, the system identification process has been presented and formulated as several stages (experiment, data preparation, model structure selection, parameter estimation, model validation). A special attention has been paid to the presentation of the most common dynamic model architectures (linear models, non-linear polynomial models, Takagi–Sugeno fuzzy models, MLP and RBF networks) and to



the discussion of their properties. Different techniques, which can be applied for each identification stage, have been described. It is thus appropriate to conclude this chapter by briefly outlining a generic working procedure for the identification of HSS models with concrete recommendations of the suitable method in each stage, depending on the model purpose.

**Experiment.** The experiment activities provide the basis for the further identification stages. In particular, the information content of the data gained in experiments is a very critical issue. The experiment should excite at least all states of the process that can be affected when the model is to be used. Clear distinction has to be made between linear models, for which PRBS are sufficient, and non-linear models that require the use of PRMS to cover as many amplitude–frequency combinations as possible (*curse of dimensionality*); see Section 5.2.1. An obvious, but very important step is to *inspect the data* visually to detect, for instance, non-linear effects (different damping of step responses at different levels and different directions, different response time constants, *etc.*)

**Model Structure Selection/Search.** This step involves the choice of a model family with the corresponding number of parameters. A good engineering principle in this stage is to *try simple things first*. Simplicity primarily means to select a model with a small number of parameters to be estimated. By searching from simpler (*i.e.*, linear) to more complex (*i.e.*, non-linear) models until a satisfactory one is found, a good trade-off between bias and variance can typically be achieved.

The selection of a model structure can also be facilitated if there is some *a priori* knowledge, *e.g.*, from physical insight (theoretical analysis), as one “should not try to estimate what is already known” (Ljung, 1993). For instance, semi-physical regressors could be incorporated in the model, as described in Section 5.7, leading to grey-box models. Only if this gives unsatisfactory results, black-box structures should be considered. Even in that case, prior knowledge is at least required for the data acquisition procedure, *e.g.*, for design of excitation signals.

Which specific kind of model structure (classical polynomial, fuzzy or neural) should be used depends on the application at hand, on the amount of prior knowledge, and on the preferences and skills of the designer or user. Each of these black-box modelling principles has strengths and weaknesses.

*Polynomial models* (Section 5.4.1) are the straightforward classical extension of linear models. Their complexity/flexibility increases with higher polynomial degree. As explained in Section 5.5.3, OLS algorithms are best suited to estimate the parameters of polynomial models with the nice by-product of automatic and efficient complexity reduction. The major drawback of polynomial models lies, however, in their tendency towards oscillatory interpolation and extrapolation behaviour, especially when high-degree polynomials are used.

Owing to their transparency and the possibility to include imprecise (expert) knowledge in terms of fuzzy rules or into form of local linear models, *fuzzy models* (especially of the Takagi–Sugeno type; see Section 5.4.2) provide a good alternative for black-box modelling. Furthermore, fast algorithms are available to estimate them.

*Artificial neural network* (ANN) models can also provide accurate predictions of process outputs, but – in contrast to Takagi–Sugeno models – they are not

transparent to interpretation. ANN models (especially MLP models; see Section 5.4.3.1) do not suffer from the curse of dimensionality, and are thus well suitable for high-dimensional problems<sup>12</sup>. The price to be paid is then the slow training speed and the high risk of convergence to bad local optima, as the parameter estimation for MLP networks requires the use of an iterative optimisation algorithm.

Whatever the model type should be, it is sometimes necessary to extract the optimum model structure in an automated way by comparing different candidate structures (with different model complexity: model orders for polynomial models, number of hidden layers and number of hidden neurons for ANN models, number of membership functions and orders for fuzzy models) in terms of some performance measures. This process is mostly known as *automatic structure selection*, *structure identification*, or *structure optimisation* (Sections 5.5.3.3, 5.5.5, 5.8.2, and 5.9.3).

**Model Parameter Estimation.** This stage deals with the task of employing some mathematical algorithms to find the appropriate parameters for the selected model structures. As already mentioned, the estimation algorithm to be used depends greatly on the selected model structure (linear vs. non-linear, parametric vs. non-parametric, discrete-time vs. continuous-time, deterministic vs. stochastic). The main distinction, however, has to be made between linear-in-the-parameters models, with parameters that can easily be estimated in one step by using a classical LS solver, and those having a more general structure, and thus requiring the application of more complicated optimisation algorithms. Of these, the Marquardt–Levenberg algorithm is the most popular and successfully applied method. Although these algorithms are thoroughly studied and well analysed, their use should not be imprudently considered, as all the nice advantageous properties of linear optimisation techniques mentioned above are lost.

To avoid non-linear optimisation, the HSS model may be linearised in a first step, which will be sufficient for many practical situations. Or linear-in-the-parameters polynomial models could be a good option.

**Model Validation.** This point is responsible for the approval of the adequacy of the model to represent the underlying system. This can be best shown in simulation. Although a number of validation methods are available (Section 5.10), a reasonable and largely accepted approach is to adopt cross-validation, *i.e.*, to use a data set that has not been utilised for identification in parallel model evaluation in order to validate the model. This is quite logical, since we are basically interested in a model that covers the process characteristics and not just reproduces a particular single time-series.

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<sup>12</sup> Note that the identification problems in hydraulic servo-systems are usually low- to medium-dimensional.

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## CHAPTER 6

# HYDRAULIC CONTROL SYSTEMS DESIGN

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The purpose of this chapter is to give a survey of basic and advanced control design methods for HSSs, including theoretical background and application to selected models derived in Chapters 4 and 5. The benefits and limitations associated with standard control design approaches are reviewed. Recent advances in the area of control theory, involving non-linear, “classical” and intelligent control techniques (specifically fuzzy control) will be utilised and applied to HSSs.

Each of the design methods – classical and modern alike – has advantages and disadvantages, strengths and limitations. It is thus the philosophy behind the presentation of this chapter that the designer needs to understand all of these to be in the best possible position to develop a satisfactory design with minimal effort. However, it has to be mentioned that this chapter cannot cover every detail of all existing linear and non-linear control strategies.

## 6.1 Introduction

Similar to other industrial control systems, the complete design process for hydraulic control systems normally encompasses the following steps:

1. *Modelling.* The determination of a mathematical model of the HSS either from physical equations (white-box modelling; see Chapter 4), or from experimental data using identification techniques (black-box modelling; see Chapter 5), or a combination of both approaches (grey-box modelling; see Chapter 5).
2. *Input–Output Controllability Analysis.* The (maximum) closed-loop performance that can be expected is examined. This helps in deciding upon an initial control structure.
3. *Control Structure Selection.* The variables to be manipulated and measured, as well as the links that should be made between them, are selected in this stage.
4. *Controller Design.* This encompasses the formulation of a mathematical design problem which captures the engineering design problem and the synthesis of a corresponding controller.
5. *Control System Analysis.* The control system is assessed by analysis and simulation against the performance specifications or the designer's experience.
6. *Controller Implementation.* In this stage the controller algorithm is coded (in software for computer control), taking care to address important issues such as anti-windup, bump-less transfer, and physical constraints.
7. *Control System Commissioning and Tuning.* Finally, the controller is brought online, to carry out onsite testing and to implement any required modifications (*i.e.*, tuning and optimisation of the controller parameters) before certifying that

the controlled HSS is fully operational, and meets the performance specifications.

In this chapter of the book, we will mainly focus on steps 4, 5 and 6. The reader interested in other steps may refer to the textbooks of Skogestad and Postlethwaite (1996) and Åström and Wittenmark (1997).

When considering HSSs, their controls have received considerable attention because they provide high-performance motion control at high power levels in operation and closed-loop position, velocity, and force control systems. Therefore, HSSs find a wide variety of applications in many industrial fields, for instance in machine tools, robotics, motions simulators, fatigue testing systems, rolling mills and so on. However, the dynamic characteristics of an HSS are complex and highly non-linear. The main non-linearities include the compressibility of the hydraulic fluid, the complex flow properties of the servo-valve and the friction in the hydraulic actuators.

Moreover, the operating conditions of, and the disturbances acting on, hydraulic systems vary in a complicated fashion; for instance the valve, oil and load parameters may vary significantly. Normally these parameters are not precisely known or time-variant for a great variety of reasons, *e.g.*, temperature-dependent behaviour. All these properties and facts make the control design and tuning difficult.

The main objectives for closed-loop control of hydraulic servo-systems are:

- Linearised input–output behaviour, which is consistent over the whole operating range.
- Sufficient damping in order to get better step response.
- Control bandwidth improvement, as much as allowed by the dynamics of the hydraulic system and the robust stability requirements imposed by unmodelled dynamics, as well as by parameter variations and disturbances.
- The size of the mechanical components and the flow rates should be kept at least unchanged.

An ideal controller would thus be robust against parameter and disturbance variations, and lead to best performance simultaneously. In practice, however, a *trade-off* has to be decided depending on the application at hand.

Many industrial controllers for an HSS achieve high bandwidth with fixed gain control laws by over-sizing the cylinder diameter in order to increase the effective stiffness of the fluid in the cylinder. This requires larger and more costly components and higher fluid flow rates in order to move a load at a given speed. A better approach to obtaining a fast response is to model the dominant dynamics of the system, and then to use this knowledge of the dynamics in the controller design. The advantage of such an approach is that, to achieve a given bandwidth, the mechanical components are smaller, the required flow rates are less, and the overall system is therefore much less expensive (Bobrow and Lum, 1996).

### 6.1.1 General Approaches

Although in most applications the control problem for the HSS can be described as the problem of controlling the piston motion, where the hydraulic actuator should exert the forces desired for this motion, a generalised control structure, shown in

Figure 6.1, can be considered, in which three parts are recognised (van Schothorst, 1997):

- **Trajectory Generation.** This part, possibly fed by some external drive signal  $r$ , generates the desired trajectory for the motion system. In many (industrial) applications, the trajectory generator (or pre-filter) creates just a desired position trajectory  $x_{\text{ref}}$ , which is to be followed by the system. More advanced trajectory generators, *e.g.*, in robot manipulators, also provide desired velocities  $\dot{x}_{\text{ref}}$  and accelerations  $\ddot{x}_{\text{ref}}$ , that can be used in the feedforward part.
- **Feedforward Control.** With the desired trajectory available, feedforward control can be used to generate signals to be supplied to the (feedback) control system, such that the controlled motion system realises the desired trajectory as good as possible. Actually, the optimal feedforward control uses an ideal inverse model of the system, to generate just those control signals that realise the desired trajectory. Although in the case where no feedback would be necessary, the feedback part is essential, because it is impossible to implement an exact inverse model of the motion system and to know all disturbances.
- **Feedback Control.** The feedback control can be seen as a corrective control, in order to cope with (modelling) errors in the feedforward path and with unknown disturbances. It combines the feedforward signals with the feedback of measured output signals of the system.

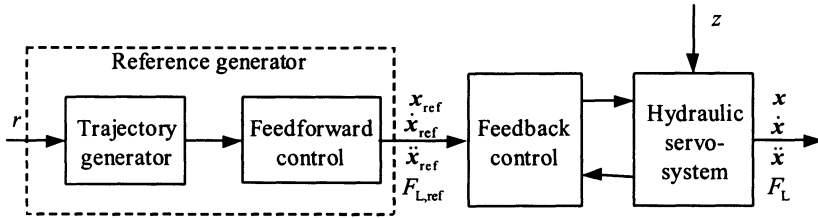


Figure 6.1. General control structure for HSSs

The trajectory generator is usually just a second-order low-pass filter, *e.g.*, Butterworth filter (see Section 5.5.3.1), driven by the external reference signal  $r(t)$  (typically a position reference), with adjustable cut-off frequency  $\omega_c$ :

$$\frac{1}{\omega_c^2} \ddot{x}_{\text{ref}}(t) + \frac{2}{\omega_c} \dot{x}_{\text{ref}}(t) + x_{\text{ref}}(t) = r(t) \quad (6.1)$$

By means of  $\omega_c$ , the frequency contents of the desired position–velocity–acceleration trajectory can be adjusted. In the high-frequency range the desired accelerations are proportional to  $\omega_c^2$ . So, in terms of physics, a high cut-off frequency  $\omega_c$  implies high demands on the system.

Considering the feedback control of hydraulic motion systems, some of the different strategies that are available are described below.

### 6.1.2 Literature Scan and Classification

In the field of hydraulic actuator control, a wide variety of control design techniques and applications are found. A complete overview and classification of control problems and control design techniques found in the literature is not an easy task.

In general, hydraulic servo-control problems are treated either as:

- *Position Control Problems* (Merritt, 1967; Viersma, 1980; von Wierchem, 1981; McClamroch, 1985; Cox and French, 1986; Kökemann, 1989; Lausch, 1990; Walters, 1991; Yun and Cho, 1991; Backé, 1992; Bona *et al.*, 1992; Heintze *et al.*, 1993; Klein, 1993; Boes, 1995; Zhao and Virvalo, 1995; Bernzen, 1999; Nissing, 2002). This is the standard application of hydraulic servo-control.
- *Velocity Control Problems* (Anderson, 1988; Backé, 1992; Chern and Wu, 1992; Jen and Lee, 1992; Lierschaft, 1993; Boes, 1995; Huang and Wang, 1995; Bernzen, 1999) or as force control problems (Pannala *et al.*, 1989; Karar and Rose, 1993; Ramachandran and Dransfield, 1993; Nissing, 2002). Velocity control is generally used for rotary drive applications, but also often included implicitly as velocity compensation in position control schemes.
- *Force Control Problems* (Sepehri *et al.*, 1990; Backé, 1992; Boes, 1995; Heintze and van der Weiden, 1995). This is typically the case in metal forming machines and testing machines.

Despite this clear distinction in different types of control problem, it is difficult to give a characterisation in terms of control design techniques that were used to solve the posed control problems. Actually, practical reasons like available computer power, experiences and preferences of the control designer, and available transducers seem to play an important role in the choice for a certain control design strategy (van Schothorst, 1997). Nevertheless, a modest classification and a brief survey of the literature of hydraulic servo-control is given in Table 6.1.

## 6.2 Classical Feedback Control Design

The standard approach to control hydraulic servo-systems is to apply linear feedback control design methods. The theoretical background of these methods can be found in a number of introductory control books such as Franklin *et al.* (1986), Föllinger (1990), Unbehauen (1993, 1994) and Lunze (2001). Refer also to Franklin *et al.* (1990), Unbehauen (1995), Skogestad and Postlethwaite (1996), Åström and Wittenmark (1997) and Lunze (2002) for advanced topics. In this section, only some selected design techniques are merely presented in a compact way.

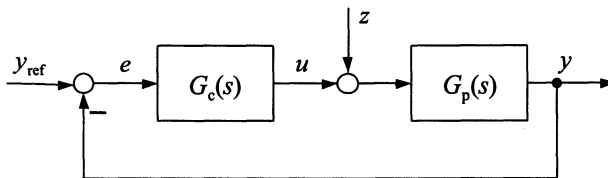
We start with recalling the principle of closed-loop control, already introduced in Section 2.4.1 (Figure 2.12), and let  $G_p(s)$  the (linear) transfer function of the system to be controlled, and  $G_c(s)$  the transfer function of the controller (Figure 6.2). Then, the transfer function of the closed-loop for the input  $y_{\text{ref}}$  is<sup>1</sup>

$$G_{\text{cl}}(s) = \frac{Y(s)}{Y_{\text{ref}}(s)} = \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)} \equiv \frac{G_o(s)}{1 + G_o(s)} \quad (6.2)$$

<sup>1</sup> Capital letters (e.g.,  $X$ ,  $Y$ ) stand for Laplace-transformed variables (e.g.,  $x$ ,  $y$ ).

**Table 6.1.** Literature overview about HSS control techniques

| Control technique                                                                | Selected references                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classical feedback control                                                       | Merritt (1967), Viersma (1980), Anderson (1988), Backé (1992)                                                                                                                                                                                                                                                                                                                                    |
| State feedback control                                                           | von Wierschem (1981), von Feuser (1984), Köckemann <i>et al.</i> (1991), Neumann <i>et al.</i> (1991a,b)                                                                                                                                                                                                                                                                                         |
| Extensions to feedback control (feedforward control, non-linearity compensation) | McClamroch (1985), Neumann <i>et al.</i> (1991), de Boer (1992), Lierschaft (1993), Heintze and van der Weiden (1995), Bobrow and Lum (1996), Polzer and Nissing (2000b), van Schothorst (1997)                                                                                                                                                                                                  |
| Predictive control                                                               | Kotzev <i>et al.</i> (1994), Stahl and Irle (1999)                                                                                                                                                                                                                                                                                                                                               |
| Adaptive control (self-tuning/learning control)                                  | Porter and Tatnall (1970), Kulkarni <i>et al.</i> (1984), Finney <i>et al.</i> (1985), Yun and Cho (1985, 1988, 1991), Saffe (1986), Hori <i>et al.</i> (1988), Goodhart <i>et al.</i> (1991), Köckemann <i>et al.</i> (1991), Boes (1992, 1995), Karar and Rose, (1993), Kotzev <i>et al.</i> (1994), Huang and Wang (1995), Bobrow and Lum (1996), Dutton and Groves (1996), Yu and Kuo (1997) |
| Variable structure control (sliding mode control)                                | Lee and Lee (1990), Chern and Wu (1992), Hwang and Lan (1994), Behmenburg (1995)                                                                                                                                                                                                                                                                                                                 |
| Feedback linearisation (input–output linearisation)                              | Hahn <i>et al.</i> (1994), Del Re and Isidori (1995), Vossoughi and Vossoughi and Donath (1995), Alleyne (1996), Bernzen (1999a), Pedersen (1999), Sohl and Bobrow (1999), Lemmen and Bröcker (1999, 2000), Lemmen <i>et al.</i> (2000), Tunay <i>et al.</i> (2001)                                                                                                                              |
| Fuzzy control                                                                    | Klein (1993), Behmenburg (1995), Boes (1995), Zhao and Virvalo (1995), Berger (1997)                                                                                                                                                                                                                                                                                                             |
| Neuro-control                                                                    | Burton <i>et al.</i> (1992), Plummer and Vaughan (1996)                                                                                                                                                                                                                                                                                                                                          |
| Bilinear control                                                                 | Naujoks and Wurmthaler (1988), Guo and Schwarz (1989), Beater (1987), Guo (1991), Yin (1992, 1994), Guo <i>et al.</i> (1994), Del Re and Isidori (1995), Schwarz <i>et al.</i> (1996)                                                                                                                                                                                                            |

**Figure 6.2.** Block diagram of a standard closed loop with input<sup>2</sup> disturbance  $z$ 

The transfer function for the disturbance  $z$  can be written as

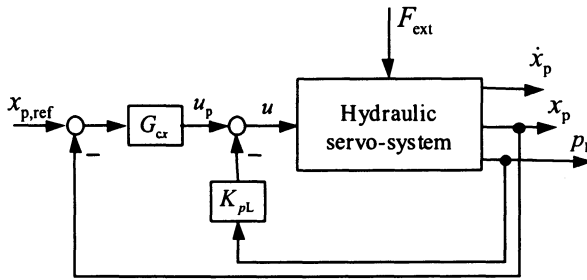
$$G_z(s) = \frac{Y(s)}{Z(s)} = \frac{G_p(s)}{1 + G_c(s)G_p(s)} \equiv \frac{G_p(s)}{1 + G_o(s)} \quad (6.3)$$

<sup>2</sup> This is only one standard assumption in linear control theory, but disturbance may also act on the output or on an internal state of the system.

A linear system allows superposition in that the output  $y$  is the combination due to both reference and disturbance inputs. The result of including both inputs is

$$Y(s) = G_{cl}(s)Y_{ref} + G_z(s)Z(s) \quad (6.4)$$

In most standard applications of hydraulic servo-control, only a position reference is available. The conventional control strategy for these types of application is proportional feedback of the actuator position to obtain servo-behaviour, while mostly a proportional pressure feedback loop is included for damping purposes (Merritt, 1967; Viersma, 1980); see Figure 6.3. The aim is to hold the position  $x(t)$  close to the position reference  $x_{ref}(t)$  in the presence of disturbances  $z$  (load or external force variations, supply pressure variations, *etc.*).



**Figure 6.3.** Position feedback including a load pressure feedback loop

To obtain more insight into the effectiveness of such a classical controller, the linearised models of a hydraulic system derived in Sections 4.4.4.3 and 4.4.4.4 are considered (neglecting the valve dynamics for simplicity). Since the model structure according to Figure 4.26 usually shows a (badly damped, *i.e.*, typically  $D_h = 0.05$ – $0.25$ ) second-order system in series with an integrator, it is clear that proportional position feedback results in tracking behaviour.

### 6.2.1 Pressure Feedback

First an analysis of the effect of a pressure feedback loop is performed. Application of the pressure feedback

$$U_l(s) = -K_{pL} P_L(s) \quad (6.5)$$

leads to the following closed-loop transfer function:

$$G_{cl}(s) = \frac{X_p(s)}{U_p(s)} = \frac{K_v K_Q^* T_h K_m}{T_h T_m s^2 + (T_h + T_m + T_m K_v K_Q^* T_h K_{pL})s + K_v K_Q^* T_h K_{pL} + T_h K_m K_d + 1} \frac{1}{s} \quad (6.6)$$

which is brought into the standard form of a  $PT_2$  in series with an integrator (see Equation 4.244). Easy calculations give the natural frequency  $\omega_{h,cl}$  and damping  $D_{h,cl}$  of the controlled system



$$\omega_{h,cl} = \sqrt{\frac{1 + T_h K_m K_d + K_v K_Q^* T_h K_{pL}}{T_h T_m}} \quad (6.7)$$

$$D_{h,cl} = \frac{T_h + T_m + T_m K_v K_Q^* T_h K_{pL}}{2\sqrt{(1 + T_h K_m K_d + K_v K_Q^* T_h K_{pL})T_h T_m}} \quad (6.8)$$

Provided the viscous friction coefficient  $\sigma$  is small (as it usually is for hydraulic systems), then Equations 6.7 and 6.8 show just a weak influence of the pressure feedback upon the natural frequency  $\omega_{h,cl}$ . On the other hand, the pressure feedback has an equivalent effect upon the damping  $D_{h,cl}$  as a *leakage term*. Consequently,  $D_{h,cl}$  of the pressure-controlled system is easily adjustable with the pressure feedback, at a very modest change of  $\omega_{h,cl}$ .

However, a hydraulic actuator exhibits a limited stiffness with respect to disturbing forces. Because the pressure feedback can be regarded as an artificially introduced leakage, pressure feedback will increase the disturbing effect of, for instance, Coulomb friction or stiction on the positioning accuracy (Heintze, 1997). For this, the disturbance transfer function

$$\begin{aligned} G_z(s) &= \frac{X_p(s)}{F_{ext}(s)} \\ &= \frac{-A_p K_m [(1 + sT_h) + K_v K_Q^* T_h K_{pL}]}{T_h T_m s^2 + (T_h + T_m + T_m K_v K_Q^* T_h K_{pL})s + K_v K_Q^* T_h K_{pL} + T_h K_m K_d + 1} \frac{1}{s} \end{aligned} \quad (6.9)$$

is considered. Thus the load stiffness can be determined as

$$\begin{aligned} c_{h,cl} &= \left. \frac{F_{ext}(s)}{\dot{X}_p(s)} \right|_{(s=0)} = \frac{1 + T_h K_m K_d + K_v K_Q^* T_h K_{pL}}{-A_p K_m [1 + K_v K_Q^* T_h K_{pL}]} \\ &= \frac{1 + T_h K_m K_d}{-A_p K_m} \frac{1 + \frac{K_v K_Q^* T_h K_{pL}}{1 + T_h K_m K_d}}{1 + K_v K_Q^* T_h K_{pL}} \end{aligned} \quad (6.10)$$

or equivalently

$$c_{h,cl} = c_{h,open} \frac{1 + 1 + T_h K_m K_d + K_v K_Q^* T_h K_{pL}}{1 + T_h K_m K_d + K_v K_Q^* T_h K_{pL} + K_v K_Q^* T_h^2 K_m K_d K_{pL}} \quad (6.11)$$

Since the second term on the right-hand side of this equation is less than unity, the load stiffness is reduced through pressure feedback.

To avoid this problem, a pressure feedback of the form

$$U_1(s) = -\frac{K_{pL} T_{pL} s}{1 + T_{pL} s} P_L(s) \quad (6.12)$$

should be used instead of the control law in Equation 6.5. The effectiveness of this kind of pressure feedback can be seen if the characteristic parameters are calculated again to give

$$\omega_{h,cl} = \sqrt{\frac{1 + T_h K_m K_d}{T_h T_m}} \quad (6.13)$$

$$D_{h,cl} = \frac{T_h + T_m + T_m K_v K_o^* T_h K_{pL}}{2\sqrt{(1 + T_h K_m K_d) T_h T_m}} \quad (6.14)$$

$$c_{h,cl} = \frac{1 + T_h K_m K_d}{-A_p K_m} \quad (6.15)$$

Thus, only the damping of the hydraulic system is affected by the pressure feedback gain  $K_{pL}$ .

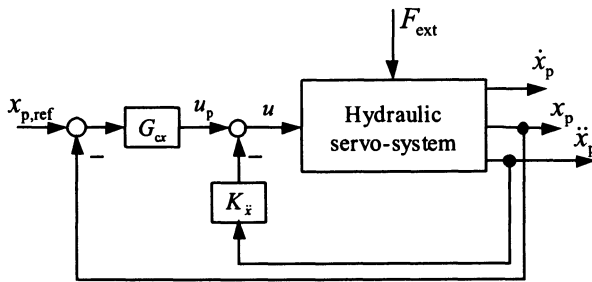


Figure 6.4. Position feedback including an acceleration feedback loop

## 6.2.2 Acceleration Feedback

As acceleration and pressure difference are closely related; see Equation 4.257, hydraulic damping  $D_{h,cl}$  can also be adjusted via acceleration feedback; see Figure 6.4. In this case, we can derive

$$D_{h,cl} = \frac{T_h + T_m + T_m K_v K_o^* T_h K_{\ddot{x}}}{2\sqrt{(1 + T_h K_m K_d) T_h T_m}} \quad (6.16)$$

$\omega_{h,cl}$  and  $c_{h,cl}$  are as in Equations 6.13 and 6.15 respectively. So, acceleration feedback also does not deteriorate the stiffness of the hydraulic drive system.

However, measuring acceleration is generally considered to be more complicated, and hence costly, than measuring  $p_L$ . Thus, in practical applications, there are often no sensors to measure the acceleration. Nevertheless, acceleration may be estimated by differentiation of position or velocity signals (see Section 6.10.1.3). However, the quality of the acceleration signal constructed will not be so impressive due to the noisy position signal and the phase delay involved through differentiation.

Acceleration feedback control has been used by many researchers to obtain high-performance hydraulic servo-mechanisms (Welch, 1962; Matsuura *et al.*, 1994; FitzSimons and Palazolo, 1996; Tafazoli *et al.*, 1998). FitzSimons and Palazolo (1996) proposed the root locus method to find the controller gains for a single-rod hydraulic actuator and presented simulation results. Welch (1962) discussed the quadratic resonance phenomenon that is generally observed in the transfer function

relating the load velocity to the servo-valve current. He then explained that acceleration feedback can be used to damp the hydraulic system so as to achieve a higher bandwidth. Welch assumed that the load is predominantly of inertial type (with no dissipation) and used the pressure readings to calculate acceleration. By using linear analysis of the system dynamics, he verified the effectiveness of adding a pressure feedback term (as a minor feedback loop) to a conventional PD controller. He also suggested high-pass filtering of the pressure feedback signal, to allow full pressure feedback in the frequency range of the load resonance. Welch's approach has been named "derivative pressure control", and experimental results using a hydraulic motor have been presented by this author.

A similar technique was employed by Matsuura *et al.* (1994) for damping out the resonance of a hydraulic cylinder in an industrial application. Another approach that combines friction estimation with an acceleration feedback law has been proposed by Tafazoli *et al.* (1996, 1998). They used the heuristic method of Studenny *et al.* (1991) to choose the controller gains and presented experimental results for an electro-hydraulic manipulator as an integral part of an automated fish processing machine.

So the trend to avoid, at all costs, the use of acceleration measurements does not seem to be appropriate. The acceleration can improve the tracking performance, it is also relatively easy and straightforward to measure, and the measurement can be done by structure-mounted devices. An increase in tracking accuracy by a factor of 1.5 may be possible in practice, as demonstrated by de Jager (1994), and by a factor 1.93, as achieved by Tafazoli *et al.* (1996), compared with a well-tuned PD controller.

### 6.2.3 Position Feedback

Next, to achieve a position tracking behaviour, feedback of position  $x_p$  is applied. This feedback mainly shifts the pole in zero into the left half plane, and has some influence upon the location of the complex conjugate pole pair, of which the damping in the first instance is adjusted by the pressure feedback. There are many options for position control, such as simple P-control.

To provide both stability and adequate damping, the position feedback gain  $K_p$  should be selected to meet the so-called  $M = 1.3$ -criterion (Viersma, 1980). This implies that the closed-loop amplitude (ratio) never exceeds 1.3 (called the frequency response resonant peak). Obeying this criterion, the closed-loop *step response* normally shows an overshoot of approximately 23% (which is exact for a second-order system). In good engineer's practice, the  $M = 1.3$ -criterion leads to the (well-known) amplitude margin of about two and phase margin (PM) of about 45°. Note that the phase margin for a second-order system is related to the damping ratio by  $D \approx \text{PM}/100$  (Franklin *et al.*, 1990).

Note further that if a  $\text{PT}_1$  controller (first-order lag), *i.e.*,

$$G_{cx}(s) = \frac{U_p(s)}{X_{p,\text{ref}}(s)} = \frac{K_p}{1+sT} \quad (6.17)$$

or a  $\text{PPT}_1$  controller, *i.e.*,

$$G_{cx}(s) = K_{p1} + \frac{K_{p2}}{1 + sT_p} \quad (6.18)$$

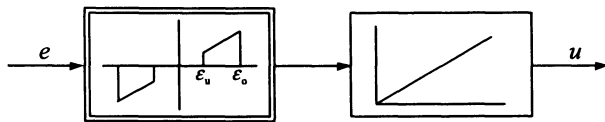
is used, *faster* closed-loop step responses are achieved. A summary of the suitability of conventional linear controllers for position control, velocity and force control is given in Table 6.2. See also Anderson (1988) for more details about classical control of electro-hydraulic systems with emphasis on velocity control.

For some applications of hydraulic servo-systems, a PI controller might be used in order to increase the open-loop gain at low frequencies and to decrease steady-state load errors as well as “acceleration” errors (Viersma, 1980). However, the integrator time constant should be considered carefully, as the transfer function of a closed loop then contains a double integrator, causing stability problems.

**Table 6.2.** Overview of linear controls performance on a scale from -- (worst) to ++ (best)

|                   | Linear controller |                   |                                        |                  |                                                |                           |                                    |
|-------------------|-------------------|-------------------|----------------------------------------|------------------|------------------------------------------------|---------------------------|------------------------------------|
|                   | P                 | I                 | PI                                     | PD               | PID                                            | PT <sub>1</sub>           | PPT <sub>1</sub>                   |
| Transfer function | $K_p$             | $\frac{1}{T_I s}$ | $K_p \left(1 + \frac{1}{T_I s}\right)$ | $K_p(1 + T_D s)$ | $K_p \left(1 + \frac{1}{T_I s} + T_D s\right)$ | $\frac{K_{p1}}{1 + sT_p}$ | $K_{p1} + \frac{K_{p2}}{1 + sT_p}$ |
| Position control  | +                 | --                | - / 0                                  | -                | 0 / -                                          | ++                        | ++                                 |
| Velocity control  | -                 | +                 | ++                                     | --               | ++                                             | + / 0                     |                                    |
| Force control     | -                 | +                 | ++                                     | --               | ++                                             |                           |                                    |

A practical compromise is to use a switching integrator that is only activated in an interval  $[\epsilon_u, \epsilon_o]$  around zero of the controller input; see Figure 6.5.



**Figure 6.5.** Inclusion of a switching integrator

## 6.2.4 Summary

The choice of feedback gains  $K_p$  and  $K_{pL}$  takes place in two steps:

1. The pressure difference feedback gain  $K_{pL}$  is tuned such that the damping ratio  $D_h$  of the resulting closed loop system (still with zero position feedback) is somewhere in the interval  $0.5 < D_h < 0.7$ , depending on the actual application and the desired system characteristics.
2. Thereafter, the position feedback gain  $K_p$  is increased to obtain position tracking, for instance until the so-called  $M = 1.3$ -criterion is met (Viersma, 1980). Other good practice is to choose the proportional gain so that the frequency  $\omega_{h,cl}$  of the controlled system lies somewhere in the region:  $1.3 \omega_{h,open} < \omega_{h,cl} < 1.7 \omega_{h,open}$ .

It is stressed here, that the main emphasis in this control strategy is on position control, where only a position reference  $x_{\text{ref}}$  is available. Pressure difference feedback is only used to modify the closed-loop dynamics (damping). The pressure feedback gain should not be too large as this would result in an over-damped system with restricted bandwidth. If available, acceleration feedback is preferable, as it improves the damping without deteriorating the stiffness with respect to force disturbances.

Note that the tuning of a controller for a reasonably performing HSS is, according to the insights stated above, a straightforward task without the need for a specific parameterised model, as is usual in practice. However, an arbitrary placement of the poles, such that a certain requested controlled behaviour of an HSS results, is not possible with this method.

## 6.3 Estimator-based State Feedback Control

It is well-known in control theory that all poles of a system can arbitrarily be placed by full state feedback to force a desired behaviour of the system. Thus, by adding velocity feedback and pressures (or pressure difference) feedback to the position control scheme of the previous subsection, the closed-loop poles of the hydraulic servo-system can be placed at desired locations. This method has been successfully employed in different applications (von Wierschem, 1981; von Feuser, 1984; Köckemann *et al.*, 1991). In an extended version of the method, references of all state variables are incorporated in the feedback loops (Neumann *et al.*, 1991a,b).

One of the attractive features of the state space design method is that the procedure consists of the following steps:

1. Design a control law under the assumptions that all elements of the state vector are available, and that temporarily the system has no reference signal ( $y_{\text{ref}} = 0$ ).
2. Design an estimator, also termed an observer, which estimates the entire (or a part of) state vector when provided with the measurement of some states.
3. Combining the control law and the estimator, where the control law calculations are based on the estimated states rather than the actual states.
4. Add a pre-filter or an integrator to remove the steady-state error in the case of a proportional state controller or to reject unknown disturbing effects acting on the system.

Figure 6.6 shows how the control law and the estimator fit together for HSSs, considering the linearised model Equation 4.234.

### 6.3.1 Computation of the State Control Law

Consider the linear time-invariant (LTI) state equations of the form

$$\begin{aligned}\dot{x}(t) &= Ax(t) + bu(t) \\ y(t) &= c^T x(t)\end{aligned}\tag{6.19}$$

There are two common possibilities to determine a state feedback law for such systems: the *pole assignment* approach and the *optimal control* approach. The latter is also well known as the *linear-quadratic regulator* (LQR) design approach.

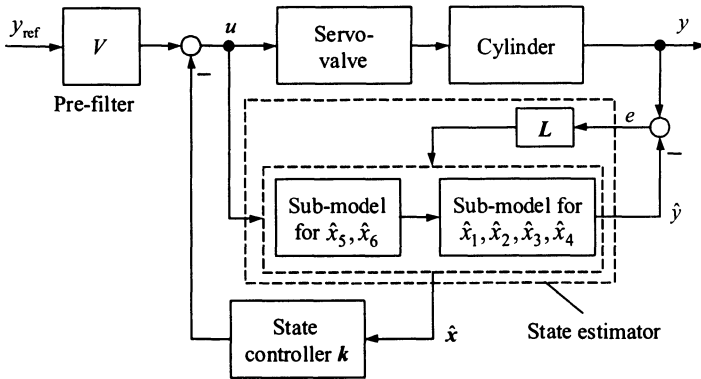


Figure 6.6. Linear observer-based state feedback control for HSSs

### 6.3.1.1 Pole Assignment

The first step is to temporarily assume  $y_{\text{ref}} = 0$  and to find a simple feedback law of the form

$$u = -\mathbf{k}^T \mathbf{x} = -[k_1 \quad k_2 \quad \dots \quad k_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (6.20)$$

Substituting the control law, Equation 6.20, into Equation 6.19 yields

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{b}\mathbf{k}^T) \mathbf{x} \equiv \mathbf{A}_c \mathbf{x} \quad (6.21)$$

and hence the characteristic equation of the closed-loop system is

$$C(s) = \det[s\mathbf{I}_n - (\mathbf{A} - \mathbf{b}\mathbf{k}^T)] = 0 \quad (6.22)$$

The control law design then consists of picking the gain vector  $\mathbf{k}$  so that the roots of Equation 6.22 are in desirable locations, *i.e.*,  $s = s_1, s_2, \dots, s_n$  such that

$$\begin{aligned} C(s) &= \prod_{i=1}^n (s - s_i) = (s - s_1)(s - s_2) \cdots (s - s_n) = 0 \\ &\equiv s^n + \alpha_1 s^{n-1} + \alpha_2 s^{n-2} + \cdots + \alpha_n \end{aligned} \quad (6.23)$$

Hence, the required elements of  $\mathbf{k}$  are determined by matching coefficients in Equations 6.22 and 6.23.

The shifting of the system poles by state feedback to any desired location is possible if, and only if, the system is (fully) controllable, *i.e.*, the controllability matrix

$$\mathbf{F} = [\mathbf{b} \quad \mathbf{A}\mathbf{b} \quad \mathbf{A}^2\mathbf{b} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{b}] \quad (6.24)$$

is of full rank  $n$ . An alternative method for solving the feedback gains is to use Ackermann formula (Ackermann, 1972)

$$\mathbf{k}^T = [0 \quad \dots \quad 0 \quad 1]FC(F) \quad (6.25)$$

where  $C(F)$  is defined as

$$C(F) = A^n + \alpha_1 A^{n-1} + \alpha_2 A^{n-2} + \dots + \alpha_n I_n \quad (6.26)$$

and the  $\alpha_i$  are the coefficients of the desired characteristic polynomial (Equation 6.23).

### 6.3.1.2 Optimal Control

The linear-quadratic regulator design (LQR) for time-continuous LTI systems (Equation 6.19) calculates the optimal gain vector  $\mathbf{k}$  such that the state-feedback law (Equation 6.20) minimises the cost function

$$J = \frac{1}{2} \int_0^\infty \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + u^T(t) R u(t) dt \quad (6.27)$$

with appropriately selected semi-definite matrix  $\mathbf{Q}$  and positive (definite matrix)  $R$ . ( $R$  is here a scalar due to the consideration of SISO systems) The optimal gain vector  $\mathbf{k}$  is given by

$$\mathbf{k}^T = \mathbf{k}_{\text{opt}}^T = R^{-1} \mathbf{b}^T \mathbf{S} \quad (6.28)$$

where  $\mathbf{S}$  is the solution of the algebraic Riccati matrix equation

$$\mathbf{S} \mathbf{A} + \mathbf{A}^T \mathbf{S} - \mathbf{S} \mathbf{b} R^{-1} \mathbf{b}^T \mathbf{S} + \mathbf{Q} = 0 \quad (6.29)$$

Design parameters are the matrices  $\mathbf{Q}$  and  $R$ .  $\mathbf{Q}$  is usually selected to be diagonal, often set to the identity matrix.  $R$  is used to penalise the control action. Changing the relative magnitude between the elements of the weighting matrices means a trade-off between the speed of the recovery and the magnitudes of the control signals. This process of finding the relative weights is difficult, and is usually done by trial-and-error.

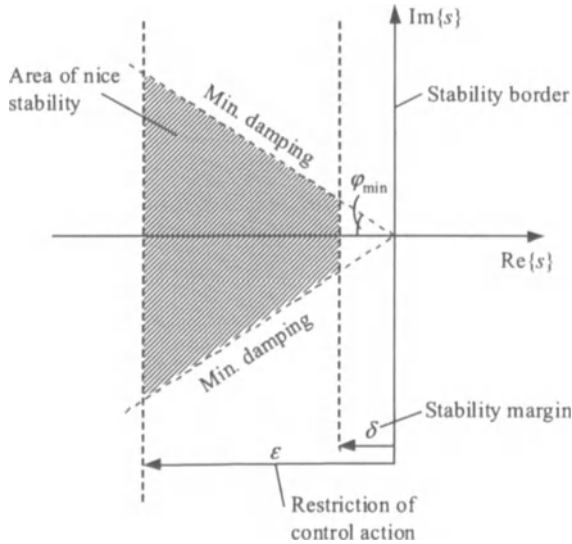
### 6.3.2 Selection of Pole Locations

The design of the state feedback gains usually takes place by standard design methods, for instance pole-placement, as described above. Thereby, the poles can be placed in arbitrary patterns to shape the closed loop dynamics of HSSs, where the main emphasis is on fast tracking of the position reference signal.

It is generally recommended to place the poles in the area of “nice stability”, which is enveloped by the following boundaries (see Figure 6.7):

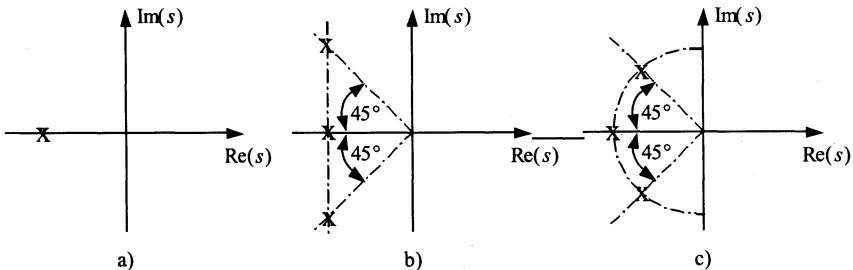
- Stability border (imaginary axis) diminished by a safety margin  $\delta > 0$ .
- Two straight lines of constant minimal damping  $D_{\min} = \cos \varphi_{\min}$ .
- Maximal distance  $\varepsilon > 0$  from stability border in order to restrict the control actions, and suppress the measurement noise.

Typically, the closed-loop poles are chosen as a desired pair of dominant second-order poles, with the rest of the poles selected to have real parts corresponding to sufficiently damped modes, so that the closed-loop system will mimic a second-order system with reasonable control effort (Franklin *et al.*, 1986). Recommended pole constellations for hydraulic systems (here for the reduced linear model in Equation 4.250 or 4.252 with  $n = 3$ ) are:



**Figure 6.7.** Definition of nice stability region

- (a) All poles are placed in the same location on the real axis (binominal form); see Figure 6.8a. The closed-loop system thus will show maximal damping  $D = 1$ , *i.e.* no overshoot (so-called non-periodic case).
- (b) All poles have the same real part, and two of them form a complex pole pair with an angle of  $45^\circ$ ; see Figure 6.8b. The second-order poles are selected to have a damping ration of  $D = \cos(45^\circ) \approx 0.7$ . The third pole has the same real part.
- (c) The poles are located on a circle; see Figure 6.8c. The second-order poles are chosen as in (b). The pole with the real part is moved a little bit to the left, so the closed-loop dynamic will accordingly be faster.



**Figure 6.8.** Typical pole constellations for state feedback of HSSs



### 6.3.3 Elimination of Steady-state Errors

If the system is Type 1 (Integral) or higher<sup>3</sup> (such as HSSs for position control) and  $y_{\text{ref}}$  is a step, there will be no steady-state error, *i.e.*,  $y(\infty) \equiv y_{\text{ss}} = y_{\text{ref,ss}} \equiv y_{\text{ref}}(\infty)$ . For Type 0 (Proportional) systems (such as HSSs for *velocity*, *pressure* or *force* control), there will be an error because some control is required to maintain the system at the desired state. To solve this problem, two common solutions are available: the design of a *pre-filter* for the reference signal or the introduction of *integral action*.

#### 6.3.3.1 Design of Pre-filter

A pre-filter can be designed as

$$V = [c^T (bk^T - A)^{-1} b]^{-1} \quad (6.30)$$

to eliminate steady state error, *i.e.*, to ensure that the steady-state values  $y_{\infty}$  and  $y_{\text{ref},\infty}$  are equal ( $y_{\infty} = y_{\text{ref},\infty}$ ). The total control law is then

$$u = Vy_{\text{ref}} - k^T x \quad (6.31)$$

Thus, the state-space description

$$\dot{x} = (A - bk^T)x + bVy_{\text{ref}} \quad (6.32)$$

is obtained for the closed-loop system (with  $y_{\text{ref}}$  as the input).

#### 6.3.3.2 Integral Control by State Extension (“PI State Feedback Control”)

The principle of this approach is to extend the model Equation 6.19 with an “artificial” integrator, thus adding an integral output to the existing output. That means to augment the state vector  $x$  with  $x_i$ , which obeys the differential equation

$$\dot{x}_i = y_{\text{ref}} - y = y_{\text{ref}} - c^T x \quad (6.33)$$

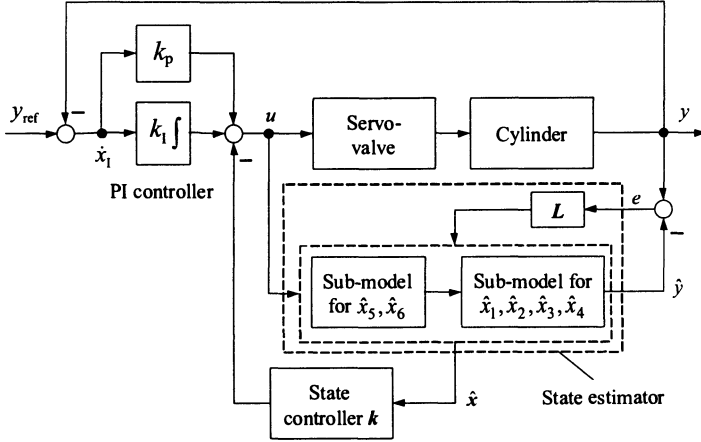
Therefore, the extended system model is obtained

$$\begin{bmatrix} \dot{x}_i \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & -c^T \\ 0 & A \end{bmatrix} \begin{bmatrix} x_i \\ x \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix} u \quad (6.34)$$

The PI state control law is

<sup>3</sup> The ability of a system to follow (reference) polynomials is given by the highest degree,  $k$ , of the polynomial, for which the (steady-state) error is finite but non-zero. The system is called type  $k$  to identify this polynomial degree. For example, a type 0 system has finite, non-zero steady-state error to a zero-degree polynomial (step), and a type 1 system has finite, non-zero error to a first-degree input (ramp). Under unity feedback, a system is said to be of type  $k$  if its transfer function has a pole of order  $k$  at the origin ( $s = 0$ ) (Franklin *et al.*, 1990).

$$u = -\begin{bmatrix} k_1 & (k + k_p c)^T \end{bmatrix} \begin{bmatrix} x_1 \\ x \end{bmatrix} \equiv -\begin{bmatrix} k_1 & k_1^T \end{bmatrix} \begin{bmatrix} x_1 \\ x \end{bmatrix} \equiv -\tilde{k}^T \tilde{x} \quad (6.35)$$



**Figure 6.9.** Linear observer-based state feedback control with augmented integral control action (PI state controller)

This will result in the control structure shown in Figure 6.9. With this extended definition of the system, the design techniques (pole placement and optimal control) already presented in Section 6.3.1 can be used in similar fashion to determine  $\tilde{k}$  and thus  $k_1$  and  $k$ . As  $k_p$  should be selected to be (Föllinger, 1990)

$$k_p = -(c^T A^{-1} b)^{-1} \quad (6.36)$$

the calculation of  $k$  is straightforward

$$k = k_1 - k_p c \quad (6.37)$$

The PI state controller is able to reject unknown disturbing effects acting on the system, *e.g.*, through parameter variations. Note that the observer (or estimator) design has to be performed based on the extended model Equation 6.34.

### 6.3.4 Application to Hydraulic Servo-system Linear Models

Formally, a state feedback controller can be designed on the basis of the linear model Equation 4.234. However, this would be much more complicated than a controller design based on the simplified model Equation 4.250 (where the valve dynamics is neglected). Then the feedback law has the form

$$u = -\tilde{k}^T \tilde{x} = -\begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = -\begin{bmatrix} K_p & K_x & K_{p_L} \end{bmatrix} \begin{bmatrix} x_p \\ \dot{x}_p \\ p_L \end{bmatrix} \quad (6.38)$$

to give the system matrix of the closed loop (Equation 6.21)

$$A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{1}{T_m} & \frac{A_p}{m_p} \\ -K_v K_Q^* K_p & -K_d - K_v K_Q^* K_i & -\frac{1}{T_h} - K_v K_Q^* K_{pL} \end{bmatrix} \quad (6.39)$$

The application of Equations 6.22 and 6.23 leads to the controller gains

$$K_p = \frac{\alpha_2}{K_v K_Q^*} \frac{m_p}{A_p} \quad (6.40)$$

$$K_{\dot{x}} = \frac{\alpha_2 - \frac{1}{T_m T_h} - \frac{\alpha_1 - \left( \frac{1}{T_m} + \frac{1}{T_h} \right)}{T_m} - K_d \frac{A_p}{m_p}}{K_v K_Q^* \frac{A_p}{m_p}} \quad (6.41)$$

$$K_{pL} = \frac{\alpha_1 - \left( \frac{1}{T_m} + \frac{1}{T_h} \right)}{K_v K_Q^*} \quad (6.42)$$

As before, the coefficients  $\alpha_i$  stand in relation to the desirable pole locations in the specific application.

## 6.4 Extensions to Linear Feedback Control

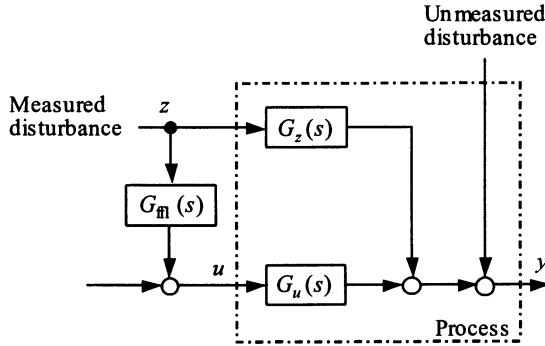
The controllers addressed up to now are all based on the principle of linear feedback. On the one hand, the main purposes of using feedback are to stabilise unstable systems and to reduce the effects from possible disturbances and model inaccuracies. However, fast reference tracking using feedback usually has the side effect that the controller becomes highly sensitive to noise, which is unavoidable in practice. On the other hand, linear control design cannot ensure the same dynamic behaviour of the closed-loop system all over the operating range. Thus linear feedback control should be combined with other techniques to achieve satisfactory results in practice.

There is a number of (non-linear) extensions to the (linear) feedback control approaches presented so far, such as the application of feedforward control, gain scheduling or adaptive methods, and the compensation for specific non-linearities. The principles and benefits of some of these methods are explained in this section.

### 6.4.1 Combined Feedback and Feedforward Control

Feedforward control is a control method that eliminates disturbances that can be measured. The basic idea is to employ the measured disturbance to anticipate the

influence of the disturbance on the process variables and to provide suitable compensating actions. The generic principle of this approach is illustrated in Figure 6.10. Compared with feedback control, feedforward control has the advantage that corrective actions may be applied *before* the disturbance has influenced the variables. Thus feedforward control allows rapid reference tracking without affecting the system stability.



**Figure 6.10.** Principle of feedforward control for disturbance compensation

Consider the transfer functions  $G_z(s)$  and  $G_u(s)$  relating the output  $y$  to the disturbance  $z$  and the control  $u$ , respectively; see Section 4.4.4.4. A “perfect” cancellation of the disturbance is obtained by means of

$$U_{ff}(s) = G_m(s) Z(s) = -G_u^{-1}(s) G_z(s) Z(s) \quad (6.43)$$

provided that it is possible to obtain and physically realise the inverse  $G_u^{-1}(s)$ . This may, of course, not be true, *e.g.*, due to time delay that occurs when the measurement of the disturbance is made in advance of its appearing to the process. In that case, a suitable approximation has to be chosen instead.

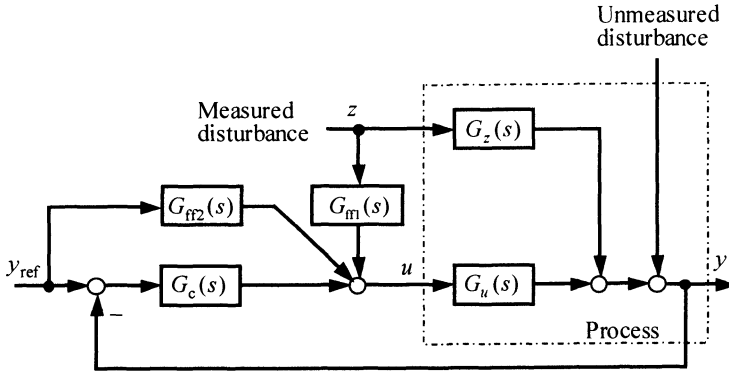
The feedforward control scheme can be extended, assuming that plant and controller are stable and that all the disturbances are known, to yield “perfect” control:

$$Y(s) = G_u(s)U(s) + G_z(s) Z(s) \stackrel{!}{=} Z_{ref}(s) \quad (6.44)$$

which gives

$$U(s) = G_u^{-1}(s) Z_{ref}(s) - G_u^{-1}(s) G_z(s) Z(s) \quad (6.45)$$

Unfortunately,  $G_u(s)$  and  $G_z(s)$  are never exact models, and the disturbances are never known exactly. In addition, a very active control signal with high amplitude is often encountered (if no filtering is used). Thus, although feedforward control can take some anticipative action for disturbances, it cannot, on its own, guarantee good control quality, and a combination with feedback control is necessary to reduce the effect of model and (disturbance) signal uncertainties, as shown in Figure 6.11.

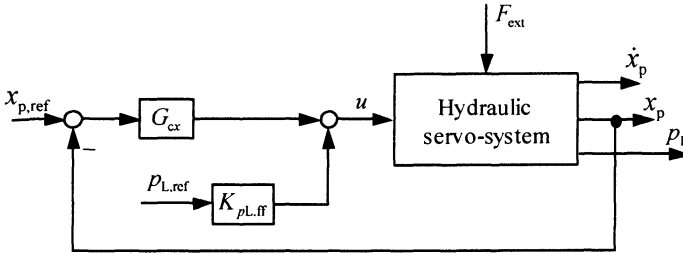


**Figure 6.11.** Universal feedforward control scheme

Feedforward control is particularly useful for disturbances generated by changes in the command or reference signals or for cascade processes when downstream disturbances are generated by variations in processes upstream. The design of feedforward compensators is often based on simple static models. Then the transfer functions  $G_{ff1}(s)$  and  $G_{ff2}(s)$  are simply static gains.

The application of such a control scheme to HSSs by supplying references for other states, *i.e.*, velocity and load difference is addressed in the following. The discussion is based on the work of van Schothorst (1997).

#### 6.4.1.1 Pressure-difference Feedforward Control



**Figure 6.12.** Position feedback control combined with load pressure feedforward control

If a pressure-difference reference is available, it can be fed forward to the pressure-difference feedback loop, as indicated in Figure 6.12. The pressure-difference feedforward gain can be calculated as the inverse of the static gain of the transfer function in Equation 4.258:

$$K_{pL,ff} = \frac{1}{G_{pL}(s=0)} = \frac{m_p \omega_h^2}{K_v K_Q^* \sigma} = \frac{\frac{\sigma}{T_h} + \frac{A_p^2}{C_h}}{A_p \sigma K_v K_Q^*} \quad (6.46)$$

Owing to the simultaneous supply of a position reference  $x_{\text{ref}}$  and a pressure-difference reference  $p_{L,\text{ref}}$ , the feedback control design gets the character of a trade-off between position control and pressure control.

Especially when the model knowledge used to generate the  $p_L$ -reference is rather accurate, pressure control can be emphasised in this trade-off. So the pressure feedback gain can be increased, at the cost of a smaller position feedback gain. Although this leads to over-damped closed-loop dynamics, the input–output behaviour can be considerably improved, especially in the high-frequency region, where acceleration forces play an important role. Moreover, when the estimated  $m_p$  is correctly adapted to varying load conditions, the high-frequency input–output behaviour will be less sensitive to these varying load conditions.

An extra advantage of  $p_L$ -reference feedforward is that it allows compensation for external forces, provided that an estimation  $F_{\text{ext}}$  is available; see Section 6.10.1.3.

#### 6.4.1.2 Velocity Feedforward Control

Considering the model structure of the hydraulic actuator, Figure 4.26, from the view of position servo-control, it can be seen that, under unity feedback, position references with constant velocity can only be tracked with a constant tracking error (a so-called type 1 system). In terms of physics, a constant velocity motion requires a constant valve flow, implying a constant tracking error in the case of feedback control (van Schothorst, 1997). An obvious method to avoid this tracking error is the application of velocity feedforward, which is of course only possible if a velocity reference  $\dot{x}_{\text{ref}}$  is available; see Figure 6.13.

The design of the velocity feedforward gain  $K_{\dot{x},\text{ff}}$  is straightforward, and should be chosen as the inverse of the static gain of the transfer function in Equation 4.255:

$$K_{\dot{x},\text{ff}} = \frac{1}{G_v(s=0)} = \frac{m_p \omega_h^2}{K_v K_\varrho^* A_p} = \frac{\frac{\sigma}{T_h} + \frac{A_p^2}{C_h}}{A_p K_v K_\varrho^*} \quad (6.47)$$

Neglecting the friction term (*i.e.*,  $\sigma = 0$ ), this can be simplified to

$$K_{\dot{x},\text{ff}} = \frac{A_p}{K_v K_\varrho^*} \left( \frac{E_A'}{V_A} + \alpha^2 \frac{E_B'}{V_B} \right) \quad (6.48)$$

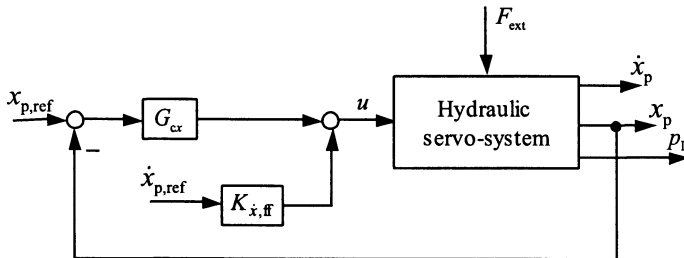
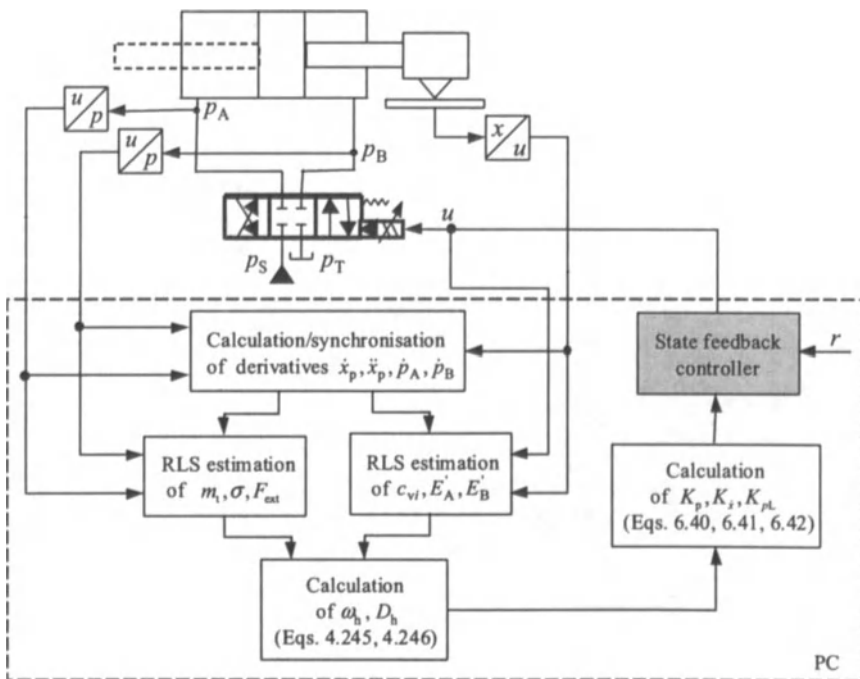


Figure 6.13. Position feedback control combined with velocity feedforward control

Thus far, velocity feedforward is applied whenever an explicit velocity reference is available. However, velocity feedforward may be also integrated implicitly by velocity compensation and outer loop feedback in a cascade control scheme. An important by-product of velocity feedforward is the compensation for the effect of the piston velocity on the mass balance such that the pressure dynamics and the load dynamics are decoupled; see also Section 6.6.2.

## 6.4.2 Adaptive Control

As explained in the previous section, the standard approach to control HSSs relies on local (Jacobian) linearisation of the dynamics around an operating point, followed by linear control design. However, the operating conditions and (load) disturbance of hydraulic systems vary in an unknown fashion; for instance the valve, oil and load parameters may vary significantly. (Note also that parameters may change due to ageing.) Thus, any conventional approach to controller design with constant feedback gains or modern control design methodology without considering the variation of the operating condition and load disturbance cannot obtain the optimum performance. Even unstable system operation can occur. To assure satisfactory performance, the controller parameters have to be redesigned repeatedly when the process is operating at different operating conditions and with different load disturbances.



**Figure 6.14.** Adaptive state feedback control scheme based on grey-box models

This leads directly to the application of adaptive control techniques, which have been proposed and explicitly applied for HSSs by many authors. To mention just a

few: Plummer and Vaughan (1996) suggested indirect adaptive control based on the linearised model around the mid-stroke of the cylinder; Bobrow and Lum (1996) presented an alternative, which used Jacobian linearisation around the desired trajectory, followed by online state model identification and linear quadratic control.

In line with these approaches, the online identification scheme of Section 5.7 (Figure 5.22) can now be extended for use in an adaptive control scheme, as can be seen in Figure 6.14; see also Boes (1992, 1995) for in-depth treatment of this control approach. Of course, a parametric black-box model, say an ARX model, can be used instead of the grey-box (linear) model; see for example Yu and Kuo (1997), who proposed a robust indirect adaptive control algorithm with self-excitation capability for position control of an electro-hydraulic servo-system subject to variations of the operating conditions and load disturbances.

### 6.4.3 Compensation of Special (Static) Non-linearities

All the design methodologies presented up to now do not take into account the non-linear effects that exist in hydraulic drives, such as Coulomb friction, backlash and non-linear valve flow characteristics. How some non-linearity compensations can be added to a linear controller to improve the performance of the closed loop system will be discussed below.

The non-linear characteristic  $\mathcal{NL}(x)$  can be compensated for by implanting an estimated inverse  $\mathcal{NL}^{-1}(x)$  of the non-linearity in the control loop, just before the control signal enters the system. This assumes that the non-linearity can be experimentally determined or estimated by any kind of model or observer.

#### 6.4.3.1 Compensation for Non-ideal Valve Geometry

An offset correction term can be added to the control effort to compensate for dead-band behaviour, *e.g.*, due to hysteresis, overlapping or radial clearance of the servo-valve (as mentioned in Section 4.2.1.1) and to reduce steady-state errors (Figure 6.15). The right value for such a correction term may be determined experimentally from inverting the flow-signal function  $Q(u)$  (Section 4.5.1.1); see also Lierschaft (1993). Otherwise, an estimate may be extracted from manufacturers' catalogue.

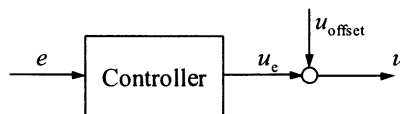


Figure 6.15. Addition of an offset correction

In particular, most servo-valves have an underlap (closed centre) or an overlap (open centre), although zero-lap is highly desirable. To circumvent this problem for controller design, the valve compensation due to Polzer and Nissing (2000b) can be used.

For this purpose, the piston velocity for an input voltage  $u$  is measured for the static case, characterised by  $\ddot{x}_p = 0$  and  $\dot{p}_A = \dot{p}_B = 0$ . By means of the measured pressures, it is possible to calculate a theoretical piston velocity that depends on the input voltage  $u$ . The required relationship can be derived from subtracting the



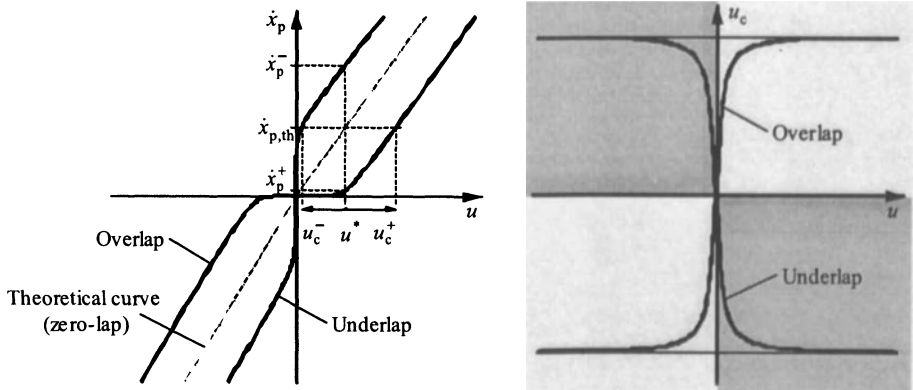
pressure dynamic Equations 4.198 and 4.199 for  $\dot{p}_A$  and  $\dot{p}_B$  from each other to give

$$\dot{x}_{p,th} = \begin{cases} \frac{c_v (\sqrt{p_s - p_A} + \sqrt{p_B - p_T})}{A_p (1 + \alpha)} u & \forall u \geq 0 \\ \frac{c_v (\sqrt{p_A - p_T} + \sqrt{p_s - p_B})}{A_p (1 + \alpha)} u & \forall u < 0 \end{cases} \quad (6.49)$$

which depends upon the direction of motion, since the piston's surface area is different for positive and negative velocities. Typical input–velocity curves are given in Figure 6.16 (left) for valves with overlap or underlap.

To compensate for the overlap, and thus ensure that Equations 4.198 and 4.199 and the controller design are valid, the difference between the two curves and the zero-lap curve  $\dot{x}_{p,th}(u)$  is established and used to represent the valve compensation  $u_c$ . The compensation plot is given in Figure 6.16 (right) and shows the valve offset. For a given input voltage  $u^{unc}$  (from the controller) an additional compensation voltage  $u_c$  has to be added so that the theoretical piston velocity can be reached by the total output

$$u = u^{unc} + u_c \quad (6.50)$$



**Figure 6.16.** Valve curve and its compensation curve (Taken from Nissing (2002), p 17 (original publication). Reprinted with kind permission of VDI Verlag.)

### 6.4.3.2 Compensation for Non-linear Flow Characteristic

Examples of this compensation technique are available in the literature, where the square-root expression is inverted and used for control. Neumann *et al.* (1991a,b) show non-linear simulation results and conclude there is considerable performance improvement. They use an observed pressure difference for the compensation, because the actual pressure difference is not measured. Lierschaft (1993), and later Heintze and van der Weiden (1995), showed experimental results, in which the non-

linear load compensation is included, where the measured pressure difference is used for compensation.

The static relationship in Equation 4.212 can be rewritten as

$$Q_L = c_v u \sqrt{2 p_{S,\max}} \sqrt{\frac{p_S - \text{sign}(u) p_L}{p_{S,\max}}} \quad (6.51)$$

where  $p_{S,\max}$  is the maximum supply pressure. From this, it can be directly seen that the non-linear load (and supply pressure changes) can be compensated for by

$$u = u^{\text{unc}} \sqrt{\frac{p_{S,\max}}{p_S - \text{sign}(u) p_L}} \quad (6.52)$$

However, most of these compensation techniques have mainly been applied to symmetric cylinders. One exception is Boes (1995), who recognised that, for load compensation of differential cylinders, the cylinder area ratio  $\alpha$  plays an important role.

For differential cylinders, the corresponding (more complicated) relations can be derived from Equations 4.196 and 4.197 (without leakage term) as

$$u = \frac{u^{\text{unc}}}{A_p \left[ \frac{E'_A}{V_A(x_p)} c_{v1} \sqrt{p_S - p_A} + \alpha \frac{E'_B}{V_B(x_p)} c_{v4} \sqrt{p_B - p_T} \right]} \quad (6.53)$$

for  $u \geq 0$  and

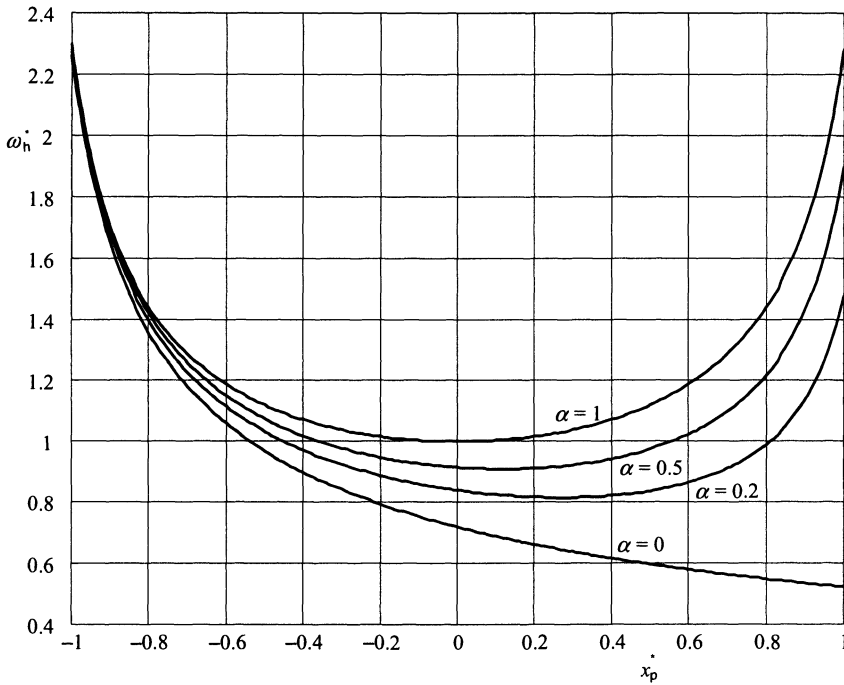
$$u = \frac{u^{\text{unc}}}{A_p \left[ \frac{E'_A}{V_A(x_p)} c_{v2} \sqrt{p_A - p_T} + \alpha \frac{E'_B}{V_B(x_p)} c_{v3} \sqrt{p_S - p_B} \right]} \quad (6.54)$$

for  $u < 0$ . A “nice” by-product of this compensation technique is that the system (velocity) behaviour becomes symmetric.

### 6.4.3.3 Switching of Controller Gains

Although less important than the flow non-linearity of the hydraulic actuator, the position dependence of the actuator dynamics has to be considered with respect to the closed-loop performance of the servo-system.

The position dependence of the parameters  $\zeta_5$  and  $\zeta_6$  in Equations 4.234 and 4.235 makes the stiffness of the actuator, and therewith its natural frequency, position dependent; see Figure 6.17. For instance, symmetric actuators have the lowest stiffness in the middle position. The situation is more complicated for differential (asymmetric) cylinders.



**Figure 6.17.** Natural frequency as a function of piston position (normalised with respect to scaled the maximum stroke) and area ratio  $\alpha$

The effect of this stiffness on the closed-loop performance depends also on the control strategy, *i.e.*, on the feedback gain setting:

- If position control is the primary aim, relatively high position feedback gains will be chosen. In this case, the position feedback gain is critical for stability; the feedback gain and the attainable performance is restricted by the actuator stiffness. A high stiffness allows high feedback gains. So a robust feedback design is based on the lowest stiffness, for instance the middle position for a symmetric actuator (Viersma, 1980).
- If pressure control is the primary objective, the pressure difference gain is to be chosen as high as possible. For the third-order (symmetric) actuator model, this pressure difference gain is not principally restricted. The effect of the position dependence of the actuator is that, for a certain pressure-difference feedback settings the bandwidth of the pressure difference control loop increases with increasing actuator stiffness. However, if this bandwidth is to be restricted for robustness reasons, the feedback design has to be based on the highest actuator stiffness occurring in the actuator stroke.

Note that switching of controller gains has to be considered if a differential cylinder is to be used in an HSS.

#### 6.4.3.4 Velocity Compensation for Differential Cylinders

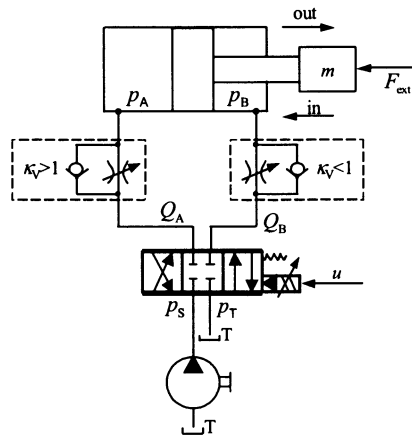
For HSSs, it is desirable to have the same (absolute) velocity independent of the sign of valve control input (but with the same absolute value); see Saffe (1986) and Lemmen (1995). As this is not inherently the case for systems including a differential cylinder, a compensation may be needed to enforce the desired (symmetric) velocity behaviour, *i.e.*,

$$r_v(p_s, p_T, p_L) = \frac{\dot{x}_{p,\text{in}}}{\dot{x}_{p,\text{out}}} = \sqrt{\frac{\alpha p_s - p_T + p_L}{p_s - \alpha p_T - p_L}} = 1 \quad (6.55)$$

This implies that a correction gain

$$\kappa_v = \frac{p_s - \alpha p_T - p_L}{\alpha p_s - p_T + p_L} \quad (6.56)$$

has to be applied.



**Figure 6.18.** Velocity compensation for differential cylinders using throttle-check valves (Taken from Lemmen (1995), p 64 (original publication). Reprinted with kind permission of VDI Verlag.)

To calculate the correction gain, the parameters  $p_s$ ,  $p_T$ , and  $p_L = F_L/A_p$  must be known or estimated with sufficient accuracy. An in-depth discussion of the influence of these parameters on the quality of the velocity compensation can be found in Lemmen (1995). The interested reader is encouraged to consult this work for details. A practical scheme to implement such a velocity compensation using throttle-check valves is shown in Figure 6.18.

Note that the compensation feature discussed here is included in the compensation for non-linear flow characteristics presented in Section 6.4.3.2.

### 6.4.4 Conclusions and Drawbacks of Classical Approaches

The common design steps of the classical control approach are: first, an integral model of actuator and load is derived from the linearisation of a non-linear model (or by means of direct identification of a linear model); then a model-based control design technique is applied to the linear model. Such an approach has a number of drawbacks (Heintze, 1997):

1. In the case of the linear model-based control design, obviously the linearised model is only valid in a small region around the operating point. Thus, linear control design methods cannot guarantee the same dynamic behaviour of the closed loop all over the operating range.
2. Full parametric knowledge is necessary in the case of an arbitrary pole placement or for optimal control. This can be particularly problematic for a number of parameters related to the hydraulic part of the system; see Chapters 4 and 5.
3. The application, *i.e.*, implementation and tuning for a specific situation, of a full state feedback controller according to Section 6.3 may often be not transparent for non-control-specialists. Therefore, the practical application in industrial situations may be difficult.

To tackle the difficulties stated in these items, some authors proposed methods of robust control design ( $H_\infty$ : H-infinity); for instance see Lambrechts (1994) and Tunay *et al.* (2001). However, as the uncertainties in the bulk modulus, for example, can be 50% or more, very conservative controllers will result from these methods. Moreover, these control design procedures and the resulting controllers are so complex for  $H_\infty$  control that the practical use of such a robust design method can be questioned (Heintze, 1997).

The next sections introduce some recent approaches for the control design for hydraulic drive systems that permits one to avoid the above-mentioned drawbacks.

## 6.5 Feedback Linearising Control

Feedback linearisation is an approach to non-linear control design that has attracted great research interest in the last two decades. The general idea of the approach is to transform non-linear system dynamics algebraically into a (fully or partly) linear one, so that linear control techniques can be applied. This differs entirely from conventional linearisation (*i.e.*, Jacobian linearisation), in that feedback linearisation is achieved by exact state transformations and feedback, rather than by linear approximations of the dynamics. In its simplest form, feedback linearisation amounts to cancelling the non-linearities in a non-linear system so that the closed-loop dynamics are in a linear form (Slotine and Li, 1991).

Full feedback linearisation is usually known as *exact linearisation* or *input-state linearisation* (via feedback). If the (restrictive) conditions required for exact linearisation are not satisfied, there is no way to obtain a linear controllable system via feedback and change of coordinates. However, it is always possible to achieve a decomposed system, in which one of the two component subsystems is linear; one may then wish at least search for a feedback and a coordinates transformation that (if possible) maximises the dimension of the linear subsystem (Isidori, 1995:194ff). This leads directly to the concept of *input-output linearisation*.

Extensive theoretical developments of the design and analysis of non-linear control systems, especially of exact linearisation, can be found in Isidori (1995) and Slotine and Li (1991), from which the definitions and theorems of Section 6.5 are adapted. The reader may also consult the books by Nijmeijer and van der Schaft (1990) and Schwarz (1991, 2000), which contain many interesting results on this topic. The basic concepts of feedback linearisation are described intuitively and illustrated with simple examples by Slotine and Li (1991).

### 6.5.1 Feedback Linearisation and the Companion Form

The simplest form of feedback linearisation is to cancel the non-linearities in a non-linear system, and then impose a desired linear dynamics of the closed-loop system. This can easily be performed if the non-linear system is in the so-called *companion form*, or *controllability canonical form*, i.e., (for SISO systems)

$$\dot{x}^{(n)}(t) = f_n(\mathbf{x}) + g_n(\mathbf{x})u(t) \quad (6.57)$$

This equation can be represented by the state-space model

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} x_2 \\ \vdots \\ x_n \\ f_n(\mathbf{x}) \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ g_n(\mathbf{x}) \end{bmatrix} u(t) \quad (6.58)$$

For systems described by the controllability canonical form, the non-linearities can be cancelled by applying the control input

$$u(t) = \frac{v(t) - f_n(\mathbf{x})}{g_n(\mathbf{x})} \quad (6.59)$$

to obtain the simple input–output relation (multiple integrator form)

$$\dot{x}^{(n)}(t) = v(t) \quad (6.60)$$

$v(t)$  is the new input  $v$ , also termed *virtual* or *artificial control input*, which can be determined by the pole placement method; see Section 6.5.3.3.

### 6.5.2 Intuitive Concept of Input–Output Linearisation

Consider non-linear systems described by the single-input single-output (SISO) state-space representation

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})u(t) \\ y(t) &= h(\mathbf{x}) \quad \mathbf{x}_0 = \mathbf{x}(t_0) = \mathbf{x}(0) \end{aligned} \quad (6.61)$$

where  $u$  denotes the input,  $y$  the output,  $\mathbf{x}$  the state vector, and  $\mathbf{f}(\mathbf{x})$ ,  $\mathbf{g}(\mathbf{x})$  and  $h(\mathbf{x})$  are non-linear functions. Systems of this structure are said to be *control affine* or *linear in control*.

The intuitive and simple basis of the so-called input–output linearisation approach to non-linear control design is to find a direct (and simple) relation between the system output  $y(t)$  and the control input  $u(t)$ .

To generate such a direct relationship, the output equation  $y = h(x)$  is differentiated so often as the input  $u$  does not explicitly appear. Once an explicit relationship

$$y^{(r)} = \phi(x, u) \quad (6.62)$$

between  $u$  and  $y$  is obtained, the corresponding output derivative is set to the new input  $v$ , and the relationship is solved with respect to  $u$  to give the control law for input–output linearisation.

In other words, and more precisely, we start by differentiating the output  $y(t)$  with respect to time. Applying the well-known chain rule, we get

$$\dot{y} = \frac{\partial h(x)}{\partial x} \dot{x} = \frac{\partial h}{\partial x} f(x) + \frac{\partial h}{\partial x} g(x)u \quad (6.63)$$

To provide a compact notation for the expressions  $\partial h(x)/\partial x f(x)$  and  $\partial h(x)/\partial x g(x)$ , as well as for higher derivatives, the so-called *Lie derivatives*, known from differential geometry, are used; see Appendix C.2. The Lie derivatives are nothing more than a simple notation for certain expressions that occur when repeatedly differentiating  $y(t)$  using the chain rule. Using this notation, Equation 6.63 can be written as

$$\dot{y} = L_f h(x) + L_g h(x)u \quad (6.64)$$

If  $L_g h(x)$  is different from zero, this equation gives the explicit relationship looked for between  $u$  and  $y$  (it is assumed that no direct feedthrough connection between  $u$  and  $y$  exists, which is usually the case for most real plant).

If  $L_g h(x)$  is equal to zero for all  $x$ , the output  $y(t)$  has to be differentiated once again to give

$$\ddot{y} = \frac{\partial L_f h(x)}{\partial x} \dot{x} = L_f^2 h(x) + L_g L_f h(x)u \quad (6.65)$$

Again, if  $L_g L_f h(x)$  is equal to zero for all  $x$ , the differentiation procedure is repeated over and over, until it leads to

$$y^{(r)} = L_f^r h(x) + L_g L_f^{r-1} h(x)u \quad (6.66)$$

for some integer  $r$  (referred to as the *relative degree*) with  $L_g L_f^{r-1} h(x) \neq 0$ . Equation 6.66 can now obviously be rearranged to yield a feedback of the form

$$u = \frac{v - L_f^r h(x)}{L_g L_f^{r-1} h(x)} \quad (6.67)$$

that cancels the non-linearities of the plant and leads to a linear input–output relationship of the form

$$y^{(r)} = v \quad (6.68)$$

(between the system output  $y$  and the new input  $v$ ) or equivalently a transfer function

$$G(s) = \frac{Y(s)}{V(s)} = \frac{1}{s^r} \quad (6.69)$$

**Example 6.1.** The first example is a synchronising cylinder with non-linear friction and non-linear flow function (assuming constant bulk modulus and constant chamber volumes); see Equation 4.213. This system is of order  $n = 3$  and described by ( $u > 0$ )<sup>4</sup>

$$\dot{x}_1 = x_2 \quad (6.70)$$

$$\dot{x}_2 = \frac{A}{m} x_3 - \frac{1}{m} F_f(x_2) \quad (6.71)$$

$$\dot{x}_3 = \alpha_1 \sqrt{p_s - p_T - x_3} u - \alpha_2 x_2 \quad (6.72)$$

$$y = x_1 \quad (6.73)$$

Differentiating the output  $y(t)$  yields:

$$y = x_1 = h(\mathbf{x}) \quad (6.74)$$

$$\dot{y} = \dot{x}_1 = x_2 = L_f h(\mathbf{x}) \quad (6.75)$$

Note that  $u$  does not appear in this equation, *i.e.*,  $L_g h(\mathbf{x}) = 0$ ; thus, Equation 6.75 is differentiated again to get

$$\ddot{y} = \dot{x}_2 = \frac{A}{m} x_3 - \frac{1}{m} F_f(x_2) = L_f^2 h(\mathbf{x}) \quad (6.76)$$

Since  $L_g L_f h(\mathbf{x})$  is equal to zero, a third differentiation is required:

$$y^{(3)} = \frac{A}{m} \dot{x}_3 - \frac{1}{m} F'_f(x_2) \dot{x}_2 = L_f^3 h(\mathbf{x}) + L_g L_f^2 h(\mathbf{x}) u \quad (6.77)$$

where

$$F'_f(x_2) \equiv \frac{\partial F_f(x_2)}{\partial x_2} \quad (6.78)$$

$$L_f^3 h(\mathbf{x}) = -\frac{A}{m} \alpha_2 x_2 - \frac{1}{m} F'_f(x_2) \left[ \frac{A}{m} x_3 - \frac{1}{m} F_f(x_2) \right] \quad (6.79)$$

$$L_g L_f^2 h(\mathbf{x}) = \frac{A}{m} \alpha_1 \sqrt{p_s - p_T - x_3} \quad (6.80)$$

---

<sup>4</sup> In this example, the valve dynamics is neglected.



Thus (as the input appears explicitly), the system is input–output linearisable with a relative degree  $r = 3$ . The canonical representation (Equation 6.77) of this system is depicted in Figure 6.19.

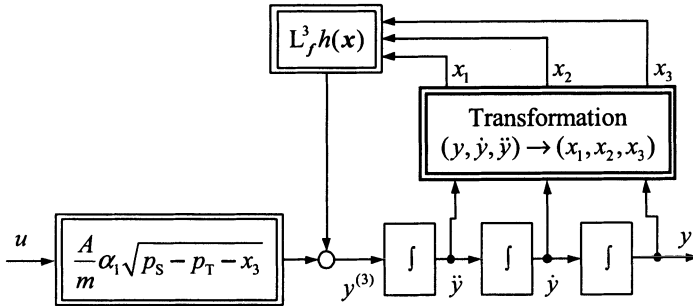
It is now obvious that a feedback of the form

$$u = \frac{v - L_f^3 h(x)}{L_g L_f^2 h(x)} = \frac{v + \frac{A}{m} \alpha_2 x_2 + \frac{1}{m} F_f'(x_2) \left[ \frac{A}{m} x_3 - \frac{1}{m} F_f(x_2) \right]}{\frac{A}{m} \alpha_1 \sqrt{p_s - p_T - x_3}} \quad (6.81)$$

cancels the non-linearities of the system and leads to a linear relationship

$$y^{(3)} = v \quad (6.82)$$

between the output  $y$  and the new (virtual) input  $v$ . Using this linear relationship, the system output can easily be made to exhibit the desired behaviour. For this, the virtual signal  $v$  can be selected to be an appropriate function of  $y$  and  $y_{\text{ref}}$  (and their time derivatives). Note that the system considered in this example is *exactly linearisable* (as  $r = n$  holds).



**Figure 6.19** Through input–output linearisation transformed synchronising cylinder model with non-linear friction and non-linear flow function □

### 6.5.3 Formalised Theory of Feedback Linearisation

The objective of this subsection is to describe the formalised concepts of feedback linearisation of non-linear systems. The readers interested in the proof of the theorems should consult the textbooks by Slotine and Li (1991) and Isidori (1995).

Note that practitioners may skip this subsection and go directly to the next subsection where the input–output linearisation concept is further applied to HSS models.

#### 6.5.3.1 Input–Output Linearisation of Single-input Single-output Systems

Consider again the state-space representation of Equation 6.61. The point of departure of the analysis of feedback linearisation is the formal definition of the relative degree:

**Definition 6.1. (Relative Degree)** The non-linear system, Equation 6.61, is said to have the *relative degree*  $r$  in a region  $\Omega$  (or at a point  $x_0$ ) if, for  $\forall x \in \Omega$  (or in a neighbourhood of  $x_0$ )

$$(i) \quad L_g L_f^i h(x) = 0 \quad \forall i < r-1$$

$$(ii) \quad L_g L_f^{r-1} h(x) \neq 0$$

□

For linear systems, the notation reduces to the familiar definition of the relative degree as the excess of poles over zeros.

Note that it is possible that the relative degree is undefined at some points  $x_0$ . This occurs, in fact, when the input coefficient  $L_g L_f^{r-1} h(x)$  is zero at  $x_0$ , but non-zero at some points  $x$  arbitrarily close to  $x_0$ .

**Theorem 6.1. (Normal Form)** When the relative degree  $r$  is (well) defined and  $r < n$ , the non-linear system, Equation 6.61, can be transformed using a coordinate transformation

$$\begin{aligned} z_1(x) &= h(x) \\ z_2(x) &= L_f h(x) \\ &\vdots \\ z_r(x) &= L_f^{r-1} h(x) \\ z_{r+1}(x) &= t_{r+1}(x) \\ &\vdots \\ z_n(x) &= t_n(x) \end{aligned} \quad (6.83)$$

with arbitrary  $t_i(x)$ , into a so-called *normal form* (sometimes referred to as Byrnes–Isidori normal form)

$$\begin{aligned} y &= z_1(t) \\ \dot{z}_1 &= z_2(t) \\ \dot{z}_2 &= z_3(t) \\ &\vdots \\ \dot{z}_{r-1} &= z_r(t) \\ \dot{z}_r &= b(z) + a(z)u(t) \\ \dot{z}_{r+1} &= q_{r+1}(z) + p_{r+1}(z)u(t) \\ &\vdots \\ \dot{z}_n &= q_n(z) + p_n(z)u(t) \end{aligned} \quad (6.84)$$

Setting the state feedback control law

$$u = \frac{v - b(z)}{a(z)} \quad (6.85)$$

where  $v$  is the external reference input, on the normal form equations, one obtains a system

$$\begin{aligned}
y &= z_1(t) \\
\dot{z}_1 &= z_2(t) \\
\dot{z}_2 &= z_3(t) \\
&\vdots \\
\dot{z}_{r-1} &= z_r(t) \\
\dot{z}_r &= v(t) \\
\dot{z}_{r+1} &= q_{r+1}(z) \\
&\vdots \\
\dot{z}_n &= q_n(z)
\end{aligned} \tag{6.86}$$

□

It becomes clear that this system is decomposed into a linear subsystem of dimension  $r$  (or a chain of  $r$  integrators), which is only responsible for the input–output behaviour, and a possibly non-linear subsystem of dimension  $n - r$ , the behaviour of which, however, is “unobservable”, *i.e.*, it does not affect the output; see Figure 6.20.

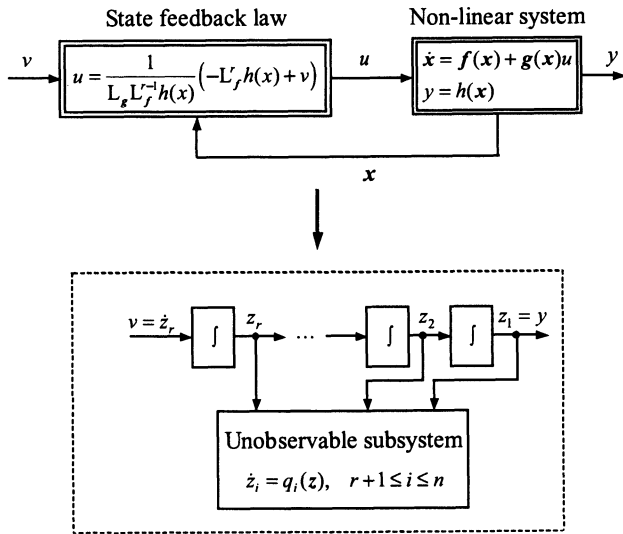


Figure 6.20. Block diagram of input–output linearisation procedure

The linearising feedback can be specified in terms of the functions  $f(x)$ ,  $g(x)$  and  $h(x)$  characterising the original description, Equation 6.61, of the system in the following theorem.

**Theorem 6.2. (Input–Output Linearisation)** When the relative degree  $r$  is (well) defined and  $r < n$ , the state feedback

$$u = \frac{v - L_f^r h(x)}{L_g L_f^{r-1} h(x)} \tag{6.87}$$

transforms the non-linear system (Equation 6.61) into a system whose input–output behaviour is identical to that of a linear system having the input–output relationship of Equation 6.68 and the transfer function of Equation 6.69.  $\square$

In the particular case of  $r = n$ , the non-linear system (Equation 6.61) is exactly linearisable (or input–state linearisable). The set of conditions for the triple  $\{f(x), g(x), h(x)\}$  to have the relative degree  $n$  is provided by the following theorem:

**Theorem 6.3. (Exact Linearisation)** The non-linear system, Equation 6.61, is *exactly linearisable* if, and only if, there exists a region  $\Omega$  (or a point  $x_0$ ) such that the following conditions hold:

- (a) The vector fields  $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-1} g\}$  are linearly independent, *i.e.*, the matrix  $\begin{bmatrix} g & \text{ad}_f g & \dots & \text{ad}_f^{n-1} g \end{bmatrix}$  is of full-rank (*controllability condition*).
- (b) The set  $\{g, \text{ad}_f g, \dots, \text{ad}_f^{n-2} g\}$  is involutive in  $\Omega$  (or in a neighbourhood of  $x_0$ ) (*involutivity condition*).  $\square$

Involutivity means that if one forms the Lie bracket of any pair of vector fields from the set  $\{f_1, f_2, \dots, f_m\}$ , then the resulting vector field can be expressed as a linear combination of the original set of vector fields; see Appendix C.3 (Theorem C.1).

A recipe to perform the exact linearisation (or input–state linearisation) can be found in Slotine and Li (1991:241–242).

### 6.5.3.2 Internal and Zero Dynamics

As mentioned above, the input–output linearisation leads to a decomposition of the non-linear system into an external (linear input–output) part and an internal (unobservable) part. Since the control design must account for the entire system, both parts have to be stable. The stability of the external part can easily be guaranteed because of the controllability canonical form (Equation 6.86). Thus, there remains to be studied the stability behaviour of the internal dynamics.

Consider Equation 6.86, and define the vectors

$$\xi \equiv \begin{bmatrix} z_1 \\ \vdots \\ z_r \end{bmatrix}, \quad \eta \equiv \begin{bmatrix} z_{r+1} \\ \vdots \\ z_n \end{bmatrix} \quad (6.88)$$

to give

$$\begin{aligned} y &= z_1(t) \\ \dot{z}_1 &= z_2(t) \\ \dot{z}_2 &= z_3(t) \\ &\vdots \\ \dot{z}_{r-1} &= z_r(t) \\ \dot{z}_r &= b(\xi, \eta) + a(\xi, \eta)u(t) \\ \dot{\eta} &= q(\xi, \eta) \end{aligned} \quad (6.89)$$

Then, the internal dynamics associated with the input–output linearisation simply corresponds to the last  $(n - r)$  equations of  $\dot{\eta} = q(\xi, \eta)$  of the normal form; Equation 6.89. When the input and initial conditions are chosen in such a way as to constrain the output to remain identically zero, the internal dynamics is called the *zero dynamics*, and is written as

$$\begin{aligned}\dot{\xi} &= 0 \\ \dot{\eta} &= q(0, \eta)\end{aligned}\tag{6.90}$$

The importance of the zero dynamics is underscored in the theorem below.

**Theorem 6.4. (Stabilisation of Input–Output Linearisable Systems)** Assume that the non-linear system (Equation 6.61) has the (well) defined relative degree  $r$  ( $r < n$ ). Further, the trivial equilibrium point of the  $(n - r)$ -dimensional zero dynamics system, (Equation 6.90) is *locally asymptotically stable*. Under these conditions, the control law (Equation 6.85) yields a *locally asymptotically stable* closed-loop system.  $\square$

This result indicates that, provided that the zero dynamics is asymptotically stable, the input–output linearising control law indeed stabilises the whole system locally (*not globally*). An approach to *global stabilisation* based on partial feedback is simply to consider the control problem as a standard *Lyapunov controller design problem*, but simplified by the fact that putting the system in normal form makes part of the dynamics linear (Slotine and Li, 1991).

### 6.5.3.3 Controller Design for Input–Output Linearisable Systems

Once the non-linear system is transformed into a system with a linear input–output relationship, the control design can be completed by using all well-known and powerful linear design techniques. It is important to mention that to design the control law there is no need to know explicitly the expression of the system in normal form. It is only required to know that the system has zero dynamics with a locally (or globally) asymptotically stable equilibrium at  $\eta = 0$ .

A simple control design method is the *linear pole-placement* method. This means that the new (artificial) input  $v$  is set to

$$v = -k_{r-1}y^{(r-1)} - \dots - k_1\dot{y} - k_0y + \tilde{v}\tag{6.91}$$

where  $\tilde{v}$  is the new external input (any function of the reference signal  $y_{\text{ref}}$  and the output  $y$ ). The coefficients  $k_i$  are chosen such that the polynomial

$$p(s) = s^r + k_{r-1}s^{r-1} + \dots + k_1s + k_0\tag{6.92}$$

has all its roots strictly in the left-half plane. The actual control input can then be written, from Equation 6.87,

$$u = \frac{\tilde{v} - L_f^r h(x) - \sum_{i=1}^r k_{i-1} L_f^{i-1} h(x)}{L_g L_f^{r-1} h(x)}\tag{6.93}$$

in which, e.g., a simple linear output feedback  $\tilde{v} = K(y_{\text{ref}} - y)$  can be included<sup>5</sup> to yield the structure shown in Figure 6.21.

Moreover, the simple pole-placement controller can be extended to asymptotic tracking control tasks, as stated by Isidori (1995):

**Theorem 6.5. (Asymptotic Output Tracking)** Assume that the non-linear system, Equation 6.61, has the (well) defined relative degree  $r$  ( $r < n$ ), and that the given desired trajectory

$$y_{\text{ref}} = \begin{bmatrix} y_{\text{ref}} & \dot{y}_{\text{ref}} & \cdots & y_{\text{ref}}^{(r-1)} \end{bmatrix}^T \quad (6.94)$$

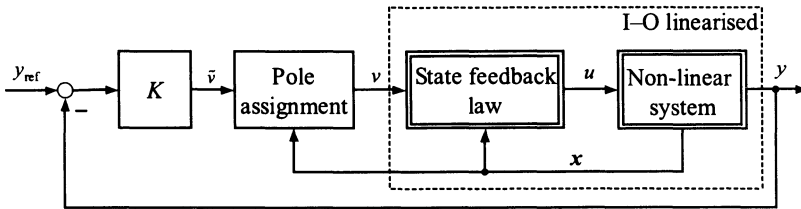
is smooth and bounded. Further assume that the solution  $\eta_{\text{ref}}$  of the equation

$$\dot{\eta}_{\text{ref}} = q(\xi_{\text{ref}}, \eta_{\text{ref}}), \quad \eta_{\text{ref}}(0) = 0 \quad (6.95)$$

exists, is bounded, and is uniformly asymptotically stable. Choose constants such that the polynomial in Equation 6.92 has all its roots strictly in the left-half plane. Then, by using the control law

$$u = \frac{-L_f^r h(x) + y_{\text{ref}}^{(r)} - \sum_{i=1}^r k_{i-1} (L_f^{(i-1)} h(x) - y_{\text{ref}}^{(i-1)})}{L_g L_f^{r-1} h(x)} \quad (6.96)$$

the whole state remains bounded and the tracking error converges to zero exponentially.  $\square$



**Figure 6.21.** Input–output linearisation with pole placement and linear feedback controller

Selecting the virtual control input  $v$  as the reference plus an appropriate linear combination of past outputs again allows for desired assignment of closed-loop poles, and thus dynamics.

As can clearly be seen in Figure 6.21, feedback linearisation results in a cascade control (or *inner/outer control*) scheme. The inner loop used for feedback linearisation transforms the dynamics of the system to a set of SISO non-coupled integrators. The outer loop is used to place the poles of the resulting linear system such that a desired behaviour of the controlled system is achieved, regardless of the original system dynamics. An example of inner/outer control well known from the robotic literature is the *computed torque control* (Asada and Slotine, 1986; An *et al.*, 1988; Spong and Vidyasagar, 1989).

<sup>5</sup> Integral action can also be included as described in Section 6.3.3.

The feedback linearisation laws described so far assume that all the state variables of the system are available. For many applications of HSSs, only the piston position can be measured directly. The other state variables must be estimated using a state observer; see Section 6.10. This leads to the complete control schemes as depicted in Figure 6.22.

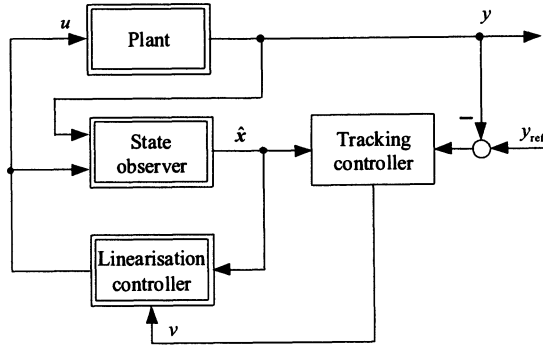


Figure 6.22. Complete control system based on feedback linearisation

## 6.5.4 Application to Hydraulic Servo-system Models

Depending on the HSS model complexity taken as basis for input–output linearisation, different (more or less complicated) control laws can be derived.

### 6.5.4.1 Application to Simplified Hydraulic Servo-system Model

Firstly, we perform an input–output linearisation for the subsystem “pressure dynamics” (Section 4.4.1, Equation 4.196, where  $n = 2$ ) with the output equation

$$y = F_L = A_p(x_3 - \alpha x_4) \quad (6.97)$$

The differentiation of this equation and substitution of the state equations for pressure dynamics yields

$$\begin{aligned} \dot{y} &= A_p \dot{x}_3 - \alpha A_p \dot{x}_4 \\ &= -A_p^2 x_2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] + \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} \right] K_{Li}(x_3 - x_4) \\ &\quad + A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v1} \sqrt{p_S - x_3} + \alpha A_p \frac{E'_B(x_4)}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \right] u \\ &=: v \end{aligned} \quad (6.98)$$

Since Equation 6.98 represents an explicit relation between  $u$  and  $y$ , it can be rearranged to give the state feedback linearising control law (for  $u \geq 0$ )

$$u = \frac{v + A_p^2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] x_2 - \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} \right] K_{Li}(x_3 - x_4)}{A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v1} \sqrt{p_S - x_3} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \right]} \quad (6.99)$$

The control law for the case  $u < 0$  follows straightforwardly as

$$u = \frac{v + A_p^2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] x_2 - \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} \right] K_{Li}(x_3 - x_4)}{A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v2} \sqrt{x_3 - p_T} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} c_{v3} \sqrt{p_S - x_4} \right]} \quad (6.100)$$

For the practical implementation of the linearising control law, the pressures  $x_3 = p_A$  and  $x_4 = p_B$ , as well as the piston velocity  $x_2 = \dot{x}_p$ , must be available through measurement or estimation.

For all operating conditions, we get an exactly linear input–output behaviour from the new input  $v$  to  $y = A_p(x_3 - \alpha x_4)$ . Thus, the load state equations take the form

$$\begin{aligned} \dot{y} &= v \\ \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{m_p} [y - \sigma x_2 - F_L] \end{aligned} \quad (6.101)$$

For controlling the piston position the control law according to Equations 6.99, 6.100 and, for instance, the simple output feedback

$$v = K(y_{p,ref} - y_p) \quad (6.102)$$

can be used.

**Interpretation of the Control Law.** The total control law consists of a direct feedback (Equation 6.102) in combination with a *velocity compensation*, a *leakage compensation*, and a *cancellation of valve flow non-linearity*:

$$u = \frac{\overbrace{K(y_{p,ref} - y_p)}^{\text{direct feedback}} + \overbrace{A_p^2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] x_2}^{\text{velocity compensation}} - \overbrace{\left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} \right] K_{Li}(x_3 - x_4)}^{\text{leakage compensation}}}{\underbrace{A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v1} \sqrt{p_S - x_3} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \right]}_{\text{cancellation of valve flow nonlinearity}}} \quad (6.103)$$

The velocity compensation is meant to compensate for the contribution of the piston velocity to the mass balances of the actuator chambers. In other words, velocity compensation eliminates the coupling between the actuator pressure dynamics on



the one hand and the dynamics of the piston with load on the other hand. The leakage compensation can, in some cases, increase the performance of the controller, but at the expense of more model knowledge and more restrictive stability conditions.

#### 6.5.4.2 Application to Simplified Hydraulic Servo-system Model with Velocity Elimination

Again, the model of the subsystem “pressure dynamics” (Section 4.4.1, Equations 4.196 and 4.197) is considered. The velocity of the piston cannot be directly measured, and an observer may fail due to the transducer and quantisation noise. Therefore, it is desirable to look for a description of the pressure dynamics where the velocity does not appear. Exploiting an idea given in Kugi *et al.* (1999), the velocity elimination can be achieved by the following transformation, assuming the bulk modulus is constant and equal in both chambers, *i.e.*,  $E'_A(x_3) = E'_B(x_4) = E'$ :

$$\begin{aligned}\tilde{x}_3 &= x_3 + E' \ln(V_A(x_1)) \\ \tilde{x}_4 &= x_4 + E' \ln(V_B(x_1))\end{aligned}\quad (6.104)$$

and then Equations 4.196 and 4.197 can be rewritten as

$$\begin{bmatrix} \dot{\tilde{x}}_3 \\ \dot{\tilde{x}}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'}{V_A(x_1)} K_{Li}(x_3 - x_4) \\ -\frac{E'}{V_B(x_1)} K_{Li}(x_3 - x_4) \end{bmatrix} + \begin{bmatrix} \frac{E'}{V_A(x_1)} c_{v1} \sqrt{p_S - x_3} \\ -\frac{E'}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \end{bmatrix} u \quad (6.105)$$

for  $u \geq 0$  and by

$$\begin{bmatrix} \dot{\tilde{x}}_3 \\ \dot{\tilde{x}}_4 \end{bmatrix} = \begin{bmatrix} \frac{E'}{V_A(x_1)} K_{Li}(x_3 - x_4) \\ -\frac{E'}{V_B(x_1)} K_{Li}(x_3 - x_4) \end{bmatrix} + \begin{bmatrix} \frac{E'}{V_A(x_1)} c_{v2} \sqrt{x_3 - p_T} \\ -\frac{E'}{V_B(x_1)} c_{v3} \sqrt{p_S - x_4} \end{bmatrix} u \quad (6.106)$$

for  $u < 0$ . Note that Equation 6.104 is nothing else than the pressure of the chambers  $p_A$  and  $p_B$  plus the deviation of the pressure due to the change of the chamber volumes.

Now, we perform again the input–output linearisation for

$$y = A_p(\tilde{x}_3 - \alpha \tilde{x}_4) \quad (6.107)$$

as described above to get the control law

$$u = \frac{v - \left[ \frac{E'}{V_A(x_1)} + \alpha \frac{E'}{V_B(x_1)} \right] K_{Li}(x_3 - x_4)}{A_p \left[ \frac{E'}{V_A(x_1)} c_{v1} \sqrt{p_S - x_3} + \alpha \frac{E'}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \right]} \quad (6.108)$$

for  $u \geq 0$  and by

$$u = \frac{v - \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} \right] K_{Li}(x_3 - x_4)}{A_p \left[ \frac{E'_A}{V_A(x_1)} c_{v2} \sqrt{x_3 - p_T} + \alpha \frac{E'_B}{V_B(x_1)} c_{v3} \sqrt{p_S - x_4} \right]} \quad (6.109)$$

for  $u < 0$ , which does not contain the piston velocity.

In this case, we also obtain an (exactly) linear input–output behaviour from the new input  $v$  to  $y = A_p(\tilde{x}_3 - \alpha\tilde{x}_4)$ , but the load state equations take the slightly different form

$$\begin{aligned} \dot{y} &= v \\ \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{m_p} [y - \sigma x_2 - F_L - f(x_1)] \end{aligned} \quad (6.110)$$

with the non-linear element

$$\begin{aligned} f(x_1) &= A_p E' \ln(V_A(x_1)) + \alpha A_p E' \ln(V_B(x_1)) \\ &= E' \ln \left[ (V_A(x_1))^{\alpha_p} (V_B(x_1))^{\alpha_p} \right] \end{aligned} \quad (6.111)$$

One can immediately see that the piston velocity is removed from the control law using the transformation of the pressures at the expense of the appearance of the non-linearity  $f(x_1)$  in Equation 6.110, compared with Equation 6.101.

It has been proved by Kugi (2001), making use of the well-known Hurwitz and Popov criteria, that the control law in Equations 6.108 and 6.109 guarantees asymptotical stability of the stationary point  $(x_{1,\text{ref}})$  throughout the whole operating domain for all possible values of the bulk modulus  $E' > E'_{\min}$ , provided that the inequality conditions

$$0 < \alpha_1 < \min \left( \frac{\sigma}{m_p}, \frac{c_k E'_{\min}}{\sigma} \right) \quad (6.112)$$

with

$$c_k = \frac{\alpha A_p \left( \sqrt{A_p} + \sqrt{\alpha A_p} \right)^2}{V_{B0} + \alpha V_{A0}} \quad (6.113)$$

are satisfied. Then the output feedback

$$v = \alpha_1 (x_{1,\text{ref}} - x_1) \quad (6.114)$$

can be used.

### 6.5.4.3 Application to a More Complicated Hydraulic Servo-system Model

Of course, input–output linearisation can be derived based on a more complicated model, Equation 4.181, as demonstrated by Lemmen and Bröcker (1999, 2000). Since the required calculations for designing such a controller are very complicated, derivation by hand is tedious and error prone. Hence, it is more convenient to use CACSD toolboxes based on symbolic packages like Maple or Mathematica. The advantages of using such symbolic packages have been demonstrated by many authors, *e.g.*, Lemmen *et al.*, (2000), from which the following equations are adapted. The input–output linearising controller for Equation 4.181, neglecting the servo-valve dynamics and the leakage flows, reads

$$u = \frac{u_{\text{num}}}{u_{\text{den}}} \quad (6.115)$$

where the numerator is calculated as

$$u_{\text{num}} = \tilde{v} + \frac{\sum_{i=1}^5 f_i(x)}{m_t^2 V_A V_B} - k_0 x_1 - k_1 x_2 - k_2 \left( \frac{-A_p x_3 + \alpha A_p x_4 + F_f + F_L}{m_t} \right) \quad (6.116)$$

with

$$\begin{aligned} f_1(x) &= (1-\alpha)A_p^2 x_2 V_A V_B (x_3 - \alpha x_4) \\ f_2(x) &= -(1-\alpha)A_p x_2 V_A V_B (F_f - F_L) \\ f_3(x) &= F_f'(x_2) V_A V_B A_p (x_3 - \alpha x_4) \\ f_4(x) &= F_f'(x_2) V_A V_B (F_f - F_L) \\ f_5(x) &= A_p^2 x_2 m_t \left[ E_A'(x_3) V_B + E_B'(x_4) V_A \right] \end{aligned} \quad (6.117)$$

The derivative  $F_f'(x_2) \equiv dF_f/dx_2$  can be gained from derivation of

$$F_f(x_2) = \sigma x_2 + \text{sign}(x_2) \left[ F_{c0} + F_{s0} \exp\left(-\frac{|x_2|}{c_s}\right) \right] \quad (6.118)$$

with the smooth approximations

$$\text{sign}(x_2) \approx \frac{2}{\pi} \arctan(\gamma x_2) \quad (6.119)$$

$$|x_2| \approx \frac{2}{\pi} x_2 \arctan(\gamma x_2) \quad (6.120)$$

The factor  $\gamma$  determines the approximation quality; see Figure 6.23.

The denominator of Equation 6.115 is computed by

$$u_{\text{den}} = \frac{A_p c_{v1} K_v E_A'(x_3) \text{sign}(p_s - x_3) \sqrt{|p_s - x_3|}}{\alpha m_t V_A V_B}$$

$$+ \frac{\alpha A_p c_{v4} K_v E'_B(x_4) \text{sign}(x_4 - p_T) \sqrt{|x_4 - p_T|}}{\alpha m_l V_A V_B} \quad (6.121)$$

for  $u \geq 0$  and by

$$u_{\text{den}} = \frac{A_p c_{v2} K_v E'_A(x_3) \text{sign}(x_3 - p_T) \sqrt{|x_3 - p_T|}}{\alpha m_l V_A V_B} + \frac{\alpha A_p c_{v3} K_v E'_B(x_4) \text{sign}(p_S - x_4) \sqrt{|p_S - x_4|}}{\alpha m_l V_A V_B} \quad (6.122)$$

for  $u < 0$ .

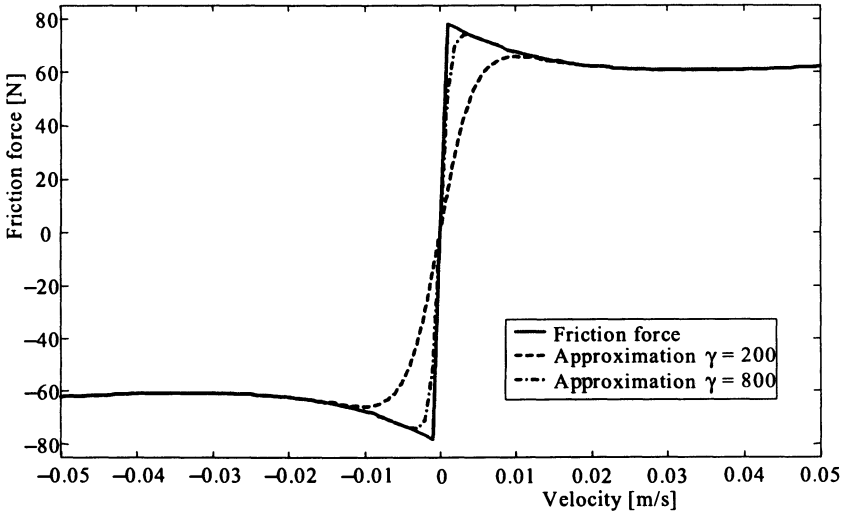


Figure 6.23. Approximation of Stribeck curve

### 6.5.5 Feedback Linearisation Based on Bilinear Models

Bilinear models are an important subclass of non-linear models and are appealing for three reasons:

1. Bilinear models are the simplest type of non-linear model.
2. They can be regarded as a suitable approximation in formulating a model for many non-linear physical systems.
3. The theory of bilinear systems has become well established (Mohler, 1973; Schwarz, 1991, 2000; Isidori, 1995).

Moreover, several attempts of modelling and control of hydraulic systems using bilinear models have been made in recent years (Beater, 1987; Naujoks and Wurmthaler, 1988; Guo and Schwarz, 1989; Guo, 1991; Guo *et al.*, 1994; Schwarz and Ingenbleek, 1994; Yin, 1994; Del Re and Isidori 1995; Schwarz *et al.*, 1996).

For instance, Yin (1992, 1994) developed a state feedback controller based on identified bilinear models in some special canonical form.

The controller consists of a feedback linearising control law and a tracking control algorithm, and thus results in a special case of the control scheme of Section 6.5.3.3. The non-measurable state variables are determined with an optimal bilinear observer from position measurements of the servo-system. This control concept (Figure 6.24) is described in more details by Yin (1992, 1994).

Applying the procedure of Section 6.5.1 for SISO bilinear systems

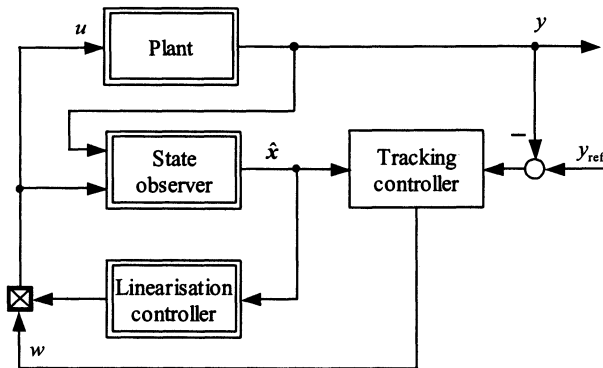
$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + (\mathbf{N}\mathbf{x} + \mathbf{b})u \\ y &= \mathbf{c}^T \mathbf{x}\end{aligned}\quad (6.123)$$

in controllability canonical form, *i.e.*,

$$\begin{aligned}\mathbf{A} &= \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ & & & & 0 \end{bmatrix} & \mathbf{N} &= \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ & & & & 0 \end{bmatrix} \\ \mathbf{a}^T &= [a_1 \ a_2 \ \dots \ a_n] & \mathbf{n}^T &= [n_1 \ n_2 \ \dots \ n_n] \\ \mathbf{b}^T &= [0 \ 0 \ \dots \ b] & \mathbf{c}^T &= [1 \ 0 \ \dots \ 0]\end{aligned}\quad (6.124)$$

the non-linearity can be cancelled using the following state feedback law (see Equation 6.59):

$$u = \frac{v - \mathbf{a}^T \mathbf{x}}{\mathbf{n}^T \mathbf{x} + b} \quad \mathbf{n}^T \mathbf{x} + b \neq 0 \quad (6.125)$$



**Figure 6.24.** Scheme of bilinear control system based on feedback linearisation (Taken from Yin (1994), p 80 (original publication). Reprinted with kind permission of VDI Verlag.)

However, in this case, where the only non-linearity lies in the input part, it is sufficient to consider (Yin, 1992, 1994)

$$u = \frac{1}{1 + k_1 x} w \quad k_1 = \frac{n^T}{b} \quad 1 + k_1 x \neq 0 \quad (6.126)$$

Indeed, substituting Equation 6.126 into Equation 6.123 yields the linear state-space description

$$\begin{aligned} \dot{x} &= Ax + bw \\ y &= c^T x \end{aligned} \quad (6.127)$$

Thus, the task of the controller design for the bilinear systems considered turns out to be the controller design for a linear system; Equation 6.127.

For HSSs (with normalised input),  $|k_1| < 1$  holds because of the constraint  $|u| < 1$ , so that the feedback linearisation law has the form

$$u = \begin{cases} \frac{1}{1 + k_1 x} w & \text{for } w \geq 0 \\ \frac{1}{1 - k_1 x} w & \text{for } w < 0 \end{cases} \quad \text{and } 1 \pm k_1 x \neq 0 \quad (6.128)$$

However, in reality it is impossible to linearise the non-linear plant dynamics exactly in the whole range of input signal variation due to the constraint on  $u$ . Experimental results for the valve–cylinder combination considered in Chapter 7.1 showed that the above feedback linearisation law is efficient only for  $|w| < 0.5$  (Yin, 1994).

## 6.6 Approaches Similar to Feedback Linearisation

Non-linear controllers obtained from *feedback linearisation*, an *input–output linearisation* framework or similar approaches have been developed by many researchers, *e.g.*, Hahn *et al.* (1994), Del Re and Isidori (1995), Vossoughi and Donath (1995), Alleyne (1996), Heintze (1997), Bernzen (1999a), Sohl and Bobrow (1999), Tunay *et al.* (2001). Some of these approaches will be presented and discussed in this section.

### 6.6.1 Direct Inverse Control

One of the first well-known options for controlling non-linear systems is to use the inverse dynamics model to act as a controller. If this model represents the *exact inverse*, the control input produced by it will drive the system output to the reference value in one sampling interval, *i.e.*, a dead-beat controller is obtained; see Åström and Wittenmark (1990).

In general, however, the practical relevance of direct inverse control is considered limited due to a number of serious inconveniences. Using such control to achieve a fast response to reference changes will typically result in a poor robustness with a high sensitivity to noise and high-frequency disturbances. In addition, one will often encounter a very active control signal. If the system is linear, this occurs when the poles are situated close to the imaginary axis in the continuous-time

domain (or unit circle in the discrete-time domain). In the non-linear case there is no unique set of zeros, but of course a similar phenomenon exists.

If the inverse model is unstable (corresponding to poles in the RHP in the linear case), one must anticipate that the closed-loop system becomes unstable. Unfortunately, this situation occurs quite frequently in practice, as is the case for HSSs; see the analysis of direct inverse control by Heintze (1997). He pointed out that this controller will try to perform a pole-zero cancellation that, however, does not take place due to parameter uncertainties (which leads, for instance, to imperfect velocity compensation). Thus, a zero near the imaginary axis and a rather undamped complex conjugate pole pair may result.

## 6.6.2 Cascade Load Pressure (Load Force) Control

The strategy of cascade  $F_L$  (or equivalently  $p_L$ ) control of hydraulic actuators is relatively new, and was first presented by Sepehri *et al.* (1990). The method is elaborated and formalised by Heintze and van der Weiden (1995), Heintze *et al.* (1995) and van Schothorst (1997). More recently, similar methods (based on input–output linearisation) have been derived by Bernzen (1999a) and by Sohl and Bobrow (1999).

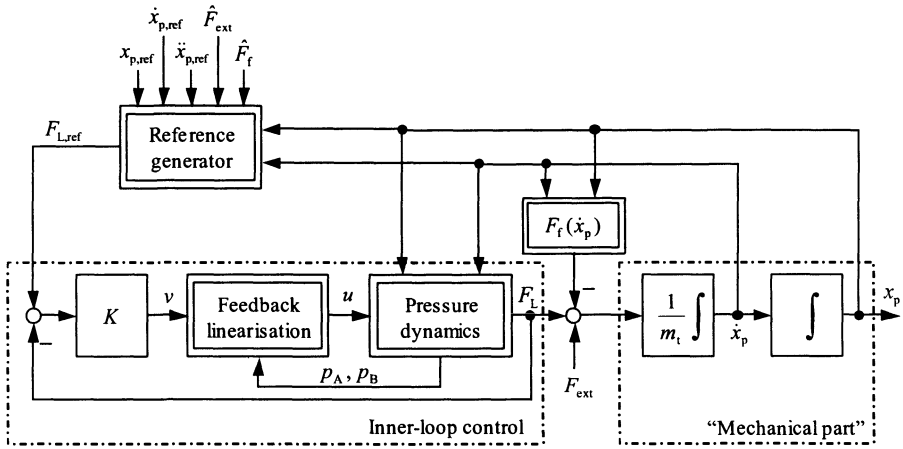
### 6.6.2.1 Introduction to Cascade Load Force ( $F_L$ ) Control

The following presentation is mainly based on the studies of Heintze (1997) and van Schothorst (1997), and Bernzen (1999), which contain in-depth treatments and theoretical analysis of this controller.

**Basic Control Structure.** As depicted in Figure 6.25, the cascade  $F_L$  control consists of two loops. First, the inner-loop control, which is the actual actuator control loop, has the objective of controlling the actuator pressure-difference-force  $F_L$  using the input–output linearisation, independent of the resulting motions of the load. This means that it gives the actuator the character of a force generator. Second, the outer-loop control is concerned with the (stabilising) control of the actuator load and the compensation of external forces  $F_{\text{ext}}$ .

**Inner-loop Control.** As already stated in Section 6.5.4.1 (see Equation 6.103), the inner-loop control, *i.e.*, the input–output linearising feedback, consists of a velocity compensation, a leakage compensation, and a cancellation of valve flow non-linearity.

In the case of perfect velocity compensation, a complete decoupling of pressure and load dynamics is achieved. For the simple linear (third-order) model, this pressure dynamics is of first order, where the bandwidth is determined by the proportional feedback gain  $K_{pL}$ . Subsequently, high-gain pressure difference feedback can be applied, in order to shape the pressure dynamics. Thus, according to Figure 6.25, the inner-loop cascade  $F_L$  controller gives the actuator the character of a force generator, where the actuator pressure-difference-force  $F_L$  follows the desired pressure-difference-force  $F_{L,\text{ref}}$  with first-order dynamics, independent of the piston motion.



**Figure 6.25.** Cascade load force ( $F_L$ ) control structure (Taken from Bernzen (1999), p 24 (original publication). Reprinted with kind permission of VDI Verlag.)

With the piston motion decoupled from the controlled pressure dynamics by the velocity compensation in the inner loop, a marginally stable system is obtained. The target pressure difference is followed by the actuator pressure difference, which together with the external forces  $F_{ext}$  drives the load dynamics (the equation of motion of the piston) in an open-loop manner. This requires a stabilising feedback of the load dynamics, including a compensation for the external forces  $F_{ext}$ . This part of the actuator control is taken care of by the outer loop.

**Outer-loop Control.** In fact, the outer-loop design is just a standard mechanical motion control problem, where the actuator acts as force generator over a certain frequency range. In the case of general motion control problems, this outer-loop control will be a rather complex high-level control.

The use of the control law according to Equation 6.102 is only one option to construct the artificial input  $v$ . A rather general, though simple, structure of the (outer) control law is to use a kind of *force generator* that consists of position and velocity feedback, including a feedforward term to compensate for any disturbance effects.

In order to achieve position tracking, the desired force should be chosen to be (Bernzen, 1999a)

$$F_{L,ref} = m_t \ddot{x}_{p,ref} - k_v (\dot{x}_p - \dot{x}_{p,ref}) - k_p (x_p - x_{p,ref}) + \hat{F}_f(\dot{x}_p) \quad (6.129)$$

where  $\hat{F}_f(\dot{x}_p)$  is an estimate of friction forces in the cylinder, combined with the simple proportional feedback

$$v = K(F_{L,ref} - F_L) \quad (6.130)$$

to give a first-order closed-loop behaviour

$$\frac{1}{K} \dot{F}_L + F_L = F_{L,ref} \quad (6.131)$$



A feedback of the form

$$v = \dot{F}_{L,\text{ref}} + K(F_{L,\text{ref}} - F_L) \quad (6.132)$$

has also been proposed by Sohl and Bobrow (1999) that guarantees exponential force stabilisation, and leads to the closed-loop description

$$\frac{1}{K}\dot{F}_L + F_L = \frac{1}{K}\dot{F}_{L,\text{ref}} + F_{L,\text{ref}} \quad (6.133)$$

However, not only such simple feedback schemes, but also more sophisticated control techniques can be utilised in the outer-loop control design. For instance, Heintze *et al.* (1995) and Heintze (1997) successfully applied non-linear sliding mode control as an outer-loop control strategy.

The whole algorithm of cascade load force control based on input–output linearisation is summarised in Figure 6.26.

$$u = \frac{v + A_p^2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] x_2}{A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v1} \sqrt{p_s - x_3} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} c_{v4} \sqrt{x_4 - p_T} \right]} \quad (\text{control law for the case } u \geq 0)$$

$$u = \frac{v + A_p^2 \left[ \frac{E'_A(x_3)}{V_A(x_1)} + \alpha^2 \frac{E'_B(x_4)}{V_B(x_1)} \right] x_2}{A_p \left[ \frac{E'_A(x_3)}{V_A(x_1)} c_{v2} \sqrt{x_3 - p_T} + \alpha \frac{E'_B(x_4)}{V_B(x_1)} c_{v3} \sqrt{p_s - x_4} \right]} \quad (\text{control law for the case } u < 0)$$

$$v = K(F_{L,\text{ref}} - F_L) \quad (\text{output proportional feedback})$$

$$F_{L,\text{ref}} = m_1 \ddot{x}_{p,\text{ref}} - k_v(\dot{x}_p - \dot{x}_{p,\text{ref}}) - k_p(x_p - x_{p,\text{ref}}) + \hat{F}_t(\dot{x}_p) \quad (\text{reference load force generator})$$

**Figure 6.26.** Summarised algorithm of cascade load force control based on input–output linearisation

### 6.6.2.2 Linear Cascade $F_L$ Control

The derivation of such a controller is usually based on a *linearised third-order model* of hydraulic servo-systems with *symmetric actuator* of the form as in Equation 4.252. For differential actuators the model Equation 4.250 can be used. Considering the controlled actuator as a force generator, and the load as a slightly damped inertia, the outer-loop feedback shown gives the system the character of a mass–spring–damper system. The poles of this system can be placed by the gains  $K_1$  and  $K_2$ , representing stiffness and damping constant respectively. Thus, from an interaction point of view, the controlled actuator responds on external inputs as a

physical system, of which the characteristics have been modified by control. This is generally referred to as impedance control (Heintze, 1997).

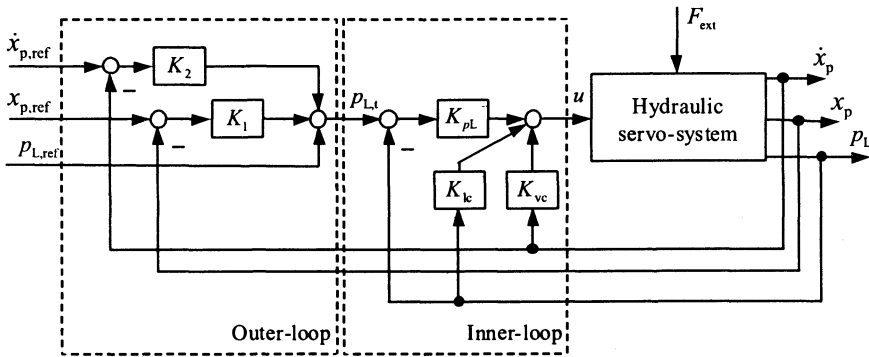
### 6.6.2.3 Comparison of Some Control Schemes

Compared with a direct inverse control for the inner-loop, such a cascade  $F_L$  controller is easier to design and has better robustness properties, since an inverse dynamic controller is based upon the cancellation of the (uncertain and position-dependent) dynamic of the hydraulic part, and is thus sensitive to an inaccurate model used at the control design stage.

Comparing the linear cascade  $F_L$  control strategy (with outer-loop control according to Figure 6.27) with the controller of Tafazoli *et al.* (1996), and with the state feedback strategy of Figure 6.6, *all strategies appear to be almost equivalent*. This can be further elucidated by studying the feedback law of the cascade  $F_L$  control strategy with outer-loop feedback (van Schothorst, 1997:217):

$$\begin{aligned} u &= K_{vc} \dot{x} + K_{pL} \left[ K_1 (x_{ref} - x) + K_2 (\dot{x}_{ref} - \dot{x}) + (p_{L,ref} - p_L) \right] + K_{lc} p_L \\ &= K_p (x_{ref} - x) + K_i (\dot{x}_{ref} - \dot{x}) + K_{pL} (p_{L,ref} - p_L) + K_{vc} \dot{x}_{ref} + K_{lc} p_L \end{aligned} \quad (6.134)$$

where  $K_p = K_{pL} K_1$  and  $K_i = K_{pL} K_2 - K_{vc}$ . Apparently, the only difference is an additional velocity feedforward term  $K_{vc} \dot{x}_{ref}$  and a leakage compensation term  $K_{lc} p_L$ , which has briefly been discussed in Section 6.4.1.

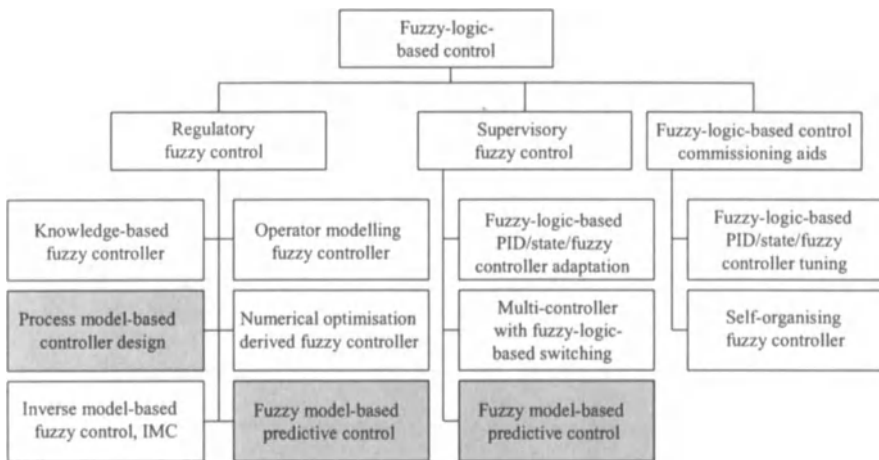


**Figure 6.27.** Complete cascade  $p_L$  control with inner-loop control of actuator pressure difference and outer-loop control of load motion (Heintze, 1997; van Schothorst, 1997)

## 6.7 Fuzzy Control

Fuzzy logic can be utilised in various ways for control purposes. Figure 6.28 presents some alternatives. Of these, fuzzy state feedback control and fuzzy model predictive control will be discussed further, as both particularly use the advantages of Sugeno-type fuzzy models. The fuzzy state feedback controller directly builds upon the local model concept by using and extending the idea of the “parallel distributed compensator” (Wang *et al.*, 1995). Fuzzy model predictive control

(FMPC) exploits the excellent prediction capabilities of Sugeno-type models. Several applications of this kind are reported (Babuška *et al.*, 1996; Fischer *et al.*, 1997; Babuška, 1998; Kroll, 2000). A similarly excellent prediction capability has also been achieved with model predictive control concepts that use (recurrent) artificial neural networks as prediction model (e.g. Babuška *et al.*, 1996; Gomm *et al.*, 1997; Latrille *et al.*, 1997; Su and McAvoy, 1997).



**Figure 6.28.** Some examples for utilising fuzzy logic in control (This section discusses examples corresponding to the shaded blocks.)

## 6.7.1 Fuzzy State Control

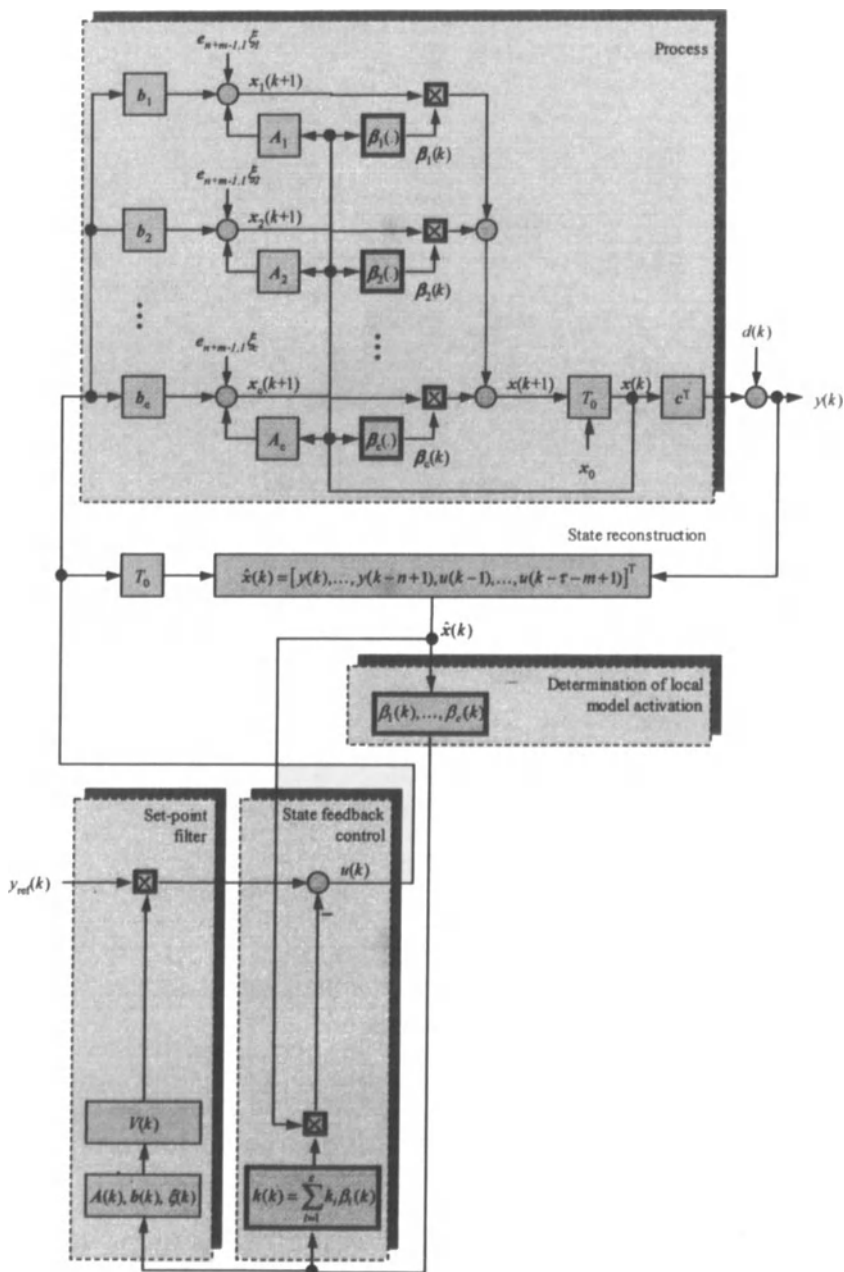
### 6.7.1.1 Introduction

The idea of regime-based system decomposition of Sugeno-type models can also be used for control. Each regime is considered as a separate dynamic system for which a local controller is designed. Then, in each instance, all local controllers are weighted and superposed to deliver the global control action.

Proportional-integral-derivative (PID) controllers (Tanaka and Sugeno, 1992; Domanski *et al.*, 1997) and state feedback controllers have been proposed as local controllers (Tanaka and Sano, 1994; Wang *et al.*, 1995; Korba and Frank, 1997; Kroll *et al.*, 2000). In the case of state feedback controllers the regime-based decompositional approach is also referred to as the “parallel distributed compensator”. State feedback controllers are considered further because of their advantages:

- methods from linear state feedback controller design can be transferred.
- dead time can easily be represented within the state-space model.
- a uniform control design procedure is possible, even for systems with different characteristics.

- stability analysis and stable control design tools have been developed (Lyapunov or LMI-based methods (*e.g.*, Tanaka and Sugeno, 1992 and Tanaka and Sano, 1994).



**Figure 6.29.** Control concept with adaptive set-point filter and fuzzy proportional state controller for a plant described by a fuzzy state-space model

In the following sections, a non-linear fuzzy state feedback controller is introduced. This controller builds on the structure of the Sugeno-type models (in fuzzy state-space form) that were introduced in Section 5.4.2.4. The constant term in each local model requires particular consideration, as it prevents a direct transfer of linear design methods to the local systems. An adaptive set-point filter is proposed to remove the steady-state error of the closed-loop characteristics for non-integral processes. Methods for designing such a controller are presented. The approach bases on the concept proposed in Kroll *et al.* (2000).

### 6.7.1.2 Control Algorithm

The proposed fuzzy state controller consists of four elements, which are also sketched in Figure 6.29:

1. **State reconstruction module.** The process state is reconstructed from process inputs and outputs.
2. **Local model activation estimator.** The current activation of each of the  $c$  local systems is determined by evaluating the according membership functions.
3. **Fuzzy state feedback controller.** The resulting global feedback gain is calculated from the given local gains weighted by their current activation.
4. **Set-point filter.** A set-point filter eliminates the steady-state error, which remains due to the proportional character of the state feedback controller in the case of a non-integral process.

The fuzzy state feedback controller uses the process state to calculate the control signal:

$$u(k) = V(k) y_{\text{ref}}(k) - \mathbf{k}^T(k) \hat{\mathbf{x}}(k) \quad (6.135)$$

The second term,  $\mathbf{k}^T(k) \hat{\mathbf{x}}(k)$ , is the actual state feedback and the first term,  $V(k)y_{\text{ref}}(k)$ , is the scaled reference value.

The process state can, for example, be estimated by a fuzzy observer or it can be reconstructed. A fuzzy state observer uses a dynamic non-linear fuzzy model to estimate the global process state out of past plant inputs (and outputs for correction). The concept of regime-wise modelling opens several alternatives for handling the error feedback. The methods considered in the literature (Ma *et al.*, 1998; Cao *et al.*, 1999) use local error feedback gain matrices by which the output error is fed back to the local system.

The process state can also be calculated from past inputs and outputs. No additional dynamical system (which goes along with an according dynamic design task) is used for the state estimation as in the case of an observer. This approach will be used in the following. It is referred to as “state reconstruction” (Isermann, 1987). If the state vector is composed of delayed inputs and outputs, a non-minimal realisation can be derived that permits one to construct the state directly out of past inputs and outputs without any further calculation:

$$\hat{\mathbf{x}}(k) = [y(k), \dots, y(k-n+1), u(k-1), \dots, u(k-\tau-m+1)]^T \quad (6.136)$$

This also means that each local system starts from the same state and is re-initialised in each instance.

The gain of the global feedback controller is composed of the  $c$  local gains weighted with the activation  $\beta_i(k)$  of the according  $i$ th regime:

$$\mathbf{k}(k) = \sum_{i=1}^c \mathbf{k}_i \beta_i(k) \quad (6.137)$$

The  $c$  fuzzy basis functions  $\beta_i(k)$  are identically chosen as those of the underlying identified fuzzy model. This identity is one of the key elements of the “parallel distributed compensator” concept.

The proposed proportional fuzzy state feedback controller implies a steady-state error in the case of processes with non-integral characteristics. Different extensions can be implemented to remove the error (see also Section 6.3.3 for the classical case): the local systems can be augmented to implement local integral actions in order to remove the steady-state error (Cao *et al.*, 1999). However, it is not obvious how to design the integral gains in each local system. Alternatively, the global system can be linearised in each instance and a single steady-state error-free linear controller can be recalculated in each instance. Instead, an adaptive set-point filter will be used in the following, which scales the reference such that a possible steady-state error is removed.

The derivation of the set-point filter starts from the equivalent description of the Sugeno-type fuzzy system as a time-variant affine system:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}(k)\mathbf{x}(k) + \mathbf{b}(k)u(k) + \xi(k)e \\ y(k+1) &= \mathbf{c}^T \mathbf{x}(k+1) \end{aligned} \quad (6.138)$$

with

$$\mathbf{A}(k) = \sum_{i=1}^c \mathbf{A}_i \beta_i(k); \quad \mathbf{b}(k) = \sum_{i=1}^c \mathbf{b}_i \beta_i(k); \quad \xi(k) = \sum_{i=1}^c \xi_i \beta_i(k) \quad (6.139)$$

Note that the vector  $\mathbf{c}$  is time-invariant in the chosen representation. Similar to linear theory (see also Section 6.3.3.1), a set-point filter, enforcing

$$\mathbf{c}^T \mathbf{x}_\infty = y_\infty = y_{\text{ref},\infty} \quad (6.140)$$

within a steady-state situation  $\mathbf{A}_\infty, \mathbf{b}_\infty, \mathbf{k}_\infty, \xi_\infty, \mathbf{x}_\infty, y_{\text{ref},\infty}$ , follows as

$$V_\infty = \frac{1 - 1/y_{\text{ref},\infty} \mathbf{c}^T [\mathbf{I} - \mathbf{A}_\infty + \mathbf{b}_\infty \mathbf{k}_\infty^T]^{-1} \xi_\infty e}{\mathbf{c}^T [\mathbf{I} - \mathbf{A}_\infty + \mathbf{b}_\infty \mathbf{k}_\infty^T]^{-1} \mathbf{b}_\infty} \quad (6.141)$$

This filter requires knowledge of  $\mathbf{A}_\infty, \mathbf{b}_\infty, \mathbf{k}_\infty, \xi_\infty, \mathbf{x}_\infty, y_{\text{ref},\infty}$ , which in turn requires the availability of  $u_\infty$  and  $y_\infty$  for a given arbitrary reference  $y_{\text{ref},\infty}$ . The determination requires solving a non-linear implicit equation. Instead of doing this, an adaptive set-point filter can be used, which calculates the correction factor  $V(k)$  out of the directly available values  $\mathbf{A}(k), \mathbf{b}(k), \mathbf{k}(k), \xi(k)$  for  $u(k)$  and  $y(k)$  in each instant:

$$V(k) = \frac{1 - 1/y_{\text{ref},\infty} c^T [I - A(k) + b(k)k^T(k)]^{-1} \xi(k)e}{c^T [I - A(k) + b(k)k^T(k)]^{-1} b(k)} \quad (6.142)$$

The practical experience shows that  $V(k)$  finally converges to  $V_\infty$  for the given  $y_{\text{ref},\infty}$ . Figure 6.30 summarises the control algorithm.

|                                                                                                                             |                                              |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| $\hat{x}(k) = [y(k), \dots, y(k-n+1), u(k-1), \dots, u(k-\tau-m+1)]^T$                                                      | (state reconstruction)                       |
| $k(k) = \sum_{i=1}^c k_i \beta_i(k)$                                                                                        | (state feedback controller gain calculation) |
| $V(k) = \frac{1 - 1/y_{\text{ref},\infty} c^T [I - A(k) + b(k)k(k)^T]^{-1} \xi(k)e}{c^T [I - A(k) + b(k)k(k)^T]^{-1} b(k)}$ | (set-point scaling factor computation)       |
| $u(k) = V(k)y_{\text{ref}}(k) - k^T(k)\hat{x}(k)$                                                                           | (control signal calculation)                 |

**Figure 6.30.** Summarised algorithm of the fuzzy state feedback controller with state reconstruction and adaptive set-point filter for steady-state error removal

### 6.7.1.3 Design Parameter Choice

The controller design is carried out in two phases. The first phase follows the ideas of the “parallel distributed compensator” and considers all local systems isolated from each other. A uniform pole-placement-based state feedback control design is carried out. In the second phase the controllers are retuned under consideration of the couplings between the local systems and by permitting non-uniform closed-loop characteristics of the local system. Numerical optimisation methods are used to carry out the retuning.

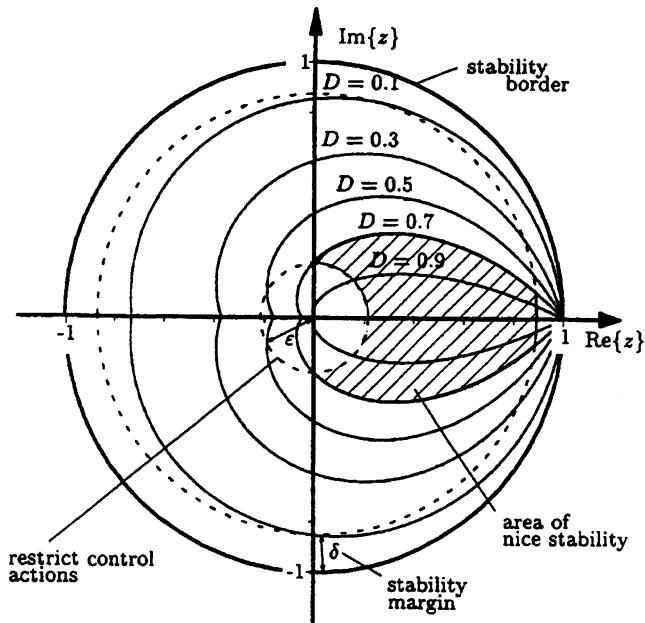
In order to place the discrete-time closed-loop poles in the first phase, linear theory can be consulted. Ackermann (1985) recommends placing the poles in the “area of nice stability”. It is enveloped by the following boundaries:

- Stability border (unit circle around origin) diminished by a safety margin  $\delta > 0$ .
- Two logarithmic spirals of constant minimal damping  $D_{\min}$ :  $z_{1,2} = r \exp(\pm j\varphi_p)$ ;  
 $r = \exp\left(-\varphi_p D_{\min} / \sqrt{1 - D_{\min}^2}\right)$ .
- Minimal distance  $\varepsilon > 0$  around the origin in order to restrict the control actions.

Figure 6.31 sketches these boundaries and the resulting “area of nice stability”. (Section 6.3.2 provides corresponding discussion for the continuous-time case.)

The closed-loop poles of each local system are placed as follows. The  $\tau$  poles at the origin, which stem from modelling the dead time by a corresponding increase of system order, remain unchanged. Also the  $m - 1$  poles at the origin, which follow from the input-related part of the state, are not altered. They are due to the chosen non-minimal realisation of the system. The  $n$  poles of the (kernel) dynamic system are placed as follows. Following Ackermann’s (1985) suggestion, a dominant pole

pair is placed on the spiral of minimal damping  $D_{\min} \approx 0.7$ . Real parts are chosen according to the dynamic requirements regarding closed-loop behaviour. For simplicity, the  $n - 2$  recessive poles are all placed at the origin. The same dominant closed-loop poles are set for all local systems. Requiring locally varying closed-loop poles is also possible, though it is difficult to specify how the poles should vary locally. The affine term is neglected during the local controller design by pole placement.



**Figure 6.31.** Area of “nice stability” of discrete-time systems (Kroll *et al.*, 2000 © 2000 IEEE)

In the second phase the *global* closed-loop system behaviour is tuned by numerical methods. The improvement potential lies within the neglected couplings of the local systems and the removal of the uniform closed-loop pole placement used. Good results were obtained with the objective of minimising the mean-squared-error between controlled variable and reference, with the latter chosen as a pseudo-random multilevel signal with sufficient duration for each period. Only the dominant pole pairs of each local system are optimised. In contrast to phase one, the local systems can have different dominant closed-loop poles in phase 2. The optimisation is carried out with consideration of the closed-loop behaviour including the active set-point filter.



## 6.7.2 Fuzzy Model Predictive Control

### 6.7.2.1 Introduction

Model (-based) predictive control (“MPC”, “MBPC”) concerns a class of automatic control algorithms that use an explicit process model to calculate an optimal future control sequence within a finite horizon. It is also referred to as “receding horizon control” or “moving horizon control”, as will be explained later. The control problem is treated in the time domain.

The idea dates back to the late 1970s, and was born in industry: within the Shell Oil Company, the use of “dynamic matrix control (DMC)” (Cutler and Ramaker, 1980; Prett and Gillete, 1980), within ADERSA/GERBIOS the use of “model predictive heuristic control (MPHC)” (Richalet *et al.*, 1978), and within Scientific Systems/ADERSA/GERBIOS the use of “model algorithmic control (MAC)” (Richalet *et al.*, 1976; Rouhani and Mehra, 1982) was invented. Later, the “predictive functional control (PFC)” was proposed by Richalet *et al.* (1987) for fast processes.

Once the optimal future control sequence is calculated within one control interval, only the first element of the optimal control sequence is applied to the process. The procedure is repeated in the next control interval, and so on. The recalculation for each cycle time permits one to use the current feedforward and feedback information in each time step in order to compensate for model mismatch and (measurable and not measurable) disturbances. Consequently, the finite horizon moves one step (control interval) towards the future with each new control sequence calculation. For this reason, MPC is sometimes referred to as “receding/moving horizon control”.

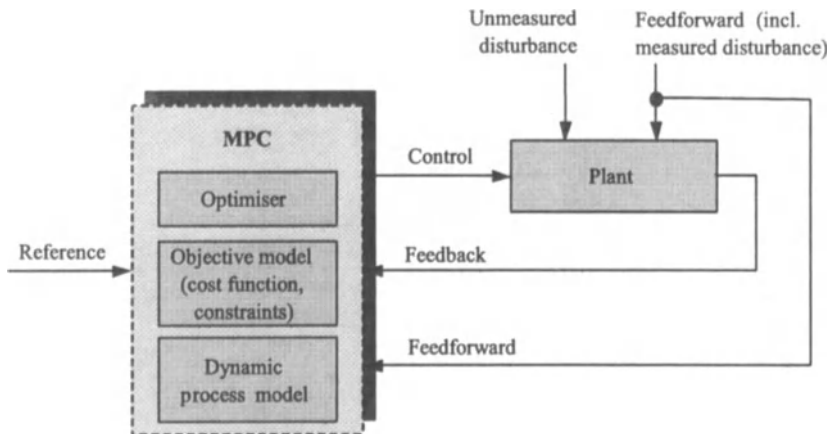
Figure 6.32 sketches a basic MPC control loop and the kernel modules of an MPC. MPCs applied in practice have additional modules, which depend on MPC type and possible extensions. For instance, a state-space-type MPC typically features a state estimator. The reference signal may be filtered, *e.g.*, to damp step-type reference signal changes.

Particularly attractive features of MPCs are:

- Constraints regarding control signal, controlled and state quantities can be considered.
- MPC is applicable to:
  - processes with dead-time,
  - processes with non-minimal phase characteristics, and
  - multivariable and unstable processes.
- Knowledge about the future course of the reference signal can be used directly to reduce sluggish servo-control.

From an industrial point of view, the availability of many commercial (linear model-based) MPC packages makes this control approach interesting. Linear model-based MPC is considered as state-of-the-art in some process industries such as refining, whereas non-linear model-based MPC is comparatively seldom applied industrially. Commercial MPC packages typically contain tools for model identification and analysis, controller design and tuning, as well as controller

validation in closed-loop simulation. Quin and Badgwell (1997) give an overview on applications and commercially available packages.



**Figure 6.32.** Basic structure of an MPC loop

As the name suggests, the “model” is one of the key parts of an MPC. The performance of an MPC depends significantly on the prediction quality of the internal model. A variety of models have been proposed in the literature:

- Linear input–output and state-space models, *e.g.*, Soeterboeck (1992) and Camacho and Bordons (1999),
- Hybrid models (or block models), such as Wiener or Hammerstein models with a static non-linearity, *e.g.*, Fruzzetti *et al.* (1997),
- Volterra series models, *e.g.*, Doyle *et al.* (1992) and Genceli and Nikolaou (1995),
- Neural networks, *e.g.*, Babuška *et al.* (1996), Latrille *et al.* (1997), and Nørgaard *et al.* (2000),
- Fuzzy models, *e.g.*, Babuška and Verbruggen (1996), Fischer *et al.* (1997), Babuška (1998) and Kroll (2000),
- Physically motivated non-linear models, *e.g.*, Allgöwer and Zheng (2000),
- Multiple interpolated linear model-based GPC (Chow *et al.*, 1995).

Also, algorithms using a linear model for optimisation and a non-linear model as prediction corrector (ExDMC, ExGPC) are proposed; *e.g.*, see Peterson *et al.* (1989), Draeger and Engell (1994) and Arahall *et al.* (1997).

For an introduction to MPC methodology, the reader is referred to textbooks such as those by Soeterboeck (1992) or Camacho and Bordons (1999). More recent results and discussions of selected aspects of non-linear MPC can be found in Allgöwer and Zheng (2000), Maciejowski (2001), and the references stated therein.

### 6.7.2.2 Mathematical Problem Formulation and Design Parameter Choice

Within each optimisation cycle, *i.e.* each  $T_0$ , an (I/O model-based) MPC solves the optimisation problem:

$$\min_{\mathbf{u}(0), \dots, \mathbf{u}(N_u-1)} J(k) = \sum_{j=N_1}^{N_2} \|\hat{\mathbf{y}}(k+j) - \mathbf{y}_{\text{ref}}(k+j)\|_Q^2 + \sum_{j=0}^{N_u-1} \|\mathbf{u}(k+j) - \mathbf{u}(k-1+j)\|_R^2 + \sum_{j=0}^{N_u-1} \|\mathbf{u}(k+j) - \mathbf{u}_s\|_S^2 \quad (6.143)$$

subject to a (I/O) model constraint (state-space case accordingly):

$$\hat{\mathbf{y}}(k) = \mathbf{f}(\hat{\mathbf{y}}(k-1), \mathbf{u}(k-1)) \quad (6.144)$$

and subject to inequality constraints:

$$\begin{aligned} \mathbf{u}_{\min} &\leq \mathbf{u}(k) \leq \mathbf{u}_{\max} \\ \Delta \mathbf{u}_{\min} &\leq \Delta \mathbf{u}(k) \leq \Delta \mathbf{u}_{\max} \\ \mathbf{y}_{\min} &\leq \hat{\mathbf{y}}(k) \leq \mathbf{y}_{\max} \end{aligned} \quad (6.145)$$

within the corresponding horizons. (Only rarely is another metric than the Euclidian used in the cost function.)  $\mathbf{u}(k)$  is the vector of control signals.<sup>6</sup>  $\hat{\mathbf{y}}(k)$  is the vector of predicted controlled quantities.<sup>7</sup>  $\mathbf{y}_{\text{ref}}(k)$  is the vector of reference values.  $\mathbf{u}_s$  is the vector of desired steady-state values of the control signals.  $\|\cdot\|$  is an inner product type vector norm (with form matrices  $\mathbf{Q}$ ,  $\mathbf{R}$  or  $\mathbf{S}$  as specified in the subscript).  $N_1$  ( $N_2$ ) is the minimum (maximum) costing horizon, and  $N_u$  is the control horizon. Typically, the control horizon is, for computational reasons, shorter than the maximum costing horizon ( $N_u < N_2$ ). Therefore, the last calculated control value  $\mathbf{u}(k + N_u - 1)$  is kept constant through the remaining costing horizon up to  $N_2$ .

The result of the optimisation is the control sequence  $\mathbf{u}(k), \dots, \mathbf{u}(k + N_u - 1)$ , of which only the first element  $\mathbf{u}(k)$  is applied to the process. A new optimal control sequence is calculated in the next time step  $k + 1$ : it considers the process outputs as feedback and the measured disturbances as feedforwards, as they actually occur at time  $k + 1$ .

The cost function  $J$  is composed of three terms. The first term punishes the deviation of the controlled variable from the reference value. This impacts the control quality. The second term assigns a cost to the rate of change in order to punish fast changes of manipulated variable(s): too rapid a change may stimulate the process in unwanted manner and may decrease the life expectancy of the actuators. The third term punishes the deviation of the control variable from the desired steady-state value  $\mathbf{u}_s$ . This is used to punish deviations from an economic optimum (energy, utilities, raw material consumption). The steady-state values are sometimes optimised within a steady-state optimiser. In de Nicolao *et al.* (1997) and Chen and Allgöwer (1998), an additional terminal constraint term is proposed to guarantee the stability of the closed loop. In order to guarantee the feasibility of the solutions, the hard constraints regarding the controlled values  $\mathbf{y}$  are often exchanged by soft constraints. Soft constraints can be violated, but this is punished with a high cost.

Figure 6.33 sketches the concept of the “receding horizon”. The SISO case is considered, as it is sufficient to explain the concept. At time  $k$ , a control sequence

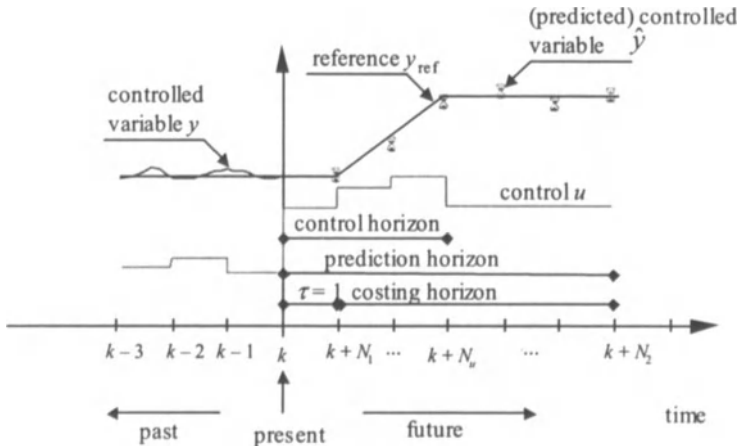
<sup>6</sup> In MPC context a control signal is also referred to as a “manipulated variable (MV)”.

<sup>7</sup> In MPC context a controlled quantity is also referred to as a “controlled variable (CV)”.

$u(k), \dots, u(k + N_u - 1)$  is calculated, such that the controlled quantity  $\hat{y}$  optimally follows the reference sequence  $y_{\text{ref}}$ .

The design parameters for a given model are the horizons  $N_1$ ,  $N_2$  and  $N_u$ , and the weights  $\mathbf{Q}$ ,  $\mathbf{R}$  and  $\mathbf{S}$  within the cost function. The constraints typically follow from the process and the instrumentation.

The minimum costing horizon  $N_1$  follows straight from the process dead-time:  $N_1 = 1 + \tau$ . For systems without dead time,  $N_1 = 1$  is used. The minimum costing horizon is increased by the dead time. By doing this, only the part of the future process behaviour is punished, which is a reaction to the controls that are subject to the current optimisation cycle. The maximum costing horizon  $N_2$  depends on the dominant time constant of the process. A rule of thumb is that it should reflect the settling time. Note that the computational effort increases with  $N_2$ . The choice of the control horizon  $N_u$  also means the determination of the number of optimisation variables. Therefore, it should be chosen as large as necessary, but as small as possible to limit the computational effort.



**Figure 6.33.** Concept of moving (receding) horizon and optimal future control profile

Typically, diagonal matrixes  $\mathbf{Q}$ ,  $\mathbf{R}$  and  $\mathbf{S}$  are used to weight the cost contribution of each time step. The matrices also adjust the weighting of the costs associated with deviation from reference, control action and rate of change of control action relative to each other. The choice of the elements is part of the design task. The weights can also be time-variant. A simple choice of the initial control sequence is a zero vector (“cold start”). Alternatively, the results of the previous optimisation run can be used as initialisation (“hot start”). The tuning of linear MPC (DMC, GPC, UPC) is discussed in more detail, for example, by Soeterboeck (1992), Rani and Unbehauen (1997) and Maciejowski (2001).

### 6.7.2.3 Implementation Details for Model Predictive Control with Fuzzy Prediction Model

A non-linear fuzzy model-based MPC has been selected for further elaboration due to the non-linear nature of hydraulic drives. It utilises an enhanced Sugeno-type

fuzzy model (of input–output type) as described in Section 5.4.2.3 as a prediction model. After the MPC concept was introduced in the previous sections, this section summarises the details required to implement such an MPC practically. The handling of issues like model mismatch is explained. The concept was proposed by Kroll and Bernd (2000).

The objective of applying an MPC to hydraulic drives in this context is to provide for a reference of control performance, permitting one to assess the performance of other approaches. The use of a non-linear model to calculate the optimal control sequence requires applying computationally costly non-linear constraint optimisation methods. Not being targeting at an algorithm, which achieves real-time performance on a standard industrial control platform, it was regarded as sufficient to use the standard SQP solver provided by MATLAB<sup>TM</sup> Optimisation Toolbox (Grace, 1990). Regulatory control of hydraulic drives typically requires sampling times in the range of 1–2ms. In order to meet such cycle-time requirements, tailored approaches are required. Such cycle times are more than challenging for commercial linear MPC products. Nevertheless, even shorter cycle times have been reported for a special algorithm for mechatronic applications.

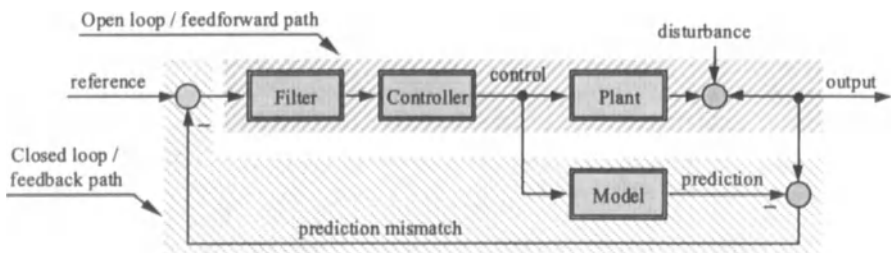
Several contributions to improve the computational performance by algorithmic changes have been proposed. For example, Babuška (1998) proposes a combination of inverse model control and branch-and-bound optimisation for SISO and a combination of branch-and-bound optimisation and SQP for MIMO systems. Branch-and-bound optimisation is combined with another method, as it uses amplitude-discretised control signals only, which can cause oscillations around the reference trajectory. Much research has been done on improving numerical solution methods for differential algebraic equation (DAE) model-based NMPCs; see the contributions in Allgöwer and Zheng (2000). Camacho and Bordons (1999) propose a “feasible direction [optimisation] method”, which guarantees improvements even if the iteration has to be stopped as the MPC runs out of time in real-time applications. Others simplify the modelling concept, in order to solve only a linear model-based problem, while using non-linear prediction components: Fischer *et al.* (1997) linearise a Sugeno-type model in each instant and solve a linear model-based MPC problem. Extensions of DMC and GPC algorithms have been proposed (ExDMC, ExGPC) that use a linear model for optimisation and a non-linear model as prediction corrector, *e.g.*, see Peterson *et al.* (1989), Draeger and Engell (1994) and Arahal *et al.* (1997).

The MPC uses feedback to compensate for unmodelled/unmeasured disturbances and model error (mismatch). The steady-state error can be removed by augmentation of the kernel MPC algorithm with an error model, or by applying the MPC within an internal model control (IMC) structure. Both can be shown to be equivalent under some assumptions. In contrast to IMC, an error model can also be of higher order, which is difficult to derive, though.

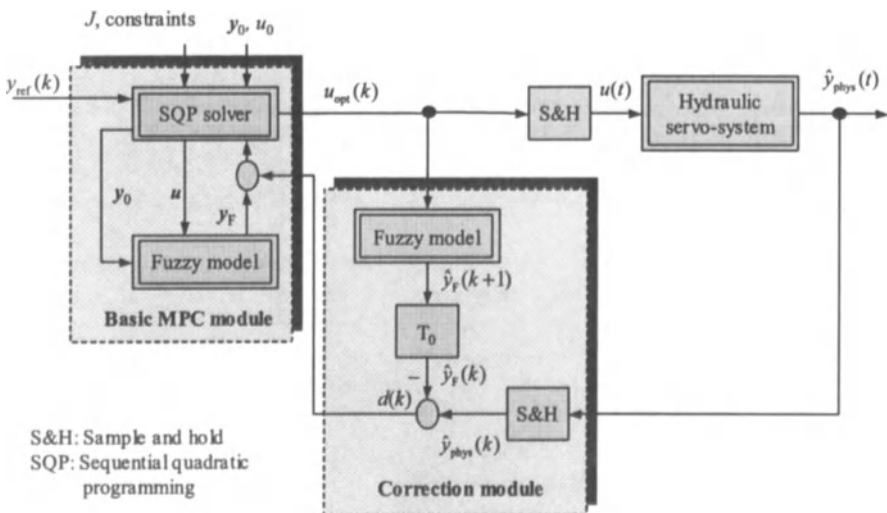
What does “IMC” mean? Ideal IMC uses an inverse plant model as controller but feeds back the prediction error to compensate for disturbances and model mismatch. It features stability, perfect control and zero offset (no steady-state error) in the linear (Garcia and Morari, 1982) and the non-linear case (Economou *et al.*, 1986) for a perfect plant model. Figure 6.34 shows the basic structure of a non-linear IMC scheme. The perfect control means infinite gain of a corresponding feedback controller and assumes a perfect model. Hence, for practical applications, a filter is

typically introduced to increase robustness and avoid sensitivity and stability problems (Economou *et al.*, 1986). Doyle *et al.* (1992) proposed replacing the inverse plant-type controller by an MPC to avoid the problems of plant model inversion. By doing this, one obtains a control structure that is equivalent to a zero-order error model as described in the following paragraph.

Another concept to remove steady-state error is the use of a prediction error model. The simplest case is a zero-order error model, which corresponds to the zero-order “disturbance model” concept in Genceli and Nikolaou (1995): by comparing the process output due to the applied control action with the model prediction due to the same control action, the mismatch  $d(k) = y(k) - \hat{y}(k)$  can be calculated. It is used in the objective function to compensate the offset in the MPC prediction model (which is a copy of this model).



**Figure 6.34.** Non-linear IMC scheme (Economou *et al.*, 1986): the controller is assumed to be the inverse plant, which defines the open-loop part with the plant. The parallel plant model with residual calculation and feedback defines the closed loop.



**Figure 6.35.** Structure of fuzzy MPC with basic module and steady-state model mismatch correction module (developed out of figure in Kroll and Bernd (2000), p 334 (original publication). Reprinted with kind permission of VDI Verlag.)

Figure 6.35 illustrates this concept: a sampled data system is shown, as the process is simulated by a continuous-time physical plant model. Sample and hold (S&H) blocks are used to connect the discrete-time MPC and the continuous-time process simulator. A zero-order hold is assumed. The single delay  $T_0$  is introduced in the correction module as the model provides for a one-step-ahead prediction of  $\hat{y}(k+1)$  for a given input, while the process output reflects current time  $y(k)$ .

Further on, a SISO MPC problem with the zero-order error model and diagonal weight matrices with time-invariant weights will be used. This results in the following objective function:

$$\begin{aligned} \min_{\Delta u(k), \dots, \Delta u(k+N_u-1)} J(k) = & \sum_{j=N_1}^{N_2} \alpha [y_{\text{ref}}(k+j) - \hat{y}(k+j) - d(k)]^2 \\ & + \sum_{j=0}^{N_u-1} \beta \Delta u^2(k+j) + \sum_{j=0}^{N_s-1} \gamma [u(k+j) - u_s]^2 \end{aligned} \quad (6.146)$$

subject to

$$u(k) = u(N_u) \quad \forall N_2 \geq k > N_u$$

$$u(k) = u(k-1) + \Delta u(k)$$

$$y_{\min} \leq \hat{y}(k) \leq y_{\max}$$

$$u_{\min} \leq u(k) \leq u_{\max}$$

$$\Delta u_{\min} \leq \Delta u(k) \leq \Delta u_{\max}$$

The entire algorithm is summarised in Figure 6.36. Note that choosing  $\Delta u_0=0$  as initial value is a cold start regarding  $\Delta u$  but a warm start regarding  $u$ , which is the physically relevant quantity.

|                                                                                                                                                                                                                         |                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| $\min_{\Delta u(k), \dots, \Delta u(k+N_u-1)} J(k) = \sum_{j=N_1}^{N_2} \alpha [y_{\text{ref}}(k+j) - \hat{y}(k+j) - d(k)]^2$ $+ \sum_{j=0}^{N_u-1} \beta \Delta u^2(k+j) + \sum_{j=0}^{N_s-1} \gamma [u(k+j) - u_s]^2$ | (objective function)                                 |
| subject to:                                                                                                                                                                                                             |                                                      |
| $u(k) = u(k-1) + \Delta u(k)$                                                                                                                                                                                           | (change of control is optimised)                     |
| $u(k) = u(N_u) \quad \forall N_2 \geq k > N_u$                                                                                                                                                                          | (control not optimised for $k > N_u$ )               |
| $\hat{y}(k) = f(\hat{y}(k-1), u(k-1))$                                                                                                                                                                                  | (fuzzy prediction model)                             |
| $d(k+j) = y_{\text{process}}(k) - \hat{y}_F(k) \quad \forall j \in \{N_1; N_2\}$                                                                                                                                        | (error model)                                        |
| $y_{\min} \leq \hat{y}(k) \leq y_{\max}$                                                                                                                                                                                | (constraint w.r.t. controlled variable)              |
| $u_{\min} \leq u(k) \leq u_{\max}$                                                                                                                                                                                      | (constraint w.r.t. control action)                   |
| $\Delta u_{\min} \leq \Delta u(k) \leq \Delta u_{\max}$                                                                                                                                                                 | (constraint w.r.t. rate of change of control action) |

**Figure 6.36.** Summarised fuzzy MPC algorithm for an input–output SISO model

## 6.8 Neural-network-based Control

Most control design approaches, which can be used for fuzzy control, are also suitable to be applied in neural-network-based control schemes (neuro-control). However, it is not within the scope of this section to discuss all these control design methods in detail. The reader should refer to Narendra and Parthasarathy (1990), Hunt *et al.* (1992) and the more recent book of Nørgaard *et al.* (2000) for more complete surveys of the subject. In the following, only selected neural-network-based control design methods will be briefly described.

There are two fundamentally different approaches to neuro-control design (Nørgaard *et al.*, 2000):

1. **Direct Control System Design.** The controller itself is a neural network. Typical examples are
  - direct inverse control (see Section 6.6.1) and specialised inverse control
  - internal model control (see Section 6.7.2)
  - feedback linearisation (see Section 6.5 and 6.8.1)
  - feedforward control (see Section 6.4.1)
  - optimal control (see Section 6.3).

The ANN-based versions of such controllers are easy to implement, but difficult to design and to tune, implying a retraining of the network every time a design parameter is modified.

2. **Indirect Control System Design.** A neural network (trained in advance) is used as a system model in a more “conventional” control design. Therefore, the processes of modelling/identification and control design can be performed separately. Prominent methods of this category are
  - control based on instantaneous linearisation (see Section 6.8.2)
  - non-linear predictive control (see Section 6.7.2).

The indirect control design approach is very flexible, and thus appropriate for the majority of common control problems.

### 6.8.1 Neural-network-based Feedback Linearisation

The concept of feedback (and input–output) linearisation has been described in Section 6.5 based on “classical” (non-linear) continuous-time state-space models of the form in Equation 6.61. This concept is also widely used in the ANN community, and a number of publications have appeared on the application of feedback linearisation using neural networks. The reader can consult Sanner and Slotine (1992), and Tzirkel-Hancock and Fallside (1992), Jagannathan and Lewis (1996) and Nørgaard *et al.* (2000).

Based on the canonical representation of Equations 5.119 and 5.120 (and keeping in mind the presentation in Section 6.5.1), the principle of applying feedback linearisation based on neural networks is quite easy. The counterpart to Equation 6.85 follows straightforwardly as

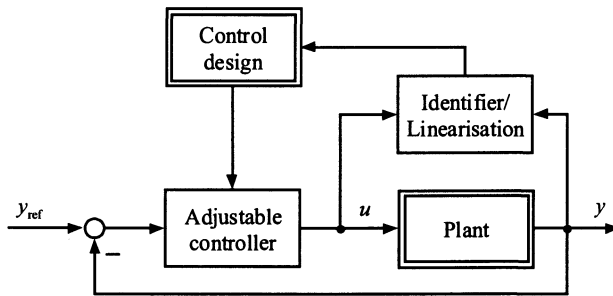
$$u(k) = \frac{v(k) - \hat{f}(y(k), \dots, y(k-n+1), u(k-1), \dots, y(k-m+1), \boldsymbol{\theta}_f)}{\hat{g}(y(k), \dots, y(k-n+1), u(k-1), \dots, y(k-m+1), \boldsymbol{\theta}_g)} \quad (6.147)$$



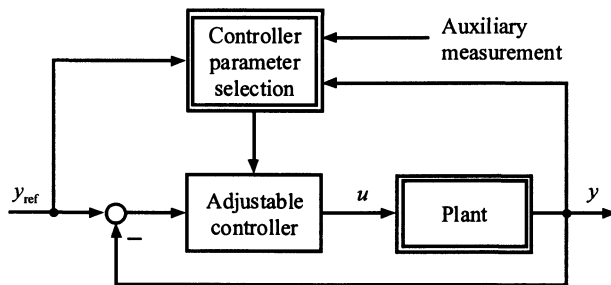
Once the two separate neural networks approximating the functions  $f$  and  $g$  have been trained (which, of course, implies extensive computation for training and network selection), the implementation of the controller is relatively simple.

### 6.8.2 Control Based on Instantaneous Linearisation

An intuitive approach to designing non-linear controllers is to linearise the non-linear system model (classical, fuzzy, neuro) at each sampling instant, and then to (online) design an appropriate linear controller. The main advantage of this approach lies in the fact that all well-established linear control design methods can be applied. The drawback, however, is that it leads to control schemes that require relatively high computational burden for real-time implementation. This non-linear control approach reminds of the scheme of model-based adaptive control (Figure 6.37) or the gain scheduling control (Figure 6.38) approach.



**Figure 6.37.** Equivalent structure for a self-regulator control and a control based on instantaneous linearisation



**Figure 6.38.** Structure of gain scheduling controller

Indeed, the structural equivalence with indirect self-tuning regulators described in standard textbooks on adaptive control (Wang, 1994; Åström and Wittenmark, 1995; Unbehauen, 1995) is obvious. Formally, only the block "identifier" in an indirect self-tuning regulator has to be replaced by the block "linearisation" to obtain a controller based on instantaneous linearisation. Loosely speaking, "identification" can be seen as a kind of "linearisation through online parameter estimation".

Consequently, many considerations from the self-tuning regulator can be directly carried over when selecting the controller design. In addition, the required computational effort as well as the related problems for the two methods are widely comparable.

On the other hand, a control approach, which switches between predefined (local) linear controllers, each having been tuned (offline) for a specific operating point/regime, is known as gain scheduling control (Rugh, 1991; Shamma and Athans, 1990, 1992). If the switching progresses continuously, a gain scheduling control scheme with *infinite schedule* is obtained. From this fact, the similarity to the control based on instantaneous linearisation is evident.

As mentioned above, control based on instantaneous linearisation is appealing, since any linear control design can be incorporated in the design block. Typical examples are pole placement design, optimal control, and predictive control. More or less detailed information about this control approach (combined with different model types and different linear control designs) can be found in Sørensen (1993, 1996), Ahmed and Tasadduq (1994), Lightbody and Irwin (1995), Fischer *et al.* (1998), Fink *et al.* (1999), Nørgaard *et al.* (2000), for instance.

## 6.9 Vibration Damping Control

### 6.9.1 Introduction

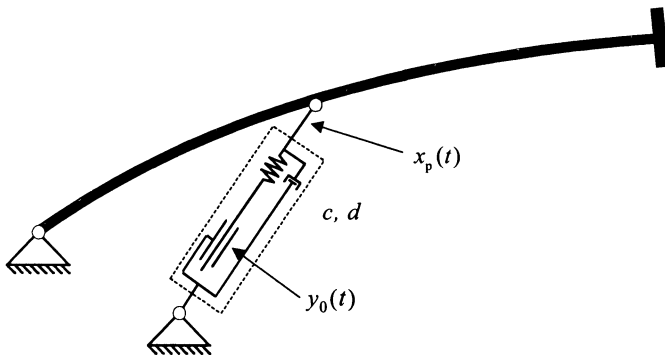
Generally, all mechanical systems are subject to deformation and hence are compliant. The speed, operating range, and/or load of rigid link robots are low, as mechanical links are assumed to be rigid, and so compliance is neglected. Two cases are obvious, where elastic deformations have to be taken into account: heavy loads, and very wide operating ranges. A pump-manipulator for distributing concrete in the construction industry (Figure 2.13) is an example of a very wide operating range. In such applications, actuators with a significant power density are required to be able to produce high forces. (Light actuators are necessary because they significantly determine the weight of the total link.) This can be achieved profitably by the use of *hydraulic actuators*.

During the last decade, robots with flexible links have become a popular research object for control engineers owing to their sophisticated properties (Book, 1993). The majority of researchers have concentrated on robots with serial chains of joints and flexible links. Usually, only one or two link manipulators are treated, based on a schematic arbitrary formulation for the Lagrangian dynamics of serial chains of flexible links and joints (Book, 1984).

Although it is possible to obtain a linear model for a single-link flexible robot (Fraser and Daniel, 1991; Canadus de Wit *et al.*, 1996), massive problems occur when trying to obtain an accurate analytical model for multi-link flexible robots in practice, as the extent of the equations grows rapidly with each additional link. In the case of closed kinematic loops and, in addition, actuators with significant eigendynamics, like hydraulic actuators (Heintze and van der Weiden, 1995; Bernzen *et al.*, 1997), it is almost impossible to obtain an analytical model suitable for controller design and especially a verified model. Numerical simulation models can be derived using multi-body system toolkits (Hiller, 1996).

There are different passive and *active* methods for *vibration damping* of such mechanical systems. Whereas active vibration damping approaches for flexible robots are generally model based, passive strategies try to increase the damping properties of the system using, for example, suitable materials for the links or surface treatment with visco-elastic layers (Alberts *et al.*, 1992; Canadus de Wit *et al.*, 1996). Another well-known passive strategy is the use of passive mechanical vibration absorbers (mass–spring–damper elements). This development results in the use of pure actuators with fully tuneable but passive properties (Juang and Phan, 1992). This unites the advantages of passive absorbers with the flexibility of active ones.

The approach of active damping control for multi-link flexible robots utilised in this section uses the actuator of the robot (a hydrostatic differential cylinder) as a virtual passive mechanical element by controlling the actuator in a suitable way (Bernzen, 1997, 1999a,b). The spring–damper element is a passive energy dissipative device that can never destabilise a system, and thus the control concept is not sensitive to modelling errors. To satisfy performance requirements, the parameters spring stiffness and damping constant have to be adjusted experimentally or by using a system model (Nissing, 2002). The more that is known about the system, the better these parameters may be adjusted to meet the performance requirements. The new input of the controlled system is a virtual variation of the spring base (see Figure 6.39).



**Figure 6.39.** Flexible robot with a spring–damper element and variation of the spring base (Taken from Bernzen (1999), p 46 (original publication). Reprinted with kind permission of VDI Verlag.)

## 6.9.2 Vibration Damping Concept

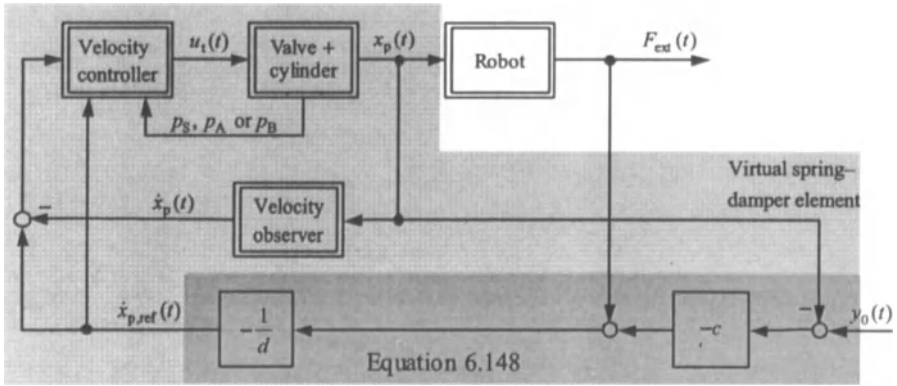
The main aspect of vibration damping is to reduce the vibration energy in the system, which can only be done actively by the actuator. The simplest (passive) mechanical system to realise this is a spring–damper element, which generates a force proportional to piston position and velocity by the factors spring stiffness  $c$  and damping constant  $d$  respectively (see Figure 6.39). Treating the actuator as a virtual passive spring–damper element results in a new system with new properties depending on the spring stiffness and the damping constant. In reality, the velocity

$\dot{x}_p$  of the hydraulic cylinder has to be controlled depending on the position  $x_p$  of the cylinder and the force  $F_{\text{ext}}$  acting on it and measured by a force sensor.

The robot arm can then be driven by an assumed variation of the spring base  $y_0$ , as is illustrated in Figure 6.39. This results in a reference piston velocity for the cylinder of

$$\dot{x}_{p,\text{ref}}(t) = -\frac{F_{\text{ext}}(t) - c(y_0(t) - x_p(t))}{d} \quad (6.148)$$

The damper base is assumed to be absolutely fixed, *i.e.*,  $\dot{y}_0 = 0$ ; see Bernzen (1997 1999a) for more details. The whole active vibration damping control concept is shown in Figure 6.40.



**Figure 6.40.** Block diagram of the vibration damping controller structure (Taken from Nissing (2002), p 20 (original publication). Reprinted with kind permission of VDI Verlag.)

### 6.9.3 Integrated Velocity Control

Hyrostatic differential cylinders have strong non-linear static and dynamic effects. To realise a simple and effective controller for practical implementation, a simplified version of the cascade non-linear control scheme presented in Section 6.6.2 is used. The velocity feedback controller for the hydrostatic differential cylinder is designed using the simple model of Equations 4.198 and 4.199.

In the first step to practical implementation the pressure dynamics are neglected. Equations 4.198 and 4.199 are used as a basis for velocity control with  $\dot{p}_A = \dot{p}_B = 0$  (neglecting the leakage flows), which results in

$$u = \begin{cases} \frac{A_p \dot{x}_{p,\text{ref}}}{c_v \sqrt{p_s - p_A}} & \forall \dot{x}_{p,\text{ref}} \geq 0 \\ \frac{A_p \dot{x}_{p,\text{ref}}}{c_v \sqrt{p_A - p_T}} & \forall \dot{x}_{p,\text{ref}} < 0 \end{cases} \quad (6.149)$$

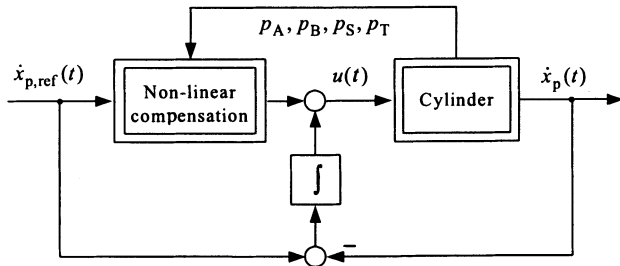
and

$$u = \begin{cases} \frac{\alpha A_p \dot{x}_{p,ref}}{c_v \sqrt{p_B - p_T}} & \forall \dot{x}_{p,ref} \geq 0 \\ \frac{\alpha A_p \dot{x}_{p,ref}}{c_v \sqrt{p_S - p_B}} & \forall \dot{x}_{p,ref} < 0 \end{cases} \quad (6.150)$$

That means that only one chamber pressure ( $p_A$  or  $p_B$ ) has to be measured. An additional (output error) integrating feedback path is added to the non-linear velocity compensation, as depicted in Figure 6.41.

## 6.10 State Estimation

In most feedback control applications of hydraulic actuators presented above, the use of a state estimator is common (von Feuser, 1984; Köckemann *et al.*, 1991; Neumann *et al.*, 1991a, 1991b; Jaritz and Spong, 1996). Restricting attention to the basic dynamics of the HSS, while the pressures and the position are assumed to be measured, the state estimator basically serves as a velocity or acceleration estimator, since the application of such transducers is often considered to be a too expensive hydraulic servo-technique.



**Figure 6.41.** Velocity controller (Taken from Bernzen (1999), p 66 (original publication). Reprinted with kind permission of VDI Verlag.)

### 6.10.1 Velocity Estimation

Different techniques for estimating the velocity are described in this section.

#### 6.10.1.1 Velocity Estimation Using Finite Differences Method

The best direct way to estimate velocity is the application of a digital filter, with differentiating characteristics up to a certain frequency and high-frequency roll-off to avoid the high sensitivity to noise. This is due to the fact that it is often difficult to achieve good signal validity of the position signal. Even after application of an analogue anti-aliasing filter with high cut-off frequency the signal can be rather noisy.

The differentiating characteristics are basically obtained by applying some approximate difference method. The simplest method to compute velocity from position measurements is to apply the Euler approximation

$$v(k) = \frac{x(k) - x(k-1)}{T_0} \quad (6.151)$$

This approximation works well as long as the sampling interval is not too low, as the differentiation includes a division by the sample time. For this reason, several alternative schemes to estimate the velocity have been proposed and tested.

The trade-off lies between noise reduction and using the most recent values for the control. One solution is to simply average the last  $n_{sa}$  (typically  $n_{sa} = 3$ ) velocity estimations,  $v(k), \dots, v(k - n_{sa} + 1)$ , with  $v(l)$  calculated from the Euler approximation (Equation 6.151)

$$v_{sa}(k) = \sum_{l=0}^{n_{sa}-1} v(k-l) = \frac{1}{nT_0} [x(k) - x(k - n_{sa})] \quad (6.152)$$

This equation (known as sliding average approximation) is comparable to using an  $n_{sa}$ -times longer sampling time for the velocity estimation to avoid the velocity degradation, which comes with such high sampling frequency.

Alternatively, velocity can easily be estimated in real time by using a first-order low-pass filtered differentiator with the transfer function (Tafazoli *et al.*, 1998)

$$H(s) = \frac{\kappa s}{s + \kappa} \quad (6.153)$$

where  $\kappa$  is the filter gain. Equation 6.153 can be transformed, using  $s \approx (z-1)/(T_0 z)$ , to

$$H(z) = \frac{\kappa(z-1)}{(1 + \kappa T_0)z - 1} \quad (6.154)$$

which is equivalent to the recursive form

$$v(k) = \frac{1}{1 + \kappa T_0} v(k-1) + \frac{\kappa T_0}{1 + \kappa T_0} \frac{x(k) - x(k-1)}{T_0} \quad (6.155)$$

The application of filters according to Equations 6.152 and 6.153 obviously involves a trade-off between the noise level and the phase lag in the estimated velocity signal. Thus the filter parameters  $n_{sa}$  and  $\kappa$  should not be chosen too high or too low respectively.

Note that a high-order delay-free digital FIR filter (*i.e.*, zero-phase forward and reverse digital filter) can be used to avoid this problem completely, but only if off-line estimation is considered, as this kind of filter is *non-realisable*.

### 6.10.1.2 Linear State Estimator (Kalman Filter)

A more sophisticated method to estimate the velocities is by using a Kalman filter (rather than by placing poles in a deterministic setting) for the system model in the form

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{v} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{w}\end{aligned}\quad (6.156)$$

with the initial state  $\mathbf{x}_0 = \mathbf{x}(t_0)$ , a white process term  $\mathbf{v}$ , and a measurement noise term  $\mathbf{w}$ , which are assumed to be independent and Gaussian, *i.e.*,

$$\mathbb{E}\{\mathbf{v}(t)\mathbf{v}^T(\tau)\} = \mathbf{Q}\delta(t-\tau) \quad \mathbb{E}\{\mathbf{v}(t)\} \equiv 0 \quad (6.157)$$

$$\mathbb{E}\{\mathbf{w}(t)\mathbf{w}^T(\tau)\} = \mathbf{R}\delta(t-\tau) \quad \mathbb{E}\{\mathbf{w}(t)\} \equiv 0 \quad (6.158)$$

Then, an online unbiased estimate  $\hat{\mathbf{x}}$  of  $\mathbf{x}$  is provided by the equation (Anderson and Moore, 1989)

$$\dot{\hat{\mathbf{x}}} = \mathbf{F}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} - \mathbf{L}\mathbf{y}(t) \quad (6.159)$$

or

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}[\mathbf{C}^T\hat{\mathbf{x}} - \mathbf{y}(t)] \quad (6.160)$$

where the filter gain  $\mathbf{L}$  and the estimate matrix  $\mathbf{F}$  are given from

$$\mathbf{F} = \mathbf{A} + \mathbf{L}\mathbf{C}^T \quad (6.161)$$

and

$$\mathbf{L} = -\mathbf{P}\mathbf{C}\mathbf{R}^{-1} \quad (6.162)$$

$\mathbf{P}$  is the solution of the matrix Ricatti equation

$$\dot{\mathbf{P}} = \mathbf{P}\mathbf{A}^T + \mathbf{A}\mathbf{P} - \mathbf{P}\mathbf{C}\mathbf{R}^{-1}\mathbf{C}^T\mathbf{P} + \mathbf{Q} \quad (6.163)$$

In order to keep the estimator design transparent, the covariance matrices should be chosen diagonal, as follows:

$$\mathbf{Q} = \mathbb{E}\{\mathbf{v}\mathbf{v}^T\} = \text{diag}[q_{11} \quad q_{22} \quad \dots \quad q_{nn}] \quad \mathbf{R} = \mathbb{E}\{\mathbf{w}\mathbf{w}^T\} = \rho\mathbf{I}_p \quad (6.164)$$

Thus, the Kalman filter design is established by the choice of process noise variances  $q_{ii}$ , and the weighting factor  $\rho$ , based on the (linear) third-order actuator model (Equation 4.250 or 4.252, with the state  $\mathbf{x}$  constituted by the state variables  $x_p$ ,  $\dot{x}$  and  $p_L$  (with measurable  $x_p$  and  $p_L$ )).

**Interpretation of Variances and Weighting Factor.** The weighting factor  $\rho$  can be interpreted as a measure for the reliability of the measured output signals, relative to the quality of the model used in the estimator. For small  $\rho$ , representing small measurement noise, the measured outputs are “reliable”, and high-gain feedback is allowed. In other words, by reducing  $\rho$ , the estimator dynamics can be made faster,

resulting in fast decay of estimation errors with respect to the measured output signals. Obviously, the cost is increased noise sensitivity.

The process noise variances  $q_{ii}$  can be interpreted as a measure of the reliability of the individual state equations of the system model. For instance, if the third state equation of the actuator model (Equation 4.250 or 4.252), which should describe the dynamic behaviour of the actuator pressure difference, is a bad approximation of the real pressure difference dynamics,  $q_{33}$  can take large values. The effect on the Kalman filter design will be a high-gain feedback of the difference between the estimated and the measured pressure difference. In other words, the estimated pressure difference  $p_L$  is based on the measured pressure difference rather than on the model describing the dynamic behaviour of the pressure difference as a result of the control input (van Schothorst, 1997).

**Full Order vs. Reduced Order State Estimator.** The advantage of full order estimators, using the complete plant model, is that they have better noise reduction properties. This is due to the fact that there is no direct feed-through of measured output signals to the estimated state signals. The reduced order state estimator, usually of the order of the number of unmeasured states, may have direct feed-through of measured output signals to the estimated state signals. Hence, it is more sensitive to measurement noise, and will not be discussed here. Nevertheless, the interested reader could consult van Schothorst (1997) on this topic, where a comparison of both concepts is made.

### 6.10.1.3 Non-linear Observers/Filters

The Kalman filter is a linear state observer, in which gains are optimised for the assumed noise. Another possible solution to the velocity estimation problem would be to design a state observer that includes the non-linearities of the system, *i.e.*, non-linear observers. If the properties of process and measurement noise as well as parameter uncertainties have to be taken into account, non-linear filters can be designed, such as the extended Kalman filter (Jazwinski, 1970) or the strong tracking filter (Zhou and Sun, 1994; Bai, 1998). However, the design, implementation and tuning of such observers/filters is much more difficult than for the linear case, and thus will not be treated here.

## 6.10.2 Estimation of Acceleration and Friction Forces

As friction forces are difficult to measure directly, the use of an observer may be helpful to estimate them. Researchers have developed friction observers of varying simplicity (*e.g.*, Friedland and Park, 1992; Friedland and Mentzelopoulou, 1992; Tafazoli *et al.*, 1998), some of which will be discussed in the following.

**Friedland–Park Coulomb Friction Observer.** One simple non-linear observer was proposed by Friedland and Park (1992), which uses the force applied by the actuator and calculated from the pressure readings, *i.e.*,  $F_L$ .

The friction force is assumed to be of classical Coulomb type, as given by

$$F_c = m\alpha_c \operatorname{sign}(\dot{x}) \quad (6.165)$$



To estimate the Coulomb friction parameter  $\alpha_c$ , Friedland and Park proposed the following non-linear observer:

$$\begin{aligned}\hat{\alpha}_c &= z - k_a |\dot{x}|^\mu \\ \dot{z} &= k_a \mu |\dot{x}|^{\mu-1} (F_L - \hat{F}_c) \text{sign}(\dot{x})\end{aligned}\quad (6.166)$$

where  $\hat{\alpha}_c$  is the estimated Coulomb friction parameter,  $z$  denotes the state variable of the friction observer, and  $\hat{F}_c$  the estimated Coulomb friction, *i.e.*,

$$\hat{F}_c = m \hat{\alpha}_c \text{sign}(\dot{x}) \quad (6.167)$$

The design parameters are  $k_a > 0$  and the exponent  $\mu > 0$ .

The approximate pole of the observer is given by

$$s_{\text{FPO}} = -k_a \mu |\dot{x}|^{\mu-1} \quad (6.168)$$

which can be used to choose the gain  $k_a$  and the exponent  $\mu$ . The latter should lie in the range  $1 < \mu < 2$  to obtain a high speed of convergence (Tafazoli *et al.*, 1998).

**Modification of the Friedland–Park Observer.** Tafazoli *et al.* (1998) suggested the empirical function

$$F_f = F_c + (F_L - F_c) \exp\left[-(\dot{x}/D_v)^2\right] \quad (6.169)$$

in which  $F_L$  stands for the actuator force,  $F_f$  for the net friction force,  $F_c$  for the Coulomb friction force, and  $D_v$  for the threshold velocity. A small neighbourhood of zero velocity is defined by  $D_v$ . Outside this neighbourhood, friction is mainly dynamic in nature (a function of velocity). Within the neighbourhood, velocity is considered negligible, and static friction is dominant (a function of the applied force). Thus the net friction force can be computed from the following equation:

$$\hat{F}_f = \hat{F}_c + (F_L - \hat{F}_c) \exp\left[-(\hat{x}/D_v)^2\right] \quad (6.170)$$

Likewise, the estimated acceleration would be

$$\hat{a} = \frac{F_L - \hat{F}_f}{m} = \frac{(F_L - \hat{F}_c) - (F_L - \hat{F}_c) \exp\left[-(\hat{x}/D_v)^2\right]}{m} \quad (6.171)$$

Figure 6.42 depicts the combined observer for velocity, friction and acceleration.

With the estimated acceleration (taking into account friction effects), acceleration feedback control can now be realised, as described in Section 6.2 (Figure 6.3) (but without direct measurement of acceleration) to obtain high-performance HSSs. For this, both velocity and friction observer should be made a few times faster than the closed-loop system; see Tafazoli *et al.* (1998) for a detailed study.

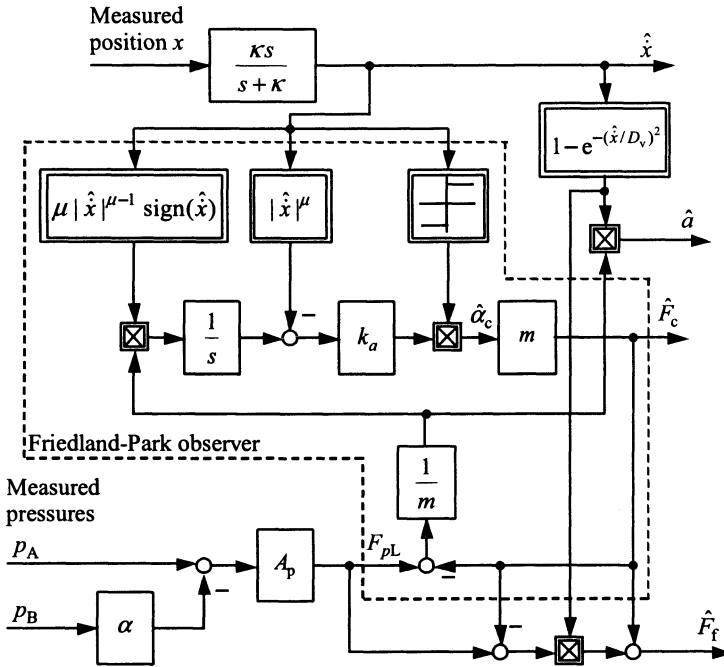


Figure 6.42. Block diagram of the combined observer (Tafazoli *et al.*, 1998 © 2000 IEEE)

### 6.10.3 Estimation of External Forces

It is sometimes useful to estimate the (external) forces acting on the piston rod. For instance, the availability of these forces is essential for vibration damping control; see Section 6.9.

#### 6.10.3.1 Estimation from Pressure Difference Force

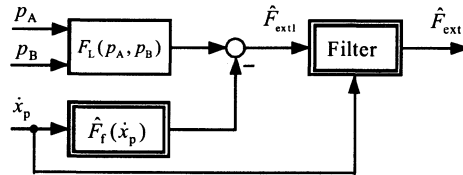
The following easy way to estimate the pressure-difference force has been proposed by Polzer and Nissing (2000a). With an estimate of the friction force (Equation 4.264) and with Equation 4.269 for the load force and by neglecting the acceleration force, which is reasonable for small masses and/or accelerations, it is possible to calculate an estimated force  $\hat{F}_{\text{ext}}$  acting on the piston rod by (rearrangement of Equation 4.56)

$$\hat{F}_{\text{extl}}(p_A, p_B, \dot{x}_p) = F_L(p_A, p_B) - \hat{F}_f(\dot{x}_p) \quad (6.172)$$

A non-linear  $PT_2$  filter is used to

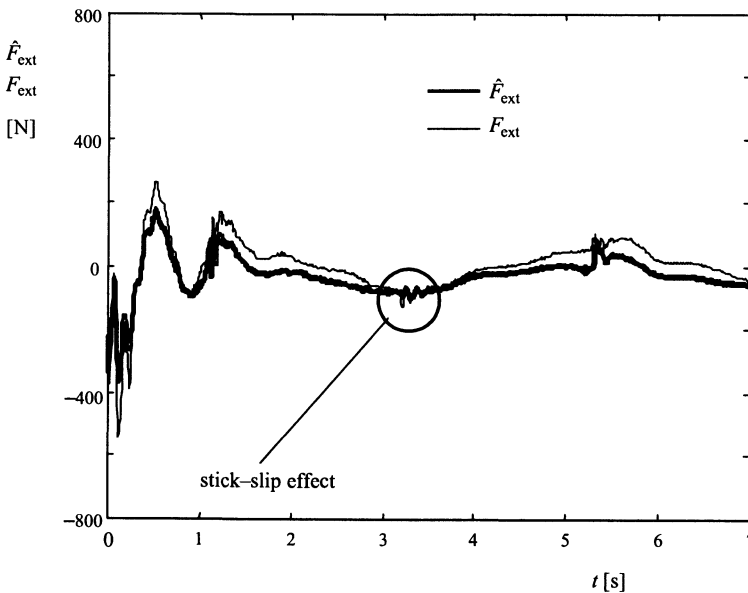
$$\hat{F}_{\text{ext}} = \begin{cases} \hat{F}_{\text{extl}} & \text{for } |\dot{x}_p| \geq D_v \\ \hat{F}_{\text{extl}} - T_1 \dot{\hat{F}}_{\text{extl}} - T_2 \ddot{\hat{F}}_{\text{extl}} & \text{for } |\dot{x}_p| < D_v \end{cases} \quad (6.173)$$

compensate for pressure vibrations and to reduce the forces evoked by the stick-slip effect (for low velocity), so that the output is the approximation  $\hat{F}_{\text{ext}}$  for the measured force acting on the piston rod  $F_{\text{ext}}$ . The block diagram in Figure 6.43 shows the total estimator with pressures  $p_A$  and  $p_B$  and the piston velocity  $\dot{x}_p$  as inputs, and the output  $\hat{F}_{\text{ext}}$ .



**Figure 6.43.** Block diagram of force estimator

Figure 6.44 depicts the comparison between the estimated and the measured force during a test run. These results have been gained by Polzer and Nissing (2000a) from experiments with the hydraulically driven flexible robot test-bed of Section 7.2. One can see the good quality of the force estimator.



**Figure 6.44.** Measured and estimated cylinder force (Polzer and Nissing, 2000a)

### 6.10.3.2 Estimation Using State Observers

As with the velocity estimation, state observers can be used to estimate the external forces, given some pressure measurements. Nissing (2002) proposed and compared some linear and non-linear observers with this aim. The reader is encouraged to consult the detailed study presented in this reference.

## 6.11 Implementation and Software Tools

To support the design and implementation of controllers, many researchers and organisations have developed user-friendly toolboxes. Some of these control software packages are mentioned in the following. For more information, the interested user should contact the according homepage.

Commercially available MATLAB toolboxes (<http://www.mathworks.com/products>) are:

- *Control System Toolbox* for classical controller analysis and synthesis of linear systems.
- *$\mu$ -Analysis and Synthesis Toolbox* for design of multivariable feedback controllers for systems with model uncertainty.
- *Robust Control Toolbox* for design of robust multivariable feedback control systems.
- *LMI Control Toolbox* for design of robust controllers using convex optimisation techniques.
- *Nonlinear Control Design Blockset* for optimising design parameters in nonlinear control systems.
- *Model Predictive Control Toolbox* for control of large, multivariable processes in the presence of constraints. The toolbox functions have been modified slightly by Maciejowski (2001) to enhance the functionality; see <http://www-control.eng.cam.ac.uk/jmm/mpcbook/software/software.html>.
- *System/Observer/Controller ID Toolbox* for advanced data analysis for modal testing and controller design of multivariable and large dimensional systems.

Free-of-charge toolboxes are:

- *NNCTRL: Neural Network Based Control System Design Toolkit* (Nørgaard, 2000b), <http://www.iau.dtu.dk/research/control/nnctrl.html> for different methods of non-linear control based on neural network models, identified by the NNSYSID toolbox.
- *NEFCON-WIN* for neuro-fuzzy control design and optimisation, <http://fuzzy.cs.uni-magdeburg.de/nefcon/nefcon.html>.
- *Fuzzy Model-based Predictive Control Toolbox*: Simulink library for different methods of non-linear model-based predictive control. The fuzzy models are identified using the FMID Toolbox, <http://dutera.et.tudelft.nl/~crweb/software>.

## 6.12 Rapid Prototyping Tools for Control

The conventional paradigm for design and implementing control systems typically follows the route:

1. **Initial design phase.** During this phase, the initial control idea is born, and a mathematical description of the system is derived. Simulations are then used to verify and adjust the chosen design. Also, the hardware is selected, designed, and checked.
2. **Detailed design phase, including the production of specification documents (Software Requirement Specification).** In this phase there is often further detailed simulation. Also, the designs, which are often most precisely defined in the simulations, are translated into text and generic block diagram descriptions. Concepts for functionality and safety or diagnostic possibilities are selected.
3. **Coding phase of the design for the chosen target environment.** The target environment for the control system usually comprises one or a number of PLCs and minicomputers. During this phase the paper design is translated into the target code, typically special PLC languages (block diagram and ladder) and high-level computer languages (C and FORTRAN). This process requires qualified staff, specialised knowledge and time.
4. **Testing phase for the target code.** Extensive testing is undertaken to ensure the target code meets the design specification. During this step, results from the target implementation are compared with results from the initial and detailed design-phase simulations. Possibly, the entire iterative design process must be started again.
5. **Plant commissioning phase.** The target system is installed commissioned and tuned. In this stage, problems can arise that require the intervention of the designer and, in particular, require simulation to validate the design changes. The outcome of the design changes must go through the detailed design, coding, and design phases. A large co-ordination effort between programming and online testing is essential during this phase.

This design route is used successfully to implement industrial control systems. However, the procedure requires the translation of the same design into different media: design simulations to design documents to target code. The weak point in this procedure is the medium of exchange between the designer and implementer, *i.e.*, the design document. This change of medium is often a cause of error because of ambiguity, incompleteness, mistranslation, etc. Elimination of the need for the several media will result in a major improvement of the efficiency of implementation.

New implementation paradigms allow one to remove migration through the different design stages, and make the design and implementation process much more efficient and less error prone (rapid control prototyping); see also Harms and Lang (1999). Such routes use commercially available hardware and software packages that enable:

1. **graphic-based programming** in the form of block diagrams (*e.g.*, with Simulink®), including modular libraries for A/D and D/A systems, logical control operations, sensor signals, *etc.*;
2. **automatic generation of target code** directly from control description of the system (*e.g.*, automatic translation of Matlab®/Simulink® code into C-code with the RealTime Workshop®);

3. **downloading of the generated code** into a real-time operating environment (e.g., Digital Signal Processors) that allows input–output communication with the plant (e.g., RealTime Interface<sup>®</sup>);
4. **interactive controlling of experiment parameters and data logging** that gathers dynamic data from the operational control system and recovers it into the design and simulation environment (e.g., ControlDesk<sup>®</sup>)\*.

The experimental verification in real-time can be performed using rapid control prototyping (RCP) tools and faster and more flexible experimental software and hardware, as provided by dSPACE<sup>®</sup>. Using this method, weak design points can be identified and corrected in the basic design phase. The hardware-in-the-loop simulation is a further important component in the design process. The software and hardware configuration implemented for a new solution is experimentally tested in a simulated environment under real-time conditions. The advantages of this pre-test performed on the actual plant are the enormous saving of time and costs and better reproducibility of the tests, even in critical situations.

## 6.13 Section Summary

A number of different approaches to the design of controllers for HSSs have now been presented. It is thus appropriate to recapitulate the overall issues of Chapter 6. The purpose is to outline a set of rules of thumb for selecting the most suitable control design method for a specific application. However, it must be stressed that the definitive decision lies with the user or designer, who has to take into account the characteristics of and the requirements on the application at hand.

One central conclusion of the analysis of different control strategies is that, besides position and possibly velocity feedback, pressure (difference) or acceleration feedback is very important to achieve a high-performance servo-control (Section 6.2). Velocity feedforward should be applied whenever an explicit velocity reference is available. Also, velocity feedforward may also be integrated implicitly by velocity compensation and outer-loop feedback in a cascade control scheme (Section 6.4.1).

As position, velocity and pressures (or load pressure) are usually considered as internal states, linear state feedback control including a state estimator is a good option for controlling HSSs (Section 6.3). Tools for adaptive control may also be applied if the change of system parameters and/or load situation is significant.

Linear controllers are easy to design and implement, and will bring good control performance in situations where the system is working in the neighbourhood of the operating point considered, and therefore a linear representation is adequate. In situations where this does not occur, or if the system has severe non-linearities, the performance of a linear controller will probably be poor.

A supplementary measure to improve linear control design is the inclusion of static compensation terms for severe non-linearities, such as valve flow non-linearity (Section 6.4.3). This technique is quite effective and can lead to substantial enhancement of control performance, and is thus often used in recent practice.

In good engineering practice, with the state-of-the-art software and hardware available, controller implementation issues do not form any limitation for the

\* RealTime Interface and ControlDesk are products of dSPACE GmbH.

<sup>9</sup> Contact <http://www.dspace.de> for more information.

application of advanced (usually model-based) control for HSSs. High performance levels are achieved, for instance, using the powerful concept of feedback linearisation control (Section 6.5) in conjunction with the cascade load force control introduced in Section 6.6.2.1.

Although the behaviour of HSSs is well understood using a physically based model, control design can alternatively be performed based on black-box models. This leads to the application of fuzzy control and neuro-control, provided a model has been identified in a first step using the techniques presented in Chapter 5 (especially in Sections 5.8 and 5.9). With the increased availability of software packages, which enable the user to perform the identification and control design steps automatically, black-box model-based control (Sections 6.7 and 6.8) is becoming increasingly realistic alternative for practical implementations of hydraulic servo-control.

For industrial applications, criteria such as the availability of development tools and individual experiences, as well as marketing aspects, product strategies and customers' demands, are significant decision criteria for the right approach. Today, there are modern total development environments that can significantly shorten the development phase of control design or control software building. Therefore, the possibilities of faster transference of advanced control strategies into the practical installations are increasing.

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## CHAPTER 7

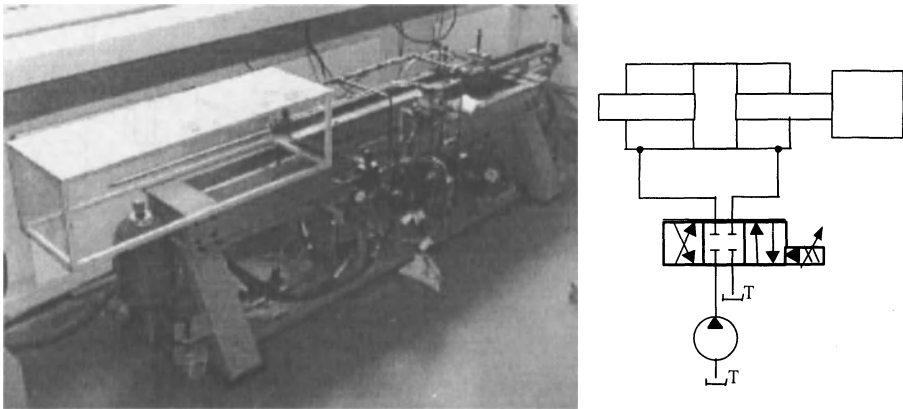
# CASE STUDIES AND EXPERIMENTAL RESULTS

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In this chapter, some case studies of the use of selected methods for modelling and control of HSSs are described to demonstrate the performance and different aspects of the methodologies presented in this book. Emphasis is placed on the comparison of widely used classical (mostly linear) approaches with advanced (usually non-linear) strategies concerning development time, closed-loop performance and computational burden.

### 7.1 Identification and Control of a Synchronising Cylinder

The first case study deals with modelling and control of the electro-hydraulic drive shown in Figure 7.1. Different techniques (classical, fuzzy logic, neural network) will be applied to identify black-box models of the system and to control it.



**Figure 7.1.** “Long cylinder test bed”: actual construction and schematic diagram

#### 7.1.1 System Description

This system consists mainly of a synchronising cylinder (with long piston rod) and a servo-valve (see Appendix B.1 for overview of drive parameters). Figure 7.2 illustrates typical open-loop velocity responses of the system to step changes of the input signal  $u$ , the servo-valve voltage. The step responses clearly show input-signal-dependent damping: oscillatory behaviour for small and damped behaviour for large



input signals. Thus, a linear model can obviously not approximate such behaviour over the whole operating range.

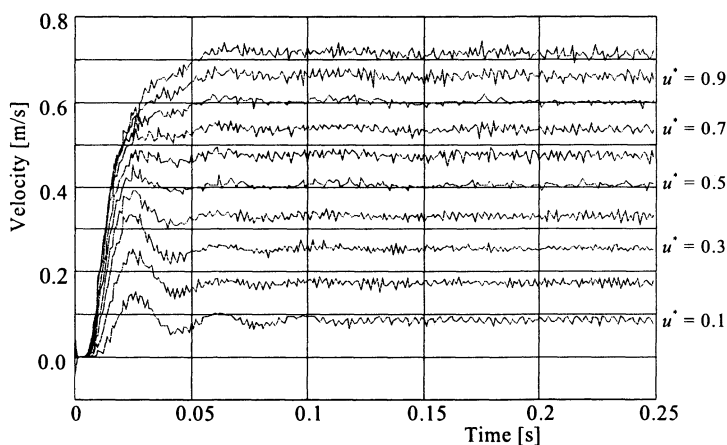


Figure 7.2. Open-loop velocity step responses

Owing to the integral behaviour of the system regarding the piston position, the piston velocity calculated using a simple finite difference method (*i.e.*, Euler approximation) will be used as the output signal  $y$  for identification. The input signal is scaled  $u^* = u / 10$  [V] (10V is the maximum servo-valve voltage).

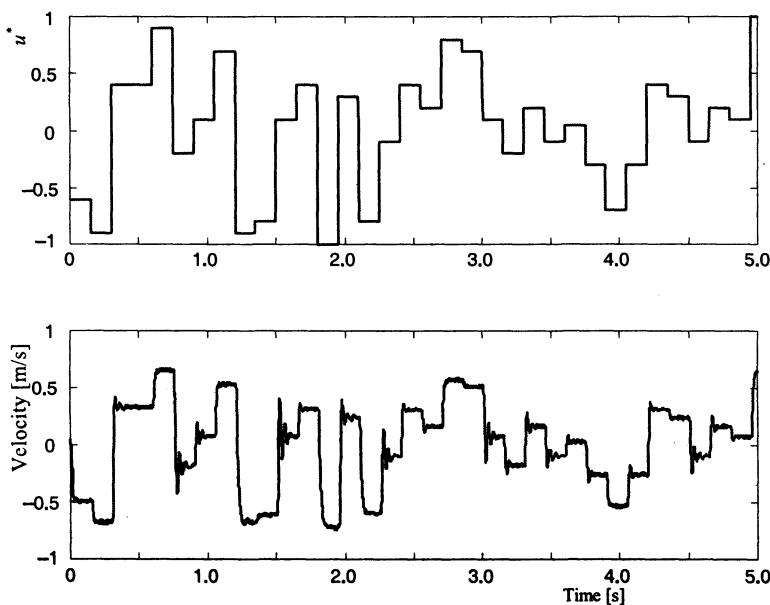


Figure 7.3. An identification data set: input signal (top) and resulting piston velocity (bottom)

The hydraulic servo-system was excited with a pseudo-random multi-level signal. 5000 pairs of input and output data records sampled at a rate of 1 kHz were collected for the identification of the system parameters. The input and output data records are illustrated in Figure 7.3. In similar fashion, several data sets (pseudo-random multi-level signals and family of step responses) have been produced for the validation process.

### 7.1.2 Continuous-time Model in Canonical Form

We start this case study with the derivation of a continuous-time model based on the model structure introduced in Section 5.4.1 and using the OLS method in combination with delayed state variable filters, as described in Section 5.5.3.

A sixth-order Butterworth filter equalised with two second-order all-pass filters and having a cut-off frequency of 400 rad/s was selected as the delayed state variable filter to produce the delayed filtered input, output and associated higher order derivatives for system identification. A fifth-order model ( $n = 5$ ) in non-linear observability canonical form (see Equations 5.64 and 5.67) with a second degree of non-linearity ( $l = 2$ ) was specified for the parameter estimation (NOBCF(5,2)). The specification contains 21 candidate terms and the sum of ERRs (Equation 5.138) reached 99.95, indicating that the selected terms adequately represent the system. Figures 7.4 and 7.5 show the cross-validation results with a (PRMS) test data set and the ten step responses for the valve input with amplitudes  $u^* = 0.1, 0.2, \dots, 1$ , respectively. It is clear that the identified (NOBCF) description provides a high-quality model of the drive in terms of small steady-state deviations and well-captured variable damping.

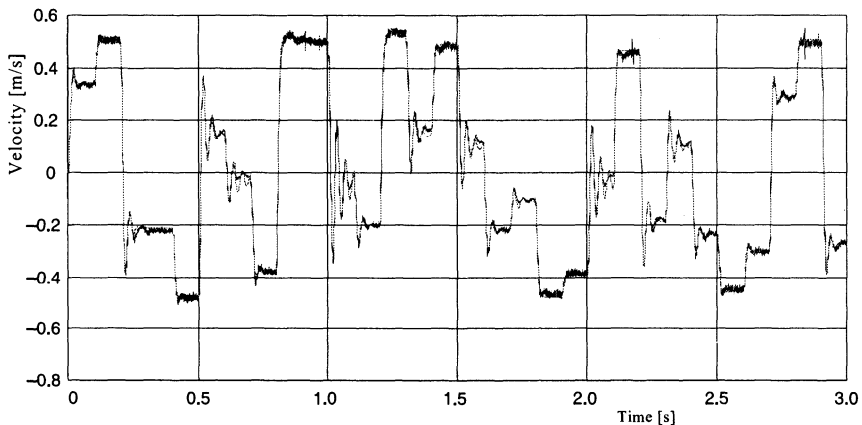


Figure 7.4. Evaluation of the NOBCF(5,2) model on the validation data

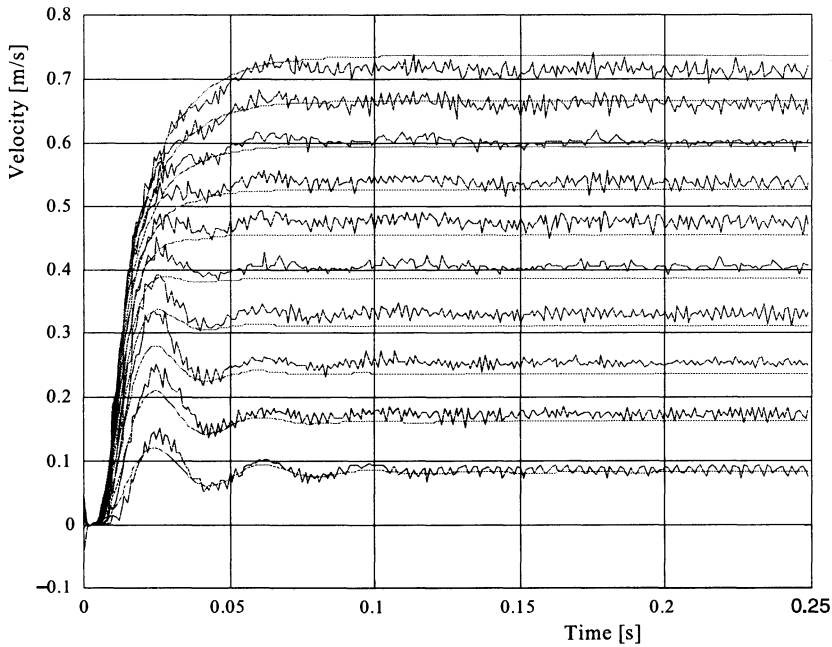
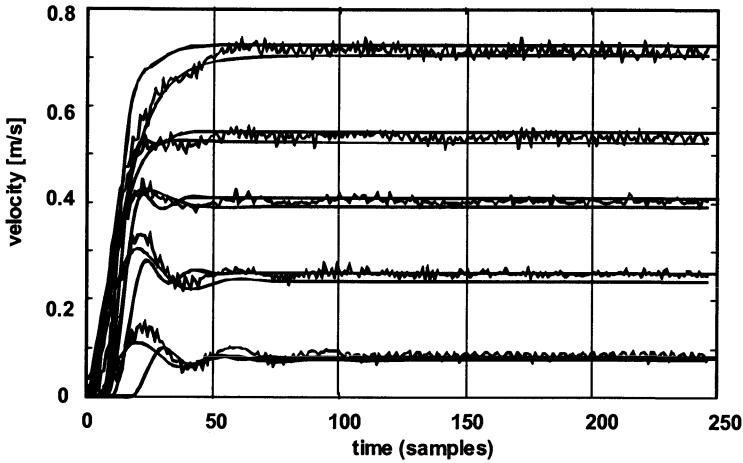


Figure 7.5. Step responses of the NOBCF(4,2) model (simulation)

### 7.1.3 Fuzzy Model Identification

The application of the structure search algorithm of Kortmann (1989), as described in Section 5.8.2.1, identified regressor and feature vector to be composed of  $y(k-1)$ ,  $y(k-2)$ ,  $y(k-3)$ ,  $y(k-4)$ , and  $u(k-4)$  (in order to predict  $y(k)$ ). Reuter (1995) determined a suitable sampling time to lie between 0.86 and 1.3 ms. However, a sampling time of  $T_0 = 2$  ms was chosen as tests indicated that this would permit the fuzzy state feedback control algorithm to be executable in real time on a standard personal computer at the time of implementation (PC 486-66DX). A cycle time of 1 ms did not permit to execute the non-linear control laws in real time. A fuzzy model with these inputs,  $c=4$  rules,  $L_2$ -norm, and  $\nu=1.5$  was identified and optimised. Figure 7.6 shows the cross-validation with the different step responses. Additionally, the simulation results for a physical model are depicted. The fuzzy model performs better than the simplified first-principles-based model, but took magnitudes less time to develop.



**Figure 7.6.** Step responses for different inputs (parallel evaluation): measurement (noisy curves), predictions of the fuzzy model (smooth curves close to measurements) and predictions of physical model (smooth curves differing from measurements)

### 7.1.4 Fuzzy Model Predictive Controller and Fuzzy State Feedback Controller

The servo-control of the linear electro-hydraulic drive is studied in the following. The control objective is defined as:

- an arbitrary step response of the velocity reference should be followed as fast as possible given a constrained control signal and a constrained rate of change of the control signal;
- the closed loop characteristics should be independent of the control signal amplitude;
- the overshoot should be tolerable, say  $\sim 15\%$ .

All signals are normalised:  $y^* = y/1$  [m/s],  $y_{\text{ref}}^* = y_{\text{ref}}/1$  [m/s],  $u^* = u/10$  [V]. All results presented in this section were gained in simulation studies.

#### 7.1.4.1 Implementation Details of Fuzzy Model Predictive Controller

The fuzzy MPC employed has been described in Section 6.7.2. It uses a Sugeno-type fuzzy model as identified in Section 7.1.3 as internal prediction model. Owing to the identified dead time of  $\tau = 3$  the minimum costing horizon is chosen as  $N_1 = 1 + \tau = 4$ . The maximum costing horizon  $N_2$  is set according to the dominant time constant of the process:  $N_2 = 8$  corresponds approximately to the settling time for a step response with amplitude  $y_{\text{ref}} = 0.4$ , which is half of the maximal possible amplitude.  $N_u = 2$  was selected due to a clear improvement compared with  $N_u = 1$  and a negligible improvement for choosing  $N_u > 2$ .

Several constraints had to be acknowledged:  $|u^*| \leq 1$  for the scaled control signal follows from the control range of the servo-valve.  $|\Delta u^*| \leq 0.8$  was chosen arbitrarily.

$|y^*| \leq 0.8$  was physically given (maximal achievable velocity), but never violated. Hence it could be removed from the optimisation task. The  $u^2$  term causes steady-state errors when the steady-state references  $u_s$  are not chosen correctly, *e.g.*, if  $u_s = 0$  is chosen for any reference. The permitted arbitrary velocity reference would require having a module for calculating arbitrary steady-state controls. To avoid this additional module, the  $u^2$  term was not considered in the cost function ( $\gamma = 0$ ).  $\beta$  is set to unity.  $\alpha = 5$  is chosen, as this provided for a reasonable trade-off of control performance versus control signal activity.  $\Delta u_0 = 0$  is chosen as the initial value for the optimisation.

The process was simulated with a simplified version of the physically based continuous-time model described in Chapter 4.

#### 7.1.4.2 Implementation Details of Fuzzy State Feedback Controller

The fuzzy state feedback controller was designed as described in Section 6.7.1. In the first phase the dominant poles of each local model are chosen such that the corresponding continuous-time second-order delay system has a damping of  $D = 0.7$  and a characteristic frequency of  $\omega_0 = 500 \text{ s}^{-1}$ . The control algorithm can be expected to be executable in real time on standard industrial control platforms; the calculation of one control action took less than 2 ms on a PC 486-66DX.

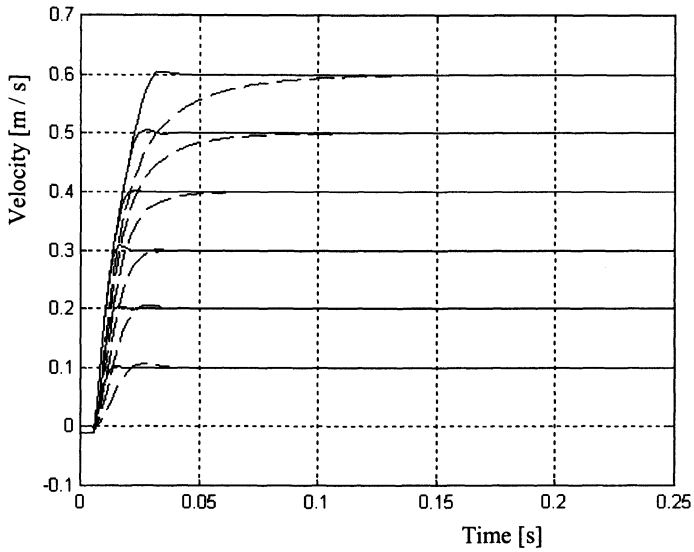
The process was simulated with a Sugeno-type fuzzy model. This has been described in Section 5.4.2.3 and has been identified in Section 7.1.3.

#### 7.1.4.3 Simulation Results

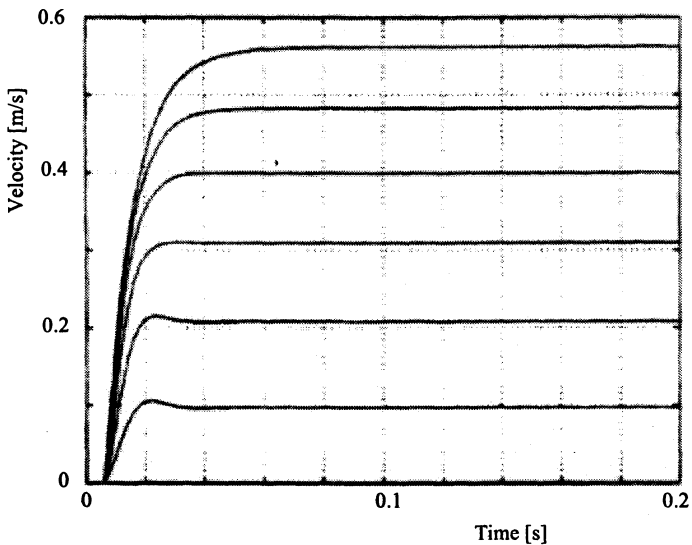
Figure 7.7 shows the step responses for the fuzzy MPC and the fuzzy state space controller to a set of step changes of the reference velocity to  $y_{\text{ref}} = 0.1, \dots, 0.6 \text{ m/s}$  (in 0.1 m/s increments). The simulations start from an initial velocity of  $y_0 = -0.0112 \text{ m/s}$ , as that was determined as numerical equilibrium for  $u = 0$ . The set of step responses permits one to assess the dependency of closed-loop dynamics on the step magnitude.

The dynamics of the step responses are independent of the magnitude of the reference value change for the fuzzy MPC (Figure 7.7). Moreover, the settling time is significantly shorter than for the non-linear fuzzy state feedback controller: after 20 ms (small reference amplitude) to 40 ms (large reference amplitude) the controlled variable remains with the reference. No steady-state error remained, even though physical and fuzzy model differed in steady-state gain. However, as mentioned, the fuzzy MPC is not real-time capable for this application if executed on state-of-the-art industrial automation systems.

The fuzzy state feedback controller has compensated the strongly varying open-loop dynamics to a minimum. The adaptive set-point filter fully removed the steady-state error, which occurs in the case of a (linear/fuzzy) proportional state feedback controller without or with a constant set-point filter (Figure 7.8). The settling time is (slightly) less than in the case of a linear state feedback controller. The fuzzy state feedback controller can be expected to be real-time capable on standard industrial hardware, as mentioned above.



**Figure 7.7.** Fuzzy state controller vs. fuzzy NMPC for reference steps (simulation): fuzzy state space controller with dashed line; MPC with full line (Taken from Kroll and Bernd (2000), p 334 (original publication). Reprinted with kind permission of VDI Verlag.)



**Figure 7.8.** Step responses with a linear state feedback controller with time-invariant set-point filter (Kroll *et al.*, 2000 © 2000 IEEE)

A linear state feedback controller with adaptive set-point filter yields results nearly equal to those of the fuzzy state feedback controller, though the resulting

local closed-loop systems turned out to be unstable. Furthermore, LMI-based controller design with guaranteed stability has also been tested. However, the closed-loop performance was significantly worse than that presented, due to the conservativeness of the design.

Figure 7.9 shows the step responses for a linear PID controller as a comparison. The PID controller was designed using numerical optimisation in order to provide for fair comparison: a pseudo-random multilevel signal with sufficient length of each step was used as reference sequence. The three controller parameters are chosen to minimise the sum of the squared deviations of the controlled variable and the reference regarding the full reference sequence. As expected, the integral part eliminates steady-state error, but the dynamics of the closed-loop system clearly vary with the step amplitude of the reference signal. Also, the settling time is significantly longer. With a time-invariant PID controller it is not possible to compensate the varying damping of the hydraulic drive such that the closed-loop system has similar dynamics for all reference signals.

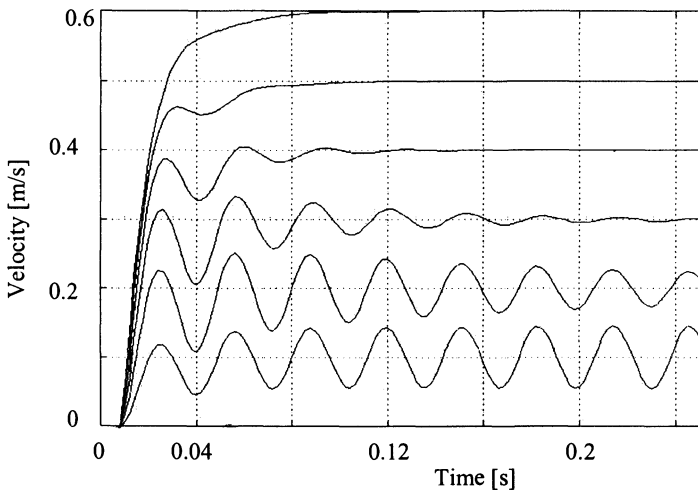


Figure 7.9. Step responses with a PID controller (Kroll *et al.*, 2000 © 2000 IEEE)

### 7.1.5 Neural Network (Multi-layer Perceptron) Identification

A (fully connected) neural network (MLP) model is trained for a comparison. The model structure is selected as an NNOE(4,1,4) with six neurons in the single hidden layer. The data set is scaled to zero mean and unity variance prior to the training (Equations 5.10 and 5.11). After the training, the weights in the network have been rescaled to allow the model to be applied to unscaled data.

In Figure 7.10, the observed output is compared with the parallel model evaluation (*i.e.*, a pure simulation of the model) of a few input steps. The simulations show that good prediction quality of the ANN model.

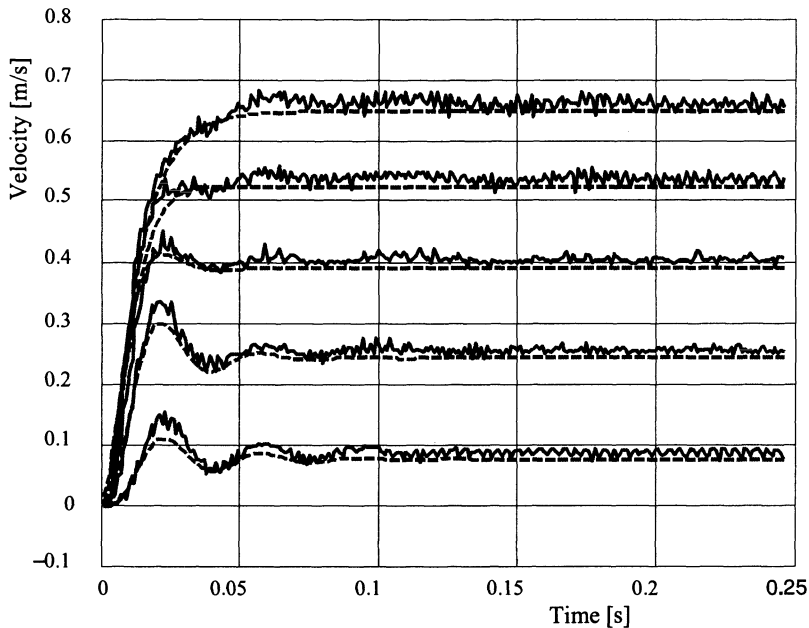


Figure 7.10. Step responses of the NNOE(4,1,4) model (simulation)

### 7.1.6 Section Summary

From identification results presented in this section, it seems that the hydraulic drive considered can be modelled properly with all models considered (*i.e.*, polynomial models, fuzzy models and ANN models). Other black-box identification results for the hydraulic system considered in this section can be found in Yin (1994); Jelali and Schwarz (1995a–c); Reuter (1995), Jelali (1997), Bernd *et al.* (1999) and Otto (2000).

The controller simulation study has proved that fuzzy state feedback controllers with adaptive set-point filters yield superior results than a linear PID controller or even a linear state feedback controller. The best control performance is achieved by using a fuzzy model predictive controller. However the latter is still far from realisable on industrial applications on standard automation platforms.

## 7.2 Modelling and Control of a Small Differential Cylinder

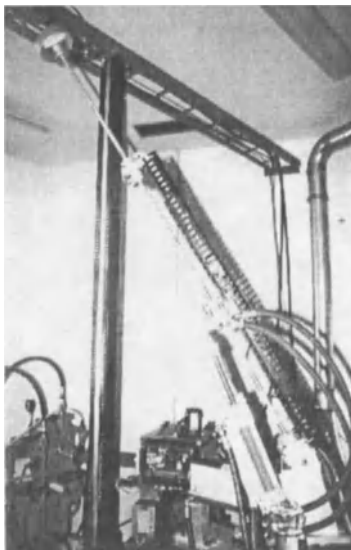
The second case study is about modelling and control of a small differential cylinder servo-system (see Appendix B.1 for an overview of system parameters) for controlling a robot arm.



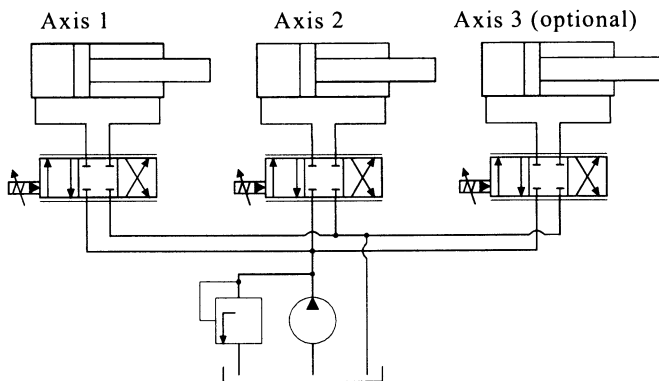
### 7.2.1 System Description

In the following, experimental results from the laboratory test-bed, as depicted in Figures 7.11 and 7.12, are presented. The test-bed was designed for realistic laboratory-scale experiments and is equipped with links manufactured of spring steel to realise a significant flexibility.

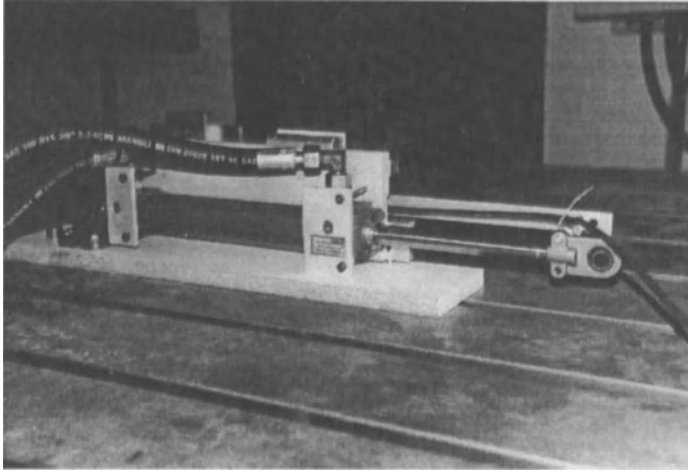
The two/three rotary joints are driven by small hydrostatic differential cylinders within closed kinematic loops to transform the translation of the drives into a rotation of the joints. For the following investigations, the cylinder was mounted separately on a plate as depicted in Figure 7.13



**Figure 7.11.** Laboratory test-bed “flexible robot”



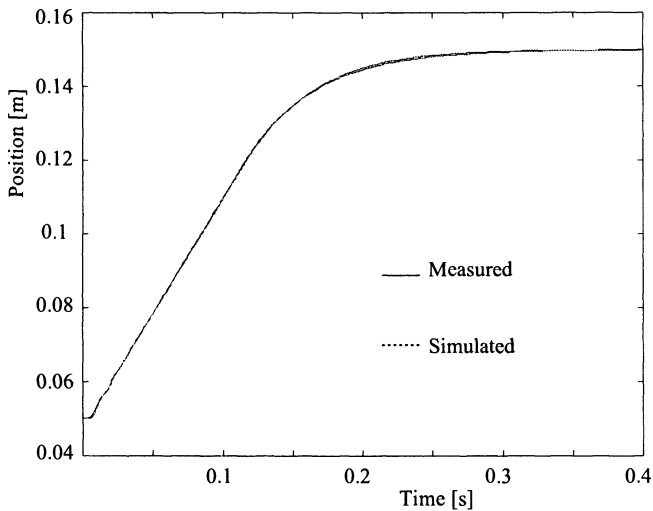
**Figure 7.12.** Hydraulic circuit actuating the robot (Taken from Nissing (2002), p 9 (original publication). Reprinted with kind permission of VDI Verlag.)



**Figure 7.13.** Hydraulic differential cylinder

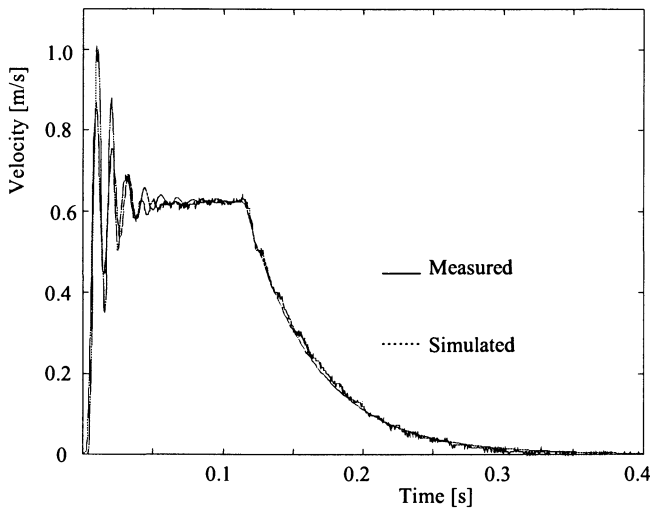
### 7.2.2 Physically Based Model

To obtain a model for simulation and control design, the differential Equations 4.181 have been implemented in MATLAB. For simplicity, the leakage flows, the fluid mass, and the valve friction were neglected. Also, bulk modulus (starting with the parameters as in Equation 3.21) and chamber volumes has been assumed to be equal at both sides of the cylinder. The initial friction model parameters  $\sigma = 175 \text{ N s/m}$ ,  $F_{c0} = 120 \text{ N}$ ,  $F_{s0} = 185 \text{ N}$  and  $c_s = 0.0174 \text{ m/s}$  have been used.

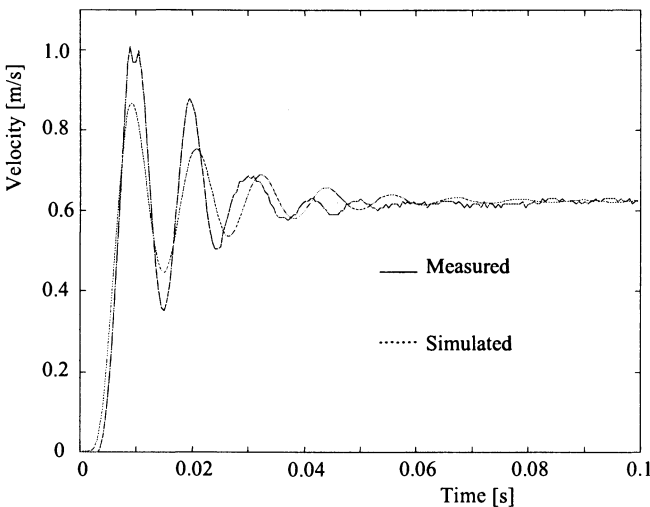


**Figure 7.14.** Comparison of measured and simulated piston position step responses (proportional position controller)

In order to fit this model to the real system, the calculated and measured position step responses are compared. To avoid integrating the error, system and model are controlled by a simple proportional controller with a gain of  $K = 30$  V/m. Figure 7.14 shows a step response for a change in reference position from 0.05 to 0.15 m. Here, only a small deviation is visible. Figure 7.15 shows the corresponding piston velocity to the piston position signal in Figure 7.14.



**Figure 7.15.** Comparison of measured and simulated velocity step responses (under position P-controller)



**Figure 7.16.** Comparison of measured and simulated piston velocity step responses (open-loop system)

The plots show good correspondence between measured and estimated signals until the velocity drops below 0.05 m/s after 0.25 s. For additional tuning information, the piston velocity step response of the open-loop system is used for a change in servo-valve input from 0 to 100%. It was necessary to adjust the bulk modulus function (Equation 3.21) as the pressure pipelines are hose pipelines. Also, some of the friction force (see Figure 4.32) parameters had to be tuned (*i.e.*, adapted to measurements). The result is depicted in Figure 7.16.

### 7.2.3 Linear vs. Non-linear Control

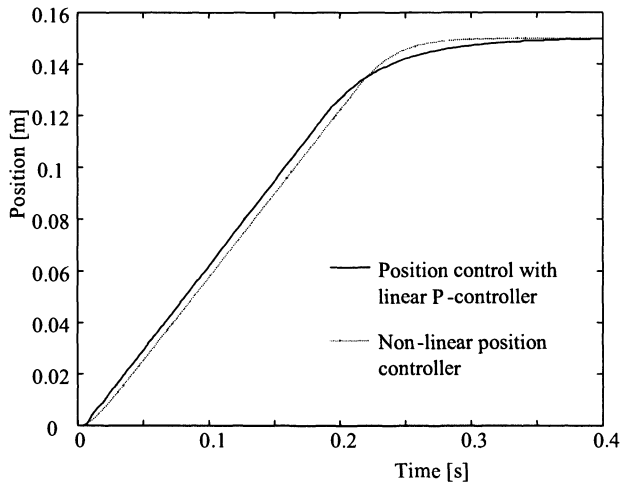
The cascade pressure-difference-force control concept, which is based on input-output linearisation (see Sections 6.5 and 6.6, Figures 6.25 and 6.26), has been applied to the differential cylinder. Simulation and experimental results are presented in the following.

#### 7.2.3.1 Simulation Results

The reference force generator

$$F_{L,ref} = K(x_{p,ref} - x_p) + \hat{F}_c + \hat{F}_s + F_{ext} \quad (7.1)$$

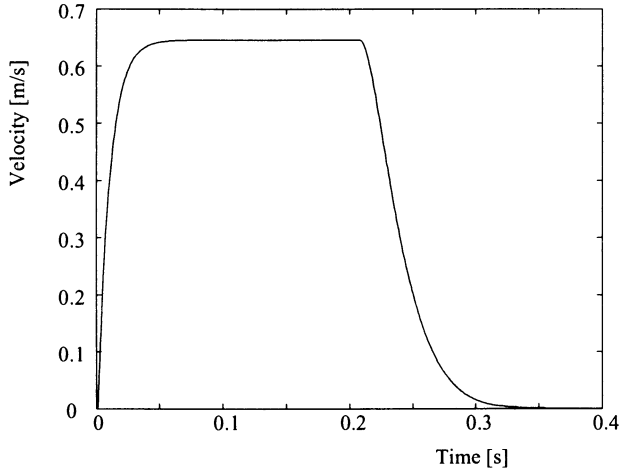
consists of a proportional feedback of the position deviation and two feedforward terms to compensate for the (measured) external force and for the friction force. Only static and Coulomb friction is taken into account; viscous friction is neglected. Simulation results are presented in the following.



**Figure 7.17.** Simulation: Piston position step response (linear proportional vs. non-linear controller)

Figure 7.17 shows the position step response of the closed-loop system (reference step from 0 to 0.15 m), obtained by applying a P-controller and the cascade non-linear controller. From this figure, it seems that no significant

improvement can be achieved using the non-linear controller. However, on taking a look at the velocity step responses (from 0 to 0.65 m/s and back to zero) in Figures 7.15 and 7.18, one can clearly see the big difference in the results of the two controllers. The non-linear controller performs far better than the linear controller.



**Figure 7.18.** Simulation: Piston velocity step response (for non-linear position controller)

### 7.2.3.2 Experimental Results

In contrast to the easy application of the non-linear controller in simulation, the implementation on the real system is more difficult. Owing the fact that the cylinder is small, its dynamics are relatively fast, such that the time constant is not too far from those of the neglected subsystems (*i.e.*, valve and pipeline dynamics). Indeed, from Figure 7.16 it can be concluded that the natural frequency of the unloaded cylinder is about 100 Hz, which lies in a region where servo-valves and pipeline dynamics should be considered. Also, the natural frequency of the pressure dynamics may be in the region of 100 Hz, as can be experimentally checked or approximately be determined as described in Section 4.4.2.

Taking into account Equation 3.82 and the numerical values

$$E' \approx 10^9 \frac{\text{N}}{\text{m}^2}, \quad c_v^* \approx 10^{-7} \frac{\text{m}^4}{\text{s}\sqrt{\text{N}}}, \quad V \approx 10^{-4} \text{ m}^3, \quad p_s = 15 \times 10^6 \frac{\text{N}}{\text{m}^2}$$

for the cylinder considered leads to the approximation of the time constant  $T_p$ :

$$T_p = \frac{V\sqrt{p_s}}{E'c_v^*} \frac{1}{\sqrt{1-p_0^*}} \Rightarrow T_p < 10 \text{ ms} \quad \text{for} \quad 0 \leq p_0^* \leq 0.85$$

which is in good agreement with the experimental results (see the figures in Section 4.4.2.1).

Another problem is due to the limitation of the used PC at that test-bed, where a sampling time smaller than 1 ms cannot be realised (a sampling time in the region of 10–50  $\mu\text{s}$  would be required).

To solve these problems, and thus apply the concept on the test-bed, the control laws in Equation 6.99 (with the assumptions:  $K_{Li} = 0$ ,  $E'_A = E'_B = E'$ ,  $V_A = V_B = V$ ) and Equation 6.129–6.130 are written in the form (Bernzen and Riege, 1996)

$$u = K_{FL}(x_3, x_4, p_s, p_T)(F_{L,ref} - F_L) + K_{x2}(x_3, x_4, p_s, p_T)x_2 \quad (7.2)$$

The pressure dynamic compensation term  $K_{FL}$  contains only fast pressure-dependent parts, and is thus assumed to be constant, *i.e.*,

$$K_{FL} = \frac{KV}{A_p E' c_v (\sqrt{p_s - x_3} + \alpha \sqrt{x_4 - p_T})} \approx \text{const.} \quad (7.3)$$

The velocity compensation term  $K_{x2}$  can be approximated by

$$K_{x2} = \frac{A_p^2 E' (1 + \alpha^2)}{A_p E' c_v (\sqrt{p_s - x_3} + \alpha \sqrt{x_4 - p_T})} \approx \frac{A_p}{2c_v} \left( \frac{1}{\sqrt{p_s - x_3}} + \frac{\alpha}{\sqrt{x_4 - p_T}} \right) \quad (7.4)$$

With these approximations, the much simplified control law

$$u = K(F_{L,ref} - F_L) + \frac{A_p}{2c_v} \left( \frac{1}{\sqrt{p_s - x_3}} + \frac{\alpha}{\sqrt{x_4 - p_T}} \right) x_2 \quad (7.5)$$

for  $u \geq 0$  is obtained, and can easily be implemented. The control law for the case  $u < 0$  follows analogously by combining Equations 6.99, 6.129 and 6.130 (see Figure 7.19 for summary).

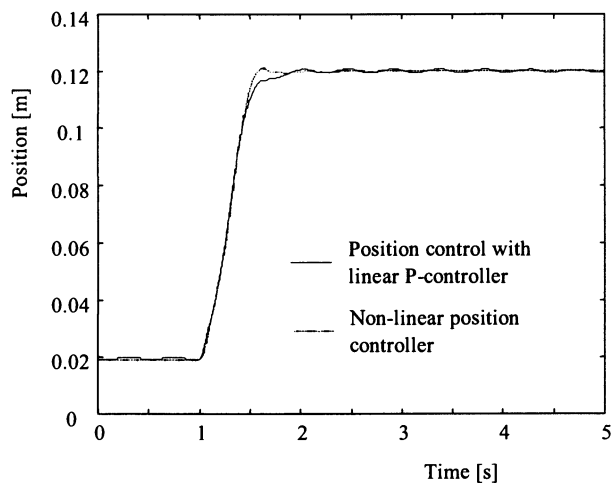
Finally, the control law of Figure 7.19 has been applied to the small differential cylinder, now remounted in the robot. Figures 7.20 and 7.21 show typical results, obtained using a (linear) P-controller and the cascade (non-linear) controller. Better control performance (smaller rise time and less steady-state position errors) is achieved with the non-linear controller.

|                                                                                                                                                                     |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| $u = v + \frac{A_p}{2c_v} \left( \frac{1}{\sqrt{p_s - x_3}} + \frac{\alpha}{\sqrt{x_4 - p_T}} \right) x_2$                                                          | (control law for the case $u \geq 0$ ) |
| $u = v + \frac{A_p}{2c_v} \left( \frac{1}{\sqrt{x_3 - p_T}} + \frac{\alpha}{\sqrt{p_s - x_4}} \right) x_2$                                                          | (control law for the case $u < 0$ )    |
| $v = K(F_{L,ref} - F_L)$                                                                                                                                            | (output proportional feedback)         |
| $F_{L,ref} = m_t \ddot{x}_{p,ref} - k_v (\dot{x}_p - \dot{x}_{p,ref}) - k_p (x_p - x_{p,ref}) + \hat{F}_t(\dot{x}_p) \quad \text{(reference load force generator)}$ |                                        |

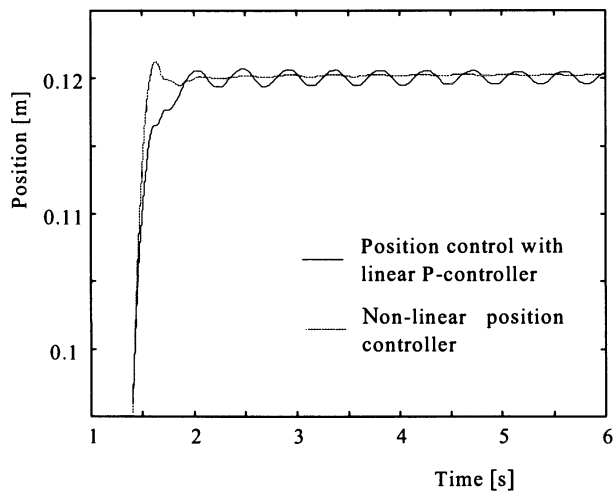
**Figure 7.19.** Simplified algorithm of cascade load force control based on input–output linearisation for HSSs with small cylinder

The reaction of the controlled system using the linear concept is very sensitive to disturbances caused by the elastic arm; see Figure 7.21. Position errors of more than

1.2 mm can be seen for the application of the P-controller, whereas deviations less than 0.25 mm are achieved using the non-linear controller. Furthermore, Figure 7.22 shows that the load-pressure-force tracking for the non-linear controller works well; some deviation is only seen near fast step-change regions.



**Figure 7.20.** Piston position step response (proportional linear vs. non-linear controller)



**Figure 7.21.** Piston position step response (zoomed Figure 7.21)

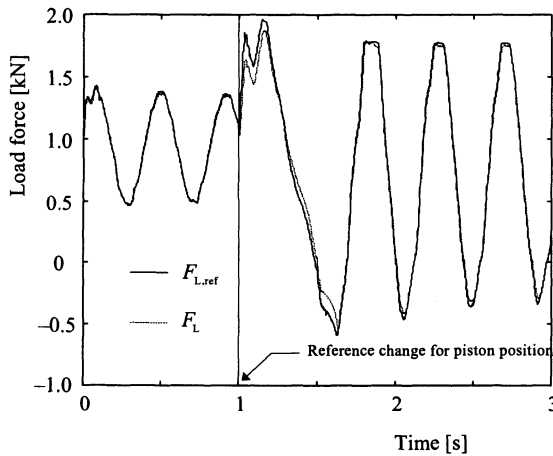


Figure 7.22. Load force tracking when applying the non-linear controller

## 7.3 Control of a Big Differential Cylinder

This case study, taken from Bernzen (1999:36–39), deals with the position control of a big differential cylinder servo-system (see Appendix B.1 for overview of drive parameters). The performance of the cascade pressure-difference-force control presented in Section 6.6.2.1, Figures 6.25 and 6.26, is investigated experimentally once again within this example, and compared with the performance achieved by using a proportional position controller.

### 7.3.1 System Description

In this application, a test-bed that consists of a (big) differential cylinder coupled with a load mass is considered. Figure 7.23 shows the actual construction and the simplified schematic diagram of the test-bed. The main instrument of the control design is the selection of the reference generator for the pressure-difference force, since the pressure dynamic has been linearised exactly, as described in Section 6.6.2.1, and so only the controller gain  $K$  has to be chosen.

### 7.3.2 Linear vs. Non-linear Control

The state feedback linearising control law of Figure 6.26 has been used. Good results have been achieved using  $K = 110 \text{ s}^{-1}$ . The system pressure  $p_s$  and the tank pressure  $p_T$  were measured and have been used in the control law. For the reference generator, the following relationship

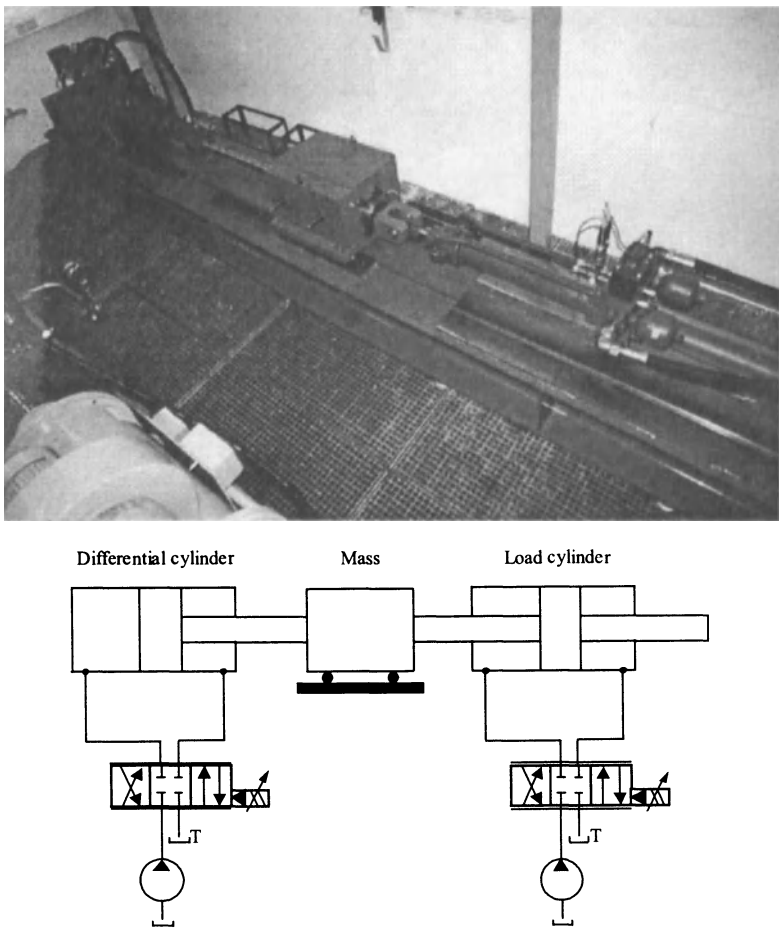
$$F_{L,\text{ref}} = K_p (x_p - x_{p,\text{ref}}) - K_v (\dot{x}_p - \dot{x}_{p,\text{ref}}) + F_f + F_{\text{ext}} \quad (7.6)$$



with  $K_p = 200 \text{ kN s/m}$  and  $K_v = 120 \text{ kN s/m}$  has been employed. The external force is given by the acceleration force of the load mass

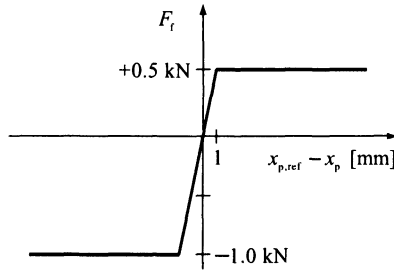
$$F_{\text{ext}} = m_L \ddot{x}_{p,\text{ref}} \tag{7.7}$$

The acceleration has been estimated by finite differentiation of the position signal.



**Figure 7.23.** “Big cylinder test-bed”

If the main emphasis is on position tracking, the static friction force is the important factor. To compensate for this force, the simplified curve (function of the position deviation) shown Figure 7.24 has been used. The function parameters have been determined experimentally. The position deviation region of 1 mm around zero aims at avoiding limit cycles. The main objective is to yield a good dynamic behaviour of the hydraulic drive with a high position control performance.

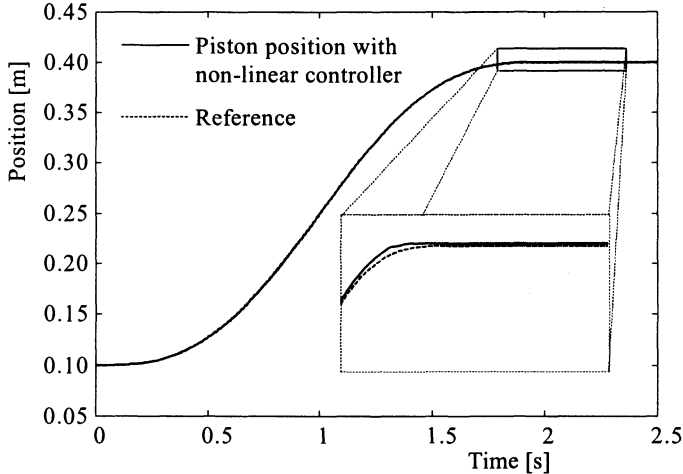


**Figure 7.24.** Friction force compensation curve (Taken from Bernzen (1999), p 37 (original publication). Reprinted with kind permission of VDI Verlag.)

The reference position is given by the function

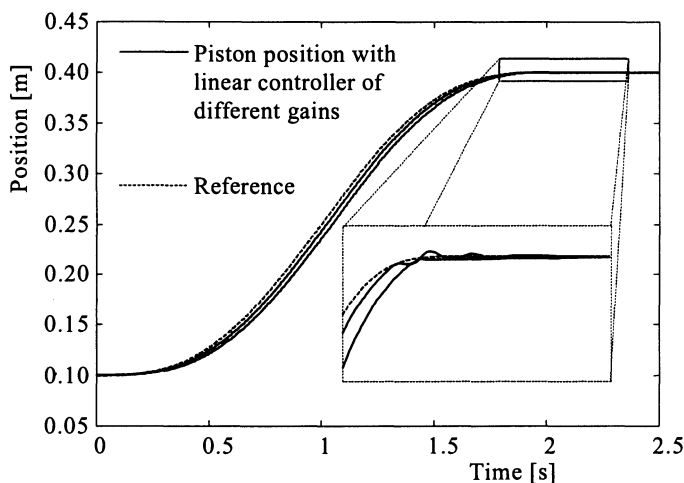
$$x_{p,\text{ref}}(t) = \frac{0.15}{\pi} \sin(\pi t) + 0.15t + 0.1 \quad (7.8)$$

which means to position the system from 0.1 to 0.4 m within 2 s. The position reference signal is differentiated to obtain the reference velocity  $\dot{x}_{p,\text{ref}}(t)$  and reference acceleration  $\ddot{x}_{p,\text{ref}}(t)$ . Figure 7.25 demonstrates the high performance of the non-linear controller, *i.e.*, fast response without overshoot. The control results obtained using a (linear) P-controller are shown in Figure 7.26.

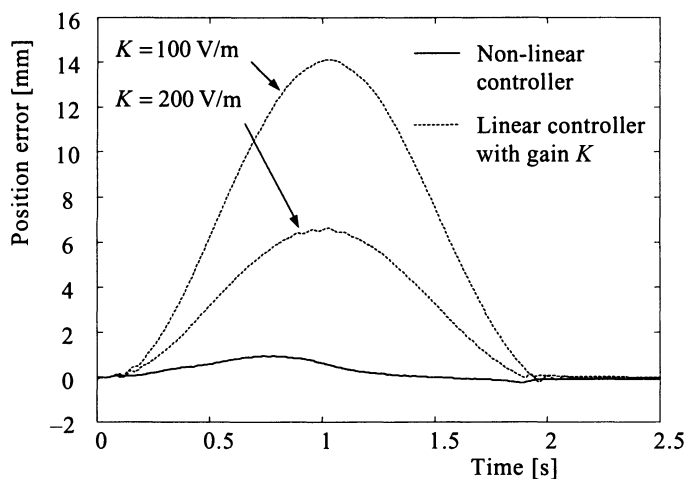


**Figure 7.25.** Position control performance of the non-linear controller (Taken from Bernzen (1999), p 38 (original publication). Reprinted with kind permission of VDI Verlag.)

The inspection of Figures 7.25 and 7.26 may, at the first glance, lead to the conclusion that the non-linear controller is not superior to the linear one. However, considering the positioning error presented in Figure 7.27, significant performance improvement using the non-linear concept is shown. The application of the linear controller yields a remaining position deviation that can be reduced by a higher gain, but only at the expense of more overshoot in the position signal.

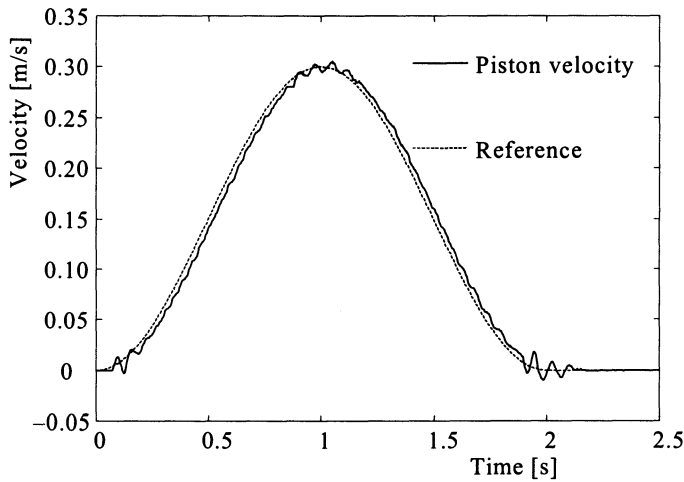


**Figure 7.26.** Position performance of a proportional (linear) controller with different gains (Taken from Bernzen (1999), p 38 (original publication). Reprinted with kind permission of VDI Verlag.)

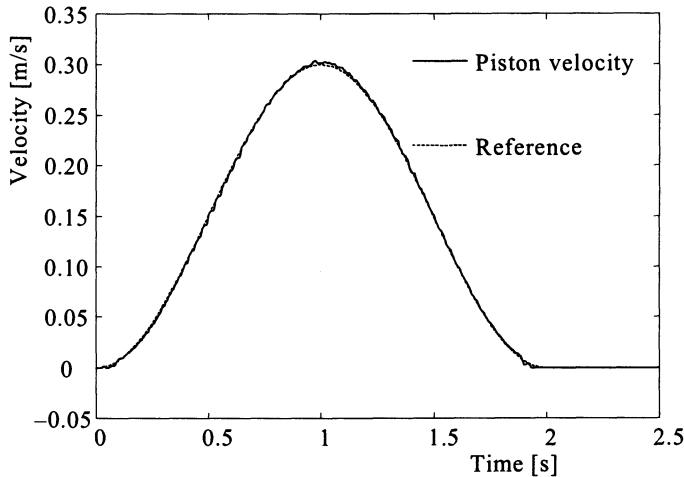


**Figure 7.27.** Position deviation (zoomed) corresponding to Figures 7.26 and 7.27 (Taken from Bernzen (1999), p 39 (original publication). Reprinted with kind permission of VDI Verlag.)

The vibrations can also be clearly seen in the velocity signals (Figure 7.28). In contrast to the linear controller, the application of the non-linear one leads to very small position deviation ( $< 1\text{ mm}$ ) and a vibration-free velocity signal (see Figure 7.29).



**Figure 7.28.** Velocity for linear control with  $K = 200$  V/m (Taken from Bernzen (1999), p 39 (original publication). Reprinted with kind permission of VDI Verlag.)



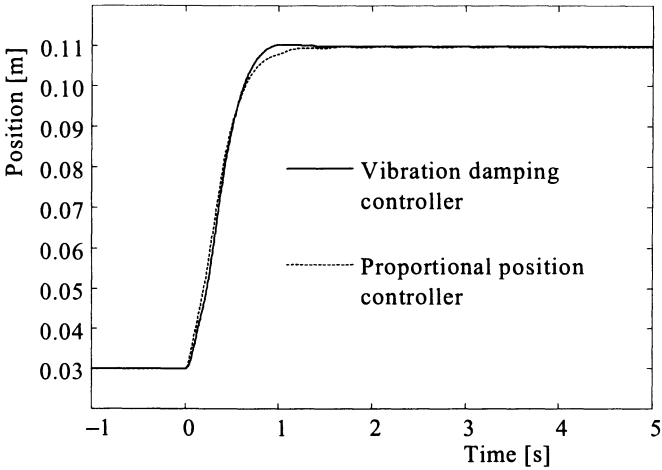
**Figure 7.29.** Velocity for non-linear control (Taken from Bernzen (1999), p 39 (original publication). Reprinted with kind permission of VDI Verlag.)

## 7.4 Vibration Damping Control for a Flexible Robot

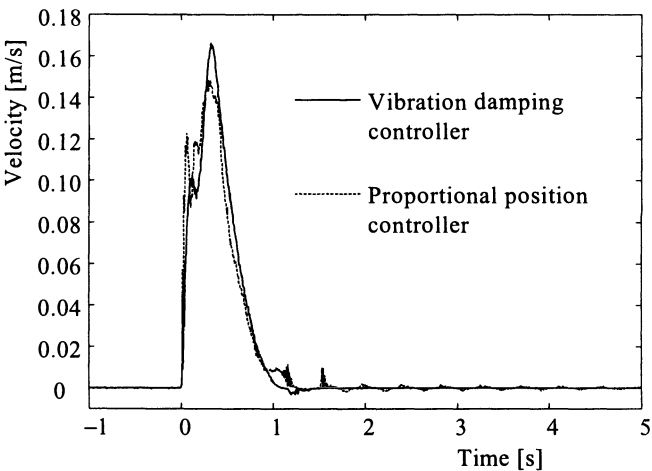
In this case study, adapted from Nissing *et al.* (1999a), the laboratory test-bed of Figure 7.11 is considered again (with remounted actuator). Figures 7.30–7.32 show a comparison between a simple P-controller and the new vibration damping controller as introduced in Section 6.9. The actuator is following a step in the reference position from 0.03 to 0.11 m, which is approximately 50% of the operating range, but for the position (Figure 7.30) and even for the velocity (Figure 7.31) only

small differences are visible, the actuator force in Figure 7.32 clearly shows the excellent performance of the new control concept. The vibration damping controller only needs approximately 1 s to damp out the vibrations without deteriorating the position control performance.

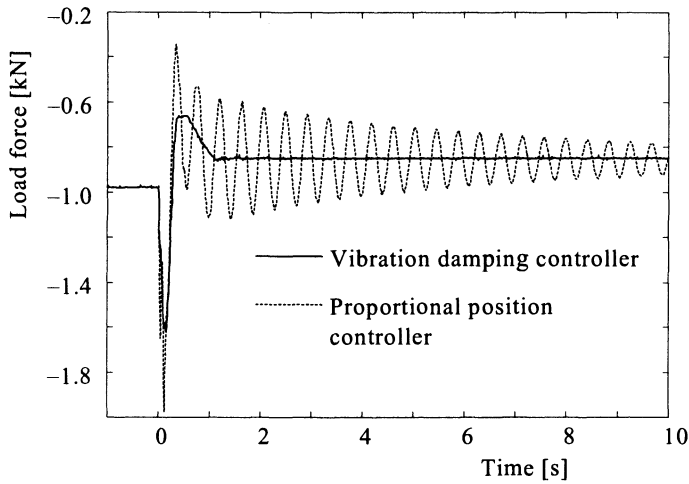
In contrast to this, the proportional position controller causes strong vibrations that fade away only after more than 30 s. In order to be fair, it must be mentioned that the linear concept inherently does not undertake any direct effort to avoid vibrations.



**Figure 7.30.** Piston position (Taken from Bernzen (1999), p 67 (original publication). Reprinted with kind permission of VDI Verlag.)



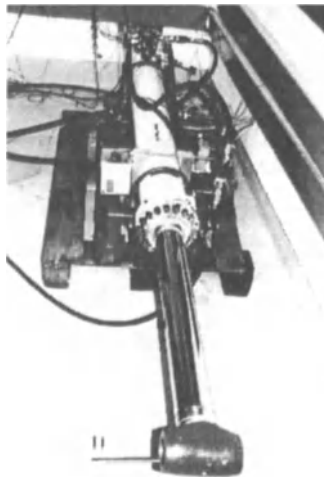
**Figure 7.31.** Piston velocity (Taken from Bernzen (1999), p 67 (original publication). Reprinted with kind permission of VDI Verlag.)



**Figure 7.32.** Measured load force (Taken from Bernzen (1999), p 68 (original publication). Reprinted with kind permission of VDI Verlag.)

## 7.5 Vibration Damping Control for a Concrete Pump

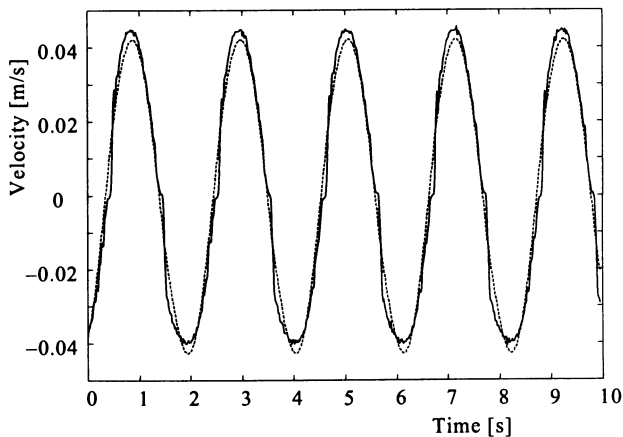
The last case study deals with the active vibration control of a real concrete pump robot (see Figure 2.13), and is adapted from Nissing *et al.* (1999a,b). As a first step to implementing the control concept for this system, one of the hydrostatic differential cylinders was dismantled and set up in the laboratory (see Figure 7.33) for experimental velocity controller design. Several measurement devices were installed, including sensors for pressure and for position/velocity measurement.



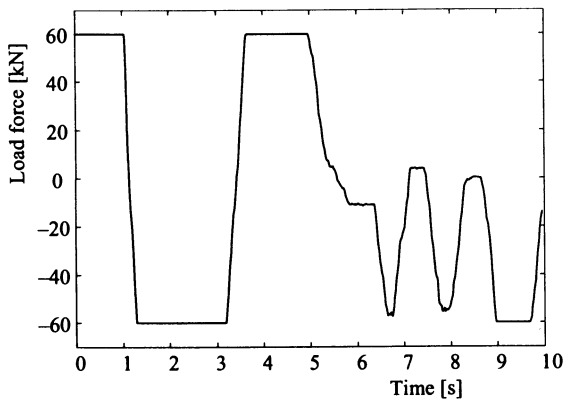
**Figure 7.33.** Dismounted hydrostatic differential cylinder of the concrete pump

A simplified non-linear velocity controller was designed and implemented according to Section 6.9.3. Figures 7.34–7.36 show the experimental results. In the first experiment, a sinusoidal disturbance force of amplitude 60 kN (which typically occurs in a real situation) is assumed. From this disturbance force the reference piston velocity is calculated using the spring–damper element (Equation 6.148. Figure 7.34) shows that tracking quality of the velocity controller is sufficient.

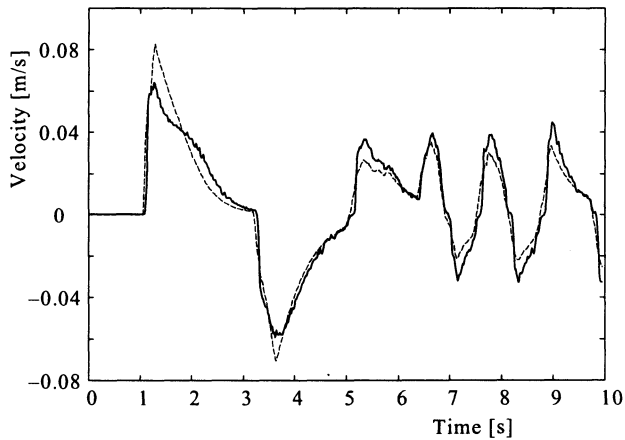
For the second experiment, again the reference piston velocity is calculated using the spring–damper element (Equation 6.148). Figure 7.35 shows the force signal generated, and Figure 7.36 the corresponding piston velocity and its reference.



**Figure 7.34.** Actual and reference piston velocity (Taken from Nissing (2002), p 21 (original publication). Reprinted with kind permission of VDI Verlag.)

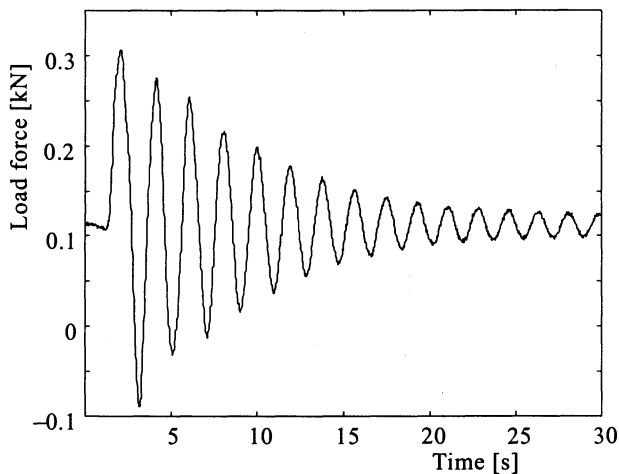


**Figure 7.35.** Load force (Taken from Nissing (2002), p 22 (original publication). Reprinted with kind permission of VDI Verlag.)



**Figure 7.36.** Piston velocity and its reference (Taken from Nissing (2002), p 22 (original publication). Reprinted with kind permission of VDI Verlag.)

Finally, the hydrostatic differential cylinder was reinstalled in the concrete pump, and the control concept was proved on the real system. The optimum parameters spring stiffness  $c$  and damping constant  $d$  were determined experimentally (Nissing, 2000, 2002). Figure 7.37 shows the standardised actuator force alteration of the cylinder without vibration control when the end-effector is excited manually at  $t = 0$  and all links were horizontally aligned. One can clearly see the elastic link vibrations fading away in more than 30 s.

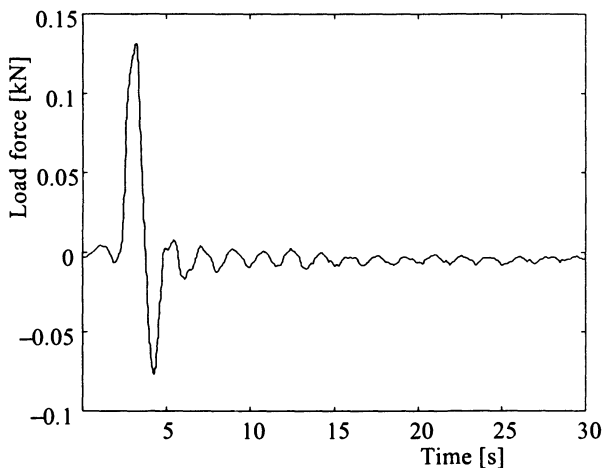


**Figure 7.37.** Load force without vibration control (Bernzen *et al.*, 1999a)

In comparison, Figure 7.38 shows the actuator force with active vibration damping concept. It is clearly visible that the vibrations fade away within 5 s. In



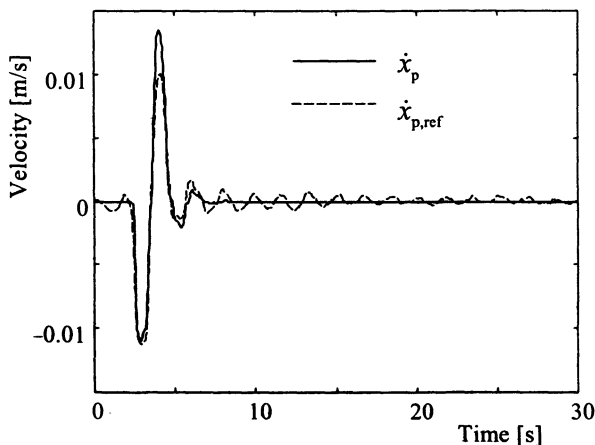
addition, the absolute force alteration without the vibration control concept is more than two times greater than the piston force with the use of the control concept.



**Figure 7.38.** Load force with vibration control (Bernzen *et al.*, 1999a)

Figure 7.39 illustrates the good quality of the velocity controller. The piston velocity follows the reference velocity calculated by Equation 6.148 satisfactorily. It can also be seen that the actuator yields when a force acts on the piston rod. The differential cylinder is extending and retracting once.

When the arm is manually excited at its natural frequency, the force amplitude, and consequently the position of the end-effector, is large in the case without vibration damping. This does not occur when the vibration damping control (Section 6.9, Figure 6.40) is active: the force amplitude remains low at the same level at each impulse.



**Figure 7.39.** Actual and reference piston velocity (Bernzen *et al.*, 1999a)

APPENDIX A

FLUID POWERS SYMBOLS

The symbols listed below are used to describe hydraulic system layouts (or circuits), and are based upon the German standard DIN-ISO 1219 (1978) and the international standard ISO 1219-1 (1991) for fluid power symbols.

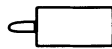
Pumps, Motors, and Drives

|                                                             | Fixed | Variable |                                              |  |
|-------------------------------------------------------------|-------|----------|----------------------------------------------|--|
| Single direction pump                                       |       |          | Double acting actuator with double-ended rod |  |
| Double direction pump                                       |       |          | Piston with adjustable end cushioning        |  |
| Single direction motor                                      |       |          | Piston with fixed cushioning                 |  |
| Double direction motor                                      |       |          | Telescopic, single acting actuator           |  |
| Single direction pump/motor with reversal of flow direction |       |          | Telescopic, double acting actuator           |  |
| Single direction pump/motor with single flow direction      |       |          | Pressure intensifier                         |  |
| Double direction pump/motor with two directions of flow     |       |          | Double acting actuator                       |  |
| Hydrostatic drive, split system type                        |       |          | Differential actuator with oversize rod      |  |
| Hydrostatic drive, compact, reversal output                 |       |          |                                              |  |
| Semi rotary actuator                                        |       |          |                                              |  |

Linear Actuators (Cylinders)

|                                                    |  |                               |  |
|----------------------------------------------------|--|-------------------------------|--|
| Single acting ram (load returns the ram)           |  | Undefined control             |  |
| Single acting actuator (load returns the piston)   |  | Hand lever (rotary or linear) |  |
| Single acting actuator (spring returns the piston) |  | Push button                   |  |
|                                                    |  | Foot lever                    |  |
|                                                    |  | Cam roller                    |  |

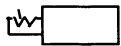
Plunger  
(piston or ball)



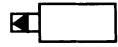
Spring



Detent mechanism



Pressure relief



Pressure applied



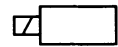
Pneumatic pilot



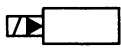
Hydraulic pilot



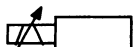
Solenoid



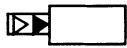
Solenoid/hydraulic pilot  
(electro-hydraulic)



Torque motor



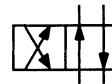
Pneumatic/hydraulic pilot



Spring centred



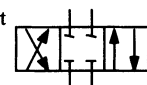
Two-position, four-port  
(4/2) valve



Two-position, five-port  
(5/2) valve

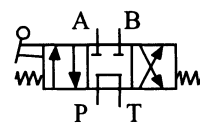
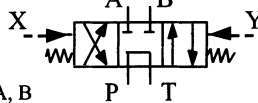


Three-position, four-port  
(4/3) valve with fully  
closed centre  
configuration



Port labelling:

- working lines A, B
- pilot lines X, Y
- pressure line P
- tank line T



Check valve



Spring-loaded check valve



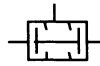
Pilot-loaded check valve



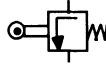
OR function valve



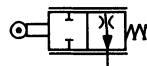
AND function valve



Deceleration valve



Deceleration valve

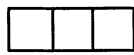


## Directional Control Valves

Directional control valve  
with two discrete positions



Directional control valve  
with three discrete positions



Directional control valve  
with significant cross-over positions



Valve with two discrete positions  
and an infinite number of intermediate  
throttling positions



Valve with three discrete positions  
and an infinite number of intermediate  
throttling positions



Two-position, two-port (2/2) valve

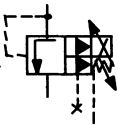


Two-position, three-port (3/2) valve

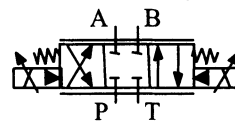


## Servo and Proportional Valves

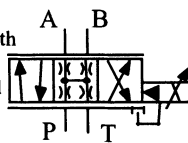
Proportional control pressure  
relief valve (with integral max.  
pressure limitation)



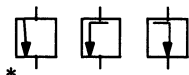
Pilot-operated  
directional  
proportional valve



Four-way servo-valve with  
mechanical feedback,  
standard overlapping and  
hydraulic zero



Pressure-relief valve (fixed)



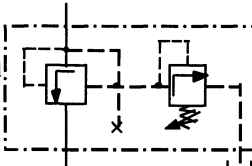
Pressure-relief valve (adjustable)



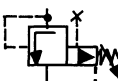
Detailed symbol of Pilot-operated Pressure-relief valve (compounded relief valve)



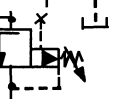
Simplified detailed symbol of compounded relief valve (pilot flow externally drained)



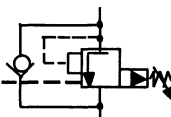
(pilot flow internally drained)



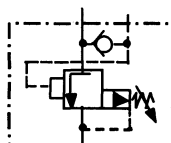
Brake valve



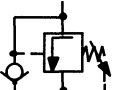
Unloading valve (accumulator charging valve)



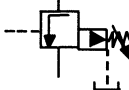
Counter balance valve (back pressure valve)



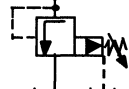
Sequence valve with remote control (external pilot)



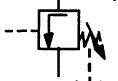
Sequence valve with direct control (internal pilot)



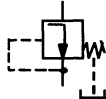
Offloading valve



Pressure reducing valve (fixed)



Pressure reducing valve (adjustable)

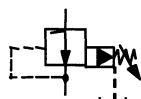


## Pressure Control Valves

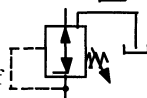
Throttling orifice normally closed or normally open (\*: optional)



Pilot-operated pressure reducing valve



Pressure reducing valve with secondary system relief



## Flow Control Valves

Throttle valve not affected by viscosity



Throttle valve (fixed)



Throttle valve (adjustable)



Flow control valve, pressure and temperature compensated



Flow control valve with reverse free flow check



By-pass flow control valve



Flow divider



## Fluid Plumbing and Storage

Fluid flow direction



Adjustability



Pressure source



Working line, return line, feed line



Pilot control line



Drain line



Enclosure line



|                                        |  |                                    |  |
|----------------------------------------|--|------------------------------------|--|
| Flexible line                          |  | Cooler with coolant lines          |  |
| Electric line                          |  | Heater                             |  |
| Pipeline connections                   |  |                                    |  |
| Crossing pipelines (not connected)     |  |                                    |  |
| Air vent                               |  | Pressure gauge, pressure indicator |  |
| Reservoir with inlet below fluid level |  | Flow meter                         |  |
| Reservoir with inlet above fluid level |  | Thermometer                        |  |

**Miscellaneous Symbols**

|                                             |  |                              |  |
|---------------------------------------------|--|------------------------------|--|
| Electric motor                              |  | Pressure switch (electrical) |  |
| Heat engine                                 |  | Shut-off valve               |  |
| Electric motor with pump and drive coupling |  |                              |  |
| Plugged line                                |  |                              |  |
| Plugged line with take-off line             |  |                              |  |
| Quick connect coupling                      |  |                              |  |
| Rotary connection                           |  |                              |  |
| Accumulator                                 |  |                              |  |
| Filter, strainer                            |  |                              |  |

Note that the symbols for pressure valves are a little bit different between the DIN-ISO 1219 and ISO 1219-1 versions:

| DIN-ISO 1219 | ISO 1219-1 |
|--------------|------------|
|              |            |
|              |            |

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## APPENDIX B

### DATA AND CATALOGUE SHEETS

---

#### B.1 Parameter Sets for Experimental Setups

The lists below give an overview of the parameters of the components in the experimental setups used in Chapter 7.

##### Synchronising Cylinder Test-bed “Long Cylinder”

- **Synchronising Cylinder Bosch Rexroth<sup>1</sup>** (CG70F 25/16/1000 ZY7592)

|                          |                   |
|--------------------------|-------------------|
| $d_p = 25 \text{ mm}$    | piston diameter   |
| $d_{pr} = 16 \text{ mm}$ | rod diameter      |
| $A_r = 290 \text{ mm}^2$ | ring area         |
| $S = 1000 \text{ mm}$    | stroke            |
| $m_p = 6.6 \text{ kg}$   | total moving mass |

- **Friction Parameters**

|                              |                                        |
|------------------------------|----------------------------------------|
| $\sigma = 800 \text{ N s/m}$ | viscous friction parameter for $v > 0$ |
| $F_{c0} = 30 \text{ N}$      | Coulomb friction for $v > 0$           |
| $F_{s0} = 140 \text{ N}$     | static friction for $v > 0$            |
| $c_s = 0.015 \text{ m/s}$    | Stribeck velocity for $v > 0$          |

- **Servo-valve Bosch Rexroth** (4WS2EM10-42/20B2ET315Z8DM)

|                                    |                             |
|------------------------------------|-----------------------------|
| $\Delta p_N = 70 \text{ bar}$      | nominal pressure difference |
| $Q_N = 20 \text{ dm}^3/\text{min}$ | nominal flow                |

- **Oil Supply**

|                         |                         |
|-------------------------|-------------------------|
| $p_s = 50 \text{ bar}$  | working supply pressure |
| $p_T = 1.8 \text{ bar}$ | tank pressure           |

- **PC Interface** (dSPACE System)

Processor board DS 1003 (TI TMS320C40 60 MHz)

DS 2201 Multi-I/O Board with 20 A/D- und 8 D/A-Interfaces

---

<sup>1</sup> Former Mannesmann–Rexroth

## DS 3002 Incremental Encoder Interface Board with 6 Channels

$T_0 = 1-4 \text{ ms}$  sampling time

**Flexible Robot Test-bed**

- **Differential Cylinder** Bosch Rexroth (CD160B 25/18)

$d_p = 25 \text{ mm}$  piston diameter  
 $d_{pr} = 18 \text{ mm}$  rod diameter  
 $A_p = 49.9 \text{ mm}^2$  piston area  
 $A_r = 236.4 \text{ mm}^2$  ring area  
 $\alpha = 1/2.08$  piston area ratio  $\alpha = A_r/A_p$   
 $S = 200/250 \text{ mm}$  stroke  
 $m_l = 2 \text{ kg}$  total moving mass  
 $V_{pl,A} = V_{pl,B} = 94.25 \text{ cm}^3$  inefficient volumes in chamber A / B

- **Friction Parameters (Nissing, 2002)**

$\sigma^+ = 220 \text{ N s/m}$  viscous friction parameter for  $v > 0$   
 $F_{c0}^+ = 50 \text{ N}$  Coulomb friction for  $v > 0$   
 $F_{s0}^+ = 30 \text{ N}$  static friction for  $v > 0$   
 $c_s^+ = 0.015 \text{ m/s}$  Stribeck velocity for  $v > 0$   
 $\sigma^- = 180 \text{ N s/m}$  viscous friction parameter for  $v < 0$   
 $F_{c0}^- = 50 \text{ N}$  Coulomb friction for  $v < 0$   
 $F_{s0}^- = 20 \text{ N}$  static friction for  $v < 0$   
 $c_s^- = 0.007 \text{ m/s}$  Stribeck velocity constant for  $v < 0$

- **Servo-valve** Bosch Rexroth (4WSE2EE10-45/10B9ET210Z9EM)

$\Delta p_N = 70 \text{ bar}$  nominal pressure difference  
 $Q_N = 10.8 \text{ dm}^3/\text{min}$  nominal flow  
 with critical centre

- **Pressure Supply**

$p_s = 50 \text{ bar}$  supply pressure

- **Flexible Arm**

$d_l = 14 \text{ mm}$  diameter of spring steel  
 $l_{arm} = 1.5 \text{ m}$  arm length  
 $m_{end} = 3 \text{ kg}$  end mass

- **PC Interface** (dSPACE System)

Processor board DS 1003 (TI TMS320C40 60 MHz)

DS 2201 Multi-I/O Board with 20 A/D- und 8 D/A-Interfaces

DS 3002 Incremental Encoder Interface Board with 6 Channels

$T_0 = 1 \text{ ms}$  sampling time

**“Big Cylinder” Test-bed.**

- **Differential Cylinder** Bosch Rexroth (CYH3MP563/45/500A10/B1CSDTD574)
 

|                  |                 |
|------------------|-----------------|
| $d_p = 63$ mm    | piston diameter |
| $d_{pr} = 45$ mm | rod diameter    |
| $S = 500$ mm     | stroke          |
- **Control Valve** Bosch Rexroth (4WRSE10E50-30/G24KO/A1VR)
 

|                                 |                             |
|---------------------------------|-----------------------------|
| $Q_N = 50$ dm <sup>3</sup> /min | nominal flow                |
| $\Delta p_N = 10$ bar           | nominal pressure difference |
| $x_o = 11\text{--}16$ %         | overlap                     |
- **Pressure Supply**

|                                     |                                               |
|-------------------------------------|-----------------------------------------------|
| $p_{s,max} = 315$ bar               | max. supply pressure                          |
| $p_s = 200$ bar                     | supply pressure for the experiments performed |
| $Q_{max} = 67$ dm <sup>3</sup> /min | max. flow                                     |
- **Synchronising Cylinder (Loading Unit)** Bosch Rexroth (CXSA280B70/56-0500X10/16S01A00127)
 

|                  |                 |
|------------------|-----------------|
| $d_p = 80$ mm    | piston diameter |
| $d_{pr} = 56$ mm | rod diameter    |
| $S = 554$ mm     | stroke          |
- **Control Valve** Bosch Rexroth (4WS2EM16-21/150B12 ET315Z8EM)
 

|                                  |                             |
|----------------------------------|-----------------------------|
| $\Delta p_N = 70$ bar            | nominal pressure difference |
| $Q_N = 150$ dm <sup>3</sup> /min | nominal flow                |
| $x_u = 0\text{--}0.5$ %          | underlap                    |
- **Pressure Supply**

|                                     |                      |
|-------------------------------------|----------------------|
| $p_{s,max} = 350$ bar               | max. supply pressure |
| $Q_{max} = 92$ dm <sup>3</sup> /min | max. flow            |
- **PC Interface** (dSPACE System)
 

|                               |               |
|-------------------------------|---------------|
| Single Board Solution DS 1102 |               |
| $T_0 = 1$ ms                  | sampling time |

**Concrete Pump Robot.**

- **Differential Cylinder** Bosch Rexroth (D140/90 × 1173)
 

|                                 |                                      |
|---------------------------------|--------------------------------------|
| $d_p = 140$ mm                  | piston diameter                      |
| $d_{pr} = 90$ mm                | rod diameter                         |
| $A_p = 15393.8$ mm <sup>2</sup> | piston area                          |
| $A_r = 9032.1$ mm <sup>2</sup>  | ring area                            |
| $\alpha = 1/1.704$              | piston area ratio $\alpha = A_r/A_p$ |



- $S = 1173 \text{ mm}$  stroke
- $m_t = 2 \text{ kg}$  cylinder mass (empty weight)
- **Servo-valve MOOG (D 633-265 R 16K01M0N6M)**
- 4/3 directional control valve, zero lap
- $\Delta p_N = 70 \text{ bar}$  nominal pressure difference
- $Q_N = 40 \text{ dm}^3/\text{min}$  nominal flow
- $p_{\max} = 70 \text{ bar}$  max. supply pressure
- **Power Supply System Bosch Rexroth (Axial piston pump A7V Series 2.0)**
- $p_N = 350 \text{ bar}$  nominal pressure
- $p_{\max} = 400 \text{ bar}$  max. pressure
- $n_E = 1450 \text{ 1/min}$  nominal (rotary) speed
- $Q_E = 39.5 \text{ dm}^3/\text{min}$  flow at  $n_E$  and  $p_N$
- $P_E = 24 \text{ kW}$  power at  $n_E$  and  $p_N$
- **PC Interface (dSPACE System)**
- DS 1102 DSP Controller Board (TMS320C31 60 MHz)
- 2 A/D-Interfaces 16 bit, 2 A/D-Interfaces 12 bit, 4 D/A- Interfaces 12bit
- Incremental Encoder Interface with 2 Channels
- $T_0 = 15 \text{ ms}$  sampling time

## B.2 Typical Parameter Values for Simulation of Servo-valves

Feedback spring constant  $K_{fb} = 120 \text{ N m/m}$  (van Schothorst, 1997)

Gain of torque motor  $\beta = 0.985 \text{ N m/A}$  (Hayase *et al.*, 2000)

Magnetomotive force  $M_0 = 1000$  A (van Schothorst, 1997)

Permeability of magnetic circuit  $\mu_0 = 4\pi \times 10^{-7}$  V s/A m (van Schothorst, 1997)

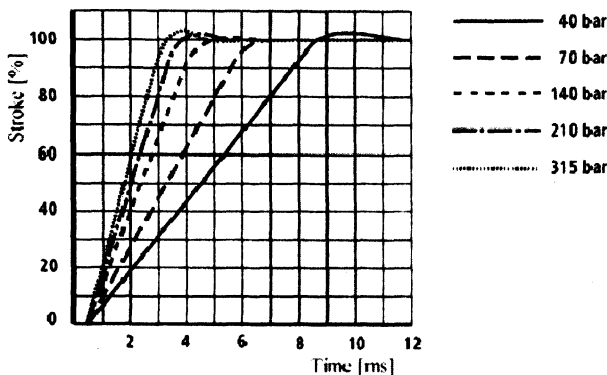
Viscous friction coefficient for flapper  $\sigma_f = 0.025 \text{ N m s/m}$  (van Schothorst, 1997)

Viscous friction coefficient for spool  $\sigma_s = 100 \text{ N s/m}$  (Hayase *et al.*, 2000)

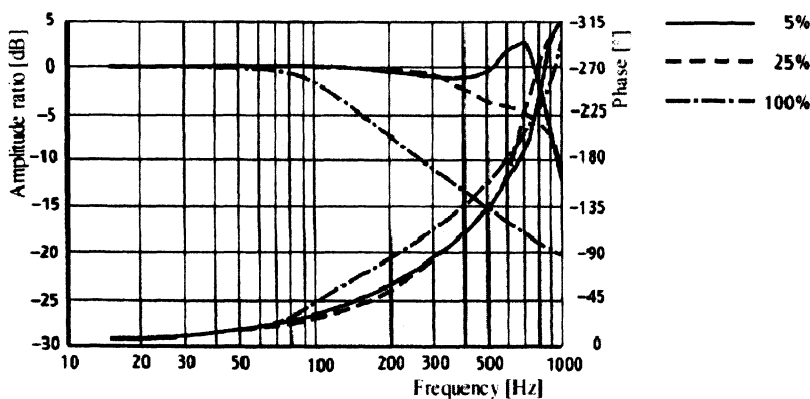
$$\sigma_s = 20 \text{ N s/m (van Schothorst, 1997)}$$

### B.3 Some Catalogue Diagrams

Step responses at working pressure = 340 bar



Frequency diagram at working pressure = 340 bar



Frequency dependence

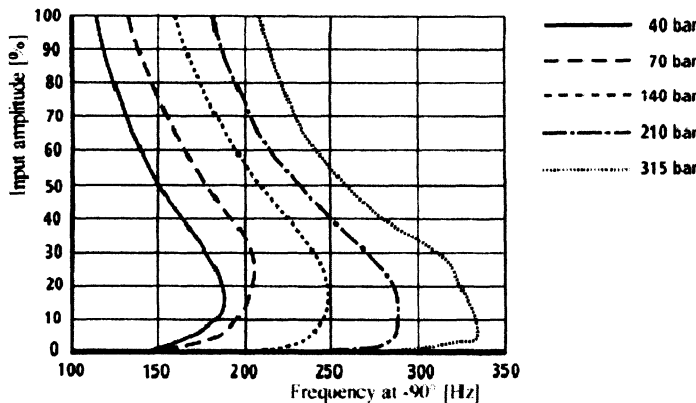
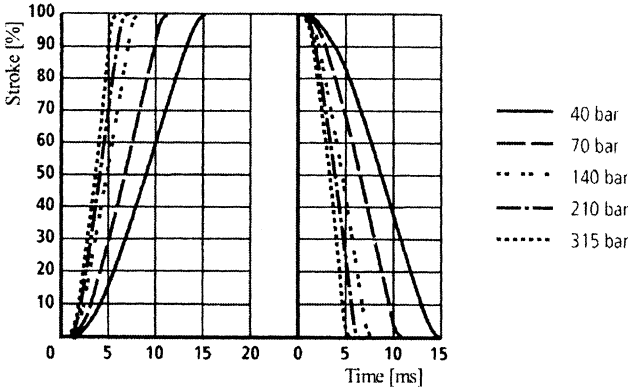
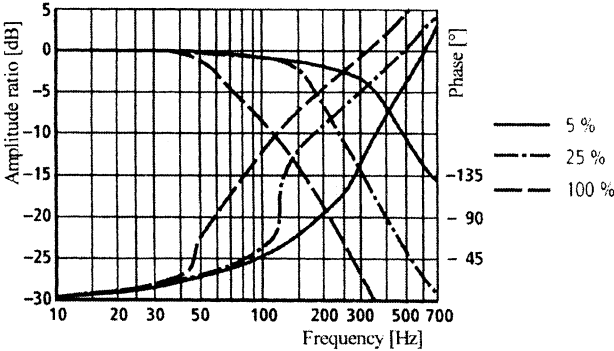


Figure B.1. Characteristics of the servo-valve of type Bosch Rexroth 4WS2EM (Bosch Rexroth, 2000)

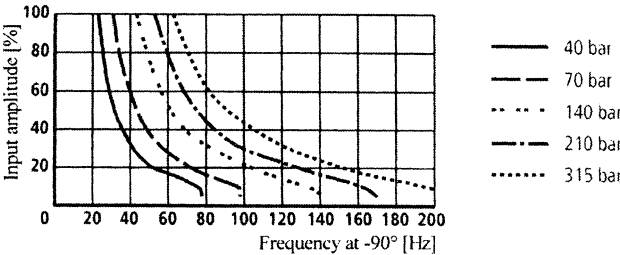
Step responses at working pressure = 340 bar



Frequency diagram at working pressure = 340 bar



Frequency dependence



**Figure B.2.** Characteristics of the servo-valve of type Bosch Rexroth 4WSE3EE 16 (Bosch Rexroth, 2000)

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## APPENDIX C

### NON-LINEAR CONTROL BACKGROUND

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In this section, fundamental results of advanced matrix calculus and mathematical tools from differential geometry and topology theory are briefly reviewed. Non-linear systems, described by the affine SISO state-space model

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})u(t) \\ y(t) &= h(\mathbf{x})\end{aligned}\tag{C.1}$$

are considered, where  $h(\mathbf{x})$  is a smooth scalar function, and  $\mathbf{f}(\mathbf{x})$ ,  $\mathbf{g}(\mathbf{x})$  are smooth vector fields.

#### C.1 Kronecker Product/Matrix Operations

##### Definition C.1. (Kronecker Product)

- The Kronecker product of two matrices  $\mathbf{A} = (a_{ij})$  and  $\mathbf{B} = (b_{kl})$ , of dimensions  $m \times n$  and  $p \times q$  respectively, is denoted  $\mathbf{A} \otimes \mathbf{B}$ , and is defined as

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \dots & a_{1n}\mathbf{B} \\ \vdots & & \vdots \\ a_{m1}\mathbf{B} & \dots & a_{mn}\mathbf{B} \end{bmatrix} \in \mathbb{R}^{mp \times nq}\tag{C.2}$$

- For the special case of two vectors  $\mathbf{a}$  and  $\mathbf{b}$ , of dimensions  $m$  and  $n$  respectively, we have

$$\mathbf{a} \otimes \mathbf{b} = \begin{bmatrix} a_1\mathbf{b}^T & a_2\mathbf{b}^T & \dots & a_m\mathbf{b}^T \end{bmatrix} \in \mathbb{R}^{mn}\tag{C.3}$$

- The Kronecker power of order  $i$ ,  $\mathbf{x}^{(i)}$ , of the vector  $\mathbf{x}$  is defined by

$$\mathbf{x}^{(i)} = \underbrace{\mathbf{x} \otimes \mathbf{x} \otimes \dots \otimes \mathbf{x}}_{i\text{-times}}\tag{C.4}$$

□

**Definition C.2. (Reduced Kronecker Product)** The reduced Kronecker product (power of order  $i$ ),  $\mathbf{x}^{[i]}$ , is defined recursively as

$$\mathbf{x}^{[i]} = \begin{pmatrix} x_1 \mathbf{b}_1^{i-2} \\ x_2 \mathbf{b}_2^{i-2} \\ \vdots \\ x_n \mathbf{b}_n^{i-2} \end{pmatrix} \quad i > 1 \quad (\text{C.5})$$

where

$$\mathbf{b}_j^0 = \begin{pmatrix} x_j \\ x_{j+1} \\ \vdots \\ x_n \end{pmatrix} \quad \begin{pmatrix} x_j \mathbf{b}_j^{i-1} \\ x_{j+1} \mathbf{b}_{j+1}^{i-1} \\ \vdots \\ x_n \mathbf{b}_n^{i-1} \end{pmatrix} \quad i > 0, \quad j = 1, 2, \dots, n \quad (\text{C.6})$$

□

### Definition C.3. (Derivatives)

- The derivative of scalar-valued function  $\lambda(\mathbf{x})$  with respect to the vector  $\mathbf{x}$  is defined as

$$\frac{\partial \lambda(\mathbf{x})}{\partial \mathbf{x}} = \left[ \frac{\partial \lambda(\mathbf{x})}{\partial x_1} \quad \frac{\partial \lambda(\mathbf{x})}{\partial x_2} \quad \dots \quad \frac{\partial \lambda(\mathbf{x})}{\partial x_n} \right] \quad (\text{C.7})$$

- The Jacobian (matrix) of a vector field  $\mathbf{f}(\mathbf{x})$  is taken to be

$$\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1 & \frac{\partial}{\partial x_2} f_1 & \dots & \frac{\partial}{\partial x_n} f_1 \\ \frac{\partial}{\partial x_1} f_2 & \frac{\partial}{\partial x_2} f_2 & \dots & \frac{\partial}{\partial x_n} f_2 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m & \frac{\partial}{\partial x_2} f_m & \dots & \frac{\partial}{\partial x_n} f_m \end{bmatrix} \quad (\text{C.8})$$

- The matrix derivative is defined by

$$\begin{aligned} \frac{\partial \mathbf{G}(\mathbf{x})}{\partial \mathbf{x}} &= \left[ \frac{\partial \mathbf{G}(\mathbf{x})}{\partial x_1} \quad \frac{\partial \mathbf{G}(\mathbf{x})}{\partial x_2} \quad \dots \quad \frac{\partial \mathbf{G}(\mathbf{x})}{\partial x_n} \right] \\ &= \sum_{i=1}^n \mathbf{e}_i^\top \otimes \frac{\partial \mathbf{G}(\mathbf{x})}{\partial x_i} \end{aligned} \quad (\text{C.9})$$

where  $\mathbf{e}_i$  is the unit vector, which is “1” in the  $i$ th component and zero elsewhere.

□

## C.2 Lie Derivatives and Lie Brackets

**Definition C.4. (Lie Derivative)** The Lie derivative of  $h$  with respect to  $f$  is the scalar function defined by

$$L_f h = \frac{\partial h}{\partial x} f = \nabla h \cdot f \quad (\text{C.10})$$

Repeated Lie derivatives are defined recursively

$$L_f^i h = L_f (L_f^{i-1} h) = \nabla (L_f^{i-1} h) \cdot f \quad (\text{C.11})$$

with  $L_f^0 h = h$ . Similarly, the scalar function  $L_g L_f h$  is

$$L_g L_f h = \nabla (L_f h) \cdot g \quad (\text{C.12})$$

□

Another important mathematical operator on vector fields is the Lie bracket.

**Definition C.5. (Lie Bracket)** The Lie bracket of  $f$  and  $g$  is the vector field

$$[f, g] = \frac{\partial g}{\partial x} f - \frac{\partial f}{\partial x} g \quad (\text{C.13})$$

The Lie bracket is commonly written as  $\text{ad}_f g$  (where “ad” stands for “adjoint”). The recursive operation is defined by

$$\text{ad}_f^i g = [f, \text{ad}_f^{i-1} g] \quad (\text{C.14})$$

with  $\text{ad}_f^i g = g$ .

□

The following lemma (Slotine and Li, 1991) on Lie bracket manipulation is useful.

**Lemma C.1. (Lie Bracket Properties)** Lie brackets have the following properties.

(a) Bilinearity:

$$\begin{aligned} [\alpha_1 f_1 + \alpha_2 f_2, g] &= \alpha_1 [f_1, g] + \alpha_2 [f_2, g] \\ [f, \alpha_1 g_1 + \alpha_2 g_2] &= \alpha_1 [f, g_1] + \alpha_2 [f, g_2] \end{aligned} \quad (\text{C.15})$$

where  $f, f_1, f_2, g_1, g_2$  are smooth vector fields and  $\alpha_1, \alpha_2$  are constants.

(b) Skew-commutativity:

$$[f, g] = -[g, f] \quad (\text{C.16})$$

(c) Jacobi-identity:

$$L_{\text{ad}_f g} h = L_f L_g h - L_g L_f h \quad (\text{C.17})$$

□

One can easily see the relevance of Lie derivatives to dynamic systems by considering Equation C.1. The derivatives of the output are

$$\begin{aligned} \dot{y} &= \frac{\partial h}{\partial \mathbf{x}} \dot{\mathbf{x}} = L_f h \\ \ddot{y} &= \frac{\partial (L_f h)}{\partial \mathbf{x}} \dot{\mathbf{x}} = L_f^2 h \\ &\vdots \\ y^{(n)} &= \frac{\partial (L_f^{n-1} h)}{\partial \mathbf{x}} \dot{\mathbf{x}} = L_f^n h \end{aligned} \quad (\text{C.18})$$

### C.3 Diffeomorphisms and State Transformations

In order to define non-linear changes of coordinates, the following concept is needed.

**Definition C.6. (Diffeomorphism)** A function  $\Phi(\mathbf{x})$  is said to be a diffeomorphism in a region  $\Omega$  if it is smooth, and if its inverse  $\Phi^{-1}(\mathbf{x})$  exists and is also smooth. □

A sufficient condition for a smooth function  $\Phi(\mathbf{x})$  to be a diffeomorphism in a neighbourhood of the origin is that the Jacobian  $\partial \Phi / \partial \mathbf{x}$  is non-singular at zero. The conditions for feedback linearisability of a non-linear system are strongly related to the following theorem.

**Theorem C.1. (Frobenius)** Let  $\{f_1, f_2, \dots, f_m\}$  be a set of linearly independent vector fields. Then the following statements are equivalent (Slotine and Li, 1991):

(i) **Complete Integrability.** There exists  $n - m$  scalar functions  $h_i$  such that

$$L_{f_j} h_i = 0 \quad 1 \leq i \leq n - m, \quad 1 \leq j \leq m \quad (\text{C.19})$$

where  $\partial h_i / \partial \mathbf{x}$  are linearly independent.

(ii) **Involutivity.** There exist scalar functions  $\alpha_{ijk} : \mathbb{R}^n \rightarrow \mathbb{R}$  such that

$$[f_i, f_j] = \sum_{k=1}^m \alpha_{ijk}(\mathbf{x}) f_k(\mathbf{x}) \quad \forall i, j \quad (\text{C.20})$$

□

A diffeomorphism can be used to transform a non-linear system into another non-linear system in terms of a new set of states, as is commonly done in the analysis of linear systems. Consider again the dynamic systems described by Equation C.1, and let a new set of states be defined by

$$\mathbf{z} = \Phi(\mathbf{x}) \quad (\text{C.21})$$

Differentiation of Equation C.21 gives

$$\dot{\mathbf{z}} = \frac{\partial \Phi}{\partial \mathbf{x}} \dot{\mathbf{x}} = \frac{\partial \Phi}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u}] \quad (\text{C.22})$$

One can easily write the new state-space representation as

$$\begin{aligned} \dot{\mathbf{z}}(t) &= \mathbf{f}^*(\mathbf{z}) + \mathbf{g}^*(\mathbf{z})\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{h}^*(\mathbf{z}) \end{aligned} \quad (\text{C.23})$$

where  $\mathbf{x} = \Phi^{-1}(\mathbf{z})$  has been used, and the functions  $\mathbf{f}^*(\mathbf{x})$ ,  $\mathbf{g}^*(\mathbf{x})$  and  $\mathbf{h}^*(\mathbf{x})$  are defined obviously.

## C.4 Approximation of Non-linear Systems

The physically based modelling of technical systems usually leads to state-space models of the form

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ \mathbf{y}(t) &= \mathbf{h}(\mathbf{x}) \quad \mathbf{x}_0 = \mathbf{x}(t_0) \end{aligned} \quad (\text{C.24})$$

The functions  $\mathbf{f}(\mathbf{x}, \mathbf{u})$  and  $\mathbf{h}(\mathbf{x})$  are assumed to be continuously differentiable. For control design purposes, local (*e.g.*, linear) approximations of these descriptions are needed. Such approximations can be derived by applying the Taylor expansion (Vetter, 1973)

$$\mathbf{f}(\mathbf{z}) = \mathbf{f}(\mathbf{z}_0) + \sum_{j=1}^l \frac{1}{j!} \frac{\partial^j \mathbf{f}(\mathbf{z})}{\partial \mathbf{z}^j} (\mathbf{z} - \mathbf{z}_0)^{(j)} + \mathbf{R}_{l+1}(\mathbf{z}, \mathbf{z}_0) \quad (\text{C.25})$$

where

$$\mathbf{R}_{l+1}(\mathbf{z}, \mathbf{z}_0) = \frac{1}{j!} \int_{\mathbf{z}_0}^{\mathbf{z}} \frac{\partial^{j+1} \mathbf{f}(\xi)}{\partial \xi^{j+1}} \left[ \mathbf{I}_n \otimes (\mathbf{z} - \xi)^{(j)} \right] d\xi \quad (\text{C.26})$$

Considering Equation C.24 for a working point  $\mathbf{P}_0 = (\mathbf{x}_0, \mathbf{u}_0)$  of the state and control variables, linear, bilinear, quadratic, and polynomial models are calculated using Equation C.25 as follows (Jelali, 1997).

**Linear Systems:**

$$\begin{aligned} \dot{\mathbf{x}} &= \left. \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} \right|_{\mathbf{P}_0} \mathbf{x} + \left. \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{P}_0} \mathbf{u} \\ &= \mathbf{A}_1 \mathbf{x} + \mathbf{B}_0 \mathbf{u} \end{aligned} \quad (\text{C.27})$$



**Bilinear Systems:**

$$\begin{aligned}\dot{x} &= \frac{\partial f(x, u)}{\partial x} \Big|_{r_0} x + \frac{\partial f(x, u)}{\partial u} \Big|_{r_0} u + \frac{\partial^2 f(x, u)}{\partial x \partial u} \Big|_{r_0} x \otimes u \\ &= A_1 x + B_0 u + B_1 x \otimes u\end{aligned}\quad (C.28)$$

**Quadratic Systems:**

$$\begin{aligned}\dot{x} &= \frac{\partial f(x, u)}{\partial x} \Big|_{r_0} x + \frac{1}{2} \frac{\partial^2 f(x, u)}{\partial x^2} \Big|_{r_0} x \otimes x + \frac{\partial f(x, u)}{\partial u} \Big|_{r_0} u + \frac{\partial^2 f(x, u)}{\partial x \partial u} \Big|_{r_0} x \otimes u \\ &= A_1 x + A_2 x \otimes x + B_0 u + B_1 x \otimes u\end{aligned}\quad (C.29)$$

**Polynomial Systems:**

$$\begin{aligned}\dot{x} &= \frac{\partial f(x, u)}{\partial x} \Big|_{r_0} x + \sum_{j=2}^r \frac{1}{j!} \frac{\partial^j f(x, u)}{\partial x^j} \Big|_{r_0} x^{(j)} \\ &\quad + \frac{\partial f(x, u)}{\partial u} \Big|_{r_0} u + \sum_{j=1}^{r-1} \frac{1}{j!} \frac{\partial^{j+1} f(x, u)}{\partial x^j \partial u} \Big|_{r_0} x^{(j)} \otimes u \\ &= A_1 x + \sum_{i=2}^r A_i x^{(i)} + B_0 u + \sum_{j=1}^{r-1} B_j x^{(j)} \otimes u\end{aligned}\quad (C.30)$$

The calculations (of the derivatives) that have to be carried out are cumbersome and error prone. Thus, they may better be done with the help of symbolic or computational algebra packages like Maple (van Essen and de Jager, 1993; Lemmen *et al.* 1995; Spielmann and Jelali, 1996).

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